

# **HANDBOOK STRUCTURAL STEEL**



**YICK HOE**

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All technical data, weights and measures shown in this handbook will be governed by standards of the respective specifications so offered. Care has been taken to ensure that all data and information herein is factual and the numerical values are accurate to the best of our knowledge. Yick Hoe Companies assumes no responsibility for errors in or misinterpretation of the information contained in this handbook or in its use.

# COMPANY PROFILE



- **Yick Hoe Ferrous Steel Sdn. Bhd.** (37855-P)
- **Yick Hoe Steel Industries Sdn. Bhd.** (556274-P)  
(Formerly known as Yick Hoe Special Steel & Engineering Sdn Bhd)
- **Sandakan Metal Sdn. Bhd.** (879779-K)
- **Lahad Datu Metal Sdn. Bhd.** (730343-A)

## Yick Hoe Ferrous Steel Sdn. Bhd.

### Head Quarters:

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Importer, Exporter & Distribution of Mild Steel, Stainless Steel materials like plates, structural steel, Long products, operating a steel coil center and manufacturing of GI Purlin products likes:

- GI COILS
- COLD ROLLED COILS
- EG COILS
- PURLIN
- PLATES
- BEAM
- PIPES
- DEFORMED BARS
- SECTIONS
- PIPE FITTINGS
- TUBES

## COIL SERVICING CENTER



**Slitting Machine**

Capability : Thickness : 0.30 ~ 3.0mm  
Width : 22 ~ 1250mm



**Shearing Machine**

Capability : Thickness : 0.30 ~ 3.0mm  
Width : 300 ~ 1600mm  
Length : 300 ~ 4000mm

## GI PURLING MANUFACTURING



| Thickness | Width | Grade           | Strength               |
|-----------|-------|-----------------|------------------------|
| 1.60mm    | 75mm  | HT GI JIS G3302 | Tensile : min 500N/MM2 |
| 2.00mm    | 100mm | SGC490M1        | Yield : Min 500N/MM2   |
| 2.5mm     | 125mm |                 |                        |
|           | 150mm |                 |                        |
|           | 200mm |                 |                        |
|           | 250mm |                 |                        |
|           | 300mm |                 |                        |

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YICK HOE

# EXPLANATORY NOTES

## MATERIAL

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The structural components referred to in this handbook are of steel to BS 4360, 'Weldable structural steels' and/or its related equivalents.

## SECTIONS

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The universal beams and columns and tees cut therefrom, the joists, channels, universal bearing piles and rolled tees are generally as listed in BS 4: Part 1: 1980. Some other joists, universal bearing piles and rolled tees in regular production are also included.

N.B. The universal beams listed in pages 48 and 49 all have 2° 52' tapered flanges.

Column core. This section, although listed separately, is rolled in the serial size 356 mm x 406 mm. It is the most appropriate one to be used as the core of a built-up stanchion if a heavier section than those rolled is required.

## DIMENSIONAL UNITS

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The dimensions of sections are given in millimetres (mm) and the calculated properties (centroidal distances, cross-sectional areas, radii of gyration, moments of inertia, elastic and plastic moduli) are given in centimetre (cm) units. Surface areas are in square metres (m<sup>2</sup>). For tolerances on dimensions and shape, reference should be made to BS 4848: Part 2.

## OTHER UNITS

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The units of force, mass and acceleration are those of the Systeme International (SI). They are the Newton (N), the kilogramme (kg) and the metre per second per second (m/s<sup>2</sup>) so that 1N = 1kg x 1m/s<sup>2</sup>. The acceleration due to gravity varies slightly from place to place and for convenience a 'standard' value of 9.80665 m/s<sup>2</sup> has become generally accepted in structural engineering. With this convention, the force exerted by a mass of 1kg under the action of gravity is the 'technical unit' of 9.80665N. In the same way 9.80665 kilonewtons (kN) is the force exerted by a mass of 1 tonne (1000 kg) under gravity, and 1kN the force from a mass of 0.102 tonne.

# EXPLANATORY NOTES

## COMPARABLE SPECIFICATIONS AND CHEMICAL ANALYSIS

### Specifications

The following specifications are normally readily available, but any offer will depend upon acceptance of full specification details and/or any other specifications not listed below.

#### Structural Sections

| Quality    | Grade     | Tensile strength   |                   |                    | Min. yield stress  |                   |                    |
|------------|-----------|--------------------|-------------------|--------------------|--------------------|-------------------|--------------------|
|            |           | lb/in <sup>2</sup> | N/mm <sup>2</sup> | kg/mm <sup>2</sup> | lb/in <sup>2</sup> | N/mm <sup>2</sup> | kg/mm <sup>2</sup> |
| BS4360     | 43A       | 62300/84000        | 430/580           | 44/59              | 38500              | 265               | 27                 |
|            | 50A       | 72500/93000        | 490/640           | 50/65              | 50000              | 345               | 36                 |
| ASTM       | A36       | 58000/80000        | 400/550           | 41/56              | 36000              | 250               | 25                 |
| ASTM A572  | 50        | 65000 min          | 450 min           | 46 min             | 50000              | 345               | 36                 |
| Din 17100  | RSt. 37.2 | 52000/74000        | 360/510           | 36/52              | 32500              | 225               | 23                 |
|            | St 44.2   | 62300/83500        | 430/580           | 44/59              | 38500              | 265               | 27                 |
|            | St 52.3   | 74000/98500        | 510/680           | 52/69              | 50000              | 345               | 36                 |
| JIS G 3101 | SS 41     | 58000/74000        | 402/510           | 41/52              | 34000              | 235               | 24                 |
|            | SS 50     | 72500/88500        | 490/608           | 50/62              | 40000              | 275               | 28                 |
| JIS G 3106 | SM 41     | 58000/74000        | 402/510           | 41/52              | 34000              | 235               | 24                 |
|            | SM 50     | 72500/88500        | 490/608           | 50/62              | 45000              | 314               | 32                 |

Notes: Sections over 5/8 inch (16mm) up to and including 1 inch (25mm) thick  
All figures are approximate and for guidance only  
Other specifications are obtainable subject to acceptance of specification (eg. UNI 7070-72 and NFA 35-501)

### Chemical analyses

| Specification    | Grade   | Composition (Maxima unless stated) |      |             |      |      |            |            |    |               |
|------------------|---------|------------------------------------|------|-------------|------|------|------------|------------|----|---------------|
| Quality          |         | C                                  | Si   | Mn          | P    | S    | Nb         | V          | N1 | Others        |
| BS4360 (1986)    | 43A     | 0.25                               | 0.50 | 1.60        | 0.05 | 0.05 | --         | --         | -- | --            |
|                  | 50B     | 0.20                               | 0.50 | 1.50        | 0.05 | 0.05 | 0.003/0.10 | 0.003/0.10 | -- | --            |
| ASTM A36 (1984)  |         | 0.26                               | --   | --          | 0.04 | 0.05 | --         | --         | -- | Cu 0.20 (min) |
| ASTM A572 (1984) | 50      | 0.23                               | 0.40 | 1.35        | 0.04 | 0.05 | --         | --         | -- | Cu 0.20 (min) |
| JIS 3101 (1976)  | SS 41   | --                                 | --   | --          | 0.05 | 0.05 | --         | --         | -- | --            |
|                  | SS 50   | --                                 | --   | --          | 0.05 | 0.05 | --         | --         | -- | --            |
| JIS 3106 (1976)  | SM 41 A | 0.25                               | --   | 2.5 x C min | 0.04 | 0.04 | --         | --         | -- | --            |
|                  | SM 41 B | 0.22                               | 0.35 | 0.6 - 1.20  | 0.04 | 0.04 | --         | --         | -- | --            |
|                  | SM 41 C | 0.18                               | 0.35 | 1.40        | 0.04 | 0.04 | --         | --         | -- | --            |
|                  | SM 50 A | 0.22                               | 0.55 | 1.50        | 0.04 | 0.04 | --         | --         | -- | --            |
|                  | SM 50 B | 0.20                               | 0.55 | 1.50        | 0.04 | 0.04 | --         | --         | -- | --            |
|                  | SM 50 C | 0.18                               | 0.55 | 1.50        | 0.04 | 0.04 | --         | --         | -- | --            |



# EXPLANATORY NOTES

## AVAILABLE SPECIFICATIONS

| Application                          |                                                 | Standard                                                                                                        |                                                                                                                 |                                                   |                                                                                                                            |
|--------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|                                      |                                                 | KS                                                                                                              | JIS                                                                                                             | ASTM                                              | Others                                                                                                                     |
| Steel for General Structure          |                                                 | KS D3503<br>SS 34, 41, 50, 55                                                                                   | JIS G3101<br>SS 34, 41, 50, 55                                                                                  | A36, A113, A242<br>A233, A284, A441<br>A529, A573 | BS 1449 Part 1<br>50/35 HR, HS<br>DIN 17100<br>ST 33-1,2<br>ST 37-2,3<br>ST 52-3<br>RST 37-2<br>UST 37-2<br>IS 226 ST 42-S |
| Steel for Welded Structure           |                                                 | KS D3515<br>SWS 41A, B, C<br>SWS 50A, B, C<br>SWS 50YA, YB<br>SWS 538, C                                        | JIS G3106<br>SM 41A, B, C<br>SM 50A, B, C<br>SM 50YA, YB<br>SM 53B, C                                           | A36, A242, A283<br>A284, A440, A529<br>A573       | BS 4360<br>43A, B<br>IS 2062<br>ST 42-W<br>AS 250                                                                          |
| Hot Rolled Mild Steel Sheets & Strip |                                                 | KS D3501<br>SHP 1, 2, 3                                                                                         | JIS G3131<br>SPHC, SPHD<br>SPHE                                                                                 | A569<br>A570<br>A621                              | BS 1449<br>HR 15<br>HR 4<br>HR 3                                                                                           |
| Pipes & Tubes                        | Hot Rolled Carbon Steel Strip for Pipes & Tubes | KS D3555<br>HRS 1, 2, 3, 4<br>KS D3566<br>SPS 30, 41, 50, 51                                                    | JIS G3132<br>SPHT 1, 2, 3, 4<br>JIS G3444<br>STK 30, 41, 50, 51                                                 | A53<br>A252-2,3                                   |                                                                                                                            |
|                                      | Boiler & Heat Exchanger Tubes                   | KS D3563<br>STBH 33, 35, 42<br>KS D3571<br>STLT 39<br>KS D3572<br>STHA 12, 13, 20, 22                           | JIS G3461<br>STB 33, 35, 42<br>JIS G3462<br>STBA12, 13, 20, 22<br>JIS G3464<br>STBL 39                          |                                                   |                                                                                                                            |
|                                      | Carbon Steel Pipes for High Temperature Service | KS D3570<br>SPHT 38, 42                                                                                         | JIS G3456<br>STPT 38, 42                                                                                        |                                                   |                                                                                                                            |
|                                      | Steel Pipes for Low Temperature Service         | KS D3571<br>SPLT 39                                                                                             | JIS G3460<br>STPL 39                                                                                            |                                                   |                                                                                                                            |
|                                      | Carbon Steel Tubes for Machine Structure        | KS D3517<br>STKM 11 A<br>STKM 12A, B, C<br>STKM 13A, B, C<br>STKM 14A, B, C<br>STKM 15A, B, C<br>STKM 18A, B, C | JIS G3445<br>STKM 11A<br>STKM 12A, B, C<br>STKM 13A, B, C<br>STKM 14A, B, C<br>STKM 15A, B, C<br>STKM 18A, B, C |                                                   |                                                                                                                            |

# EXPLANATORY NOTES

## Specifications

| Application                                         |                                        | Standard                        |                                  |                                                   |                                                                                        |
|-----------------------------------------------------|----------------------------------------|---------------------------------|----------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------|
|                                                     |                                        | KS                              | JIS                              | ASTM                                              | Others                                                                                 |
| Pipes & Tubes                                       | Pipes for Oil Well Tubing & Line Pipes |                                 |                                  |                                                   | API 5L(S)-A,B<br>API 5LX(S)-X42, X46, X56, X60, X65, X70<br>API 5A- H40, J55, K55, N80 |
| Ordinary Strength Hull Structural Steel             |                                        |                                 |                                  |                                                   | KR - RA, RB, RD<br>AB - A<br>LR - A,B<br>NK- KA<br>NV -A                               |
| Superior Atmospheric Corrosion Resisting Steel      |                                        |                                 | JIS G3125<br>SPA-H               | A441<br>A588                                      |                                                                                        |
| Hot Rolled Sheets & Strips for Automobile Structure |                                        | KS D3519<br>SAPH 32, 38, 41, 45 | JIS G3113<br>SAPH 32, 38, 41, 45 |                                                   | SAE 950B                                                                               |
| Spring Steel                                        |                                        | KS D3701<br>SPS 7               | JIS G4801<br>SUP 9               |                                                   |                                                                                        |
| Chromium Molybdenum Steels                          |                                        | KS D3711<br>SCM 430             | JIS G4106<br>SCM 430             |                                                   |                                                                                        |
| Manganese Steels                                    |                                        |                                 | JIS G4106<br>SMn 420             |                                                   |                                                                                        |
| Carbon Steel for Machine Structure                  |                                        | KS D3752<br>SM10C ~ SM58C       | JIS G4051<br>S10C ~ S58C         |                                                   | SAE 1006 ~ SAE 1050<br>AISI 1006 ~ AISI 1050                                           |
| Low Alloy High Strength Steels for Structure        |                                        |                                 |                                  | A607 - 45, 50, 55<br>A715 - 50TY7<br>A572<br>A242 |                                                                                        |
| Steel for Boilers & Other Pressure Vessels          |                                        | KS D3560<br>SBB 42, 46, 49      | JIS G3103<br>SB 42, 46, 49       | A285<br>A442<br>A455<br>A516                      |                                                                                        |
| Steel Sheets & Strip for Gas Cylinders              |                                        | KS D3533<br>SG 26, 30, 33, 37   | JIS G3116<br>SG 26, 30, 33, 37   | A285-C<br>A455                                    |                                                                                        |

# EXPLANATORY NOTES

## COMPARISON OF EN10025 & BS4360

| EN 10025: 1993 |                |                  |                                              |                                               |                                           |           |                  | BS 4360: 1986 |                                         |                                              |                                               |                                           |           |
|----------------|----------------|------------------|----------------------------------------------|-----------------------------------------------|-------------------------------------------|-----------|------------------|---------------|-----------------------------------------|----------------------------------------------|-----------------------------------------------|-------------------------------------------|-----------|
|                |                |                  |                                              |                                               | Impact Energy (J °C)<br>Nominal Thickness |           |                  |               |                                         |                                              |                                               | Impact Energy (J °C)<br>Nominal Thickness |           |
| Grade          | Former Grade   | Tensile Strength | Min Yield Strength at 16mm N/mm <sup>2</sup> | Max Thk for Specified Yield N/mm <sup>2</sup> | Temp °C                                   | <150mm -1 | >150mm <250mm -1 | Grade         | Tensile Strength >3mm N/mm <sup>2</sup> | Min Yield Strength at 16mm N/mm <sup>2</sup> | Max thk Specified Yield N/mm <sup>2</sup> (2) | Temp °C                                   | <100mm -3 |
| S185 (4)       | Fe 310-0(4)    | 290/510          | 185                                          | 25                                            | -                                         | -         | -                | -             | -                                       | -                                            | -                                             | -                                         | -         |
| S235 (5)       | Fe 360A (5)    | 340/470          | 235                                          | 250                                           | -                                         | -         | -                | 40A           | 340/500                                 | 235                                          | 150                                           | -                                         | -         |
| S235JR (4)     | Fe 360B (4)    | 340/470          | 235                                          | 25                                            | 20                                        | 27        | -                | -             | -                                       | -                                            | -                                             | -                                         | -         |
| (4)            | Fe 360(FU) (4) | 340/470          | 235                                          | 25                                            | +20 (6)                                   | 27        | -                | -             | -                                       | -                                            | -                                             | -                                         | -         |
| S235JRG2       | Fe 360B (FN)   | 340/470          | 235                                          | 250                                           | +20 (6)                                   | 27        | 23               | 40B           | 340/500                                 | 235                                          | 150                                           | +20 (6)                                   | 27        |
| S235JO         | Fe 360C        | 340/470          | 235                                          | 250                                           | 0                                         | 27        | 23               | 40C           | 340/500                                 | 235                                          | 150                                           | 0                                         | 27        |
| S235J2G3       | Fe 360D1       | 340/470          | 235                                          | 250                                           | -20                                       | 27        | 23               | 40D           | 340/500                                 | 235                                          | 150                                           | -20                                       | 27        |
| S235J2G4       | Fe 360D2       | 340/470          | 235                                          | 250                                           | -20                                       | 27        | 23               | 40D           | 340/500                                 | 235                                          | 150                                           | -20                                       | 27        |
| S275 (5)       | Fe 430A (5)    | 410/560          | 275                                          | 250                                           | -                                         | -         | -                | 43A           | 430/580                                 | 275                                          | 150                                           | -                                         | -         |
| S275JR         | Fe 430B        | 410/560          | 275                                          | 250                                           | +20 (6)                                   | 27        | 23               | 43B           | 430/580                                 | 275                                          | 150                                           | +20 (6)                                   | 27        |
| S275JO         | Fe 430C        | 410/560          | 275                                          | 250                                           | 0                                         | 27        | 23               | 43C           | 430/580                                 | 275                                          | 150                                           | 0                                         | 27        |
| S275J2G3       | Fe 430D1       | 410/560          | 275                                          | 250                                           | -20                                       | 27        | 23               | 43D           | 430/580                                 | 275                                          | 150                                           | -20                                       | 27        |
| S275J2G4       | Fe 430D2       | 410/560          | 275                                          | 250                                           | -20                                       | 27        | 23               | 43D           | 430/580                                 | 275                                          | 150                                           | -20                                       | 27        |
| S355 (5)       | Fe 510A        | 490/630          | 355                                          | 250                                           | -                                         | -         | -                | 50A           | 490/640                                 | 355                                          | 150                                           | -                                         | -         |
| S355JR         | Fe 510B        | 490/630          | 355                                          | 250                                           | +20 (6)                                   | 27        | 23               | 50B           | 490/640                                 | 355                                          | 150                                           | +20 (6)                                   | 27        |
| S355JO         | Fe 510C        | 490/630          | 355                                          | 250                                           | 0                                         | 27        | 23               | 50C           | 490/640                                 | 355                                          | 150                                           | 0                                         | 27        |
| S355J2G3       | Fe 510D1       | 490/630          | 355                                          | 250                                           | -20                                       | 27        | 23               | 50D           | 490/640                                 | 355                                          | 150                                           | -20                                       | 27        |
| S355J2G4       | Fe 510D2       | 490/630          | 355                                          | 250                                           | -20                                       | 27        | 23               | 50D           | 490/640                                 | 355                                          | 150                                           | -20                                       | 27        |
| S355K2G3       | Fe 510DD1      | 490/630          | 355                                          | 250                                           | -20                                       | 40        | 33               | 50DD          | 490/640                                 | 355                                          | 150                                           | -30                                       | 27        |
| S355K2G4       | Fe 510DD2      | 490/630          | 355                                          | 250                                           | -20                                       | 40        | 33               | 50DD          | 490/640                                 | 355                                          | 150                                           | -30                                       | 27        |
| E295           | Fe 490-2       | 470/610          | 295                                          | 250                                           | -                                         | -         | -                | -             | -                                       | -                                            | -                                             | -                                         | -         |
| E335           | Fe 590-2       | 570/710          | 335                                          | 250                                           | -                                         | -         | -                | -             | -                                       | -                                            | -                                             | -                                         | -         |
| E360           | Fe 690-2       | 670/830          | 350                                          | 250                                           | -                                         | -         | -                | -             | -                                       | -                                            | -                                             | -                                         | -         |

- (1) For sections up to and including 100mm only
- (2) For wide flats and Sections up to and including 63mm and 100mm respectively
- (3) For wide flats up to and including 50mm and sections no limit is stated
- (4) Only available up to and including 25mm thick
- (5) The Steel grades S235 (Fe 360A), S275 (Fe 430A), and S355 (Fe 510A) appear only in the English language version (BS EN 10025) as non conflicting additions and do not appear in European versions
- (6) Verification of the specified impact value is only carried out when agreed at time of enquiry and order

# EXPLANATORY NOTES

| Designation:                  |                 | Equivalent Former Designations in |                         |                        |                  |                |             |              |             |              |               |                   |              |               |                         |
|-------------------------------|-----------------|-----------------------------------|-------------------------|------------------------|------------------|----------------|-------------|--------------|-------------|--------------|---------------|-------------------|--------------|---------------|-------------------------|
| EN 10025: 1993<br>Designation |                 | According<br>EN 10025:<br>1990    | Germany<br>Din<br>17100 | France<br>NFA<br>35501 | UK<br>BS<br>4360 | Spain          | Italy       | Belgium      | Sweden      | Portugal     | Austria       | Norway<br>NS12101 | USA<br>ASTM  | USA<br>API    | Japan<br>JIS<br>G3101/6 |
| Steel<br>Name                 | Steel<br>Number |                                   |                         |                        |                  |                |             |              |             |              |               |                   |              |               |                         |
| S185                          | 1.0035          | Fe 310-0                          | St 33                   | A33                    |                  | A3 10-0        | Fe 320      | A 320        | 13<br>00-00 | Fe<br>310-00 | St 320        |                   |              |               |                         |
| S235JR                        | 1.0037          | Fe 360B                           | St 37-2                 | E24-2                  |                  |                | Fe<br>360 B | AE<br>235-B  | 13<br>11-00 | Fe 360-B     |               | NS<br>12 120      | A283<br>GrC  |               |                         |
| S235JRG1                      | 1.0036          | Fe 360BFU                         | U St<br>37-2            |                        |                  | AE 235<br>B-FU |             |              |             |              | B             | NS<br>12 122      | A283<br>GrD  |               | SM41A                   |
| S235JRG2                      | 1.0038          | Fe 360BFN                         | R St<br>37-2            |                        | 40B              | AE 235<br>B-FN |             |              | 13<br>12-00 |              | R St<br>360 B | NS<br>12 123      |              |               | SM41B                   |
| S235JO                        | 1.0014          | Fe 360c                           | St<br>37-3U             | E 24-3                 | 40C              | AE 235<br>D    | Fe<br>360 C | Ae<br>235-C  |             | Fe<br>360-C  | St<br>360 C   | St<br>430 B       |              |               |                         |
|                               |                 |                                   |                         |                        |                  |                |             |              |             |              | St<br>360 CE  | NS<br>12 142      |              |               |                         |
| S235J2G3                      | 1.0116          | S Fe 360<br>D1                    | St<br>37-3N             | E 24-4                 | 40D              |                | Fe<br>360 D | AE<br>236-D  |             | Fe<br>360-D  | St<br>360 D   | NS<br>12 124      |              |               |                         |
| S234J2G4                      | 1.0117          | Fe 360 D2                         |                         |                        |                  |                |             |              |             |              |               |                   |              |               |                         |
| S275JR                        | 1.0044          | Fe 430B                           | ST 44-2                 | E 28-2                 | 43B              | AE 275<br>B    | Fe<br>430 B | AE<br>255-B  | 14<br>12-00 | Fe<br>430-B  | St<br>430 B   | NS<br>12-142      | Gr65         |               | SS400                   |
| S275JO                        | 1.0143          | Fe 430C                           | ST<br>44-3U             | E 28-3                 | 43C              | AE 275<br>C    | Fe<br>430 C | AE<br>255-C  |             | Fe<br>430-C  | St<br>430 C   | NS<br>12-143      |              |               |                         |
|                               |                 |                                   |                         |                        |                  |                |             |              |             |              | ST<br>430 CE  |                   |              |               |                         |
| S275J2G3                      | 1.0144          | Fe 430D1                          | ST<br>430 D1            | E 28-4                 | 43D              | AE 275<br>D    | Fe<br>430 D | AE<br>255-D  | 14<br>14-00 | Fe<br>430-D  | ST<br>430 D   | NS<br>12-143      | Gr42         | 2H<br>GR42    |                         |
| S275J2G4                      | 1.0145          | Fe 430D2                          |                         |                        |                  |                |             |              | 14<br>14-01 |              |               |                   |              |               |                         |
| S355JR                        | 1.0045          | Fe 510 B                          |                         | E 36-2                 | 50B              | AE 355<br>B    | Fe<br>310 B | AE<br>355B   |             | Fe<br>510-B  |               |                   | A572<br>Gr50 |               | SS490                   |
| S355JO                        | 1.0553          | Fe 510 C                          | ST<br>52-3U             | E 36-3                 | 50C              | AE 355<br>C    | Fe<br>310 C | AE<br>355C   |             | Fe<br>510 C  | St<br>510 C   | Ns<br>12 153      | A790<br>Gr50 | 2H<br>Gr50    |                         |
| S355J2G3                      | 1.057           | Fe 510 D1                         | ST<br>52-3N             |                        | 50D              | AE 355<br>D    | Fe<br>310 D | AE<br>355D   |             | Fe<br>510- D | St<br>510-D   | NS<br>12 153      | Gr50         | 2 W/<br>YGr50 |                         |
| S355J2G4                      | 1.0577          | Fe 510 D2                         |                         |                        |                  |                |             |              |             |              |               |                   |              |               |                         |
| S355K2G3                      | 1.0595          | Fe 510 DD1                        |                         | E36- 4                 | 50DD             |                |             | AE<br>355 DD |             | Fe<br>510DD  |               |                   | A533<br>GrD  |               |                         |
| S355K2G4                      | 1.0596          | Fe 510 DD2                        |                         |                        | 50DD             |                |             |              |             |              |               |                   |              |               |                         |
| E295                          | 1.005           | Fe 490-2                          | ST 50-2                 | A 50-2                 |                  | A 490          | Fe<br>480   | A 490-2      | 15<br>50-00 | FE<br>490-2  | St 490        |                   |              |               |                         |
|                               |                 |                                   |                         |                        |                  |                |             |              | 15<br>50-01 |              |               |                   |              |               |                         |
| E355                          | Fe 590-2        | ST 590-2                          | St 60-2                 | A 60-2                 |                  | A 590          | Fe 580      | A590-2       | 16<br>50-00 | Fe<br>590-2  | St 590        |                   |              |               |                         |
|                               |                 |                                   |                         |                        |                  |                |             |              | 16<br>50-01 |              |               |                   |              |               |                         |
| E 360                         | 1.007           | Fe 690-2                          | St 70-2                 | A 70-2                 |                  | A 690          | Fe 650      | A690-2       | 16<br>55-00 | Fe<br>690-2  | St 590        |                   |              |               |                         |
|                               |                 |                                   |                         |                        |                  |                |             |              | 16<br>55-01 |              |               |                   |              |               |                         |

# EXPLANATORY NOTES

## HIGH STRENGTH STEEL

OFFSHORE PLATFORM, F.P.S.O., BRIDGE GIRDERS, STRUCTURAL TUBULARS & FABRICATIONS. LOW TEMP CHARPY TESTED, + M = T.M.C.P QUALITY

| National Standard                 |                  | Grade            | Min Yield<br>N/mm² | UTS<br>N/mm² | Charpy<br>°C | Impacts<br>(L)<br>Joules | Former National Specifications |               |                |     |                 |  |
|-----------------------------------|------------------|------------------|--------------------|--------------|--------------|--------------------------|--------------------------------|---------------|----------------|-----|-----------------|--|
|                                   |                  |                  |                    |              |              |                          | EU                             |               | GB             |     | DE              |  |
| Year-ASTM + Shipbuilding = Latest |                  |                  |                    |              |              |                          | EN 10025: 1993                 |               | BS 43601: 1986 |     | DIN 17100: 1983 |  |
| JP                                | JIS G3106 :2004  | SM490B           | 325                | 490/610      | 0            | 27                       |                                |               |                |     |                 |  |
| ANS                               | AS/NZS 3678:1996 | 350              | 340                | 480 min      | N/R          | N/R                      |                                |               |                |     |                 |  |
|                                   |                  | 350 L0           | 340                | 480 min      | 0            | 27                       |                                |               |                |     |                 |  |
| USASTM                            | A572M Gr 50      | 345              | 450 min            | N/R          | N/R          |                          |                                |               |                |     |                 |  |
|                                   |                  | A633M Gr C       | 345                | 450 min      | 0            | 26                       | Equivalent to                  | Min Yield     | UTS            | °C  | J               |  |
|                                   |                  | A588 Gr B        | 345                | 485 min      | N/R          | N/R                      |                                |               |                |     |                 |  |
|                                   |                  |                  |                    |              |              |                          | COR-TEN A                      | 355           | 470-630        | -20 | 27              |  |
|                                   |                  |                  |                    |              |              |                          | COR-TEN B                      | 355           | 470-630        | -20 | 27              |  |
| CN                                | GB/T 1591:1994   | Q345 A           | 345                | 470/630      | N/R          | N/R                      |                                |               |                |     |                 |  |
|                                   |                  | Q345 B           | 345                | 470/630      | +20          | 27                       |                                |               |                |     |                 |  |
|                                   |                  | Q345 C           | 345                | 470/630      | 0            | 27                       |                                |               |                |     |                 |  |
|                                   |                  | Q345 D           | 345                | 470/630      | -20          | 27                       |                                |               |                |     |                 |  |
| IN                                | IS 2062:2006     | E350 A           | 350                | 490 min      | N/R          | N/R                      | IS 8500: 1991                  |               |                |     |                 |  |
|                                   |                  | E350 BR          | 350                | 490 min      | +20          | 27                       |                                |               |                |     |                 |  |
|                                   |                  | E350 B0          | 350                | 490 min      | 0            | 27                       |                                |               |                |     |                 |  |
|                                   |                  | E350 C           | 350                | 490 min      | -20          | 27                       |                                |               |                |     |                 |  |
| EUEN 10025-2:2004                 |                  | S355 JR          | 355                | 470/630      | +20          | 27                       | S355 JR                        | 50B           | St.52-3        |     |                 |  |
|                                   |                  | S355 JO          | 355                | 470/630      | 0            | 27                       | S355 JO                        | 50C           | St.52-3        |     |                 |  |
|                                   |                  | S355 J2+N        | 355                | 470/630      | -20          | 27                       | S355 J2G3                      | 50D           | St.52-3 N      |     |                 |  |
|                                   |                  | S355 J2          | 355                | 470/630      | -20          | 27                       | S355 J2G4                      | 50EE          |                |     |                 |  |
|                                   |                  | S355 K2/<br>K2+N | 355                | 470/630      | -20          | 40                       | S355 K2G3                      | 50DD          |                |     |                 |  |
| EUEN 10025-3:2004                 |                  | S355N            | 355                | 470/630      | -20          | 40                       | EN 10113:<br>1993 S355 N       | 50            |                |     |                 |  |
|                                   |                  | S355NL           | 355                | 470/630      | -50          | 27                       | EN 10113:<br>1993 S355 NL      | 50EE          |                |     |                 |  |
| EUEN 10025-4:2004                 |                  | S355M            | 355                | 520/680      | -20          | 40                       | EN 10113:<br>1993 S355 M       |               |                |     |                 |  |
|                                   |                  | S355ML           | 355                | 520/680      | -50          | 27                       | EN 10113:<br>1993 S355 ML      |               |                |     |                 |  |
| EUEN 10225:2004                   |                  | S355 G2/G3+N     | 355                | 470/630      | -20/-40      | 50                       | API: 1993                      | BS 7191: 1989 |                |     |                 |  |
|                                   |                  | S355 G5/G6+M     | 355                | 470/630      | -20/-40      | 50                       |                                | 355D/E        |                |     |                 |  |
|                                   |                  | S355 G7+M/N      | 355                | 470/630      | -40          | 50                       |                                | 355D/E        |                |     |                 |  |
|                                   |                  | S355 G8+M/N      | 355                | 470/630      | -40          | 50                       | API - 2H Gr50                  | 355EM         |                |     |                 |  |
|                                   |                  | S355 G9+M/N      | 355                | 470/630      | -40          | 50                       | API - 2W Gr50                  | 355MZ         |                |     |                 |  |
|                                   |                  | S355 G10+M/N     | 355                | 470/630      | -40          | 50                       | API - 2W Gr50T                 |               |                |     |                 |  |
| JP                                | JIS G3106:2004   | SM490YB          | 365                | 490/610      | 0            | 27                       |                                |               |                |     |                 |  |
|                                   |                  | SM520B           | 365                | 520/640      | 0            | 27                       |                                |               |                |     |                 |  |
| JP                                | JIS G3101:2004   | SS540            | 390                | 540 min      | N/R          | N/R                      |                                |               |                |     |                 |  |

## ROLLED TUBULARS

Availability: 20" O/dia + over

API-2B/ 2H: 1 or more "Cans" can be fabricated from above S355 plates to suit req'd OD x WT x Length

## PRODUCTS STOCK SIZES

|           |                  |                                                          |                                              |
|-----------|------------------|----------------------------------------------------------|----------------------------------------------|
| AVAILABLE |                  | S355JR - S355G10+N/M                                     | ASTM A572-50-A588 GrB                        |
| Plates    |                  | Wide: 1.25m - 2.0m, 2.5m, 3.05m & 3.5m                   | Wide: 48" - 120" (1219 - 3048mm)             |
|           |                  | Long: 3.0m - 6.0m, 8.0m, 10.0m, 12.0m & 15.0m            | Long: 3' - 40' > 60' (915mm - 12.2m > 18.3m) |
|           |                  | THK: 5mm - 350mm                                         | THK: 3/16" - 12" (4.75mm - 305mm)            |
|           |                  | S355G7 - G10+M = 100mm MAX<br>S355G7 - G10+N = 150mm MAX |                                              |
| Coils     | S355JR<br>S355J2 | Wide: 1.0m, 1.25m, 1.5m - 2.0m<br>THK: 1mm - 12mm        |                                              |

# PRODUCT LIST

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# I-BEAMS

## DESIGN OF BEAMS

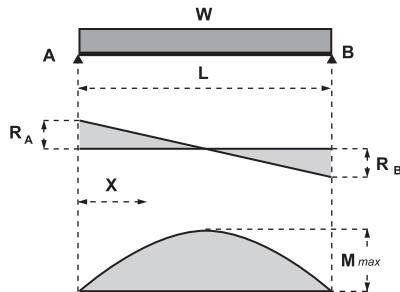
### Design formulae for beams

|                      | Notation                                                                                                                                                            |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $L$                  | = length of span in millimetres                                                                                                                                     |
| $W$                  | = total distributed or point load in Newtons                                                                                                                        |
| $W_1$ or $W_2$       | = point load in Newtons                                                                                                                                             |
| $\Sigma$             | = resultant of point loads in Newtons                                                                                                                               |
| $R_A, R_B, R_C$ etc. | = reaction at A, B, or C etc, in Newtons                                                                                                                            |
| $F$                  | = shearing force in Newtons                                                                                                                                         |
| $m$                  | = applied moment in Newton millimetres                                                                                                                              |
| $M_x$                | = bending moment in Newton millimetres at distance X from the left hand support A                                                                                   |
| $\delta_x$           | = deflection in millimetres                                                                                                                                         |
| $i_x$                | = slope in radians                                                                                                                                                  |
| $M_A$ or $M_B$       | = end fixing moment in Newton millimetres                                                                                                                           |
| $M_{max}$            | = maximum bending moment in Newton millimetres                                                                                                                      |
| $M_{max\ max}$       | = absolute maximum bending moment in Newton millimetres                                                                                                             |
| $M_{load}$           | = bending moment under the load in Newton millimetres                                                                                                               |
| $\delta_{max}$       | = maximum deflection in millimetres                                                                                                                                 |
| $\delta_{max\ max}$  | = absolute maximum deflection in millimetres                                                                                                                        |
| $\delta_{negative}$  | = negative (i.e. upward) deflection in millimetres                                                                                                                  |
| $i_A$ or $i_B$       | = slope at A or at B in radians                                                                                                                                     |
| $E$                  | = modulus of elasticity, $2.1 \times 10^5 \text{ N/mm}^2$                                                                                                           |
| $I$                  | = constant moment of inertia of uniform section beam in $\text{mm}^4$                                                                                               |
| Sign convention      |                                                                                                                                                                     |
| Loads                | + positive when acting downward                                                                                                                                     |
| Support Reaction     | + positive when acting upward                                                                                                                                       |
| Shearing Force       | + positive on a section where the upward left hand support reaction is greater than the algebraic sum of external loads located left of that section                |
| Bending Moment       | + positive (shown above base line on diagrams) when causing convexity downward                                                                                      |
| Deflection           | + positive when downward                                                                                                                                            |
| Slope                | appropriate values in radians are given, but the signs depend upon which support or which section is being considered, and can be readily ascertained by inspection |

Where space permits, general equations for  $M_x$  and  $i_x$  at any point of the beam, and also the equation to the elastic line ( $\delta_x$ ), have been included.

Values for Slope. These may be used in evaluating the angle of rotation for rubber bearings and similar constructional elements.

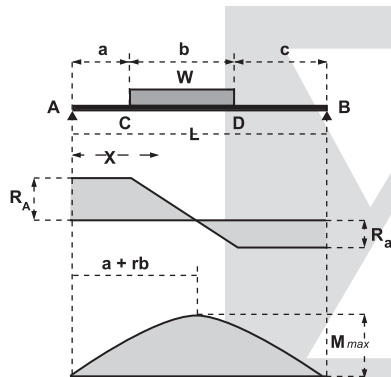
## Simply supported beam



## Uniform load on full span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total uniform load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_{\max} &= \frac{WL^2}{8} \\ \delta_{\max} &= \frac{5}{384} \cdot \frac{WL^3}{EI} \end{aligned} \right. \\
 i_A = i_B &= \frac{WL^2}{24EI} \\
 \text{at } X \text{ from A} &\left\{ \begin{aligned} M_x &= \frac{WX}{2L}(L-X) \\ \delta_x &= \frac{WX}{24EI}(X^3-2X^2L+L^3) \\ i_x &= \frac{W}{24EI}(4X^3-6X^2L+L^3) \end{aligned} \right.
 \end{aligned}$$

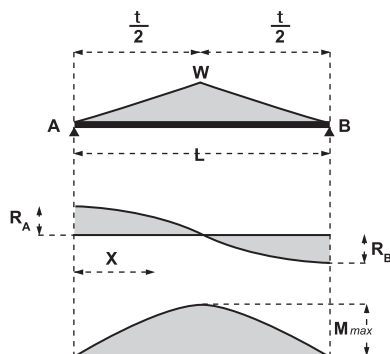
## Simply supported beam



## Uniform load on part of span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total uniform load} &= W \\
 \text{Let } r &= \frac{0.5b+c}{L} \\
 R_A = Wr; \quad R_B &= W(l-r) \\
 \text{at } X = a + rb, M_{\max} &= Wr(a+0.5rb) \\
 i_A = \frac{Wr(L^2-c^2-Lbr)}{6EI}; \quad i_B &= \frac{W(l-r)}{6EI}[L^2-a^2-Lb(l-r)] \\
 \text{Equation to elastic line between C and D, i.e. } a \leq X \leq a+b & \\
 \delta_x &= \frac{W}{24EIb}[X^4-4(a+rb)X^3+6a^2X^2+4\{rb(L^2-c^2-cb-\frac{b^2}{2})-a^2\}X+a^4]
 \end{aligned}$$

## Simply supported beam



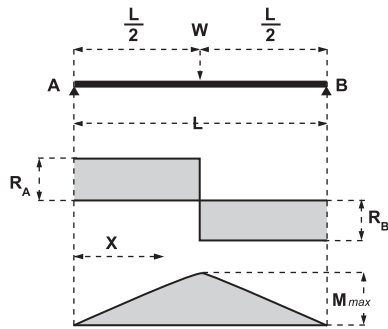
## Triangular load on full span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_{\max} &= \frac{WL^2}{6} \\ \delta_{\max} &= \frac{WL^3}{60EI} \end{aligned} \right. \\
 i_A = i_B &= \frac{5WL^2}{96EI} \\
 \text{at } X \text{ from A} &\left\{ \begin{aligned} M_x &= \frac{WX}{6L^2}(3L^2-4X^2) \\ \delta_x &= \frac{WX}{480EI}(16X^4-40X^2L^2+25L^4) \\ i_x &= \frac{W}{96EI}(16X^4-24X^2L^2+5L^4) \end{aligned} \right.
 \end{aligned}$$



# I-BEAMS

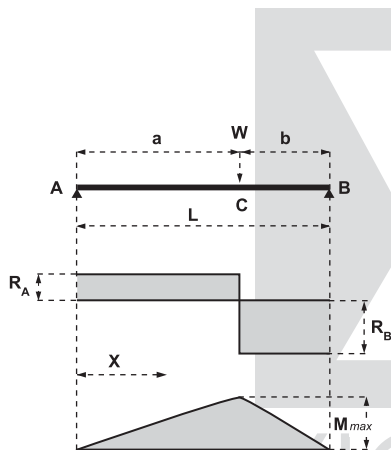
## Simply supported beam



## Point load at mid-span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_{\max} &= \frac{WL}{4} \\ \delta_{\max} &= \frac{1}{48} \cdot \frac{WL^3}{EI} \\ &= \frac{WL^2}{16EI} \end{aligned} \right. \\
 i_A = i_B & \\
 \text{at } X \text{ from A} &\left\{ \begin{aligned} M_x &= \frac{WX}{2} \\ \delta_x &= \frac{WX}{48EI} (3L^2 - 4X^2) \\ i_x &= \frac{W}{16EI} (L^2 - 4X^2) \end{aligned} \right.
 \end{aligned}$$

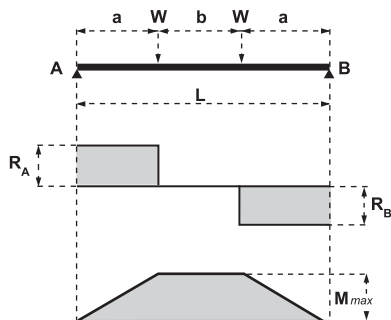
## Simply supported beam



## Point load at any position

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A = \frac{Wb}{L} & \quad R_B = \frac{Wa}{L} \\
 \text{at C under load} &\left\{ \begin{aligned} M_{\max} &= \frac{Wab}{L} \\ \delta_c &= \frac{Wa^2b^2}{3EIL} \\ i_A &= \frac{Wab}{6EIL} (L+b); \quad i_B = \frac{Wab}{6EIL} (L+a) \end{aligned} \right. \\
 \text{When } a > b &\left\{ \begin{aligned} \delta_{\max} &= \frac{Wab(L+b)}{27EIL} \sqrt{3a(L+b)} \\ X &= \sqrt{\frac{a(L+b)}{3}} \end{aligned} \right.
 \end{aligned}$$

## Simply supported beam

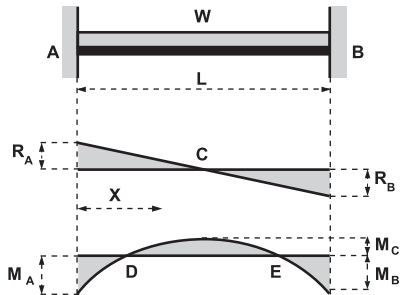


## Two equal symmetrical point loads

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Two point loads, each} &= W \\
 R_A = R_B &= W \\
 M_{\max} \text{ over length } b &= Wa \\
 \delta_{\max} \text{ at mid-span} &= \frac{Wa}{24EI} (3L^2 - 4a^2) \\
 \delta \text{ under either load} &= \frac{Wa^2}{6EI} (3L - 4a) \\
 i_A = i_B &= \frac{Wa}{2EI} (L - a) \\
 \text{If } a = b = \frac{L}{3}, \delta_{\max} &= \frac{23}{648} \frac{WL^3}{EI}
 \end{aligned}$$

## Beam fixed at both ends

## Uniform load on full span

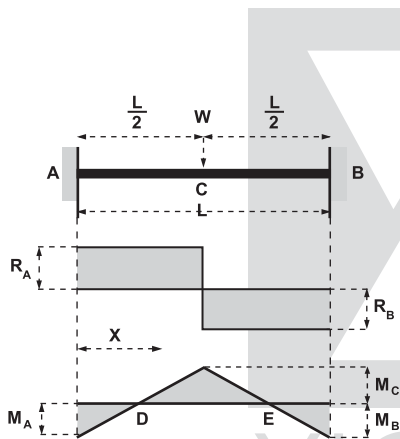


$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total Uniform load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 M_A = M_B &= -\frac{WL}{12} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_C &= \frac{WL}{24} \\ \delta_{\max} &= \frac{WL^3}{384EI} \end{aligned} \right. \\
 \text{at X from A} &\left\{ \begin{aligned} M_X &= \frac{W}{12L}(L^2 - 6LX + 6X^2) \\ \delta_X &= \frac{WX^2}{24EI}(L-X)^2 \\ i_X &= \frac{WX}{12EI}(L^2 - 3LX + 2X^2) \end{aligned} \right.
 \end{aligned}$$

at 0.211L from either end  $M_D = M_E = 0$

## Beam fixed at both ends

## Point load at mid-span

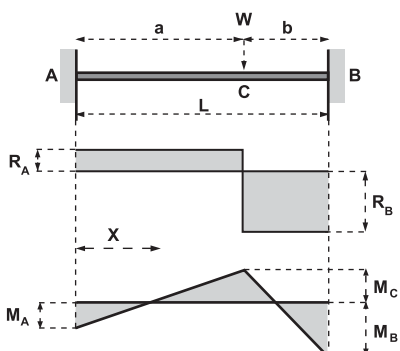


$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 M_A = M_B &= -\frac{WL}{8} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_C &= \frac{WL}{8} \\ \delta_{\max} &= \frac{WL^3}{192EI} \end{aligned} \right. \\
 \text{at X from A between A and C} &\left\{ \begin{aligned} M_X &= \frac{W}{8}(4X - L) \\ \delta_X &= \frac{WX^2}{48EI}(3L - 4X) \\ i_X &= \frac{WX}{8EI}(L - 2X) \end{aligned} \right.
 \end{aligned}$$

at 0.25L from either end  $M_D = M_E = 0$

## Beam fixed at both ends

## Point load at any position



$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A &= \frac{Wb^2(L+2a)}{L^3} & R_B &= \frac{Wa^2(L+2b)}{L^3} \\
 M_A &= -\frac{Wab^2}{L^2} & M_B &= -\frac{Wa^2b}{L^2} \\
 \text{at C, under load, } M_C &= \frac{2Wa^2b^2}{L^3} \\
 \text{at X from A between A and C} &\left\{ \begin{aligned} M_X &= \frac{Wab^2}{L^2} \frac{Wb^2(L+2a)X}{L^3} + \\ \delta_X &= \frac{Wb^2X^2[3La - (L+2a)X]}{6EIL^3} \\ i_X &= \frac{Wb^2X[2La - (L+2a)X]}{2EIL^3} \\ \delta_{\max} &= \frac{2Wa^3b^2}{3EI(L+2a)^2} \end{aligned} \right.
 \end{aligned}$$

when  $a > b$  the maximum deflection is at  $X = \frac{2La}{L+2a}$

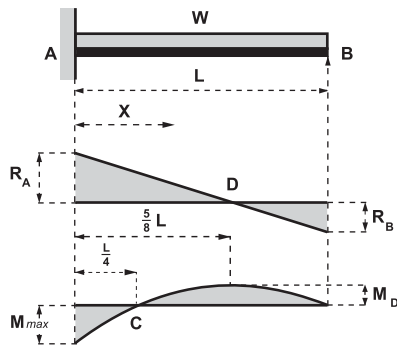
# I-BEAMS

| Cantilever | Uniform load on part of span                                                       | Special case: Uniform load on full span |
|------------|------------------------------------------------------------------------------------|-----------------------------------------|
|            | Span = L                                                                           | Span = L = b                            |
|            |                                                                                    | a = c = 0                               |
|            | Uniform load = W                                                                   | Uniform load = W                        |
|            | $R_A = W$                                                                          | $R_A = W$                               |
|            | $M_A = -W(a + \frac{b}{2})$                                                        | $M_A = -\frac{WL}{2}$                   |
|            | $i_C = i_D = \frac{W}{6EI}(3a^2 + 3ab + b^2)$                                      | $i_D = \frac{WL^2}{6EI}$                |
|            |                                                                                    | $\delta_D = \frac{WL^3}{8EI}$           |
|            | $\delta_D = \frac{WL}{24EI}[8a^3 + 18a^2b + 12ab^2 + 3b^3 + 4c(3a^2 + 3ab + b^2)]$ |                                         |

| Cantilever | Triangular load on part of span                 | Special case: Triangular load on full span |
|------------|-------------------------------------------------|--------------------------------------------|
|            | Span = L                                        | Span = L = a                               |
|            |                                                 | b = 0                                      |
|            | Triangular load = W                             | Triangular load = W                        |
|            | $R_A = W$                                       | $R_A = W$                                  |
|            | $M_A = -\frac{Wa}{3}$                           | $M_A = -\frac{WL}{3}$                      |
|            | $\delta_C = \frac{Wa^2}{15EI}(L + \frac{b}{4})$ | $\delta_C = \frac{WL^3}{15EI}$             |
|            | $i_B = i_C = \frac{Wa^2}{12EI}$                 | $i_C = \frac{WL^2}{12EI}$                  |

| Cantilever | Point load at any position                     | Special case: Point load at free end |
|------------|------------------------------------------------|--------------------------------------|
|            | Span = L                                       | Span = L = a                         |
|            |                                                | b = 0                                |
|            | Point load = W                                 | Point load = W                       |
|            | $R_A = W$                                      | $R_A = W$                            |
|            | $M_A = -Wa$                                    | $M_A = -WL$                          |
|            | $\delta_C = \frac{Wa^2}{3EI}(L + \frac{b}{2})$ | $\delta_C = \frac{WL^3}{3EI}$        |
|            | $i_B = i_C = \frac{Wa^2}{2EI}$                 | $i_C = \frac{WL^2}{2EI}$             |

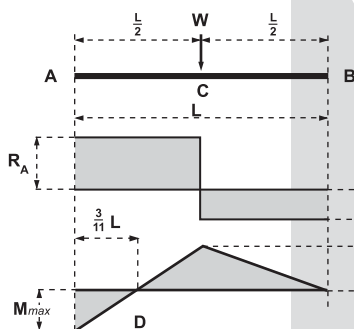
## Propped cantilever



## Uniform load on full span

|                                     |                                                   |     |
|-------------------------------------|---------------------------------------------------|-----|
| Span                                | =                                                 | $L$ |
| Total uniform load                  | =                                                 | $W$ |
| $R_A = \frac{5}{8}W$                | $R_B = \frac{3}{8}W$                              |     |
| at A                                | $M_{\max} = -\frac{WL}{8}$                        |     |
| at $\frac{5}{8}L$ from A            | $M_D = \frac{9}{128}WL$                           |     |
| at 0.5785L from A                   | $\delta_{\max} = \frac{WL^3}{185EI}$              |     |
| at B                                | $i_B = \frac{WL^2}{48EI}$                         |     |
| at X                                | $M_X = \frac{W}{8L}(L^2 - 5LX + 4X^2)$            |     |
| from A                              | $\delta_X = \frac{WX^2}{48EI}(3L^2 - 5LX + 2X^2)$ |     |
|                                     | $i_X = \frac{WX}{48EI}(6L^2 - 15LX + 8X^2)$       |     |
| at $\frac{1}{4}L$ from A, $M_C = 0$ |                                                   |     |

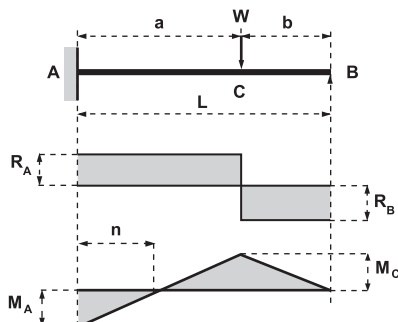
## Propped cantilever



## Point load at mid-span

|                                    |                                      |     |
|------------------------------------|--------------------------------------|-----|
| Span                               | =                                    | $L$ |
| Point load                         | =                                    | $W$ |
| $R_A = \frac{11}{16}W$             | $R_B = \frac{5}{16}W$                |     |
| at A                               | $M_{\max} = -\frac{3}{16}WL$         |     |
| at mid-span                        | $M_C = \frac{5}{32}WL$               |     |
| under load                         | $\delta_C = \frac{7WL^3}{768EI}$     |     |
| at 0.5528L from A                  | $\delta_{\max} = \frac{WL^3}{107EI}$ |     |
| at B                               | $i_B = \frac{WL^2}{32EI}$            |     |
| at $\frac{3}{11}L$ from A, $M = 0$ |                                      |     |

## Propped cantilever

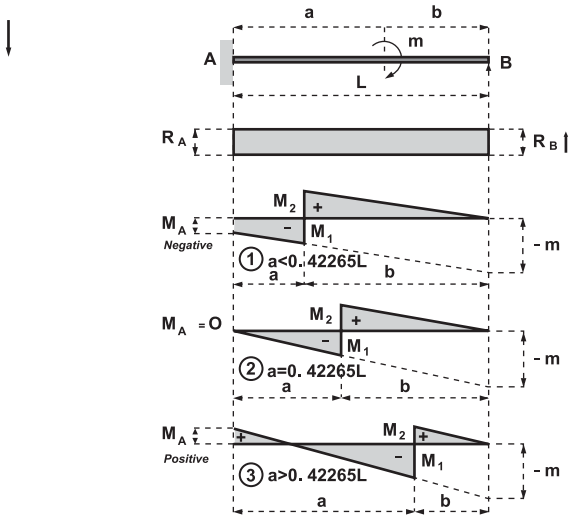


## Point load at any position

|                                                                      |                                                                  |     |
|----------------------------------------------------------------------|------------------------------------------------------------------|-----|
| Span                                                                 | =                                                                | $L$ |
| Point load                                                           | =                                                                | $W$ |
| $R_A = \frac{Wb(3L^2 - b^2)}{2L^3}$                                  | $R_B = \frac{Wa^2(2L+b)}{2L^3}$                                  |     |
| $M_A = -\frac{Wab(L+b)}{2L^2}$                                       | $M_C = \frac{Wa^2b(2L+b)}{2L^3}$                                 |     |
| $i_B = \frac{Wa^2b}{4EI}$                                            |                                                                  |     |
| Absolute max deflection is under the load when $a=b\sqrt{2}=0.5858L$ | $\delta_{\max \max} = \frac{WL^3}{102EI}$                        |     |
| When $a>b\sqrt{2}$ max deflection is between A and C                 | $\delta_{\max} = \frac{Wa^3b}{3EI} \frac{(L+b)^3}{(3L^2-b^2)^2}$ |     |
| When $a<b\sqrt{2}$ max deflection is between C and B                 | $\delta_{\max} = \frac{Wa^2b}{6EI} \sqrt{\frac{b}{2L+b}}$        |     |
| at $n=aL \frac{L+b}{3L^2-b^2}$ from A, $M = 0$                       |                                                                  |     |

# I-BEAMS

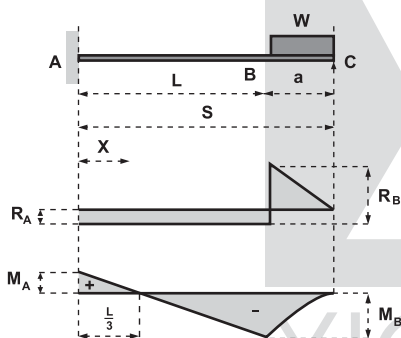
## Propped cantilever



## Moment applied at any point

$$\begin{aligned} \text{Span} &= L \\ \text{Applied moment} &= m \\ M_A &= \frac{L^2 - 3b^2}{2L^2} m & i_B &= \frac{ma}{4EI}(2b-a) \\ R_A = -R_B &= -\frac{3(L^2 - b^2)}{2L^3} m = -\frac{m + M_A}{L} \\ \begin{cases} M_1 &= -\frac{m}{L^3}(a^3 + \frac{3}{2}ab^2 + b^3) \\ M_2 &= \frac{3mab}{L^3}(b + \frac{a}{2}) = m + M_1 \end{cases} \\ \begin{cases} M_1 &= -0.42265m \\ M_2 &= 0.57735m \end{cases} \\ \begin{cases} M_1 &= \\ M_2 &= \end{cases} & \text{as for ①} \end{aligned}$$

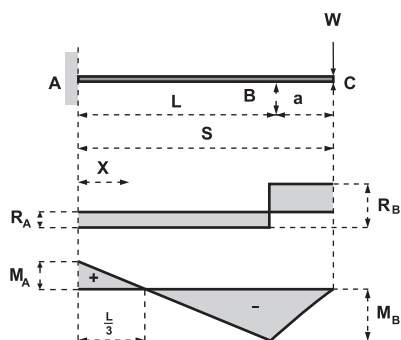
## Propped cantilever



## Uniform load on length beyond prop

$$\begin{aligned} \text{Span} &= L & \text{Full length} &= S \\ \text{Uniform load} &= W \\ R_A &= -\frac{3Wa}{4L} & R_B &= \frac{W}{L}(S - \frac{a}{4}) \\ M_A &= \frac{Wa}{4} & M_B &= -\frac{Wa}{2} \\ \text{Deflection at C} &= \Delta_{\max} & &= \frac{Wa^2S}{BEI} \\ \text{Max. negative deflection} &= \Delta_{\text{neg}} & &= -\frac{WL^2a}{54EI} \\ \text{at } X = \frac{2}{3}L \\ \text{Slope at C} &= i_C & &= \frac{Wa}{8EI}(S + \frac{a}{3}) \\ \text{at } X = \frac{L}{3} \text{ from A, } M &= 0 \end{aligned}$$

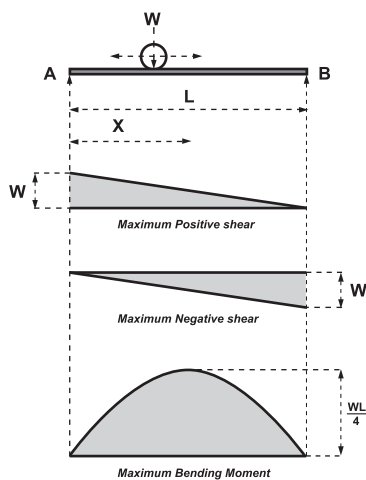
## Propped cantilever



## Point load at free end

$$\begin{aligned} \text{Span} &= L & \text{Full length} &= S \\ \text{Point load} &= W \\ R_A &= -\frac{3Wa}{2L} & R_B &= \frac{W}{L}(S + \frac{a}{2}) \\ M_A &= \frac{Wa}{2} & M_B &= -Wa \\ \text{Deflection at C} &= \Delta_{\max} & &= \frac{Wa^2}{4EI}(S + \frac{a}{3}) \\ \text{Max. negative deflection} &= \Delta_{\text{neg}} & &= -\frac{WL^2a}{27EI} \\ \text{at } X = \frac{2}{3}L \\ \text{Slope at C} &= i_C & &= \frac{Wa}{4EI}(S + a) \\ \text{at } X = \frac{L}{3} \text{ from A, } M &= 0 \end{aligned}$$

## Simply supported beam



## Single concentrated moving load

Maximum Positive Shear at any section occurs when the load is immediately to the right of the section. Similarly, Maximum Negative Shear occurs when the load is to the left. For a section distance X from A:

$$\text{Positive } F_{x \max} = W \frac{L-X}{L} \quad \text{Negative } F_{x \max} = -W \frac{X}{L}$$

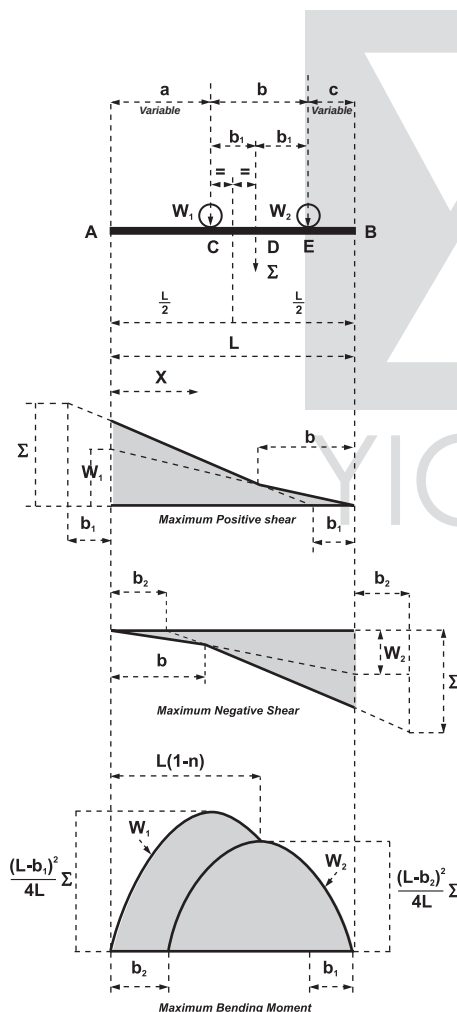
Maximum Bending Moment at any section occurs when the load is over the section. For a section distance X from A:

$$M_{x \max} = W \frac{X(L-X)}{L}$$

The Absolute Maximum Bending Moment and Deflection occur under the load at mid-span

$$\left. \begin{array}{l} M_{\max \max} = \frac{WL}{4} \\ \delta_{\max \max} = \frac{WL^3}{48EI} \end{array} \right\} \quad \left. \begin{array}{l} \text{Maximum end slope at A occurs} \\ \text{with the load at } X=0.42265L \\ \text{from A.} \end{array} \right\} i_{A \max} = 0.06415 \frac{WL^2}{EI}$$

## Simply supported beam



## Two concentrated moving load

Assume:  $W_1 > W_2$   $W_1 + W_2 = \Sigma$   $W_2 = n\Sigma$   
Fixed Distance  $b = mL$

$$b_1 = \frac{W_2 b}{W_1 + W_2} = nmL \quad b_2 = (m-n)L$$

Maximum Reaction at A and Absolute Maximum Positive Shear occur when  $W_1$  is immediately to the right of A:

$$R_{A \max} = \text{Positive } F_{\max \max} = W_1 + W_2 \quad \frac{L-b}{L}$$

For a section distance X from A:

$$X \leq L-b \quad : \quad L-b \leq X$$

$$\text{Positive } F_{\max} = \frac{L-X}{L} m W_2 \quad : \quad \text{Positive } F_{\max} = \frac{L-X}{L} W_1$$

Note: 1. For  $R_{B \max}$ , interchange values of  $W_1$  and  $W_2$  in the formula for  $R_{A \max}$

2. For Negative Shear, interchange  $W_1$  and  $W_2$  in formulae for Positive Shear, measuring X from B towards A

3. If  $m > \frac{n}{1-n}$  calculate  $R_{B \max}$  and Negative Shear values for  $W_1$  only as single load.

Absolute Maximum Bending Moment occurs under  $W_1$  when that load and the resultant of both loads are equidistant from mid-span (see loading diagram):

$$M_{\max \max} = \frac{(L-b_1)^2}{4L} \Sigma$$

If  $m < n$ , the Maximum Bending Moment at any section occurs under one of the loads. For a section distance X from A:

$$X \leq L(1-n)$$

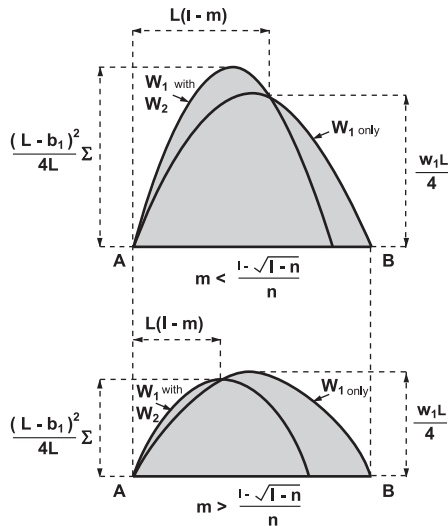
$$M_{\max} \text{ under } W_1 = \frac{(L-b_1-X)X}{L}$$

$$L(1-n) \leq X$$

$$M_{\max} \text{ under } W_2 = \frac{(X-b_2)(L-X)}{L}$$

# I-BEAMS

## Simply supported beam



## Two concentrated moving loads (continued)

If  $m > n$ , the Maximum Bending Moment at any section always occurs under  $W_1$  (the heavier load), whether  $W_2$  is on or off the span.

For a section distance X from A:

$$X \leq L(1 - m) \quad : \quad L(1 - m) \leq X$$

$$M_{\max} = \frac{(L - b_1 - X)X}{L} \quad \Sigma \quad : \quad M_{\max} = \frac{(L - X)X}{L} W_1$$

If  $n < m < \frac{L - \sqrt{L - n}}{n}$  the Absolute Maximum Bending

Moment occurs under  $W_1$  with  $W_2$  on the span.

If  $n < m > \frac{L - \sqrt{L - n}}{n}$  the Absolute Maximum Bending

Moment occurs under  $W_1$  at mid-span with  $W_2$  off the span.

Note: When the two loads are equal ( $W_1 = W_2$  and  $n = 1/2$ )

the critical value of  $\frac{L - \sqrt{L - n}}{n} = 0.5858$ .

## Simply supported beams carrying several moving concentrated loads

The Maximum Reaction and the Maximum Shear due to several moving concentrated loads occur at one support with one of the loads at that support. The location producing the Absolute Maximum must be found by trial.

The Maximum Bending Moment due to several moving concentrated loads occurs under one of the loads when that load and the gravity centre of all loads are equidistant from mid-span. The Absolute Maximum must be determined by trial.

## STRUCTURE PROPERTIES

### FORMULAS FOR GEOMETRICAL PROPERTIES OF SECTIONS

Cross-section area

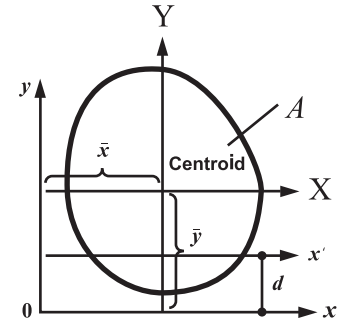
$$A = \int_A dA$$

First moment

$$S_x = \int_A y dA, S_y = \int_A x dA$$

Position of centroid

$$\bar{x} = \frac{S_y}{A}, \bar{y} = \frac{S_x}{A}$$



Moment of inertia

$$J_x = \int_A y^2 dA, J_y = \int_A x^2 dA$$

Product of inertia

$$J_{xy} = \int_A xy dA$$

Polar moment of inertia

$$J_p = \int_A r^2 dA = J_x + J_y$$

Radius of gyration

$$i_x = \sqrt{\frac{J_x}{A}}, i_y = \sqrt{\frac{J_y}{A}}$$

Polar radius of gyration

$$i_p = \sqrt{\frac{J_p}{A}}$$

Section modulus

$$Z_1 = \frac{J}{y_1}, Z_2 = \frac{J}{y_2}$$

Transition of axis

$$S_x' = S_x - Ad, S_y' = Ay - Ad$$

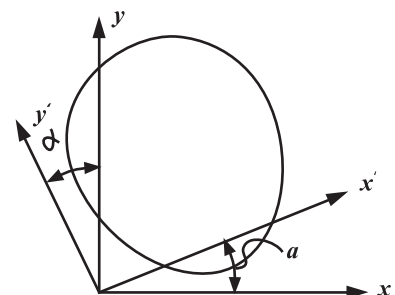
$$J_x' = J_x + Ay^2, J_y' = J_y + Ax^2, J_{xy}' = J_{xy} + xAy$$

Rotation of axis

$$S_x' = S_x \cos \alpha - S_y \sin \alpha$$

$$J_x' = J_x \sin^2 \alpha + J_y \cos^2 \alpha + J_{xy} \sin 2\alpha$$

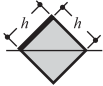
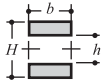
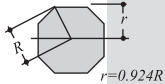
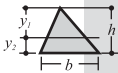
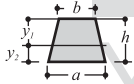
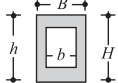
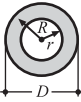
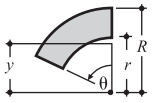
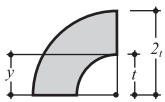
$$J_x' + J_y' = J_x + J_y = J_p$$



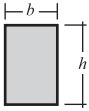
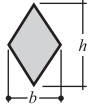
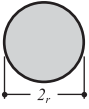
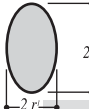
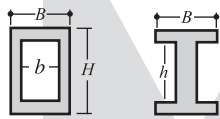
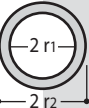


# I-BEAMS

## GEOMETRICAL PROPERTIES OF SECTIONS

| Section                                                                             | Centroid axis<br>from edge: y<br>cm                                  | Moment of inertia: J<br><br>cm <sup>4</sup>                                                                                                 | Section modulus: Z<br><br>cm <sup>3</sup>                                                            |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|    | $\frac{h}{2}$                                                        | $\frac{bh^3}{12}$                                                                                                                           | $\frac{bh^2}{6}$                                                                                     |
|    | $\frac{h}{\sqrt{2}}$                                                 | $\frac{h^4}{12}$                                                                                                                            | $\frac{\sqrt{2}h^3}{12}$                                                                             |
|    | $\frac{H}{2}$                                                        | $\frac{b}{12} (H^3 - h^3)$                                                                                                                  | $\frac{b}{6H} (H^3 - h^3)$                                                                           |
|    | $\frac{D}{2}$                                                        | $\frac{\pi D^4}{64}$                                                                                                                        | $\frac{\pi D^3}{32}$                                                                                 |
|  | 0.924 R                                                              | 0.6381 R <sup>4</sup>                                                                                                                       | 0.6906 R <sup>3</sup>                                                                                |
|  | $y_1 = \frac{2h}{3}$<br>$y_2 = \frac{h}{3}$                          | $\frac{bh^3}{36}$                                                                                                                           | $Z_1 = \frac{bh^2}{24}$<br>$Z_2 = \frac{bh^2}{12}$                                                   |
|  | $y_1 = \frac{1}{3} \frac{2a+b}{a+b} h$                               | $\frac{a^2+4ab+b^2}{36(a+b)} h^3$                                                                                                           | $Z_1 = \frac{a^2+4ab+b^2}{12(2a+b)} h^2$                                                             |
|  | $\frac{H}{2}$                                                        | $\frac{BH^3 - bh^3}{12}$                                                                                                                    | $\frac{BH^3 - bh^3}{6H}$                                                                             |
|  | $R = \frac{D}{2}$                                                    | $\frac{\pi}{4} (R^4 - r^4)$                                                                                                                 | $\frac{\pi}{4} \cdot \frac{R^4 - r^4}{R}$<br>Around Upper Section                                    |
|  | $\frac{2(R^3 - r^3)}{3(R^2 - r^2)} \cdot \frac{\sin \theta}{\theta}$ | $\frac{R^4 - r^4}{8} (\theta + \sin \theta \cdot \cos \theta)$<br>$\frac{-2(R^3 - r^3)^2}{9(R^2 - r^2)} \cdot \frac{\sin^2 \theta}{\theta}$ | Around Upper Section $Z = \frac{J}{R - y}$<br>Around Lower Section $Z = \frac{J}{y - r \cos \theta}$ |
|  | 0.9903 t                                                             | 0.6364 t <sup>4</sup>                                                                                                                       | Around Upper Section<br>0.6303 t <sup>3</sup>                                                        |

## DISTRIBUTION DIAGRAM OF SHEAR STRESS IN BEAM SECTIONS

| Section                                                                             | Shear stress diagram                                                                | $T_{\max}$                                                                                   |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|    |    | $\tau_{\max} = \frac{3Q}{2bh} = \frac{3}{2} \frac{Q}{A}$                                     |
|    |    | $\tau_{\max} = \frac{9}{4} \frac{Q}{bh} = \frac{9}{8} \frac{Q}{A}$                           |
|    |    | $\tau_{\max} = \frac{4}{3} \frac{Q}{\pi r^2} = \frac{4}{3} \frac{Q}{A}$                      |
|    |    | $\tau_{\max} = \frac{4}{3} \frac{Q}{\pi r r^1} = \frac{4}{3} \frac{Q}{A}$                    |
|  |  | $\tau_{\max} = \frac{3}{2} \frac{(BH^2 - bh^2)(BH - bh)}{2(BH^3 - bh^3)(B - b)} \frac{Q}{A}$ |
|  |  | $\tau_{\max} = \frac{4(r_2^2 + r_2 r_1 + r_1^2)}{3(r_2^2 + r_1^2)} \frac{Q}{A}$              |

YICK HOE

# I-BEAMS

## STRESS AND DEFLECTION OF BEAMS

| No. | Load condition | Bending moment                                               | Reaction force and shear stress                            | Deflection                                                                                                                                                       |
|-----|----------------|--------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   |                | <br>$M_x = Px$<br>$M_{max} = Pa$                             | <br>$R_A = P$                                              | <br>$dc = \frac{Pa^3}{3EI}$<br>$d_{max} = \frac{Pa^3}{3EI} (1 + \frac{3b}{2a})$                                                                                  |
| 2   |                | <br>$M_{max} = M_x = M_r$                                    | <br>$R_A = 0$                                              | <br>$dc = \frac{Ma^2}{2EI}$<br>$d_{max} = \frac{Ma^2}{2EI} (1 + \frac{2b}{a})$                                                                                   |
| 3   |                | <br>$M_x = \frac{Wx^2}{2a}$<br>$M_{max} = \frac{Wa}{2}$      | <br>$R_A = W$                                              | <br>$dc = \frac{wa^3}{8EI}$<br>$d_{max} = \frac{wa^3}{8EI} (1 + \frac{4b}{3a})$                                                                                  |
| 4   |                | <br>$M_{max} = W(a + \frac{b}{2})$                           | <br>$R_A = W$                                              | <br>$d_{max} = \frac{w}{24EI} \times (8a^3 + 18a^2b + 12ab^2 + 3b^3 + 12a^2c + 12abc + 4b^3c)$                                                                   |
| 5   |                | <br>$M_x = \frac{Wx^2}{3a^2}$<br>$M_A = \frac{Wa}{3}$        | <br>$R_A = W$                                              | <br>$dc = \frac{Wa^3}{15EI}$<br>$d_{max} = \frac{Wa^3}{15EI} (1 + \frac{5b}{4a})$                                                                                |
| 6   |                | <br>$M_{max} = W(a + \frac{2b}{3})$                          | <br>$R_A = W$                                              | <br>$d_{max} = \frac{W(20a^3 + 50a^2b + 40ab^2 + 11b^3)}{60EI}$                                                                                                  |
| 7   |                | <br>$M_{max} = \frac{Pl}{4}$                                 | <br>$R_A = R_B = \frac{P}{2}$                              | <br>$d_{max} = \frac{Pl^3}{48EI}$                                                                                                                                |
| 8   |                | <br>$M_{max} = \frac{Pab}{l}$                                | <br>$R_A = P*b/l$ $R_B = P*a/l$                            | <br>$d_{max} \text{ takes place in the range of } 0.0774l \text{ from the span center.}$<br>$d_{center} = \frac{Pl^3}{48EI} [ \frac{3a}{l} - 4(\frac{a}{l})^3 ]$ |
| 9   |                | <br>$M_C = \frac{Pa(b+2c)}{l}$<br>$M_D = \frac{Pc(b+2a)}{l}$ | <br>$R_A = \frac{P(b+2c)}{l}$<br>$R_B = \frac{P(b+2a)}{l}$ | <p>Deflection of span center is calculated by applying formula given in No.8 to each load.</p>                                                                   |
| 10  |                | <br>$M_{max} = \frac{Pl}{3}$                                 | <br>$R_A = R_B = P$                                        | <br>$d_{max} = \frac{23Pl^3}{648EI}$                                                                                                                             |

In case of beam subject to distributed loads, load symbol, W, indicates the total load of the beam. For example, in-case of a beam subject to uniformly distributed load, w, W equals  $w\ell$ , where  $\ell$  is span length.

| No. | Load condition | Bending moment                                                                             | Reaction force and shear stress                                                | Deflection                                                     |
|-----|----------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------|
| 11  |                | $M_C = M_E = \frac{Pl}{4}$ $M_D = \frac{5Pl}{12}$                                          | $R_A = R_B = \frac{3P}{2}$                                                     | $d_{max} = \frac{53Pl^3}{1296EI}$                              |
| 12  |                | $M_C = M_E = \frac{3Pl}{8}$ $M_D = \frac{Pl}{2}$                                           | $R_A = R_B = \frac{3P}{2}$                                                     | $d_{max} = \frac{19Pl^3}{348EI}$                               |
| 13  |                | $M_x = \frac{Wx}{2} (l - \frac{x}{l})$<br>$M_{max} = \frac{Wl}{8}$                         | $R_A = R_B = \frac{W}{2}$                                                      | $d_{max} = \frac{5}{348} \cdot \frac{Wl^3}{EI}$                |
| 14  |                | $M_{max} = \frac{W}{b} (\frac{x_1^2}{2} - a^2)$<br>where $x_1 = a + \frac{Rab}{W}$         | $R_A = \frac{W}{l} (\frac{b}{2} + c)$<br>$R_B = \frac{W}{l} (\frac{b}{2} + a)$ | $d_{max} = \frac{W}{384EI} (8l^3 + 4lb^2b^3)$<br>where $a = c$ |
| 15  |                | $M_{max} = \frac{Wa}{4}$                                                                   | $R_A = R_B = \frac{W}{2}$                                                      | $d_{max} = \frac{Wa(3l^3 - 2a^2)}{96EI}$                       |
| 16  |                | $M_x = \frac{Wx}{3} (l - \frac{x^2}{l^2})$<br>$M_{max} = 0.128Wl$<br>where $x_1 = 0.5774l$ | $R_A = \frac{W}{3}$ $R_B = \frac{2W}{3}$                                       | $d_{max} = \frac{0.01304 Wl^3}{EI}$<br>where $x = 0.5193l$     |
| 17  |                | $M_x = Wx (\frac{l}{2} - \frac{2x^2}{3l})$<br>$M_{max} = Wl/6$                             | $R_A = R_B = \frac{W}{2}$                                                      | $d_{max} = \frac{Wl^3}{60EI}$                                  |
| 18  |                | $M_{max} = \frac{W}{4} (l - \frac{b}{3})$                                                  | $R_A = R_B = \frac{W}{2}$                                                      | $d_{max} = \frac{W}{480EI} (8l^3 + 7al^2 + 4a^2l - 4a^3)$      |
| 19  |                | $M_{max} = \frac{Wa}{6}$                                                                   | $R_A = R_B = \frac{W}{2}$                                                      | $d_{max} = \frac{Wa}{240EI} (18a^3 + 20ab + 5b^3)$             |
| 20  |                | $-M_A = -M_B = M_C = \frac{Pl}{8}$                                                         | $R_A = R_B = \frac{P}{2}$                                                      | $d_{max} = \frac{Pl^3}{192EI}$                                 |

In case of beam subject to distributed loads, load symbol, W, indicates the total load of the beam. For example, incase of a beam subject to uniformly distributed load, w, W equals  $w\ell$ , where  $\ell$  is span length.

# I-BEAMS

| No. | Load condition | Bending moment                                                                                                                                                                                           | Reaction force and shear stress                                                                                                            | Deflection                                                                                           |
|-----|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 21  |                | $M_A = -\frac{Pab^2}{l^2}$ $M_B = -\frac{Pa^2b}{l^2}$ $M_C = \frac{2Pa^2b^2}{l^3}$                                                                                                                       | $R_A = P \left(\frac{b}{l}\right)^2 \left(1 + 2\frac{a}{l}\right)$ $R_B = P \left(\frac{a}{l}\right)^2 \left(1 + 2\frac{b}{l}\right)$      | $d_{max} = \frac{Pa^3b^3}{3EI l^3}$ $d_{max} = \frac{2Pa^2b^3}{3EI(3l-2a)} \times \frac{l^2}{3l-2a}$ |
| 22  |                | $M_A = M_B = -2Pl/9$ $M_C = M_D = Pl/9$                                                                                                                                                                  | $R_A = R_B = P$                                                                                                                            | $d_{max} = \frac{5Pl^3}{648EI}$                                                                      |
| 23  |                | $M_A = M_B = -19Pl/72$ $M_D = 11Pl/72$                                                                                                                                                                   | $R_A = R_B = 3P/2$                                                                                                                         | $d_{max} = \frac{41Pl^3}{5184EI}$                                                                    |
| 24  |                | $M_A = M_B = -5Pl/16$ $M_D = 3Pl/16$                                                                                                                                                                     | $R_A = R_B = 3P/2$                                                                                                                         | $d_{max} = \frac{Pl^3}{96EI}$                                                                        |
| 25  |                | $M_A = M_B = -\frac{Wl^2}{12}$ $M_C = \frac{Wl^2}{24}$                                                                                                                                                   | $R_A = R_B = W/2$                                                                                                                          | $d_{max} = \frac{Wl^3}{384EI}$                                                                       |
| 26  |                | $M_A = -\frac{W}{12Pl^2} [e^3(4l-3e) - c^3(4l-3c)]$ $M_B = -\frac{W}{12Pl^2} [d^3(4l-3d) - a^3(4l-3a)]$                                                                                                  | $R_A = r_A + \frac{M_A - M_B}{l}$ $R_B = r_B + \frac{M_B - M_A}{l}$ <p><math>r</math> indicates the reaction force of the single span.</p> | $d_{max} = \frac{W}{384EI} (l^3 + 2l^2a + 4la^2 - 8a^3)$                                             |
| 27  |                | $M_X = -\frac{Wl}{30} \left( \frac{10x^3}{l^3} - \frac{9x}{l} + 2 \right)$ <p>where <math>+M_{max} = Wl/23.3</math>, <math>x = 0.55l</math><br/><math>M_A = -Wl/15</math>, <math>M_B = -Wl/10</math></p> | $R_A = 0.3W$ , $R_B = 0.7W$                                                                                                                | $d_{max} = \frac{Wl^3}{382EI}$ <p>where <math>x_1 = 0.525l</math></p>                                |
| 28  |                | $M_A = M_B = -\frac{5Wl}{48}$ $M_C = Wl/16$                                                                                                                                                              | $R_A = R_B = W/2$                                                                                                                          | $d_{max} = \frac{1.4Wl^3}{384EI}$                                                                    |
| 29  |                | $M_A = M_B = -\frac{W}{48l} x$ <p><math>(5l^2 + 4al - 4a^2)</math></p>                                                                                                                                   | $R_A = R_B = W/2$                                                                                                                          | $d_{max} = \frac{W}{1920EI} (7l^3 + 8al^2 + 4a^2l - 16a^3)$                                          |
| 30  |                | $M_A = M_B = -\frac{Wa}{12l} (2l - a)$                                                                                                                                                                   | $R_A = R_B = W/2$                                                                                                                          | $d_{max} = \frac{Wa^2}{480EI} (5l - 4a)$                                                             |

In case of beam subject to distributed loads, load symbol, W, indicates the total load of the beam. For example, in case of a beam subject to uniformly distributed load, w, W equals  $w\ell$ , where  $\ell$  is span length.

## WORKING STRESSES

The values given in the safe load tables are based on the allowable stresses in BS 449 as follows:

| Thickness of material<br>(mm) | Grade<br>of steel | Allowable stresses (N/mm <sup>2</sup> ) |         |         |
|-------------------------------|-------------------|-----------------------------------------|---------|---------|
|                               |                   | Bending                                 | Shear*1 | Bearing |
| Up to and including 40        | 43                | 165                                     | 100     | 190     |
| Over 40                       |                   | 150                                     | 90      |         |
| Up to and including 65        | 50                | 230                                     | 140     | 260     |
| Over 65                       |                   | $Y_g/1.52^{*2}$                         |         |         |
| Up to and including 40        | 55                | 280                                     | 170     | 320     |
| Over 40                       |                   | 260                                     | 160     |         |

Note: \*1 On unstiffened web.

\*2  $Y_g$  = yield stress, to be agreed with the manufacturer and not to exceed 350N/mm<sup>2</sup>. This applies only to universal column sections and plated compound sections and, in this handbook, affects certain universal column stanchions only for which a value of  $Y_g = 325\text{N/mm}^2$  has been assumed.

### Web buckling

The buckling values for the unstiffened webs have been calculated in accordance with the formula in Clause 28a(i) of BS 449, namely  $W = p_c t B$  where

- $W$  = allowable reaction or concentrated load in N, but tabulated in kN.
- $p_c$  = allowable axial stress for struts as given in BS 449, Clause 30a for a slenderness ratio of  $(d/3)/t$ , in which
- $d$  = clear depth of web between root fillets
- $t$  = web thickness
- $B$  =  $\frac{1}{2}D + t_p + L_b$  for end bearings of simply supported beams
- =  $D + 2t_p + L_b$  for intermediate bearings, in which
- $D$  = overall depth of beam
- $t_p$  = thickness of bearing and/or flange plate
- $L_b$  = length of stiff portion of bearing which shall not be taken as greater than half depth of beam for simply supported beams and the full depth of beam for an intermediate bearing over which the beam is continuous, unless the web is stiffened.

### Web crushing

To prevent local crushing of the web, the direction bearing stress at the root of the web is limited to the values given in BS 449, and included in the table above.

The length of web resisting crushing is determined on the assumption that the load is dispersed through the flange and the bearing and/or flange plate at an angle of 30° (BS 449, Clause 27e).

## COMBINED STRESSES

Where bending and shear stresses, or bearing, bending and shear stresses, are co-existent, the beam should be checked in accordance with BS 449, Clauses 14.c and d.

## DEFLECTION COEFFICIENTS

Clause 15 to BS 449 includes the requirement that the maximum deflection due to loads other than the weight of the structural floors or roof, steelwork and casing, if any, shall not exceed 1/360th of the span. For a simply supported beam, the uniformly distributed load  $W_D$  in kN to produce this deflection is:

$$W_D = \frac{384EI}{5 \times 360 L^2} \times \frac{Cl}{1000}$$

where C = deflection coefficient tabulated for span L in m  
 E = modulus of elasticity = 21000kN/cm<sup>2</sup> (=210000N/mm<sup>2</sup>)  
 I = moment of inertia of beam, in cm<sup>4</sup>

The load  $W_D$  will be less than the tabular load if the span exceeds 12.17, 16.97 or 10.00 times the beam depth for grades 50, 43 or 55 steel respectively. For such cases, it may be necessary to confirm that the total load is within the capacity of the beam and that the loads to be considered for deflection purposes do not exceed  $W_D$ .

The table below gives limiting values of the span/depth ratio for uniformly loaded simply supported beams for different grades of steel for various ratios of  $W_d/W_t$  where  $W_d$  = load considered for deflection purposes and  $W_t$  = total load on beam. If the appropriate span/ depth ratio is exceeded, then the relevant deflection will exceed 1 /360th of the span unless the bending stress is reduced.

| Material | Limiting values of span to depth ratio for $W_d/W_t$ |       |       |       |       |       |
|----------|------------------------------------------------------|-------|-------|-------|-------|-------|
|          | 1.0                                                  | 0.9   | 0.8   | 0.7   | 0.6   | 0.5   |
| Grade 50 | 12.17                                                | 13.53 | 15.22 | 17.39 | 20.29 | 24.35 |
| Grade 43 | 16.97                                                | 18.86 | 21.21 | 24.24 | 28.28 | 33.94 |
| Grade 55 | 10.00                                                | 11.11 | 12.50 | 14.29 | 16.67 | 20.00 |

## SAFE LOADS (for as-rolled sections)

These are the safe uniformly distributed loads (including self weight) which can be carried by simply supported beams with adequate lateral restraint to the compression flange. They have therefore generally been calculated in accordance with the allowable bending stresses given in BS 449.

The loads are printed in three different type faces to draw attention to particular criteria which may affect the load-carrying capacity of a beam. Bold, italic and ordinary type faces have been used in that sequence for maximum clarity and the significance of each is explained below.

**Bold face.** Loads printed in this type are greater than the web buckling capacity of the beam (UB, joist or channel) alone. If sufficient additional capacity is not provided by the bearing, web stiffeners will be necessary to realise the full web capacity.

*Italic face.* Loads printed in this type are within the buckling capacity of the unstiffened web and produce a maximum deflection of less than 1/360th of the span.

Ordinary face. Loads printed in this type produce a maximum deflection exceeding 1/360th of the span. Such cases should therefore be examined as indicated above under 'Deflection coefficients'.

In addition to the criteria covered by the use of different type faces, two other matters require comment.

- Shear capacity. Where the shear capacity of the unstiffened web is less than the bending capacity of a beam, the safe load is calculated on the allowable average shear stress given in BS 449.
- Web crushing. Many of the loads tabulated, in the various type faces, exceed the beam component of the direct bearing capacity of the web at its junction with the flange. The designer should therefore ensure that the length of stiff bearing and thickness of flange plate or packing (if any) provide a sufficient additional length of web in bearing.

## BEAMS WITHOUT LATERAL SUPPORT

The allowable compressive stress due to bending about the x-x axis for universal beams and columns, joists and channels is the lesser of the values of  $p_{bc}$  given in Tables 2 and 3 a, b or c (depending on the grade of steel) in BS 449. The values in Table 2 apply when the compression flange is so supported that lateral instability is obviated. When the compression flange is unsupported or insufficiently supported laterally, the appropriate Table 3 applies. The maximum effective length of compression flange which may be used without reduction of the allowable bending stresses in Table 2 is given as  $L_c$  with the safe load tables.

Example: Find the allowable stress,  $p_{bc}$  and the safe uniformly distributed load,  $W$ , for a 533 = 210 UB 82 in grade 43 steel spanning 7m without lateral restraint and having the ends of the compression flange partially restrained.

Effective length =  $0.85 \times 700 = 595$  cm (see BS 449, Clause 26)

$l/r_y 595/4.38 = 136$ ;  $D/T = 40$ .

From BS 449, Table 3a,  $p_{bc} = 119 \text{ N/mm}^2$

The safe uniformly distributed load,  $W$

$$\begin{aligned}
 &= 8fZ/L \\
 &= (8 \times 119 \times 1798 \times 10^3 / 7000) \text{ N} \\
 &= 245 \text{ kN}
 \end{aligned}$$

Lateral instability is not a criterion for these sections when bent about the y-y axis and the allowable stresses in Table 2 apply.





# I-BEAMS

## STRESS AND SAFE LOAD TABLES

Table 3a of BS 449. Allowable stress  $p_{bc}$  in bending (N/mm<sup>2</sup>) for beams of Grade 43 steel

| $l/r_y$ | D/T |     |     |     |     |     |     |     |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|
|         | 10  | 15  | 20  | 25  | 30  | 35  | 40  | 50  |
| 90      | 165 | 165 | 165 | 165 | 165 | 165 | 165 | 165 |
| 95      | 165 | 165 | 165 | 163 | 163 | 163 | 163 | 163 |
| 100     | 165 | 165 | 165 | 157 | 157 | 157 | 157 | 157 |
| 105     | 165 | 165 | 160 | 152 | 152 | 152 | 152 | 152 |
| 110     | 165 | 165 | 156 | 147 | 147 | 147 | 147 | 147 |
| 115     | 165 | 165 | 152 | 141 | 141 | 141 | 141 | 141 |
| 120     | 165 | 162 | 148 | 136 | 136 | 136 | 136 | 136 |
| 130     | 165 | 155 | 139 | 126 | 126 | 126 | 126 | 126 |
| 140     | 165 | 149 | 130 | 115 | 115 | 115 | 115 | 115 |
| 150     | 165 | 143 | 122 | 104 | 104 | 104 | 104 | 104 |
| 160     | 163 | 136 | 113 | 95  | 94  | 94  | 94  | 94  |
| 170     | 159 | 130 | 104 | 91  | 85  | 82  | 82  | 82  |
| 180     | 155 | 124 | 96  | 87  | 80  | 76  | 72  | 71  |
| 190     | 151 | 118 | 93  | 83  | 77  | 72  | 68  | 62  |
| 200     | 147 | 111 | 89  | 80  | 73  | 68  | 64  | 59  |
| 210     | 143 | 105 | 87  | 77  | 70  | 65  | 61  | 55  |
| 220     | 139 | 99  | 84  | 74  | 67  | 62  | 58  | 52  |
| 230     | 134 | 95  | 81  | 71  | 64  | 59  | 55  | 49  |
| 240     | 130 | 92  | 78  | 69  | 61  | 56  | 52  | 47  |
| 250     | 126 | 90  | 76  | 66  | 59  | 54  | 50  | 44  |
| 260     | 122 | 88  | 74  | 64  | 57  | 52  | 48  | 42  |
| 270     | 118 | 86  | 72  | 62  | 55  | 50  | 46  | 40  |
| 280     | 114 | 84  | 70  | 60  | 53  | 48  | 44  | 39  |
| 290     | 110 | 82  | 68  | 58  | 51  | 46  | 42  | 37  |
| 300     | 106 | 80  | 66  | 56  | 49  | 44  | 41  | 36  |

Intermediate values may be obtained by linear interpolation.

Note : For materials over 40 mm thick the stress shall not exceed 150 N/mm<sup>2</sup>.

## AXIAL STRESSES

The safe loads on stanchions and struts have been calculated in accordance with the allowable axial stresses specified in BS 449, Clause 30, and shown in the tables that follow :

# I-BEAMS

## DESIGN OF STANCHIONS AND STRUTS

Table 17a of BS 449. Allowable stress  $p_c$  on gross section for axial compression for Grade 43 steel

| l/r | $p_c$ (N/mm <sup>2</sup> ) for Grade 43 steel |     |     |     |     |     |     |     |     |     |
|-----|-----------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|     | 0                                             | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   |
| 0   | 155                                           | 155 | 154 | 154 | 153 | 153 | 153 | 152 | 152 | 151 |
| 10  | 151                                           | 151 | 150 | 150 | 149 | 149 | 148 | 148 | 148 | 147 |
| 20  | 147                                           | 146 | 146 | 146 | 145 | 145 | 144 | 144 | 144 | 143 |
| 30  | 143                                           | 142 | 142 | 142 | 141 | 141 | 141 | 140 | 140 | 139 |
| 40  | 139                                           | 138 | 138 | 137 | 137 | 136 | 136 | 136 | 135 | 134 |
| 50  | 133                                           | 133 | 132 | 131 | 130 | 130 | 129 | 128 | 127 | 126 |
| 60  | 126                                           | 125 | 124 | 123 | 122 | 121 | 120 | 119 | 118 | 117 |
| 70  | 115                                           | 114 | 113 | 112 | 111 | 110 | 108 | 107 | 106 | 105 |
| 80  | 104                                           | 102 | 101 | 100 | 99  | 97  | 96  | 95  | 94  | 92  |
| 90  | 91                                            | 90  | 89  | 87  | 86  | 85  | 84  | 83  | 81  | 80  |
| 100 | 79                                            | 78  | 77  | 76  | 75  | 74  | 73  | 72  | 71  | 70  |
| 110 | 69                                            | 68  | 67  | 66  | 65  | 64  | 63  | 62  | 61  | 61  |
| 120 | 60                                            | 59  | 58  | 57  | 56  | 56  | 55  | 54  | 53  | 53  |
| 130 | 52                                            | 51  | 51  | 50  | 49  | 49  | 48  | 48  | 47  | 46  |
| 140 | 46                                            | 45  | 45  | 44  | 43  | 43  | 42  | 42  | 41  | 41  |
| 150 | 40                                            | 40  | 39  | 39  | 38  | 38  | 38  | 37  | 37  | 36  |
| 160 | 36                                            | 35  | 35  | 35  | 34  | 34  | 33  | 33  | 33  | 32  |
| 170 | 32                                            | 32  | 31  | 31  | 31  | 30  | 30  | 30  | 29  | 29  |
| 180 | 29                                            | 28  | 28  | 28  | 28  | 27  | 27  | 27  | 26  | 26  |
| 190 | 26                                            | 26  | 25  | 25  | 25  | 25  | 24  | 24  | 24  | 24  |
| 200 | 24                                            | 23  | 23  | 23  | 23  | 22  | 22  | 22  | 22  | 22  |
| 210 | 21                                            | 21  | 21  | 21  | 21  | 20  | 20  | 20  | 20  | 20  |
| 220 | 20                                            | 19  | 19  | 19  | 19  | 19  | 19  | 18  | 18  | 18  |
| 230 | 18                                            | 18  | 18  | 18  | 17  | 17  | 17  | 17  | 17  | 17  |
| 240 | 17                                            | 16  | 16  | 16  | 16  | 16  | 16  | 16  | 16  | 15  |
| 250 | 15                                            |     |     |     |     |     |     |     |     |     |
| 300 | 11                                            |     |     |     |     |     |     |     |     |     |
| 350 | 8                                             |     |     |     |     |     |     |     |     |     |

Intermediate values may be obtained by linear interpolation.

Note : For material over 40 mm thick, other than rolled I-beams or channels, and for universal columns of thicknesses exceeding 40 mm, the limiting stress is 140 N/mm<sup>2</sup>.

# UNIVERSAL BEAMS AND COLUMNS

## SAFE LOADS FOR GRADE 43 STEEL

| Section size | Mass per metre | Safe distributed loads in kilonewtons for spans in metres and deflection coefficients |               |               |               |               |               |               |               |               |               |               |               |                | Critical span | Lc   |
|--------------|----------------|---------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|----------------|---------------|------|
| mm           | kg             | 2.00<br>112.0                                                                         | 2.50<br>71.68 | 3.00<br>49.78 | 3.50<br>36.57 | 4.00<br>28.00 | 4.50<br>22.12 | 5.00<br>17.92 | 5.50<br>14.81 | 6.00<br>12.44 | 7.00<br>9.143 | 8.00<br>7.000 | 9.00<br>5.531 | 10.00<br>4.480 |               | m    |
| 203 x 133    | 25             | 153                                                                                   | 122           | 102           | 87            | 77            | 68            | 61            | 56            | 51            |               |               |               |                |               | 2.80 |
|              | 30             | 184                                                                                   | 147           | 123           | 105           | 92            | 82            | 74            | 67            | 61            | 53            |               |               |                |               | 3.03 |
| 254 x 102    | 22             | 149                                                                                   | 119           | 99            | 85            | 74            | 66            | 60            | 54            | 50            | 43            | 37            |               |                |               | 1.75 |
|              | 25             | 175                                                                                   | 140           | 117           | 100           | 88            | 78            | 70            | 64            | 58            | 50            | 37            |               |                |               | 1.87 |
|              | 28             | 203                                                                                   | 163           | 135           | 116           | 102           | 90            | 81            | 74            | 68            | 58            | 51            |               |                |               | 2.01 |
| 254 x 146    | 31             | 233                                                                                   | 186           | 155           | 133           | 117           | 104           | 93            | 85            | 78            | 67            | 58            |               |                |               | 2.96 |
|              | 37             | 286                                                                                   | 229           | 191           | 164           | 143           | 127           | 115           | 104           | 95            | 82            | 72            |               |                |               | 3.22 |
|              | 43             | 333                                                                                   | 267           | 222           | 191           | 167           | 148           | 133           | 121           | 111           | 95            | 83            |               |                |               | 3.41 |
| 305 x 102    | 25             | 190                                                                                   | 152           | 127           | 109           | 95            | 84            | 76            | 69            | 63            | 54            | 47            | 42            | 38             |               | 1.64 |
|              | 28             | 232                                                                                   | 185           | 154           | 132           | 116           | 103           | 93            | 84            | 77            | 66            | 58            | 51            | 46             |               | 1.79 |
|              | 33             | 274                                                                                   | 219           | 183           | 156           | 137           | 122           | 110           | 100           | 91            | 78            | 68            | 61            | 55             |               | 1.90 |
| 305 x 127    | 37             | 311                                                                                   | 249           | 207           | 178           | 156           | 138           | 124           | 113           | 104           | 89            | 78            | 69            | 62             |               | 2.37 |
|              | 42             | 351                                                                                   | 280           | 234           | 200           | 175           | 156           | 140           | 127           | 117           | 100           | 88            | 78            | 70             |               | 2.45 |
|              | 48             | 404                                                                                   | 323           | 269           | 231           | 202           | 180           | 162           | 147           | 135           | 115           | 101           | 90            | 81             |               | 2.59 |
| 305 x 165    | 40             | 370                                                                                   | 296           | 247           | 212           | 185           | 165           | 148           | 135           | 123           | 106           | 93            | 82            | 74             |               | 3.38 |
|              | 46             | 412†                                                                                  | 342           | 285           | 244           | 214           | 190           | 171           | 155           | 143           | 122           | 107           | 95            | 86             |               | 3.53 |
|              | 54             | 479†                                                                                  | 398           | 331           | 284           | 249           | 221           | 199           | 181           | 166           | 142           | 124           | 110           | 99             |               | 3.69 |
| 356 x 127    | 33             | 311                                                                                   | 248           | 207           | 177           | 155           | 138           | 124           | 113           | 104           | 89            | 78            | 69            | 62             |               | 2.18 |
|              | 39             | 377                                                                                   | 302           | 252           | 216           | 189           | 168           | 151           | 137           | 126           | 108           | 94            | 84            | 75             |               | 2.33 |
| 356 x 171    | 45             | 453                                                                                   | 363           | 302           | 259           | 227           | 201           | 181           | 165           | 151           | 130           | 113           | 101           | 91             |               | 3.23 |
|              | 51             | 519†                                                                                  | 420           | 350           | 300           | 263           | 234           | 210           | 191           | 175           | 150           | 131           | 117           | 105            |               | 3.38 |
|              | 57             | 574†                                                                                  | 473           | 394           | 338           | 296           | 263           | 237           | 215           | 197           | 169           | 148           | 131           | 118            |               | 3.50 |
|              | 67             | 662†                                                                                  | 567           | 472           | 405           | 354           | 315           | 283           | 258           | 236           | 202           | 177           | 157           | 142            |               | 3.72 |
| 406 x 140    | 39             | 414                                                                                   | 331           | 276           | 236           | 207           | 184           | 165           | 150           | 138           | 118           | 103           | 92            | 83             |               | 2.41 |
|              | 46             | 513                                                                                   | 411           | 342           | 293           | 257           | 228           | 205           | 187           | 171           | 147           | 128           | 114           | 103            |               | 2.58 |
| 406 x 178    | 54             | 611                                                                                   | 489           | 407           | 349           | 305           | 271           | 244           | 222           | 204           | 174           | 153           | 136           | 122            |               | 3.28 |
|              | 60             | 634†                                                                                  | 559           | 466           | 399           | 349           | 310           | 279           | 254           | 233           | 200           | 175           | 155           | 140            |               | 3.45 |
|              | 67             | 721†                                                                                  | 627           | 523           | 448           | 392           | 348           | 314           | 285           | 261           | 224           | 196           | 174           | 157            |               | 3.54 |
|              | 74             | 801†                                                                                  | 699           | 583           | 499           | 437           | 388           | 350           | 318           | 291           | 250           | 218           | 194           | 175            |               | 3.65 |
| 457 x 152    | 52             | 626                                                                                   | 501           | 418           | 358           | 313           | 278           | 251           | 228           | 209           | 179           | 157           | 139           | 125            |               | 2.62 |
|              | 60             | 728†                                                                                  | 591           | 493           | 422           | 370           | 329           | 296           | 269           | 246           | 211           | 185           | 164           | 148            |               | 2.78 |
|              | 67             | 825                                                                                   | 660           | 550           | 471           | 413           | 367           | 330           | 300           | 275           | 236           | 206           | 183           | 165            |               | 2.81 |
|              | 74             | 913†                                                                                  | 742           | 619           | 530           | 464           | 412           | 371           | 337           | 309           | 265           | 282           | 206           | 186            |               | 2.92 |
|              | 82             | 995†                                                                                  | 822           | 685           | 587           | 514           | 457           | 411           | 374           | 343           | 294           | 257           | 228           | 206            |               | 3.03 |
| 457x191      | 67             | 771†                                                                                  | 684           | 570           | 489           | 428           | 380           | 342           | 311           | 285           | 244           | 214           | 190           | 171            |               | 3.53 |
|              | 74             | 832†                                                                                  | 771           | 643           | 551           | 482           | 429           | 386           | 351           | 321           | 276           | 241           | 214           | 193            |               | 3.65 |
|              | 82             | 911†                                                                                  | 851           | 709           | 608           | 532           | 473           | 426           | 387           | 355           | 304           | 266           | 236           | 213            |               | 3.74 |
|              | 89             | 983†                                                                                  | 935           | 779           | 668           | 584           | 519           | 467           | 425           | 389           | 334           | 292           | 260           | 234            |               | 3.86 |
|              | 98             | 1066†                                                                                 | 1033          | 861           | 738           | 645           | 574           | 516           | 469           | 430           | 369           | 323           | 287           | 258            |               | 4.00 |

Loads printed in italic type do not cause overloading of the unstiffened web, and do not cause deflection exceeding span/360.

Loads printed in ordinary type should be checked for deflection

† Load is based on allowable shear of web and is less than allowable load in bending.

# UNIVERSAL BEAMS AND COLUMNS

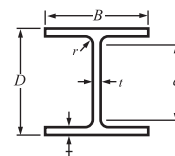
## SAFE LOADS FOR GRADE 43 STEEL

| Section size | Mass per metre | Safe distributed loads in kilonewtons for spans in metres and deflection coefficients |               |               |               |               |               |                |                |                |                |                |                |                | Critical span L <sub>c</sub><br>m |
|--------------|----------------|---------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-----------------------------------|
|              |                | 4.00<br>28.00                                                                         | 5.00<br>17.92 | 6.00<br>12.44 | 7.00<br>9.143 | 8.00<br>7.000 | 9.00<br>5.531 | 10.00<br>4.480 | 11.00<br>3.702 | 12.00<br>3.111 | 13.00<br>2.651 | 14.00<br>2.286 | 15.00<br>1.991 | 16.00<br>1.750 |                                   |
| 533 x 210    | 82             | 593                                                                                   | 475           | 396           | 339           | 297           | 264           | 237            | 216            | 198            | 183            | 170            | 158            | 148            | 3.70                              |
|              | 92             | 685                                                                                   | 548           | 457           | 391           | 343           | 304           | 274            | 249            | 228            | 211            | 196            | 183            | 171            | 3.88                              |
|              | 101            | 758                                                                                   | 607           | 506           | 433           | 379           | 337           | 303            | 276            | 253            | 233            | 217            | 202            | 190            | 3.99                              |
|              | 109            | 816                                                                                   | 653           | 544           | 467           | 408           | 363           | 327            | 297            | 272            | 251            | 233            | 218            | 204            | 4.07                              |
|              | 122            | 924                                                                                   | 739           | 616           | 528           | 462           | 411           | 369            | 336            | 308            | 284            | 264            | 246            | 231            | 4.24                              |
| 610 x 229    | 101            | 830                                                                                   | 664           | 553           | 474           | 415           | 369           | 332            | 302            | 277            | 255            | 237            | 221            | 207            | 4.01                              |
|              | 113            | 950                                                                                   | 760           | 633           | 543           | 475           | 422           | 380            | 345            | 317            | 292            | 271            | 253            | 238            | 4.18                              |
|              | 125            | 1063                                                                                  | 851           | 709           | 608           | 532           | 473           | 425            | 387            | 354            | 327            | 304            | 284            | 266            | 4.33                              |
|              | 140            | 1197                                                                                  | 957           | 798           | 684           | 598           | 532           | 479            | 435            | 399            | 368            | 342            | 319            | 299            | 4.48                              |
| 610 x 305    | 149            | 1350                                                                                  | 1080          | 900           | 771           | 675           | 600           | 540            | 491            | 450            | 415            | 386            | 360            | 337            | 6.10                              |
|              | 179            | 1621                                                                                  | 1297          | 1080          | 926           | 810           | 720           | 648            | 589            | 540            | 499            | 463            | 432            | 405            | 6.40                              |
|              | 238            | 2164                                                                                  | 1732          | 1443          | 1237          | 1082          | 962           | 866            | 787            | 721            | 666            | 618            | 577            | 541            | 7.03                              |
| 686 x 254    | 125            | 1149                                                                                  | 919           | 766           | 656           | 574           | 511           | 459            | 418            | 383            | 353            | 328            | 306            | 287            | 4.41                              |
|              | 140            | 1316                                                                                  | 1053          | 877           | 752           | 658           | 585           | 526            | 479            | 439            | 405            | 376            | 351            | 329            | 4.60                              |
|              | 152            | 1443                                                                                  | 1154          | 962           | 824           | 721           | 641           | 577            | 525            | 481            | 444            | 412            | 385            | 361            | 4.73                              |
|              | 170            | 1621                                                                                  | 1297          | 1080          | 926           | 810           | 720           | 648            | 589            | 540            | 499            | 463            | 432            | 405            | 4.88                              |
| 762 x 267    | 147            | 1479                                                                                  | 1184          | 986           | 845           | 740           | 658           | 592            | 538            | 493            | 455            | 423            | 395            | 370            | 4.52                              |
|              | 173            | 1777                                                                                  | 1422          | 1185          | 1015          | 889           | 790           | 711            | 646            | 592            | 547            | 508            | 474            | 444            | 4.77                              |
|              | 197            | 2057                                                                                  | 1646          | 1371          | 1176          | 1029          | 914           | 823            | 748            | 686            | 633            | 588            | 549            | 514            | 5.01                              |
| 838 x 292    | 176            | 1945                                                                                  | 1556          | 1297          | 1111          | 973           | 864           | 778            | 707            | 648            | 598            | 556            | 519            | 486            | 4.94                              |
|              | 194            | 2194                                                                                  | 1755          | 1463          | 1254          | 1097          | 975           | 878            | 798            | 731            | 675            | 627            | 585            | 548            | 5.14                              |
|              | 226            | 2635                                                                                  | 2108          | 1757          | 1506          | 1318          | 1171          | 1054           | 958            | 879            | 811            | 753            | 703            | 659            | 5.45                              |
| 914 x 305    | 201            | 2379                                                                                  | 1903          | 1586          | 1360          | 1190          | 1057          | 952            | 865            | 793            | 732            | 680            | 634            | 595            | 5.07                              |
|              | 224            | 2725                                                                                  | 2180          | 1817          | 1557          | 1363          | 1211          | 1090           | 991            | 908            | 839            | 779            | 727            | 681            | 5.32                              |
|              | 253            | 3137                                                                                  | 2510          | 2092          | 1793          | 1569          | 1394          | 1255           | 1141           | 1046           | 965            | 896            | 837            | 784            | 5.55                              |
|              | 289            | 3594                                                                                  | 2875          | 2396          | 2054          | 1797          | 1597          | 1438           | 1307           | 1198           | 1106           | 1027           | 958            | 899            | 5.75                              |
| 914 x 419    | 343            |                                                                                       | 3536†         | 3019          | 2588          | 2264          | 2013          | 1811           | 1647           | 1509           | 1393           | 1294           | 1208           | 1132           | 8.39                              |
|              | 388            |                                                                                       | 3958†         | 3436          | 2945          | 2577          | 2290          | 2061           | 1874           | 1718           | 1586           | 1472           | 1374           | 1288           | 8.73                              |
| 686 x 254    | 125            | 1146                                                                                  | 917           | 764           | 655           | 573           | 509           | 458            | 417            | 382            | 353            | 327            | 306            | 286            | 4.20                              |
|              | 140            | 1313                                                                                  | 1050          | 875           | 750           | 657           | 584           | 525            | 477            | 438            | 404            | 375            | 350            | 328            | 4.43                              |
|              | 152            | 1440                                                                                  | 1152          | 960           | 823           | 720           | 640           | 576            | 524            | 480            | 443            | 411            | 384            | 360            | 4.57                              |
|              | 170            | 1618                                                                                  | 1294          | 1078          | 924           | 809           | 719           | 647            | 588            | 539            | 498            | 462            | 431            | 404            | 4.73                              |
| 762 x 267    | 147            | 1475                                                                                  | 1180          | 984           | 843           | 738           | 656           | 590            | 537            | 492            | 454            | 422            | 393            | 369            | 4.33                              |
|              | 173            | 1773                                                                                  | 1419          | 1182          | 1013          | 887           | 788           | 709            | 645            | 591            | 546            | 507            | 473            | 443            | 4.61                              |
|              | 197            | 2054                                                                                  | 1643          | 1369          | 1173          | 1027          | 913           | 821            | 747            | 685            | 632            | 587            | 548            | 513            | 4.86                              |
| 838 x 292    | 176            | 1940                                                                                  | 1552          | 1293          | 1109          | 970           | 862           | 776            | 705            | 647            | 597            | 554            | 517            | 485            | 4.72                              |
|              | 194            | 2189                                                                                  | 1751          | 1459          | 1251          | 1094          | 973           | 876            | 796            | 730            | 674            | 625            | 584            | 547            | 4.94                              |
|              | 226            | 2630                                                                                  | 2104          | 1754          | 1503          | 1315          | 1169          | 1052           | 957            | 877            | 809            | 752            | 701            | 658            | 5.29                              |
| 914 x 305    | 201            | 2373                                                                                  | 1899          | 1582          | 1356          | 1187          | 1055          | 949            | 863            | 791            | 730            | 678            | 633            | 593            | 4.86                              |
|              | 224            | 2720                                                                                  | 2176          | 1813          | 1554          | 1360          | 1209          | 1088           | 989            | 907            | 837            | 777            | 725            | 680            | 5.14                              |
|              | 253            | 3132                                                                                  | 2505          | 2088          | 1789          | 1566          | 1392          | 1253           | 1139           | 1044           | 964            | 895            | 835            | 783            | 5.39                              |
|              | 289            | 3588                                                                                  | 2871          | 2392          | 2051          | 1794          | 1595          | 1435           | 1305           | 1196           | 1104           | 1025           | 957            | 897            | 5.60                              |
| 914 x 419    | 343            |                                                                                       | 3536†         | 3012          | 2582          | 2259          | 2008          | 1807           | 1643           | 1506           | 1390           | 1291           | 1205           | 1130           | 8.08                              |
|              | 388            |                                                                                       | 3958†         | 3429          | 2939          | 2572          | 2286          | 2057           | 1870           | 1714           | 1583           | 1470           | 1372           | 1286           | 8.44                              |

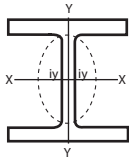
The loads listed are based on a bending stress of 165 N/mm<sup>2</sup> and assume adequate lateral support. Without such support the span must not exceed L<sub>c</sub> unless the compressive stress is reduced in accordance with clause 19a of BS 449.

Loads printed in bold type may cause overloading of the unstiffened web, the capacity of which should be checked.

# UNIVERSAL BEAMS AND COLUMNS



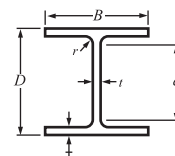
| Section size            | Unit weight<br>M |       | Section area<br>A |       | Section depth<br>D |       | Flange width<br>B |       | Flange thickness<br>T |       | Web thickness<br>t |       | Corner radius<br>r |      |
|-------------------------|------------------|-------|-------------------|-------|--------------------|-------|-------------------|-------|-----------------------|-------|--------------------|-------|--------------------|------|
| in (mm)                 | lb/ft            | kg/m  | in²               | cm²   | in                 | mm    | in                | mm    | in                    | mm    | in                 | mm    | in                 | mm   |
| W4 4 x 4 (102 x 102)    | 13               | 19.35 | 3.83              | 24.71 | 4.16               | 105.7 | 4.06              | 103.1 | 0.345                 | 8.76  | 0.28               | 7.11  | 0.25               | 6.35 |
| W5 5 x 3 (127 x 76)     | 9                | 13.00 | 2.60              | 16.80 | 5.00               | 127.0 | 3.00              | 76.2  | 0.299                 | 7.60  | 0.165              | 4.20  | 0.30               | 6.35 |
| W5 5 x 5 (127 x 127)    | 16               | 23.81 | 4.71              | 30.39 | 5.01               | 127.3 | 5.00              | 127.0 | 0.36                  | 9.14  | 0.24               | 6.10  | 0.30               | 7.62 |
|                         | 19               | 28.28 | 5.56              | 35.87 | 5.15               | 130.8 | 5.03              | 127.8 | 0.43                  | 10.90 | 0.27               | 6.86  | 0.30               | 7.62 |
| W6 6 x 3½ (152 x 89)    | 11               | 16.00 | 3.18              | 20.50 | 6.00               | 152.4 | 3.50              | 88.9  | 0.303                 | 7.70  | 0.181              | 4.60  | 0.25               | 6.35 |
| W6 6 x 4 (152 x 102)    | 9                | 13.39 | 2.68              | 17.29 | 5.90               | 149.9 | 3.94              | 100.1 | 0.215                 | 5.46  | 0.17               | 4.32  | 0.25               | 6.35 |
|                         | 12               | 17.86 | 3.55              | 22.90 | 6.03               | 153.2 | 4.00              | 101.6 | 0.28                  | 7.11  | 0.23               | 5.84  | 0.25               | 6.35 |
|                         | 16               | 23.81 | 4.74              | 30.58 | 6.28               | 159.5 | 4.03              | 102.4 | 0.405                 | 10.20 | 0.26               | 6.60  | 0.25               | 6.35 |
| W6 6 x 6 (152 x 152)    | 15               | 22.50 | 4.43              | 28.58 | 5.99               | 152.1 | 5.99              | 152.1 | 0.26                  | 6.60  | 0.23               | 5.84  | 0.25               | 6.35 |
|                         | 15.7             | 23.00 | 4.62              | 29.80 | 6.00               | 152.4 | 6.00              | 152.4 | 0.269                 | 6.80  | 0.24               | 6.10  | 0.25               | 6.35 |
|                         | 20               | 30.00 | 5.87              | 37.87 | 6.20               | 157.5 | 6.02              | 152.9 | 0.365                 | 9.27  | 0.26               | 6.60  | 0.25               | 6.35 |
|                         | 25               | 37.20 | 7.34              | 47.35 | 6.38               | 162.1 | 6.08              | 154.4 | 0.455                 | 11.50 | 0.32               | 8.13  | 0.25               | 6.35 |
| W7 7 x 4 (178 x 102)    | 13               | 19.00 | 3.75              | 24.20 | 7.00               | 177.8 | 4.00              | 101.6 | 0.311                 | 7.90  | 0.185              | 4.70  | 0.30               | 7.62 |
| W8 8 x 4 (203 x 102)    | 10               | 14.88 | 2.96              | 19.10 | 7.89               | 200.4 | 3.94              | 100.1 | 0.205                 | 5.21  | 0.17               | 4.32  | 0.30               | 7.62 |
|                         | 13               | 19.35 | 3.84              | 24.77 | 7.99               | 202.9 | 4.00              | 101.6 | 0.255                 | 6.48  | 0.23               | 5.84  | 0.30               | 7.62 |
|                         | 15               | 22.32 | 4.44              | 28.65 | 8.11               | 206.0 | 4.015             | 102.0 | 0.315                 | 8.00  | 0.245              | 6.22  | 0.30               | 7.62 |
|                         | 15.5             | 23.00 | 4.50              | 29.00 | 8.00               | 203.2 | 4.00              | 101.6 | 0.368                 | 9.30  | 0.205              | 5.20  | 0.30               | 7.62 |
| W8 8 x 5¼ (203 x 133)   | 17               | 25.30 | 5.01              | 32.30 | 8.00               | 203.2 | 5.25              | 133.4 | 0.308                 | 7.80  | 0.23               | 5.84  | 0.30               | 7.62 |
|                         | 18               | 26.79 | 5.26              | 33.94 | 8.14               | 206.8 | 5.25              | 133.3 | 0.33                  | 8.38  | 0.23               | 5.84  | 0.30               | 7.62 |
|                         | 20               | 30.00 | 5.89              | 38.00 | 8.14               | 206.8 | 5.268             | 133.8 | 0.378                 | 9.60  | 0.248              | 6.30  | 0.30               | 7.62 |
|                         | 21               | 31.30 | 6.16              | 39.74 | 8.28               | 210.3 | 5.27              | 133.9 | 0.40                  | 10.10 | 0.25               | 6.35  | 0.30               | 7.62 |
| W8 8 x 6½ (203 x 165)   | 24               | 35.72 | 7.08              | 45.68 | 7.93               | 201.4 | 6.495             | 165.0 | 0.40                  | 10.10 | 0.245              | 6.22  | 0.40               | 10.2 |
|                         | 28               | 41.67 | 8.25              | 53.23 | 8.06               | 204.7 | 6.535             | 166.0 | 0.465                 | 11.80 | 0.285              | 7.24  | 0.40               | 10.2 |
| W8 8 x 8 (203 x 203)    | 31               | 46.13 | 9.12              | 58.84 | 8.00               | 203.2 | 7.995             | 203.1 | 0.435                 | 11.00 | 0.285              | 7.24  | 0.40               | 10.2 |
|                         | 35               | 52.09 | 10.3              | 66.45 | 8.12               | 206.2 | 8.02              | 203.7 | 0.495                 | 12.50 | 0.31               | 7.87  | 0.40               | 10.2 |
|                         | 40               | 60.00 | 11.7              | 75.48 | 8.25               | 209.5 | 8.07              | 205.0 | 0.56                  | 14.20 | 0.36               | 9.14  | 0.40               | 10.2 |
|                         | 48               | 71.43 | 14.1              | 90.97 | 8.50               | 215.9 | 8.11              | 206.0 | 0.685                 | 17.40 | 0.40               | 10.16 | 0.40               | 10.2 |
|                         | 58               | 86.31 | 17.1              | 110.3 | 8.75               | 222.2 | 8.22              | 208.8 | 0.81                  | 20.57 | 0.51               | 12.95 | 0.40               | 10.2 |
|                         | 67               | 100.0 | 19.7              | 127.1 | 9.00               | 228.6 | 8.28              | 210.3 | 0.935                 | 23.75 | 0.57               | 14.48 | 0.40               | 10.2 |
| W10 10 x 4 (254 x 102)  | 12               | 17.86 | 3.54              | 22.84 | 9.87               | 250.7 | 3.96              | 100.6 | 0.21                  | 5.33  | 0.19               | 4.83  | 0.30               | 7.62 |
|                         | 15               | 22.32 | 4.41              | 28.45 | 9.99               | 253.7 | 4.00              | 101.6 | 0.27                  | 6.86  | 0.23               | 5.84  | 0.30               | 7.62 |
|                         | 17               | 25.30 | 4.99              | 32.19 | 10.11              | 256.8 | 4.01              | 101.9 | 0.33                  | 8.38  | 0.24               | 6.10  | 0.30               | 7.62 |
|                         | 19               | 28.28 | 5.62              | 36.26 | 10.24              | 260.1 | 4.02              | 102.1 | 0.395                 | 10.00 | 0.25               | 6.35  | 0.30               | 7.62 |
| W10 10 x 5¾ (254 x 146) | 21               | 31.25 | 6.20              | 40.00 | 9.90               | 251.5 | 5.75              | 146.1 | 0.34                  | 8.60  | 0.24               | 6.10  | 0.30               | 7.62 |
|                         | 22               | 32.74 | 6.49              | 41.87 | 10.17              | 258.3 | 5.75              | 146.0 | 0.36                  | 9.14  | 0.24               | 6.10  | 0.30               | 7.62 |
|                         | 25               | 37.20 | 7.36              | 47.50 | 10.08              | 256.0 | 5.762             | 146.4 | 0.43                  | 10.90 | 0.252              | 6.40  | 0.30               | 7.62 |
|                         | 26               | 38.69 | 7.61              | 49.10 | 10.33              | 262.4 | 5.77              | 146.6 | 0.44                  | 11.10 | 0.26               | 6.60  | 0.30               | 7.62 |
|                         | 29               | 43.00 | 8.54              | 55.10 | 10.22              | 259.6 | 5.799             | 147.3 | 0.50                  | 12.70 | 0.289              | 7.30  | 0.30               | 7.62 |
|                         | 30               | 44.80 | 8.85              | 57.10 | 10.47              | 265.9 | 5.819             | 147.8 | 0.51                  | 12.90 | 0.30               | 7.62  | 0.30               | 7.62 |
| W10 10 x 8 (254 x 203)  | 33               | 49.11 | 9.71              | 62.64 | 9.73               | 247.1 | 7.96              | 202.2 | 0.435                 | 11.00 | 0.29               | 7.37  | 0.50               | 12.7 |
|                         | 39               | 58.04 | 11.5              | 74.19 | 9.92               | 252.0 | 7.985             | 202.8 | 0.53                  | 13.40 | 0.315              | 8.00  | 0.50               | 12.7 |
|                         | 45               | 66.97 | 13.3              | 85.81 | 10.10              | 256.5 | 8.02              | 203.7 | 0.62                  | 15.70 | 0.35               | 8.89  | 0.50               | 12.7 |



# UNIVERSAL BEAMS AND COLUMNS

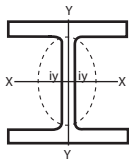
| Section<br>Size            | Moment of inertia |                 |                 |                 | Radius of gyration |      |                |      | Modulus of section |                 |                 |                 |
|----------------------------|-------------------|-----------------|-----------------|-----------------|--------------------|------|----------------|------|--------------------|-----------------|-----------------|-----------------|
|                            | I <sub>x</sub>    |                 | I <sub>y</sub>  |                 | i <sub>x</sub>     |      | i <sub>y</sub> |      | Z <sub>x</sub>     |                 | Z <sub>y</sub>  |                 |
| in (mm)                    | in <sup>4</sup>   | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in                 | cm   | in             | cm   | in <sup>3</sup>    | cm <sup>3</sup> | in <sup>3</sup> | cm <sup>3</sup> |
| W4 4 x 4 (102 x 102)       | 11.3              | 470             | 3.86            | 161             | 1.72               | 4.37 | 1.00           | 2.54 | 5.46               | 89.5            | 1.90            | 31.1            |
| W5 5 x 3 (127 x 76)        | 11.5              | 477             | 1.30            | 56              | 2.10               | 5.33 | 0.72           | 1.83 | 4.60               | 75              | 0.90            | 15.0            |
| W5 5 x 5 (127 x 127)       | 21.4              | 891             | 7.51            | 313             | 2.13               | 5.41 | 1.26           | 3.20 | 8.55               | 140             | 3.00            | 49.2            |
|                            | 26.3              | 1,090           | 9.13            | 380             | 2.17               | 5.51 | 1.28           | 3.25 | 10.2               | 167             | 3.63            | 59.5            |
| W6 6 x 3 1/2 (152 x 89)    | 20.1              | 838             | 2.20            | 90              | 2.52               | 6.40 | 0.83           | 2.10 | 6.70               | 110             | 1.20            | 20.0            |
| W6 6 x 4 (152 x 102)       | 16.4              | 683             | 2.20            | 91.6            | 2.47               | 6.27 | 0.905          | 2.30 | 5.56               | 91.1            | 1.11            | 18.2            |
|                            | 22.1              | 920             | 2.99            | 124             | 2.49               | 6.32 | 0.918          | 2.33 | 7.31               | 120             | 1.50            | 24.6            |
|                            | 32.1              | 1,340           | 4.43            | 184             | 2.60               | 6.60 | 0.966          | 2.45 | 10.2               | 167             | 2.20            | 36.1            |
| W6 6 x 6 (152 x 152)       | 29.1              | 1,210           | 9.32            | 388             | 2.56               | 6.50 | 1.45           | 3.68 | 9.72               | 159             | 3.11            | 51.0            |
|                            | 30.3              | 1,263           | 9.7             | 403             | 2.56               | 6.51 | 1.45           | 3.68 | 10.1               | 165.7           | 3.20            | 52.95           |
|                            | 41.4              | 1,720           | 13.3            | 554             | 2.66               | 6.76 | 1.50           | 3.81 | 13.4               | 220             | 4.41            | 72.3            |
|                            | 53.4              | 2,220           | 17.1            | 712             | 2.70               | 6.86 | 1.52           | 3.86 | 16.7               | 274             | 5.61            | 91.9            |
| W7 7 x 4 (178 x 102)       | 32.6              | 1,357           | 3.30            | 138             | 2.95               | 7.49 | 0.940          | 2.39 | 9.30               | 153             | 1.60            | 27.0            |
| W8 8 x 4 (203 x 102)       | 30.8              | 1,280           | 2.09            | 87              | 3.22               | 8.18 | 0.841          | 2.14 | 7.81               | 128             | 1.06            | 17.4            |
|                            | 39.6              | 1,650           | 2.73            | 114             | 3.21               | 8.15 | 0.843          | 2.14 | 9.91               | 162             | 1.37            | 22.4            |
|                            | 48.0              | 2,000           | 3.41            | 142             | 3.29               | 8.36 | 0.876          | 2.22 | 11.8               | 193             | 1.70            | 27.9            |
|                            | 50.2              | 2,091           | 3.90            | 163             | 3.34               | 8.49 | 0.93           | 2.37 | 12.6               | 206             | 2.00            | 32.0            |
| W8 8 x 5 1/4 (203 x 133)   | 56.4              | 2,348           | 6.72            | 280             | 3.36               | 8.53 | 1.16           | 2.95 | 14.1               | 231             | 2.60            | 42.6            |
|                            | 61.9              | 2,580           | 7.97            | 332             | 3.43               | 8.71 | 1.23           | 3.12 | 15.2               | 249             | 3.04            | 49.8            |
|                            | 69.2              | 2,981           | 8.50            | 354             | 3.43               | 8.71 | 1.20           | 3.05 | 17.0               | 279             | 3.20            | 52.4            |
|                            | 75.3              | 3,130           | 9.77            | 407             | 3.49               | 8.86 | 1.26           | 3.20 | 18.2               | 298             | 3.71            | 60.8            |
| W8 8 x 6 1/2 (203 x 165)   | 82.8              | 3,450           | 18.3            | 762             | 3.42               | 8.69 | 1.61           | 4.09 | 20.9               | 342             | 5.63            | 92.3            |
|                            | 98.0              | 4,080           | 21.7            | 903             | 3.45               | 8.76 | 1.62           | 4.11 | 24.3               | 398             | 6.63            | 109             |
| W8 8 x 8 (203 x 203)       | 110               | 4,580           | 37.1            | 1,540           | 3.47               | 8.81 | 2.02           | 5.13 | 27.5               | 451             | 9.27            | 152             |
|                            | 127               | 5,290           | 42.6            | 1,770           | 3.51               | 8.92 | 2.03           | 5.16 | 31.2               | 511             | 10.6            | 174             |
|                            | 146               | 6,080           | 49.1            | 2,040           | 3.53               | 8.97 | 2.04           | 5.18 | 35.5               | 582             | 12.2            | 200             |
|                            | 184               | 7,660           | 60.9            | 2,530           | 3.61               | 9.17 | 2.08           | 5.28 | 43.3               | 710             | 15.0            | 246             |
|                            | 228               | 9,490           | 75.1            | 3,130           | 3.65               | 9.27 | 2.10           | 5.33 | 52.0               | 852             | 18.3            | 300             |
|                            | 272               | 11,300          | 88.6            | 3,690           | 3.72               | 9.45 | 2.12           | 5.38 | 60.4               | 990             | 21.4            | 351             |
| W10 10 x 4 (254 x 102)     | 53.8              | 2,240           | 2.18            | 90.7            | 3.90               | 9.91 | 0.785          | 1.99 | 10.9               | 179             | 1.10            | 18.0            |
|                            | 68.9              | 2,870           | 2.89            | 120             | 3.95               | 10.0 | 0.81           | 2.06 | 13.8               | 226             | 1.45            | 23.8            |
|                            | 81.9              | 3,410           | 3.56            | 148             | 4.05               | 10.3 | 0.845          | 2.15 | 16.2               | 265             | 1.78            | 29.2            |
|                            | 96.3              | 4,010           | 4.29            | 179             | 4.14               | 10.5 | 0.874          | 2.22 | 18.8               | 308             | 2.14            | 35.1            |
| W10 10 x 5 3/4 (254 x 146) | 106               | 4,425           | 9.7             | 404             | 4.14               | 10.5 | 1.25           | 3.18 | 21.5               | 352             | 3.40            | 55.7            |
|                            | 118               | 4,910           | 11.4            | 474             | 4.27               | 10.8 | 1.33           | 3.38 | 23.2               | 380             | 3.97            | 65.1            |
|                            | 133               | 5,545           | 12.7            | 529             | 4.26               | 10.8 | 1.31           | 3.33 | 26.4               | 433             | 4.40            | 72.1            |
|                            | 144               | 5,990           | 14.1            | 587             | 4.35               | 11.0 | 1.36           | 3.45 | 27.9               | 457             | 4.89            | 80.1            |
|                            | 157               | 6,548           | 15.2            | 633             | 4.29               | 10.9 | 1.34           | 3.40 | 30.8               | 505             | 5.20            | 85.2            |
|                            | 170               | 7,080           | 16.8            | 699             | 4.38               | 11.1 | 1.38           | 3.51 | 32.5               | 533             | 5.76            | 94.4            |
| W10 10 x 8 (254 x 203)     | 170               | 7,080           | 36.6            | 1,520           | 4.19               | 10.6 | 1.94           | 4.93 | 35.0               | 574             | 9.20            | 151             |
|                            | 209               | 8,700           | 45.0            | 1,870           | 4.27               | 10.8 | 1.98           | 5.03 | 42.1               | 690             | 11.3            | 185             |
|                            | 248               | 10,300          | 53.4            | 2,220           | 4.32               | 11.0 | 2.01           | 5.11 | 49.1               | 805             | 13.3            | 218             |

# UNIVERSAL BEAMS AND COLUMNS



| Section size               | Unit weight<br>M |       | Section area<br>A |                 | Section depth<br>D |       | Flange width<br>B |       | Flange thickness<br>T |       | Web thickness<br>t |       | Corner radius<br>r |      |
|----------------------------|------------------|-------|-------------------|-----------------|--------------------|-------|-------------------|-------|-----------------------|-------|--------------------|-------|--------------------|------|
| in (mm)                    | lb/ft            | kg/m  | in <sup>2</sup>   | cm <sup>2</sup> | in                 | mm    | in                | mm    | in                    | mm    | in                 | mm    | in                 | mm   |
| W10 10 x 10 (254 x 254)    | 49               | 73.00 | 14.4              | 92.9            | 9.98               | 253.5 | 10.00             | 254.0 | 0.56                  | 14.20 | 0.34               | 8.64  | 0.50               | 12.7 |
|                            | 54               | 80.36 | 15.8              | 101.9           | 10.09              | 256.3 | 10.03             | 254.8 | 0.615                 | 15.60 | 0.37               | 9.40  | 0.50               | 12.7 |
|                            | 60               | 89.29 | 17.6              | 113.5           | 10.22              | 259.6 | 10.08             | 256.0 | 0.68                  | 17.27 | 0.42               | 10.67 | 0.50               | 12.7 |
|                            | 68               | 101.2 | 20.0              | 129.0           | 10.40              | 264.2 | 10.13             | 257.3 | 0.77                  | 19.56 | 0.47               | 11.94 | 0.50               | 12.7 |
|                            | 72               | 107.0 | 21.2              | 136.6           | 10.50              | 266.7 | 10.17             | 258.3 | 0.808                 | 20.50 | 0.51               | 13.00 | 0.50               | 12.7 |
|                            | 77               | 115.0 | 22.6              | 145.8           | 10.60              | 269.2 | 10.19             | 258.8 | 0.87                  | 22.10 | 0.53               | 13.46 | 0.50               | 12.7 |
|                            | 88               | 131.0 | 25.9              | 167.1           | 10.84              | 275.3 | 10.265            | 260.7 | 0.99                  | 25.15 | 0.605              | 15.37 | 0.50               | 12.7 |
|                            | 89               | 132.0 | 26.2              | 168.9           | 10.88              | 276.4 | 10.275            | 261.0 | 0.998                 | 25.30 | 0.615              | 15.60 | 0.50               | 12.7 |
|                            | 100              | 149.0 | 29.4              | 189.7           | 11.11              | 282.2 | 10.34             | 262.6 | 1.12                  | 28.45 | 0.68               | 17.27 | 0.50               | 12.7 |
|                            | 112              | 167.0 | 32.9              | 212.3           | 11.36              | 288.5 | 10.415            | 264.5 | 1.25                  | 31.75 | 0.755              | 19.18 | 0.50               | 12.7 |
| W12 12 x 4 (305 x 102)     | 14               | 20.83 | 4.16              | 26.84           | 11.91              | 302.5 | 3.97              | 100.8 | 0.225                 | 5.71  | 0.20               | 5.08  | 0.30               | 7.62 |
|                            | 16               | 23.81 | 4.71              | 30.39           | 11.99              | 304.5 | 3.99              | 101.3 | 0.265                 | 6.73  | 0.22               | 5.59  | 0.30               | 7.62 |
|                            | 16.5             | 25.00 | 4.87              | 31.40           | 12.00              | 304.8 | 4.00              | 101.6 | 0.269                 | 6.80  | 0.23               | 5.80  | 0.30               | 7.62 |
|                            | 19               | 28.28 | 5.57              | 35.94           | 12.16              | 308.9 | 4.005             | 101.7 | 0.35                  | 8.89  | 0.235              | 5.97  | 0.30               | 7.62 |
|                            | 22               | 33.00 | 6.48              | 41.81           | 12.31              | 312.7 | 4.03              | 102.4 | 0.425                 | 10.80 | 0.26               | 6.60  | 0.30               | 7.62 |
| W12 12 x 5 (305 x 127)     | 25               | 37.00 | 7.36              | 47.50           | 11.96              | 303.8 | 4.864             | 123.5 | 0.421                 | 10.70 | 0.284              | 7.20  | 0.30               | 7.62 |
|                            | 28               | 42.00 | 8.25              | 53.20           | 12.07              | 306.6 | 4.893             | 124.3 | 0.476                 | 12.10 | 0.313              | 8.00  | 0.30               | 7.62 |
|                            | 32               | 48.00 | 9.42              | 60.80           | 12.22              | 310.4 | 4.93              | 125.2 | 0.551                 | 14.00 | 0.350              | 8.90  | 0.30               | 7.62 |
| W12 12 x 6 1/2 (305 x 165) | 26               | 38.70 | 7.65              | 49.35           | 12.22              | 310.4 | 6.49              | 164.8 | 0.38                  | 9.65  | 0.23               | 5.84  | 0.30               | 7.62 |
|                            | 27               | 40.00 | 7.98              | 51.50           | 11.96              | 303.8 | 6.50              | 165.1 | 0.40                  | 10.20 | 0.24               | 6.10  | 0.30               | 7.62 |
|                            | 30               | 44.64 | 8.79              | 56.71           | 12.34              | 313.4 | 6.52              | 165.6 | 0.44                  | 11.10 | 0.26               | 6.60  | 0.30               | 7.62 |
|                            | 31               | 46.00 | 9.13              | 58.90           | 12.09              | 307.1 | 6.525             | 165.7 | 0.465                 | 11.80 | 0.265              | 6.70  | 0.30               | 7.62 |
|                            | 35               | 52.09 | 10.3              | 66.45           | 12.50              | 317.5 | 6.56              | 166.6 | 0.52                  | 13.20 | 0.30               | 7.62  | 0.30               | 7.62 |
|                            | 36               | 54.00 | 10.6              | 68.40           | 12.24              | 310.9 | 6.565             | 166.8 | 0.54                  | 13.70 | 0.305              | 7.70  | 0.30               | 7.62 |
| W12 12 x 8 (305 x 203)     | 40               | 59.53 | 11.8              | 76.13           | 11.94              | 303.3 | 8.005             | 203.3 | 0.515                 | 13.00 | 0.295              | 7.49  | 0.60               | 15.2 |
|                            | 45               | 66.97 | 13.2              | 85.16           | 12.06              | 306.3 | 8.045             | 204.3 | 0.575                 | 14.60 | 0.335              | 8.51  | 0.60               | 15.2 |
|                            | 50               | 74.41 | 14.7              | 94.84           | 12.19              | 309.6 | 8.08              | 205.2 | 0.64                  | 16.20 | 0.37               | 9.40  | 0.60               | 15.2 |
| W12 12 x 10 (305 x 254)    | 53               | 78.87 | 15.6              | 100.6           | 12.06              | 306.3 | 9.995             | 253.9 | 0.575                 | 14.60 | 0.345              | 8.76  | 0.60               | 15.2 |
|                            | 58               | 86.31 | 17.0              | 109.7           | 12.19              | 309.6 | 10.01             | 254.3 | 0.64                  | 16.20 | 0.36               | 9.14  | 0.60               | 15.2 |
| W12 12 x 12 (305 x 305)    | 65               | 97.0  | 19.1              | 123.2           | 12.12              | 307.8 | 12.00             | 304.8 | 0.605                 | 15.30 | 0.39               | 9.91  | 0.60               | 15.2 |
|                            | 72               | 107.1 | 21.1              | 136.1           | 12.25              | 311.1 | 12.04             | 305.8 | 0.67                  | 17.02 | 0.43               | 10.92 | 0.60               | 15.2 |
|                            | 79               | 118.0 | 23.2              | 149.7           | 12.38              | 314.5 | 12.08             | 306.8 | 0.735                 | 18.67 | 0.47               | 11.94 | 0.60               | 15.2 |
|                            | 87               | 129.5 | 25.6              | 165.2           | 12.53              | 318.3 | 12.125            | 308.0 | 0.81                  | 20.57 | 0.515              | 13.08 | 0.60               | 15.2 |
|                            | 92               | 137.0 | 27.1              | 174.6           | 12.62              | 320.4 | 12.155            | 308.7 | 0.856                 | 21.70 | 0.545              | 13.80 | 0.60               | 15.2 |
|                            | 96               | 143.0 | 28.2              | 182.0           | 12.71              | 322.8 | 12.16             | 308.9 | 0.90                  | 22.86 | 0.55               | 13.97 | 0.60               | 15.2 |
|                            | 106              | 158.0 | 31.2              | 201.3           | 12.89              | 327.4 | 12.22             | 310.4 | 0.99                  | 25.15 | 0.61               | 15.49 | 0.60               | 15.2 |
|                            | 120              | 179.0 | 35.3              | 227.7           | 13.12              | 333.2 | 12.32             | 312.9 | 1.10                  | 28.07 | 0.71               | 18.03 | 0.60               | 15.2 |
|                            | 133              | 198.0 | 39.1              | 252.3           | 13.38              | 339.9 | 12.365            | 314.1 | 1.236                 | 31.40 | 0.755              | 19.20 | 0.60               | 15.2 |
|                            | 136              | 202.4 | 39.9              | 257.4           | 13.41              | 340.6 | 12.40             | 315.0 | 1.25                  | 31.75 | 0.79               | 20.07 | 0.60               | 15.2 |
|                            | 152              | 226.2 | 44.7              | 288.4           | 13.71              | 348.2 | 12.48             | 317.0 | 1.40                  | 35.56 | 0.87               | 22.10 | 0.60               | 15.2 |
|                            | 161              | 240.0 | 47.4              | 305.6           | 13.88              | 352.6 | 12.515            | 317.9 | 1.486                 | 37.70 | 0.905              | 23.00 | 0.60               | 15.2 |
|                            | 170              | 253.0 | 50.0              | 322.6           | 14.03              | 356.4 | 12.57             | 319.3 | 1.56                  | 39.62 | 0.96               | 24.38 | 0.60               | 15.2 |
|                            | 190              | 283.0 | 55.8              | 360.0           | 14.38              | 365.3 | 12.67             | 321.8 | 1.735                 | 44.07 | 1.06               | 26.92 | 0.60               | 15.2 |
|                            | 210              | 312.5 | 61.8              | 398.7           | 14.71              | 373.6 | 12.79             | 324.9 | 1.90                  | 48.26 | 1.18               | 29.97 | 0.60               | 15.2 |
|                            | 230              | 342.3 | 67.7              | 436.8           | 15.05              | 382.3 | 12.895            | 327.5 | 2.07                  | 52.58 | 1.285              | 32.64 | 0.60               | 15.2 |
|                            | 252              | 375.0 | 74.0              | 477.4           | 15.41              | 391.4 | 13.005            | 330.3 | 2.25                  | 57.15 | 1.395              | 35.43 | 0.60               | 15.2 |
|                            | 278              | 413.7 | 81.9              | 528.4           | 15.85              | 402.6 | 13.14             | 333.8 | 2.47                  | 62.74 | 1.53               | 38.86 | 0.60               | 15.2 |
|                            | 305              | 453.9 | 89.6              | 578.1           | 16.32              | 414.5 | 13.235            | 336.2 | 2.705                 | 68.71 | 1.625              | 41.27 | 0.60               | 15.2 |
|                            | 336              | 500.0 | 98.8              | 637.4           | 16.82              | 427.2 | 13.385            | 340.0 | 2.955                 | 75.06 | 1.775              | 45.08 | 0.60               | 15.2 |
| W14 14 x 5 (356 x 127)     | 22               | 33.0  | 6.49              | 41.87           | 13.74              | 349.0 | 5.00              | 127.0 | 0.335                 | 8.51  | 0.23               | 5.84  | 0.40               | 10.2 |
|                            | 26               | 39.0  | 7.69              | 49.61           | 13.91              | 353.3 | 5.025             | 127.6 | 0.42                  | 10.60 | 0.255              | 6.48  | 0.40               | 10.2 |



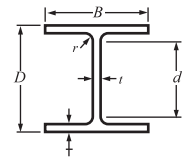


# UNIVERSAL BEAMS AND COLUMNS

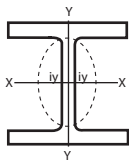
| Section<br>Size            | Moment of inertia |                 |                 |                 | Radius of gyration |      |       |      | Modulus of section |                 |                 |                 |
|----------------------------|-------------------|-----------------|-----------------|-----------------|--------------------|------|-------|------|--------------------|-----------------|-----------------|-----------------|
|                            | Ix                |                 | Iy              |                 | ix                 |      | iy    |      | Zx                 | Zy              |                 |                 |
| in (mm)                    | in <sup>4</sup>   | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in                 | cm   | in    | cm   | in <sup>3</sup>    | cm <sup>3</sup> | in <sup>3</sup> | cm <sup>3</sup> |
| W10 10 x 10 (254 x 254)    | 272               | 11,300          | 93.4            | 3,890           | 4.35               | 11.0 | 2.54  | 6.45 | 54.6               | 895             | 18.7            | 306             |
|                            | 303               | 12,600          | 103             | 4,290           | 4.37               | 11.1 | 2.56  | 6.50 | 60.0               | 983             | 20.6            | 338             |
|                            | 341               | 14,200          | 116             | 4,830           | 4.39               | 11.2 | 2.57  | 6.53 | 66.7               | 1,090           | 23.0            | 377             |
|                            | 394               | 16,400          | 134             | 5,580           | 4.44               | 11.3 | 2.58  | 6.55 | 75.7               | 1,240           | 26.4            | 433             |
|                            | 421               | 17,510          | 142             | 5,903           | 4.46               | 11.3 | 2.59  | 6.58 | 80.1               | 1,312           | 27.9            | 457             |
|                            | 455               | 18,900          | 154             | 6,410           | 4.49               | 11.4 | 2.60  | 6.60 | 85.9               | 1,410           | 30.1            | 493             |
|                            | 534               | 22,200          | 179             | 7,450           | 4.54               | 11.5 | 2.63  | 6.68 | 98.5               | 1,610           | 34.8            | 570             |
|                            | 542               | 22,580          | 181             | 7,518           | 4.55               | 11.6 | 2.63  | 6.68 | 99.7               | 1,634           | 35.2            | 577             |
|                            | 624               | 26,000          | 207             | 8,620           | 4.61               | 11.7 | 2.65  | 6.73 | 112.0              | 1,840           | 40.0            | 655             |
|                            | 716               | 29,800          | 236             | 9,820           | 4.66               | 11.8 | 2.68  | 6.81 | 126.0              | 2,060           | 45.3            | 742             |
| W12 12 x 4 (305 x 102)     | 88.6              | 3,690           | 2.36            | 98.2            | 4.62               | 11.7 | 0.753 | 1.91 | 14.9               | 244             | 1.19            | 19.5            |
|                            | 103               | 4,290           | 2.82            | 117             | 4.67               | 11.9 | 0.773 | 1.96 | 17.1               | 280             | 1.41            | 23.1            |
|                            | 105.4             | 4,387           | 2.9             | 120             | 4.65               | 11.8 | 0.770 | 1.96 | 17.6               | 288             | 1.40            | 23.6            |
|                            | 130               | 5,410           | 3.76            | 156             | 4.82               | 12.2 | 0.822 | 2.09 | 21.3               | 349             | 1.88            | 30.8            |
|                            | 156               | 6,490           | 4.66            | 194             | 4.91               | 12.5 | 0.848 | 2.15 | 25.4               | 416             | 2.31            | 37.9            |
| W12 12 x 5 (305 x 127)     | 172               | 7,162           | 8.1             | 337             | 4.83               | 12.3 | 1.05  | 2.67 | 28.8               | 472             | 3.30            | 54.6            |
|                            | 195.6             | 8,143           | 9.3             | 388             | 4.87               | 12.4 | 1.06  | 2.70 | 32.4               | 531             | 3.80            | 62.5            |
|                            | 228.3             | 9,504           | 11.1            | 460             | 4.92               | 12.5 | 1.08  | 2.75 | 37.4               | 612             | 4.50            | 73.5            |
| W12 12 x 6 1/2 (305 x 165) | 204               | 8,490           | 17.3            | 720             | 5.17               | 13.1 | 1.51  | 3.84 | 33.4               | 547             | 5.34            | 87.5            |
|                            | 204.1             | 8,496           | 16.6            | 691             | 5.06               | 12.8 | 1.44  | 3.66 | 34.1               | 559             | 5.10            | 83.6            |
|                            | 238               | 9,910           | 20.3            | 845             | 5.21               | 13.2 | 1.52  | 3.86 | 38.6               | 633             | 6.24            | 102             |
|                            | 238.4             | 9,924           | 19.8            | 824             | 5.11               | 12.9 | 1.47  | 3.73 | 39.4               | 646             | 6.10            | 100             |
|                            | 280.8             | 11,690          | 23.7            | 986.6           | 5.15               | 13.1 | 1.50  | 3.81 | 45.9               | 752             | 7.20            | 118             |
|                            | 285               | 11,900          | 24.5            | 1,020           | 5.25               | 13.3 | 1.54  | 3.91 | 45.6               | 747             | 7.47            | 122             |
|                            |                   |                 |                 |                 |                    |      |       |      |                    |                 |                 |                 |
| W12 12 x 8 (305 x 203)     | 310               | 12,900          | 44.1            | 1,840           | 5.13               | 13.0 | 1.94  | 4.93 | 51.9               | 850             | 11.0            | 180             |
|                            | 350               | 14,600          | 50.0            | 2,080           | 5.15               | 13.1 | 1.94  | 4.93 | 58.1               | 952             | 12.4            | 203             |
|                            | 394               | 16,400          | 56.3            | 2,340           | 5.18               | 13.2 | 1.96  | 4.98 | 64.7               | 1,060           | 13.9            | 228             |
| W12 12 x 10 (305 x 254)    | 425               | 17,700          | 95.8            | 3,990           | 5.23               | 13.3 | 2.48  | 6.30 | 70.6               | 1,160           | 19.2            | 315             |
|                            | 475               | 19,800          | 107             | 4,450           | 5.28               | 13.4 | 2.51  | 6.38 | 78.0               | 1,280           | 21.4            | 351             |
| W12 12 x 12 (305 x 305)    | 533               | 22,200          | 174             | 7,240           | 5.28               | 13.4 | 3.02  | 7.67 | 87.9               | 1,440           | 29.1            | 477             |
|                            | 597               | 24,800          | 195             | 8,120           | 5.31               | 13.5 | 3.04  | 7.72 | 97.4               | 1,600           | 32.4            | 531             |
|                            | 662               | 27,600          | 216             | 8,990           | 5.34               | 13.6 | 3.05  | 7.75 | 107                | 1,750           | 35.8            | 587             |
|                            | 740               | 30,800          | 241             | 10,000          | 5.38               | 13.7 | 3.07  | 7.80 | 118                | 1,930           | 39.7            | 651             |
|                            | 789               | 32,840          | 256             | 10,670          | 5.40               | 13.7 | 3.08  | 7.82 | 125                | 2,084           | 42.2            | 691             |
|                            | 833               | 34,700          | 270             | 11,200          | 5.44               | 13.8 | 3.09  | 7.85 | 131                | 2,150           | 44.4            | 728             |
|                            | 933               | 38,800          | 301             | 12,500          | 5.47               | 13.9 | 3.11  | 7.90 | 145                | 2,380           | 49.3            | 808             |
|                            | 1,070             | 44,500          | 345             | 14,400          | 5.51               | 14.0 | 3.13  | 7.95 | 163                | 2,670           | 56.0            | 918             |
|                            | 1,221             | 50,840          | 390             | 16,230          | 5.59               | 14.2 | 3.16  | 8.03 | 183                | 2,991           | 63.1            | 1,034           |
|                            | 1,240             | 51,600          | 398             | 16,600          | 5.58               | 14.2 | 3.16  | 8.03 | 186                | 3,050           | 64.2            | 1,050           |
|                            | 1,430             | 59,500          | 454             | 18,900          | 5.66               | 14.4 | 3.19  | 8.10 | 209                | 3,420           | 72.8            | 1,190           |
|                            | 1,542             | 64,180          | 486             | 20,240          | 5.70               | 14.5 | 3.20  | 8.13 | 222                | 3,641           | 77.1            | 1,273           |
|                            | 1,650             | 68,700          | 517             | 21,500          | 5.74               | 14.6 | 3.22  | 8.18 | 235                | 3,850           | 82.3            | 1,350           |
|                            | 1,890             | 78,700          | 589             | 24,500          | 5.82               | 14.8 | 3.25  | 8.25 | 263                | 4,310           | 93.0            | 1,520           |
|                            | 2,140             | 89,100          | 664             | 27,600          | 5.89               | 15.0 | 3.28  | 8.33 | 292                | 4,780           | 104             | 1,700           |
|                            | 2,420             | 101,000         | 742             | 30,900          | 5.97               | 15.2 | 3.31  | 8.41 | 321                | 5,260           | 115             | 1,880           |
|                            | 2,720             | 113,000         | 828             | 34,500          | 6.06               | 15.4 | 3.34  | 8.48 | 353                | 5,780           | 127             | 2,080           |
|                            | 3,110             | 129,000         | 937             | 39,000          | 6.16               | 15.6 | 3.38  | 8.59 | 393                | 6,440           | 143             | 2,340           |
|                            | 3,550             | 148,000         | 1,050           | 43,700          | 6.29               | 16.0 | 3.42  | 8.69 | 435                | 7,130           | 159             | 2,610           |
|                            | 4,060             | 169,000         | 1,190           | 49,500          | 6.41               | 16.3 | 3.47  | 8.81 | 483                | 7,910           | 177             | 2,900           |
|                            |                   |                 |                 |                 |                    |      |       |      |                    |                 |                 |                 |
| W14 14 x 5 (356 x 127)     | 199               | 8,280           | 7.00            | 291             | 5.54               | 14.1 | 1.04  | 2.64 | 29.0               | 475             | 2.80            | 45.9            |
|                            | 245               | 10,200          | 8.91            | 371             | 5.65               | 14.4 | 1.08  | 2.74 | 35.3               | 578             | 3.55            | 58.2            |



# UNIVERSAL BEAMS AND COLUMNS



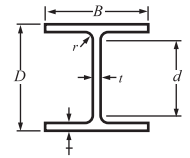
| Section size                                             | Unit weight<br>M |         | Section area<br>A |                 | Section depth<br>D |        | Flange width<br>B |       | Flange thickness<br>T |       | Web thickness<br>t |       | Corner radius<br>r |      |
|----------------------------------------------------------|------------------|---------|-------------------|-----------------|--------------------|--------|-------------------|-------|-----------------------|-------|--------------------|-------|--------------------|------|
| in (mm)                                                  | lb/ft            | kg/m    | in <sup>2</sup>   | cm <sup>2</sup> | in                 | mm     | in                | mm    | in                    | mm    | in                 | mm    | in                 | mm   |
| W14 14 x 6 <sup>3</sup> / <sub>4</sub><br>(356 x 171)34  | 30               | 45.0    | 8.85              | 57.10           | 13.84              | 351.5  | 6.73              | 170.9 | 0.385                 | 9.78  | 0.27               | 6.86  | 0.40               | 10.2 |
|                                                          | 51.0             | 10.0    | 64.52             | 13.98           | 355.1              | 6.745  | 171.3             | 0.455 | 11.5                  | 0.285 | 7.24               | 0.40  | 10.2               |      |
|                                                          | 38               | 57.0    | 11.2              | 72.26           | 14.10              | 358.1  | 6.77              | 172.0 | 0.515                 | 13.0  | 0.31               | 7.87  | 0.40               | 10.2 |
|                                                          | 45               | 67.0    | 13.2              | 85.40           | 14.33              | 364.0  | 6.82              | 173.2 | 0.618                 | 15.7  | 0.357              | 9.10  | 0.40               | 10.2 |
| W14 14 x 8<br>(356 x 203)48                              | 43               | 63.99   | 12.6              | 81.29           | 13.66              | 347.0  | 7.995             | 203.1 | 0.53                  | 13.4  | 0.305              | 7.75  | 0.60               | 15.2 |
|                                                          | 71.43            | 14.1    | 90.97             | 13.79           | 350.3              | 8.03   | 204.0             | 0.595 | 15.1                  | 0.34  | 8.64               | 0.60  | 15.2               |      |
|                                                          | 53               | 78.87   | 15.6              | 100.6           | 13.92              | 353.6  | 8.06              | 204.7 | 0.66                  | 16.7  | 0.37               | 9.40  | 0.60               | 15.2 |
| W14 14 x 10<br>(356 x 254)68                             | 61               | 90.78   | 17.9              | 115.5           | 13.89              | 352.8  | 9.995             | 253.9 | 0.645                 | 16.30 | 0.375              | 9.52  | 0.60               | 15.2 |
|                                                          | 101.2            | 20.0    | 129.0             | 14.04           | 356.6              | 10.035 | 254.9             | 0.723 | 18.36                 | 0.415 | 10.54              | 0.60  | 15.2               |      |
|                                                          | 74               | 110.1   | 21.8              | 140.6           | 14.17              | 359.9  | 10.07             | 255.8 | 0.785                 | 19.94 | 0.45               | 11.43 | 0.60               | 15.2 |
|                                                          | 82               | 122.0   | 24.1              | 155.5           | 14.31              | 363.5  | 10.13             | 257.3 | 0.855                 | 21.72 | 0.51               | 12.95 | 0.60               | 15.2 |
| W14 14 x 14 <sup>1</sup> / <sub>2</sub><br>(356 x 368)90 | 87               | 129.0   | 25.6              | 164.9           | 14.00              | 355.6  | 14.50             | 368.3 | 0.688                 | 17.50 | 0.42               | 10.70 | 0.60               | 15.2 |
|                                                          | 134.0            | 26.5    | 171.0             | 14.02           | 356.1              | 14.52  | 368.8             | 0.71  | 18.03                 | 0.44  | 11.18              | 0.60  | 15.2               |      |
|                                                          | 99               | 147.3   | 29.1              | 187.7           | 14.16              | 359.7  | 14.565            | 370.0 | 0.78                  | 19.81 | 0.485              | 12.32 | 0.60               | 15.2 |
|                                                          | 103              | 153.0   | 30.3              | 195.2           | 14.25              | 362.0  | 14.575            | 370.2 | 0.813                 | 20.70 | 0.495              | 12.60 | 0.60               | 15.2 |
|                                                          | 109              | 162.2   | 32.0              | 206.5           | 14.32              | 363.7  | 14.605            | 371.0 | 0.86                  | 21.84 | 0.525              | 13.33 | 0.60               | 15.2 |
|                                                          | 119              | 177.0   | 35.0              | 225.7           | 14.50              | 368.3  | 14.65             | 372.1 | 0.938                 | 23.80 | 0.57               | 14.50 | 0.60               | 15.2 |
|                                                          | 120              | 179.0   | 35.3              | 227.7           | 14.48              | 367.8  | 14.67             | 372.6 | 0.94                  | 23.88 | 0.59               | 14.99 | 0.60               | 15.2 |
|                                                          | 132              | 196.4   | 38.8              | 250.3           | 14.66              | 372.4  | 14.725            | 374.0 | 1.03                  | 26.16 | 0.645              | 16.38 | 0.60               | 15.2 |
|                                                          | 136              | 202.0   | 40.0              | 257.9           | 14.75              | 374.7  | 14.74             | 374.4 | 1.063                 | 27.00 | 0.66               | 16.80 | 0.60               | 15.2 |
| W14 14 x 16<br>(356 x 406)158                            | 145              | 216.0   | 42.7              | 275.5           | 14.78              | 375.4  | 15.50             | 393.7 | 1.09                  | 27.69 | 0.68               | 17.27 | 0.60               | 15.2 |
|                                                          | 235.0            | 46.5    | 299.8             | 15.00           | 381.0              | 15.55  | 395.0             | 1.188 | 30.20                 | 0.73  | 18.50              | 0.60  | 15.2               |      |
|                                                          | 159              | 237.0   | 46.7              | 301.3           | 14.98              | 380.5  | 15.565            | 395.4 | 1.19                  | 30.23 | 0.745              | 18.92 | 0.60               | 15.2 |
|                                                          | 176              | 262.0   | 51.8              | 334.2           | 15.22              | 386.6  | 15.65             | 397.5 | 1.31                  | 33.27 | 0.83               | 21.08 | 0.60               | 15.2 |
|                                                          | 193              | 287.2   | 56.8              | 366.4           | 15.48              | 393.2  | 15.71             | 399.0 | 1.44                  | 36.58 | 0.89               | 22.61 | 0.60               | 15.2 |
|                                                          | 211              | 314.0   | 62.0              | 400.0           | 15.72              | 399.3  | 15.80             | 401.3 | 1.56                  | 39.62 | 0.98               | 24.89 | 0.60               | 15.2 |
|                                                          | 228              | 340.0   | 67.1              | 432.7           | 16.00              | 406.4  | 15.865            | 403.0 | 1.688                 | 42.90 | 1.045              | 26.50 | 0.60               | 15.2 |
|                                                          | 233              | 347.0   | 68.5              | 441.9           | 16.04              | 407.4  | 15.89             | 403.6 | 1.72                  | 43.69 | 1.07               | 27.18 | 0.60               | 15.2 |
|                                                          | 257              | 382.5   | 75.6              | 487.7           | 16.38              | 416.1  | 15.995            | 406.3 | 1.89                  | 48.01 | 1.175              | 29.84 | 0.60               | 15.2 |
|                                                          | 264              | 393.0   | 77.6              | 500.9           | 16.50              | 419.1  | 16.025            | 407.0 | 1.938                 | 49.20 | 1.205              | 30.60 | 0.60               | 15.2 |
|                                                          | 283              | 421.1   | 83.3              | 537.4           | 16.74              | 425.2  | 16.11             | 409.2 | 2.07                  | 52.58 | 1.29               | 32.77 | 0.60               | 15.2 |
|                                                          | 311              | 463.0   | 91.4              | 589.7           | 17.12              | 434.8  | 16.23             | 412.2 | 2.26                  | 57.40 | 1.41               | 35.81 | 0.60               | 15.2 |
|                                                          | 314              | 467.0   | 92.3              | 595.5           | 17.19              | 436.6  | 16.235            | 412.4 | 2.283                 | 58.00 | 1.415              | 35.90 | 0.60               | 15.2 |
|                                                          | 342              | 509.0   | 101               | 651.6           | 17.54              | 445.5  | 16.36             | 415.5 | 2.47                  | 62.74 | 1.54               | 39.12 | 0.60               | 15.2 |
|                                                          | 370              | 551.0   | 109               | 703.2           | 17.92              | 455.2  | 16.475            | 418.5 | 2.66                  | 67.56 | 1.655              | 42.04 | 0.60               | 15.2 |
|                                                          | 398              | 592.3   | 117               | 754.8           | 18.29              | 464.6  | 16.59             | 421.4 | 2.845                 | 72.26 | 1.77               | 44.96 | 0.60               | 15.2 |
|                                                          | 426              | 634.0   | 125               | 806.4           | 18.67              | 474.2  | 16.695            | 424.1 | 3.035                 | 77.09 | 1.875              | 47.62 | 0.60               | 15.2 |
|                                                          | 455              | 677.1   | 134               | 864.5           | 19.02              | 483.1  | 16.835            | 427.6 | 3.21                  | 81.53 | 2.015              | 51.18 | 0.60               | 15.2 |
|                                                          | 500              | 744.1   | 147               | 948.4           | 19.60              | 497.8  | 17.01             | 432.1 | 3.50                  | 88.90 | 2.19               | 55.63 | 0.60               | 15.2 |
|                                                          | 550              | 818.5   | 162               | 1,045           | 20.24              | 514.1  | 17.20             | 436.9 | 3.82                  | 97.03 | 2.38               | 60.45 | 0.60               | 15.2 |
|                                                          | 605              | 900.3   | 178               | 1,148           | 20.92              | 531.4  | 17.415            | 442.3 | 4.16                  | 105.6 | 2.595              | 65.91 | 0.60               | 15.2 |
|                                                          | 665              | 989.6   | 196               | 1,265           | 21.64              | 549.7  | 17.65             | 448.3 | 4.52                  | 114.8 | 2.83               | 71.88 | 0.60               | 15.2 |
|                                                          | 730              | 1,086.0 | 215               | 1,387           | 22.42              | 569.5  | 17.89             | 454.4 | 4.91                  | 124.7 | 3.07               | 77.98 | 0.60               | 15.2 |
| W16 16 x 5 <sup>1</sup> / <sub>2</sub><br>(406 x 140)31  | 26               | 39.0    | 7.68              | 49.55           | 15.69              | 398.5  | 5.50              | 139.7 | 0.345                 | 8.76  | 0.25               | 6.35  | 0.40               | 10.2 |
|                                                          | 46.13            | 9.12    | 58.84             | 15.88           | 403.4              | 5.525  | 140.3             | 0.44  | 11.1                  | 0.275 | 6.98               | 0.40  | 10.2               |      |



# UNIVERSAL BEAMS AND COLUMNS

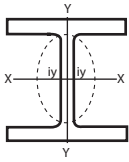
| Section<br>Size                                     | Moment of inertia |                 |                 |                 | Radius of gyration |      |      |      | Modulus of section |                 |                 |                 |
|-----------------------------------------------------|-------------------|-----------------|-----------------|-----------------|--------------------|------|------|------|--------------------|-----------------|-----------------|-----------------|
|                                                     | Ix                |                 | Iy              |                 | ix                 |      | iy   |      | Zx                 |                 | Zy              |                 |
| in (mm)                                             | in <sup>4</sup>   | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in                 | cm   | in   | cm   | in <sup>3</sup>    | cm <sup>3</sup> | in <sup>3</sup> | cm <sup>3</sup> |
| W14 14 x 6 <sup>3</sup> / <sub>4</sub> (356 x 171)  | 291               | 12,100          | 19.6            | 816             | 5.73               | 14.6 | 1.49 | 3.78 | 42.0               | 688             | 5.82            | 95.4            |
|                                                     | 340               | 14,200          | 23.3            | 970             | 5.83               | 14.8 | 1.53 | 3.89 | 48.6               | 796             | 6.91            | 113             |
|                                                     | 385               | 16,000          | 26.7            | 1,110           | 5.88               | 14.9 | 1.55 | 3.94 | 54.6               | 895             | 7.88            | 129             |
|                                                     | 469               | 19,522          | 32.7            | 1,362           | 5.95               | 15.1 | 1.57 | 3.99 | 65.5               | 1,073           | 9.60            | 157             |
| W14 14 x 8 (356 x 203)                              | 428               | 17,800          | 45.2            | 1,880           | 5.82               | 14.8 | 1.89 | 4.80 | 62.7               | 1,030           | 11.3            | 185             |
|                                                     | 485               | 20,200          | 51.4            | 2,140           | 5.85               | 14.9 | 1.91 | 4.85 | 70.3               | 1,150           | 12.8            | 210             |
|                                                     | 542               | 22,600          | 57.7            | 2,400           | 5.89               | 15.0 | 1.92 | 4.88 | 77.8               | 1,270           | 14.3            | 234             |
| W14 14 x 10 (356 x 254)                             | 640               | 26,600          | 107             | 4,450           | 5.98               | 15.2 | 2.45 | 6.22 | 92.2               | 1,510           | 21.5            | 352             |
|                                                     | 725               | 30,200          | 122             | 5,080           | 6.01               | 15.3 | 2.47 | 6.27 | 103                | 1,690           | 24.3            | 398             |
|                                                     | 796               | 33,100          | 134             | 5,580           | 6.04               | 15.3 | 2.48 | 6.30 | 112                | 1,840           | 26.6            | 436             |
|                                                     | 882               | 36,700          | 148             | 6,160           | 6.05               | 15.4 | 2.48 | 6.30 | 123                | 2,020           | 29.3            | 480             |
| W14 14 x 14 <sup>1</sup> / <sub>2</sub> (356 x 368) | 966               | 40,250          | 350             | 14,560          | 6.15               | 15.6 | 3.70 | 9.40 | 138                | 2,263           | 48.2            | 790             |
|                                                     | 999               | 41,600          | 362             | 15,100          | 6.14               | 15.6 | 3.70 | 9.40 | 143                | 2,340           | 49.9            | 818             |
|                                                     | 1,110             | 46,200          | 402             | 16,700          | 6.17               | 15.7 | 3.71 | 9.42 | 157                | 2,570           | 55.2            | 905             |
|                                                     | 1,166             | 48,530          | 420             | 17,470          | 6.21               | 15.8 | 3.72 | 9.45 | 164                | 2,681           | 57.6            | 944             |
|                                                     | 1,240             | 51,600          | 447             | 18,600          | 6.22               | 15.8 | 3.73 | 9.47 | 173                | 2,830           | 61.2            | 1,000           |
|                                                     | 1,373             | 57,160          | 492             | 20,470          | 6.26               | 15.9 | 3.75 | 9.52 | 189                | 3,104           | 67.1            | 1,099           |
|                                                     | 1,380             | 57,400          | 495             | 20,600          | 6.24               | 15.8 | 3.74 | 9.50 | 190                | 3,110           | 67.5            | 1,110           |
|                                                     | 1,530             | 63,700          | 548             | 22,800          | 6.28               | 16.0 | 3.76 | 9.55 | 209                | 3,420           | 74.5            | 1,220           |
|                                                     | 1,593             | 66,310          | 568             | 23,630          | 6.31               | 16.0 | 3.77 | 9.58 | 216                | 3,540           | 77.0            | 1,262           |
| W14 14 x 16 (356 x 406)                             | 1,710             | 71,200          | 677             | 28,200          | 6.33               | 16.1 | 3.98 | 10.1 | 232                | 3,800           | 87.3            | 1,430           |
|                                                     | 1,900             | 79,100          | 745             | 31,010          | 6.40               | 16.2 | 4.00 | 10.2 | 253                | 4,153           | 95.8            | 1,570           |
|                                                     | 1,900             | 79,100          | 748             | 31,100          | 6.38               | 16.2 | 4.00 | 10.2 | 254                | 4,160           | 96.2            | 1,580           |
|                                                     | 2,140             | 89,100          | 838             | 34,900          | 6.43               | 16.3 | 4.02 | 10.2 | 281                | 4,600           | 107             | 1,750           |
|                                                     | 2,400             | 99,900          | 931             | 38,800          | 6.50               | 16.5 | 4.05 | 10.3 | 310                | 5,080           | 119             | 1,950           |
|                                                     | 2,660             | 111,000         | 1,030           | 42,900          | 6.55               | 16.6 | 4.07 | 10.3 | 338                | 5,540           | 130             | 2,130           |
|                                                     | 2,942             | 122,500         | 1,124           | 46,820          | 6.62               | 16.8 | 4.10 | 10.4 | 368                | 6,028           | 141             | 2,324           |
|                                                     | 3,010             | 125,000         | 1,150           | 47,900          | 6.63               | 16.8 | 4.10 | 10.4 | 375                | 6,150           | 145             | 2,380           |
|                                                     | 3,400             | 142,000         | 1,290           | 53,700          | 6.71               | 17.0 | 4.13 | 10.5 | 415                | 6,800           | 161             | 2,640           |
|                                                     | 3,526             | 146,800         | 1,331           | 55,420          | 6.74               | 17.1 | 4.14 | 10.5 | 427                | 7,005           | 166             | 2,772           |
|                                                     | 3,840             | 160,000         | 1,440           | 59,900          | 6.79               | 17.2 | 4.17 | 10.6 | 459                | 7,520           | 179             | 2,930           |
|                                                     | 4,330             | 180,000         | 1,610           | 67,000          | 6.88               | 17.5 | 4.20 | 10.7 | 506                | 8,290           | 199             | 3,260           |
|                                                     | 4,399             | 183,100         | 1,631           | 67,910          | 6.90               | 17.5 | 4.20 | 10.7 | 512                | 8,390           | 201             | 3,294           |
|                                                     | 4,900             | 204,000         | 1,810           | 75,300          | 6.98               | 17.7 | 4.24 | 10.8 | 559                | 9,160           | 221             | 3,620           |
|                                                     | 5,440             | 226,000         | 1,990           | 82,800          | 7.07               | 18.0 | 4.27 | 10.8 | 607                | 9,950           | 241             | 3,950           |
|                                                     | 6,000             | 250,000         | 2,170           | 90,300          | 7.16               | 18.2 | 4.31 | 10.9 | 656                | 10,700          | 262             | 4,290           |
|                                                     | 6,600             | 275,000         | 2,360           | 98,200          | 7.26               | 18.4 | 4.34 | 11.0 | 706                | 11,600          | 283             | 4,640           |
|                                                     | 7,190             | 299,000         | 2,560           | 107,000         | 7.33               | 18.6 | 4.38 | 11.1 | 756                | 12,400          | 304             | 4,980           |
|                                                     | 8,210             | 342,000         | 2,880           | 120,000         | 7.48               | 19.0 | 4.43 | 11.3 | 838                | 13,700          | 339             | 5,560           |
|                                                     | 9,430             | 392,000         | 3,250           | 135,000         | 7.63               | 19.4 | 4.49 | 11.4 | 931                | 15,300          | 378             | 6,190           |
|                                                     | 10,800            | 450,000         | 3,680           | 153,000         | 7.80               | 19.8 | 4.55 | 11.6 | 1,040              | 17,000          | 423             | 6,930           |
| W16 16 x 5 <sup>1</sup> / <sub>2</sub> (406 x 140)  | 301               | 12,500          | 9.59            | 399             | 6.26               | 15.9 | 1.12 | 2.84 | 38.4               | 629             | 3.49            | 57.2            |
|                                                     | 375               | 15,600          | 12.4            | 516             | 6.41               | 16.3 | 1.17 | 2.97 | 47.2               | 773             | 4.49            | 73.6            |

# UNIVERSAL BEAMS AND COLUMNS



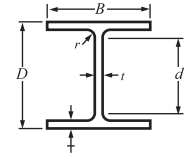
| Section size                                              | Unit weight<br>M |       | Section area<br>A |                 | Section depth<br>D |        | Flange width<br>B |       | Flange thickness<br>T |       | Web thickness<br>t |       | Corner radius<br>r |      |
|-----------------------------------------------------------|------------------|-------|-------------------|-----------------|--------------------|--------|-------------------|-------|-----------------------|-------|--------------------|-------|--------------------|------|
| in (mm)                                                   | lb/ft            | kg/m  | in <sup>2</sup>   | cm <sup>2</sup> | in                 | mm     | in                | mm    | in                    | mm    | in                 | mm    | in                 | mm   |
| W16 16 x 7 (406 x 178)                                    | 36               | 54.0  | 10.6              | 68.39           | 15.86              | 402.8  | 6.985             | 177.4 | 0.43                  | 10.90 | 0.295              | 7.49  | 0.40               | 10.2 |
|                                                           | 40               | 60.0  | 11.8              | 76.13           | 16.01              | 406.7  | 6.995             | 177.7 | 0.505                 | 12.80 | 0.305              | 7.75  | 0.40               | 10.2 |
|                                                           | 45               | 67.0  | 13.3              | 85.81           | 16.13              | 409.7  | 7.035             | 178.7 | 0.565                 | 14.30 | 0.345              | 8.76  | 0.40               | 10.2 |
|                                                           | 50               | 74.41 | 14.7              | 94.84           | 16.26              | 413.0  | 7.07              | 179.6 | 0.63                  | 16.00 | 0.38               | 9.65  | 0.40               | 10.2 |
|                                                           | 57               | 85.0  | 16.8              | 108.4           | 16.43              | 417.3  | 7.12              | 180.8 | 0.715                 | 18.16 | 0.43               | 10.92 | 0.40               | 10.2 |
| W16 16 x 10 <sup>1</sup> / <sub>4</sub><br>(406 x 260)77  | 67               | 99.71 | 19.7              | 127.1           | 16.33              | 414.8  | 10.235            | 260.0 | 0.665                 | 16.89 | 0.395              | 10.03 | 0.40               | 10.2 |
|                                                           | 114.6            | 22.6  | 145.8             | 16.52           | 419.6              | 10.295 | 261.5             | 0.76  | 19.30                 | 0.455 | 11.56              | 0.40  | 10.2               | 10.2 |
|                                                           | 89               | 132.4 | 26.2              | 169.0           | 16.75              | 425.4  | 10.365            | 263.3 | 0.875                 | 22.22 | 0.525              | 13.33 | 0.40               | 10.2 |
|                                                           | 100              | 148.8 | 29.4              | 189.7           | 16.97              | 431.0  | 10.425            | 264.8 | 0.985                 | 25.02 | 0.585              | 14.86 | 0.40               | 10.2 |
| W18 18 x 6 (457 x 152)                                    | 35               | 52.09 | 10.3              | 66.45           | 17.70              | 449.6  | 6.00              | 152.4 | 0.425                 | 10.80 | 0.30               | 7.62  | 0.40               | 10.2 |
|                                                           | 40               | 60.00 | 11.8              | 76.13           | 17.90              | 454.7  | 6.015             | 152.8 | 0.525                 | 13.30 | 0.315              | 8.00  | 0.40               | 10.2 |
|                                                           | 45               | 67.00 | 13.2              | 85.4            | 18.00              | 457.2  | 5.982             | 151.9 | 0.589                 | 15.00 | 0.36               | 9.10  | 0.40               | 10.2 |
|                                                           | 46               | 68.46 | 13.5              | 87.1            | 18.06              | 458.7  | 6.06              | 153.9 | 0.605                 | 15.30 | 0.36               | 9.14  | 0.40               | 10.2 |
|                                                           | 50               | 74.00 | 14.7              | 95.0            | 18.16              | 461.3  | 6.011             | 152.7 | 0.669                 | 17.00 | 0.389              | 9.90  | 0.40               | 10.2 |
|                                                           | 55               | 82.00 | 16.2              | 104.5           | 18.31              | 465.1  | 6.042             | 153.5 | 0.744                 | 18.90 | 0.42               | 10.70 | 0.40               | 10.2 |
| W18 18 x 7 <sup>1</sup> / <sub>2</sub><br>(457 x 191)50   | 45               | 67.0  | 13.2              | 85.4            | 17.86              | 453.6  | 7.476             | 189.9 | 0.50                  | 12.70 | 0.334              | 8.50  | 0.40               | 10.2 |
|                                                           | 74.41            | 14.7  | 94.84             | 17.99           | 456.9              | 7.495  | 190.4             | 0.57  | 14.40                 | 0.355 | 9.02               | 0.40  | 10.2               | 10.2 |
|                                                           | 55               | 82.0  | 16.2              | 104.5           | 18.11              | 460.0  | 7.53              | 191.3 | 0.63                  | 16.00 | 0.39               | 9.91  | 0.40               | 10.2 |
|                                                           | 60               | 89.29 | 17.6              | 113.5           | 18.24              | 463.3  | 7.555             | 191.9 | 0.695                 | 17.65 | 0.415              | 10.54 | 0.40               | 10.2 |
|                                                           | 65               | 97.0  | 19.1              | 123.2           | 18.35              | 466.1  | 7.59              | 192.8 | 0.75                  | 19.05 | 0.45               | 11.43 | 0.40               | 10.2 |
|                                                           | 66               | 98.0  | 19.4              | 125.3           | 18.40              | 467.4  | 7.592             | 192.8 | 0.77                  | 19.60 | 0.45               | 11.40 | 0.40               | 10.2 |
|                                                           | 71               | 106.0 | 20.8              | 134.2           | 18.47              | 469.1  | 7.635             | 193.9 | 0.81                  | 20.57 | 0.495              | 12.57 | 0.40               | 10.2 |
| W18 18 x 11 (457 x 279)                                   | 76               | 113.1 | 22.3              | 143.9           | 18.21              | 462.5  | 11.035            | 280.3 | 0.68                  | 17.27 | 0.425              | 10.80 | 0.40               | 10.2 |
|                                                           | 86               | 128.0 | 25.3              | 163.2           | 18.39              | 467.1  | 11.09             | 281.7 | 0.77                  | 19.56 | 0.48               | 12.19 | 0.40               | 10.2 |
|                                                           | 97               | 144.4 | 28.5              | 183.9           | 18.59              | 472.2  | 11.145            | 283.1 | 0.87                  | 22.10 | 0.535              | 13.59 | 0.40               | 10.2 |
|                                                           | 106              | 157.7 | 31.1              | 200.6           | 18.73              | 475.7  | 11.20             | 284.5 | 0.94                  | 23.88 | 0.59               | 14.99 | 0.40               | 10.2 |
|                                                           | 119              | 177.1 | 35.1              | 226.5           | 18.97              | 481.8  | 11.265            | 286.1 | 1.06                  | 26.92 | 0.655              | 16.64 | 0.40               | 10.2 |
| W21 21 x 6 <sup>1</sup> / <sub>2</sub><br>(533 x 165)50   | 44               | 66.0  | 13.0              | 83.87           | 20.66              | 524.8  | 6.50              | 165.1 | 0.45                  | 11.40 | 0.35               | 8.89  | 0.50               | 12.7 |
|                                                           | 74.41            | 14.7  | 94.84             | 20.83           | 529.1              | 6.53   | 165.9             | 0.535 | 13.50                 | 0.38  | 9.65               | 0.50  | 12.7               | 12.7 |
|                                                           | 57               | 85.0  | 16.7              | 107.7           | 21.06              | 534.9  | 6.555             | 166.5 | 0.65                  | 16.51 | 0.405              | 10.29 | 0.50               | 12.7 |
| W21 21 x 8 <sup>1</sup> / <sub>4</sub><br>(533 x 210)62   | 55               | 82.0  | 16.2              | 104.4           | 20.80              | 528.3  | 8.216             | 208.7 | 0.52                  | 13.20 | 0.376              | 9.60  | 0.50               | 12.7 |
|                                                           | 92.27            | 18.3  | 118.1             | 20.99           | 533.1              | 8.24   | 209.3             | 0.615 | 15.62                 | 0.40  | 10.16              | 0.50  | 12.7               | 12.7 |
|                                                           | 68               | 101.2 | 20.0              | 129.0           | 21.13              | 536.7  | 8.27              | 210.1 | 0.685                 | 17.40 | 0.43               | 10.92 | 0.50               | 12.7 |
|                                                           | 73               | 109.0 | 21.5              | 138.7           | 21.24              | 539.5  | 8.295             | 210.7 | 0.74                  | 18.80 | 0.455              | 11.56 | 0.50               | 12.7 |
|                                                           | 82               | 122.0 | 24.1              | 155.8           | 21.44              | 544.6  | 8.342             | 211.9 | 0.84                  | 21.30 | 0.502              | 12.80 | 0.50               | 12.7 |
|                                                           | 83               | 123.5 | 24.3              | 156.8           | 21.43              | 544.3  | 8.355             | 212.2 | 0.835                 | 21.21 | 0.515              | 13.08 | 0.50               | 12.7 |
|                                                           | 93               | 138.4 | 27.3              | 176.1           | 21.62              | 549.1  | 8.42              | 213.9 | 0.93                  | 23.62 | 0.58               | 14.73 | 0.50               | 12.7 |
| W21 21 x 12 <sup>1</sup> / <sub>4</sub><br>(533 x 311)111 | 101              | 150.3 | 29.8              | 192.3           | 21.36              | 542.5  | 12.29             | 312.2 | 0.80                  | 20.32 | 0.50               | 12.70 | 0.50               | 12.7 |
|                                                           | 165.2            | 32.7  | 211.0             | 21.51           | 546.4              | 12.34  | 313.4             | 0.875 | 22.22                 | 0.55  | 13.97              | 0.50  | 12.7               | 12.7 |
|                                                           | 122              | 181.6 | 35.9              | 231.6           | 21.68              | 550.7  | 12.39             | 314.7 | 0.96                  | 24.38 | 0.60               | 15.24 | 0.50               | 12.7 |
|                                                           | 132              | 196.4 | 38.8              | 250.3           | 21.83              | 554.5  | 12.44             | 316.0 | 1.03                  | 26.29 | 0.65               | 16.51 | 0.50               | 12.7 |
|                                                           | 147              | 218.8 | 43.2              | 278.7           | 22.06              | 560.3  | 12.51             | 317.8 | 1.15                  | 29.21 | 0.72               | 18.29 | 0.50               | 12.7 |
| W24 24 x 7 (610 x 178)                                    | 55               | 82    | 16.2              | 104.5           | 23.57              | 598.7  | 7.005             | 177.9 | 0.505                 | 12.83 | 0.395              | 10.03 | 0.50               | 12.7 |
|                                                           | 61               | 91    | 17.9              | 116.5           | 23.72              | 602.5  | 7.023             | 178.4 | 0.591                 | 15.00 | 0.419              | 10.60 | 0.50               | 12.7 |
|                                                           | 62               | 92.27 | 18.2              | 117.4           | 23.74              | 603.0  | 7.040             | 178.8 | 0.59                  | 14.99 | 0.43               | 10.92 | 0.50               | 12.7 |

# UNIVERSAL BEAMS AND COLUMNS

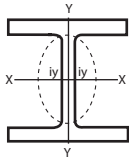


| Section<br>Size                                     | Moment of inertia |                 |                 |                 | Radius of gyration |      |      |      | Modulus of section |                 |                 |                 |
|-----------------------------------------------------|-------------------|-----------------|-----------------|-----------------|--------------------|------|------|------|--------------------|-----------------|-----------------|-----------------|
|                                                     | Ix                |                 | Iy              |                 | ix                 |      | iy   |      | Zx                 |                 | Zy              |                 |
| in (mm)                                             | in <sup>4</sup>   | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in                 | cm   | in   | cm   | in <sup>3</sup>    | cm <sup>3</sup> | in <sup>3</sup> | cm <sup>3</sup> |
| W16 16 x 7 (406 x 178)                              | 448               | 18,600          | 24.5            | 1,020           | 6.51               | 16.5 | 1.52 | 3.86 | 56.5               | 926             | 7.00            | 115             |
|                                                     | 518               | 21,600          | 28.9            | 1,200           | 6.63               | 16.8 | 1.57 | 3.99 | 64.7               | 1,060           | 8.25            | 135             |
|                                                     | 586               | 24,400          | 32.8            | 1,370           | 6.65               | 16.9 | 1.57 | 3.99 | 72.7               | 1,190           | 9.34            | 153             |
|                                                     | 659               | 27,400          | 37.2            | 1,550           | 6.68               | 17.0 | 1.59 | 4.04 | 81.0               | 1,330           | 10.5            | 172             |
|                                                     | 758               | 31,600          | 43.1            | 1,790           | 6.72               | 17.1 | 1.60 | 4.06 | 92.2               | 1,510           | 12.1            | 198             |
| W16 16 x 10 <sup>1</sup> / <sub>4</sub> (406 x 260) | 954               | 39,700          | 119             | 4,950           | 6.96               | 17.7 | 2.46 | 6.25 | 117                | 1,920           | 23.2            | 380             |
|                                                     | 1,100             | 46,200          | 138             | 5,740           | 7.00               | 17.8 | 2.47 | 6.27 | 134                | 2,200           | 26.9            | 441             |
|                                                     | 1,300             | 54,100          | 163             | 6,780           | 7.05               | 17.9 | 2.49 | 6.32 | 155                | 2,540           | 31.4            | 515             |
|                                                     | 1,490             | 62,000          | 186             | 7,740           | 7.10               | 18.0 | 2.51 | 6.38 | 175                | 2,870           | 35.7            | 585             |
| W18 18 x 6 (457 x 152)                              | 510               | 21,200          | 15.3            | 637             | 7.04               | 17.9 | 1.22 | 3.10 | 57.6               | 944             | 5.12            | 83.9            |
|                                                     | 612               | 25,500          | 19.1            | 795             | 7.21               | 18.3 | 1.27 | 3.23 | 68.4               | 1,120           | 6.35            | 104             |
|                                                     | 686               | 28,577          | 21.1            | 878             | 7.20               | 18.3 | 1.26 | 3.21 | 76.3               | 1,250           | 7.00            | 116             |
|                                                     | 712               | 29,600          | 22.5            | 937             | 7.25               | 18.4 | 1.29 | 3.28 | 78.8               | 1,290           | 7.43            | 122             |
|                                                     | 779               | 32,435          | 24.3            | 1,012           | 7.28               | 18.5 | 1.28 | 3.26 | 85.8               | 1,406           | 8.10            | 133             |
|                                                     | 870               | 36,215          | 27.5            | 1,143           | 7.33               | 18.6 | 1.30 | 3.31 | 95.0               | 1,557           | 9.10            | 149             |
| W18 18 x 7 <sup>1</sup> / <sub>4</sub> (457 x 191)  | 704               | 29,330          | 31.9            | 1,328           | 7.30               | 18.5 | 1.55 | 3.94 | 78.9               | 1,293           | 8.5             | 139             |
|                                                     | 800               | 33,300          | 40.1            | 1,670           | 7.38               | 18.7 | 1.65 | 4.19 | 88.9               | 1,460           | 10.7            | 175             |
|                                                     | 890               | 37,000          | 44.9            | 1,870           | 7.41               | 18.8 | 1.67 | 4.24 | 98.3               | 1,610           | 11.9            | 195             |
|                                                     | 984               | 41,000          | 50.1            | 2,090           | 7.47               | 19.0 | 1.68 | 4.27 | 108                | 1,770           | 13.3            | 218             |
|                                                     | 1,070             | 44,500          | 54.8            | 2,280           | 7.49               | 19.0 | 1.69 | 4.29 | 117                | 1,920           | 14.4            | 236             |
|                                                     | 1,098             | 45,717          | 56.3            | 2,343           | 7.52               | 19.1 | 1.70 | 4.33 | 119                | 1,956           | 14.8            | 243             |
|                                                     | 1,170             | 48,700          | 60.3            | 2,510           | 7.50               | 19.0 | 1.70 | 4.32 | 127                | 2,080           | 15.8            | 259             |
| W18 18 x 11 (457 x 279)                             | 1,330             | 55,400          | 152             | 6,330           | 7.73               | 19.6 | 2.61 | 6.63 | 146                | 2,390           | 27.6            | 452             |
|                                                     | 1,530             | 63,700          | 175             | 7,280           | 7.77               | 19.7 | 2.63 | 6.68 | 166                | 2,720           | 31.6            | 518             |
|                                                     | 1,750             | 72,800          | 201             | 8,370           | 7.82               | 19.9 | 2.65 | 6.73 | 188                | 3,080           | 36.1            | 592             |
|                                                     | 1,910             | 79,500          | 220             | 9,170           | 7.84               | 19.9 | 2.66 | 6.76 | 204                | 3,340           | 39.4            | 646             |
|                                                     | 2,190             | 91,200          | 253             | 10,500          | 7.90               | 20.1 | 2.69 | 6.83 | 231                | 3,790           | 44.9            | 736             |
| W21 21 x 6 <sup>1</sup> / <sub>2</sub> (533 x 165)  | 843               | 35,100          | 20.7            | 862             | 8.06               | 20.5 | 1.26 | 3.20 | 81.6               | 1,340           | 6.37            | 104             |
|                                                     | 984               | 41,000          | 24.9            | 1,040           | 8.18               | 20.8 | 1.30 | 3.30 | 94.5               | 1,550           | 7.64            | 125             |
|                                                     | 1,170             | 48,700          | 30.6            | 1,270           | 8.36               | 21.2 | 1.35 | 3.43 | 111                | 1,820           | 9.35            | 153             |
| W21 21 x 8 <sup>1</sup> / <sub>4</sub> (533 x 210)  | 1,140.7           | 47,490          | 44.0            | 1,832           | 8.40               | 21.3 | 1.65 | 4.19 | 109.7              | 1,798           | 10.7            | 175.4           |
|                                                     | 1,330             | 55,400          | 57.5            | 2,390           | 8.54               | 21.7 | 1.77 | 4.50 | 127                | 2,080           | 14.0            | 229             |
|                                                     | 1,480             | 61,600          | 64.7            | 2,690           | 8.60               | 21.8 | 1.80 | 4.57 | 140                | 2,290           | 15.7            | 257             |
|                                                     | 1,600             | 66,600          | 70.6            | 2,940           | 8.64               | 21.9 | 1.81 | 4.60 | 151                | 2,470           | 17.0            | 279             |
|                                                     | 1,830             | 76,200          | 81.4            | 3,390           | 8.67               | 22.0 | 1.83 | 4.65 | 170.8              | 2,799           | 19.5            | 320             |
|                                                     | 1,830.7           | 76,207          | 81.5            | 3,393           | 8.71               | 22.1 | 1.84 | 4.67 | 171                | 2,800           | 19.5            | 320.2           |
|                                                     | 2,070             | 86,200          | 92.9            | 3,870           | 8.70               | 22.1 | 1.84 | 4.67 | 192                | 3,150           | 22.1            | 362             |
|                                                     |                   |                 |                 |                 |                    |      |      |      |                    |                 |                 |                 |
| W21 21 x 12 <sup>1</sup> / <sub>4</sub> (533 x 311) | 2,420             | 101,000         | 248             | 10,300          | 9.02               | 22.9 | 2.89 | 7.34 | 227                | 3,720           | 40.3            | 660             |
|                                                     | 2,670             | 111,000         | 274             | 11,400          | 9.05               | 23.0 | 2.90 | 7.37 | 249                | 4,080           | 44.5            | 729             |
|                                                     | 2,960             | 123,000         | 305             | 12,700          | 9.09               | 23.1 | 2.92 | 7.42 | 273                | 4,470           | 49.2            | 806             |
|                                                     | 3,220             | 134,000         | 333             | 13,900          | 9.12               | 23.2 | 2.93 | 7.44 | 295                | 4,830           | 53.5            | 877             |
|                                                     | 3,630             | 151,000         | 376             | 15,600          | 9.17               | 23.3 | 2.95 | 7.49 | 329                | 5,390           | 60.1            | 985             |
| W24 24 x 7 (610 x 178)                              | 1,350             | 56,200          | 29.1            | 1,210           | 9.11               | 23.1 | 1.34 | 3.40 | 114                | 1,870           | 8.30            | 136             |
|                                                     | 1,540             | 64,085          | 34.3            | 1,432           | 9.25               | 23.5 | 1.38 | 3.51 | 130                | 2,124           | 9.76            | 159.5           |
|                                                     | 1,550             | 64,500          | 34.5            | 1,440           | 9.23               | 23.4 | 1.38 | 3.51 | 131                | 2,150           | 9.80            | 161             |

# UNIVERSAL BEAMS AND COLUMNS



| Section size                                                  | Unit weight<br>M |       | Section area<br>A |                 | Section depth<br>D |        | Flange width<br>B |       | Flange thickness<br>T |       | Web thickness<br>t |       | Corner radius<br>r |      |
|---------------------------------------------------------------|------------------|-------|-------------------|-----------------|--------------------|--------|-------------------|-------|-----------------------|-------|--------------------|-------|--------------------|------|
| in (mm)                                                       | lb/ft            | kg/m  | in <sup>2</sup>   | cm <sup>2</sup> | in                 | mm     | in                | mm    | in                    | mm    | in                 | mm    | in                 | mm   |
| W24 24 x 9 (610 x 229)                                        | 68               | 101.2 | 20.1              | 129.7           | 23.73              | 602.7  | 8.965             | 227.7 | 0.585                 | 14.86 | 0.415              | 10.54 | 0.50               | 12.7 |
|                                                               | 76               | 113.1 | 22.4              | 144.5           | 23.92              | 607.6  | 8.99              | 228.3 | 0.680                 | 17.27 | 0.44               | 11.18 | 0.50               | 12.7 |
|                                                               | 84               | 125   | 24.7              | 159.4           | 24.10              | 612.1  | 9.02              | 229.1 | 0.770                 | 19.56 | 0.47               | 11.94 | 0.50               | 12.7 |
|                                                               | 94               | 140   | 27.7              | 178.7           | 24.31              | 617.5  | 9.065             | 230.2 | 0.875                 | 22.22 | 0.515              | 13.08 | 0.50               | 12.7 |
| W24 24 x 12 (610 x 305)                                       | 100              | 149   | 29.4              | 190.1           | 24.00              | 609.6  | 12.00             | 304.8 | 0.775                 | 19.70 | 0.468              | 11.90 | 0.70               | 17.8 |
|                                                               | 120              | 179   | 35.3              | 227.9           | 24.31              | 617.5  | 12.088            | 307.0 | 0.93                  | 23.60 | 0.556              | 14.10 | 0.70               | 17.8 |
|                                                               | 160              | 238   | 47.0              | 303.8           | 24.92              | 633.0  | 12.264            | 311.5 | 1.235                 | 31.40 | 0.732              | 18.60 | 0.70               | 17.8 |
| W24 24 x 12 <sup>3</sup> / <sub>4</sub><br>(610 x 324)117     | 104              | 154.8 | 30.6              | 197.4           | 24.06              | 611.1  | 12.75             | 323.8 | 0.75                  | 19.05 | 0.50               | 12.70 | 0.50               | 12.7 |
|                                                               | 174.1            | 34.4  | 221.9             | 24.26           | 616.2              | 12.80  | 325.1             | 0.85  | 21.59                 | 0.55  | 13.97              | 0.50  | 12.7               |      |
|                                                               | 131              | 194.9 | 38.5              | 248.4           | 24.48              | 621.8  | 12.855            | 326.5 | 0.96                  | 24.38 | 0.605              | 15.37 | 0.50               | 12.7 |
|                                                               | 146              | 217.3 | 43.0              | 277.4           | 24.74              | 628.4  | 12.90             | 327.7 | 1.09                  | 27.69 | 0.65               | 16.51 | 0.50               | 12.7 |
|                                                               | 162              | 241.1 | 47.7              | 307.7           | 25.00              | 635.0  | 12.955            | 329.1 | 1.22                  | 30.99 | 0.705              | 17.91 | 0.50               | 12.7 |
| W27 27 x 10 (686 x 254)                                       | 84               | 125   | 24.8              | 160.0           | 26.71              | 678.4  | 9.96              | 253.0 | 0.64                  | 16.26 | 0.46               | 11.68 | 0.60               | 15.2 |
|                                                               | 94               | 140   | 27.7              | 178.7           | 26.92              | 683.8  | 9.99              | 253.7 | 0.745                 | 18.92 | 0.49               | 12.45 | 0.60               | 15.2 |
|                                                               | 102              | 152   | 30.0              | 193.5           | 27.09              | 688.1  | 10.015            | 254.4 | 0.83                  | 21.08 | 0.515              | 13.08 | 0.60               | 15.2 |
|                                                               | 114              | 170   | 33.5              | 216.1           | 27.29              | 693.2  | 10.07             | 255.8 | 0.93                  | 23.62 | 0.57               | 14.48 | 0.60               | 15.2 |
| W27 27 x 14 (686 x 356)                                       | 146              | 217.3 | 42.9              | 276.8           | 27.38              | 695.5  | 13.965            | 354.7 | 0.975                 | 24.76 | 0.605              | 15.37 | 0.60               | 15.2 |
|                                                               | 161              | 239.6 | 47.4              | 305.8           | 27.59              | 700.8  | 14.02             | 356.1 | 1.08                  | 27.43 | 0.66               | 16.76 | 0.60               | 15.2 |
|                                                               | 178              | 264.9 | 52.3              | 337.4           | 27.81              | 706.4  | 14.085            | 357.8 | 1.19                  | 30.23 | 0.725              | 18.41 | 0.60               | 15.2 |
| W30 30 x 10 <sup>1</sup> / <sub>2</sub><br>(762 x 267)108     | 99               | 147.3 | 29.1              | 187.7           | 29.65              | 753.1  | 10.45             | 265.4 | 0.67                  | 17.02 | 0.52               | 13.21 | 0.65               | 16.5 |
|                                                               | 161              | 31.7  | 204.5             | 29.83           | 757.7              | 10.475 | 266.1             | 0.76  | 19.30                 | 0.545 | 13.84              | 0.65  | 16.5               |      |
|                                                               | 116              | 173   | 34.2              | 220.6           | 30.01              | 762.3  | 10.495            | 266.6 | 0.85                  | 21.59 | 0.565              | 14.35 | 0.65               | 16.5 |
|                                                               | 124              | 185   | 36.5              | 235.5           | 30.17              | 766.3  | 10.515            | 267.1 | 0.93                  | 23.62 | 0.585              | 14.86 | 0.65               | 16.5 |
|                                                               | 132              | 197   | 38.9              | 251.0           | 30.31              | 769.9  | 10.545            | 267.8 | 1.00                  | 25.40 | 0.615              | 15.62 | 0.65               | 16.5 |
| W30 30 x 15 (762 x 381)                                       | 173              | 257.5 | 50.8              | 327.7           | 30.44              | 773.2  | 14.985            | 380.6 | 1.06                  | 27.05 | 0.655              | 16.64 | 0.65               | 16.5 |
|                                                               | 191              | 284.2 | 56.1              | 361.9           | 30.68              | 779.3  | 15.04             | 382.0 | 1.18                  | 30.10 | 0.71               | 18.03 | 0.65               | 16.5 |
|                                                               | 211              | 314   | 62.0              | 400.0           | 30.94              | 785.9  | 15.105            | 383.7 | 1.31                  | 33.40 | 0.775              | 19.68 | 0.65               | 16.5 |
| W33 33 X 11 <sup>1</sup> / <sub>2</sub> 118<br>(838 x 292)130 | 176              | 34.7  | 223.9             | 32.86           | 834.6              | 11.48  | 291.6             | 0.74  | 18.80                 | 0.55  | 13.97              | 0.70  | 17.8               |      |
|                                                               | 194              | 38.3  | 247.1             | 33.09           | 840.5              | 11.51  | 292.4             | 0.855 | 21.72                 | 0.58  | 14.73              | 0.70  | 17.8               |      |
|                                                               | 141              | 210   | 41.6              | 268.4           | 33.30              | 845.8  | 11.535            | 293.0 | 0.96                  | 24.38 | 0.605              | 15.37 | 0.70               | 17.8 |
|                                                               | 152              | 226.2 | 44.7              | 288.4           | 33.49              | 850.6  | 11.565            | 293.8 | 1.05                  | 26.80 | 0.635              | 16.13 | 0.70               | 17.8 |
| W33 33 x 15 <sup>3</sup> / <sub>4</sub><br>(838 x 400)221     | 201              | 299.1 | 59.1              | 381.3           | 33.68              | 855.5  | 15.745            | 399.9 | 1.15                  | 29.21 | 0.715              | 18.16 | 0.70               | 17.8 |
|                                                               | 328.9            | 65.0  | 419.4             | 33.93           | 861.8              | 15.805 | 401.4             | 1.27  | 32.38                 | 0.775 | 19.68              | 0.70  | 17.8               |      |
|                                                               | 241              | 358.6 | 70.9              | 457.4           | 34.18              | 868.2  | 15.86             | 402.8 | 1.40                  | 35.56 | 0.83               | 21.08 | 0.70               | 17.8 |
| W36 36 x 12<br>(914 x 305)150                                 | 135              | 201   | 39.7              | 256.1           | 35.55              | 903.0  | 11.95             | 303.5 | 0.79                  | 20.07 | 0.60               | 15.24 | 0.75               | 19.0 |
|                                                               | 224              | 44.2  | 285.2             | 35.85           | 910.6              | 11.975 | 304.2             | 0.94  | 23.88                 | 0.625 | 15.87              | 0.75  | 19.0               |      |
|                                                               | 160              | 238.1 | 47.0              | 303.2           | 36.01              | 914.7  | 12.00             | 304.8 | 1.02                  | 25.91 | 0.65               | 16.51 | 0.75               | 19.0 |
|                                                               | 170              | 253   | 50.0              | 322.6           | 36.17              | 918.7  | 12.03             | 305.6 | 1.10                  | 27.94 | 0.68               | 17.27 | 0.75               | 19.0 |
|                                                               | 182              | 271   | 53.6              | 345.8           | 36.33              | 922.8  | 12.075            | 306.7 | 1.18                  | 29.97 | 0.725              | 18.41 | 0.75               | 19.0 |
|                                                               | 194              | 289   | 57.0              | 367.7           | 36.49              | 926.8  | 12.115            | 307.7 | 1.26                  | 32.00 | 0.765              | 19.43 | 0.75               | 19.0 |
|                                                               | 210              | 313   | 61.8              | 398.7           | 36.69              | 931.9  | 12.18             | 309.4 | 1.36                  | 34.54 | 0.83               | 21.08 | 0.75               | 19.0 |
| W36 36 x 16 <sup>1</sup> / <sub>2</sub><br>(914 x 419)245     | 230              | 343   | 67.6              | 436.1           | 35.90              | 911.9  | 16.47             | 418.3 | 1.26                  | 32.00 | 0.76               | 19.30 | 0.95               | 24.1 |
|                                                               | 365              | 72.1  | 465.2             | 36.08           | 916.4              | 16.51  | 419.4             | 1.35  | 34.29                 | 0.80  | 20.32              | 0.95  | 24.1               |      |
|                                                               | 260              | 388   | 76.5              | 493.5           | 36.26              | 921.0  | 16.55             | 420.4 | 1.44                  | 36.58 | 0.84               | 21.34 | 0.95               | 24.1 |
|                                                               | 280              | 417   | 82.4              | 531.6           | 36.52              | 927.6  | 16.595            | 421.5 | 1.57                  | 39.88 | 0.885              | 22.48 | 0.95               | 24.1 |
|                                                               | 300              | 447   | 88.3              | 569.7           | 36.74              | 933.2  | 16.655            | 423.0 | 1.68                  | 42.67 | 0.945              | 24.00 | 0.95               | 24.1 |

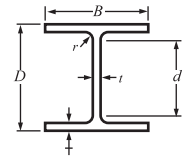


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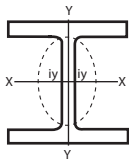
| Section<br>Size                                     | Moment of inertia |                 |                 |                 | Radius of gyration |       |      |      | Modulus of section |                 |                 |                 |
|-----------------------------------------------------|-------------------|-----------------|-----------------|-----------------|--------------------|-------|------|------|--------------------|-----------------|-----------------|-----------------|
|                                                     | Ix                |                 | Iy              |                 | ix                 |       | iy   |      | Zx                 |                 | Zy              |                 |
| in (mm)                                             | in <sup>4</sup>   | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in                 | cm    | in   | cm   | in <sup>3</sup>    | cm <sup>3</sup> | in <sup>3</sup> | cm <sup>3</sup> |
| W24 24 x 9 (610 x 229)                              | 1,830             | 76,200          | 70.4            | 2,930           | 9.55               | 24.3  | 1.87 | 4.75 | 154                | 2,520           | 15.7            | 257             |
|                                                     | 2,100             | 87,400          | 82.5            | 3,430           | 9.69               | 24.6  | 1.92 | 4.88 | 176                | 2,880           | 18.4            | 302             |
|                                                     | 2,370             | 98,600          | 94.4            | 3,930           | 9.79               | 24.9  | 1.95 | 4.95 | 196                | 3,210           | 20.9            | 342             |
|                                                     | 2,700             | 112,000         | 109.0           | 4,540           | 9.87               | 25.1  | 1.98 | 5.03 | 222                | 3,640           | 24.0            | 393             |
| W24 24 x 12 (610 x 305)                             | 2,987.3           | 124,400         | 203.5           | 8,471           | 10.08              | 25.6  | 2.63 | 6.68 | 248.9              | 4,079.5         | 33.9            | 555.6           |
|                                                     | 3,635.3           | 151,300         | 254.0           | 10,570          | 10.15              | 25.78 | 2.68 | 6.81 | 299.1              | 4,902.2         | 42.0            | 688.4           |
|                                                     | 4,986.3           | 207,571         | 380.5           | 15,838          | 10.28              | 26.1  | 2.84 | 7.22 | 400.2              | 6,559           | 62.0            | 1,017           |
| W24 24 x 12 <sup>3</sup> / <sub>4</sub> (610 x 324) | 3,100             | 129,000         | 259             | 10,800          | 10.1               | 25.7  | 2.91 | 7.39 | 258                | 4,230           | 40.7            | 667             |
|                                                     | 3,540             | 147,000         | 297             | 12,400          | 10.1               | 25.7  | 2.94 | 7.47 | 291                | 4,770           | 46.5            | 762             |
|                                                     | 4,020             | 167,000         | 340             | 14,200          | 10.2               | 25.9  | 2.97 | 7.54 | 329                | 5,390           | 53.0            | 869             |
|                                                     | 4,580             | 191,000         | 391             | 16,300          | 10.3               | 26.2  | 3.01 | 7.65 | 371                | 6,080           | 60.5            | 991             |
|                                                     | 5,170             | 215,000         | 443             | 18,400          | 10.4               | 26.4  | 3.05 | 7.75 | 414                | 6,780           | 68.4            | 1,120           |
| W27 27 x 10 (686 x 254)                             | 2,850             | 119,000         | 106             | 4,410           | 10.7               | 27.2  | 2.07 | 5.26 | 213                | 3,490           | 21.2            | 347             |
|                                                     | 3,270             | 136,000         | 124             | 5,160           | 10.9               | 27.7  | 2.12 | 5.38 | 243                | 3,980           | 24.8            | 406             |
|                                                     | 3,620             | 151,000         | 139             | 5,790           | 11.0               | 27.9  | 2.15 | 5.46 | 267                | 4,380           | 27.8            | 456             |
|                                                     | 4,090             | 170,000         | 159             | 6,620           | 11.0               | 27.9  | 2.18 | 5.54 | 299                | 4,900           | 31.5            | 516             |
| W27 27 x 14 (686 x 356)                             | 5,630             | 234,000         | 443             | 18,400          | 11.4               | 29.0  | 3.21 | 8.15 | 411                | 6,740           | 63.5            | 1,040           |
|                                                     | 6,280             | 261,000         | 497             | 20,700          | 11.5               | 29.2  | 3.24 | 8.23 | 455                | 7,460           | 70.9            | 1,160           |
|                                                     | 6,990             | 291,000         | 555             | 23,100          | 11.6               | 29.5  | 3.26 | 8.28 | 502                | 8,230           | 78.8            | 1,290           |
| W30 30 x 10 <sup>1</sup> / <sub>2</sub> (762 x 267) | 4,000             | 166,000         | 128             | 5,330           | 11.7               | 29.7  | 2.10 | 5.33 | 269                | 4,410           | 24.5            | 401             |
|                                                     | 4,470             | 186,000         | 146             | 6,080           | 11.9               | 30.2  | 2.15 | 5.46 | 299                | 4,900           | 27.9            | 457             |
|                                                     | 4,930             | 205,000         | 164             | 6,830           | 12.0               | 30.5  | 2.19 | 5.56 | 329                | 5,390           | 31.3            | 513             |
|                                                     | 5,360             | 223,000         | 181             | 7,530           | 12.1               | 30.7  | 2.23 | 5.66 | 355                | 5,820           | 34.4            | 564             |
|                                                     | 5,770             | 240,000         | 196             | 8,160           | 12.2               | 31.0  | 2.25 | 5.71 | 380                | 6,230           | 37.2            | 610             |
| W30 30 x 15 (762 x 381)                             | 8,200             | 341,000         | 598             | 24,900          | 12.7               | 32.3  | 3.43 | 8.71 | 539                | 8,830           | 79.8            | 1,310           |
|                                                     | 9,170             | 382,000         | 673             | 28,000          | 12.8               | 32.5  | 3.46 | 8.79 | 598                | 9,800           | 89.5            | 1,470           |
|                                                     | 10,300            | 429,000         | 757             | 31,500          | 12.9               | 32.8  | 3.49 | 8.86 | 663                | 10,900          | 100             | 1,640           |
| W33 33 x 11 <sup>1</sup> / <sub>2</sub> (838 x 292) | 5,900             | 246,000         | 187             | 7,780           | 13.0               | 33.0  | 2.32 | 5.89 | 359                | 5,880           | 32.6            | 534             |
|                                                     | 6,710             | 279,000         | 218             | 9,070           | 13.2               | 33.5  | 2.38 | 6.05 | 406                | 6,650           | 37.9            | 621             |
|                                                     | 7,450             | 310,000         | 246             | 10,200          | 13.4               | 34.0  | 2.43 | 6.17 | 448                | 7,340           | 42.7            | 700             |
|                                                     | 8,160             | 340,000         | 273             | 11,400          | 13.5               | 34.3  | 2.47 | 6.27 | 487                | 7,980           | 47.2            | 773             |
| W33 33 x 15 <sup>3</sup> / <sub>4</sub> (838 x 400) | 11,500            | 479,000         | 749             | 31,200          | 14.0               | 35.6  | 3.56 | 9.04 | 684                | 11,200          | 95.2            | 1,560           |
|                                                     | 12,800            | 533,000         | 840             | 35,000          | 14.1               | 35.8  | 3.59 | 9.12 | 757                | 12,400          | 106             | 1,740           |
|                                                     | 14,200            | 591,000         | 932             | 38,800          | 14.1               | 35.8  | 3.63 | 9.22 | 829                | 13,600          | 118             | 1,930           |
| W36 36 x 12 (914 x 305)                             | 7,800             | 325,000         | 225             | 9,370           | 14.0               | 35.6  | 2.38 | 6.05 | 439                | 7,190           | 37.7            | 618             |
|                                                     | 9,040             | 376,000         | 270             | 11,200          | 14.3               | 36.3  | 2.47 | 6.27 | 504                | 8,260           | 45.1            | 739             |
|                                                     | 9,760             | 406,000         | 295             | 12,300          | 14.4               | 36.6  | 2.50 | 6.35 | 542                | 8,880           | 49.1            | 805             |
|                                                     | 10,500            | 437,000         | 320             | 13,300          | 14.5               | 36.8  | 2.53 | 6.43 | 580                | 9,500           | 53.2            | 872             |
|                                                     | 11,300            | 470,000         | 347             | 14,400          | 14.5               | 36.8  | 2.55 | 6.48 | 623                | 10,200          | 57.6            | 944             |
|                                                     | 12,100            | 504,000         | 375             | 15,600          | 14.6               | 37.1  | 2.56 | 6.50 | 664                | 10,900          | 61.9            | 1,010           |
|                                                     | 13,200            | 549,000         | 411             | 17,100          | 14.6               | 37.1  | 2.58 | 6.55 | 719                | 11,800          | 67.5            | 1,110           |
| W36 36 x 16 <sup>1</sup> / <sub>2</sub> (914 x 419) | 15,000            | 624,000         | 940             | 39,100          | 14.9               | 37.8  | 3.73 | 9.47 | 837                | 13,700          | 114             | 1,870           |
|                                                     | 16,100            | 670,000         | 1,010           | 42,000          | 15.0               | 38.1  | 3.75 | 9.52 | 895                | 14,700          | 123             | 2,020           |
|                                                     | 17,300            | 720,000         | 1,090           | 45,400          | 15.0               | 38.1  | 3.78 | 9.60 | 953                | 15,600          | 132             | 2,160           |
|                                                     | 18,900            | 787,000         | 1,200           | 49,900          | 15.1               | 38.4  | 3.81 | 9.68 | 1,030              | 16,900          | 144             | 2,360           |
|                                                     | 20,300            | 845,000         | 1,300           | 54,100          | 15.2               | 38.6  | 3.83 | 9.73 | 1,110              | 18,200          | 156             | 2,560           |



# UNIVERSAL BEAMS AND COLUMNS



| Section size | Unit weight<br>M | Section depth<br>D | Flange width<br>B | Thickness |            | Corner radius<br>r | Section area<br>A | Moment of inertia |                 | Radius of gyration |                | Modulus of section |                 |
|--------------|------------------|--------------------|-------------------|-----------|------------|--------------------|-------------------|-------------------|-----------------|--------------------|----------------|--------------------|-----------------|
|              |                  |                    |                   | t (web)   | T (flange) |                    |                   | I <sub>x</sub>    | I <sub>y</sub>  | i <sub>x</sub>     | i <sub>y</sub> | Z <sub>x</sub>     | Z <sub>y</sub>  |
| mm           | kg/m             | mm                 | mm                | mm        | mm         | mm                 | cm <sup>2</sup>   | cm <sup>4</sup>   | cm <sup>4</sup> | cm                 | cm             | cm <sup>3</sup>    | cm <sup>3</sup> |
| 100 x 50     | 9.3              | 100                | 50                | 5         | 7          | 8                  | 11.85             | 187               | 14.8            | 3.98               | 1.12           | 37.5               | 5.91            |
| 100 x 100    | 17.2             | 100                | 100               | 6         | 8          | 10                 | 21.90             | 333               | 134             | 4.18               | 2.47           | 76.5               | 26.7            |
| 125 x 60     | 13.2             | 125                | 60                | 6         | 8          | 9                  | 16.84             | 413               | 29.2            | 4.95               | 1.32           | 66.1               | 9.73            |
| 125 x 125    | 23.8             | 125                | 125               | 6.5       | 9          | 10                 | 30.31             | 847               | 293             | 5.29               | 3.11           | 136                | 47.0            |
| 150 x 75     | 14.0             | 150                | 75                | 5         | 7          | 8                  | 17.85             | 666               | 49.5            | 6.11               | 1.66           | 88.8               | 13.2            |
| 150 x 100    | 21.1             | 148                | 100               | 6         | 9          | 11                 | 26.84             | 1,020             | 151             | 6.17               | 2.37           | 138                | 30.1            |
| 150 x 150    | 31.5             | 150                | 150               | 7         | 10         | 11                 | 40.14             | 1,640             | 563             | 6.39               | 3.75           | 219                | 75.1            |
| 175 x 90     | 18.1             | 175                | 90                | 5         | 8          | 9                  | 23.04             | 1,210             | 97.5            | 7.26               | 2.06           | 139                | 21.7            |
| 175 x 125    | 23.3             | 169                | 125               | 5.5       | 8          | 12                 | 29.65             | 1,530             | 261             | 7.18               | 2.97           | 181                | 41.8            |
| 175 x 175    | 40.2             | 175                | 175               | 7.5       | 11         | 12                 | 51.21             | 2,880             | 984             | 7.50               | 4.38           | 330                | 112             |
| 200 x 100    | 18.2             | 198                | 99                | 4.5       | 7          | 11                 | 23.18             | 1,580             | 114             | 8.26               | 2.21           | 160                | 23.0            |
|              | 21.3             | 200                | 100               | 5.5       | 8          | 11                 | 27.16             | 1,840             | 134             | 8.24               | 2.22           | 184                | 26.8            |
| 200 x 150    | 30.6             | 194                | 150               | 6         | 9          | 13                 | 39.01             | 2,690             | 507             | 8.30               | 3.61           | 277                | 67.6            |
| 200 x 200    | 49.9             | 200                | 200               | 8         | 12         | 13                 | 63.53             | 4,720             | 1,600           | 8.62               | 5.02           | 472                | 160             |
|              | 56.2             | 200                | 204               | 12        | 12         | 13                 | 71.53             | 4,980             | 1,700           | 8.35               | 4.88           | 498                | 167             |
|              | 65.7             | 208                | 202               | 10        | 16         | 13                 | 83.69             | 6,530             | 2,200           | 8.83               | 5.13           | 628                | 218             |
| 250 x 125    | 25.7             | 248                | 124               | 5         | 8          | 12                 | 32.68             | 3,540             | 255             | 10.4               | 2.79           | 285                | 41.1            |
|              | 29.6             | 250                | 125               | 6         | 9          | 12                 | 37.66             | 4,050             | 294             | 10.4               | 2.79           | 324                | 47.0            |
| 250 x 175    | 44.1             | 244                | 175               | 7         | 11         | 16                 | 56.24             | 6,120             | 984             | 10.4               | 4.18           | 502                | 113             |
| 250 x 250    | 64.4             | 244                | 252               | 11        | 11         | 16                 | 82.06             | 8,780             | 2,940           | 10.3               | 5.98           | 720                | 233             |
|              | 66.5             | 248                | 249               | 8         | 13         | 16                 | 84.70             | 9,930             | 3,350           | 10.8               | 6.29           | 801                | 269             |
|              | 72.4             | 250                | 250               | 9         | 14         | 16                 | 92.18             | 10,800            | 3,650           | 10.8               | 6.29           | 867                | 292             |
|              | 82.2             | 250                | 255               | 14        | 14         | 16                 | 104.7             | 11,500            | 3,880           | 10.5               | 6.09           | 919                | 304             |
| 300 x 150    | 32.0             | 298                | 149               | 5.5       | 8          | 13                 | 40.80             | 6,320             | 442             | 12.4               | 3.29           | 424                | 59.3            |
|              | 36.7             | 300                | 150               | 6.5       | 9          | 13                 | 46.78             | 7,210             | 508             | 12.4               | 3.29           | 481                | 67.7            |
| 300 x 200    | 56.8             | 294                | 200               | 8         | 12         | 18                 | 72.38             | 11,300            | 1,600           | 12.5               | 4.71           | 771                | 160             |
|              | 65.4             | 298                | 201               | 9         | 14         | 18                 | 83.36             | 13,300            | 1,900           | 12.6               | 4.77           | 893                | 189             |
| 300 x 300    | 84.5             | 294                | 302               | 12        | 12         | 18                 | 107.7             | 16,900            | 5,520           | 12.5               | 7.16           | 1,150              | 365             |
|              | 87               | 298                | 299               | 9         | 14         | 18                 | 110.8             | 18,800            | 6,240           | 13.0               | 7.51           | 1,270              | 417             |
|              | 94               | 300                | 300               | 10        | 15         | 18                 | 119.8             | 20,400            | 6,750           | 13.1               | 7.51           | 1,360              | 450             |
|              | 106              | 300                | 305               | 15        | 15         | 18                 | 134.8             | 21,500            | 7,100           | 12.6               | 7.26           | 1,440              | 466             |
|              | 106              | 304                | 301               | 11        | 17         | 18                 | 134.8             | 23,400            | 7,730           | 13.2               | 7.57           | 1,540              | 514             |
| 350 x 175    | 41.4             | 346                | 174               | 6         | 9          | 14                 | 52.68             | 11,100            | 798             | 14.5               | 3.88           | 641                | 91              |
|              | 49.6             | 350                | 175               | 7         | 11         | 14                 | 63.14             | 13,600            | 984             | 14.7               | 3.95           | 775                | 112             |
| 350 x 250    | 69.2             | 336                | 249               | 8         | 12         | 20                 | 88.15             | 18,500            | 3,090           | 14.5               | 5.92           | 1,100              | 248             |
|              | 79.7             | 340                | 250               | 9         | 14         | 20                 | 101.5             | 21,700            | 3,650           | 14.6               | 6.00           | 1,280              | 292             |

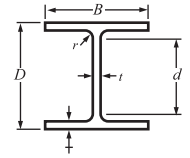


# UNIVERSAL BEAMS AND COLUMNS

| Section size | Unit weight<br>M | Section depth<br>D | Flange width<br>B | Thickness |            | Corner radius<br>r | Section area<br>A | Moment of inertia |                 | Radius of gyration |                | Modulus of section |                 |
|--------------|------------------|--------------------|-------------------|-----------|------------|--------------------|-------------------|-------------------|-----------------|--------------------|----------------|--------------------|-----------------|
|              |                  |                    |                   | t (web)   | T (flange) |                    |                   | I <sub>x</sub>    | I <sub>y</sub>  | i <sub>x</sub>     | i <sub>y</sub> | Z <sub>x</sub>     | Z <sub>y</sub>  |
| mm           | kg/m             | mm                 | mm                | mm        | mm         | mm                 | cm <sup>2</sup>   | cm <sup>4</sup>   | cm <sup>4</sup> | cm                 | cm             | cm <sup>3</sup>    | cm <sup>3</sup> |
| 350 x 350    | 106              | 338                | 351               | 13        | 13         | 20                 | 135.3             | 28,200            | 9,380           | 14.4               | 8.33           | 1,670              | 534             |
|              | 115              | 344                | 348               | 10        | 16         | 20                 | 146.0             | 33,300            | 11,200          | 15.1               | 8.78           | 1,940              | 646             |
|              | 131              | 344                | 354               | 16        | 16         | 20                 | 166.6             | 35,300            | 11,800          | 14.6               | 8.43           | 2,050              | 669             |
|              | 137              | 350                | 350               | 12        | 19         | 20                 | 173.9             | 40,300            | 13,600          | 15.2               | 8.84           | 2,300              | 776             |
|              | 156              | 350                | 357               | 19        | 19         | 20                 | 198.4             | 42,800            | 14,400          | 14.7               | 8.53           | 2,450              | 809             |
|              | 159              | 356                | 352               | 14        | 22         | 20                 | 202.0             | 47,600            | 16,000          | 15.3               | 8.90           | 2,670              | 909             |
| 400 x 200    | 56.6             | 396                | 199               | 7         | 11         | 16                 | 72.16             | 20,000            | 1,450           | 16.7               | 4.48           | 1,010              | 145             |
|              | 66.0             | 400                | 200               | 8         | 13         | 16                 | 84.12             | 23,700            | 1,740           | 16.8               | 4.54           | 1,190              | 174             |
| 400 x 300    | 94.3             | 386                | 299               | 9         | 14         | 22                 | 120.1             | 33,700            | 6,240           | 16.7               | 7.21           | 1,740              | 418             |
|              | 107              | 390                | 300               | 10        | 16         | 22                 | 136.0             | 38,700            | 7,210           | 16.9               | 7.28           | 1,980              | 481             |
| 400 x 400    | 140              | 388                | 402               | 15        | 15         | 22                 | 178.5             | 49,000            | 16,300          | 16.6               | 9.54           | 2,250              | 809             |
|              | 147              | 394                | 398               | 11        | 18         | 22                 | 186.8             | 56,100            | 18,900          | 17.3               | 10.10          | 2,850              | 951             |
|              | 168              | 394                | 405               | 18        | 18         | 22                 | 214.4             | 49,700            | 20,000          | 16.7               | 9.65           | 3,030              | 985             |
|              | 172              | 400                | 400               | 13        | 21         | 22                 | 218.7             | 66,600            | 22,400          | 17.5               | 10.10          | 3,330              | 1,120           |
|              | 197              | 400                | 408               | 21        | 21         | 22                 | 250.7             | 70,900            | 23,800          | 16.8               | 9.75           | 3,540              | 1,170           |
|              | 200              | 406                | 403               | 16        | 24         | 22                 | 254.9             | 78,000            | 26,200          | 17.5               | 10.10          | 3,840              | 1,300           |
|              | 232              | 414                | 405               | 18        | 28         | 22                 | 295.5             | 92,800            | 31,000          | 17.7               | 10.20          | 4,480              | 1,530           |
|              | 283              | 428                | 407               | 20        | 35         | 22                 | 360.7             | 119,000           | 39,400          | 18.2               | 10.40          | 5,570              | 1,930           |
|              | 415              | 458                | 417               | 30        | 50         | 22                 | 528.6             | 187,000           | 60,500          | 18.8               | 10.70          | 8,170              | 2,900           |
|              | 605              | 498                | 432               | 45        | 70         | 22                 | 770.1             | 298,000           | 94,400          | 19.7               | 11.10          | 12,000             | 4,370           |
| 450 x 200    | 66.2             | 446                | 199               | 8         | 12         | 18                 | 84.30             | 28,700            | 1,580           | 18.5               | 4.33           | 1,290              | 159             |
|              | 76.0             | 450                | 200               | 9         | 14         | 18                 | 96.76             | 33,500            | 1,870           | 18.6               | 4.40           | 1,490              | 187             |
| 450 x 300    | 106              | 434                | 299               | 10        | 15         | 24                 | 135.0             | 46,800            | 6,690           | 18.6               | 7.04           | 2,160              | 448             |
|              | 124              | 440                | 300               | 11        | 18         | 24                 | 157.4             | 56,100            | 8,110           | 18.9               | 7.18           | 2,550              | 541             |
| 500 x 200    | 79.5             | 496                | 199               | 9         | 14         | 20                 | 101.3             | 41,900            | 1,840           | 20.3               | 4.27           | 1,690              | 185             |
|              | 89.6             | 500                | 200               | 10        | 16         | 20                 | 114.2             | 47,800            | 2,140           | 20.5               | 4.33           | 1,910              | 214             |
|              | 103              | 506                | 201               | 11        | 19         | 20                 | 131.3             | 56,500            | 2,580           | 20.7               | 4.43           | 2,230              | 257             |
| 500 x 300    | 114              | 482                | 300               | 11        | 15         | 26                 | 145.5             | 60,400            | 6,760           | 20.4               | 6.82           | 2,500              | 451             |
|              | 128              | 488                | 300               | 11        | 18         | 26                 | 163.5             | 71,000            | 8,110           | 20.8               | 7.04           | 2,910              | 541             |
| 600 x 200    | 94.6             | 596                | 199               | 10        | 15         | 22                 | 120.5             | 68,700            | 1,980           | 23.9               | 4.05           | 2,310              | 199             |
|              | 106              | 600                | 200               | 11        | 17         | 22                 | 134.4             | 77,600            | 2,280           | 24.0               | 4.12           | 2,590              | 228             |
|              | 120              | 606                | 201               | 12        | 20         | 22                 | 152.5             | 90,400            | 2,720           | 24.3               | 4.22           | 2,980              | 271             |
|              | 134              | 612                | 202               | 13        | 23         | 22                 | 107.7             | 103,000           | 3,180           | 24.6               | 4.31           | 3,380              | 314             |
| 600 x 300    | 137              | 582                | 300               | 12        | 17         | 28                 | 174.5             | 103,000           | 7,670           | 24.3               | 6.63           | 3,530              | 511             |
|              | 151              | 588                | 300               | 12        | 20         | 28                 | 192.5             | 118,000           | 9,020           | 24.8               | 6.85           | 4,020              | 601             |
|              | 175              | 594                | 302               | 14        | 23         | 28                 | 222.4             | 137,000           | 10,600          | 28.6               | 6.90           | 4,620              | 701             |
| 700 x 300    | 166              | 692                | 300               | 13        | 20         | 28                 | 211.5             | 172,000           | 9,020           | 28.6               | 6.53           | 4,980              | 602             |
|              | 185              | 700                | 300               | 13        | 24         | 28                 | 235.5             | 201,000           | 10,800          | 29.3               | 6.78           | 5,760              | 722             |
|              | 215              | 708                | 302               | 14        | 28         | 28                 | 273.6             | 237,000           | 12,900          | 29.4               | 6.86           | 6,700              | 853             |
| 800 x 300    | 191              | 792                | 300               | 14        | 22         | 28                 | 243.4             | 254,000           | 9,930           | 32.3               | 6.39           | 6,410              | 662             |
|              | 210              | 800                | 300               | 14        | 26         | 28                 | 267.4             | 292,000           | 11,700          | 33.0               | 6.62           | 7,290              | 782             |
|              | 241              | 808                | 302               | 16        | 30         | 28                 | 307.6             | 339,000           | 13,800          | 33.2               | 6.70           | 8,400              | 915             |
| 900 x 300    | 213              | 890                | 299               | 15        | 23         | 28                 | 270.9             | 345,000           | 10,300          | 35.7               | 6.16           | 7,760              | 688             |
|              | 243              | 900                | 300               | 16        | 28         | 28                 | 309.8             | 411,000           | 12,600          | 36.4               | 6.39           | 9,140              | 843             |
|              | 286              | 912                | 302               | 18        | 34         | 28                 | 364.0             | 498,000           | 15,700          | 37.0               | 6.46           | 10,900             | 1,040           |



# LIGHT BEAMS AND JOISTS

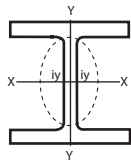


| Section size                           | Unit weight<br>M |        | Section depth<br>D |      | Flange width<br>B |      | Web thickness<br>t |       | Flange thickness<br>T |       | Corner radius<br>r |       | Section area<br>A |                 |
|----------------------------------------|------------------|--------|--------------------|------|-------------------|------|--------------------|-------|-----------------------|-------|--------------------|-------|-------------------|-----------------|
|                                        | lb/ft            | kg/m   | in                 | mm   | in                | mm   | in                 | mm    | in                    | mm    | in                 | mm    | in <sup>2</sup>   | cm <sup>2</sup> |
| 6 x 4 (J)                              | 8.5              | 12.65  | 5.83               | 148  | 3.94              | 100  | 0.17               | 4.32  | 0.194                 | 4.93  | 0.25               | 6.35  | 2.50              | 16.1            |
| 6 x 4 (L)                              | 12               | 17.86  | 6.00               | 152  | 4.00              | 102  | 0.23               | 5.84  | 0.279                 | 7.09  | 0.25               | 6.35  | 3.53              | 22.8            |
|                                        | 16               | 23.81  | 6.25               | 159  | 4.03              | 102  | 0.26               | 6.60  | 0.404                 | 10.26 | 0.25               | 6.35  | 4.72              | 30.5            |
| 8 x 4 (J)                              | 10               | 14.88  | 7.90               | 201  | 3.94              | 100  | 0.17               | 4.32  | 0.204                 | 5.18  | 0.30               | 7.62  | 2.95              | 19.0            |
| 8 x 4 (L)                              | 13               | 19.35  | 8.00               | 203  | 4.00              | 102  | 0.23               | 5.84  | 0.254                 | 6.45  | 0.30               | 7.62  | 3.83              | 24.7            |
|                                        | 15               | 22.32  | 8.12               | 206  | 4.015             | 102  | 0.245              | 6.22  | 0.314                 | 7.98  | 0.30               | 7.62  | 4.43              | 28.6            |
| 10 x 4 (J)                             | 11.5             | 17.11  | 9.87               | 251  | 3.95              | 100  | 0.18               | 4.57  | 0.204                 | 5.18  | 0.30               | 7.62  | 3.39              | 21.9            |
| 10 x 4 (L)                             | 15               | 22.32  | 10.00              | 254  | 4.00              | 102  | 0.23               | 5.84  | 0.269                 | 6.83  | 0.30               | 7.62  | 4.40              | 28.4            |
|                                        | 17               | 25.30  | 10.12              | 257  | 4.01              | 102  | 0.24               | 6.10  | 0.329                 | 8.36  | 0.30               | 7.62  | 4.98              | 32.1            |
|                                        | 19               | 28.28  | 10.25              | 260  | 4.02              | 102  | 0.25               | 6.35  | 0.394                 | 10.01 | 0.30               | 7.62  | 5.61              | 36.2            |
| 12 x 4 (J)                             | 14               | 20.83  | 11.91              | 303  | 3.97              | 101  | 0.20               | 5.08  | 0.224                 | 5.69  | 0.30               | 7.62  | 4.14              | 26.7            |
| 12 x 4 (L)                             | 16.5             | 24.55  | 12.00              | 305  | 4.00              | 102  | 0.23               | 5.84  | 0.269                 | 6.83  | 0.30               | 7.62  | 4.86              | 31.4            |
|                                        | 19               | 28.28  | 12.16              | 309  | 4.01              | 102  | 0.24               | 6.10  | 0.349                 | 8.86  | 0.30               | 7.62  | 5.62              | 36.3            |
|                                        | 22               | 32.74  | 12.31              | 313  | 4.03              | 102  | 0.26               | 6.60  | 0.424                 | 10.77 | 0.30               | 7.62  | 6.47              | 41.7            |
| 14 x 4 (L)                             | 17.2*            | 25.60* | 14.00*             | 356* | 4.00*             | 102* | 0.21*              | 5.33* | 0.272                 | 6.91* | 0.43*              | 10.9* | 5.05*             | 32.6*           |
| 14 x 5 (L)                             | 22               | 32.74  | 13.72              | 348  | 5.00              | 127  | 0.23               | 5.84  | 0.335                 | 8.51  | 0.43               | 10.9  | 6.47              | 41.7            |
|                                        | 26               | 38.69  | 13.89              | 353  | 5.025             | 128  | 0.255              | 6.48  | 0.418                 | 10.62 | 0.43               | 10.9  | 7.65              | 49.4            |
| 16 x 5 <sup>1</sup> / <sub>2</sub> (L) | 26               | 38.69  | 15.65              | 398  | 5.50              | 140  | 0.25               | 6.35  | 0.345                 | 8.76  | 0.43               | 10.9  | 7.65              | 49.4            |
|                                        | 31               | 46.13  | 15.84              | 402  | 5.525             | 140  | 0.275              | 6.99  | 0.442                 | 11.23 | 0.43               | 10.9  | 9.12              | 58.8            |

L = Light beams, J = Joists

Note:

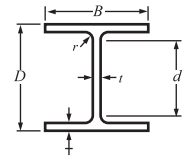
Sizes indicated by an asterisk (\*) are not included in regular rolling schedules.



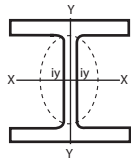
# LIGHT BEAMS AND JOISTS

| Section<br>Size | Moment of inertia |                 |                 |                 | Radius of gyration |        |       |       | Modulus of section |                 |                 |                 |
|-----------------|-------------------|-----------------|-----------------|-----------------|--------------------|--------|-------|-------|--------------------|-----------------|-----------------|-----------------|
|                 | Ix                |                 | Iy              |                 | ix                 |        | iy    |       | Zx                 |                 | Zy              |                 |
|                 | in <sup>4</sup>   | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in                 | cm     | in    | cm    | in <sup>3</sup>    | cm <sup>3</sup> | in <sup>3</sup> | cm <sup>3</sup> |
| 6 x 4 (J)       | 14.8              | 616             | 1.89            | 78.66           | 2.43               | 6.17   | 0.87  | 2.21  | 5.07               | 83.1            | 0.96            | 15.73           |
| 6 x 4 (L)       | 21.7              | 903.3           | 2.89            | 120.3           | 2.48               | 6.30   | 0.90  | 2.29  | 7.24               | 118.7           | 1.44            | 23.60           |
|                 | 31.7              | 1,320           | 4.32            | 179.8           | 2.59               | 6.58   | 0.96  | 2.44  | 10.1               | 165.5           | 2.14            | 35.07           |
| 8 x 4 (J)       | 30.8              | 1,282           | 1.99            | 82.82           | 3.23               | 8.20   | 0.82  | 2.08  | 7.79               | 127.7           | 1.01            | 16.55           |
| 8 x 4 (L)       | 39.5              | 1,644           | 2.62            | 109.1           | 3.21               | 8.15   | 0.83  | 2.11  | 9.88               | 161.9           | 1.31            | 21.47           |
|                 | 48.0              | 1,994           | 3.30            | 137.4           | 3.29               | 8.36   | 0.86  | 2.18  | 11.8               | 193.4           | 1.65            | 27.04           |
| 10 x 4 (J)      | 51.9              | 2,160           | 2.01            | 83.66           | 3.92               | 9.96   | 0.77  | 1.96  | 10.5               | 172.1           | 1.02            | 16.72           |
| 10 x 4 (L)      | 68.8              | 2,864           | 2.79            | 116.1           | 3.95               | 10.03  | 0.80  | 2.03  | 13.8               | 226.2           | 1.39            | 22.78           |
|                 | 81.8              | 3,405           | 3.45            | 143.6           | 4.05               | 10.29  | 0.83  | 2.11  | 16.2               | 265.5           | 1.72            | 28.19           |
|                 | 96.2              | 4,005           | 4.19            | 174.4           | 4.14               | 10.52  | 0.86  | 2.18  | 18.8               | 308.1           | 2.08            | 34.09           |
| 12 x 4 (J)      | 88.2              | 3,671           | 2.25            | 93.65           | 4.61               | 11.71  | 0.74  | 1.88  | 14.8               | 242.5           | 1.13            | 18.52           |
| 12 x 4 (L)      | 105.3             | 4,383           | 2.79            | 116.1           | 4.65               | 11.81  | 0.76  | 1.93  | 17.5               | 286.8           | 1.39            | 22.78           |
|                 | 130.1             | 5,416           | 3.67            | 152.8           | 4.81               | 12.22  | 0.81  | 2.06  | 21.4               | 350.7           | 1.83            | 29.99           |
|                 | 155.7             | 6,482           | 4.55            | 189.4           | 4.91               | 12.47  | 0.84  | 2.13  | 25.3               | 414.7           | 2.26            | 37.04           |
| 14 x 4 (L)      | 147.3*            | 6,131*          | 2.65*           | 110.3 *         | 5.40*              | 13.72* | 0.72* | 1.83* | 21.0 *             | 344.2*          | 1.32*           | 21.63*          |
| 14 x 5 (L)      | 197.4             | 8,212           | 6.40            | 266.4           | 5.52               | 14.02  | 0.99  | 2.51  | 28.8               | 472.0           | 2.56            | 41.96           |
|                 | 242.6             | 10,100          | 8.26            | 343.8           | 5.63               | 14.30  | 1.04  | 2.64  | 34.9               | 572.0           | 3.29            | 53.92           |
| 16 x 5 1/2 (L)  | 298.1             | 12,410          | 8.71            | 362.5           | 6.24               | 15.85  | 1.07  | 2.72  | 38.1               | 624.5           | 3.17            | 51.96           |
|                 | 372.5             | 15,500          | 11.57           | 481.5           | 6.39               | 16.23  | 1.13  | 2.87  | 47.0               | 770.3           | 4.19            | 68.67           |

# LIGHT BEAMS AND JOISTS



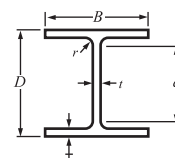
| Section size            | Unit weight<br>M |       | Web thickness<br>t |       | Flange thickness<br>T |       | Radius Root<br>r <sub>1</sub> |      | Toe<br>r <sub>2</sub> |       | Section Area<br>A |                 |
|-------------------------|------------------|-------|--------------------|-------|-----------------------|-------|-------------------------------|------|-----------------------|-------|-------------------|-----------------|
|                         | kg/m             | lb/ft | mm                 | in    | mm                    | in    | mm                            | in   | mm                    | in    | cm <sup>2</sup>   | in <sup>2</sup> |
| 76 x 51 (3 x 2)         | 6.71             | 4.5   | 3.81               | 0.15  | 5.69                  | 0.224 | 6.90                          | 0.27 | 2.4                   | 0.094 | 8.5               | 1.32            |
| 102 x 64 (4 x 2 1/2)    | 9.65             | 6.5   | 4.09               | 0.161 | 6.63                  | 0.261 | 6.90                          | 0.27 | 2.4                   | 0.094 | 12.3              | 1.91            |
| 102 x 76 (4 x 3)        | 14.90            | 10    | 6.10               | 0.24  | 8.81                  | 0.347 | 9.40                          | 0.37 | 4.6                   | 0.18  | 18.9              | 2.94            |
| 127 x 76 (5 x 3)        | 13.36            | 9     | 4.52               | 0.178 | 7.62                  | 0.30  | 7.90                          | 0.31 | 2.4                   | 0.094 | 17.0              | 2.64            |
| 152 x 76 (6 x 3)        | 17.88            | 12    | 5.84               | 0.23  | 9.57                  | 0.377 | 9.40                          | 0.37 | 4.6                   | 0.18  | 22.8              | 3.53            |
| 152 x 89 (6 x 3 1/2)    | 17.09            | 11.5  | 4.93               | 0.194 | 8.28                  | 0.326 | 7.90                          | 0.31 | 2.4                   | 0.094 | 21.8              | 3.38            |
| 152 x 127 (6 x 5)       | 37.25            | 25    | 10.41              | 0.41  | 13.21                 | 0.520 | 13.50                         | 0.53 | 6.6                   | 0.26  | 47.5              | 7.37            |
| 178 x 102 (7 x 4)       | 21.54            | 14.5  | 5.33               | 0.21  | 9.07                  | 0.357 | 9.40                          | 0.37 | 3.2                   | 0.13  | 27.4              | 4.26            |
| 178 x 102 (7 x 4)       | 23.84            | 16    | 6.35               | 0.25  | 9.83                  | 0.387 | 11.40                         | 0.45 | 5.6                   | 0.22  | 30.6              | 4.75            |
| 203 x 102 (8 x 4)       | 25.33            | 17    | 5.84               | 0.23  | 10.36                 | 0.408 | 9.40                          | 0.37 | 3.2                   | 0.13  | 32.3              | 5.00            |
| 203 x 102 (8 x 4)       | 26.82            | 18    | 7.11               | 0.28  | 10.11                 | 0.398 | 11.40                         | 0.45 | 5.6                   | 0.22  | 34.2              | 5.30            |
| 203 x 152 (8 x 6)       | 52.16            | 35    | 8.89               | 0.35  | 16.46                 | 0.648 | 15.50                         | 0.61 | 7.6                   | 0.30  | 66.5              | 10.30           |
| 228 x 102 (9 x 4)       | 31.29            | 21    | 7.62               | 0.30  | 11.61                 | 0.457 | 11.40                         | 0.45 | 5.6                   | 0.22  | 39.9              | 6.18            |
| 254 x 127 (10 x 5)44.71 | 30               | 9.14  | 0.36               | 14.02 | 0.552                 | 13.50 | 0.53                          | 6.6  | 0.26                  | 57.1  | 8.85              |                 |
| 254 x 152 (10 x 6)59.61 | 40               | 9.14  | 0.36               | 18.01 | 0.709                 | 15.50 | 0.61                          | 7.6  | 0.30                  | 75.9  | 11.77             |                 |
| 254 x 203 (10 x 8)81.97 | 55               | 10.16 | 0.40               | 19.89 | 0.783                 | 19.60 | 0.77                          | 9.7  | 0.38                  | 104.4 | 16.18             |                 |
| 305 x 127 (12 x 5)47.69 | 32               | 8.89  | 0.35               | 11.61 | 0.550                 | 13.50 | 0.53                          | 6.6  | 0.26                  | 60.9  | 9.45              |                 |
| 305 x 152 (12 x 6)65.57 | 44               | 10.16 | 0.40               | 18.21 | 0.717                 | 15.50 | 0.61                          | 7.6  | 0.30                  | 83.9  | 13.00             |                 |



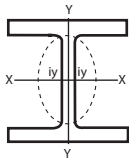
# LIGHT BEAMS AND JOISTS

| Section<br>Size    | Moment of inertia |                 |                 |                 | Radius of gyration |      |      |      | Modulus of section |                 |                 |                 |
|--------------------|-------------------|-----------------|-----------------|-----------------|--------------------|------|------|------|--------------------|-----------------|-----------------|-----------------|
|                    | Ix                |                 | Iy              |                 | ix                 |      | iy   |      | Zx                 |                 | Zy              |                 |
| mm (inch)          | cm <sup>4</sup>   | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm                 | in   | cm   | in   | cm <sup>3</sup>    | in <sup>3</sup> | cm <sup>3</sup> | in <sup>3</sup> |
| 76 x 51 (3 x 2)    | 83                | 2.0             | 11.1            | 0.27            | 3.12               | 1.23 | 1.14 | 0.45 | 21.67              | 1.33            | 4.37            | 0.27            |
| 102 x 64 (4 x 2½)  | 218               | 5.2             | 25.3            | 0.61            | 4.21               | 1.66 | 1.43 | 0.57 | 42.84              | 2.62            | 7.97            | 0.49            |
| 102 x 76 (4 x 3)   | 325               | 7.8             | 55.4            | 1.33            | 4.14               | 1.63 | 1.70 | 0.67 | 63.7               | 3.89            | 14.4            | 0.88            |
| 127 x 76 (5 x 3)   | 476               | 11.5            | 50.2            | 1.21            | 5.29               | 2.08 | 1.72 | 0.68 | 74.9               | 4.58            | 13.17           | 0.81            |
| 152 x 76 (6 x 3)   | 874               | 21.5            | 60.8            | 1.46            | 6.20               | 2.44 | 1.63 | 0.64 | 114.7              | 7.00            | 15.9            | 0.97            |
| 152 x 89 (6 x 3½)  | 881               | 21.7            | 86.0            | 2.07            | 6.36               | 2.51 | 1.99 | 0.78 | 115.6              | 7.07            | 19.34           | 1.18            |
| 152 x 127 (6 x 5)  | 1,819             | 43.7            | 378.7           | 9.10            | 6.20               | 2.44 | 2.82 | 1.11 | 239.1              | 14.56           | 59.6            | 3.64            |
| 178 x 102 (7 X 4)  | 1,519             | 36.6            | 139.2           | 3.36            | 7.44               | 2.93 | 2.25 | 0.89 | 170.9              | 10.45           | 27.41           | 1.68            |
| 178 x 102 (7 x 4)  | 1,644             | 39.5            | 140.3           | 3.37            | 7.34               | 2.89 | 2.13 | 0.84 | 185.0              | 11.29           | 27.7            | 1.69            |
| 203 x 102 (8 x 4)  | 2,294             | 55.1            | 162.6           | 3.91            | 8.43               | 3.32 | 2.25 | 0.88 | 225.8              | 13.78           | 32.02           | 1.95            |
| 203 x 102 (8 x 4)  | 2,314             | 55.6            | 146.1           | 3.51            | 8.23               | 3.24 | 2.06 | 0.81 | 227.9              | 13.91           | 28.7            | 1.75            |
| 203 x 152 (8 x 6)  | 4,790             | 115.1           | 813.3           | 19.54           | 8.48               | 3.34 | 3.51 | 1.38 | 471.3              | 28.76           | 106.7           | 6.51            |
| 228 x 102 (9 x 4)  | 3,375             | 81.1            | 172.7           | 4.15            | 9.19               | 3.62 | 2.08 | 0.82 | 295.5              | 18.03           | 33.9            | 2.07            |
| 254 x 127 (10 x 5) | 6,085             | 146.2           | 405.0           | 9.73            | 10.31              | 4.06 | 2.67 | 1.05 | 479.3              | 29.25           | 63.7            | 3.89            |
| 254 x 152 (10 x 6) | 8,524             | 204.8           | 905.6           | 21.76           | 10.59              | 4.17 | 3.45 | 1.36 | 671.2              | 40.96           | 118.8           | 7.25            |
| 254 x 203 (10 x 8) | 12,016            | 288.7           | 2,278.3         | 54.74           | 10.72              | 4.22 | 4.67 | 1.84 | 946.2              | 57.74           | 224.3           | 13.69           |
| 305 x 127 (12 x 5) | 9,202             | 221.1           | 403.3           | 9.69            | 12.29              | 4.84 | 2.56 | 1.01 | 603.7              | 36.84           | 63.6            | 3.88            |
| 305 x 152 (12 x 6) | 13,181            | 316.7           | 920.6           | 22.12           | 12.55              | 4.94 | 3.30 | 1.30 | 865.1              | 52.79           | 120.8           | 7.37            |

# BEARING PILES



| Section size | Unit weight<br>M |       | Section depth<br>D |       | Flange width<br>B |       | Web & Flange thickness<br>t T |      | Corner radius<br>r |      | Section area<br>A |                 |
|--------------|------------------|-------|--------------------|-------|-------------------|-------|-------------------------------|------|--------------------|------|-------------------|-----------------|
|              | lb/ft            | kg/m  | in                 | mm    | in                | mm    | in                            | mm   | in                 | mm   | in <sup>2</sup>   | cm <sup>2</sup> |
| 8 x 8        | 30               | 45.0  | 7.88               | 200.2 | 8.085             | 205.4 | 0.373                         | 9.5  | 0.4                | 10.2 | 8.83              | 57.0            |
|              | 36               | 54.0  | 8.026              | 203.9 | 8.158             | 207.2 | 0.446                         | 11.3 | 0.4                | 10.2 | 10.6              | 68.4            |
| 10 x 10      | 42               | 63.0  | 9.72               | 246.9 | 10.078            | 256.0 | 0.418                         | 10.6 | 0.5                | 12.7 | 12.4              | 79.7            |
|              | 48               | 71.0  | 9.838              | 249.9 | 10.137            | 257.5 | 0.477                         | 12.1 | 0.5                | 12.7 | 14.1              | 91.0            |
|              | 57               | 85.0  | 10.012             | 254.3 | 10.224            | 259.7 | 0.564                         | 14.3 | 0.5                | 12.7 | 16.8              | 108.1           |
| 12 x 12      | 53               | 79.0  | 11.78              | 299.2 | 12.046            | 306.0 | 0.436                         | 11.1 | 0.6                | 15.2 | 15.5              | 100.4           |
|              | 59               | 88.0  | 11.878             | 301.7 | 12.095            | 307.2 | 0.485                         | 12.3 | 0.6                | 15.2 | 17.36             | 112.0           |
|              | 64               | 95.0  | 11.96              | 303.8 | 12.136            | 308.3 | 0.526                         | 13.4 | 0.6                | 15.2 | 18.4              | 121.0           |
|              | 74               | 110.0 | 12.122             | 307.9 | 12.217            | 310.3 | 0.607                         | 15.4 | 0.6                | 15.2 | 21.8              | 140.4           |
|              | 85               | 126.0 | 12.308             | 312.6 | 12.31             | 312.7 | 0.70                          | 17.8 | 0.6                | 15.2 | 24.6              | 161.0           |
|              | 100              | 149.0 | 12.528             | 318.2 | 12.42             | 315.5 | 0.81                          | 20.6 | 0.6                | 15.2 | 29.4              | 190.0           |
|              | 125              | 186.0 | 12.928             | 328.4 | 12.62             | 320.5 | 1.01                          | 25.7 | 0.6                | 15.2 | 36.73             | 237.0           |
|              | 150              | 223.0 | 13.308             | 338.0 | 12.81             | 325.4 | 1.20                          | 30.5 | 0.6                | 15.2 | 44.17             | 285.0           |
| 14 x 14 1/2  | 73               | 109.0 | 13.636             | 346.4 | 14.586            | 370.5 | 0.506                         | 12.9 | 0.6                | 15.2 | 21.4              | 138.4           |
|              | 89               | 133.0 | 13.856             | 351.9 | 14.696            | 373.3 | 0.616                         | 15.6 | 0.6                | 15.2 | 26.1              | 169.0           |
|              | 102              | 152.0 | 14.032             | 356.4 | 14.784            | 375.5 | 0.704                         | 17.9 | 0.6                | 15.2 | 30.0              | 193.6           |
|              | 117              | 174.1 | 14.234             | 361.5 | 14.885            | 378.1 | 0.805                         | 20.4 | 0.6                | 15.2 | 34.4              | 222.2           |



# BEARING PILES

| Section Size           |       | Moment of inertia |                 |                 |                 | Radius of gyration |      |      |      | Modulus of section |                 |                 |                 |
|------------------------|-------|-------------------|-----------------|-----------------|-----------------|--------------------|------|------|------|--------------------|-----------------|-----------------|-----------------|
|                        |       | Ix                |                 | Iy              |                 | ix                 |      | iy   |      | Zx                 |                 | Zy              |                 |
| in (mm)                |       | in <sup>4</sup>   | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in                 | cm   | in   | cm   | in <sup>3</sup>    | cm <sup>3</sup> | in <sup>3</sup> | cm <sup>3</sup> |
| 8 x 8                  |       | 98.0              | 4,079           | 37.0            | 1,539           | 3.33               | 8.46 | 1.93 | 4.90 | 24.9               | 408             | 8.1             | 133             |
|                        |       | 119.8             | 4,987           | 40.4            | 1,683           | 3.36               | 8.54 | 1.95 | 4.95 | 29.8               | 489             | 9.9             | 162             |
| 10 x 10                | 210.8 | 8,775             | 71.4            | 2,971           | 4.13            | 10.5               | 2.41 | 6.11 | 43.4 | 711                | 14.2            | 232             |                 |
|                        |       | 243.9             | 10,153          | 82.9            | 3,451           | 4.17               | 10.6 | 2.40 | 6.10 | 49.6               | 813             | 16.4            | 268             |
|                        |       | 294.6             | 12,264          | 100.6           | 4,188           | 4.21               | 10.7 | 2.45 | 6.22 | 58.9               | 965             | 19.7            | 323             |
| 12 x 12                | 394.0 | 16,400            | 127.1           | 5,292           | 5.04            | 12.8               | 2.86 | 7.26 | 66.9 | 1,096              | 21.1            | 346             |                 |
|                        |       | 442.1             | 18,402          | 143.2           | 5,959           | 5.04               | 12.8 | 2.87 | 7.30 | 74.4               | 1,220           | 23.7            | 388             |
|                        |       | 483.2             | 20,111          | 156.9           | 6,529           | 5.08               | 12.9 | 2.87 | 7.30 | 80.8               | 1,324           | 25.9            | 424             |
|                        |       | 566.5             | 23,580          | 184.7           | 7,689           | 5.12               | 13.0 | 2.91 | 7.40 | 93.5               | 1,532           | 30.3            | 496             |
|                        |       | 661.3             | 27,526          | 216.5           | 9,013           | 5.16               | 13.1 | 2.95 | 7.50 | 107.6              | 1,763           | 35.2            | 577             |
|                        |       | 793.8             | 33,040          | 261.1           | 10,869          | 5.20               | 13.2 | 2.99 | 7.60 | 126.6              | 2,075           | 42.0            | 689             |
|                        |       | 1,024.1           | 42,625          | 338.9           | 14,108          | 5.28               | 13.4 | 3.03 | 7.70 | 158.5              | 2,597           | 53.8            | 881             |
|                        |       | 1,268.9           | 52,817          | 422.1           | 17,570          | 5.35               | 13.6 | 3.07 | 7.80 | 190.8              | 3,126           | 65.9            | 1,080           |
| 14 x 14 <sup>1/2</sup> |       | 733.1             | 30,515          | 261.9           | 10,901          | 5.83               | 14.8 | 3.49 | 8.87 | 107.5              | 1,762           | 35.9            | 588             |
|                        |       | 909.1             | 37,840          | 326.2           | 13,576          | 5.91               | 15.0 | 3.53 | 8.96 | 131.2              | 2,150           | 44.4            | 727             |
|                        |       | 1,055.1           | 43,916          | 379.6           | 15,799          | 5.94               | 15.1 | 3.56 | 9.03 | 150.4              | 2,464           | 51.3            | 841             |
|                        |       | 1,228.5           | 51,134          | 443.1           | 18,444          | 5.98               | 15.2 | 3.59 | 9.11 | 172.6              | 2,829           | 59.6            | 976             |



## STEEL PILES

LARSEN AND FRODINGHAM SHEET PILES

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U-TYPE

61

Z-TYPE / STRAIGHT WEB-TYPE

63



YICK HOE



# STEEL PILES

Larssen and Frodingham profiles of sheet piling are well known to civil engineers throughout the world.

Their usefulness and reliability in all types of earth and water retaining structures has been proven on countless occasions over many years. However, with the benefit of experience and research, the range and efficiency of the sections has continually been improved.

The following information contains full details of a major development - the new range of Larssen profiles recently developed.

The new range provides sections which are markedly superior to those previously available in terms of strength to weight ratios, but which have also been carefully proportioned to retain the traditionally highly valued driving characteristics.

More comprehensive information on the design and installation of all sections is available in the Piling Handbook, available from the publisher of this handbook - CHYE HIN HARDWARE PTE LTD.

## STEEL QUALITIES

|                                                  | Ultimate stress    |                   | Minimum yield stress            |                   |                                            |                   | Minimum elongation on 200 mm   |                |
|--------------------------------------------------|--------------------|-------------------|---------------------------------|-------------------|--------------------------------------------|-------------------|--------------------------------|----------------|
|                                                  |                    |                   | Up to and including 16 mm thick |                   | Over 16 mm up to and including 40 mm thick |                   | Up to and including 9 mm thick | Over 9mm thick |
|                                                  | kg/mm <sup>2</sup> | N/mm <sup>2</sup> | kg/mm <sup>2</sup>              | N/mm <sup>2</sup> | kg/mm <sup>2</sup>                         | N/mm <sup>2</sup> | %                              | %              |
| BS.4360:1986 (mild steel)<br>Grade FE 430A       | 43.8/59.0          | 430/580           | 28.0                            | 275               | 27.0                                       | 265               | 16                             | 20             |
| BS.4360:1986 (High yield steel)<br>Grade FE 510A | 49.9/65.2          | 490/640           | 36.2                            | 355               | 35.2                                       | 345               | 15                             | 18             |
| Similar to ASTM-A328                             | 49.2 min.          | 482.6 min.        | 27.0                            | 265               | 27.0                                       | 265               | --                             | 17             |

Notes:

COPPER BEARING STEEL - all the grades of steel are available with 0.20% up to 0.35% or over 0.35% up to 0.50% Copper content.

1N/mm<sup>2</sup> = 0.101972 kg/mm<sup>2</sup>

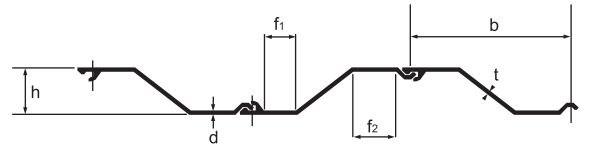
## RECOMMENDED WORKING STRESSES FOR STEEL SHEET PILING

| Class of work | BS.4360<br>Grade 43A<br>ASTM A328 and CSA G40.7<br>(mild steel) |                   | BS.4360<br>Grade 50A<br>(high yield steel) |                   |
|---------------|-----------------------------------------------------------------|-------------------|--------------------------------------------|-------------------|
|               | kg/mm <sup>2</sup>                                              | N/mm <sup>2</sup> | kg/mm <sup>2</sup>                         | N/mm <sup>2</sup> |
| Permanent     | 14.2                                                            | 140               | 18.3                                       | 180               |
| Temporary     | 17.8                                                            | 175               | 22.9                                       | 225               |

Note:

Stresses in temporary conditions occurring during construction may be varied at the discretion of the client's engineer.

## Sheet



## Frodingham Steel Sheet Piling - Dimensions and Properties

| Section | b<br>mm<br>(nom.) | h<br>mm<br>(nom.) | d<br>mm | t<br>mm<br>(nom.) | f <sub>1</sub><br>mm<br>(nom.) | f <sub>2</sub><br>mm<br>(nom.) | Section<br>area<br>sq. cm<br>per metre<br>of wall | Unit<br>weight<br>kg per<br>linear<br>metre |       | Moment<br>of inertia<br>cm <sup>4</sup><br>per<br>metre | Modulus<br>of section<br>cm <sup>3</sup><br>per<br>metre |
|---------|-------------------|-------------------|---------|-------------------|--------------------------------|--------------------------------|---------------------------------------------------|---------------------------------------------|-------|---------------------------------------------------------|----------------------------------------------------------|
| 1BXN    | 476               | 143               | 12.7    | 12.7              | 78                             | 123                            | 166.5                                             | 62.1                                        | 130.4 | 4,919                                                   | 688                                                      |
| 1N      | 483               | 170               | 9.0     | 9.0               | 105                            | 137                            | 126.0                                             | 47.8                                        | 99.1  | 6,048                                                   | 713                                                      |
| 2N      | 483               | 235               | 9.7     | 8.4               | 97                             | 149                            | 143.0                                             | 54.2                                        | 112.3 | 13,513                                                  | 1,150                                                    |
| 3N      | 483               | 283               | 11.7    | 8.9               | 89                             | 145                            | 175.0                                             | 66.2                                        | 137.1 | 23,885                                                  | 1,688                                                    |
| 3NA     | 483               | 305               | 9.7     | 9.5               | 96                             | 146                            | 165.0                                             | 62.6                                        | 129.8 | 25,687                                                  | 1,690                                                    |
| 4N      | 483               | 330               | 14.0    | 10.4              | 77                             | 127                            | 218.0                                             | 82.4                                        | 170.8 | 39,831                                                  | 2,414                                                    |
| 5       | 425               | 311               | 17.0    | 11.9              | 89                             | 118                            | 302.0                                             | 100.8                                       | 236.9 | 49,262                                                  | 3,168                                                    |

Rolling margin is within 4 per cent over and 2.5 per cent under theoretical mass; 75 mm over and 50 mm under on length.

\*Please check on availability before specifying.

### Special Sections

Sections may be "rolled up" (thickened) or "rolled down" (thinned) by special arrangement to increase or decrease the thickness of both webs and flanges by a maximum of 0.8 mm.

## INTERLOCKING SECTIONS

| Section<br>(Tongue) | Interlocks with<br>(Groove) | Section<br>(Groove) | Interlocks with<br>(Tongue) |
|---------------------|-----------------------------|---------------------|-----------------------------|
| 1N                  | 2N, 3NA                     | 1N                  | NONE                        |
| 1BXN                | 3N, 4N                      | 1BXN                | 2N, 3NA                     |
| 2N                  | 1BXN, 3N, 3NA               | 2N                  | 1N, 3NA                     |
| 3N                  | 4N                          | 3N                  | 1BXN, 2N, 3NA               |
| 3NA                 | 3N, 1BXN, 2N                | 3NA                 | 1N, 2N                      |
| 4N                  | 5                           | 4N                  | 1BXN, 3N                    |
| 5                   | NONE                        | 5                   | 4N                          |

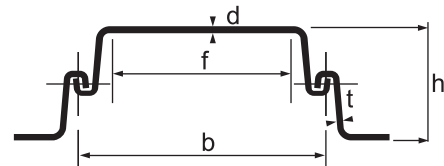
## RECOMMENDED MAXIMUM LENGTHS FOR DRIVING

The maximum length for each piling section depends upon the type of strata encountered, penetration required, and the type of construction for which the piling is designed. The following table is provided as a guide only.

In hard driving conditions it may be necessary to move up a section size to achieve the required penetration. Alternatively Grade 50A may be used.

| Section | Approx. max.<br>length (metres) | Section | Approx. max.<br>length (metres) | Section | Approx. max.<br>length (metres) |
|---------|---------------------------------|---------|---------------------------------|---------|---------------------------------|
| 6W      | 9                               | 32W     | 26                              | 2N      | 14                              |
| 9W      | 14                              | 3       | 18                              | 3N      | 18                              |
| 12W     | 17                              | 4A      | 23                              | 3NA     | 18                              |
| 16W     | 20                              | 6       | 30                              | 4N      | 23                              |
| 20W     | 23                              | 1N      | 11                              | 5       | 24                              |
| 25W     | 23                              | 1BXN    | 14                              |         |                                 |

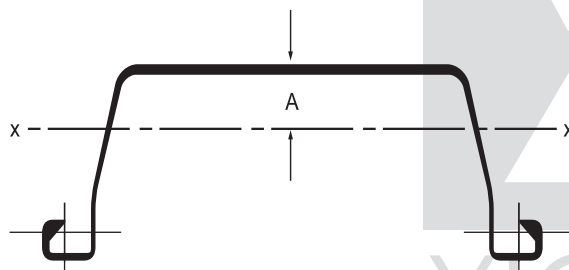
# STEEL PILES



## LARSEN STEEL SHEET PILING - Dimensions and Properties

| Section | b<br>mm<br>(nom.) | h<br>mm<br>(nom.) | d<br>mm | t<br>mm<br>(nom.) | f<br>Flat of<br>pan<br>mm | Section<br>area<br>sq. cm<br>per metre<br>of wall | Unit<br>weight            |                                | Combined<br>moment<br>of inertia<br>cm <sup>4</sup><br>per<br>metre | Modulus<br>of<br>section<br>cm <sup>3</sup><br>per<br>metre |
|---------|-------------------|-------------------|---------|-------------------|---------------------------|---------------------------------------------------|---------------------------|--------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------|
|         |                   |                   |         |                   |                           |                                                   | kg per<br>linear<br>metre | kg per<br>sq. metre<br>of wall |                                                                     |                                                             |
| 6W      | 525               | 212               | 7.8     | 6.4               | 331                       | 108                                               | 44.7                      | 85.1                           | 6,371                                                               | 601                                                         |
| 9W      | 525               | 260               | 8.9     | 6.4               | 343                       | 124                                               | 51.0                      | 97.1                           | 11,726                                                              | 902                                                         |
| 12W     | 525               | 306               | 9.0     | 8.5               | 343                       | 147                                               | 60.4                      | 115.1                          | 18,345                                                              | 1,199                                                       |
| 16W     | 525               | 348               | 10.5    | 8.9               | 341                       | 166                                               | 68.3                      | 130.1                          | 27,857                                                              | 1,601                                                       |
| 20W     | 525               | 400               | 11.3    | 9.4               | 333                       | 188                                               | 77.3                      | 147.2                          | 40,180                                                              | 2,009                                                       |
| 25W     | 525               | 454               | 12.1    | 10.5              | 317                       | 213                                               | 87.9                      | 167.4                          | 56,727                                                              | 2,499                                                       |
| 32W     | 525               | 454               | 17.0    | 10.5              | 317                       | 252                                               | 103.6                     | 197.4                          | 73,003                                                              | 3,216                                                       |
| 3       | 400               | 250               | 14.1    | 8.5               | 248                       | 198                                               | 62.2                      | 155.5                          | 16,980                                                              | 1,360                                                       |
| 4A      | 400               | 381               | 15.7    | 9.6               | 219                       | 236                                               | 74.0                      | 185.1                          | 44,916                                                              | 2,360                                                       |
| 6       | 420               | 440               | 22.0    | 14.0              | 248                       | 370                                               | 122.0                     | 290.5                          | 92,452                                                              | 4,200                                                       |
| 6       | 420               | 440               | 25.4    | 14.0              | 251                       | 397                                               | 131.0                     | 311.8                          | 102,861                                                             | 4,675                                                       |
| 6       | 420               | 440               | 28.6    | 14.0              | 251                       | 421                                               | 138.7                     | 330.2                          | 111,450                                                             | 5,066                                                       |
| GSP3    | 400               | 250               | 13.0    | 8.6               | 271                       | 191                                               | 60.0                      | 150.0                          | 16,800                                                              | 1,340                                                       |

Rolling margin is within 4 per cent over and 2.5 per cent under theoretical mass; 75mm over and 50 mm under on length.



| Section  | Area<br>cm <sup>2</sup> | Dimension<br>A<br>mm | Moment<br>of inertia<br>about axis<br>XX<br>cm <sup>4</sup> | Section<br>modulus<br>about axis<br>XX<br>cm <sup>3</sup> |
|----------|-------------------------|----------------------|-------------------------------------------------------------|-----------------------------------------------------------|
| 6W       | 57                      | 45                   | 1,247                                                       | 155                                                       |
| 9W       | 65                      | 50                   | 2,031                                                       | 204                                                       |
| 12W      | 77                      | 62                   | 3,240                                                       | 288                                                       |
| 16W      | 87                      | 66                   | 4,495                                                       | 347                                                       |
| 20W      | 98                      | 79                   | 6,737                                                       | 467                                                       |
| 25W      | 112                     | 93                   | 9,605                                                       | 607                                                       |
| 32W      | 132                     | 82                   | 10,580                                                      | 626                                                       |
| 3        | 79                      | 49                   | 2,227                                                       | 227                                                       |
| 4A       | 94                      | 76                   | 5,650                                                       | 412                                                       |
| 6 (22.0) | 155                     | 90                   | 12,437                                                      | 770                                                       |
| 6 (25.4) | 167                     | 85                   | 12,882                                                      | 777                                                       |
| 6 (28.6) | 177                     | 82                   | 13,185                                                      | 780                                                       |
| GSP3     | 76.4                    | --                   | 2,220                                                       | 223                                                       |

## INTERLOCKING SECTIONS

| Section | Interlocks with                | Section | Interlocks with                    |
|---------|--------------------------------|---------|------------------------------------|
| 6W      | 9W, 12W, 16W, 4A               | 25W     | 12W, 16W, 20W, 32W, 3, 4A, 6       |
| 9W      | 6W, 12W, 16W, 4A               | 32W     | 12W, 16W, 20W, 25W, 3, 4A, 6       |
| 12W     | 6W, 9W, 16W, 20W, 25W, 32W, 4A | 3       | 20W, 25W, 32W, 4A                  |
| 16W     | 6W, 9W, 12W, 20W, 25W, 32W, 4A | 4A      | 6W, 9W, 12W, 16W, 20W, 25W, 32W, 3 |
| 20W     | 12W, 16W, 25W, 32W, 3, 4A, 6   | 6       | 20W, 25W, 32W                      |

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## GENERAL INFORMATION

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Steel sheet piling is used in all types of temporary works and permanent structures including cofferdams, retaining walls, river frontages, quays, wharves, dock and harbour works, permanent foundations, land reclamation and sea defence works.

The sections are designed to provide the maximum strength and durability at the lowest possible weight consistent with good driving qualities. The design of the section interlocks facilitates pitching and driving and results in a series of closely fitting joints, forming a continuous wall.

A comprehensive range of sections in widely differing sizes and weights is obtainable in various grades of steel which enables the most economical choice to be made to suit the nature and requirements of any given contract. This range can be extended if the tonnage required is economic.

Certain sections intended for use in permanent structures where stresses are not severe, such as cut-off-walls, are of uniform thickness throughout in order to provide the maximum effective useful life.

Where piles form temporary works, they can be extracted easily and without distortion for re-use.

Corner, junction and closure piles are available to cover all requirements.

Should any doubts or problems arise advice is available on application to the publisher of this handbook - CHYE HIN HARDWARE PTE LTD.

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## SPLICES

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Steel sheet piling is generally available in lengths up to 30m. Where it is necessary to increase the pile length during driving, fishplated or site-welded joints can be used. Particulars of standard fish-plated joints or weld preparations are available on application.

Because of normal rolling tolerances there may be variations in the pile profile. Every effort should be made to match pile ends before welding.

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## CIRCULAR CONSTRUCTION

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Steel sheet piling can be driven to form a complete circle without the water tightness of the locks being impaired.

The maximum angle of deviation in the locks is 9° for Larssen piles, and 3° for Frodingham piles. Deviation in the locks of Larssen Box Piles is limited to 3°.

The following table gives the approximate minimum diameters of circles which can be constructed using various sheet pile sections.

The diameters are only intended for guidance, as the possible deviation will vary according to length of piles, penetration required and section used.

Smaller diameters can be obtained if individual bent corners are introduced.

# STEEL PILES

| Type       | Section | Minimum number of piles used | Approx. minimum diameter at internal face of wall (metres) | Approx. maximum driving length at minimum diameter (metres) |
|------------|---------|------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| Larssen    | 6W      | 40                           | 6.47                                                       | 6                                                           |
| Larssen    | 9W      | 40                           | 6.42                                                       | 9                                                           |
| Larssen    | 12W     | 40                           | 6.38                                                       | 9                                                           |
| Larssen    | 16W     | 40                           | 6.34                                                       | 15                                                          |
| Larssen    | 20W     | 40                           | 6.28                                                       | 15                                                          |
| Larssen    | 25W     | 40                           | 6.23                                                       | 15                                                          |
| Larssen    | 32W     | 40                           | 6.23                                                       | 15                                                          |
| Larssen    | 3       | 40                           | 4.85                                                       | 9                                                           |
| Larssen    | 4A      | 40                           | 4.71                                                       | 15                                                          |
| Larssen    | 6       | 40                           | 4.91                                                       | 15                                                          |
| Frodingham | 1N      | 120                          | 18.36                                                      | 6                                                           |
| Frodingham | 1BXN    | 120                          | 18.10                                                      | 6                                                           |
| Frodingham | 2N      | 120                          | 18.32                                                      | 9                                                           |
| Frodingham | 3N, 3NA | 120                          | 18.30                                                      | 15                                                          |
| Frodingham | 4N      | 120                          | 18.28                                                      | 15                                                          |
| Frodingham | 5       | 120                          | 16.07                                                      | 15                                                          |

## EFFECTIVE LIFE

The effective life of unpainted or otherwise unprotected steel piling to BS 4360 Grades 43A and 50A depends upon the combined effects of imposed stresses and corrosion.

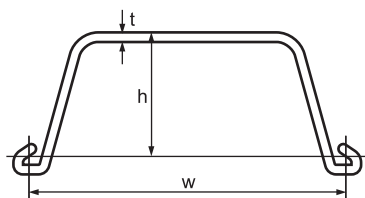
Performance is clearly optimised with combinations of high stress and low corrosion rate. Although other stress and corrosion combinations are less favourable, good design can ensure acceptable lives. The use of grade 50A steel at grade 43A stresses can also increase effective life. Copper-bearing steels are marginally advantageous but only in atmospheric exposures.

The table below indicates the minimum effective life for various environments assuming that maximum permissible design stress occurs in the zone under consideration. For regions of lower stress, the permissible loss in thickness may be increased and appropriate effective lives calculated from the corrosion data for each service environment.

### Minimum Effective Life for Maximum Stress in Various Service Environments

| Section type                                 |            | Environments                                                         |                          |                            |                                    |             |
|----------------------------------------------|------------|----------------------------------------------------------------------|--------------------------|----------------------------|------------------------------------|-------------|
| Frodingham                                   | Larssen    | Permissible loss in thickness (mm) at position of max stress in pile | Atmospheric corrosion    | Splash and low water zones | Sea water immersion and tidal zone | Underground |
|                                              |            |                                                                      | Corrosion rates, mm/year |                            |                                    |             |
|                                              |            |                                                                      | 0.05 (mean)              | 0.09 (mean)                | 0.05 (mean)                        | 0.03 (max)  |
|                                              |            |                                                                      | Effective life in years  |                            |                                    |             |
| 1N<br>2N<br>3NA<br>3N<br><br>1BXN<br>4N<br>5 | 6W, 9W     | 3.5                                                                  | 70                       | *                          | *                                  | 117         |
|                                              | 12W, 16W   | 4.5                                                                  | 90                       | 50                         | 90                                 | 120 +       |
|                                              | 20W, 3, 4A | 5.0                                                                  | 100                      | 55                         | 100                                | 120 +       |
|                                              |            | 5.2                                                                  | 104                      | 58                         | 104                                | 120 +       |
|                                              |            | 5.4                                                                  | 108                      | 60                         | 108                                | 120 +       |
|                                              |            | 5.5                                                                  | 110                      | 60                         | 110                                | 120 +       |
|                                              |            | 5.9                                                                  | 118                      | 65                         | 118                                | 120 +       |
|                                              | 25W, 32W   | 6.0                                                                  | 120                      | 67                         | 120                                | 120 +       |
|                                              | All No. 6  | 7.0                                                                  | 120 +                    | 78                         | 120 +                              | 120 +       |
|                                              |            | 7.2                                                                  | 120 +                    | 80                         | 120 +                              | 120 +       |
|                                              |            | 7.4                                                                  | 120 +                    | 82                         | 120 +                              | 120 +       |
|                                              |            | 8.9                                                                  | 120 +                    | 99                         | 120 +                              | 120 +       |

\* Not suitable for Marine Environment.



## U-TYPE

### KSP DIMENSIONS & PROPERTIES

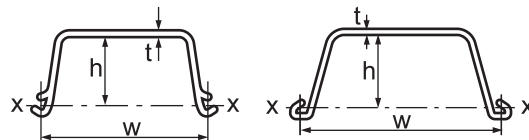
| Section number | Dimensions |         |         | Section area<br>Per pile | Unit weight      |                                               | Moment of inertia           |                                                | Modulus of section          |                                                |
|----------------|------------|---------|---------|--------------------------|------------------|-----------------------------------------------|-----------------------------|------------------------------------------------|-----------------------------|------------------------------------------------|
|                | w<br>mm    | h<br>mm | t<br>mm |                          | Per pile<br>kg/m | Per linear meter of wall<br>kg/m <sup>2</sup> | Per pile<br>cm <sup>4</sup> | Per linear meter of wall<br>cm <sup>4</sup> /m | Per pile<br>cm <sup>3</sup> | Per linear meter of wall<br>cm <sup>3</sup> /m |
| KSP-IA         | 400        | 85      | 8.0     | 45.2                     | 35.5             | 88.8                                          | 598                         | 4,500                                          | 88                          | 529                                            |
| KSP-II         | 400        | 100     | 10.5    | 61.2                     | 48.0             | 120                                           | 1,190                       | 8,700                                          | 146                         | 870                                            |
| KSP-IIA        | 400        | 120     | 9.2     | 55.0                     | 43.2             | 108                                           | 1,450                       | 10,600                                         | 162                         | 880                                            |
| KSP-III        | 400        | 130     | 13.0    | 76.4                     | 60.0             | 150                                           | 2,320                       | 17,400                                         | 232                         | 1,340                                          |
| KSP-IIIA       | 400        | 150     | 13.1    | 74.4                     | 58.4             | 146                                           | 2,840                       | 22,800                                         | 253                         | 1,520                                          |
| KSP-IV         | 400        | 160     | 16.0    | 96.9                     | 76.1             | 190                                           | 4,110                       | 34,400                                         | 334                         | 2,150                                          |
| KSP-IVA        | 400        | 180     | 16.2    | 94.2                     | 74.0             | 185                                           | 4,910                       | 40,600                                         | 374                         | 2,250                                          |
| KSP-VL         | 500        | 200     | 24.3    | 133.8                    | 105.0            | 210                                           | 7,960                       | 63,000                                         | 520                         | 3,150                                          |
| KSP-VIL        | 500        | 225     | 27.6    | 153.0                    | 120.0            | 240                                           | 11,400                      | 86,000                                         | 680                         | 3,820                                          |

## U-TYPE

### Imperial units

| Section number | Dimensions |          |          | Section area<br>Per pile | Unit weight       |                                               | Moment of inertia           |                                                | Modulus of section          |                                                |
|----------------|------------|----------|----------|--------------------------|-------------------|-----------------------------------------------|-----------------------------|------------------------------------------------|-----------------------------|------------------------------------------------|
|                | w<br>in.   | h<br>in. | t<br>in. |                          | Per pile<br>lb/ft | Per linear foot of wall<br>lb/ft <sup>2</sup> | Per pile<br>in <sup>4</sup> | Per linear foot of wall<br>in <sup>4</sup> /ft | Per pile<br>in <sup>3</sup> | Per linear foot of wall<br>in <sup>3</sup> /ft |
| KSP-IA         | 15.75      | 3.35     | 0.31     | 7.01                     | 23.9              | 18.2                                          | 14                          | 33                                             | 5.4                         | 9.8                                            |
| KSP-II         | 15.75      | 3.94     | 0.41     | 9.49                     | 32.3              | 24.6                                          | 29                          | 64                                             | 8.9                         | 16.2                                           |
| KSP-IIA        | 15.75      | 4.72     | 0.36     | 8.53                     | 29.0              | 22.1                                          | 35                          | 78                                             | 9.9                         | 16.4                                           |
| KSP-III        | 15.75      | 5.12     | 0.51     | 11.84                    | 40.3              | 30.7                                          | 56                          | 127                                            | 14.2                        | 24.9                                           |
| KSP-IIIA       | 15.75      | 5.91     | 0.52     | 11.53                    | 39.2              | 29.9                                          | 68                          | 167                                            | 15.4                        | 28.3                                           |
| KSP-IV         | 15.75      | 6.30     | 0.63     | 15.02                    | 51.1              | 38.9                                          | 99                          | 252                                            | 20.4                        | 40.0                                           |
| KSP-IVA        | 15.75      | 7.09     | 0.64     | 14.60                    | 49.7              | 37.9                                          | 118                         | 297                                            | 22.8                        | 41.9                                           |
| KSP-VL         | 19.68      | 7.87     | 0.96     | 20.74                    | 70.6              | 43.0                                          | 191                         | 461                                            | 31.7                        | 58.6                                           |
| KSP-VIL        | 19.68      | 8.86     | 1.09     | 23.72                    | 80.6              | 49.2                                          | 274                         | 630                                            | 41.5                        | 71.1                                           |

# STEEL PILES

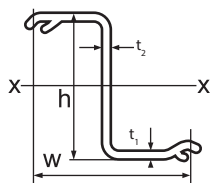


## U-TYPE

| Section  | Dimensions  |             |               | Section area<br>Per pile           | Unit weight   |                                         | Moment of inertia                  |                                           | Modulus of section                 |                                           |
|----------|-------------|-------------|---------------|------------------------------------|---------------|-----------------------------------------|------------------------------------|-------------------------------------------|------------------------------------|-------------------------------------------|
|          | w           | h           | t             |                                    | Per pile      | Per wall width                          | Per pile                           | Per wall width                            | Per pile                           | Per wall width                            |
|          | mm<br>in    | mm<br>in    | mm<br>in      | cm <sup>2</sup><br>in <sup>2</sup> | kg/m<br>lb/ft | kg/m <sup>2</sup><br>lb/ft <sup>2</sup> | cm <sup>4</sup><br>in <sup>4</sup> | cm <sup>4</sup> /m<br>in <sup>4</sup> /ft | cm <sup>3</sup><br>in <sup>3</sup> | cm <sup>3</sup> /m<br>in <sup>3</sup> /ft |
| YSP I    | 400<br>15.7 | 75<br>2.95  | 8.0<br>0.315  | 46.49<br>7.206                     | 36.5<br>24.5  | 91.2<br>18.7                            | 429<br>10.3                        | 3,820<br>28.0                             | 66.4<br>4.05                       | 509<br>9.47                               |
| YSP U-5  | 400<br>15.7 | 80<br>3.15  | 7.6<br>0.299  | 45.21<br>7.008                     | 35.5<br>23.9  | 88.8<br>18.2                            | 454<br>10.9                        | 4,220<br>30.9                             | 64.7<br>3.95                       | 527<br>9.80                               |
| FSP IA   | 400<br>15.7 | 85<br>3.35  | 8.0<br>0.315  | 45.21<br>7.008                     | 35.5<br>23.9  | 88.8<br>18.2                            | 598<br>14.4                        | 4,500<br>33.0                             | 88.0<br>5.37                       | 529<br>9.84                               |
| YSP II   | 400<br>15.7 | 100<br>3.94 | 10.5<br>0.413 | 61.18<br>9.483                     | 48.0<br>32.3  | 120<br>24.6                             | 986<br>23.7                        | 8,690<br>63.6                             | 121<br>7.38                        | 869<br>16.2                               |
| FSP II   | 400<br>15.7 | 100<br>3.94 | 10.5<br>0.413 | 61.18<br>9.483                     | 48.0<br>32.3  | 120<br>24.6                             | 1,240<br>29.8                      | 8,740<br>64.0                             | 152<br>9.28                        | 874<br>16.3                               |
| YSP U-9  | 400<br>15.7 | 110<br>4.33 | 9.3<br>0.366  | 55.01<br>8.527                     | 43.2<br>29.0  | 108<br>22.1                             | 1,070<br>25.7                      | 9,680<br>70.9                             | 120<br>7.32                        | 880<br>16.4                               |
| FSP IIA  | 400<br>15.7 | 120<br>4.72 | 9.2<br>0.362  | 55.01<br>8.527                     | 43.2<br>29.0  | 108<br>22.1                             | 1,460<br>35.1                      | 10,600<br>77.6                            | 160<br>9.76                        | 880<br>16.4                               |
| YSP III  | 400<br>15.7 | 125<br>4.92 | 13.0<br>0.512 | 76.42<br>11.85                     | 60.0<br>40.3  | 150<br>30.7                             | 1,920<br>46.1                      | 16,400<br>120                             | 196<br>12.0                        | 1,310<br>24.4                             |
| FSP III  | 400<br>15.7 | 125<br>4.92 | 13.0<br>0.512 | 76.42<br>11.85                     | 60.0<br>40.3  | 150<br>30.7                             | 2,220<br>53.3                      | 16,800<br>123                             | 223<br>13.6                        | 1,340<br>24.9                             |
| YSP U-15 | 400<br>15.7 | 150<br>5.91 | 12.2<br>0.48  | 74.40<br>11.53                     | 58.4<br>39.2  | 146<br>29.9                             | 2,700<br>64.9                      | 22,800<br>167                             | 238<br>14.5                        | 1,520<br>28.3                             |
| FSP IIIA | 400<br>15.7 | 150<br>5.91 | 13.1<br>0.516 | 74.40<br>11.53                     | 58.4<br>39.2  | 146<br>29.9                             | 2,790<br>67.0                      | 22,800<br>167                             | 250<br>15.3                        | 1,520<br>28.3                             |
| YSP IV   | 400<br>15.7 | 155<br>6.10 | 15.5<br>0.61  | 96.99<br>15.03                     | 76.1<br>51.1  | 190<br>38.9                             | 3,690<br>88.7                      | 31,900<br>234                             | 311<br>19.0                        | 2,060<br>38.3                             |
| FSP IV   | 400<br>15.7 | 170<br>6.69 | 15.5<br>0.61  | 96.99<br>15.03                     | 76.1<br>51.1  | 190<br>38.9                             | 4,670<br>112                       | 38,600<br>283                             | 362<br>22.1                        | 2,270<br>42.2                             |
| YSP U-23 | 400<br>15.7 | 175<br>6.89 | 14.7<br>0.579 | 94.21<br>14.60                     | 74.0<br>49.7  | 185<br>37.9                             | 4,380<br>105                       | 39,400<br>289                             | 330<br>20.1                        | 2,250<br>41.9                             |
| FSP IVA  | 400<br>15.7 | 185<br>7.28 | 16.1<br>0.634 | 94.21<br>14.60                     | 74.0<br>49.7  | 185<br>37.9                             | 5,300<br>127                       | 41,600<br>305                             | 400<br>24.4                        | 2,250<br>41.9                             |
| YSP V    | 420<br>16.5 | 175<br>6.89 | 22.0<br>0.866 | 134.0<br>20.77                     | 105.0<br>70.6 | 250<br>51.2                             | 5,950<br>143                       | 55,200<br>404                             | 433<br>26.4                        | 3,150<br>58.6                             |
| FSP VL   | 500<br>19.7 | 200<br>7.87 | 24.3<br>0.957 | 133.8<br>20.74                     | 105.0<br>70.6 | 210<br>43.0                             | 7,960<br>191                       | 63,000<br>461                             | 520<br>31.7                        | 3,150<br>58.6                             |
| FSP VIL  | 500<br>19.7 | 225<br>8.86 | 27.6<br>1.09  | 153.0<br>23.72                     | 120.0<br>80.6 | 240<br>49.2                             | 11,400<br>274                      | 86,000<br>630                             | 680<br>41.5                        | 3,820<br>71.1                             |

Note:  
Weight per linear meter (foot) of wall is rounded off using the JIS Z 8401-specified formula: Weight per section x

$$\frac{1,000}{W \text{ (effective width)}}$$



# STEEL PILES

## Z-TYPE

| Section  | Dimensions |          |          |          | Section area<br>Per pile           | Unit weight   |                                         | Moment of inertia                  |                                           | Modulus of section                 |                                           |
|----------|------------|----------|----------|----------|------------------------------------|---------------|-----------------------------------------|------------------------------------|-------------------------------------------|------------------------------------|-------------------------------------------|
|          | w          | h        | t1       | t2       |                                    | Per pile      | Per wall width                          | Per pile                           | Per wall width                            | Per pile                           | Per wall width                            |
|          | mm<br>in   | mm<br>in | mm<br>in | mm<br>in | cm <sup>2</sup><br>in <sup>2</sup> | kg/m<br>lb/ft | kg/m <sup>2</sup><br>lb/ft <sup>2</sup> | cm <sup>4</sup><br>in <sup>4</sup> | cm <sup>4</sup> /m<br>in <sup>4</sup> /ft | cm <sup>3</sup><br>in <sup>3</sup> | cm <sup>3</sup> /m<br>in <sup>3</sup> /ft |
| FSP Z-25 | 400        | 305      | 13.0     | 9.6      | 94.32                              | 74.0          | 185                                     | 15,300                             | 38,300                                    | 1,000                              | 2,510                                     |
|          | 15.7       | 12.0     | 0.512    | 0.378    | 14.62                              | 49.7          | 37.9                                    | 368                                | 280                                       | 61.0                               | 46.7                                      |
| FSP Z-32 | 400        | 344      | 14.2     | 10.4     | 107.7                              | 84.5          | 211                                     | 22,000                             | 55,000                                    | 1,280                              | 3,200                                     |
|          | 15.7       | 13.5     | 0.559    | 0.409    | 16.69                              | 56.8          | 43.2                                    | 529                                | 403                                       | 78.1                               | 59.5                                      |
| FSP Z-38 | 400        | 364      | 17.2     | 11.4     | 122.3                              | 96.0          | 240                                     | 27,700                             | 69,200                                    | 1,520                              | 3,800                                     |
|          | 15.7       | 14.3     | 0.677    | 0.449    | 18.96                              | 64.5          | 49.2                                    | 665                                | 507                                       | 92.8                               | 70.7                                      |
| FSP Z-45 | 400        | 367      | 21.9     | 13.2     | 148.2                              | 116           | 290                                     | 33,400                             | 83,500                                    | 1,820                              | 4,550                                     |
|          | 15.7       | 14.4     | 0.862    | 0.52     | 22.97                              | 77.9          | 59.4                                    | 802                                | 611                                       | 111                                | 84.6                                      |

## STRAIGHT WEB-TYPE



| Section | Dimensions |          |          | Section area<br>Per pile           | Unit weight   |                                         | Moment of inertia<br>Per pile      | Modulus of section<br>Per pile     | Interlock strength min. |     |
|---------|------------|----------|----------|------------------------------------|---------------|-----------------------------------------|------------------------------------|------------------------------------|-------------------------|-----|
|         | w          | h        | t        |                                    | Per pile      | Per wall width                          |                                    |                                    |                         |     |
|         | mm<br>in   | mm<br>in | mm<br>in | cm <sup>2</sup><br>in <sup>2</sup> | kg/m<br>lb/ft | kg/m <sup>2</sup><br>lb/ft <sup>2</sup> | cm <sup>4</sup><br>in <sup>4</sup> | cm <sup>3</sup><br>in <sup>3</sup> | t/m<br>lb/ft            |     |
| YSP F   | 400        | 44.5     | 9.5      | 69.07                              | 54.2          | 136                                     | 190                                | 47.8                               | 400                     |     |
|         | 15.7       | 1.75     | 0.374    | 10.71                              | 36.4          | 27.9                                    | 4.56                               | 2.92                               | 268                     | 780 |
| YSP FA  | 400        | 44.5     | 12.7     | 77.50                              | 60.8          | 152                                     | 196                                | 48.3                               | 400                     |     |
|         | 15.7       | 1.75     | 0.5      | 12.01                              | 40.9          | 31.1                                    | 4.71                               | 2.95                               | 268                     | 780 |
| YSP FX  | 400        | 47.0     | 12.7     | 85.66                              | 67.2          | 168                                     | 224                                | 56.1                               | 600                     |     |
|         | 15.7       | 1.85     | 0.5      | 13.28                              | 45.2          | 34.4                                    | 5.38                               | 3.42                               | 403                     | 170 |

Note:

- YSP FA with minimum interlock strength of 500 t/m is subject to negotiation.
- Normal interlock swing of YSP and YSP FA is at least 10° and that of YSP FX at least 8°.





## API PIPES ERW & SEAMLESS PIPES

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YICK HOE

# API PIPES ERW & SEAMLESS PIPES

## API MECHANICAL & CHEMICAL SPECIFICATION

### TENSILE AND CHEMICAL REQUIREMENTS (EXCERPTS FROM API SPECIFICATIONS, 1983 EDITION)

| Application               | Grade                                                                      | Tensile Requirements |            |                                    |                                        |                    |                           | Elongation                                   |       |
|---------------------------|----------------------------------------------------------------------------|----------------------|------------|------------------------------------|----------------------------------------|--------------------|---------------------------|----------------------------------------------|-------|
|                           |                                                                            | Yield Strength       |            |                                    | Tensile Strength                       |                    |                           |                                              | % min |
|                           |                                                                            | psi<br>min           | MPa<br>min | kg/mm <sup>2</sup><br>min          | psi<br>min                             | MPa<br>min         | kg/mm <sup>2</sup><br>min |                                              |       |
| Line<br>Pipe              | API 5L Grade A                                                             | 30,000               | 207        | 21.1                               | 48,000                                 | 331                | 33.7                      | e = 625,000 x $\frac{A^{0.2}_{*2}}{U^{0.9}}$ |       |
|                           | API 5L Grade B                                                             | 35,000               | 241        | 24.6                               | 60,000                                 | 413                | 42.2                      |                                              |       |
|                           | API 5L Grade X42                                                           | 42,000               | 289        | 29.5                               | 60,000                                 | 413                | 42.2                      |                                              |       |
|                           | API 5L Grade X46                                                           | 46,000               | 317        | 32.3                               | 63,000                                 | 434                | 44.3                      |                                              |       |
|                           | API 5L Grade X52                                                           | 52,000               | 358        | 36.6                               | (1) 66,000* <sup>1</sup><br>(2) 72,000 | (1) 455<br>(2) 496 | 46.4<br>50.6              |                                              |       |
|                           | API 5L Grade X56* <sup>6</sup> 56,000                                      | 386                  | 39.2       | (1)                                | 71,000* <sup>1</sup><br>(2) 75,000     | (1) 489<br>(2) 517 | 49.9<br>52.7              |                                              |       |
|                           | API 5L Grade X60* <sup>3</sup> 60,000<br>* <sup>6</sup>                    | 413                  | 42.2       | (1)                                | 75,000* <sup>1</sup><br>(2) 78,000     | (1) 517<br>(2) 537 | 52.7<br>54.8              |                                              |       |
|                           | * <sup>6</sup><br>API 5L Grade X65* <sup>8</sup> 65,000<br>* <sup>11</sup> | 448                  | 45.7       | (2)                                | (1) 77,000* <sup>1</sup><br>80,000     | (1) 530<br>(2) 551 | 54.1<br>56.2              |                                              |       |
|                           | API 5L Grade X70* <sup>6</sup> 70,000<br>* <sup>11</sup>                   | 482                  | 49.2       |                                    | 82,000                                 | 565                | 57.6                      |                                              |       |
|                           | API 5LU Grade X8080,000                                                    | 551                  | 56.2       |                                    | 95,000/ * <sup>4</sup><br>125,000      | 655/<br>862        | 66.8/<br>87.8             |                                              |       |
| API 5LU Grade X100100,000 | 689                                                                        | 70.3                 |            | 110,000/ * <sup>4</sup><br>135,000 | 758/<br>931                            | 77.3/<br>94.9      |                           |                                              |       |

- \*1 (1) Wall thickness > 0.375 in., OD ≥ 20 in.  
Any wall thickness, OD < 20 in.  
(2) Wall thickness ≤ 0.375 in., OD ≥ 20 in.

- \*2 The minimum elongation in 2 inches (50.8 mm) shall be determined by the following formula:

$$e = 625,000 \times \frac{A_{*2}^{0.2}}{U^{0.9}}$$

where:

e = minimum elongation in 2 inches (50.80 mm) in per cent rounded to the nearest 1/2 per cent.

A = cross sectional area of the tensile test specimen in square inches, based on specified outside diameter or nominal specimen width and specified wall thickness rounded to the nearest 0.01 sq. in., or 0.75 sq. in., whichever is smaller.

U = specified tensile strength, psi.

- \*3 The minimum tensile strength for grade X60 electric-resistance welded pipe in all sizes and wall thicknesses shall be 75,000 psi (52.7 kg/mm<sup>2</sup>).

- \*4 Tensile strength means min. value/max. value.

- \*5 Figures in parentheses are values for seamless pipe. Upper line value means non-expanded. Lowerline value means cold-expanded. In grades X42 through X65, for each reduction of 0.01 per cent below the maximum carbon content, an increase of 0.05 per cent manganese above the specified maximum is permissible, up to a maximum of 1.45 per cent for X52 and lower and up to a maximum of 1.60 per cent for grades higher than X52.

- \*6 Other chemical analyses may be furnished by agreement between purchaser and manufacturer.

- \*7 Either columbium, vanadium, titanium, or a combination thereof, shall be used at the discretion of the manufacturer.

# API PIPES ERW & SEAMLESS PIPES

## CHEMICAL REQUIREMENTS FOR LADLE ANALYSES, %

| C <sup>*5</sup><br>max     | Si<br>max | Mn <sup>*5</sup><br>max | P<br>max | S<br>max | Cb<br>min           | V<br>min           | Ti<br>min          |
|----------------------------|-----------|-------------------------|----------|----------|---------------------|--------------------|--------------------|
| 0.21 (0.22)                |           | 0.90                    | 0.04     | 0.05     |                     |                    |                    |
| 0.26 (0.27)                |           | 1.15                    | 0.04     | 0.05     |                     |                    |                    |
| 0.28 (0.29) <sup>*13</sup> |           | 1.25                    | 0.04     | 0.05     |                     |                    |                    |
| 0.30 (0.31) <sup>*13</sup> |           | 1.35                    | 0.04     | 0.05     |                     |                    |                    |
| 0.28 (0.29) <sup>*13</sup> |           | 1.25                    |          |          |                     |                    |                    |
| 0.30 (0.31) <sup>*13</sup> |           | 1.35                    | 0.04     | 0.05     |                     |                    |                    |
| 0.28 (0.29) <sup>*13</sup> |           | 1.25                    |          |          |                     |                    |                    |
| 0.26                       |           | 1.35                    | 0.04     | 0.05     | 0.005 <sup>*7</sup> | 0.02 <sup>*7</sup> | 0.03 <sup>*7</sup> |
| 0.26                       |           | 1.35                    | 0.04     | 0.05     | 0.005 <sup>*7</sup> | 0.02 <sup>*7</sup> | 0.03 <sup>*7</sup> |
| 0.26                       |           | 1.40                    | 0.04     | 0.05     | 0.005 <sup>*9</sup> | 0.02 <sup>*9</sup> |                    |
| 0.23 <sup>*10</sup>        |           | 1.60 <sup>*10</sup>     | 0.04     | 0.05     |                     |                    |                    |
| 0.26 <sup>*12</sup>        | 0.35      | 1.40 <sup>*12</sup>     | 0.04     | 0.05     |                     |                    |                    |
| 0.26 <sup>*12</sup>        | 0.35      | 1.40 <sup>*12</sup>     | 0.04     | 0.05     |                     |                    |                    |

\*8 For grades X65 in sizes 16 in. and larger with wall thickness 0.50 in. and less, the chemical composition shall be as shown or as agreed upon between the purchaser and manufacturer. For other sizes and wall thicknesses the chemical composition shall be as agreed upon between the purchaser and manufacturer. (Applicable to welded pipe only).

\*9 Either columbium or vanadium or a combination of both shall be used at the discretion of the manufacturer.

\*10 For each reduction of 0.01 per cent below the specified maximum carbon content, an increase of 0.05 per cent manganese above the specified maximum is permissible.

\*11 For seamless pipe of grades X65 and X70, the chemical composition shall be agreed upon between the purchaser and manufacturer.

\*12 For each reduction of 0.01 per cent below the specified maximum carbon content, an increase of 0.05 per cent manganese above the specified maximum is permissible, up to a maximum of 1.50 per cent.

\*13 For cold expanded seamless pipe in size 20 in. and larger, the maximum carbon content shall be 0.28 per cent.

# API PIPES ERW & SEAMLESS PIPES

COMPARATIVE TABLES OF STEEL QUALITIES

| MECHANICAL VALUES       |                                       |                         |                      |             | CHEMICAL ANALYSIS |             |             |             |             |            |             |                              |
|-------------------------|---------------------------------------|-------------------------|----------------------|-------------|-------------------|-------------|-------------|-------------|-------------|------------|-------------|------------------------------|
| Material                | Tensile strength<br>N/mm <sup>2</sup> | Yield strength<br>(min) | Elongation<br>% min. | C<br>%      | Si<br>%           | Mn<br>%     | P<br>% max. | S<br>% max. | Mo<br>%     | Cr<br>%    | Ni<br>%     | Div<br>%                     |
| A53 - Grade A           | min. 330                              | 205                     | variable             | max. 0.25   | --                | max. 0.95   | 0.05        | 0.60        |             |            |             |                              |
| A53 - Grade B           | min. 415                              | 240                     | variable             | max. 0.30   | --                | max. 1.20   | 0.05        | 0.60        |             |            |             |                              |
| A106 - Grade A          | min. 330                              | 205                     | 35                   | max. 0.25   | min 0.1           | 0.27 - 0.93 | 0.048       | 0.058       |             |            |             |                              |
| A106 - Grade B          | min. 415                              | 240                     | 30                   | max. 0.30   | min 0.1           | 0.29 - 1.06 | 0.048       | 0.058       |             |            |             |                              |
| A179                    | Hardness Rb. max. 72                  |                         |                      | 0.06 - 0.18 | --                | 0.27 - 0.63 | 0.048       | 0.058       |             |            |             |                              |
| A333 - Grade 1          | min. 379                              | 207                     | 35                   | max. 0.30   | --                | 0.40 - 1.06 | 0.05        | 0.60        |             |            |             |                              |
| A333 - Grade 6          | min. 414                              | 241                     | 30                   | max. 0.30   | min. 0.1          | 0.29 - 1.06 | 0.048       | 0.058       |             |            |             |                              |
| A333 - Grade 3          | min. 448                              | 241                     | 30                   | max. 0.19   | 0.18 - 0.37       | 0.31 - 0.64 | 0.05        | 0.05        |             |            | 3.18 - 3.82 |                              |
| A335 - Grade P1         | min. 379                              | 207                     | 30                   | 0.1 - 0.2   | 0.1 - 0.5         | 0.3 - 0.8   | 0.045       | 0.045       | 0.44 - 0.65 |            |             |                              |
| A335 - Grade P5         | min. 414                              | 207                     | 30                   | max. 0.15   | max. 0.5          | 0.3 - 0.6   | 0.03        | 0.03        | 0.44 - 0.65 | 4.0 - 6.0  |             |                              |
| A335 - Grade P11        | min. 414                              | 207                     | 30                   | max. 0.15   | 0.5 - 1.0         | 0.3 - 0.6   | 0.03        | 0.03        | 0.44 - 0.65 | 1.0 - 1.5  |             |                              |
| A335 - Grade P12        | min. 414                              | 207                     | 30                   | max. 0.15   | max. 0.5          | 0.3 - 0.6   | 0.045       | 0.045       | 0.44 - 0.65 | 0.8 - 1.25 |             |                              |
| A335 - Grade P22        | min. 414                              | 207                     | 30                   | max. 0.15   | max. 0.5          | 0.3 - 0.6   | 0.03        | 0.03        | 0.87 - 1.13 | 1.9 - 2.6  |             |                              |
| API-5L - Grade A        | min. 331                              | 207                     | variable             | max. 0.22   | --                | max. 0.9    | 0.04        | 0.05        |             |            |             |                              |
| API-5L - Grade B        | min. 413                              | 241                     | variable             | max. 0.27   | --                | max. 1.15   | 0.04        | 0.05        |             |            |             |                              |
| API-5L - Grade X42      | min. 413                              | 289                     | variable             | max. 0.29   | --                | max. 1.25   | 0.04        | 0.05        |             |            |             |                              |
| API-5L - Grade X46      | min. 434                              | 317                     | variable             | max. 0.29   | --                | max. 1.35   | 0.04        | 0.05        |             |            |             |                              |
| API-5L - Grade X52      | min. 455                              | 358                     | variable             | max. 0.31   | --                | max. 1.35   | 0.04        | 0.05        |             |            |             |                              |
| API-5L - Grade X56      | min. 489                              | 386                     | variable             | max. 0.26   | --                | max. 1.35   | 0.04        | 0.05        |             |            |             |                              |
| API-5L - Grade X60      | min. 517                              | 413                     | variable             | max. 0.26   | --                | max. 1.35   | 0.04        | 0.05        |             |            |             |                              |
| BS3059/1 - 320          | 320-480                               | 195                     | 25                   | max. 0.16   | --                | min. 0.3    | 0.05        | 0.05        |             |            |             |                              |
| BS3059/2 - 360          | 360-500                               | 215                     | 24                   | max. 0.17   | max. 0.35         | 0.4 - 0.8   | 0.045       | 0.045       |             |            |             |                              |
| BS3601 - ERW320         | 320-440                               | 195                     | 25                   | max. 0.16   | --                | 0.3 - 0.7   | 0.05        | 0.05        |             |            |             |                              |
| BS3601 - ERW360         | 360-480                               | 215                     | 24                   | max. 0.17   | max. 0.35         | 0.4 - 0.8   | 0.045       | 0.045       |             |            |             |                              |
| BS3602 - ERW/<br>SAW410 | 410-530                               | 235                     | 22                   | max. 0.21   | max. 0.35         | 0.4 - 1.2   | 0.05        | 0.05        |             |            |             |                              |
| BS3601-S360             | 360-480                               | 215                     | 24                   | max. 0.17   | max. 0.35         | 0.4 - 0.8   | 0.05        | 0.05        |             |            |             |                              |
| BS3601-S410             | 410-530                               | 235                     | 22                   | max. 0.21   | max. 0.35         | 0.4 - 1.2   | 0.05        | 0.05        |             |            |             |                              |
| BS3602/1-360            | 360-500                               | 215                     | 24                   | max. 0.17   | max. 0.35         | 0.4 - 0.8   | 0.045       | 0.045       |             |            |             |                              |
| BS3602/1-410            | 410-550                               | 245                     | 22                   | max. 0.21   | max. 0.35         | 0.4 - 1.2   | 0.045       | 0.045       |             |            |             |                              |
| BS3602/1-490Nb          | 490-630                               | 340                     | 20                   | max. 0.23   | max. 0.35         | 0.8 - 1.5   | 0.045       | 0.045       |             |            |             | Nb 0.01-0.1                  |
| BS3602/2-410            | 410-550                               | 245                     | 24                   | max. 0.2    | max. 0.35         | 0.5 - 1.3   | 0.04        | 0.04        |             |            |             |                              |
| BS3603-410              | 410-530                               | 235                     | 22                   | max. 0.2    | max. 0.35         | 0.6 - 1.2   | 0.045       | 0.045       |             |            |             |                              |
| BS3603-503              | 440-590                               | 245                     | 16                   | max. 0.15   | 0.15 - 0.35       | 0.3 - 0.8   | 0.025       | 0.02        |             |            | 3.25 - 3.75 |                              |
| BS3604-620              | 440-590                               | 290                     | 22                   | 0.1 - 0.18  | 0.10 - 0.35       | 0.4 - 0.7   | 0.05        | 0.05        | 0.45 - 0.65 | 0.7 - 1.1  |             |                              |
| BS3604-622              | 440-640                               | 275                     | 20                   | 0.08 - 0.15 | max. 0.5          | 0.4 - 0.7   | 0.04        | 0.04        | 0.9 - 1.2   | 2.0 - 2.5  |             |                              |
| BS3604-625              | 450-600                               | 170                     | 20                   | max. 0.15   | max. 0.5          | 0.3 - 0.6   | 0.04        | 0.04        | 0.45 - 0.65 | 4.0 - 6.0  |             |                              |
| BS4360-50D              | 490-640                               | 355                     | 20                   | max. 0.2    | max. 0.4          | max. 1.4    | 0.04        | 0.04        |             |            |             | Nb 0.003-0.1<br>V 0.0003-0.1 |

# API PIPES ERW & SEAMLESS PIPES

| TECHNICAL STANDARDS |                        |                      | COMPARABLE QUALITIES |             |                     |                   |                        |                          |
|---------------------|------------------------|----------------------|----------------------|-------------|---------------------|-------------------|------------------------|--------------------------|
| Material            | Specification standard | Dimensional standard | ASTM                 | API         | BS                  | DIN               | Euro Standard EU 25-72 | DIV                      |
| A53                 | A53                    | ANSI-B 36.10         |                      | API-5L-Gr.A | 3601-320/360        | St. 37.0          | FE 310                 | BS4360-43B/C             |
| A53                 | A53                    | ANSI-B 36.10         |                      | API-5L-Gr.B | 3601-410            | St. 44.0          | Fe 360B/430B           |                          |
| A106/A530           | A106/A530              | ANSI-B 36.10         |                      | API-5L-Gr.A | 3602-S360           | St. 35.8          |                        |                          |
| A106/A530           | A106/A530              | ANSI-B 36.10         |                      | API-5L-Gr.B | 3602-S410           | St. 45.8          |                        |                          |
| A179/A450           | A179/A450              | ANSI-B 36.10         |                      |             | 3059-320/360        | St. 35.8          |                        |                          |
| A333/A530           | A333/A530              | ANSI-B 36.10         |                      |             | 3601HFS27LT50       | TT St. 35N        |                        |                          |
| A333/A530           | A333/A530              | ANSI-B 36.10         |                      |             | 3603HFS503LT100     | TT St. 35V        |                        |                          |
| A333/A530           | A333/A530              |                      |                      |             |                     | 10 Ni 14          |                        |                          |
| A335/A530           | A335/A530              | ANSI-B 36.10         |                      |             |                     | 15 Mo 3/16 Mo 5   |                        |                          |
| A335/A530           | A335/A530              | ANSI-B 36.10         |                      |             | 3604-625            | 12 Cr Mo 19.5     |                        |                          |
| A335/A530           | A335/A530              | ANSI-B 36.10         |                      |             | 3604-620-440-460    | 13 Cr Mo 44       |                        |                          |
| A335/A530           | A335/A530              | ANSI-B 36.10         |                      |             | 3604-620-440-460    | 13 Cr Mo 44       |                        |                          |
| A335/A530           | A335/A530              | ANSI-B 36.10         |                      |             | 3604-620-440        | 10 Cr Mo 910      |                        |                          |
| API-5L              | API-5L                 | API-5L               | A53 Gr.A             |             | 3601-320/360        | St. 37.0          | Fe 310                 | St.E 210.7               |
| API-5L              | API-5L                 | API-5L               | A53 Gr.B             |             | 3601-410-4360-43B/C | St. 44.0          | Fe 360B/430B           | St.E 240.7               |
| API-5L              | API-5L                 | API-5L               |                      |             | 4360-43 B/C         | St. E 290.7       | Fe 430B                | St.E 355                 |
| API-5L              | API-5L                 | API-5L               |                      |             |                     | St. E 320.7       |                        |                          |
| API-5L              | API-5L                 | API-5L               |                      |             | 4360-50 B/C/D       | St. E 360.7       | Fe 510B/C/D            |                          |
| API-5L              | API-5L                 | API-5L               |                      |             |                     | St. E 385.7       |                        |                          |
| API-5L              | API-5L                 | API-5L               |                      |             |                     | St. E 415.7       |                        |                          |
| BS3059 part1        | BS3059 part1           | BS3059 part1         | A179                 |             |                     | St. 35.8/St. 37.8 |                        | A672-B60<br>A672-B65/C65 |
| BS3059 part2        | BS3059 part2           | BS3059 part2         | A179                 |             |                     | St. 35.8/St. 37.8 |                        |                          |
| BS3601              | BS3601                 | BS1600/BS3600        | A53 Gr.A             | 5L-Gr.A     |                     | St. 33            | Fe 310                 |                          |
| BS3601              | BS3601                 | BS1600/BS3600        | A53 Gr.B             | 5L-Gr.B     |                     | St. 37.0/St. 37.2 | Fe 360B                |                          |
| BS3601              | BS3601                 | BS1600/BS3600        | A53 Gr.B             | 5L-Gr.B     | 4360-43 B/C         | St. 44.0/St. 44.3 | Fe 430B/C              |                          |
| BS3601              | BS3601                 | BS1600/BS3600        | A106 Gr.B            | 5L-Gr.B     |                     | St. 37.0/St. 37.2 | Fe 360B                |                          |
| BS3601              | BS3601                 | BS1600/BS3600        | A106 Gr.B            | 5L-Gr.B     |                     | St. 44.0/St. 44.3 |                        |                          |
| BS3602 part1        | BS3602 part1           | BS1600/BS3600        | A106 Gr.B            | 5L-Gr.B     |                     | St. 35.8/St. 37.8 |                        |                          |
| BS3602 part1        | BS3602 part1           | BS1600/BS3600        | A106 Gr.B            | 5L-Gr.B     |                     | St. 45.8          |                        |                          |
| BS3602 part1        | BS3602 part1           | BS1600/BS3600        |                      | 5L-Gr.X52   | 4360-50 B/C/D       | St. 52/St 52.3    | Fe 510B/C/D            |                          |
| BS3602 part2        | BS3602 part2           | BS1600/BS3600        | A672 B65/C65         | 5L-Gr.B     | 4360-43 B/C         | St. E240.7        | Fe 430B/C              |                          |
| BS3603              | BS3603                 | BS1600/BS3600        | A333 Gr.6            |             |                     | TT St. 35V        |                        |                          |
| BS3603              | BS3603                 | BS1600/BS3600        | A333 Gr.3            |             |                     | 10 Ni 14          |                        |                          |
| BS3604              | BS3604                 | BS1600/BS3600        | A335 P12             |             |                     | 13 Cr Mo 44       |                        |                          |
| BS3604              | BS3604                 | BS1600/BS3600        | A335 P22             |             |                     | 10 Cr Mo 910      |                        |                          |
| BS3604              | BS3604                 | BS1600/BS3600        | A335 P5              |             |                     | 12 Cr Mo 195      |                        |                          |
| BS4360              | BS4360                 | BS1600               | A671 CC70            | 5L-X52      |                     | St. 52-3          | Fe 510D                | St. E 555                |

# API PIPES ERW & SEAMLESS PIPES

## TOLERANCES (EXCERPTS FROM API SPECIFICATIONS)

| Pipe Specification                   |                             | API 5L                                              |                                              |  |                    | API 5LU                                         |                                                   |                                               |  |
|--------------------------------------|-----------------------------|-----------------------------------------------------|----------------------------------------------|--|--------------------|-------------------------------------------------|---------------------------------------------------|-----------------------------------------------|--|
| Outside Diameter                     | Pipe Body                   | 1.90 in. & smaller                                  | +0.016 in. -0.031 in.                        |  |                    |                                                 | Less than 20 in.                                  | +0.75% -0.75%                                 |  |
|                                      |                             | 2-3/8 in. through 18 in.                            | +0.75% -0.75%                                |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | 20 in. through 36 in.                               |                                              |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | Non-expanded                                        | + 1.00% -1.00%                               |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | Cold-expanded                                       | +0.75% -0.25%                                |  |                    |                                                 | 20 in. & larger                                   | +1.00% -1.00%                                 |  |
|                                      |                             | Larger than 36 in.                                  |                                              |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | Non-expanded                                        | +1.00% -1.00%                                |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | Cold -expanded*2                                    | +1/4 in. -1/8 in.                            |  |                    |                                                 |                                                   |                                               |  |
|                                      | Pipe Ends *1                | 10-3/4 in. & smaller                                | +1/16 in. -1/64in. (measured with ring gage) |  |                    |                                                 | 10-3/4 in & smaller                               | +1/16 in. -1/64 in. (measured with ring gage) |  |
| 12-3/4 in. through 20 in.            |                             | +3/32 in. -1/32 in. (measured with ring gage)       |                                              |  |                    | 12-3/4 in. through 20in.                        | +3/32 in. -1/32 in. (measured with ring gage)     |                                               |  |
| Larger than 20 in. (22 in. & larger) |                             | +3/32 in. -1/32 in. (measured with diameter tape)   |                                              |  |                    | Larger than 20 in.                              | +3/32 in. -1/32 in. (measured with diameter tape) |                                               |  |
| Wall Thickness                       |                             | Welded Pipe:                                        | Grade A, B X42 through X70                   |  |                    |                                                 | Welded Pipe:                                      | +15.0% -12.5%                                 |  |
|                                      |                             | 2.875 in. OD & smaller                              | +20.0% -12.5% +15.0% -12.5%                  |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | 3-1/2 in. OD                                        | +18.0% -12.5% +15.0% -12.5%                  |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | 4 in. through 18 in. OD                             | +15.0% -12.5% +15.0% -12.5%                  |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | 20 in. OD & larger                                  | +17.5% -10.0% +19.5% -8.0%                   |  |                    |                                                 | 20 in. OD & larger                                | +19.5% -8.0%                                  |  |
|                                      |                             | Seamless Pipe:                                      | Grade A,B X42 through X70                    |  |                    |                                                 | Seamless Pipe:                                    | +15.0% -12.5%                                 |  |
|                                      |                             | 2.875 in. OD & smaller                              | +20.0% -12.5% +15.0% -12.5%                  |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | 3-1/2 in. OD                                        | +18.0% -12.5% +15.0% -12.5%                  |  |                    |                                                 |                                                   |                                               |  |
| 4 in. through 18 in. OD              | +15.0% -12.5% +15.0% -12.5% |                                                     |                                              |  |                    |                                                 |                                                   |                                               |  |
| 20 in. OD & larger                   | +15.0% -12.5% +17.5% -10.0% |                                                     |                                              |  | 20 in. OD & larger | +17.5% -10.0%                                   |                                                   |                                               |  |
| Weight                               | Single Length               | Standard weight                                     |                                              |  |                    |                                                 | Regular weight                                    | +10.0% -3.5%                                  |  |
|                                      |                             | Regular weight                                      | +10.0% -3.5%                                 |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | Extra strong weight                                 |                                              |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | Double extra strong weight                          |                                              |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | Special plain end weight                            | +10.0% -5.0%                                 |  |                    |                                                 | Special light weight                              | +10.0% -5.0%                                  |  |
|                                      | Carload Lots                | -1.75%                                              |                                              |  |                    | -1.75%                                          |                                                   |                                               |  |
| Straightness                         |                             | For pipe 4-1/2 in OD & larger max. 0.2%             |                                              |  |                    | max. 0.2%                                       |                                                   |                                               |  |
| Out of Roundness                     |                             | Max. OD - specified OD ≤ 1% x specified OD          |                                              |  |                    | Max. OD - specified OD ≤ 1% x specified_OD      |                                                   |                                               |  |
| -pipe larger than 20in. OD           |                             | Specified OD - min. OD ≤ 1% x specified OD          |                                              |  |                    | Specified OD - min. OD ≤ 1% x specified_OD      |                                                   |                                               |  |
| Pipe End Preparation                 |                             | For pipe 2-3/8 in. OD & larger                      |                                              |  |                    |                                                 |                                                   |                                               |  |
|                                      |                             | Bevel angle 30°, +5° - 0°                           |                                              |  |                    | Bevel angle 30°, +5° - 0°                       |                                                   |                                               |  |
|                                      |                             | Root face 1/16 in. ± 1/32 in.                       |                                              |  |                    | Root face 1/16 in. ± 1/32 in.                   |                                                   |                                               |  |
|                                      |                             | Squareness for 10-3/4 in. OD & larger max. 1/16 in. |                                              |  |                    | Squareness 10-3/4 in. OD & larger max. 1/16 in. |                                                   |                                               |  |

Note:

\*1 Pipe and tolerances are applicable 4 in. from either end.

\*2 Not applicable to pipe hydrostatically tested to pressures in excess of standard test pressures.

# API PIPES ERW & SEAMLESS PIPES

| Size             |       |      |                |       |       | Weight |       |       | Hydrostatic Test Pressure |    |      |    |         |      |      |      |      |      |      |
|------------------|-------|------|----------------|-------|-------|--------|-------|-------|---------------------------|----|------|----|---------|------|------|------|------|------|------|
| Outside Diameter |       |      | Wall Thickness |       |       |        |       |       | API 5L                    |    |      |    | API 5LX |      |      |      |      |      |      |
| Nominal Size     | in    | mm   | Sch No         | in    | mm    | lb/ft  | kg/m  | kg/ft | A                         |    | B    |    | X42     | X46  | X52  | X56  | X60  | X65  | X70  |
| 1/8              | 0.405 | 10.3 | 40 (std)       | 0.068 | 1.73  | 0.24   | 0.36  | --    | 700                       | -  | 700  | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 80 (XS)        | 0.095 | 2.41  | 0.31   | 0.46  | --    | 850                       | -- | 850  | -- | --      | --   | --   | --   | --   | --   | --   |
| 1/4              | 0.54  | 13.7 | 40 (Std)       | 0.088 | 2.24  | 0.42   | 0.63  | --    | 700                       | -- | 700  | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 80 (XS)        | 0.119 | 3.02  | 0.54   | 0.80  | --    | 850                       | -- | 850  | -- | --      | --   | --   | --   | --   | --   | --   |
| 3/8              | 0.675 | 17.1 | 40 (std)       | 0.091 | 2.31  | 0.57   | 0.85  | --    | 700                       | -- | 700  | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 80 (XS)        | 0.126 | 3.20  | 0.74   | 1.10  | --    | 850                       | -- | 850  | -- | --      | --   | --   | --   | --   | --   | --   |
| 1/2              | 0.84  | 21.3 | 40 (std)       | 0.109 | 2.77  | 0.85   | 1.27  | --    | 700                       | -- | 700  | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 80 (XS)        | 0.147 | 3.73  | 1.09   | 1.62  | --    | 850                       | -- | 850  | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | (XXS)          | 0.294 | 7.47  | 1.71   | 2.55  | --    | 1000                      | -- | 1000 | -- | --      | --   | --   | --   | --   | --   | --   |
| 3/4              | 1.05  | 26.7 | 40 (std)       | 0.113 | 2.87  | 1.13   | 1.68  | --    | 700                       | -- | 700  | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 80 (XS)        | 0.154 | 3.91  | 1.47   | 2.19  | --    | 850                       | -- | 850  | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | (XXS)          | 0.308 | 7.82  | 2.44   | 3.63  | --    | 1000                      | -- | 1000 | -- | --      | --   | --   | --   | --   | --   | --   |
| 1                | 1.315 | 33.4 | 40 (std)       | 0.133 | 3.38  | 1.68   | 2.50  | 0.76  | 700                       | -- | 700  | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 80 (XS)        | 0.179 | 4.55  | 2.17   | 3.23  | 0.99  | 850                       | -- | 850  | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 160            | 0.250 | 6.35  | 2.84   | 4.23  | 1.29  | --                        | -- | --   | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | (XXS)          | 0.358 | 9.09  | 3.66   | 5.45  | 1.66  | 1000                      | -- | 1000 | -- | --      | --   | --   | --   | --   | --   | --   |
| 1-1/4            | 1.66  | 42.2 | 40 (std)       | 0.140 | 3.56  | 2.27   | 3.38  | 1.03  | 1200                      | -- | 1300 | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 80 (XS)        | 0.191 | 4.85  | 3.00   | 4.47  | 1.36  | 1800                      | -- | 1900 | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 160            | 0.250 | 6.35  | 3.76   | 5.60  | 1.71  | --                        | -- | --   | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | (XXS)          | 0.382 | 9.70  | 5.21   | 7.76  | 2.37  | 2200                      | -- | 2300 | -- | --      | --   | --   | --   | --   | --   | --   |
| 1-1/2            | 1.90  | 48.3 | 40 (std)       | 0.145 | 3.68  | 2.72   | 4.05  | 1.23  | 1200                      | -- | 1300 | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 80 (XS)        | 0.200 | 5.08  | 3.63   | 5.41  | 1.65  | 1800                      | -- | 1900 | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 160            | 0.281 | 7.14  | 4.86   | 7.24  | 2.21  | --                        | -- | --   | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | (XXS)          | 0.400 | 10.16 | 6.41   | 9.55  | 2.91  | 2200                      | -- | 2300 | -- | --      | --   | --   | --   | --   | --   | --   |
| 2                | 2-3/8 | 60.3 | 40 (std)       | 0.083 | 2.11  | 2.03   | 3.02  | 0.92  | 1260                      | -- | 1470 | -- | 1760    | 1930 | 2180 | 2350 | 2520 | 2730 | 2940 |
|                  |       |      |                | 0.109 | 2.77  | 2.64   | 3.93  | 1.20  | --                        | -- | --   | -- | 2310    | 2530 | 2860 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.125 | 3.18  | 3.00   | 4.47  | 1.36  | --                        | -- | --   | -- | 2650    | 2910 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.141 | 3.58  | 3.36   | 5.00  | 1.53  | --                        | -- | --   | -- | 2990    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.154 | 3.91  | 3.65   | 5.44  | 1.66  | 2330                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.172 | 4.37  | 4.05   | 6.03  | 1.84  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.188 | 4.78  | 4.39   | 6.54  | 1.99  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      | 80 (XS)        | 0.218 | 5.54  | 5.02   | 7.48  | 2.28  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.250 | 6.35  | 5.67   | 8.45  | 2.57  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.281 | 7.14  | 6.28   | 9.35  | 2.85  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.344 | 8.74  | 7.46   | 11.11 | 3.39  | --                        | -- | --   | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      | 160            | 0.436 | 11.07 | 9.03   | 13.45 | 4.10  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      | 40 (std)       | 0.083 | 2.11  | 2.47   | 3.68  | 1.12  | 1040                      | -- | 1040 | -- | 1460    | 1590 | 1800 | 1940 | 2080 | 2250 | 2430 |
|                  |       |      |                | 0.109 | 2.77  | 3.22   | 4.80  | 1.46  | --                        | -- | --   | -- | 1910    | 2090 | 2370 | 2550 | 2730 | 2960 | 3000 |
|                  |       |      |                | 0.125 | 3.18  | 3.67   | 5.47  | 1.67  | --                        | -- | --   | -- | 2190    | 2400 | 2710 | 2920 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.141 | 3.58  | 4.12   | 6.14  | 1.87  | --                        | -- | --   | -- | 2470    | 2710 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.156 | 3.96  | 4.53   | 6.75  | 2.06  | 1950                      | -- | 2280 | -- | 2730    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.172 | 4.37  | 4.97   | 7.40  | 2.26  | 2150                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.188 | 4.78  | 5.40   | 8.04  | 2.45  | 2350                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.203 | 5.16  | 5.79   | 8.62  | 2.63  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.216 | 5.49  | 6.13   | 9.13  | 2.78  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.250 | 6.35  | 7.01   | 10.44 | 3.18  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
| 2-1/2            | 2-7/8 | 73.0 | 80 (XS)        | 0.276 | 7.01  | 7.66   | 11.41 | 3.48  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | 0.375 | 9.52  | 10.01  | 14.91 | 4.54  | --                        | -- | --   | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      |                | (XXS) | 0.552 | 14.02  | 13.69 | 6.22  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      | 160            | 0.375 | 9.52  | 10.01  | 14.91 | 4.54  | --                        | -- | --   | -- | --      | --   | --   | --   | --   | --   | --   |
|                  |       |      |                | (XXS) | 0.552 | 14.02  | 13.69 | 6.22  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |
|                  |       |      |                | (XXS) | 0.552 | 14.02  | 13.69 | 6.22  | 2500                      | -- | 2500 | -- | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |



# API PIPES ERW & SEAMLESS PIPES

| Size                              |        |       |                      |                             |       | Weight   |       |       | Hydrostatic Test Pressure |      |      |      |         |      |      |      |      |      |      |      |  |  |
|-----------------------------------|--------|-------|----------------------|-----------------------------|-------|----------|-------|-------|---------------------------|------|------|------|---------|------|------|------|------|------|------|------|--|--|
| Outside Diameter                  |        |       | Wall Thickness       |                             |       |          |       |       | API 5L                    |      |      |      | API 5LX |      |      |      |      |      |      |      |  |  |
|                                   |        |       |                      |                             |       |          |       |       | A                         |      | B    |      |         |      |      |      |      |      |      |      |  |  |
| Nominal Size                      | in     | mm    | Sch No               | in                          | mm    | lb/ft    | kg/m  | kg/ft | Std.                      | Alt. | Std. | Alt. | X42     | X46  | X52  | X56  | X60  | X65  | X70  |      |  |  |
| 3                                 | 3-1/2  | 88.9  | 40 (std)             | 0.083                       | 2.11  | 3.03     | 4.51  | 1.38  | 850                       | --   | 1000 | --   | 1200    | 1310 | 1480 | 1590 | 1710 | 1850 | 1990 |      |  |  |
|                                   |        |       |                      | 0.109                       | 2.77  | 3.95     | 5.88  | 1.79  | --                        | --   | --   | --   | 1570    | 1720 | 1940 | 2090 | 2240 | 2430 | 2620 |      |  |  |
|                                   |        |       |                      | 0.125                       | 3.18  | 4.51     | 6.72  | 2.05  | 1290                      | --   | 1500 | --   | 1800    | 1970 | 2230 | 2400 | 2570 | 2790 | 3000 |      |  |  |
|                                   |        |       |                      | 0.141                       | 3.58  | 5.06     | 7.54  | 2.30  | --                        | --   | --   | --   | 2030    | 2220 | 2510 | 2710 | 2900 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.156                       | 3.96  | 5.57     | 8.30  | 2.53  | 1600                      | --   | 1870 | --   | 2250    | 2460 | 2780 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.172                       | 4.37  | 6.11     | 9.10  | 2.77  | 1770                      | --   | 2060 | --   | 2480    | 2710 | 3000 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.188                       | 4.78  | 6.65     | 9.91  | 3.02  | 1930                      | --   | 2260 | --   | 2710    | 2970 | 3000 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.216                       | 5.49  | 7.58     | 11.29 | 3.44  | 2220                      | --   | 2500 | --   | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.250                       | 6.35  | 8.68     | 12.93 | 3.94  | 2500                      | --   | 2500 | --   | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.281                       | 7.14  | 9.66     | 14.39 | 4.39  | 2500                      | --   | 2500 | --   | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       | 80 (XS)<br>160 (XXS) | 0.300                       | 7.62  | 10.25    | 15.27 | 4.65  | 2500                      | --   | 2500 | --   | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |  |  |
|                                   |        |       |                      | 0.438                       | 11.13 | 14.32    | 21.33 | 6.50  | --                        | --   | --   | --   | --      | --   | --   | --   | --   | --   | --   | --   |  |  |
|                                   |        |       |                      | 0.600                       | 15.24 | 18.58    | 27.67 | 8.44  | 2500                      | --   | 2500 | --   | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |  |  |
|                                   |        |       |                      |                             |       |          |       |       |                           |      |      |      |         |      |      |      |      |      |      |      |  |  |
| 3-1/2                             | 4      | 101.6 | 40 (std)             | 0.083                       | 2.11  | 3.47     | 5.17  | 1.58  | 750                       | --   | 870  | --   | 1050    | 1150 | 1290 | 1390 | 1490 | 1620 | 1740 |      |  |  |
|                                   |        |       |                      | 0.109                       | 2.77  | 4.53     | 6.75  | 2.06  | 980                       | --   | 1140 | --   | 1370    | 1500 | 1700 | 1830 | 1960 | 2130 | 2290 |      |  |  |
|                                   |        |       |                      | 0.125                       | 3.18  | 5.17     | 7.70  | 2.35  | 1120                      | --   | 1310 | --   | 1580    | 1720 | 1950 | 2100 | 2250 | 2440 | 2630 |      |  |  |
|                                   |        |       |                      | 0.141                       | 3.58  | 5.81     | 8.65  | 2.64  | 1270                      | --   | 1480 | --   | 1780    | 1950 | 2200 | 2370 | 2540 | 2750 | 2960 |      |  |  |
|                                   |        |       |                      | 0.156                       | 3.96  | 6.40     | 9.53  | 2.91  | 1400                      | --   | 1640 | --   | 1970    | 2150 | 2430 | 2620 | 2810 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.172                       | 4.37  | 7.03     | 10.47 | 3.19  | 1550                      | --   | 1810 | --   | 2170    | 2370 | 2680 | 2890 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.188                       | 4.78  | 7.65     | 11.39 | 3.47  | 1690                      | --   | 1970 | --   | 2370    | 2590 | 2930 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.226                       | 5.74  | 9.11     | 13.57 | 4.14  | 2030                      | --   | 2370 | --   | 2850    | 3000 | 3000 | 3000 | 3000 | 3000 | 3300 |      |  |  |
|                                   |        |       |                      | 0.250                       | 6.35  | 10.01    | 14.91 | 4.54  | 2250                      | --   | 2620 | --   | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      | 0.281                       | 7.14  | 11.16    | 16.62 | 5.07  | 2530                      | --   | 2800 | --   | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       | 80 (XS)              | 0.318                       | 8.08  | 12.50    | 18.62 | 5.68  | 2800                      | --   | 2800 | --   | 3000    | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |      |  |  |
|                                   |        |       |                      |                             |       |          |       |       |                           |      |      |      |         |      |      |      |      |      |      |      |  |  |
|                                   |        |       | 4                    | 4-1/2                       | 114.3 | 40 (std) | 0.083 | 2.11  | 3.92                      | 5.84 | 1.78 | 660  | --      | 770  | --   | 930  | 1020 | 1150 | 1240 |      |  |  |
|                                   |        |       |                      |                             |       |          | 0.109 | 2.77  | 5.11                      | 7.61 | 2.32 | 870  | --      | 1020 | --   | --   | --   | --   | --   |      |  |  |
| 0.125                             | 3.18   | 5.84  |                      |                             |       |          | 8.70  | 2.65  | 1000                      | --   | 1170 | --   | 1400    | 1530 | 1730 | 1870 |      |      |      |      |  |  |
| 0.141                             | 3.58   | 6.56  |                      |                             |       |          | 9.77  | 2.98  | 1130                      | --   | 1320 | --   | 1580    | 1730 | 1960 | 2110 |      |      |      |      |  |  |
| 0.156                             | 3.96   | 7.24  |                      |                             |       |          | 10.78 | 3.29  | 1250                      | --   | 1460 | --   | 1750    | 1910 | 2160 | 2330 |      |      |      |      |  |  |
| 0.172                             | 4.37   | 7.95  |                      |                             |       |          | 11.84 | 3.61  | 1380                      | --   | 1610 | --   | 1930    | 2110 | 2390 | 2570 |      |      |      |      |  |  |
| 0.188                             | 4.78   | 8.66  |                      |                             |       |          | 12.90 | 3.93  | 1500                      | --   | 1750 | --   | 2110    | 2310 | 2610 | 2810 |      |      |      |      |  |  |
| 0.203                             | 5.16   | 9.32  |                      |                             |       |          | 13.88 | 4.23  | 1620                      | --   | 1890 | --   | 2270    | 2490 | 2810 | 3000 |      |      |      |      |  |  |
| 0.219                             | 5.56   | 10.01 |                      |                             |       |          | 14.91 | 4.54  | 1750                      | --   | 2040 | --   | 2450    | 2690 | 3000 | 3000 |      |      |      |      |  |  |
| 0.237                             | 6.02   | 10.79 |                      |                             |       |          | 16.07 | 4.90  | 1900                      | --   | 2210 | --   | 2650    | 2910 | 3000 | 3000 |      |      |      |      |  |  |
| 60<br>80 (XS)<br>120<br>160 (XXS) | 0.25   | 6.35  |                      |                             |       | 11.35    | 16.91 | 5.15  | 2000                      | --   | 2330 | --   | 2800    | 3000 | 3000 | 3000 |      |      |      |      |  |  |
|                                   | 0.281  | 7.14  |                      |                             |       | 12.66    | 18.86 | 5.75  | 2250                      | --   | 2620 | --   | 3000    | 3000 | 3000 | 3000 |      |      |      |      |  |  |
|                                   | 0.312  | 7.92  |                      |                             |       | 13.96    | 20.79 | 6.34  | 2500                      | --   | 2800 | --   | 3000    | 3000 | 3000 | 3000 |      |      |      |      |  |  |
|                                   | 0.337  | 8.56  |                      |                             |       | 14.98    | 22.31 | 6.80  | 2700                      | --   | 2800 | --   | 3000    | 3000 | 3000 | 3000 |      |      |      |      |  |  |
|                                   |        |       |                      |                             |       |          |       |       |                           |      |      |      |         |      |      |      |      |      |      |      |  |  |
| 5                                 | 5-9/16 | 141.3 | 40 (std)             | 0.083                       | 2.11  | 4.86     | 7.24  | 2.21  | 540                       | --   | 630  | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       |                      | 0.125                       | 3.18  | 7.26     | 10.81 | 3.30  | 810                       | --   | 940  | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       |                      | 0.156                       | 3.96  | 9.01     | 13.42 | 4.09  | 1010                      | --   | 1180 | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       |                      | 0.188                       | 4.78  | 10.79    | 16.07 | 4.90  | 1220                      | --   | 1420 | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       |                      | 0.219                       | 5.56  | 12.50    | 18.62 | 5.68  | 1420                      | --   | 1650 | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       |                      | 0.258                       | 6.55  | 14.62    | 21.76 | 6.63  | 1670                      | --   | 1950 | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       |                      | 0.281                       | 7.14  | 15.85    | 23.61 | 7.20  | 1820                      | --   | 2120 | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       |                      | 0.312                       | 7.92  | 17.50    | 26.05 | 7.94  | 2020                      | --   | 2360 | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       |                      | 0.344                       | 8.74  | 19.17    | 28.55 | 8.70  | 2230                      | --   | 2600 | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       |                      | 80 (XS)<br>120<br>160 (XXS) | 0.375 | 9.52     | 20.78 | 30.94 | 9.43                      | 2430 | --   | 2800 | --      | --   | --   | --   | --   |      |      |      |  |  |
|                                   |        |       | 0.50                 |                             | 12.70 | 27.03    | 40.26 | 12.27 | 2800                      | --   | 2800 | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       | 0.625                |                             | 15.88 | 32.96    | 49.08 | 14.96 | 2800                      | --   | 2800 | --   | --      | --   | --   | --   |      |      |      |      |  |  |
|                                   |        |       | 0.75                 |                             | 19.05 | 38.55    | 57.41 | 17.50 | 2800                      | --   | 2800 | --   | --      | --   | --   | --   |      |      |      |      |  |  |

# API PIPES ERW & SEAMLESS PIPES

| Size             |        |       |                |          |        | Weight |        |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |      |
|------------------|--------|-------|----------------|----------|--------|--------|--------|--------|---------------------------|------|------|------|---------|------|------|------|------|
| Outside Diameter |        |       | Wall Thickness |          |        |        |        |        | API 5L                    |      |      |      | API 5LX |      |      |      |      |
|                  |        |       |                |          |        |        |        |        | A                         |      | B    |      |         |      |      |      |      |
| Nominal Size     | in     | mm    | Sch No         | in       | mm     | lb/ft  | kg/m   | kg/ft  | Std.                      | Alt. | Std. | Alt. | X42     | X46  | X52  | X56  |      |
| 6                | 6-5/8  | 168.3 | 30             | 0.083    | 2.11   | 5.80   | 8.64   | 2.63   | 450                       | 560  | 530  | 660  | 790     | 860  | 980  | 1050 |      |
|                  |        |       |                | 0.109    | 2.77   | 7.59   | 11.31  | 3.45   | 590                       | 740  | 690  | 860  | 1040    | 1140 | 1280 | 1380 |      |
|                  |        |       |                | 0.125    | 3.18   | 8.68   | 12.93  | 3.94   | 680                       | 850  | 790  | 990  | 1190    | 1300 | 1470 | 1580 |      |
|                  |        |       |                | 0.141    | 3.58   | 9.76   | 14.54  | 4.43   | 770                       | 960  | 890  | 1120 | 1340    | 1470 | 1660 | 1790 |      |
|                  |        |       |                | 0.156    | 3.96   | 10.78  | 16.06  | 4.89   | 850                       | 1060 | 990  | 1240 | 1480    | 1620 | 1840 | 1980 |      |
|                  |        |       |                | 0.172    | 4.37   | 11.85  | 17.65  | 5.38   | 930                       | 1170 | 1090 | 1360 | 1640    | 1790 | 2030 | 2180 |      |
|                  |        |       |                | 0.188    | 4.78   | 12.92  | 19.24  | 5.87   | 1020                      | 1280 | 1190 | 1490 | 1790    | 1960 | 2210 | 2380 |      |
|                  |        |       |                | 0.203    | 5.16   | 13.92  | 20.73  | 6.32   | 1100                      | 1380 | 1290 | 1610 | 1930    | 2110 | 2390 | 2570 |      |
|                  |        |       |                | 0.219    | 5.56   | 14.98  | 22.31  | 6.80   | 1190                      | 1490 | 1390 | 1740 | 2080    | 2280 | 2580 | 2780 |      |
|                  |        |       |                | 0.250    | 6.35   | 17.02  | 25.35  | 7.73   | 1360                      | 1700 | 1580 | 1980 | 2380    | 2600 | 2940 | 3000 |      |
|                  |        |       |                | 40 (std) | 0.280  | 7.11   | 18.97  | 28.26  | 8.61                      | 1520 | 1900 | 1780 | 2220    | 2660 | 2920 | 3000 | 3000 |
|                  |        |       |                | 0.312    | 7.92   | 21.04  | 31.34  | 9.55   | 1700                      | 2120 | 1980 | 2470 | 2970    | 3000 | 3000 | 3000 |      |
|                  |        |       | 80 (XS)        | 0.344    | 8.74   | 23.08  | 34.38  | 10.48  | 1870                      | 2340 | 2180 | 2730 | 3000    | 3000 | 3000 | 3000 |      |
|                  |        |       |                | 0.375    | 9.52   | 25.03  | 37.28  | 11.36  | 2040                      | 2550 | 2380 | 2800 | 3000    | 3000 | 3000 | 3000 |      |
|                  |        |       |                | 0.432    | 10.97  | 28.57  | 42.56  | 12.97  | 2350                      | 2800 | 2740 | 2800 | 3000    | 3000 | 3000 | 3000 |      |
|                  |        |       |                | 0.500    | 12.70  | 32.71  | 48.72  | 14.85  | 2720                      | 2800 | 2800 | 2800 | 3000    | 3000 | 3000 | 3000 |      |
|                  |        |       |                | 120      | 0.562  | 14.27  | 36.39  | 54.20  | 16.52                     | 2800 | 2800 | 2800 | 2800    | 3000 | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.625    | 15.88  | 40.05  | 59.65  | 18.18  | 2800                      | 2800 | 2800 | 2800 | 3000    | 3000 | 3000 | 3000 |      |
|                  |        |       |                | 160      | 0.719  | 18.26  | 45.35  | 67.55  | 20.59                     | 2800 | 2800 | 2800 | 2800    | 3000 | 3000 | 3000 | 3000 |
|                  |        |       |                | (XXS)    | 0.864  | 21.95  | 53.16  | 79.18  | 24.13                     | 2800 | 2800 | 2800 | 2800    | --   | --   | --   | --   |
| 8                | 8-5/8  | 219.1 | 20             | 0.125    | 3.18   | 11.35  | 16.91  | 5.15   | 520                       | 650  | 610  | 760  | 910     | 1000 | 1130 | 1220 |      |
|                  |        |       |                | 0.156    | 3.96   | 14.11  | 21.02  | 6.41   | 650                       | 810  | 760  | 950  | 1140    | 1250 | 1410 | 1520 |      |
|                  |        |       |                | 0.188    | 4.78   | 16.94  | 25.23  | 7.69   | 780                       | 980  | 920  | 1140 | 1370    | 1500 | 1700 | 1830 |      |
|                  |        |       |                | 0.203    | 5.16   | 18.26  | 27.20  | 8.29   | --                        | --   | --   | --   | 1480    | 1620 | 1840 | 2000 |      |
|                  |        |       |                | 0.219    | 5.56   | 19.66  | 29.28  | 8.93   | 910                       | 1140 | 1070 | 1330 | 1600    | 1750 | 1980 | 2130 |      |
|                  |        |       |                | 0.250    | 6.35   | 22.36  | 33.31  | 10.15  | 1040                      | 1300 | 1220 | 1520 | 1830    | 2000 | 2260 | 2430 |      |
|                  |        |       | 30             | 0.277    | 7.04   | 24.70  | 36.79  | 11.21  | 1160                      | 1450 | 1350 | 1690 | 2020    | 2220 | 2510 | 2700 |      |
|                  |        |       |                | 0.312    | 7.92   | 27.70  | 41.26  | 12.58  | 1300                      | 1630 | 1520 | 1900 | 2280    | 2500 | 2820 | 3000 |      |
|                  |        |       |                | 40 (std) | 0.322  | 8.18   | 28.55  | 42.53  | 12.96                     | 1340 | 1680 | 1570 | 1960    | 2350 | 2580 | 2910 | 3000 |
|                  |        |       | 0.344          | 8.74     | 30.42  | 45.31  | 13.81  | 1440   | 1790                      | 1680 | 2090 | 2510 | 2750    | 3000 | 3000 |      |      |
|                  |        |       | 60             | 0.375    | 9.52   | 33.04  | 49.21  | 15.00  | 1570                      | 1960 | 1830 | 2280 | 2740    | 3000 | 3000 | 3000 |      |
|                  |        |       |                | 0.406    | 10.31  | 35.64  | 53.09  | 16.18  | --                        | --   | --   | --   | --      | --   | --   | --   |      |
|                  |        |       |                | 0.438    | 11.13  | 38.30  | 57.05  | 17.39  | 1830                      | 2290 | 2130 | 2670 | 3000    | 3000 | 3000 | 3000 |      |
|                  |        |       |                | 80 (XS)  | 0.500  | 12.70  | 43.39  | 64.63  | 19.70                     | 2090 | 2610 | 2430 | 2800    | 3000 | 3000 | 3000 | 3000 |
|                  |        |       | 100            | 0.562    | 14.27  | 48.40  | 72.09  | 21.97  | 2350                      | 2800 | 2740 | 2800 | 3000    | 3000 | 3000 | 3000 |      |
|                  |        |       |                | 0.594    | 15.09  | 50.95  | 75.89  | 23.13  | --                        | --   | --   | --   | --      | --   | --   | --   |      |
|                  |        |       | 120            | 0.625    | 15.88  | 53.40  | 79.54  | 24.24  | 2610                      | 2800 | 2800 | 2800 | 3000    | 3000 | 3000 | 3000 |      |
|                  |        |       |                | 0.719    | 18.26  | 60.71  | 90.43  | 27.56  | 2800                      | 2800 | 2800 | 2800 | 3000    | 3000 | 3000 | 3000 |      |
|                  |        |       | 140            | 0.812    | 20.62  | 67.76  | 100.93 | 30.76  | --                        | --   | --   | --   | --      | --   | --   | --   |      |
|                  |        |       |                | (XXS)    | 0.875  | 22.22  | 72.42  | 107.87 | 32.88                     | 2800 | 2800 | 2800 | 2800    | --   | --   | --   | --   |
| 160              | 0.906  | 23.01 |                | 74.69    | 111.25 | 33.91  | --     | --     | --                        | --   | --   | --   | --      |      |      |      |      |
| 10               | 10-3/4 | 273.0 |                | 20       | 0.156  | 3.96   | 17.65  | 26.29  | 8.01                      | 520  | 650  | 610  | 760     | 1040 | 1130 | 1280 | 1380 |
|                  |        |       | 0.188          |          | 4.78   | 21.21  | 31.59  | 9.63   | 630                       | 790  | 730  | 920  | 1250    | 1370 | 1550 | 1660 |      |
|                  |        |       | 0.203          |          | 5.16   | 22.87  | 34.06  | 10.38  | --                        | --   | --   | --   | 1350    | 1480 | 1670 | 1800 |      |
|                  |        |       | 0.219          |          | 5.56   | 24.63  | 36.69  | 11.18  | 730                       | 920  | 860  | 1070 | 1450    | 1590 | 1800 | 1940 |      |
|                  |        |       | 30             | 0.250    | 6.35   | 28.04  | 41.77  | 12.73  | 840                       | 1050 | 980  | 1220 | 1660    | 1820 | 2060 | 2210 |      |
|                  |        |       |                | 0.279    | 7.09   | 31.20  | 46.47  | 14.16  | 930                       | 1170 | 1090 | 1360 | 1850    | 2030 | 2290 | 2470 |      |
|                  |        |       |                | 0.307    | 7.80   | 34.24  | 51.00  | 15.54  | 1030                      | 1290 | 1200 | 1500 | 2040    | 2230 | 2520 | 2720 |      |
|                  |        |       |                | 0.344    | 8.74   | 38.23  | 56.94  | 17.36  | 1150                      | 1440 | 1340 | 1680 | 2280    | 2500 | 2830 | 3000 |      |

# API PIPES ERW & SEAMLESS PIPES

| Size             |        |       |                |             |       | Weight |        |       | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|--------|-------|----------------|-------------|-------|--------|--------|-------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |        |       | Wall Thickness |             |       |        |        |       | API 5L                    |      |      |      | API 5LX |      |      |      |
| Nominal Size     | in     | mm    | Sch No         | in          | mm    | lb/ft  | kg/m   | kg/ft | A                         |      | B    |      | X42     | X46  | X52  | X56  |
|                  |        |       |                |             |       |        |        |       | Std.                      | Alt. | Std. | Alt. |         |      |      |      |
| 10               | 10-3/4 | 273.0 | 40 (std)       | 0.365       | 9.27  | 40.48  | 60.29  | 18.38 | 1220                      | 1530 | 1430 | 1780 | 2420    | 2660 | 3000 | 3000 |
|                  |        |       |                | 0.438       | 11.13 | 48.24  | 71.85  | 21.90 | 1470                      | 1830 | 1710 | 2140 | 2910    | 3000 | 3000 | 3000 |
|                  |        |       | 60 (XS)        | 0.500       | 12.70 | 54.74  | 81.54  | 24.85 | 1670                      | 2090 | 1950 | 2440 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.562       | 14.27 | 61.15  | 91.08  | 27.76 | 1880                      | 2350 | 2200 | 2740 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       | 80             | 0.594       | 15.09 | 64.43  | 95.97  | 29.25 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |        |       |                | 0.625       | 15.88 | 67.58  | 100.66 | 30.68 | 2090                      | 2620 | 2440 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       | 100            | 0.719       | 18.26 | 77.03  | 114.74 | 34.97 | 2410                      | 2800 | 2800 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.812       | 20.62 | 86.18  | 128.37 | 39.13 | 2720                      | 2800 | 2800 | 2800 | --      | --   | --   | --   |
|                  |        |       | 120            | 0.844       | 21.44 | 89.29  | 133.00 | 40.54 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |        |       |                | 1.000       | 25.40 | 104.13 | 155.10 | 47.28 | --                        | --   | --   | --   | --      | --   | --   | --   |
| 12               | 12-3/4 | 323.8 | 20             | 0.172       | 4.37  | 23.11  | 34.42  | 10.49 | 490                       | 610  | 570  | 710  | 960     | 1050 | 1190 | 1280 |
|                  |        |       |                | 0.188       | 4.78  | 25.22  | 37.57  | 11.45 | 530                       | 660  | 620  | 770  | 1050    | 1150 | 1300 | 1400 |
|                  |        |       |                | 0.203       | 5.16  | 27.20  | 40.51  | 12.35 | --                        | --   | --   | --   | 1140    | 1250 | 1410 | 1520 |
|                  |        |       |                | 0.219       | 5.56  | 29.31  | 43.66  | 13.31 | 620                       | 770  | 720  | 900  | 1230    | 1340 | 1520 | 1640 |
|                  |        |       |                | 0.250       | 6.35  | 33.38  | 49.72  | 15.15 | 710                       | 880  | 820  | 1030 | 1400    | 1530 | 1730 | 1870 |
|                  |        |       |                | 0.281       | 7.14  | 37.42  | 55.74  | 16.99 | 790                       | 990  | 930  | 1160 | 1570    | 1720 | 1950 | 2100 |
|                  |        |       | 30             | 0.312       | 7.92  | 41.45  | 61.74  | 18.82 | 880                       | 1100 | 1030 | 1280 | 1750    | 1910 | 2160 | 2330 |
|                  |        |       |                | 0.330       | 8.38  | 43.77  | 65.20  | 19.87 | 930                       | 1160 | 1090 | 1360 | 1850    | 2020 | 2290 | 2460 |
|                  |        |       |                | 0.344       | 8.74  | 45.58  | 67.89  | 20.69 | 970                       | 1210 | 1130 | 1420 | 1930    | 2110 | 2390 | 2570 |
|                  |        |       |                | (std) 0.375 | 9.52  | 49.56  | 73.82  | 22.50 | 1060                      | 1320 | 1240 | 1540 | 2100    | 2300 | 2600 | 2800 |
|                  |        |       | 40             | 0.406       | 10.31 | 53.52  | 79.72  | 24.30 | --                        | --   | --   | --   | 2270    | 2490 | 2810 | 3000 |
|                  |        |       |                | 0.438       | 11.13 | 57.59  | 85.78  | 26.15 | 1240                      | 1550 | 1440 | 1800 | 2450    | 2690 | 3000 | 3000 |
|                  |        |       | (XS) 60        | 0.500       | 12.70 | 65.42  | 97.44  | 29.70 | 1410                      | 1760 | 1650 | 2060 | 2800    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.562       | 14.27 | 73.15  | 108.96 | 33.21 | 1590                      | 1980 | 1850 | 2310 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       | 80             | 0.625       | 15.88 | 80.93  | 120.55 | 36.74 | 1760                      | 2210 | 2060 | 2570 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.688       | 17.48 | 88.63  | 132.01 | 40.24 | 1940                      | 2430 | 2270 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.750       | 19.05 | 96.12  | 143.17 | 43.64 | 2120                      | 2650 | 2470 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.812       | 20.62 | 103.53 | 154.21 | 47.00 | 2290                      | 2800 | 2670 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       | 100            | 0.844       | 21.44 | 107.32 | 159.85 | 48.72 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |        |       |                | 0.875       | 22.22 | 110.97 | 165.29 | 50.38 | 2470                      | 2800 | 2800 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       | 120            | 1.000       | 25.40 | 125.49 | 186.92 | 56.97 | --                        | --   | --   | --   | --      | --   | --   | --   |
| 14               | 14     | 355.6 | 10             | 0.188       | 4.78  | 27.73  | 41.30  | 12.59 | 480                       | 600  | 560  | 700  | 960     | 1050 | 1190 | 1280 |
|                  |        |       |                | 0.203       | 5.16  | 29.91  | 44.55  | 13.58 | 520                       | 650  | 610  | 760  | --      | --   | --   | --   |
|                  |        |       |                | 0.210       | 5.33  | 30.93  | 46.07  | 14.04 | --                        | --   | --   | --   | 1070    | 1170 | 1330 | 1430 |
|                  |        |       |                | 0.219       | 5.56  | 32.23  | 48.01  | 14.63 | --                        | --   | --   | --   | 1120    | 1220 | 1380 | 1490 |
|                  |        |       |                | 0.250       | 6.35  | 36.71  | 54.68  | 16.67 | 640                       | 800  | 750  | 940  | 1280    | 1400 | 1580 | 1700 |
|                  |        |       |                | 0.281       | 7.14  | 41.17  | 61.32  | 18.69 | 720                       | 900  | 840  | 1050 | 1430    | 1570 | 1770 | 1910 |
|                  |        |       | 20             | 0.312       | 7.92  | 45.61  | 67.94  | 20.71 | 800                       | 1000 | 940  | 1170 | 1590    | 1740 | 1970 | 2120 |
|                  |        |       |                | 0.344       | 8.74  | 50.17  | 74.73  | 22.78 | 880                       | 1110 | 1030 | 1290 | 1750    | 1920 | 2170 | 2340 |
|                  |        |       |                | (std) 0.375 | 9.52  | 54.57  | 81.28  | 24.77 | 960                       | 1210 | 1120 | 1410 | 1910    | 2090 | 2370 | 2550 |
|                  |        |       |                | 0.406       | 10.31 | 58.94  | 87.79  | 26.76 | --                        | --   | --   | --   | 2070    | 2270 | 2560 | 2760 |
|                  |        |       | 40             | 0.438       | 11.13 | 63.44  | 94.49  | 28.80 | 1130                      | 1410 | 1310 | 1640 | 2230    | 2450 | 2770 | 2980 |
|                  |        |       |                | 0.469       | 11.91 | 67.78  | 100.96 | 30.77 | --                        | --   | --   | --   | 2390    | 2620 | 2960 | 3000 |
|                  |        |       | (XS) 60        | 0.500       | 12.70 | 72.09  | 107.38 | 32.73 | 1290                      | 1610 | 1500 | 1880 | 2550    | 2790 | 3000 | 3000 |
|                  |        |       |                | 0.562       | 14.27 | 80.66  | 120.14 | 36.62 | 1450                      | 1810 | 1690 | 2110 | 2870    | 3000 | 3000 | 3000 |
|                  |        |       | 80             | 0.594       | 15.09 | 85.05  | 126.68 | 38.61 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |        |       |                | 0.625       | 15.88 | 89.28  | 132.98 | 40.53 | 1610                      | 2010 | 1880 | 2340 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.688       | 17.48 | 97.81  | 145.69 | 44.41 | 1770                      | 2210 | 2060 | 2580 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.750       | 19.05 | 106.13 | 158.08 | 48.18 | 1930                      | 2410 | 2250 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       | 100            | 0.812       | 20.62 | 114.37 | 170.35 | 51.92 | 2090                      | 2610 | 2440 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.875       | 22.22 | 122.65 | 182.69 | 55.68 | 2250                      | 2800 | 2620 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 0.938       | 23.83 | 130.85 | 194.90 | 59.41 | 2410                      | 2800 | 2800 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |        |       |                | 1.000       | 25.40 | 138.84 | 206.80 | 63.03 | --                        | --   | --   | --   | --      | --   | --   | --   |

# API PIPES ERW & SEAMLESS PIPES

| Size             |    |       |                |       |       | Weight |        |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|----|-------|----------------|-------|-------|--------|--------|--------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |    |       | Wall Thickness |       |       |        |        |        | API 5L                    |      |      |      | API 5LX |      |      |      |
|                  |    |       |                |       |       |        |        |        | A                         |      | B    |      |         |      |      |      |
| Nominal Size     | in | mm    | Sch No         | in    | mm    | lb/ft  | kg/m   | kg/ft  | Std.                      | Alt. | Std. | Alt. | X42     | X46  | X52  | X56  |
| 16               | 16 | 406.4 | 10             | 0.188 | 4.78  | 31.75  | 47.29  | 14.40  | 420                       | 530  | 490  | 620  | 840     | 920  | 1040 | 1120 |
|                  |    |       |                | 0.203 | 5.16  | 34.25  | 51.02  | 15.54  | 460                       | 570  | 530  | 670  | 910     | 990  | 1120 | 1210 |
|                  |    |       |                | 0.219 | 5.56  | 36.91  | 54.98  | 16.74  | 490                       | 620  | 570  | 720  | 980     | 1070 | 1210 | 1300 |
|                  |    |       |                | 0.250 | 6.35  | 42.05  | 62.63  | 19.07  | 560                       | 700  | 660  | 820  | 1120    | 1220 | 1380 | 1490 |
|                  |    |       |                | 0.281 | 7.14  | 47.17  | 70.26  | 21.40  | 630                       | 790  | 740  | 920  | 1250    | 1370 | 1550 | 1670 |
|                  |    |       | 20             | 0.312 | 7.92  | 52.27  | 77.86  | 23.71  | 700                       | 880  | 820  | 1020 | 1390    | 1520 | 1720 | 1860 |
|                  |    |       |                | 0.344 | 8.74  | 57.52  | 85.68  | 26.09  | 770                       | 970  | 900  | 1130 | 1540    | 1680 | 1900 | 2050 |
|                  |    |       |                | 0.375 | 9.52  | 62.58  | 93.21  | 28.39  | 840                       | 1050 | 980  | 1230 | 1670    | 1830 | 2070 | 2230 |
|                  |    |       | 30 (std)       | 0.406 | 10.31 | 67.62  | 100.72 | 30.67  | --                        | --   | --   | --   | 1810    | 1980 | 2240 | 2420 |
|                  |    |       |                | 0.438 | 11.13 | 72.80  | 108.44 | 33.02  | 990                       | 1230 | 1150 | 1440 | 1950    | 2140 | 2420 | 2610 |
|                  |    |       |                | 0.469 | 11.91 | 77.79  | 115.87 | 35.29  | --                        | --   | --   | --   | 2090    | 2290 | 2590 | 2790 |
|                  |    |       |                | 0.500 | 12.70 | 82.77  | 123.29 | 37.54  | 1120                      | 1410 | 1310 | 1640 | 2230    | 2440 | 2760 | 2980 |
|                  |    |       |                | 0.562 | 14.27 | 92.66  | 138.02 | 42.03  | 1260                      | 1580 | 1480 | 1840 | 2510    | 2750 | 3000 | 3000 |
|                  |    |       |                | 0.625 | 15.88 | 102.63 | 152.87 | 46.55  | 1410                      | 1760 | 1640 | 2050 | 2790    | 3000 | 3000 | 3000 |
|                  |    |       | 60             | 0.656 | 16.66 | 107.50 | 160.12 | 48.80  | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |       |                | 0.688 | 17.48 | 112.51 | 167.58 | 51.03  | 1550                      | 1940 | 1810 | 2260 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 0.750 | 19.05 | 122.15 | 181.94 | 55.41  | 1690                      | 2110 | 1970 | 2460 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       | 80             | 0.812 | 20.62 | 131.71 | 196.18 | 59.74  | 1830                      | 2280 | 2130 | 2660 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 0.844 | 21.44 | 136.61 | 203.48 | 62.02  | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |       |                | 0.875 | 22.22 | 141.34 | 210.53 | 64.11  | 1970                      | 2460 | 2300 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 0.938 | 23.83 | 150.89 | 224.75 | 68.44  | 2110                      | 2640 | 2460 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.000 | 25.40 | 160.20 | 238.62 | 72.67  | 2250                      | 2800 | 2620 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.062 | 26.97 | 169.43 | 252.37 | 76.85  | 2390                      | 2800 | 2790 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.125 | 28.58 | 178.72 | 266.20 | 81.07  | 2530                      | 2800 | 2800 | 2800 | 3000    | 3000 | 3000 | 3000 |
| 18               | 18 | 457.2 | 10             | 0.188 | 4.78  | 35.76  | 53.26  | 16.22  | 380                       | 470  | 440  | 550  | 750     | 820  | 920  | 990  |
|                  |    |       |                | 0.219 | 5.56  | 41.59  | 61.95  | 18.87  | 440                       | 550  | 510  | 640  | 870     | 950  | 1080 | 1160 |
|                  |    |       |                | 0.250 | 6.35  | 47.39  | 70.59  | 21.50  | 500                       | 620  | 580  | 730  | 990     | 1090 | 1230 | 1320 |
|                  |    |       |                | 0.281 | 7.14  | 53.18  | 79.21  | 24.12  | 560                       | 700  | 660  | 820  | 1110    | 1220 | 1380 | 1490 |
|                  |    |       | 20             | 0.312 | 7.92  | 58.94  | 87.79  | 26.74  | 620                       | 780  | 730  | 910  | 1240    | 1360 | 1530 | 1650 |
|                  |    |       |                | 0.344 | 8.74  | 64.87  | 96.62  | 29.43  | 690                       | 860  | 800  | 1000 | 1360    | 1490 | 1690 | 1820 |
|                  |    |       |                | 0.375 | 9.52  | 70.59  | 105.14 | 32.02  | 750                       | 940  | 880  | 1090 | 1490    | 1630 | 1840 | 1980 |
|                  |    |       | 30             | 0.406 | 10.31 | 76.29  | 113.63 | 34.61  | --                        | --   | --   | --   | 1610    | 1760 | 1990 | 2150 |
|                  |    |       |                | 0.438 | 11.13 | 82.15  | 122.36 | 37.26  | 880                       | 1100 | 1020 | 1280 | 1740    | 1900 | 2150 | 2320 |
|                  |    |       |                | 0.469 | 11.91 | 87.81  | 130.79 | 39.83  | --                        | --   | --   | --   | 1860    | 2040 | 2300 | 2480 |
|                  |    |       | 40             | 0.500 | 12.70 | 93.45  | 139.19 | 42.39  | 1000                      | 1250 | 1170 | 1460 | 1980    | 2170 | 2460 | 2640 |
|                  |    |       |                | 0.562 | 14.27 | 104.67 | 155.91 | 47.48  | 1120                      | 1400 | 1310 | 1640 | 2230    | 2440 | 2760 | 2970 |
|                  |    |       |                | 0.625 | 15.88 | 115.98 | 172.75 | 52.61  | 1250                      | 1560 | 1460 | 1820 | 2480    | 2720 | 3000 | 3000 |
|                  |    |       |                | 0.688 | 17.48 | 127.21 | 189.48 | 57.70  | 1380                      | 1720 | 1610 | 2010 | 2730    | 2990 | 3000 | 3000 |
|                  |    |       | 60             | 0.750 | 19.05 | 138.17 | 205.80 | 62.67  | 1500                      | 1880 | 1750 | 2190 | 2980    | 3000 | 3000 | 3000 |
|                  |    |       |                | 0.812 | 20.62 | 149.06 | 222.02 | 67.61  | 1620                      | 2030 | 1890 | 2370 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 0.875 | 22.22 | 160.03 | 238.36 | 72.59  | 1750                      | 2190 | 2040 | 2550 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       | 80             | 0.938 | 23.83 | 170.92 | 254.59 | 77.53  | 1880                      | 2340 | 2190 | 2740 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.000 | 25.40 | 181.56 | 270.43 | 82.36  | 2000                      | 2500 | 2330 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.062 | 26.97 | 192.11 | 286.15 | 87.14  | 2120                      | 2660 | 2480 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.125 | 28.58 | 202.75 | 302.00 | 91.97  | 2250                      | 2800 | 2620 | 2800 | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.187 | 30.15 | 213.14 | 317.47 | 96.68  | 2370                      | 2800 | 2770 | 2800 | --      | --   | --   | --   |
|                  |    |       |                | 1.188 | 30.18 | 213.31 | 317.73 | 96.76  | --                        | --   | --   | --   | 3000    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.250 | 31.75 | 223.61 | 333.07 | 101.43 | 2500                      | 2800 | 2800 | 2800 | 3000    | 3000 | 3000 | 3000 |

# API PIPES ERW & SEAMLESS PIPES

| Size             |    |       |                |       |       | Weight |        |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|----|-------|----------------|-------|-------|--------|--------|--------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |    |       | Wall Thickness |       |       |        |        |        | API 5L                    |      |      |      | API 5LX |      |      |      |
| Nominal Size     | in | mm    | Sch No         | in    | mm    | lb/ft  | kg/m   | kg/ft  | A                         |      | B    |      | X42     | X46  | X52  | X56  |
| 20               | 20 | 508.0 | 10             | 0.219 | 5.56  | 46.27  | 68.92  | 20.99  | 390                       | 490  | 460  | 570  | 830     | 910  | 1020 | 1100 |
|                  |    |       |                | 0.250 | 6.35  | 52.73  | 78.54  | 23.92  | 450                       | 560  | 520  | 660  | 940     | 1040 | 1170 | 1260 |
|                  |    |       |                | 0.281 | 7.14  | 59.18  | 88.15  | 26.84  | 510                       | 630  | 590  | 740  | 1060    | 1160 | 1320 | 1420 |
|                  |    |       |                | 0.312 | 7.92  | 65.60  | 97.71  | 29.76  | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1570 |
|                  |    |       | 20             | 0.344 | 8.74  | 72.21  | 107.56 | 32.75  | 620                       | 770  | 720  | 900  | 1300    | 1420 | 1610 | 1730 |
|                  |    |       |                | 0.375 | 9.52  | 78.60  | 117.07 | 35.65  | 680                       | 840  | 790  | 980  | 1420    | 1550 | 1760 | 1890 |
|                  |    |       |                | 0.406 | 10.31 | 84.96  | 126.55 | 38.54  | --                        | --   | --   | --   | 1530    | 1680 | 1900 | 2050 |
|                  |    |       |                | 0.438 | 11.13 | 91.51  | 136.30 | 41.51  | 790                       | 990  | 920  | 1150 | 1660    | 1810 | 2050 | 2210 |
|                  |    |       | 30             | 0.469 | 11.91 | 97.83  | 145.72 | 44.38  | --                        | --   | --   | --   | 1770    | 1940 | 2190 | 2360 |
|                  |    |       |                | 0.500 | 12.70 | 104.13 | 151.10 | 47.23  | 900                       | 1120 | 1050 | 1310 | 1890    | 2070 | 2340 | 2520 |
|                  |    |       |                | 0.562 | 14.27 | 116.67 | 173.78 | 52.92  | 1010                      | 1260 | 1180 | 1480 | 2120    | 2330 | 2630 | 2750 |
|                  |    |       |                | 0.625 | 15.88 | 129.33 | 192.64 | 58.66  | 1120                      | 1410 | 1310 | 1640 | 2360    | 2590 | 2750 | 2750 |
|                  |    |       | 60             | 0.688 | 17.48 | 141.90 | 211.36 | 64.37  | 1240                      | 1550 | 1440 | 1810 | 2600    | 2750 | 2750 | 2750 |
|                  |    |       |                | 0.750 | 19.05 | 154.19 | 229.67 | 69.94  | 1350                      | 1690 | 1580 | 1970 | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 0.812 | 20.62 | 166.40 | 247.85 | 75.48  | 1460                      | 1830 | 1710 | 2130 | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 0.875 | 22.22 | 178.72 | 266.20 | 81.07  | 1580                      | 1970 | 1840 | 2300 | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 0.938 | 23.83 | 190.96 | 284.43 | 86.62  | 1690                      | 2110 | 1970 | 2460 | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 1.000 | 25.40 | 202.92 | 302.25 | 92.04  | 1800                      | 2250 | 2100 | 2620 | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 1.062 | 26.97 | 214.80 | 319.94 | 97.43  | 1910                      | 2390 | 2230 | 2750 | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 1.125 | 28.58 | 226.78 | 337.79 | 102.87 | 2020                      | 2530 | 2360 | 2750 | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 1.187 | 30.15 | 238.50 | 355.25 | 108.18 | 2140                      | 2670 | 2490 | 2750 | --      | --   | --   | --   |
|                  |    |       |                | 1.188 | 30.15 | 238.68 | --     | --     | --                        | --   | --   | --   | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 1.250 | 31.75 | 250.31 | 372.84 | 113.54 | 2250                      | 2750 | 2620 | 2750 | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 1.312 | 33.32 | 261.86 | 390.04 | 118.78 | 2360                      | 2750 | 2750 | 2750 | 2750    | 2750 | 2750 | 2750 |
|                  |    |       |                | 1.375 | 34.92 | 273.51 | 407.39 | 124.06 | 2480                      | 2750 | 2750 | 2750 | 2750    | 2750 | 2750 | 2750 |
| 22               | 22 | 558.8 | 10             | 0.219 | 5.56  | 50.94  | 75.88  | 23.11  | 360                       | 450  | 420  | 520  | 750     | 820  | 930  | 1000 |
|                  |    |       |                | 0.250 | 6.35  | 58.07  | 86.50  | 26.34  | 410                       | 510  | 480  | 600  | 860     | 940  | 1060 | 1150 |
|                  |    |       |                | 0.281 | 7.14  | 65.18  | 97.09  | 29.57  | 460                       | 570  | 540  | 670  | 970     | 1060 | 1200 | 1290 |
|                  |    |       |                | 0.312 | 7.92  | 72.27  | 107.65 | 32.78  | 510                       | 640  | 600  | 740  | 1070    | 1170 | 1330 | 1430 |
|                  |    |       | 20             | 0.344 | 8.74  | 79.56  | 118.50 | 36.09  | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1580 |
|                  |    |       |                | 0.375 | 9.52  | 86.61  | 129.01 | 39.29  | 610                       | 770  | 720  | 890  | 1290    | 1410 | 1600 | 1720 |
|                  |    |       |                | 0.406 | 10.31 | 93.63  | 139.46 | 42.47  | --                        | --   | --   | --   | 1400    | 1530 | 1730 | 1860 |
|                  |    |       |                | 0.438 | 11.13 | 100.86 | 150.23 | 45.75  | 720                       | 900  | 840  | 1050 | 1510    | 1650 | 1860 | 2010 |
|                  |    |       | 30             | 0.469 | 11.91 | 107.85 | 160.64 | 48.92  | --                        | --   | --   | --   | 1610    | 1770 | 2000 | 2150 |
|                  |    |       |                | 0.500 | 12.70 | 114.81 | 171.01 | 52.08  | 820                       | 1020 | 950  | 1190 | 1720    | 1880 | 2130 | 2290 |
|                  |    |       |                | 0.562 | 14.27 | 128.67 | 191.65 | 58.36  | 920                       | 1150 | 1070 | 1340 | 1930    | 2120 | 2390 | 2500 |
|                  |    |       |                | 0.625 | 15.88 | 142.68 | 212.52 | 64.72  | 1020                      | 1280 | 1190 | 1490 | 2150    | 2350 | 2500 | 2500 |
|                  |    |       | 60             | 0.688 | 17.48 | 156.60 | 233.26 | 71.03  | 1130                      | 1410 | 1310 | 1640 | 2360    | 2500 | 2500 | 2500 |
|                  |    |       |                | 0.750 | 19.05 | 170.21 | 253.53 | 77.21  | 1230                      | 1530 | 1430 | 1790 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 0.812 | 20.62 | 183.75 | 273.70 | 83.35  | 1330                      | 1660 | 1550 | 1940 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 0.875 | 22.22 | 197.41 | 294.04 | 89.55  | 1430                      | 1790 | 1670 | 2090 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 0.938 | 23.83 | 211.00 | 314.28 | 95.71  | 1530                      | 1920 | 1790 | 2240 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 1.000 | 25.40 | 224.28 | 334.07 | 101.73 | 1640                      | 2050 | 1910 | 2390 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 1.062 | 26.97 | 237.48 | 353.73 | 107.72 | 1740                      | 2170 | 2030 | 2500 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       | 80             | 1.125 | 28.58 | 250.81 | 373.58 | 113.77 | 1840                      | 2300 | 2150 | 2500 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 1.188 | 30.15 | 263.85 | 393.00 | 119.68 | 1940                      | 2430 | 2270 | 2500 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 1.250 | 31.75 | 277.01 | 412.61 | 125.65 | 2050                      | 2500 | 2390 | 2500 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 1.312 | 33.32 | 289.88 | 431.78 | 129.22 | 2150                      | 2500 | 2500 | 2500 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       | 100            | 1.375 | 34.92 | 302.88 | 451.14 | 137.39 | 2250                      | 2500 | 2500 | 2500 | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 1.437 | 36.50 | 315.58 | 470.06 | 143.15 | 2350                      | 2500 | 2500 | 2500 | --      | --   | --   | --   |
|                  |    |       |                | 1.438 | 36.53 | --     | --     | --     | --                        | --   | --   | --   | 2500    | 2500 | 2500 | 2500 |
|                  |    |       |                | 1.500 | 38.10 | 328.41 | 489.17 | 148.97 | 2450                      | 2500 | 2500 | 2500 | 2500    | 2500 | 2500 | 2500 |

# API PIPES ERW & SEAMLESS PIPES

| Size             |    |       |                |       |       | Weight |        |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|----|-------|----------------|-------|-------|--------|--------|--------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |    |       | Wall Thickness |       |       |        |        |        | API 5L                    |      |      |      | API 5LX |      |      |      |
| Nominal Size     | in | mm    | Sch No         | in    | mm    | lb/ft  | kg/m   | kg/ft  | A                         |      | B    |      | X42     | X46  | X52  | X56  |
| 24               | 24 | 609.6 | 10             | 0.250 | 6.35  | 63.41  | 94.45  | 28.76  | 380                       | 470  | 440  | 550  | 790     | 860  | 980  | 1050 |
|                  |    |       |                | 0.281 | 7.14  | 71.18  | 106.02 | 32.29  | 420                       | 530  | 490  | 610  | 890     | 970  | 1100 | 1180 |
|                  |    |       |                | 0.312 | 7.92  | 78.93  | 117.57 | 35.80  | 470                       | 580  | 550  | 680  | 980     | 1080 | 1220 | 1310 |
|                  |    |       | 20             | 0.344 | 8.74  | 86.91  | 129.45 | 39.42  | 520                       | 640  | 600  | 750  | 1080    | 1190 | 1340 | 1440 |
|                  |    |       |                | 0.375 | 9.52  | 94.62  | 140.94 | 42.92  | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1580 |
|                  |    |       |                | 0.406 | 10.31 | 102.31 | 152.39 | 46.41  | --                        | --   | --   | --   | 1280    | 1400 | 1580 | 1710 |
|                  |    |       |                | 0.438 | 11.13 | 110.22 | 164.17 | 50.00  | 660                       | 820  | 770  | 960  | 1380    | 1510 | 1710 | 1840 |
|                  |    |       |                | 0.469 | 11.91 | 117.86 | 175.55 | 53.46  | --                        | --   | --   | --   | 1480    | 1620 | 1830 | 1970 |
|                  |    |       | 30             | 0.500 | 12.70 | 125.49 | 186.92 | 56.92  | 750                       | 940  | 880  | 1090 | 1580    | 1720 | 1950 | 2100 |
|                  |    |       |                | 0.562 | 14.27 | 140.68 | 209.54 | 63.81  | 840                       | 1050 | 980  | 1230 | 1770    | 1940 | 2190 | 2300 |
|                  |    |       |                | 0.625 | 15.88 | 156.03 | 232.41 | 70.78  | 940                       | 1170 | 1090 | 1370 | 1970    | 2160 | 2300 | 2300 |
|                  |    |       | 40             | 0.688 | 17.48 | 171.29 | 255.14 | 77.70  | 1030                      | 1290 | 1200 | 1500 | 2170    | 2300 | 2300 | 2300 |
|                  |    |       |                | 0.750 | 19.05 | 186.23 | 277.39 | 84.47  | 1120                      | 1410 | 1310 | 1640 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 0.812 | 20.62 | 201.09 | 299.52 | 91.21  | 1220                      | 1520 | 1420 | 1780 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 0.875 | 22.22 | 216.10 | 321.88 | 98.02  | 1310                      | 1640 | 1530 | 1910 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 0.938 | 23.83 | 231.03 | 344.12 | 104.80 | 1410                      | 1760 | 1640 | 2050 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.000 | 25.40 | 245.64 | 365.88 | 111.42 | 1500                      | 1880 | 1750 | 2190 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.062 | 26.97 | 260.17 | 387.52 | 118.01 | 1590                      | 1990 | 1860 | 2300 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.125 | 28.58 | 274.84 | 409.37 | 124.65 | 1690                      | 2110 | 1970 | 2300 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.187 | 30.15 | 289.20 | 430.76 | 131.18 | 1780                      | 2230 | 2080 | 2300 | --      | --   | --   | --   |
|                  |    |       |                | 1.188 | 30.18 | 289.44 | --     | --     | --                        | --   | --   | --   | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.250 | 31.75 | 303.71 | 452.38 | 137.76 | 1880                      | 2300 | 2190 | 2300 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.312 | 33.32 | 317.91 | 473.53 | 144.20 | 1970                      | 2300 | 2300 | 2300 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.375 | 34.92 | 332.25 | 494.89 | 150.71 | 2060                      | 2300 | 2300 | 2300 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.437 | 36.50 | 346.28 | 515.78 | 157.07 | 2160                      | 2300 | 2300 | 2300 | --      | --   | --   | --   |
|                  |    |       |                | 1.438 | 36.53 | 346.50 | --     | --     | --                        | --   | --   | --   | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.500 | 38.10 | 360.45 | 536.89 | 163.50 | 2250                      | 2300 | 2300 | 2300 | 2300    | 2300 | 2300 | 2300 |
|                  |    |       |                | 1.562 | 39.67 | 374.31 | 557.53 | 169.79 | 2300                      | 2300 | 2300 | 2300 | 2300    | 2300 | 2300 | 2300 |
| 26               | 26 | 660.4 | 10             | 0.250 | 6.35  | 68.75  | 102.40 | 31.19  | 350                       | 430  | 400  | 500  | 730     | 800  | 900  | 970  |
|                  |    |       |                | 0.281 | 7.14  | 77.18  | 114.96 | 35.01  | 390                       | 490  | 450  | 570  | 820     | 890  | 1010 | 1090 |
|                  |    |       |                | 0.312 | 7.92  | 85.60  | 127.50 | 38.83  | 430                       | 540  | 500  | 630  | 910     | 990  | 1120 | 1210 |
|                  |    |       |                | 0.344 | 8.74  | 94.26  | 140.40 | 42.76  | 480                       | 600  | 560  | 690  | 1000    | 1100 | 1240 | 1330 |
|                  |    |       |                | 0.375 | 9.52  | 102.63 | 152.87 | 46.55  | 520                       | 650  | 610  | 760  | 1090    | 1190 | 1350 | 1450 |
|                  |    |       |                | 0.406 | 10.31 | 110.98 | 165.30 | 50.34  | --                        | --   | --   | --   | 1180    | 1290 | 1460 | 1570 |
|                  |    |       | 20             | 0.438 | 11.13 | 119.57 | 178.10 | 54.24  | 610                       | 760  | 710  | 880  | 1270    | 1390 | 1580 | 1700 |
|                  |    |       |                | 0.469 | 11.91 | 127.88 | 190.48 | 58.01  | --                        | --   | --   | --   | 1360    | 1490 | 1690 | 1820 |
|                  |    |       |                | 0.500 | 12.70 | 136.17 | 202.83 | 61.77  | 690                       | 870  | 810  | 1010 | 1450    | 1590 | 1800 | 1940 |
|                  |    |       |                | 0.562 | 14.27 | 152.68 | 227.42 | 69.26  | 780                       | 970  | 910  | 1130 | 1630    | 1790 | 2000 | 2000 |
|                  |    |       |                | 0.625 | 15.88 | 169.38 | 252.29 | 76.83  | 870                       | 1080 | 1010 | 1260 | 1820    | 1990 | 2000 | 2000 |
|                  |    |       |                | 0.688 | 17.48 | 185.99 | 277.03 | 84.37  | 950                       | 1190 | 1110 | 1390 | 2000    | 2000 | 2000 | 2000 |
|                  |    |       |                | 0.750 | 19.05 | 202.25 | 301.25 | 91.74  | 1040                      | 1300 | 1210 | 1510 | 2000    | 2000 | 2000 | 2000 |
|                  |    |       |                | 0.812 | 20.62 | 218.43 | 325.35 | 99.08  | 1120                      | 1410 | 1310 | 1640 | 2000    | 2000 | 2000 | 2000 |
|                  |    |       |                | 0.875 | 22.22 | 234.79 | 349.72 | 106.50 | 1210                      | 1510 | 1410 | 1770 | 2000    | 2000 | 2000 | 2000 |
|                  |    |       |                | 0.938 | 23.83 | 251.07 | 373.97 | 113.89 | 1300                      | 1620 | 1520 | 1890 | 2000    | 2000 | 2000 | 2000 |
|                  |    |       |                | 1.000 | 25.40 | 267.00 | 397.70 | 121.11 | 1380                      | 1730 | 1620 | 2000 | 2000    | 2000 | 2000 | 2000 |
| 28               | 28 | 711.2 | 10             | 0.250 | 6.35  | 74.09  | 110.36 | 33.61  | 320                       | 400  | 370  | 470  | 680     | 740  | 840  | 900  |
|                  |    |       |                | 0.281 | 7.14  | 83.19  | 123.91 | 37.73  | 360                       | 450  | 420  | 530  | 760     | 830  | 940  | 1010 |
|                  |    |       |                | 0.312 | 7.92  | 92.26  | 137.42 | 41.85  | 400                       | 500  | 470  | 580  | 840     | 920  | 1040 | 1120 |
|                  |    |       |                | 0.344 | 8.74  | 101.61 | 151.35 | 46.09  | --                        | --   | --   | --   | 930     | 1020 | 1150 | 1240 |
|                  |    |       |                | 0.375 | 9.52  | 110.64 | 164.80 | 50.19  | 480                       | 600  | 560  | 700  | 1010    | 1110 | 1250 | 1350 |

# API PIPES ERW & SEAMLESS PIPES

| Size             |    |       |                |       |       | Weight |        |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|----|-------|----------------|-------|-------|--------|--------|--------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |    |       | Wall Thickness |       |       |        |        |        | API 5L                    |      |      |      | API 5LX |      |      |      |
|                  |    |       |                |       |       |        |        |        | A                         |      | B    |      |         |      |      |      |
| Nominal Size     | in | mm    | Sch No         | in    | mm    | lb/ft  | kg/m   | kg/ft  | Std.                      | Alt. | Std. | Alt. | X42     | X46  | X52  | X56  |
| 28               | 28 | 711.2 | 20             | 0.406 | 10.31 | 119.65 | 178.22 | 54.27  | --                        | --   | --   | --   | 1100    | 1200 | 1360 | 1460 |
|                  |    |       |                | 0.438 | 11.13 | 128.93 | 192.04 | 58.48  | 560                       | 700  | 660  | 820  | 1180    | 1300 | 1460 | 1580 |
|                  |    |       |                | 0.469 | 11.91 | 137.90 | 205.40 | 62.55  | --                        | --   | --   | --   | 1270    | 1390 | 1570 | 1690 |
|                  |    |       |                | 0.500 | 12.70 | 146.85 | 218.73 | 66.61  | 640                       | 800  | 750  | 940  | 1350    | 1480 | 1670 | 1800 |
|                  |    |       |                | 0.562 | 14.27 | 164.69 | 245.31 | 74.70  | 720                       | 900  | 840  | 1050 | 1520    | 1660 | 1880 | 2020 |
|                  |    |       |                | 0.625 | 15.88 | 182.73 | 272.18 | 82.89  | 800                       | 1000 | 940  | 1170 | 1690    | 1850 | 2090 | 2250 |
|                  |    |       | 30             | 0.688 | 17.48 | 200.68 | 298.91 | 91.03  | 880                       | 1100 | 1030 | 1290 | 1860    | 2030 | 2300 | 2480 |
|                  |    |       |                | 0.750 | 19.05 | 218.27 | 325.11 | 99.01  | 960                       | 1210 | 1120 | 1410 | 2020    | 2220 | 2510 | 2700 |
|                  |    |       |                | 0.812 | 20.62 | 235.78 | 351.19 | 106.95 | 1040                      | 1300 | 1220 | 1520 | 2190    | 2400 | 2710 | 2920 |
|                  |    |       |                | 0.875 | 22.22 | 253.48 | 377.56 | 114.98 | 1120                      | 1410 | 1310 | 1640 | 2360    | 2590 | 2920 | 3000 |
|                  |    |       |                | 0.938 | 23.83 | 271.10 | 403.80 | 122.97 | 1210                      | 1510 | 1410 | 1760 | 2530    | 2770 | 3000 | 3000 |
|                  |    |       |                | 1.000 | 25.40 | 288.36 | 429.51 | 130.80 | 1290                      | 1610 | 1500 | 1880 | 2700    | 2960 | 3000 | 3000 |
| 30               | 30 | 762.0 | 10             | 0.250 | 6.35  | 79.43  | 118.31 | 36.03  | 300                       | 370  | 350  | 440  | 630     | 690  | 780  | 840  |
|                  |    |       |                | 0.281 | 7.14  | 89.19  | 132.85 | 40.46  | 340                       | 420  | 390  | 490  | 710     | 780  | 880  | 940  |
|                  |    |       |                | 0.312 | 7.92  | 98.93  | 147.36 | 44.87  | 370                       | 470  | 440  | 550  | 790     | 860  | 970  | 1050 |
|                  |    |       |                | 0.344 | 8.74  | 108.95 | 162.28 | 49.42  | --                        | --   | --   | --   | 870     | 950  | 1070 | 1160 |
|                  |    |       |                | 0.375 | 9.52  | 118.65 | 176.73 | 53.82  | 450                       | 560  | 520  | 660  | 940     | 1040 | 1170 | 1260 |
|                  |    |       |                | 0.406 | 10.31 | 128.32 | 191.13 | 58.21  | --                        | --   | --   | --   | 1020    | 1120 | 1270 | 1360 |
|                  |    |       | 20             | 0.438 | 11.13 | 138.29 | 205.98 | 62.73  | 530                       | 660  | 610  | 770  | 1100    | 1210 | 1370 | 1470 |
|                  |    |       |                | 0.469 | 11.91 | 147.92 | 220.33 | 67.10  | --                        | --   | --   | --   | 1180    | 1290 | 1460 | 1580 |
|                  |    |       |                | 0.500 | 12.70 | 157.53 | 234.64 | 71.46  | 600                       | 750  | 700  | 880  | 1260    | 1380 | 1560 | 1680 |
|                  |    |       |                | 0.562 | 14.27 | 176.69 | 263.18 | 80.15  | 670                       | 840  | 790  | 980  | 1420    | 1550 | 1750 | 1890 |
|                  |    |       |                | 0.625 | 15.88 | 196.08 | 292.06 | 88.94  | 750                       | 940  | 880  | 1090 | 1580    | 1720 | 1950 | 2100 |
|                  |    |       |                | 0.688 | 17.48 | 215.38 | 320.81 | 97.70  | 830                       | 1030 | 960  | 1200 | 1730    | 1900 | 2150 | 2310 |
|                  |    |       |                | 0.750 | 19.05 | 234.29 | 348.97 | 106.27 | 900                       | 1120 | 1050 | 1310 | 1890    | 2070 | 2340 | 2520 |
|                  |    |       |                | 0.812 | 20.62 | 253.12 | 377.02 | 114.82 | 970                       | 1220 | 1140 | 1420 | 2050    | 2240 | 2530 | 2730 |
|                  |    |       |                | 0.875 | 22.22 | 272.17 | 405.40 | 123.46 | 1050                      | 1310 | 1220 | 1530 | 2200    | 2420 | 2730 | 2940 |
|                  |    |       |                | 0.938 | 23.83 | 291.14 | 433.65 | 132.06 | 1130                      | 1410 | 1310 | 1640 | 2360    | 2590 | 2930 | 3000 |
|                  |    |       |                | 1.000 | 25.40 | 309.72 | 461.33 | 140.49 | 1200                      | 1500 | 1400 | 1750 | 2520    | 2760 | 3000 | 3000 |
|                  |    |       |                | 1.062 | 26.97 | 328.22 | 488.88 | 148.88 | 1270                      | 1590 | 1490 | 1860 | 2680    | 2930 | 3000 | 3000 |
|                  |    |       |                | 1.125 | 28.58 | 346.93 | 516.75 | 157.37 | 1350                      | 1690 | 1580 | 1970 | 2840    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.188 | 30.18 | 365.56 | 544.50 | 165.82 | 1430                      | 1780 | 1660 | 2080 | 2990    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.250 | 31.75 | 383.81 | 571.68 | 174.10 | 1500                      | 1880 | 1750 | 2190 | 3000    | 3000 | 3000 | 3000 |
| 32               | 32 | 812.8 | 10             | 0.250 | 6.35  | 84.77  | 126.26 | 38.45  | 280                       | 350  | 330  | 410  | 590     | 650  | 730  | 790  |
|                  |    |       |                | 0.281 | 7.14  | 95.19  | 141.79 | 43.18  | 320                       | 400  | 370  | 460  | 660     | 730  | 820  | 890  |
|                  |    |       |                | 0.312 | 7.92  | 105.59 | 157.28 | 47.90  | 350                       | 440  | 410  | 510  | 740     | 810  | 910  | 980  |
|                  |    |       |                | 0.344 | 8.74  | 116.30 | 173.23 | 52.75  | --                        | --   | --   | --   | 810     | 890  | 1010 | 1080 |
|                  |    |       |                | 0.375 | 9.52  | 126.66 | 188.66 | 57.45  | 420                       | 530  | 490  | 620  | 890     | 970  | 1100 | 1180 |
|                  |    |       |                | 0.406 | 10.31 | 136.99 | 204.05 | 62.14  | --                        | --   | --   | --   | 960     | 1050 | 1190 | 1280 |
|                  |    |       | 20             | 0.438 | 11.13 | 147.64 | 219.91 | 66.97  | 490                       | 620  | 570  | 720  | 1030    | 1130 | 1280 | 1380 |
|                  |    |       |                | 0.469 | 11.91 | 157.94 | 235.25 | 71.64  | --                        | --   | --   | --   | 1110    | 1210 | 1370 | 1480 |
|                  |    |       |                | 0.500 | 12.70 | 168.21 | 250.55 | 76.30  | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1580 |
|                  |    |       |                | 0.562 | 14.27 | 188.70 | 281.07 | 85.59  | 630                       | 790  | 740  | 920  | 1330    | 1450 | 1640 | 1770 |
|                  |    |       |                | 0.625 | 15.88 | 209.43 | 311.95 | 95.00  | 700                       | 880  | 820  | 1030 | 1480    | 1620 | 1830 | 1970 |
|                  |    |       |                | 0.688 | 17.48 | 230.08 | 342.70 | 104.36 | 770                       | 970  | 900  | 1130 | 1630    | 1780 | 2010 | 2170 |
|                  |    |       |                | 0.750 | 19.05 | 250.31 | 372.84 | 113.54 | 840                       | 1050 | 980  | 1230 | 1770    | 1940 | 2190 | 2360 |
|                  |    |       |                | 0.812 | 20.62 | 270.47 | 402.87 | 122.69 | 910                       | 1140 | 1070 | 1330 | 1920    | 2100 | 2380 | 2560 |
|                  |    |       |                | 0.875 | 22.22 | 290.86 | 433.24 | 131.93 | 980                       | 1230 | 1150 | 1440 | 2070    | 2260 | 2560 | 2760 |
|                  |    |       |                | 0.938 | 23.83 | 311.17 | 463.49 | 141.15 | 1060                      | 1320 | 1230 | 1540 | 2220    | 2430 | 2740 | 2950 |
|                  |    |       |                | 1.000 | 25.40 | 331.08 | 493.14 | 150.18 | 1120                      | 1410 | 1310 | 1640 | 2360    | 2590 | 2920 | 3000 |
|                  |    |       |                | 1.062 | 26.97 | 350.90 | 522.74 | 159.17 | 1190                      | 1490 | 1390 | 1740 | 2510    | 2750 | 3000 | 3000 |
|                  |    |       |                | 1.125 | 28.58 | 370.96 | 552.54 | 168.27 | 1270                      | 1580 | 1480 | 1850 | 2660    | 2910 | 3000 | 3000 |
|                  |    |       |                | 1.188 | 30.18 | 390.94 | 582.31 | 177.33 | 1340                      | 1670 | 1560 | 1950 | 2810    | 3000 | 3000 | 3000 |
|                  |    |       |                | 1.250 | 31.75 | 410.51 | 611.45 | 186.21 | 1410                      | 1760 | 1640 | 2050 | 2950    | 3000 | 3000 | 3000 |



# API PIPES ERW & SEAMLESS PIPES

| Size             |    |       |                |       |       | Weight |        |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|----|-------|----------------|-------|-------|--------|--------|--------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |    |       | Wall Thickness |       |       |        |        |        | API 5L                    |      |      |      | API 5LX |      |      |      |
| Nominal Size     | in | mm    | Sch No         | in    | mm    | lb/ft  | kg/m   | kg/ft  | A                         |      | B    |      |         |      |      |      |
|                  |    |       |                |       |       |        |        |        | Std.                      | Alt. | Std. | Alt. | X42     | X46  | X52  | X56  |
| 34               | 34 | 863.6 | 10             | 0.250 | 6.35  | 90.11  | 134.22 | 40.87  | 260                       | 330  | 310  | 390  | 560     | 610  | 690  | 740  |
|                  |    |       |                | 0.281 | 7.14  | 101.19 | 150.72 | 45.90  | 300                       | 370  | 350  | 430  | 620     | 680  | 770  | 830  |
|                  |    |       |                | 0.312 | 7.92  | 112.25 | 167.20 | 50.92  | 330                       | 410  | 390  | 480  | 690     | 760  | 860  | 920  |
|                  |    |       |                | 0.344 | 8.74  | 123.65 | 184.18 | 56.09  | --                        | --   | --   | --   | 760     | 840  | 950  | 1020 |
|                  |    |       |                | 0.375 | 9.52  | 134.67 | 200.59 | 61.09  | 400                       | 500  | 460  | 580  | 830     | 910  | 1030 | 1110 |
|                  |    |       |                | 0.406 | 10.31 | 145.67 | 216.98 | 66.08  | --                        | --   | --   | --   | 900     | 990  | 1120 | 1200 |
|                  |    |       | 20             | 0.438 | 11.13 | 157.00 | 233.85 | 71.22  | 460                       | 580  | 540  | 680  | 970     | 1070 | 1210 | 1300 |
|                  |    |       |                | 0.469 | 11.91 | 167.95 | 250.16 | 76.18  | --                        | --   | --   | --   | 1040    | 1140 | 1290 | 1390 |
|                  |    |       |                | 0.500 | 12.70 | 178.89 | 266.46 | 81.14  | 530                       | 660  | 620  | 770  | 1110    | 1220 | 1380 | 1480 |
|                  |    |       |                | 0.562 | 14.27 | 200.70 | 298.94 | 91.04  | 600                       | 740  | 690  | 870  | 1250    | 1370 | 1550 | 1670 |
|                  |    |       | 30             | 0.625 | 15.88 | 222.78 | 331.83 | 101.05 | 660                       | 830  | 770  | 970  | 1390    | 1520 | 1720 | 1850 |
|                  |    |       |                | 0.688 | 17.48 | 244.77 | 364.58 | 111.03 | 730                       | 910  | 850  | 1060 | 1530    | 1680 | 1890 | 2040 |
|                  |    |       | 40             | 0.750 | 19.05 | 266.33 | 396.70 | 120.81 | 790                       | 990  | 930  | 1160 | 1670    | 1830 | 2060 | 2220 |
|                  |    |       |                | 0.812 | 20.62 | 287.81 | 428.69 | 130.55 | 860                       | 1070 | 1000 | 1250 | 1810    | 1980 | 2240 | 2410 |
|                  |    |       |                | 0.875 | 22.22 | 309.55 | 461.07 | 140.41 | 930                       | 1160 | 1080 | 1350 | 1950    | 2130 | 2410 | 2590 |
|                  |    |       |                | 0.938 | 23.83 | 331.21 | 493.34 | 150.24 | 990                       | 1240 | 1160 | 1450 | 2090    | 2280 | 2580 | 2780 |
|                  |    |       |                | 1.000 | 25.40 | 352.44 | 524.96 | 162.13 | 1060                      | 1320 | 1240 | 1540 | 2220    | 2440 | 2750 | 2960 |
|                  |    |       |                | 1.062 | 26.97 | 373.59 | 556.46 | 169.46 | 1120                      | 1410 | 1310 | 1640 | 2360    | 2590 | 2920 | 3000 |
|                  |    |       |                | 1.125 | 28.58 | 394.99 | 588.34 | 179.17 | 1190                      | 1490 | 1390 | 1740 | 2500    | 2740 | 3000 | 3000 |
|                  |    |       |                | 1.188 | 30.18 | 416.31 | 620.09 | 188.84 | 1260                      | 1570 | 1470 | 1830 | 2640    | 2890 | 3000 | 3000 |
|                  |    |       |                | 1.250 | 31.75 | 437.21 | 651.22 | 198.32 | 1320                      | 1650 | 1540 | 1930 | 2780    | 3000 | 3000 | 3000 |
| 36               | 36 | 914.4 | 10             | 0.250 | 6.35  | 95.45  | 142.17 | 43.30  | 250                       | 310  | 290  | 360  | 520     | 580  | 650  | 700  |
|                  |    |       |                | 0.281 | 7.14  | 107.20 | 159.67 | 48.63  | 280                       | 350  | 330  | 410  | 590     | 650  | 730  | 790  |
|                  |    |       |                | 0.312 | 7.92  | 118.92 | 177.13 | 53.94  | 310                       | 390  | 360  | 450  | 660     | 720  | 810  | 870  |
|                  |    |       |                | 0.344 | 8.74  | 131.00 | 195.12 | 59.42  | --                        | --   | --   | --   | 720     | 790  | 890  | 960  |
|                  |    |       |                | 0.375 | 9.52  | 142.68 | 212.52 | 64.72  | 380                       | 470  | 440  | 550  | 790     | 860  | 980  | 1050 |
|                  |    |       |                | 0.406 | 10.31 | 154.34 | 229.89 | 70.01  | --                        | --   | --   | --   | 850     | 930  | 1060 | 1140 |
|                  |    |       | 20             | 0.438 | 11.13 | 166.35 | 247.78 | 75.46  | 440                       | 550  | 510  | 640  | 920     | 1010 | 1140 | 1230 |
|                  |    |       |                | 0.469 | 11.91 | 177.97 | 265.09 | 80.73  | --                        | --   | --   | --   | 980     | 1080 | 1220 | 1310 |
|                  |    |       |                | 0.500 | 12.70 | 189.57 | 282.36 | 85.99  | 500                       | 620  | 580  | 730  | 1050    | 1150 | 1300 | 1400 |
|                  |    |       |                | 0.562 | 14.27 | 212.70 | 316.82 | 96.48  | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1570 |
|                  |    |       | 30             | 0.625 | 15.88 | 236.13 | 351.72 | 107.11 | 620                       | 780  | 730  | 910  | 1310    | 1440 | 1620 | 1750 |
|                  |    |       |                | 0.688 | 17.48 | 259.47 | 386.48 | 117.70 | 690                       | 860  | 800  | 1000 | 1440    | 1580 | 1790 | 1930 |
|                  |    |       | 40             | 0.750 | 19.05 | 282.35 | 420.56 | 128.07 | 750                       | 940  | 880  | 1090 | 1580    | 1720 | 1950 | 2100 |
|                  |    |       |                | 0.812 | 20.62 | 305.16 | 454.54 | 138.42 | 810                       | 1020 | 950  | 1180 | 1710    | 1870 | 2110 | 2270 |
|                  |    |       |                | 0.875 | 22.22 | 328.24 | 488.91 | 148.89 | 880                       | 1090 | 1020 | 1280 | 1840    | 2010 | 2280 | 2450 |
|                  |    |       |                | 0.938 | 23.83 | 351.25 | 523.19 | 159.33 | 940                       | 1170 | 1090 | 1370 | 1970    | 2160 | 2440 | 2630 |
|                  |    |       |                | 1.000 | 25.40 | 373.80 | 556.78 | 169.56 | 1000                      | 1250 | 1170 | 1460 | 2100    | 2300 | 2600 | 2800 |
|                  |    |       |                | 1.062 | 26.97 | 396.27 | 590.24 | 179.75 | 1060                      | 1330 | 1240 | 1550 | 2230    | 2440 | 2760 | 2970 |
|                  |    |       |                | 1.125 | 28.58 | 419.02 | 624.13 | 190.07 | 1130                      | 1410 | 1310 | 1640 | 2360    | 2590 | 2930 | 3000 |
|                  |    |       |                | 1.188 | 30.18 | 441.69 | 657.90 | 200.35 | 1190                      | 1480 | 1390 | 1730 | 2490    | 2730 | 3000 | 3000 |
|                  |    |       |                | 1.250 | 31.75 | 463.91 | 690.99 | 210.43 | 1250                      | 1560 | 1460 | 1820 | 2630    | 2870 | 3000 | 3000 |
| 38               | 38 | 965.2 |                | 0.312 | 7.92  | 125.58 | 187.05 | 56.96  | 300                       | 370  | 340  | 430  | 620     | 680  | 770  | 830  |
|                  |    |       |                | 0.344 | 8.74  | 138.35 | 206.07 | 62.76  | 330                       | 410  | 380  | 480  | 680     | 750  | 850  | 910  |
|                  |    |       |                | 0.375 | 9.52  | 150.69 | 224.45 | 68.35  | 360                       | 440  | 410  | 520  | 750     | 820  | 920  | 990  |
|                  |    |       |                | 0.406 | 10.31 | 163.01 | 242.80 | 73.94  | 380                       | 480  | 450  | 560  | 810     | 880  | 1000 | 1080 |
|                  |    |       |                | 0.438 | 11.13 | 175.71 | 261.72 | 79.70  | 410                       | 520  | 480  | 610  | 870     | 950  | 1080 | 1160 |
|                  |    |       |                | 0.469 | 11.91 | 187.99 | 280.01 | 85.27  | 440                       | 560  | 520  | 650  | 930     | 1020 | 1160 | 1240 |
|                  |    |       |                | 0.500 | 12.70 | 200.25 | 298.27 | 90.83  | 470                       | 590  | 550  | 690  | 990     | 1090 | 1230 | 1330 |
|                  |    |       |                | 0.562 | 14.27 | 224.71 | 334.71 | 101.93 | 530                       | 670  | 620  | 780  | 1120    | 1220 | 1380 | 1490 |
|                  |    |       |                | 0.625 | 15.88 | 249.48 | 371.60 | 113.16 | 590                       | 740  | 690  | 860  | 1240    | 1360 | 1540 | 1560 |



# API PIPES ERW & SEAMLESS PIPES

| Size             |    |        |                |       |       | Weight |        |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|----|--------|----------------|-------|-------|--------|--------|--------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |    |        | Wall Thickness |       |       |        |        |        | API 5L                    |      |      |      | API 5LX |      |      |      |
|                  |    |        |                |       |       |        |        |        | A                         |      | B    |      |         |      |      |      |
| Nominal Size     | in | mm     | Sch No         | in    | mm    | lb/ft  | kg/m   | kg/ft  | Std.                      | Alt. | Std. | Alt. | X42     | X46  | X52  | X56  |
| 38               | 38 | 965.2  |                | 0.688 | 17.48 | 274.16 | 408.36 | 124.36 | 650                       | 810  | 760  | 950  | 1370    | 1500 | 1690 | 1830 |
|                  |    |        |                | 0.750 | 19.05 | 298.37 | 444.42 | 135.34 | 710                       | 890  | 830  | 1040 | 1490    | 1630 | 1850 | 1990 |
|                  |    |        |                | 0.812 | 20.62 | 322.50 | 480.36 | 146.29 | 770                       | 960  | 900  | 1120 | 1620    | 1770 | 2000 | 2150 |
|                  |    |        |                | 0.875 | 22.22 | 346.93 | 516.75 | 157.37 | 830                       | 1040 | 970  | 1210 | 1740    | 1910 | 2160 | 2320 |
|                  |    |        |                | 0.938 | 23.83 | 371.28 | 553.02 | 168.41 | 890                       | 1110 | 1040 | 1300 | 1870    | 2040 | 2310 | 2490 |
|                  |    |        |                | 1.000 | 25.40 | 395.16 | 588.59 | 179.24 | 950                       | 1180 | 1110 | 1380 | 1990    | 2180 | 2460 | 2650 |
|                  |    |        |                | 1.062 | 26.97 | 418.96 | 624.04 | 190.04 | 1010                      | 1260 | 1170 | 1470 | 2110    | 2310 | 2620 | 2820 |
|                  |    |        |                | 1.125 | 28.58 | 443.05 | 659.92 | 200.97 | 1070                      | 1330 | 1240 | 1550 | 2240    | 2450 | 2770 | 2980 |
|                  |    |        |                | 1.188 | 30.18 | 467.06 | 695.69 | 211.86 | 1130                      | 1410 | 1310 | 1640 | 2360    | 2590 | 2930 | 3000 |
|                  |    |        |                | 1.250 | 31.75 | 490.61 | 730.76 | 222.54 | 1180                      | 1480 | 1380 | 1730 | 2490    | 2720 | 3000 | 3000 |
| 40               | 40 | 1016.0 |                | 0.312 | 7.92  | 132.25 | 196.99 | 59.99  | 280                       | 350  | 330  | 410  | 590     | 650  | 730  | 790  |
|                  |    |        |                | 0.344 | 8.74  | 145.69 | 217.01 | 66.08  | 310                       | 390  | 360  | 450  | 650     | 710  | 800  | 870  |
|                  |    |        |                | 0.375 | 9.52  | 158.70 | 236.38 | 71.99  | 340                       | 420  | 390  | 490  | 710     | 780  | 880  | 940  |
|                  |    |        |                | 0.406 | 10.31 | 171.68 | 255.72 | 77.87  | 370                       | 460  | 430  | 530  | 770     | 840  | 950  | 1020 |
|                  |    |        |                | 0.438 | 11.13 | 185.06 | 275.65 | 83.94  | 390                       | 490  | 460  | 570  | 830     | 910  | 1020 | 1100 |
|                  |    |        |                | 0.469 | 11.91 | 198.01 | 294.94 | 89.82  | 420                       | 530  | 490  | 620  | 890     | 970  | 1100 | 1180 |
|                  |    |        |                | 0.500 | 12.70 | 210.93 | 314.18 | 95.68  | 450                       | 560  | 520  | 660  | 940     | 1040 | 1170 | 1260 |
|                  |    |        |                | 0.562 | 14.27 | 236.71 | 352.58 | 107.37 | 510                       | 630  | 590  | 740  | 1060    | 1160 | 1320 | 1420 |
|                  |    |        |                | 0.625 | 15.88 | 262.83 | 391.49 | 119.22 | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1580 |
|                  |    |        |                | 0.688 | 17.48 | 288.86 | 430.26 | 131.03 | 620                       | 770  | 720  | 900  | 1300    | 1420 | 1610 | 1730 |
|                  |    |        |                | 0.750 | 19.05 | 314.39 | 468.28 | 142.61 | 680                       | 840  | 790  | 980  | 1420    | 1550 | 1760 | 1890 |
|                  |    |        |                | 0.812 | 20.62 | 339.84 | 506.19 | 154.15 | 730                       | 910  | 850  | 1070 | 1530    | 1680 | 1900 | 2050 |
|                  |    |        |                | 0.875 | 22.22 | 365.62 | 544.59 | 165.85 | 790                       | 980  | 920  | 1150 | 1650    | 1810 | 2050 | 2200 |
|                  |    |        |                | 0.938 | 23.83 | 391.32 | 582.87 | 177.50 | 840                       | 1060 | 980  | 1230 | 1770    | 1940 | 2190 | 2360 |
|                  |    |        |                | 1.000 | 25.40 | 416.52 | 620.41 | 188.93 | 900                       | 1120 | 1050 | 1310 | 1890    | 2070 | 2340 | 2520 |
|                  |    |        |                | 1.062 | 26.97 | 441.64 | 657.82 | 200.33 | 960                       | 1190 | 1120 | 1390 | 2010    | 2200 | 2490 | 2680 |
|                  |    |        |                | 1.125 | 28.58 | 467.08 | 695.72 | 211.87 | 1010                      | 1270 | 1180 | 1480 | 2130    | 2330 | 2630 | 2830 |
|                  |    |        |                | 1.188 | 30.18 | 492.44 | 733.49 | 223.37 | 1070                      | 1340 | 1250 | 1560 | 2250    | 2460 | 2780 | 2990 |
|                  |    |        |                | 1.250 | 31.75 | 517.31 | 770.53 | 234.65 | 1130                      | 1410 | 1310 | 1640 | 2360    | 2590 | 2930 | 3000 |
| 42               | 42 | 1066.8 |                | 0.344 | 8.74  | 153.04 | 227.95 | 69.42  | 290                       | 370  | 340  | 430  | 620     | 680  | 770  | 830  |
|                  |    |        |                | 0.375 | 9.52  | 166.71 | 248.31 | 75.62  | 320                       | 400  | 380  | 470  | 680     | 740  | 840  | 900  |
|                  |    |        |                | 0.406 | 10.31 | 180.35 | 268.63 | 81.81  | 350                       | 430  | 410  | 510  | 730     | 800  | 900  | 970  |
|                  |    |        |                | 0.438 | 11.13 | 194.42 | 289.59 | 88.19  | 380                       | 470  | 440  | 550  | 790     | 860  | 980  | 1050 |
|                  |    |        |                | 0.469 | 11.91 | 208.03 | 309.86 | 94.36  | 400                       | 500  | 470  | 590  | 840     | 920  | 1050 | 1130 |
|                  |    |        |                | 0.500 | 12.70 | 221.61 | 330.09 | 100.52 | 430                       | 540  | 500  | 620  | 900     | 990  | 1110 | 1200 |
|                  |    |        |                | 0.562 | 14.27 | 248.72 | 370.47 | 112.82 | 480                       | 600  | 560  | 700  | 1010    | 1110 | 1250 | 1350 |
|                  |    |        |                | 0.625 | 15.88 | 276.18 | 411.37 | 125.28 | 540                       | 670  | 620  | 780  | 1120    | 1230 | 1390 | 1500 |
|                  |    |        |                | 0.688 | 17.48 | 303.55 | 452.14 | 137.69 | 590                       | 740  | 690  | 860  | 1240    | 1360 | 1530 | 1650 |
|                  |    |        |                | 0.750 | 19.05 | 330.41 | 492.15 | 149.87 | 640                       | 800  | 750  | 940  | 1350    | 1480 | 1670 | 1800 |
|                  |    |        |                | 0.812 | 20.62 | 357.19 | 532.03 | 162.02 | 700                       | 870  | 810  | 1020 | 1460    | 1600 | 1810 | 1950 |
|                  |    |        |                | 0.875 | 22.22 | 384.31 | 572.43 | 174.32 | 750                       | 940  | 880  | 1090 | 1580    | 1720 | 1950 | 2100 |
|                  |    |        |                | 0.938 | 23.83 | 411.35 | 612.71 | 186.59 | 800                       | 1000 | 940  | 1170 | 1690    | 1850 | 2090 | 2250 |
|                  |    |        |                | 1.000 | 25.40 | 437.88 | 652.22 | 198.62 | 860                       | 1070 | 1000 | 1250 | 1800    | 1970 | 2230 | 2400 |
|                  |    |        |                | 1.062 | 26.97 | 464.32 | 691.60 | 210.62 | 910                       | 1140 | 1060 | 1330 | 1910    | 2090 | 2370 | 2550 |
|                  |    |        |                | 1.125 | 28.58 | 491.11 | 731.51 | 222.77 | 960                       | 1210 | 1130 | 1410 | 2030    | 2220 | 2510 | 2700 |
|                  |    |        |                | 1.188 | 30.18 | 517.82 | 771.29 | 234.88 | 1020                      | 1270 | 1190 | 1480 | 2140    | 2340 | 2650 | 2850 |
|                  |    |        |                | 1.250 | 31.75 | 544.01 | 810.30 | 246.76 | 1070                      | 1340 | 1250 | 1560 | 2250    | 2460 | 2790 | 3000 |

# API PIPES ERW & SEAMLESS PIPES

| Size             |    |        |                |       |       | Weight |        |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|----|--------|----------------|-------|-------|--------|--------|--------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |    |        | Wall Thickness |       |       |        |        |        | API 5L                    |      |      |      | API 5LX |      |      |      |
| Nominal          |    |        |                |       |       |        |        |        | A                         |      | B    |      |         |      |      |      |
| Size             | in | mm     | Sch No         | in    | mm    | lb/ft  | kg/m   | kg/ft  | Std.                      | Alt. | Std. | Alt. | X42     | X46  | X52  | X56  |
| 44               | 44 | 1117.6 |                | 0.344 | 8.74  | 160.39 | 238.90 | 72.75  | 280                       | 350  | 330  | 410  | 590     | 650  | 730  | 790  |
|                  |    |        |                | 0.375 | 9.52  | 174.72 | 260.25 | 79.26  | 310                       | 380  | 360  | 450  | 640     | 710  | 800  | 860  |
|                  |    |        |                | 0.406 | 10.31 | 189.03 | 281.56 | 85.74  | 330                       | 420  | 390  | 480  | 700     | 760  | 860  | 930  |
|                  |    |        |                | 0.438 | 11.13 | 203.78 | 303.53 | 92.43  | 360                       | 450  | 420  | 520  | 750     | 820  | 930  | 1000 |
|                  |    |        |                | 0.469 | 11.91 | 218.04 | 324.77 | 98.90  | 380                       | 480  | 450  | 560  | 810     | 880  | 1000 | 1070 |
|                  |    |        |                | 0.500 | 12.70 | 232.29 | 346.00 | 105.37 | 410                       | 510  | 480  | 600  | 860     | 940  | 1060 | 1150 |
|                  |    |        |                | 0.562 | 14.27 | 260.72 | 388.34 | 118.26 | 460                       | 570  | 540  | 670  | 970     | 1060 | 1200 | 1290 |
|                  |    |        |                | 0.625 | 15.88 | 289.53 | 431.25 | 131.33 | 510                       | 640  | 600  | 750  | 1070    | 1180 | 1330 | 1430 |
|                  |    |        |                | 0.688 | 17.48 | 318.25 | 474.03 | 144.36 | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1580 |
|                  |    |        |                | 0.750 | 19.05 | 346.43 | 516.01 | 157.14 | 610                       | 770  | 720  | 890  | 1290    | 1410 | 1600 | 1720 |
|                  |    |        |                | 0.812 | 20.62 | 374.53 | 557.86 | 169.89 | 660                       | 830  | 780  | 970  | 1400    | 1530 | 1730 | 1860 |
|                  |    |        |                | 0.875 | 22.22 | 403.00 | 600.27 | 182.80 | 720                       | 890  | 840  | 1040 | 1500    | 1650 | 1860 | 2000 |
|                  |    |        |                | 0.938 | 23.83 | 431.39 | 642.56 | 195.68 | 770                       | 960  | 900  | 1120 | 1610    | 1770 | 2000 | 2150 |
|                  |    |        |                | 1.000 | 25.40 | 459.24 | 684.04 | 208.31 | 820                       | 1020 | 950  | 1190 | 1720    | 1880 | 2130 | 2290 |
|                  |    |        |                | 1.062 | 26.97 | 487.01 | 725.40 | 220.91 | 870                       | 1090 | 1010 | 1270 | 1820    | 2000 | 2260 | 2430 |
|                  |    |        |                | 1.125 | 28.58 | 515.14 | 767.30 | 233.67 | 920                       | 1150 | 1070 | 1340 | 1930    | 2120 | 2390 | 2580 |
|                  |    |        |                | 1.188 | 30.18 | 543.19 | 809.08 | 246.39 | 970                       | 1210 | 1130 | 1420 | 2040    | 2240 | 2530 | 2720 |
|                  |    |        |                | 1.250 | 31.75 | 570.71 | 850.07 | 258.87 | 1020                      | 1280 | 1190 | 1490 | 2150    | 2350 | 2660 | 2860 |
| 46               | 46 | 1168.4 |                | 0.344 | 8.74  | 167.74 | 249.85 | 76.09  | 270                       | 340  | 310  | 390  | 570     | 620  | 700  | 750  |
|                  |    |        |                | 0.375 | 9.52  | 182.73 | 272.18 | 82.89  | 290                       | 370  | 340  | 430  | 620     | 680  | 760  | 820  |
|                  |    |        |                | 0.406 | 10.31 | 197.70 | 294.47 | 89.68  | 320                       | 400  | 370  | 460  | 670     | 730  | 830  | 890  |
|                  |    |        |                | 0.438 | 11.13 | 213.13 | 317.46 | 96.68  | 340                       | 430  | 400  | 500  | 720     | 790  | 890  | 960  |
|                  |    |        |                | 0.469 | 11.91 | 228.06 | 339.70 | 103.45 | 370                       | 460  | 430  | 540  | 770     | 840  | 950  | 1030 |
|                  |    |        |                | 0.500 | 12.70 | 242.97 | 361.90 | 110.21 | 390                       | 490  | 460  | 570  | 820     | 900  | 1020 | 1100 |
|                  |    |        |                | 0.562 | 14.27 | 272.73 | 406.23 | 123.71 | 440                       | 550  | 510  | 640  | 920     | 1010 | 1140 | 1230 |
|                  |    |        |                | 0.625 | 15.88 | 302.88 | 451.14 | 137.39 | 490                       | 610  | 570  | 710  | 1030    | 1120 | 1270 | 1370 |
|                  |    |        |                | 0.688 | 17.48 | 332.95 | 495.93 | 151.03 | 540                       | 670  | 630  | 790  | 1130    | 1240 | 1400 | 1510 |
|                  |    |        |                | 0.750 | 19.05 | 362.45 | 539.87 | 164.41 | 590                       | 730  | 680  | 860  | 1230    | 1350 | 1530 | 1640 |
|                  |    |        |                | 0.812 | 20.62 | 391.88 | 583.12 | 177.76 | 640                       | 790  | 740  | 930  | 1330    | 1460 | 1650 | 1780 |
|                  |    |        |                | 0.875 | 22.22 | 421.69 | 628.11 | 191.28 | 680                       | 860  | 800  | 1000 | 1440    | 1580 | 1780 | 1920 |
|                  |    |        |                | 0.938 | 23.83 | 451.42 | 672.39 | 204.76 | 730                       | 920  | 860  | 1070 | 1540    | 1690 | 1910 | 2060 |
|                  |    |        |                | 1.000 | 25.40 | 480.60 | 715.85 | 218.00 | 780                       | 980  | 910  | 1140 | 1640    | 1800 | 2030 | 2190 |
|                  |    |        |                | 1.062 | 26.97 | 509.69 | 759.18 | 231.20 | 830                       | 1040 | 970  | 1210 | 1750    | 1910 | 2160 | 2330 |
|                  |    |        |                | 1.125 | 28.58 | 539.17 | 803.09 | 244.57 | 880                       | 1100 | 1030 | 1280 | 1850    | 2020 | 2290 | 2470 |
|                  |    |        |                | 1.188 | 30.18 | 568.57 | 846.89 | 257.90 | 930                       | 1160 | 1080 | 1360 | 1950    | 2140 | 2420 | 2600 |
|                  |    |        |                | 1.250 | 31.75 | 597.41 | 889.84 | 270.99 | 980                       | 1220 | 1140 | 1430 | 2050    | 2250 | 2540 | 2740 |
| 48               | 48 | 1219.2 |                | 0.344 | 8.74  | 175.08 | 260.78 | 79.42  | 260                       | 320  | 300  | 380  | 540     | 590  | 670  | 720  |
|                  |    |        |                | 0.375 | 9.52  | 190.74 | 284.11 | 86.52  | 280                       | 350  | 330  | 410  | 590     | 650  | 730  | 790  |
|                  |    |        |                | 0.406 | 10.31 | 206.37 | 307.39 | 93.61  | 300                       | 380  | 360  | 440  | 640     | 700  | 790  | 850  |
|                  |    |        |                | 0.438 | 11.13 | 222.49 | 331.40 | 100.92 | 330                       | 410  | 380  | 480  | 690     | 760  | 850  | 920  |
|                  |    |        |                | 0.469 | 11.91 | 238.08 | 354.62 | 107.99 | 350                       | 440  | 410  | 510  | 740     | 810  | 910  | 980  |
|                  |    |        |                | 0.500 | 12.70 | 253.65 | 377.81 | 115.06 | 380                       | 470  | 440  | 550  | 790     | 860  | 980  | 1050 |
|                  |    |        |                | 0.562 | 14.27 | 284.73 | 424.11 | 129.15 | 420                       | 530  | 490  | 610  | 890     | 970  | 1100 | 1180 |
|                  |    |        |                | 0.625 | 15.88 | 316.23 | 471.02 | 143.44 | 470                       | 590  | 550  | 680  | 980     | 1080 | 1220 | 1310 |
|                  |    |        |                | 0.688 | 17.48 | 347.64 | 517.81 | 157.69 | 520                       | 640  | 600  | 750  | 1080    | 1190 | 1340 | 1440 |
|                  |    |        |                | 0.750 | 19.05 | 378.47 | 563.73 | 171.67 | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1580 |
|                  |    |        |                | 0.812 | 20.62 | 409.22 | 609.53 | 185.62 | 610                       | 760  | 710  | 890  | 1280    | 1400 | 1580 | 1710 |
|                  |    |        |                | 0.875 | 22.22 | 440.38 | 655.95 | 199.76 | 660                       | 820  | 770  | 960  | 1380    | 1510 | 1710 | 1840 |
|                  |    |        |                | 0.938 | 23.87 | 471.46 | 702.24 | 213.85 | 700                       | 880  | 820  | 1030 | 1480    | 1620 | 1830 | 1970 |
|                  |    |        |                | 1.000 | 25.40 | 501.96 | 747.67 | 227.69 | 750                       | 940  | 880  | 1090 | 1580    | 1720 | 1950 | 2100 |
|                  |    |        |                | 1.062 | 26.97 | 532.38 | 792.98 | 241.49 | 800                       | 1000 | 930  | 1160 | 1670    | 1830 | 2070 | 2230 |
|                  |    |        |                | 1.125 | 28.58 | 563.20 | 838.89 | 255.47 | 840                       | 1050 | 980  | 1230 | 1770    | 1940 | 2190 | 2360 |
|                  |    |        |                | 1.188 | 30.18 | 593.94 | 884.67 | 269.41 | 890                       | 1110 | 1040 | 1300 | 1870    | 2050 | 2320 | 2490 |
|                  |    |        |                | 1.250 | 31.75 | 624.11 | 929.61 | 283.10 | 940                       | 1170 | 1090 | 1370 | 1970    | 2160 | 2440 | 2620 |

# API PIPES ERW & SEAMLESS PIPES

| Size             |    |        |                |       |       | Weight |         |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|----|--------|----------------|-------|-------|--------|---------|--------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |    |        | Wall Thickness |       |       |        |         |        | API 5L                    |      |      |      | API 5LX |      |      |      |
|                  |    |        |                |       |       |        |         |        | A                         |      | B    |      |         |      |      |      |
| Nominal Size     | in | mm     | Sch No         | in    | mm    | lb/ft  | kg/m    | kg/ft  | Std.                      | Alt. | Std. | Alt. | X42     | X46  | X52  | X56  |
| 52               | 52 | 1320.8 |                | 0.375 | 9.52  | 206.76 | 307.97  | 93.79  | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.406 | 10.31 | 223.72 | 333.23  | 101.48 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.438 | 11.13 | 241.20 | 359.27  | 109.41 | 300                       | 380  | 350  | 440  | 640     | 700  | 790  | 850  |
|                  |    |        |                | 0.469 | 11.91 | 258.11 | 384.45  | 117.08 | 320                       | 410  | 380  | 470  | 680     | 750  | 840  | 910  |
|                  |    |        |                | 0.500 | 12.70 | 275.01 | 409.63  | 124.74 | 350                       | 430  | 400  | 500  | 730     | 800  | 900  | 970  |
|                  |    |        |                | 0.562 | 14.27 | 308.74 | 459.87  | 140.04 | 390                       | 490  | 450  | 570  | 820     | 890  | 1010 | 1090 |
|                  |    |        |                | 0.625 | 15.88 | 342.93 | 510.79  | 155.55 | 430                       | 540  | 500  | 630  | 910     | 1000 | 1130 | 1210 |
|                  |    |        |                | 0.688 | 17.48 | 377.03 | 561.59  | 171.02 | 480                       | 600  | 560  | 690  | 1000    | 1100 | 1240 | 1330 |
|                  |    |        |                | 0.750 | 19.05 | 410.51 | 611.45  | 186.21 | 520                       | 650  | 610  | 760  | 1090    | 1190 | 1350 | 1450 |
|                  |    |        |                | 0.812 | 20.62 | 443.91 | 661.20  | 201.36 | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1570 |
|                  |    |        |                | 0.875 | 22.22 | 477.76 | 711.62  | 216.71 | 610                       | 760  | 710  | 880  | 1270    | 1390 | 1580 | 1700 |
|                  |    |        |                | 0.938 | 23.83 | 511.53 | 761.92  | 232.03 | 650                       | 810  | 760  | 950  | 1360    | 1490 | 1690 | 1820 |
|                  |    |        |                | 1.000 | 25.40 | 544.68 | 811.30  | 247.07 | 690                       | 870  | 810  | 1010 | 1450    | 1590 | 1800 | 1940 |
|                  |    |        |                | 1.062 | 26.97 | 577.75 | 860.56  | 262.07 | 740                       | 920  | 860  | 1070 | 1540    | 1690 | 1910 | 2060 |
|                  |    |        |                | 1.125 | 28.58 | 611.26 | 910.47  | 277.27 | 780                       | 970  | 910  | 1140 | 1640    | 1790 | 2030 | 2180 |
|                  |    |        |                | 1.188 | 30.18 | 644.69 | 960.27  | 292.43 | 820                       | 1030 | 960  | 1200 | 1730    | 1890 | 2140 | 2300 |
|                  |    |        |                | 1.250 | 31.75 | 677.51 | 1009.15 | 307.32 | 870                       | 1080 | 1010 | 1260 | 1820    | 1990 | 2250 | 2420 |
| 56               | 56 | 1422.4 |                | 0.375 | 9.52  | 222.78 | 340.77  | 101.05 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.406 | 10.31 | 241.06 | 359.06  | 109.34 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.438 | 11.13 | 259.91 | 387.14  | 117.90 | 280                       | 350  | 330  | 410  | 590     | 650  | 730  | 790  |
|                  |    |        |                | 0.469 | 11.91 | 278.15 | 414.30  | 126.17 | 300                       | 380  | 350  | 440  | 630     | 690  | 780  | 840  |
|                  |    |        |                | 0.500 | 12.70 | 296.37 | 441.44  | 134.43 | 320                       | 400  | 380  | 470  | 680     | 740  | 840  | 900  |
|                  |    |        |                | 0.562 | 14.27 | 332.75 | 495.63  | 150.94 | 360                       | 450  | 420  | 530  | 760     | 830  | 940  | 1010 |
|                  |    |        |                | 0.625 | 15.88 | 369.63 | 550.56  | 167.66 | 400                       | 500  | 470  | 590  | 840     | 920  | 1040 | 1120 |
|                  |    |        |                | 0.688 | 17.48 | 406.42 | 605.36  | 184.35 | 440                       | 550  | 520  | 650  | 930     | 1020 | 1150 | 1240 |
|                  |    |        |                | 0.750 | 19.05 | 442.55 | 659.18  | 200.74 | 480                       | 600  | 560  | 700  | 1010    | 1110 | 1250 | 1350 |
|                  |    |        |                | 0.812 | 20.62 | 478.60 | 712.87  | 217.09 | 520                       | 650  | 610  | 760  | 1100    | 1200 | 1360 | 1460 |
|                  |    |        |                | 0.875 | 22.22 | 515.14 | 767.30  | 233.67 | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1570 |
|                  |    |        |                | 0.938 | 23.83 | 551.60 | 821.61  | 250.21 | 600                       | 750  | 700  | 880  | 1270    | 1390 | 1570 | 1690 |
|                  |    |        |                | 1.000 | 25.40 | 587.40 | 874.93  | 266.44 | 640                       | 800  | 750  | 940  | 1350    | 1480 | 1670 | 1800 |
|                  |    |        |                | 1.062 | 26.97 | 623.12 | 928.14  | 282.65 | 680                       | 850  | 800  | 1000 | 1430    | 1570 | 1780 | 1910 |
|                  |    |        |                | 1.125 | 28.58 | 659.32 | 982.06  | 299.07 | 720                       | 900  | 840  | 1050 | 1520    | 1660 | 1880 | 2020 |
|                  |    |        |                | 1.188 | 30.18 | 695.45 | 1035.87 | 315.46 | 760                       | 950  | 890  | 1110 | 1600    | 1760 | 1990 | 2140 |
|                  |    |        |                | 1.250 | 31.75 | 730.91 | 1088.69 | 331.54 | 800                       | 1000 | 940  | 1170 | 1690    | 1850 | 2090 | 2250 |
| 60               | 60 | 1524.0 |                | 0.375 | 9.52  | 238.80 | 355.69  | 108.32 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.406 | 10.31 | 258.40 | 384.89  | 117.21 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.438 | 11.13 | 278.62 | 415.00  | 126.38 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.469 | 11.91 | 298.19 | 444.15  | 135.26 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.500 | 12.70 | 317.73 | 473.26  | 144.12 | 300                       | 380  | 350  | 440  | 630     | 690  | 780  | 840  |
|                  |    |        |                | 0.562 | 14.27 | 356.76 | 531.39  | 161.83 | 340                       | 420  | 390  | 490  | 710     | 780  | 880  | 940  |
|                  |    |        |                | 0.625 | 15.88 | 396.33 | 590.33  | 179.78 | 380                       | 470  | 440  | 550  | 790     | 860  | 980  | 1050 |
|                  |    |        |                | 0.688 | 17.48 | 435.71 | 649.14  | 197.68 | 410                       | 520  | 480  | 600  | 870     | 950  | 1070 | 1160 |
|                  |    |        |                | 0.750 | 19.05 | 474.59 | 706.90  | 215.27 | 450                       | 560  | 530  | 660  | 950     | 1030 | 1170 | 1260 |
|                  |    |        |                | 0.812 | 20.62 | 513.29 | 764.55  | 232.83 | 490                       | 610  | 570  | 710  | 1020    | 1120 | 1270 | 1360 |
|                  |    |        |                | 0.875 | 22.22 | 552.52 | 822.98  | 250.62 | 530                       | 660  | 610  | 770  | 1100    | 1210 | 1370 | 1470 |
|                  |    |        |                | 0.938 | 23.83 | 591.67 | 881.29  | 268.38 | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1580 |
|                  |    |        |                | 1.000 | 25.40 | 630.12 | 938.56  | 285.82 | 600                       | 750  | 700  | 880  | 1260    | 1380 | 1560 | 1680 |
|                  |    |        |                | 1.062 | 26.97 | 668.48 | 995.70  | 303.22 | 640                       | 800  | 740  | 930  | 1340    | 1470 | 1660 | 1780 |
|                  |    |        |                | 1.125 | 28.58 | 707.38 | 1053.64 | 320.87 | 680                       | 840  | 790  | 980  | 1420    | 1550 | 1760 | 1890 |
|                  |    |        |                | 1.188 | 30.18 | 746.20 | 1111.46 | 338.48 | 710                       | 890  | 830  | 1040 | 1500    | 1640 | 1850 | 2000 |
|                  |    |        |                | 1.250 | 31.75 | 784.31 | 1168.23 | 355.76 | 750                       | 940  | 880  | 1090 | 1580    | 1720 | 1950 | 2100 |

# API PIPES ERW & SEAMLESS PIPES

| Size             |    |        |                |       |       | Weight |         |        | Hydrostatic Test Pressure |      |      |      |         |      |      |      |
|------------------|----|--------|----------------|-------|-------|--------|---------|--------|---------------------------|------|------|------|---------|------|------|------|
| Outside Diameter |    |        | Wall Thickness |       |       |        |         |        | API 5L                    |      |      |      | API 5LX |      |      |      |
|                  |    |        |                |       |       |        |         |        | A                         |      | B    |      |         |      |      |      |
| Nominal<br>Size  | in | mm     | Sch No         | in    | mm    | lb/ft  | kg/m    | kg/ft  | Std.                      | Alt. | Std. | Alt. | X42     | X46  | X52  | X56  |
| 64               | 64 | 1625.6 |                | 0.375 | 9.52  | 254.82 | 379.55  | 115.59 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.406 | 10.31 | 275.75 | 410.73  | 125.08 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.438 | 11.13 | 297.33 | 442.87  | 134.87 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.469 | 11.91 | 318.22 | 473.99  | 144.34 | --                        | --   | --   | --   | --      | --   | --   | --   |
|                  |    |        |                | 0.500 | 12.70 | 339.09 | 505.07  | 153.81 | 280                       | 350  | 330  | 410  | 590     | 650  | 730  | 790  |
|                  |    |        |                | 0.562 | 14.27 | 380.76 | 567.14  | 172.71 | 320                       | 400  | 370  | 460  | 660     | 730  | 820  | 890  |
|                  |    |        |                | 0.625 | 15.88 | 423.03 | 630.10  | 191.89 | 350                       | 440  | 410  | 510  | 740     | 810  | 910  | 980  |
|                  |    |        |                | 0.688 | 17.48 | 465.21 | 692.93  | 211.02 | 390                       | 480  | 450  | 560  | 810     | 890  | 1010 | 1080 |
|                  |    |        |                | 0.750 | 19.05 | 506.63 | 754.63  | 229.81 | 420                       | 530  | 490  | 620  | 890     | 970  | 1100 | 1180 |
|                  |    |        |                | 0.812 | 20.62 | 547.98 | 816.22  | 248.56 | 460                       | 570  | 530  | 670  | 960     | 1050 | 1190 | 1280 |
|                  |    |        |                | 0.875 | 22.22 | 589.90 | 878.66  | 267.58 | 490                       | 620  | 570  | 720  | 1030    | 1130 | 1280 | 1380 |
|                  |    |        |                | 0.938 | 23.87 | 631.74 | 940.98  | 286.56 | 530                       | 660  | 620  | 770  | 1110    | 1210 | 1370 | 1480 |
|                  |    |        |                | 1.000 | 25.40 | 672.84 | 1002.20 | 305.20 | 560                       | 700  | 660  | 820  | 1180    | 1290 | 1460 | 1570 |
|                  |    |        |                | 1.062 | 26.97 | 713.85 | 1063.28 | 323.80 | 600                       | 750  | 700  | 870  | 1250    | 1370 | 1550 | 1670 |
|                  |    |        |                | 1.125 | 28.58 | 755.44 | 1125.22 | 342.67 | 630                       | 790  | 740  | 920  | 1330    | 1460 | 1650 | 1770 |
|                  |    |        |                | 1.188 | 30.18 | 796.95 | 1187.06 | 361.50 | 670                       | 840  | 780  | 970  | 1400    | 1540 | 1740 | 1870 |
|                  |    |        |                | 1.250 | 31.75 | 837.71 | 1247.77 | 379.99 | 700                       | 880  | 820  | 1030 | 1480    | 1620 | 1830 | 1970 |





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YICK HOE

# COLD FORMED HOLLOW SECTIONS

## STANDARD SPECIFICATIONS FOR WELDED CIRCULAR & NON-CIRCULAR STEEL TUBES

| Standard Specification | Scope                                                                                | Grade of Tubes    | Chemical Composition Percentage<br>(max. unless range is given) |         |      |      | Mechanical Properties         |                            |                          |                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------|---------|------|------|-------------------------------|----------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|
|                        |                                                                                      |                   | C                                                               | Mn      | P    | S    | Condition                     | Tensile Strength<br>(min.) | Yield Strength<br>(min.) | Elongation<br>(% min.)                                                                                                             |
| 1                      | 2                                                                                    | 3                 | 4                                                               | 5       | 6    | 7    | 8                             | 9                          | 10                       | 11                                                                                                                                 |
| ASTM-A53               | E.R.W. Carbon Steel Pipes                                                            | Type E Grade A    | 0.25                                                            | 0.95    | 0.05 | 0.06 | As Rolled                     | ksi (MPa)                  | ksi (MPa)                | (G.L. = 2" or 50mm)                                                                                                                |
|                        |                                                                                      |                   |                                                                 |         |      |      |                               | 48 (331)                   | 30 (207)                 | As per Table X7                                                                                                                    |
| ASTM-A120              | E.R.W. Carbon Steel Pipes for ordinary use                                           | Standard Pipe     | --                                                              | --      | --   | --   | --                            | --                         | --                       | --                                                                                                                                 |
| ASTM-A135              | Standard Specification for ERW Steel Pipe                                            | Grade A           | 0.25                                                            | 0.95    | 0.05 | 0.06 | As Rolled                     | 48 (331)                   | 30 (207)                 | As per Table 1<br>Equation: (Elongation)<br>= 56 x (Actual Thickness of Specimen, inch) + 17.50                                    |
| ASTM-A500              | E.R.W. Carbon Steel Structural Tubing in Round & Shapes                              | Round Grade A     | 0.26                                                            | --      | 0.04 | 0.05 | As Rolled                     | 45 (310)                   | 33 (228)                 | 25                                                                                                                                 |
|                        |                                                                                      | Grade B           | 0.26                                                            | --      | 0.04 | 0.05 |                               | 58 (400)                   | 42 (290)                 | 23                                                                                                                                 |
|                        |                                                                                      | Shaped Grade A    | 0.26                                                            | --      | 0.04 | 0.05 |                               | 45 (310)                   | 39 (269)                 | 25                                                                                                                                 |
|                        |                                                                                      | Grade B           | 0.26                                                            | --      | 0.04 | 0.05 |                               | 58 (400)                   | 46 (317)                 | 23                                                                                                                                 |
| ASTM-A513              | E.R.W. Carbon Steel Mechanical Tubing                                                | MT 1010           | 0.5-0.15                                                        | 0.3-0.6 | 0.04 | 0.05 | As Rolled                     | 45 (310)                   | 32 (221)                 | 15                                                                                                                                 |
|                        |                                                                                      | MT 1015           | 0.1-0.2                                                         | 0.3-0.6 | 0.04 | 0.05 |                               | 48 (331)                   | 35 (241)                 | 15                                                                                                                                 |
|                        |                                                                                      | MT 1020           | 0.15-0.25                                                       | 0.3-0.6 | 0.04 | 0.05 |                               | 52 (359)                   | 38 (262)                 | 12                                                                                                                                 |
| BS-31                  | E.R.W. Steel Conduits                                                                | Class A Plain End | --                                                              | --      | --   | --   | As Rolled & Coated G.I. Strip | tonf/in <sup>2</sup>       | tonf/in <sup>2</sup>     | 15 (G.L. = 8")                                                                                                                     |
|                        |                                                                                      |                   | --                                                              | --      | --   | --   |                               |                            |                          |                                                                                                                                    |
| BS-980                 | E.R.W. Carbon Steel Pipes for Automobile Purposes                                    | ERW 1             | 0.20                                                            | 0.60    | 0.06 | 0.06 | As Rolled                     | 20                         | 11                       | 20                                                                                                                                 |
|                        |                                                                                      | ERW 2             | 0.30                                                            | 0.60    | 0.06 | 0.06 |                               | 25                         | 15                       | 15                                                                                                                                 |
|                        |                                                                                      | ERW 3             | 0.40                                                            | 0.60    | 0.06 | 0.06 |                               | 30                         | 20                       | 10<br>(G.L. = $4\sqrt{S_o}$ )                                                                                                      |
| BS-1139                | E.R.W. Metal Scaffolding                                                             |                   | --                                                              | --      | 0.06 | 0.06 | As Rolled (Black Tubes)       | 20 - 30                    | 13.5                     | $700/T.S.$ (ton/in <sup>2</sup> )<br>(G.L. = $4\sqrt{S_o}$ )                                                                       |
| BS-1387                | E.R.W. Carbon Steel Tubes                                                            | Light             | --                                                              | --      | 0.06 | 0.06 | As Rolled                     | 33 - 47                    | --                       | 20<br>(G.L. = $5.65\sqrt{S_o}$ )                                                                                                   |
|                        |                                                                                      | Medium            | --                                                              | --      | --   | --   |                               |                            |                          |                                                                                                                                    |
|                        |                                                                                      | Heavy             | --                                                              | --      | --   | --   |                               |                            |                          |                                                                                                                                    |
| BS-1717                | E.R.W. Carbon Steel Tubes for Bicycle and Motorcycle Purposes                        | ERW 101           | 0.10                                                            | 0.60    | 0.06 | 0.06 | As Rolled                     | 20                         | 11                       | --                                                                                                                                 |
|                        |                                                                                      | ERW 102           | 0.20                                                            | 0.60    | 0.06 | 0.06 |                               | 25                         | 15                       | --                                                                                                                                 |
|                        |                                                                                      | ERW 103           | 0.30                                                            | 0.60    | 0.06 | 0.06 |                               | 30                         | 20                       | --                                                                                                                                 |
| BS-1775                | E.R.W. Carbon Steel Tubes for Mechanical Structural and General Engineering Purposes | ERW 11            | --                                                              | --      | 0.06 | 0.06 | As Rolled                     | 20                         | 11                       | $700/T.S.$ (ton/in <sup>2</sup> )<br>(G.L. = $4\sqrt{S_o}$ )<br>or $600/T.S.$ (ton/in <sup>2</sup> )<br>(G.L. = $5.65\sqrt{S_o}$ ) |
|                        |                                                                                      | ERW 16            | --                                                              | --      | 0.06 | 0.06 |                               | 25                         | 16                       |                                                                                                                                    |
|                        |                                                                                      | ERW 20            | --                                                              | --      | 0.06 | 0.06 |                               | 30                         | 20                       |                                                                                                                                    |
|                        |                                                                                      | ERW 23            | --                                                              | --      | 0.06 | 0.06 |                               | 32                         | 23                       |                                                                                                                                    |

\*1: G.L. = Gauge Length

\*2:  $S_o$  = Original Cross-Section Area

\*3: T.S. = Tensile Strength

\*4: T.P. = Test Piece as per JIS Z2201

\*5: H = Distance between Outside Surfaces

\*6: D = Outside Diameter

\*7: t = Wall Thickness of Tubes

\*8: H' = Distance between Inside Surfaces

# COLD FORMED HOLLOW SECTIONS

| Other Tests                                                                                        |                                                                                      |                                                                                                                                                                            | Dimensional Tolerances                                                                                                                                                                                                   |                                                                                                                       |                                                |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Flattening<br>(min.)                                                                               | Bend Test                                                                            | Hydrostatic<br>Test                                                                                                                                                        | On Diameter except for<br>Non-Circular Tubes                                                                                                                                                                             |                                                                                                                       | On Wall<br>Thickness                           |
| 12                                                                                                 | 13                                                                                   | 14                                                                                                                                                                         | 15                                                                                                                                                                                                                       |                                                                                                                       | 16                                             |
| Apply for STD &<br>XS of NB>2"<br>Welded Portion: H= 2/3D<br>The Other Side: H=1/3D                | Apply fo NB ≤2"<br>12D at 90°<br>When ordered for Close<br>coiling: 8D at 180°       | As per Table X2                                                                                                                                                            | 1-1/2" & under                                                                                                                                                                                                           | + 1/64"<br>- 1/32"                                                                                                    | + not limited<br>-12.5%                        |
| --                                                                                                 | --                                                                                   | 1/2" to 1": P=700 psi<br>1-1/4" to 3": P= 1,000 psi<br>3-1/2" to 6": P=1,200 psi                                                                                           | 2" & over                                                                                                                                                                                                                | ± 1 % of OD                                                                                                           |                                                |
|                                                                                                    |                                                                                      |                                                                                                                                                                            | 1-1/2" & under                                                                                                                                                                                                           | + 1/64"<br>- 1/32"                                                                                                    | + not limited<br>-12.5%                        |
|                                                                                                    |                                                                                      |                                                                                                                                                                            | 2" & over                                                                                                                                                                                                                | ± 1% of OD                                                                                                            |                                                |
| Welded Portion:<br>H = 2/3D<br><br>The Other Side:<br>H = 1/3D                                     | --                                                                                   | P=2St/D, Max. 2,500 psi<br>S: Allowable Fiber Stress<br>(16,000-18,000 psi) for<br>Grade A. But in no case shall<br>the stress produced exceed<br>80% of the specified Y.P | ± 1% of OD                                                                                                                                                                                                               |                                                                                                                       | + not limited<br>-12.5%                        |
| Welded Portion:<br>H = 2/3D<br><br>The Other Side:<br>H = 1/2D                                     | --                                                                                   | --                                                                                                                                                                         | Up to 1.90"<br>2" & over<br>Under 2-1/2"<br>2-1/2" to 3-1/2" excluding<br>3-1/2" to 5-1/2" excluding<br>5-1/2" & over<br>Square of Radii                                                                                 | ± 0.5%<br>± 0.75%<br>± 0.02"<br>± 0.025"<br>± 0.03"<br>± 1%<br>± 2% max.                                              | ± 10%                                          |
| Welded Portion:<br>H = 2/3D<br><br>The Other Side:<br>H = 1/3D                                     |                                                                                      | P=2St/D<br>S: Allowable Fiber Stress<br>of 14,000 psi                                                                                                                      | As per Table 4 & 7                                                                                                                                                                                                       |                                                                                                                       | As per Table<br>6 & 8                          |
| --                                                                                                 | 6D at 90°                                                                            | --                                                                                                                                                                         | + 0.001"<br>- 0.005"                                                                                                                                                                                                     |                                                                                                                       | Nominal 0.04" - 0.072"<br>Min. 0.036" - 0.068" |
| Max. 3t<br>Max. 5t<br>Max. 8t                                                                      | --                                                                                   | --                                                                                                                                                                         | 1" & under<br>Over 1" to 2"<br>Over 2" to 2-1/2"<br>Over 2-1/2" to 3"<br>Over 3" to 3-1/2"<br>Over 3-1/2" to 4"<br>Over 4" to 4-1/2"                                                                                     | ±0.004"<br>±0.005"<br>±0.007"<br>±0.008"<br>±0.010"<br>±0.012"<br>±0.015"                                             | ±8%                                            |
| --                                                                                                 | 6D at 180°                                                                           | --                                                                                                                                                                         | Only 1.906"<br>±0.013"<br>-0.017"                                                                                                                                                                                        |                                                                                                                       | + 20%<br>- 10%                                 |
| Apply for NB>2"<br>Welded Portion: H=0.75D<br>The Other Side: H=0.6D                               | Apply for NB≤2"<br>Ungalvanized Tubes:<br>6D at 180°; Galvanized<br>Tubes: 8D at 90° | P=700 lbf/in <sup>2</sup><br>(50 kgf/cm <sup>2</sup> )                                                                                                                     | Light Tubes:                                                                                                                                                                                                             | Table 1                                                                                                               | Light Tubes:<br>-8%; + not limited             |
|                                                                                                    |                                                                                      |                                                                                                                                                                            | Medium, Heavy Tubes:                                                                                                                                                                                                     | Table 2                                                                                                               | Medium, Heavy Tubes:<br>-10%; + not limited    |
| Max. 3t<br>Max. 5t<br>Max. 8t                                                                      | --                                                                                   | --                                                                                                                                                                         | 1-1/4" & under<br><br>Over 1-1/4" to 1-5/8"                                                                                                                                                                              | + 0.001"<br>-0.005"<br>+0.002"<br>- 0.006"                                                                            | ±8%                                            |
| H'= 3t or 1/2D<br>H'= 6t or 3/4D<br>H'= 8t or 7/8D<br>H'= 6t or 3/4D<br>(whichever is the smaller) | --                                                                                   | --                                                                                                                                                                         | 1" & under<br>over 1" to 2"<br>over 2" to 2-1/2"<br>over 2-1/2" to 3"<br>over 3" to 3-1/2"<br>over 3-1/2" to 4"<br>over 4" to 4-1/2"<br>over 4-1/2" to 5"<br>over 5" to 5-1/2"<br>over 5-1/2" to 6"<br>over 6" to 6-5/8" | ±0.006"<br>±0.007"<br>±0.009"<br>±0.010"<br>±0.012"<br>±0.014"<br>±0.016"<br>±0.018"<br>±0.020"<br>±0.022"<br>±0.025" | ±8%                                            |

\*9: P = Test Pressure

\*10 t<sup>1</sup> = Thickness of Flat Portion



# COLD FORMED HOLLOW SECTIONS

| Standard Specification | Scope                                                                                                                 | Grade of Tubes                                                                                           | Chemical Composition Percentage<br>(max. unless range is given) |                                                          |                                          |                                          | Mechanical Properties      |                                                    |                                                    |                  |              |
|------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------|------------------------------------------|------------------------------------------|----------------------------|----------------------------------------------------|----------------------------------------------------|------------------|--------------|
|                        |                                                                                                                       |                                                                                                          | C                                                               | Mn                                                       | P                                        | S                                        | Condition<br><br>(min.)    | Tensile Strength<br>(min.)                         | Yield Strength<br>(% min.)                         | Elongation       |              |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    |                  |              |
| 1                      | 2                                                                                                                     | 3                                                                                                        | 4                                                               | 5                                                        | 6                                        | 7                                        | 8                          | 9                                                  | 10                                                 | 11               |              |
| JIS-G 3444             | E.R.W. Carbon Steel Pipes for Civil Engineer Architecture, Steel Towers, Scaffolding, Pile Struts & other Structures. | STK - 30<br>STK - 41<br>STK - 51<br>STK - 50                                                             | --<br>0.25<br>0.30<br>0.18                                      | --<br>--<br>0.3 - 1.0<br>1.50                            | 0.05<br>0.04<br>0.04<br>0.04             | 0.05<br>0.04<br>0.04<br>0.04             | As Rolled                  | kgf/mm²                                            | kgf/mm²                                            | longitudinal     | transversal  |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | No. 11,12 T.P.   | No. 5 T.P.   |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            | 30                                                 | —                                                  | 30               | 25           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            | 41                                                 | 24                                                 | 23               | 18           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            | 51                                                 | 36                                                 | 15               | 10           |
|                        |                                                                                                                       |                                                                                                          | 0.18                                                            | 1.50                                                     | 0.04                                     | 0.04                                     |                            | 50                                                 | 32                                                 | 23               | 18           |
| JIS-G 3445             | E.R.W. Carbon Steel Pipes for Machine Structural Purposes                                                             | STKM-11A<br>STKM-12A<br>STKM-12B<br>STKM-12C<br>STKM-13A<br>STKM-13B<br>STKM-13C<br>STKM-14A<br>STKM-14B | 0.12<br>0.20<br><br><br>0.25<br><br>0.30                        | 0.25-0.6<br>0.25-0.6<br><br><br>0.30-0.9<br><br>0.30-1.0 | 0.04<br>0.04<br><br><br>0.04<br><br>0.04 | 0.04<br>0.04<br><br><br>0.04<br><br>0.04 | As Rolled                  | 30<br>35<br>40<br>48<br>38<br>45<br>52<br>42<br>51 | --<br>18<br>28<br>36<br>22<br>31<br>39<br>25<br>36 | No. 4,11,12 T.P. | No. 4,5 T.P. |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 35               | 30           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 35               | 30           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 25               | 20           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 20               | 15           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 30               | 25           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 20               | 15           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 15               | 10           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 25               | 20           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 15               | 10           |
| JIS-G 3452             | E.R.W. Carbon Steel Pipes for ordinary use                                                                            | SGP                                                                                                      | --                                                              | --                                                       | 0.05                                     | 0.05                                     | As Rolled<br>(Black Tubes) | 30                                                 | --                                                 | No. 11, 12 T.P.  | No. 5 T.P.   |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 30               | 25           |
| JIS-G 3466             | E.R.W. Carbon Steel Square Tubes for General Structural Purposes                                                      | STKR-41<br>STKR-50                                                                                       | 0.25<br>0.18                                                    | --<br>1.50                                               | 0.04<br>0.04                             | 0.04<br>0.04                             | As Rolled                  | 41<br>50                                           | 25<br>33                                           | No. 5 T.P.       | --           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 23               | --           |
|                        |                                                                                                                       |                                                                                                          |                                                                 |                                                          |                                          |                                          |                            |                                                    |                                                    | 23               | --           |

\*1: G.L. = Gauge Length  
 \*2: S<sub>o</sub> = Original Cross-Section Area  
 \*3: T.S. = Tensile Strength

\*4: T.P. = Test Piece as per JIS Z2201  
 \*5: H = Distance between Outside Surfaces

\*6: D = Outside Diameter  
 \*7: t = Wall Thickness of Tubes  
 \*8: H' = Distance between Inside Surfaces

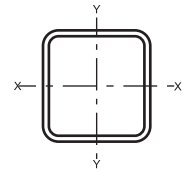
# COLD FORMED HOLLOW SECTIONS

| Other Tests                                                                                  |                                                                                                      |                          | Dimensional Tolerances                                                                                                              |                                                           |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Flattening<br>(min.)                                                                         | Bend Test                                                                                            | Hydrostatic<br>Test      | On Diameter except for<br>Non-Circular Tubes                                                                                        | On Wall<br>Thickness                                      |
| 12                                                                                           | 13                                                                                                   | 14                       | 15                                                                                                                                  | 16                                                        |
| H = 2/3D<br>H = 2/3D<br>H = 7/8D<br>H = 7/8D                                                 | 6D at 90°<br>6D at 90°<br>8D at 90°<br>6D at 90°                                                     | --<br>--<br>--           | Under 50 mm      ± 0.25 mm<br>50 mm & over      ± 0.50 %                                                                            | Under 3mm    ±0.30 mm<br>3mm - 12 mm   ± 10%<br>excluding |
| H = 1/2D<br>H = 2/3D<br>H = 2/3D<br>--<br>H = 2/3D<br>H = 3/4D<br>--<br>H = 3/4D<br>H = 7/8D | 4D at 180°<br>6D at 90°<br>6D at 90°<br>--<br>6D at 90°<br>6D at 90°<br>--<br>6D at 90°<br>8D at 90° | --                       | As per Classification (No. 1, 2, 3) of<br>Table 4                                                                                   | As per Classification<br>(No. 1,2,3) of Table 5           |
| H = 2/3D                                                                                     | 6D at 90°                                                                                            | P=25 kgf/mm <sup>2</sup> | 10.5 to 48.6 mm      ± 0.50mm<br>60.5 to 165. 2 mm      ± 1%                                                                        | + not limited<br>-12.5%                                   |
| --                                                                                           | --                                                                                                   | --                       | Up to 100 mm      ± 1.5mm<br>Over 100 mm      ± 1.5%<br>Squareness of side      ± 1.5°<br>Radii of corner      Max. 3t <sup>1</sup> | Up to 3mm    ±0.30 mm<br>3mm & over    ± 10%              |

\*9: P = Test Pressure

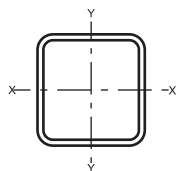
\*10 t<sup>1</sup> = Thickness of Flat Portion

# COLD FORMED HOLLOW SECTIONS



## SQUARE

| Section size | Wall thickness<br>t | Unit weight<br>M | Section area<br>A | Moment of inertia |                 | Radius of gyration |                | Modulus of section |                 |
|--------------|---------------------|------------------|-------------------|-------------------|-----------------|--------------------|----------------|--------------------|-----------------|
|              |                     |                  |                   | I <sub>x</sub>    | I <sub>y</sub>  | i <sub>x</sub>     | i <sub>y</sub> | Z <sub>x</sub>     | Z <sub>y</sub>  |
| mm           | mm                  | kg/m             | cm <sup>2</sup>   | cm <sup>4</sup>   | cm <sup>4</sup> | cm                 | cm             | cm <sup>3</sup>    | cm <sup>3</sup> |
| 13 x 13      | 1.2                 | 0.40             | 0.51              | 0.10              | 0.10            | 0.45               | 0.45           | 0.16               | 0.16            |
|              | 1.6                 | 0.50             | 0.64              | 0.12              | 0.12            | 0.43               | 0.43           | 0.19               | 0.19            |
| 16 x 16      | 1.2                 | 0.52             | 0.67              | 0.23              | 0.23            | 0.59               | 0.59           | 0.29               | 0.29            |
|              | 1.6                 | 0.67             | 0.85              | 0.27              | 0.27            | 0.57               | 0.57           | 0.34               | 0.34            |
| 19 x 19      | 1.2                 | 0.64             | 0.81              | 0.41              | 0.41            | 0.71               | 0.71           | 0.44               | 0.44            |
|              | 1.6                 | 0.82             | 1.04              | 0.50              | 0.50            | 0.69               | 0.69           | 0.53               | 0.53            |
| 25 x 25      | 1.2                 | 0.86             | 1.10              | 1.02              | 1.02            | 0.96               | 0.96           | 0.81               | 0.81            |
|              | 1.6                 | 1.12             | 1.43              | 1.26              | 1.26            | 0.94               | 0.94           | 1.01               | 1.01            |
|              | 2.3                 | 1.53             | 1.95              | 1.59              | 1.59            | 0.90               | 0.90           | 1.27               | 1.27            |
|              | 3.0                 | 1.93             | 2.48              | 1.96              | 1.96            | 0.89               | 0.89           | 1.54               | 1.54            |
| 32 x 32      | 1.2                 | 1.12             | 1.42              | 2.16              | 2.16            | 1.24               | 1.24           | 1.38               | 1.38            |
|              | 1.6                 | 1.46             | 1.87              | 2.75              | 2.75            | 1.22               | 1.22           | 1.74               | 1.74            |
|              | 2.3                 | 2.02             | 2.68              | 3.93              | 3.93            | 1.21               | 1.21           | 2.46               | 2.46            |
|              | 3.0                 | 2.53             | 3.26              | 4.29              | 4.29            | 1.15               | 1.15           | 2.70               | 2.70            |
| 38 x 38      | 1.2                 | 1.35             | 1.72              | 3.85              | 3.85            | 1.49               | 1.49           | 2.02               | 2.02            |
|              | 1.6                 | 1.77             | 2.26              | 4.91              | 4.91            | 1.47               | 1.47           | 2.58               | 2.58            |
|              | 2.3                 | 2.47             | 3.14              | 6.51              | 6.51            | 1.43               | 1.43           | 3.42               | 3.42            |
|              | 3.0                 | 3.13             | 4.04              | 7.99              | 7.99            | 1.41               | 1.41           | 4.26               | 4.26            |
| 50 x 50      | 1.6                 | 2.38             | 3.03              | 11.70             | 11.70           | 1.96               | 1.96           | 4.68               | 4.68            |
|              | 2.3                 | 3.34             | 4.25              | 15.90             | 15.90           | 1.93               | 1.93           | 6.34               | 6.34            |
|              | 3.0                 | 4.24             | 5.40              | 19.39             | 19.39           | 1.89               | 1.89           | 7.75               | 7.75            |
|              | 3.2                 | 4.50             | 5.73              | 20.40             | 20.40           | 1.89               | 1.89           | 8.16               | 8.16            |
|              | 4.5                 | 6.01             | 7.66              | 25.22             | 25.22           | 1.81               | 1.81           | 10.09              | 10.09           |
|              | 6.0                 | 7.56             | 9.63              | 28.76             | 28.76           | 1.72               | 1.72           | 11.50              | 11.50           |
| 60 x 60      | 1.6                 | 2.88             | 3.67              | 20.70             | 20.70           | 2.37               | 2.37           | 6.89               | 6.89            |
|              | 2.3                 | 4.06             | 5.17              | 28.30             | 28.30           | 2.34               | 2.34           | 9.44               | 9.44            |
|              | 3.0                 | 5.18             | 6.60              | 35.04             | 35.04           | 2.30               | 2.30           | 11.68              | 11.68           |
|              | 3.2                 | 5.50             | 7.00              | 36.90             | 36.90           | 2.30               | 2.30           | 12.30              | 12.30           |
|              | 4.5                 | 7.43             | 9.46              | 46.88             | 46.88           | 2.22               | 2.22           | 15.62              | 15.62           |
|              | 6.0                 | 9.44             | 12.03             | 55.26             | 55.26           | 2.14               | 2.14           | 18.42              | 18.42           |
| 63.5 x 63.5  | 2.3                 | 4.31             | 5.51              | 33.88             | 33.88           | 2.46               | 2.46           | 10.75              | 10.75           |
|              | 3.0                 | 5.52             | 7.16              | 42.46             | 42.46           | 2.44               | 2.44           | 13.44              | 13.44           |
|              | 4.5                 | 8.04             | 10.24             | 57.44             | 57.44           | 2.37               | 2.37           | 18.19              | 18.19           |
|              | 6.0                 | 10.57            | 13.48             | 70.34             | 70.34           | 2.28               | 2.28           | 22.12              | 22.12           |
| 75 x 75      | 2.3                 | 5.14             | 6.55              | 57.10             | 57.10           | 2.95               | 2.95           | 15.20              | 15.20           |
|              | 3.0                 | 6.60             | 8.40              | 71.50             | 71.50           | 2.91               | 2.91           | 19.06              | 19.06           |
|              | 3.2                 | 7.01             | 8.92              | 75.50             | 75.50           | 2.91               | 2.91           | 20.10              | 20.10           |
|              | 4.5                 | 9.55             | 12.16             | 98.16             | 98.16           | 2.84               | 2.84           | 26.17              | 26.17           |
|              | 6.0                 | 12.27            | 15.63             | 119.2             | 119.2           | 2.76               | 2.76           | 31.78              | 31.78           |

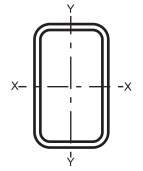


# COLD FORMED HOLLOW SECTIONS

## SQUARE

| Section size  | Wall thickness<br>t | Unit weight<br>M | Section area<br>A | Moment of inertia |                 | Radius of gyration |                | Modulus of section |                 |
|---------------|---------------------|------------------|-------------------|-------------------|-----------------|--------------------|----------------|--------------------|-----------------|
|               |                     |                  |                   | I <sub>x</sub>    | I <sub>y</sub>  | i <sub>x</sub>     | i <sub>y</sub> | Z <sub>x</sub>     | Z <sub>y</sub>  |
| mm            | mm                  | kg/m             | cm <sup>2</sup>   | cm <sup>4</sup>   | cm <sup>4</sup> | cm                 | cm             | cm <sup>3</sup>    | cm <sup>3</sup> |
| 100 x 100     | 2.3                 | 6.95             | 8.85              | 140.0             | 140             | 3.97               | 3.97           | 27.90              | 27.9            |
|               | 3.0                 | 8.95             | 11.40             | 176.9             | 176.9           | 3.93               | 3.93           | 35.38              | 35.38           |
|               | 3.2                 | 9.52             | 12.13             | 187.0             | 187             | 3.93               | 3.93           | 37.50              | 37.5            |
|               | 4.5                 | 13.1             | 16.67             | 249.0             | 249             | 3.87               | 3.87           | 49.90              | 49.9            |
|               | 6.0                 | 17.0             | 21.63             | 311.0             | 311             | 3.79               | 3.79           | 62.30              | 62.3            |
|               | 9.0                 | 24.1             | 30.67             | 408.0             | 408             | 3.65               | 3.65           | 81.60              | 81.6            |
| 112.5 x 112.5 | 2.3                 | 7.85             | 10.00             | 200.97            | 200.97          | 4.48               | 4.48           | 35.72              | 35.72           |
|               | 3.0                 | 10.13            | 12.90             | 255.43            | 255.43          | 4.44               | 4.44           | 45.41              | 45.41           |
|               | 4.5                 | 14.85            | 18.91             | 362.01            | 362.01          | 4.37               | 4.37           | 64.35              | 64.35           |
|               | 6.0                 | 19.33            | 24.63             | 455.21            | 455.21          | 4.29               | 4.29           | 80.92              | 80.92           |
| 125 x 125     | 2.3                 | 8.75             | 11.15             | 278.02            | 278.02          | 4.99               | 4.99           | 44.48              | 44.48           |
|               | 3.0                 | 11.31            | 14.40             | 354.32            | 354.32          | 4.95               | 4.95           | 56.69              | 56.69           |
|               | 3.2                 | 12.0             | 15.33             | 376               | 376             | 4.95               | 4.95           | 60.10              | 60.10           |
|               | 4.5                 | 16.61            | 21.17             | 506               | 506             | 4.89               | 4.89           | 80.90              | 80.90           |
|               | 6.0                 | 21.7             | 27.63             | 641               | 641             | 4.82               | 4.82           | 103                | 103             |
|               | 9.0                 | 31.1             | 39.67             | 865               | 865             | 4.67               | 4.67           | 138                | 138             |
| 150 x 150     | 4.5                 | 20.1             | 25.67             | 896               | 896             | 5.91               | 5.91           | 120                | 120             |
|               | 6.0                 | 26.4             | 33.63             | 1,150             | 1,150           | 5.84               | 5.84           | 153                | 153             |
|               | 9.0                 | 38.2             | 48.67             | 1,580             | 1,580           | 5.69               | 5.69           | 210                | 210             |
| 175 x 175     | 6.0                 | 31.1             | 39.63             | 1,860             | 1,860           | 6.86               | 6.86           | 213                | 213             |
|               | 9.0                 | 45.3             | 57.67             | 2,600             | 2,600           | 6.71               | 6.71           | 297                | 297             |
| 200 x 200     | 6.0                 | 35.8             | 45.63             | 2,830             | 2,830           | 7.88               | 7.88           | 283                | 283             |
|               | 8.0                 | 46.9             | 59.79             | 3,620             | 3,620           | 7.78               | 7.78           | 362                | 362             |
|               | 9.0                 | 52.3             | 66.67             | 3,990             | 3,990           | 7.73               | 7.73           | 399                | 399             |
|               | 12.0                | 67.9             | 86.53             | 4,980             | 4,980           | 7.59               | 7.59           | 498                | 498             |
| 250 x 250     | 6.0                 | 45.2             | 57.63             | 5,670             | 5,670           | 9.92               | 9.92           | 454                | 454             |
|               | 9.0                 | 66.5             | 84.67             | 8,090             | 8,090           | 9.78               | 9.78           | 647                | 647             |
|               | 12.0                | 86.8             | 110.5             | 10,300            | 10,300          | 9.63               | 9.63           | 820                | 820             |
| 300 x 300     | 6.0                 | 54.7             | 69.63             | 9,960             | 9,960           | 12.0               | 12.0           | 664                | 664             |
|               | 9.0                 | 80.6             | 102.7             | 14,300            | 14,300          | 11.8               | 11.8           | 956                | 956             |
|               | 12.0                | 106              | 134.5             | 18,300            | 18,300          | 11.7               | 11.7           | 1,220              | 1,220           |
|               | 16.0                | 138              | 175.2             | 23,100            | 23,100          | 11.5               | 11.5           | 1,540              | 1,540           |
| 350 x 350     | 9.0                 | 94.7             | 120.7             | 23,200            | 23,200          | 13.9               | 13.9           | 1,320              | 1,320           |
|               | 12.0                | 124              | 158.5             | 29,800            | 29,800          | 13.7               | 13.7           | 1,700              | 1,700           |
|               | 16.0                | 163              | 207.2             | 37,900            | 37,900          | 13.5               | 13.5           | 2,160              | 2,160           |
| 400 x 400     | 9.0                 | 109              | 138.7             | 35,100            | 35,100          | 15.9               | 15.9           | 1,750              | 1,750           |
|               | 12.0                | 143              | 182.5             | 45,300            | 45,300          | 15.8               | 15.8           | 2,270              | 2,270           |
|               | 16.0                | 188              | 239.2             | 57,900            | 57,900          | 15.6               | 15.6           | 2,900              | 2,900           |

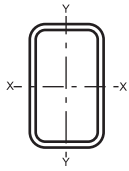
# COLD FORMED HOLLOW SECTIONS



## RECTANGULAR

| Section size | Wall thickness<br>t | Unit weight<br>M | Section area<br>A | Moment of inertia |                 | Radius of gyration |                 | Modulus of section |                 |
|--------------|---------------------|------------------|-------------------|-------------------|-----------------|--------------------|-----------------|--------------------|-----------------|
|              |                     |                  |                   | I <sub>x</sub>    | I <sub>y</sub>  | r <sub>ix</sub>    | r <sub>iy</sub> | Z <sub>x</sub>     | Z <sub>y</sub>  |
| mm           | mm                  | kg/m             | cm <sup>2</sup>   | cm <sup>4</sup>   | cm <sup>4</sup> | cm                 | cm              | cm <sup>3</sup>    | cm <sup>3</sup> |
| 19 x 9       | 1.2                 | 0.421            | 0.0833            | 0.25              | 0.07            | 0.65               | 0.35            | 0.26               | 0.16            |
|              | 1.6                 | 0.54             | 0.107             | 0.30              | 0.86            | 0.63               | 0.33            | 0.32               | 0.19            |
| 25 x 12      | 1.2                 | 0.67             | 0.85              | 0.62              | 0.21            | 0.87               | 0.50            | 0.51               | 0.33            |
|              | 1.6                 | 0.86             | 1.08              | 0.75              | 0.25            | 0.85               | 0.46            | 0.61               | 0.39            |
| 38 x 19      | 1.0                 | 0.78             | 1.104             | 2.10              | 0.78            | 1.37               | 0.81            | 1.37               | 0.80            |
|              | 1.2                 | 1.02             | 1.30              | 2.41              | 0.81            | 1.36               | 0.79            | 1.27               | 0.85            |
|              | 1.6                 | 1.34             | 1.70              | 3.09              | 1.02            | 1.35               | 0.77            | 1.62               | 1.07            |
| 50 x 25      | 1.0                 | 1.15             | 1.464             | 4.99              | 1.69            | 1.846              | 1.077           | 1.96               | 1.33            |
|              | 1.2                 | 1.36             | 1.86              | 5.99              | 2.04            | 1.84               | 1.07            | 2.36               | 1.59            |
|              | 1.6                 | 1.79             | 2.48              | 7.78              | 2.58            | 1.83               | 1.05            | 3.06               | 2.05            |
|              | 2.3                 | 2.48             | 3.19              | 9.86              | 3.23            | 1.76               | 1.01            | 3.94               | 2.59            |
|              | 3.0                 | 3.13             | 4.04              | 11.86             | 3.87            | 1.73               | 0.99            | 4.67               | 3.06            |
| 60 x 40      | 1.6                 | 2.37             | 3.03              | 15.21             | 8.14            | 2.23               | 1.63            | 5.07               | 4.07            |
|              | 2.3                 | 3.33             | 4.25              | 20.61             | 10.96           | 2.20               | 1.60            | 6.87               | 5.48            |
|              | 3.0                 | 4.24             | 5.40              | 25.29             | 13.36           | 2.16               | 1.57            | 8.43               | 6.68            |
|              | 4.5                 | 6.01             | 7.66              | 33.01             | 17.17           | 2.07               | 1.49            | 11.00              | 8.58            |
| 75 x 38      | 1.6                 | 2.70             | 3.30              | 24.97             | 8.69            | 2.72               | 1.60            | 6.55               | 4.57            |
|              | 2.3                 | 3.85             | 4.85              | 34.60             | 12.00           | 2.67               | 1.57            | 9.23               | 6.30            |
|              | 3.0                 | 4.92             | 6.35              | 45.20             | 15.15           | 2.66               | 1.54            | 11.86              | 7.95            |
| 75 x 50      | 1.9                 | 3.60             | 4.606             | 36.62             | 20.22           | 2.84               | 2.09            | 9.61               | 7.96            |
|              | 2.3                 | 4.24             | 5.40              | 41.86             | 22.36           | 2.78               | 2.03            | 11.16              | 8.94            |
|              | 3.0                 | 5.42             | 6.90              | 52.06             | 27.67           | 2.74               | 2.00            | 13.88              | 11.06           |
|              | 4.5                 | 7.78             | 9.91              | 70.20             | 36.87           | 2.66               | 1.92            | 18.72              | 14.74           |
|              | 6.0                 | 10.56            | 13.47             | 91.98             | 47.86           | 2.62               | 1.88            | 24.09              | 18.85           |
| 100 x 50     | 1.9                 | 4.36             | 5.484             | 73.67             | 25.47           | 3.65               | 2.15            | 14.50              | 10.02           |
|              | 2.3                 | 5.14             | 6.552             | 84.80             | 29.00           | 3.60               | 2.10            | 17.00              | 11.60           |
|              | 3.0                 | 6.60             | 8.40              | 106.33            | 35.95           | 3.55               | 2.06            | 21.26              | 14.38           |
|              | 3.2                 | 7.01             | 8.927             | 112.00            | 38.00           | 3.55               | 2.06            | 22.50              | 15.20           |
|              | 4.5                 | 9.55             | 12.16             | 146.18            | 48.51           | 3.46               | 1.99            | 29.23              | 19.40           |
|              | 6.0                 | 12.27            | 15.63             | 177.71            | 57.80           | 3.37               | 1.92            | 35.54              | 23.12           |
| 100 x 75     | 2.3                 | 6.04             | 7.70              | 112.22            | 72.25           | 3.81               | 3.06            | 22.44              | 19.26           |
|              | 3.0                 | 7.77             | 9.90              | 141.61            | 90.94           | 3.78               | 3.02            | 28.32              | 24.25           |
|              | 4.5                 | 11.31            | 14.41             | 197.48            | 126.12          | 3.70               | 2.95            | 39.49              | 33.63           |
|              | 6.0                 | 14.72            | 18.63             | 243.98            | 154.91          | 3.61               | 2.88            | 48.79              | 41.31           |
| 125 x 50     | 2.3                 | 6.04             | 7.70              | 148.17            | 35.44           | 4.38               | 2.14            | 23.70              | 14.17           |
|              | 3.0                 | 7.77             | 9.90              | 186.87            | 44.24           | 4.34               | 2.11            | 29.90              | 17.69           |
|              | 4.5                 | 11.31            | 14.41             | 260.18            | 60.16           | 4.24               | 2.04            | 41.62              | 24.06           |
|              | 6.0                 | 14.62            | 18.63             | 320.77            | 72.32           | 4.14               | 1.97            | 51.32              | 28.92           |

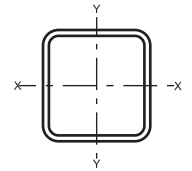
# COLD FORMED HOLLOW SECTIONS



## RECTANGULAR

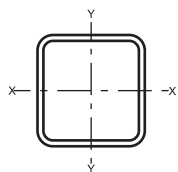
| Section size | Wall thickness<br>t | Unit weight<br>M | Section area<br>A | Moment of inertia |                 | Radius of gyration |                | Modulus of section |                 |
|--------------|---------------------|------------------|-------------------|-------------------|-----------------|--------------------|----------------|--------------------|-----------------|
|              |                     |                  |                   | I <sub>x</sub>    | I <sub>y</sub>  | i <sub>x</sub>     | i <sub>y</sub> | Z <sub>x</sub>     | Z <sub>y</sub>  |
| mm           | mm                  | kg/m             | cm <sup>2</sup>   | cm <sup>4</sup>   | cm <sup>4</sup> | cm                 | cm             | cm <sup>3</sup>    | cm <sup>3</sup> |
| 125 x 75     | 2.3                 | 6.95             | 8.852             | 192.00            | 87.50           | 4.65               | 3.14           | 30.60              | 23.30           |
|              | 3.0                 | 8.95             | 11.40             | 242.69            | 110.38          | 4.61               | 3.11           | 38.83              | 29.43           |
|              | 3.2                 | 9.52             | 12.13             | 257.00            | 117.00          | 4.60               | 3.10           | 41.10              | 31.10           |
|              | 4.5                 | 13.08            | 16.66             | 341.86            | 154.08          | 4.52               | 3.04           | 54.69              | 41.08           |
|              | 6.0                 | 16.98            | 21.63             | 426.98            | 190.62          | 4.44               | 2.96           | 68.31              | 50.83           |
| 150 x 75     | 3.0                 | 10.13            | 12.90             | 379.41            | 129.82          | 5.42               | 3.17           | 50.58              | 34.62           |
|              | 3.2                 | 10.80            | 13.73             | 402.00            | 137.00          | 5.41               | 3.16           | 53.60              | 36.60           |
|              | 4.5                 | 14.85            | 18.91             | 538.32            | 182.03          | 5.33               | 3.10           | 71.77              | 48.54           |
|              | 6.0                 | 19.33            | 24.63             | 677.58            | 226.32          | 5.24               | 3.03           | 90.34              | 60.35           |
|              | 9.0                 | 27.60            | 35.17             | 905.00            | 297.00          | 5.07               | 2.91           | 121                | 79.20           |
| 150 x 100    | 3.0                 | 11.31            | 14.40             | 460.45            | 247.47          | 5.65               | 4.14           | 61.39              | 49.49           |
|              | 3.2                 | 12.00            | 15.33             | 488.00            | 262.00          | 5.64               | 4.14           | 65.10              | 52.50           |
|              | 4.5                 | 16.60            | 21.17             | 658.00            | 352.00          | 5.58               | 4.08           | 87.70              | 70.40           |
|              | 6.0                 | 21.70            | 27.63             | 835.00            | 444.00          | 5.50               | 4.01           | 111.08             | 88.80           |
|              | 9.0                 | 31.10            | 39.67             | 1,430.00          | 595.00          | 5.33               | 3.87           | 151                | 119             |
| 200 x 100    | 4.5                 | 20.1             | 25.67             | 1,330             | 455             | 7.20               | 4.21           | 133                | 90.9            |
|              | 6.0                 | 26.4             | 33.63             | 1,700             | 577             | 7.12               | 4.14           | 170                | 115             |
|              | 9.0                 | 38.2             | 48.67             | 2,350             | 782             | 6.94               | 4.01           | 235                | 156             |
|              | 12.0                | 49.1             | 62.53             | 2,860             | 939             | 6.76               | 3.87           | 286                | 188             |
| 200 x 150    | 6.0                 | 31.1             | 39.63             | 2,270             | 1,460           | 7.56               | 6.06           | 227                | 194             |
|              | 9.0                 | 45.3             | 57.67             | 3,170             | 2,020           | 7.41               | 5.93           | 317                | 270             |
|              | 12.0                | 58.5             | 74.53             | 3,920             | 2,500           | 7.25               | 5.79           | 392                | 330             |
| 250 x 150    | 6.0                 | 35.8             | 45.63             | 3,890             | 1,770           | 9.23               | 6.23           | 311                | 236             |
|              | 9.0                 | 52.3             | 66.67             | 5,480             | 2,470           | 9.06               | 6.09           | 438                | 330             |
|              | 12.0                | 67.9             | 86.53             | 6,850             | 3,070           | 8.90               | 5.95           | 548                | 409             |
| 300 x 200    | 6.0                 | 45.2             | 57.63             | 7,370             | 3,960           | 11.3               | 8.29           | 491                | 396             |
|              | 9.0                 | 66.5             | 84.67             | 10,500            | 5,630           | 11.2               | 8.16           | 702                | 563             |
|              | 12.0                | 86.8             | 110.5             | 13,400            | 7,110           | 11.0               | 8.02           | 890                | 711             |
| 400 x 200    | 6.0                 | 54.7             | 69.63             | 14,800            | 5,090           | 14.6               | 8.55           | 739                | 509             |
|              | 9.0                 | 80.6             | 102.7             | 21,300            | 7,270           | 14.4               | 8.42           | 1,070              | 727             |
|              | 12.0                | 106.0            | 134.5             | 27,300            | 9,230           | 14.2               | 8.28           | 1,360              | 923             |

# COLD FORMED HOLLOW SECTIONS



## SQUARE

| Section size  |             | Wall thickness<br>t |     | Unit weight<br>M |        | Section area<br>A | Moment of inertia<br>I | Radius of gyration<br>i | Modulus of section<br>Z |
|---------------|-------------|---------------------|-----|------------------|--------|-------------------|------------------------|-------------------------|-------------------------|
| in            | mm          | in                  | mm  | lb/ft            | kg/m   | in <sup>2</sup>   | in <sup>4</sup>        | in                      | in <sup>3</sup>         |
| 1/2 x 1/2     | 12.7 x 12.7 | 0.047               | 1.2 | 0.272            | 0.404  | 0.082             | 0.002                  | 0.180                   | 0.010                   |
|               |             | 0.063               | 1.6 | 0.364            | 0.541  | 0.102             | 0.003                  | 0.173                   | 0.012                   |
| 5/8 x 5/8     | 15.9 x 15.9 | 0.047               | 1.2 | 0.353            | 0.525  | 0.106             | 0.005                  | 0.231                   | 0.018                   |
|               |             | 0.063               | 1.6 | 0.472            | 0.702  | 0.134             | 0.006                  | 0.223                   | 0.021                   |
| 3/4 x 3/4     | 19.1 x 19.1 | 0.047               | 1.2 | 0.434            | 0.646  | 0.140             | 0.011                  | 0.288                   | 0.030                   |
|               |             | 0.063               | 1.6 | 0.580            | 0.863  | 0.186             | 0.014                  | 0.282                   | 0.037                   |
|               |             | 0.095               | 2.4 | 0.817            | 1.230  | 0.240             | 0.017                  | 0.270                   | 0.046                   |
| 1 x 1         | 25.4 x 25.4 | 0.047               | 1.2 | 0.593            | 0.883  | 0.173             | 0.026                  | 0.385                   | 0.051                   |
|               |             | 0.063               | 1.6 | 0.770            | 1.150  | 0.226             | 0.032                  | 0.376                   | 0.064                   |
|               |             | 0.091               | 2.3 | 1.050            | 1.563  | 0.308             | 0.044                  | 0.371                   | 0.080                   |
|               |             | 0.095               | 2.4 | 1.100            | 1.640  | 0.321             | 0.045                  | 0.358                   | 0.082                   |
|               |             | 0.120               | 3.0 | 1.296            | 1.929  | 0.385             | 0.047                  | 0.350                   | 0.094                   |
| 1-1/4 x 1-1/4 | 31.8 x 31.8 | 0.047               | 1.2 | 0.754            | 1.122  | 0.220             | 0.052                  | 0.487                   | 0.084                   |
|               |             | 0.063               | 1.6 | 0.983            | 1.463  | 0.290             | 0.066                  | 0.479                   | 0.106                   |
|               |             | 0.091               | 2.3 | 1.357            | 2.020  | 0.414             | 0.082                  | 0.472                   | 0.140                   |
|               |             | 0.095               | 2.4 | 1.420            | 2.113  | 0.416             | 0.089                  | 0.461                   | 0.142                   |
|               |             | 0.120               | 3.0 | 1.698            | 2.527  | 0.505             | 0.103                  | 0.452                   | 0.165                   |
| 1-1/2 x 1-1/2 | 38.1 x 38.1 | 0.047               | 1.2 | 0.915            | 1.362  | 0.288             | 0.093                  | 0.590                   | 0.124                   |
|               |             | 0.063               | 1.6 | 1.200            | 1.785  | 0.384             | 0.120                  | 0.581                   | 0.159                   |
|               |             | 0.091               | 2.3 | 1.666            | 2.480  | 0.505             | 0.160                  | 0.573                   | 0.220                   |
|               |             | 0.095               | 2.4 | 1.740            | 2.592  | 0.526             | 0.163                  | 0.564                   | 0.220                   |
|               |             | 0.120               | 3.0 | 2.100            | 3.125  | 0.626             | 0.192                  | 0.554                   | 0.260                   |
| 2 x 2         | 50.8 x 50.8 | 0.063               | 1.6 | 1.626            | 2.424  | 0.480             | 0.300                  | 0.786                   | 0.300                   |
|               |             | 0.091               | 2.3 | 2.282            | 3.396  | 0.689             | 0.410                  | 0.776                   | 0.416                   |
|               |             | 0.095               | 2.4 | 2.380            | 3.542  | 0.700             | 0.415                  | 0.770                   | 0.415                   |
|               |             | 0.120               | 3.0 | 2.943            | 4.380  | 0.870             | 0.494                  | 0.756                   | 0.494                   |
|               |             | 0.125               | 3.2 | 3.051            | 4.540  | 0.893             | 0.508                  | 0.756                   | 0.508                   |
|               |             | 0.156               | 4.0 | 3.70             | 5.510  | 1.088             | 0.589                  | 0.740                   | 0.589                   |
|               |             | 0.180               | 4.6 | 4.173            | 6.210  | 1.227             | 0.653                  | 0.729                   | 0.653                   |
|               |             | 0.1875              | 4.8 | 4.314            | 6.420  | 1.269             | 0.668                  | 0.726                   | 0.668                   |
|               |             | 0.250               | 6.4 | 5.403            | 8.041  | 1.589             | 0.766                  | 0.694                   | 0.766                   |
| 2-1/2 x 2-1/2 | 63.5 x 63.5 | 0.091               | 2.3 | 2.898            | 4.313  | 0.86              | 0.82                   | 0.980                   | 0.656                   |
|               |             | 0.095               | 2.4 | 3.030            | 4.509  | 0.900             | 0.85                   | 0.976                   | 0.680                   |
|               |             | 0.120               | 3.0 | 3.760            | 5.590  | 1.110             | 1.02                   | 0.961                   | 0.820                   |
|               |             | 0.125               | 3.2 | 3.907            | 5.800  | 1.150             | 1.053                  | 0.958                   | 0.840                   |
|               |             | 0.156               | 4.0 | 4.760            | 7.090  | 1.400             | 1.241                  | 0.942                   | 0.993                   |
|               |             | 0.180               | 4.6 | 5.406            | 8.030  | 1.587             | 1.38                   | 0.934                   | 1.11                    |
|               |             | 0.1875              | 4.8 | 5.591            | 8.320  | 1.644             | 1.42                   | 0.930                   | 1.14                    |
|               |             | 0.250               | 6.4 | 7.122            | 10.600 | 2.089             | 1.69                   | 0.899                   | 1.35                    |



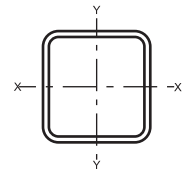
# COLD FORMED HOLLOW SECTIONS

## SQUARE

| Section size  |               | Wall thickness<br>t |      | Unit weight<br>M |        | Section area<br>A | Moment of inertia<br>I | Radius of gyration<br>i | Modulus of section<br>Z |
|---------------|---------------|---------------------|------|------------------|--------|-------------------|------------------------|-------------------------|-------------------------|
| in            | mm            | in                  | mm   | lb/ft            | kg/m   | in <sup>2</sup>   | in <sup>4</sup>        | in                      | in <sup>3</sup>         |
| 3 x 3         | 76.2 x 76.2   | 0.091               | 2.3  | 3.515            | 5.231  | 1.030             | 1.44                   | 1.19                    | 0.96                    |
|               |               | 0.095               | 2.4  | 3.670            | 5.480  | 1.080             | 1.503                  | 1.18                    | 1.002                   |
|               |               | 0.120               | 3.0  | 4.576            | 6.810  | 1.346             | 1.83                   | 1.17                    | 1.22                    |
|               |               | 0.125               | 3.2  | 4.751            | 7.070  | 1.400             | 1.89                   | 1.163                   | 1.26                    |
|               |               | 0.156               | 4.0  | 5.820            | 8.660  | 1.711             | 2.251                  | 1.147                   | 1.50                    |
|               |               | 0.180               | 4.6  | 6.620            | 9.850  | 1.947             | 2.52                   | 1.14                    | 1.68                    |
|               |               | 0.1875              | 4.8  | 6.860            | 10.200 | 2.019             | 2.60                   | 1.13                    | 1.73                    |
|               |               | 0.250               | 6.4  | 8.803            | 13.100 | 2.589             | 3.16                   | 1.10                    | 2.10                    |
| 3-1/2 x 3-1/2 | 88.9 x 88.9   | 0.120               | 3.0  | 5.390            | 8.020  | 1.585             | 2.98                   | 1.37                    | 1.70                    |
|               |               | 0.125               | 3.2  | 5.600            | 8.330  | 1.651             | 3.10                   | 1.367                   | 1.76                    |
|               |               | 0.156               | 4.0  | 6.880            | 10.240 | 2.024             | 3.71                   | 1.354                   | 2.12                    |
|               |               | 0.177               | 4.5  | 7.739            | 11.518 | 2.27              | 4.10                   | 1.34                    | 2.34                    |
|               |               | 0.180               | 4.6  | 7.862            | 11.700 | 2.31              | 4.16                   | 1.34                    | 2.38                    |
|               |               | 0.1875              | 4.8  | 8.140            | 12.113 | 2.394             | 4.29                   | 1.34                    | 2.45                    |
|               |               | 0.236               | 6.0  | 10.006           | 14.891 | 2.99              | 5.25                   | 1.33                    | 3.00                    |
|               |               | 0.250               | 6.4  | 10.500           | 15.626 | 3.10              | 5.29                   | 1.31                    | 3.02                    |
| 4 x 4         | 101.6 x 101.6 | 0.120               | 3.0  | 6.210            | 9.241  | 1.825             | 4.53                   | 1.58                    | 2.27                    |
|               |               | 0.125               | 3.2  | 6.450            | 9.681  | 1.902             | 4.70                   | 1.57                    | 2.34                    |
|               |               | 0.156               | 4.0  | 7.950            | 11.810 | 2.34              | 5.66                   | 1.56                    | 2.83                    |
|               |               | 0.177               | 4.5  | 8.945            | 13.312 | 2.63              | 6.28                   | 1.55                    | 3.14                    |
|               |               | 0.180               | 4.6  | 9.070            | 13.500 | 2.67              | 6.38                   | 1.55                    | 3.19                    |
|               |               | 0.1875              | 4.8  | 9.410            | 14.003 | 2.769             | 6.59                   | 1.54                    | 3.30                    |
|               |               | 0.236               | 6.0  | 11.614           | 17.285 | 3.39              | 7.82                   | 1.52                    | 3.91                    |
|               |               | 0.250               | 6.4  | 12.200           | 18.200 | 3.54              | 8.22                   | 1.51                    | 4.11                    |
|               |               | 0.3125              | 7.9  | 14.800           | 22.024 | 4.358             | 9.58                   | 1.48                    | 4.79                    |
|               |               | 0.375               | 9.5  | 17.300           | 25.745 | 5.075             | 10.7                   | 1.45                    | 5.35                    |
|               |               | 0.500               | 12.7 | 21.600           | 32.200 | 6.36              | 12.3                   | 1.39                    | 6.13                    |
| 5 x 5         | 127 x 127     | 0.156               | 4.0  | 10.100           | 15.000 | 2.961             | 11.43                  | 1.97                    | 4.57                    |
|               |               | 0.177               | 4.5  | 11.356           | 16.901 | 3.33              | 12.7                   | 1.97                    | 5.08                    |
|               |               | 0.180               | 4.6  | 11.500           | 17.113 | 3.39              | 12.9                   | 1.96                    | 5.18                    |
|               |               | 0.1875              | 4.8  | 12.000           | 17.857 | 3.519             | 13.4                   | 1.95                    | 5.36                    |
|               |               | 0.236               | 6.0  | 14.830           | 22.070 | 4.34              | 16.0                   | 1.92                    | 6.42                    |
|               |               | 0.250               | 6.4  | 15.656           | 23.298 | 4.54              | 16.9                   | 1.92                    | 6.78                    |
|               |               | 0.3125              | 7.9  | 19.100           | 28.424 | 5.608             | 20.1                   | 1.89                    | 8.02                    |
|               |               | 0.375               | 9.5  | 22.400           | 33.334 | 6.575             | 22.8                   | 1.86                    | 9.11                    |
|               |               | 0.500               | 12.7 | 28.400           | 42.300 | 8.356             | 27.0                   | 1.80                    | 10.8                    |
| 6 x 6         | 152.4 x 152.4 | 0.156               | 4.0  | 12.200           | 18.155 | 3.583             | 20.2                   | 2.40                    | 6.73                    |
|               |               | 0.177               | 4.5  | 13.768           | 20.490 | 4.04              | 22.5                   | 2.36                    | 7.50                    |
|               |               | 0.180               | 4.6  | 14.000           | 20.834 | 4.103             | 22.9                   | 2.36                    | 7.65                    |



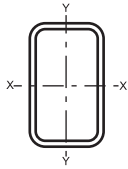
# COLD FORMED HOLLOW SECTIONS



## SQUARE

| Section size |               | Wall thickness<br>t |      | Unit weight<br>M |         | Section area<br>A | Moment of inertia<br>I | Radius of gyration<br>i | Modulus of section<br>Z |
|--------------|---------------|---------------------|------|------------------|---------|-------------------|------------------------|-------------------------|-------------------------|
| in           | mm            | in                  | mm   | lb/ft            | kg/m    | in <sup>2</sup>   | in <sup>4</sup>        | in                      | in <sup>3</sup>         |
| 6 x 6        | 152.4 x 152.4 | 0.1875              | 4.8  | 14.5             | 21.60   | 4.269             | 23.8                   | 2.36                    | 7.93                    |
|              |               | 0.236               | 6.0  | 18.045           | 26.856  | 5.28              | 28.7                   | 2.33                    | 9.58                    |
|              |               | 0.250               | 6.4  | 19.016           | 28.30   | 5.535             | 30.3                   | 2.33                    | 10.1                    |
|              |               | 0.3125              | 7.9  | 23.317           | 34.70   | 6.858             | 36.3                   | 2.30                    | 12.1                    |
|              |               | 0.375               | 9.5  | 27.5             | 40.924  | 8.075             | 41.6                   | 2.27                    | 13.9                    |
|              |               | 0.500               | 12.7 | 35.2             | 52.40   | 10.360            | 50.5                   | 2.21                    | 16.8                    |
| 7 x 7        | 177.8 x 177.8 | 0.180               | 4.6  | 16.4             | 24.405  | 4.827             | 37.1                   | 2.77                    | 10.6                    |
|              |               | 0.1875              | 4.8  | 17.1             | 25.447  | 5.019             | 38.5                   | 2.77                    | 11.0                    |
|              |               | 0.250               | 6.4  | 22.4             | 33.334  | 6.589             | 49.4                   | 2.74                    | 14.1                    |
|              |               | 0.3125              | 7.9  | 27.6             | 41.073  | 8.108             | 59.5                   | 2.71                    | 17.0                    |
|              |               | 0.375               | 9.5  | 32.0             | 48.00   | 9.575             | 68.7                   | 2.68                    | 19.6                    |
|              |               | 0.500               | 12.7 | 42.0             | 62.502  | 12.360            | 84.6                   | 2.62                    | 24.2                    |
| 8 x 8        | 203.2 x 203.2 | 0.180               | 4.6  | 18.9             | 28.126  | 5.547             | 56.1                   | 3.18                    | 14.0                    |
|              |               | 0.1875              | 4.8  | 19.6             | 29.20   | 5.769             | 58.2                   | 3.18                    | 14.6                    |
|              |               | 0.250               | 6.4  | 25.8             | 38.40   | 7.589             | 75.1                   | 3.15                    | 18.8                    |
|              |               | 0.3125              | 7.9  | 31.8             | 47.323  | 9.358             | 90.9                   | 3.12                    | 22.7                    |
|              |               | 0.375               | 9.5  | 37.7             | 56.103  | 11.08             | 106                    | 3.09                    | 26.4                    |
|              |               | 0.500               | 12.7 | 48.8             | 72.622  | 14.36             | 131                    | 3.03                    | 32.9                    |
| 10 x 10      | 254 x 254     | 0.250               | 6.4  | 32.5             | 48.512  | 9.589             | 151                    | 3.96                    | 30.1                    |
|              |               | 0.3125              | 7.9  | 40.3             | 60.00   | 11.86             | 183                    | 3.93                    | 36.7                    |
|              |               | 0.375               | 9.5  | 47.9             | 71.282  | 14.08             | 214                    | 3.90                    | 42.9                    |
|              |               | 0.500               | 12.7 | 62.4             | 92.90   | 18.36             | 271                    | 3.84                    | 54.2                    |
|              |               | 0.625               | 15.9 | 76.3             | 113.546 | 22.40             | 321                    | 3.78                    | 64.2                    |
| 12 x 12      | 304.8 x 304.8 | 0.250               | 6.4  | 39.4             | 58.63   | 11.59             | 265                    | 4.78                    | 44.1                    |
|              |               | 0.3125              | 7.9  | 48.8             | 72.62   | 14.36             | 324                    | 4.75                    | 54.0                    |
|              |               | 0.375               | 9.5  | 58.0             | 86.40   | 17.08             | 380                    | 4.72                    | 63.4                    |
|              |               | 0.500               | 12.7 | 76.0             | 113.10  | 22.36             | 485                    | 4.66                    | 80.9                    |
|              |               | 0.625               | 15.9 | 93.3             | 139.00  | 27.43             | 580                    | 4.60                    | 96.7                    |
| 14 x 14      | 355.6 x 355.6 | 0.375               | 9.5  | 68.2             | 102.00  | 20.08             | 615                    | 5.54                    | 87.9                    |
|              |               | 0.500               | 12.7 | 89.6             | 33.33   | 26.36             | 791                    | 5.48                    | 113                     |
|              |               | 0.625               | 15.9 | 110.0            | 164.00  | 32.43             | 952                    | 5.42                    | 136                     |
| 16 x 16      | 406.4 x 406.4 | 0.500               | 12.7 | 103.0            | 154.00  | 30.36             | 1,200                  | 6.29                    | 150                     |
|              |               | 0.625               | 15.9 | 127.0            | 189.00  | 37.43             | 1,450                  | 6.23                    | 182                     |

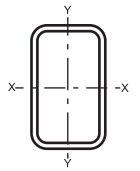
# COLD FORMED HOLLOW SECTIONS



## RECTANGULAR

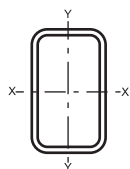
| Section size |              | Wall thickness t |     | Unit weight M |        | Section area A  | Moment of inertia |                 | Radius of gyration |                | Modulus of section |                 |
|--------------|--------------|------------------|-----|---------------|--------|-----------------|-------------------|-----------------|--------------------|----------------|--------------------|-----------------|
|              |              |                  |     |               |        |                 | I <sub>x</sub>    | I <sub>y</sub>  | i <sub>x</sub>     | i <sub>y</sub> | Z <sub>x</sub>     | Z <sub>y</sub>  |
| in           | mm           | in               | mm  | lb/ft         | kg/m   | in <sup>2</sup> | in <sup>4</sup>   | in <sup>4</sup> | in                 | in             | in <sup>3</sup>    | in <sup>3</sup> |
| 1 x ½        | 25.4 x 12.7  | 0.039            | 1.0 | 0.373         | 0.555  | 0.110           | 0.013             | 0.004           | 0.355              | 0.204          | 0.027              | 0.018           |
|              |              | 0.047            | 1.2 | 0.448         | 0.670  | 0.131           | 0.015             | 0.005           | 0.344              | 0.198          | 0.031              | 0.020           |
|              |              | 0.063            | 1.6 | 0.580         | 0.863  | 0.167           | 0.018             | 0.006           | 0.335              | 0.191          | 0.037              | 0.024           |
| 1½ x 1       | 38.1 x 25.4  | 0.039            | 1.0 | 0.638         | 0.949  | 0.188           | 0.059             | 0.031           | 0.564              | 0.412          | 0.079              | 0.063           |
|              |              | 0.047            | 1.2 | 0.767         | 1.141  | 0.229           | 0.070             | 0.037           | 0.555              | 0.405          | 0.094              | 0.075           |
|              |              | 0.063            | 1.6 | 1.010         | 1.500  | 0.297           | 0.088             | 0.047           | 0.546              | 0.398          | 0.118              | 0.094           |
| 2 x 1        | 50.8 x 25.4  | 0.047            | 1.2 | 0.915         | 1.362  | 0.288           | 0.144             | 0.049           | 0.726              | 0.422          | 0.144              | 0.097           |
|              |              | 0.0598           | 1.5 | 1.128         | 1.683  | 0.365           | 0.179             | 0.060           | 0.721              | 0.416          | 0.179              | 0.120           |
|              |              | 0.063            | 1.6 | 1.200         | 1.785  | 0.384           | 0.187             | 0.062           | 0.720              | 0.415          | 0.187              | 0.125           |
|              |              | 0.091            | 2.3 | 1.666         | 2.480  | 0.505           | 0.257             | 0.083           | 0.713              | 0.407          | 0.257              | 0.167           |
|              |              | 0.095            | 2.4 | 1.740         | 2.592  | 0.526           | 0.260             | 0.085           | 0.702              | 0.401          | 0.260              | 0.170           |
|              |              | 0.120            | 3.0 | 2.100         | 3.125  | 0.626           | 0.285             | 0.093           | 0.680              | 0.390          | 0.285              | 0.187           |
| 2½ x 1½      | 63.5 x 38.1  | 0.063            | 1.6 | 1.626         | 2.424  | 0.480           | 0.416             | 0.189           | 0.920              | 0.620          | 0.333              | 0.252           |
|              |              | 0.091            | 2.3 | 2.282         | 3.396  | 0.692           | 0.55              | 0.248           | 0.912              | 0.611          | 0.439              | 0.331           |
|              |              | 0.095            | 2.4 | 2.380         | 3.542  | 0.700           | 0.572             | 0.257           | 0.903              | 0.606          | 0.457              | 0.343           |
|              |              | 0.120            | 3.0 | 2.904         | 4.322  | 0.870           | 0.685             | 0.306           | 0.889              | 0.595          | 0.548              | 0.409           |
| 3 x 1½       | 76.2 x 38.1  | 0.063            | 1.6 | 1.860         | 2.70   | 0.527           | 0.60              | 0.209           | 1.07               | 0.63           | 0.40               | 0.279           |
|              |              | 0.075            | 1.9 | 2.166         | 3.224  | 0.629           | 0.72              | 0.246           | 1.06               | 0.623          | 0.48               | 0.328           |
|              |              | 0.091            | 2.3 | 2.590         | 3.855  | 0.762           | 0.86              | 0.293           | 1.06               | 0.62           | 0.573              | 0.391           |
|              |              | 0.095            | 2.4 | 2.700         | 4.018  | 0.795           | 0.901             | 0.304           | 1.06               | 0.619          | 0.601              | 0.406           |
|              |              | 0.120            | 3.0 | 3.306         | 4.920  | 0.985           | 1.086             | 0.364           | 1.050              | 0.607          | 0.724              | 0.485           |
| 3 x 2        | 76.2 x 50.8  | 0.075            | 1.9 | 2.421         | 3.603  | 0.714           | 0.88              | 0.486           | 1.12               | 0.825          | 0.587              | 0.486           |
|              |              | 0.091            | 2.3 | 2.898         | 4.313  | 0.863           | 1.06              | 0.578           | 1.112              | 0.818          | 0.707              | 0.578           |
|              |              | 0.095            | 2.4 | 3.030         | 4.509  | 0.900           | 1.102             | 0.589           | 1.112              | 0.813          | 0.734              | 0.589           |
|              |              | 0.120            | 3.0 | 3.760         | 5.560  | 1.110           | 1.33              | 0.707           | 1.097              | 0.800          | 0.886              | 0.707           |
|              |              | 0.125            | 3.2 | 3.910         | 5.819  | 1.150           | 1.372             | 0.730           | 1.094              | 0.797          | 0.915              | 0.730           |
|              |              | 0.156            | 4.0 | 4.760         | 7.090  | 1.400           | 1.62              | 0.860           | 1.08               | 0.782          | 1.080              | 0.860           |
|              |              | 0.180            | 4.6 | 5.400         | 8.036  | 1.587           | 1.81              | 0.952           | 1.07               | 0.775          | 1.200              | 0.952           |
|              |              | 0.1875           | 4.8 | 5.590         | 8.320  | 1.644           | 1.86              | 0.977           | 1.06               | 0.771          | 1.240              | 0.977           |
|              |              | 0.250            | 6.4 | 7.100         | 10.600 | 2.089           | 2.21              | 1.150           | 1.03               | 0.742          | 1.47               | 1.150           |
|              |              |                  |     |               |        |                 |                   |                 |                    |                |                    |                 |
| 4 x 2        | 101.6 x 50.8 | 0.075            | 1.9 | 2.930         | 4.361  | 0.85            | 1.77              | 0.612           | 1.44               | 0.848          | 0.885              | 0.612           |
|              |              | 0.091            | 2.3 | 3.515         | 5.231  | 1.03            | 2.14              | 0.732           | 1.44               | 0.843          | 1.07               | 0.732           |
|              |              | 0.095            | 2.4 | 3.670         | 5.480  | 1.080           | 2.233             | 0.761           | 1.44               | 0.840          | 1.116              | 0.761           |
|              |              | 0.120            | 3.0 | 4.511         | 6.714  | 1.346           | 2.72              | 0.920           | 1.421              | 0.830          | 1.360              | 0.919           |
|              |              | 0.125            | 3.2 | 4.790         | 7.129  | 1.400           | 2.81              | 0.950           | 1.42               | 0.824          | 1.405              | 0.950           |
|              |              | 0.156            | 4.0 | 5.820         | 8.660  | 1.711           | 3.35              | 1.121           | 1.40               | 0.809          | 1.674              | 1.121           |
|              |              | 0.180            | 4.6 | 6.620         | 9.851  | 1.947           | 3.75              | 1.25            | 1.39               | 0.802          | 1.88               | 1.25            |
|              |              | 0.1875           | 4.8 | 6.860         | 10.209 | 2.019           | 3.87              | 1.29            | 1.38               | 0.798          | 1.93               | 1.29            |
|              |              | 0.250            | 6.4 | 8.800         | 13.096 | 2.589           | 4.69              | 1.54            | 1.35               | 0.771          | 2.35               | 1.54            |
| 4 x 3        | 101.6 x 76.2 | 0.120            | 3.0 | 5.390         | 8.021  | 1.582           | 3.63              | 2.33            | 1.51               | 1.21           | 1.81               | 1.55            |
|              |              | 0.125            | 3.2 | 5.647         | 8.404  | 1.651           | 3.75              | 2.41            | 1.51               | 1.21           | 1.88               | 1.61            |
|              |              | 0.156            | 4.0 | 6.880         | 10.240 | 2.024           | 4.52              | 2.90            | 1.50               | 1.20           | 2.26               | 1.93            |

# COLD FORMED HOLLOW SECTIONS



## RECTANGULAR

| Section size |              | Wall thickness<br>t |     | Unit weight<br>M |        | Section area<br>A | Moment of inertia |                 | Radius of gyration |                | Modulus of section |                 |
|--------------|--------------|---------------------|-----|------------------|--------|-------------------|-------------------|-----------------|--------------------|----------------|--------------------|-----------------|
|              |              |                     |     |                  |        |                   | I <sub>x</sub>    | I <sub>y</sub>  | i <sub>x</sub>     | i <sub>y</sub> | Z <sub>x</sub>     | Z <sub>y</sub>  |
| in           | mm           | in                  | mm  | lb/ft            | kg/m   | in <sup>2</sup>   | in <sup>4</sup>   | in <sup>4</sup> | in                 | in             | in <sup>3</sup>    | in <sup>3</sup> |
| 4 x 3        | 101.6 x 76.2 | 0.177               | 4.5 | 7.739            | 11.517 | 2.27              | 4.99              | 3.19            | 1.48               | 1.19           | 2.50               | 2.13            |
|              |              | 0.180               | 4.6 | 7.850            | 11.682 | 2.310             | 5.07              | 3.24            | 1.48               | 1.19           | 2.53               | 2.16            |
|              |              | 0.1875              | 4.8 | 8.140            | 12.113 | 2.394             | 5.23              | 3.34            | 1.48               | 1.18           | 2.62               | 2.23            |
|              |              | 0.236               | 6.0 | 10.006           | 14.891 | 2.99              | 6.43              | 4.07            | 1.46               | 1.16           | 3.21               | 2.71            |
|              |              | 0.250               | 6.4 | 10.500           | 15.630 | 3.089             | 6.45              | 4.10            | 1.45               | 1.15           | 3.23               | 2.74            |
| 5 x 2        | 127 x 50.8   | 0.120               | 3.0 | 5.390            | 8.021  | 1.585             | 4.79              | 1.14            | 1.74               | 0.846          | 1.92               | 1.14            |
|              |              | 0.125               | 3.2 | 5.647            | 8.404  | 1.647             | 4.96              | 1.17            | 1.73               | 0.844          | 1.98               | 1.17            |
|              |              | 0.156               | 4.0 | 6.880            | 10.240 | 2.024             | 5.96              | 1.40            | 1.72               | 0.83           | 2.38               | 1.40            |
|              |              | 0.177               | 4.5 | 7.739            | 11.518 | 2.27              | 6.57              | 1.52            | 1.70               | 0.82           | 2.63               | 1.52            |
|              |              | 0.180               | 4.6 | 7.850            | 11.682 | 2.307             | 6.67              | 1.55            | 1.70               | 0.82           | 2.67               | 1.55            |
|              |              | 0.1875              | 4.8 | 8.140            | 12.100 | 2.394             | 6.89              | 1.60            | 1.70               | 0.816          | 2.75               | 1.60            |
|              |              | 0.236               | 6.0 | 10.006           | 14.891 | 2.99              | 8.50              | 1.91            | 1.68               | 0.799          | 3.39               | 1.91            |
|              |              | 0.250               | 6.4 | 10.400           | 15.626 | 3.089             | 8.48              | 1.92            | 1.66               | 0.789          | 3.40               | 1.92            |
| 5 x 3        | 127 x 76.2   | 0.120               | 3.0 | 6.210            | 9.241  | 1.825             | 6.22              | 2.83            | 1.85               | 1.25           | 2.49               | 1.89            |
|              |              | 0.125               | 3.2 | 6.505            | 9.781  | 1.902             | 6.44              | 2.93            | 1.84               | 1.24           | 2.58               | 1.95            |
|              |              | 0.156               | 4.0 | 7.950            | 11.831 | 2.340             | 7.76              | 3.14            | 1.82               | 1.23           | 3.11               | 2.34            |
|              |              | 0.177               | 4.5 | 8.945            | 13.312 | 2.63              | 8.63              | 3.90            | 1.81               | 1.22           | 3.45               | 2.60            |
|              |              | 0.180               | 4.6 | 9.070            | 13.498 | 2.667             | 8.77              | 3.96            | 1.81               | 1.22           | 3.51               | 2.64            |
|              |              | 0.1875              | 4.8 | 9.410            | 14.004 | 2.769             | 9.06              | 4.08            | 1.81               | 1.21           | 3.62               | 2.72            |
|              |              | 0.236               | 6.0 | 11.614           | 17.284 | 3.47              | 11.2              | 4.98            | 1.79               | 1.19           | 4.48               | 3.32            |
|              |              | 0.250               | 6.4 | 12.200           | 18.156 | 3.589             | 11.3              | 5.05            | 1.77               | 1.19           | 4.52               | 3.37            |
|              |              | 0.3125              | 8.0 | 14.800           | 22.025 | 4.358             | 13.2              | 5.85            | 1.74               | 1.16           | 5.27               | 3.90            |
| 6 x 2        | 152.4 x 50.8 | 0.375               | 9.5 | 17.300           | 25.745 | 5.075             | 14.7              | 6.48            | 1.70               | 1.13           | 5.89               | 4.32            |
|              |              | 0.120               | 3.0 | 6.210            | 9.241  | 1.825             | 7.64              | 1.35            | 2.05               | 0.859          | 2.55               | 1.35            |
|              |              | 0.125               | 3.2 | 6.505            | 9.680  | 1.897             | 7.92              | 1.39            | 2.04               | 0.857          | 2.64               | 1.39            |
|              |              | 0.156               | 4.0 | 7.950            | 11.831 | 2.340             | 9.56              | 1.66            | 2.02               | 0.843          | 3.19               | 1.66            |
|              |              | 0.177               | 4.5 | 8.945            | 13.312 | 2.63              | 10.5              | 1.82            | 2.01               | 0.838          | 3.50               | 1.82            |
|              |              | 0.180               | 4.6 | 9.070            | 13.500 | 2.667             | 10.7              | 1.85            | 2.01               | 0.833          | 3.58               | 1.85            |
|              |              | 0.1875              | 4.8 | 9.410            | 14.003 | 2.769             | 11.1              | 1.90            | 2.00               | 0.829          | 3.70               | 1.90            |
|              |              | 0.236               | 6.0 | 11.614           | 17.284 | 3.47              | 13.1              | 2.21            | 1.96               | 0.818          | 4.36               | 2.21            |
| 6 x 3        | 152.4 x 76.2 | 0.250               | 6.4 | 12.230           | 18.200 | 3.589             | 13.8              | 2.31            | 1.96               | 0.802          | 4.60               | 2.31            |
|              |              | 0.125               | 3.2 | 7.320            | 10.893 | 2.130             | 10.06             | 3.44            | 2.17               | 1.27           | 3.35               | 2.30            |
|              |              | 0.156               | 4.0 | 9.020            | 13.423 | 2.652             | 12.2              | 4.15            | 2.15               | 1.25           | 4.07               | 2.76            |
|              |              | 0.177               | 4.5 | 10.150           | 15.105 | 2.98              | 13.6              | 4.60            | 2.13               | 1.24           | 4.53               | 3.07            |
|              |              | 0.180               | 4.6 | 10.300           | 15.330 | 3.027             | 13.8              | 4.67            | 2.13               | 1.24           | 4.60               | 3.12            |
|              |              | 0.1875              | 4.8 | 10.700           | 15.923 | 3.144             | 14.3              | 4.83            | 2.13               | 1.24           | 4.76               | 3.22            |
|              |              | 0.236               | 6.0 | 13.222           | 19.676 | 3.87              | 17.0              | 5.72            | 2.10               | 1.22           | 5.67               | 3.81            |
|              |              | 0.250               | 6.4 | 13.700           | 20.685 | 4.089             | 17.9              | 6.00            | 2.09               | 1.21           | 5.98               | 4.00            |
|              |              | 0.3125              | 8.0 | 16.933           | 25.200 | 4.983             | 21.1              | 6.98            | 2.06               | 1.18           | 7.03               | 4.65            |
| 6 x 3        | 152.4 x 76.2 | 0.375               | 9.5 | 19.823           | 29.500 | 5.825             | 23.8              | 7.78            | 2.02               | 1.16           | 7.92               | 5.19            |



# COLD FORMED HOLLOW SECTIONS

## RECTANGULAR

| Section size |               | Wall thickness<br>t |      | Unit weight<br>M |        | Section area<br>A | Moment of inertia<br>Ix Iy |                 | Radius of gyration<br>ix iy |      | Modulus of section<br>Zx Zy |                 |
|--------------|---------------|---------------------|------|------------------|--------|-------------------|----------------------------|-----------------|-----------------------------|------|-----------------------------|-----------------|
| in           | mm            | in                  | mm   | lb/ft            | kg/m   | in <sup>2</sup>   | in <sup>4</sup>            | in <sup>4</sup> | in                          | in   | in <sup>3</sup>             | in <sup>3</sup> |
| 6 x 4        | 152.4 x 101.6 | 0.156               | 4.0  | 10.10            | 15.03  | 2.961             | 14.86                      | 7.96            | 2.24                        | 1.64 | 4.96                        | 3.98            |
|              |               | 0.177               | 4.5  | 11.356           | 16.90  |                   | 16.5                       | 8.87            | 2.23                        | 1.63 | 5.50                        | 4.44            |
|              |               | 0.180               | 4.6  | 11.50            | 17.114 | 3.387             | 16.8                       | 9.01            | 2.23                        | 1.63 | 5.62                        | 4.51            |
|              |               | 0.1875              | 4.8  | 12.00            | 17.858 | 3.519             | 17.4                       | 9.32            | 2.23                        | 1.63 | 5.81                        | 4.66            |
|              |               | 0.236               | 6.0  | 14.83            | 22.07  | 4.34              | 21.0                       | 11.1            | 2.20                        | 1.60 | 7.00                        | 5.55            |
|              |               | 0.250               | 6.4  | 15.60            | 23.215 | 4.589             | 22.1                       | 11.7            | 2.19                        | 1.60 | 7.36                        | 5.87            |
|              |               | 0.3125              | 8.0  | 19.10            | 28.424 | 5.608             | 26.2                       | 13.8            | 2.16                        | 1.57 | 8.72                        | 6.92            |
|              |               | 0.375               | 9.5  | 22.40            | 33.335 | 6.575             | 29.7                       | 15.6            | 2.13                        | 1.54 | 9.9                         | 7.82            |
|              |               | 0.500               | 12.7 | 28.424           | 42.30  | 8.356             | 35.3                       | 18.4            | 2.06                        | 1.48 | 11.8                        | 9.21            |
| 8 x 4        | 203.2 x 101.6 | 0.156               | 4.0  | 12.20            | 18.155 | 3.583             | 30.0                       | 10.27           | 2.90                        | 1.70 | 7.50                        | 5.13            |
|              |               | 0.177               | 4.5  | 13.768           | 20.49  | 3.98              | 32.0                       | 10.9            | 2.89                        | 1.69 | 8.12                        | 5.55            |
|              |               | 0.180               | 4.6  | 14.00            | 20.834 | 4.107             | 34.1                       | 11.6            | 2.88                        | 1.68 | 8.52                        | 5.82            |
|              |               | 0.1875              | 4.8  | 14.515           | 21.60  | 4.269             | 35.3                       | 12.0            | 2.88                        | 1.68 | 8.83                        | 6.02            |
|              |               | 0.236               | 6.0  | 18.045           | 26.854 | 5.21              | 40.8                       | 13.9            | 2.84                        | 1.66 | 10.4                        | 7.02            |
|              |               | 0.250               | 6.4  | 19.017           | 28.30  | 5.589             | 45.1                       | 15.3            | 2.84                        | 1.65 | 11.3                        | 7.63            |
|              |               | 0.3125              | 8.0  | 23.317           | 34.70  | 6.858             | 53.9                       | 18.1            | 2.80                        | 1.62 | 13.5                        | 9.05            |
|              |               | 0.375               | 9.5  | 27.50            | 40.924 | 8.075             | 61.9                       | 20.6            | 2.77                        | 1.60 | 15.5                        | 10.3            |
|              |               | 0.500               | 12.7 | 35.211           | 52.40  | 10.36             | 75.1                       | 24.6            | 2.69                        | 1.54 | 18.8                        | 12.3            |
| 8 x 6        | 203.2 x 152.4 | 0.180               | 4.6  | 16.40            | 24.406 | 4.827             | 45.1                       | 29.0            | 3.06                        | 2.45 | 11.3                        | 9.68            |
|              |               | 0.1875              | 4.8  | 17.10            | 25.448 | 5.019             | 46.8                       | 30.1            | 3.05                        | 2.45 | 11.7                        | 10.0            |
|              |               | 0.250               | 6.4  | 22.40            | 33.335 | 6.589             | 60.1                       | 38.6            | 3.02                        | 2.42 | 15.0                        | 12.9            |
|              |               | 0.3125              | 8.0  | 27.60            | 41.073 | 8.108             | 72.4                       | 46.4            | 2.99                        | 2.39 | 18.1                        | 15.5            |
|              |               | 0.375               | 9.5  | 32.60            | 48.514 | 9.575             | 83.7                       | 53.5            | 2.96                        | 2.36 | 20.9                        | 17.8            |
|              |               | 0.500               | 12.7 | 42.00            | 62.503 | 12.36             | 103                        | 65.7            | 2.89                        | 2.31 | 25.8                        | 21.9            |
| 10 x 6       | 254 x 152.4   | 0.180               | 4.6  | 18.90            | 28.126 | 5.547             | 76.9                       | 35.1            | 3.72                        | 2.52 | 15.4                        | 11.7            |
|              |               | 0.1875              | 4.8  | 19.621           | 29.20  | 5.769             | 79.8                       | 36.5            | 3.72                        | 2.51 | 16.0                        | 12.2            |
|              |               | 0.250               | 6.4  | 25.804           | 38.40  | 7.589             | 103                        | 46.9            | 3.69                        | 2.49 | 20.6                        | 15.6            |
|              |               | 0.3125              | 8.0  | 31.80            | 47.324 | 9.358             | 125                        | 56.5            | 3.65                        | 2.46 | 25.0                        | 18.8            |
|              |               | 0.375               | 9.5  | 37.70            | 56.104 | 11.08             | 145                        | 65.4            | 3.62                        | 2.43 | 29.0                        | 21.8            |
|              |               | 0.500               | 12.7 | 48.80            | 72.622 | 14.36             | 181                        | 80.8            | 3.55                        | 2.37 | 36.2                        | 26.9            |
| 12 x 8       | 304.8 X 203.2 | 0.250               | 6.4  | 32.60            | 48.514 | 9.589             | 196                        | 105             | 4.52                        | 3.31 | 32.6                        | 26.3            |
|              |               | 0.3125              | 8.0  | 40.30            | 60.00  | 11.86             | 239                        | 128             | 4.49                        | 3.28 | 39.8                        | 32.0            |
|              |               | 0.375               | 9.5  | 47.90            | 71.283 | 14.08             | 279                        | 149             | 4.45                        | 3.26 | 46.5                        | 37.3            |
|              |               | 0.500               | 12.7 | 62.426           | 92.90  | 18.36             | 353                        | 188             | 4.39                        | 3.20 | 58.9                        | 46.9            |
| 14 x 6       | 355.6 x 152.4 | 0.250               | 6.4  | 32.60            | 48.514 | 9.589             | 237                        | 63.4            | 4.97                        | 2.57 | 33.8                        | 21.1            |
|              |               | 0.3125              | 8.0  | 40.30            | 60.00  | 11.86             | 288                        | 76.7            | 4.93                        | 2.54 | 41.2                        | 25.6            |
|              |               | 0.375               | 9.5  | 47.90            | 71.283 | 14.08             | 337                        | 89.1            | 4.89                        | 2.52 | 48.1                        | 29.7            |
|              |               | 0.500               | 12.7 | 62.426           | 92.90  | 18.36             | 426                        | 111             | 4.82                        | 2.46 | 60.8                        | 37.1            |
| 16 x 8       | 406.4 x 203.2 | 0.250               | 6.4  | 39.40            | 58.634 | 11.59             | 393                        | 135             | 5.82                        | 3.42 | 49.1                        | 33.8            |
|              |               | 0.3125              | 8.0  | 48.80            | 72.622 | 14.36             | 481                        | 165             | 5.79                        | 3.39 | 60.1                        | 41.2            |
|              |               | 0.375               | 9.5  | 58.058           | 86.40  | 17.08             | 565                        | 193             | 5.75                        | 3.36 | 70.6                        | 48.2            |
|              |               | 0.500               | 12.7 | 76.00            | 113.10 | 22.36             | 722                        | 244             | 5.68                        | 3.30 | 90.2                        | 61.0            |

# HOT FORMED HOLLOW SECTIONS

(excerpts from BRITISH STEEL TUBES)

The size range of Structural Hollow Sections manufactured by British Steel Tubes, is in accordance with BS 4848 Hot Rolled Structural Steel Sections: Part 2, Hollow Sections.

The Structural Hollow Section size range published in BS 4848: Part 2 relates generally to ISO 657 Hot Rolled Steel Sections: Part 14, Hot Finished Structural Hollow Sections, Dimensional and Sectional Properties.

## STEEL

The steel is made by the basic oxygen or electric furnace process in accordance with BS 4360. For SHS there are three basic grades, 43, 50 and 55, each with sub-grades giving improved impact properties. For economic reasons, production has been standardised on grades 43C and 50C. Other grades and sub-grades can be supplied against mill orders and subject to minimum mill quantities.

## CHEMICAL COMPOSITION AND MECHANICAL PROPERTIES

The ladle analyses and mechanical properties, in accordance with BS 4360 for hot finished sections are as follows:

|            | Grade 43C | Grade 50C  |                                       | Grade 43C        | Grade 50C        |
|------------|-----------|------------|---------------------------------------|------------------|------------------|
|            | %max      | % max      |                                       |                  |                  |
| Carbon     | 0.21      | 0.20       | Tensile Strength N/mm <sup>2</sup>    | 430/580          | 490/640          |
| Silicon    | 0.50      | 0.50       | Yield Strength min. N/mm <sup>2</sup> | 275 <sup>†</sup> | 355 <sup>†</sup> |
| Manganese  | 1.30      | 1.50       | Min. Elongation on gauge              |                  |                  |
| Sulphur    | 0.05      | 0.045      | length $5.65\sqrt{S_0}$               | 22%              | 20%              |
| Phosphorus | 0.05      | 0.045      | Charpy V-Notch Impact Value           |                  |                  |
| Niobium    | -         | 0.003/0.10 | 10 x 10mm specimen at                 |                  |                  |
| Vanadium   | -         | 0.003/0.15 | 0°C. Min. Ave Energy Joules           | 27*              | 27               |

<sup>†</sup> up to and including 16mm thick

\* only by special arrangement

## OVERSEAS SPECIFICATIONS

Other specifications, comparable to BS 4360 can be supplied; those most frequently used are as follows:

| Ordinary Yield           | High Yield               |
|--------------------------|--------------------------|
| DIN 17100 St 44-2        | DIN 17100 St 52-3        |
| Euronorm 25 - 72 Fe 430C | Euronorm 25 - 72 Fe 510C |
| NBN A21-101 AE 255C      | NBN A21-101 AE 355C      |
| Onorm M3116 St 430C      | Onorm M3116 St 510C      |
| SS 14 14 12              | SS 14 21 34              |
| ASTM A36                 | UNI 7806 Fe 510C         |
| UNI 7806 Fe 430C         |                          |

Other sub-grades are available

## TENSILE TEST

The tensile strength, yield strength and elongation are determined from standard test pieces cut longitudinally from the sections, excluding the weld areas and away from the corners. The test pieces and testing comply with BS 18. Method for Tensile Testing of Metals.

# HOT FORMED HOLLOW SECTIONS

## IMPACT TEST

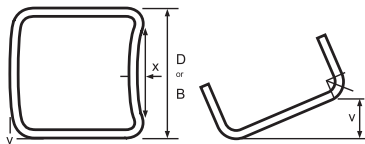
Standard impact specimens are cut longitudinally from the sections and machined on each face to 10mm x 10mm in cross section and a V notch, 2 mm deep, is cut into one face. The specimen is then tested at the appropriate temperature in accordance with BS 131: Part 2, The Charpy V-Notch Impact Test. Due to the machining, standard specimens can only be taken from material 11 mm thick, or thicker. For thinner material, tests are made on the next subsidiary standard test piece with average values in accordance with Table 15 of BS 4360.

| Average value for standard test piece | Equivalent average value for subsidiary standard test piece |                |
|---------------------------------------|-------------------------------------------------------------|----------------|
| 10mm x 10mm                           | 10mm x 7.5mm                                                | 10 mm x 5.0 mm |
| J                                     | J                                                           | J              |
| 27                                    | 22                                                          | 19             |

Unless agreed otherwise, impact tests are not carried out on Grade 43C material or where the section is less than 6 mm thick.

## DIMENSIONAL TOLERANCES

The dimensional tolerances are in accordance with BS 4848, Part 2, as follows:



|                                                  |                                                                                                                                                                                                                                                              |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) Circular Hollow Sections<br>Outside diameter | $\pm 0.5\text{mm}$ or $\pm 1\%$ whichever is the greater                                                                                                                                                                                                     |
| (b) Rectangular Hollow Sections                  |                                                                                                                                                                                                                                                              |
| (i) Outside dimensions of sides                  | $\pm 0.5\text{mm}$ or $\pm 1\%$ whichever is the greater                                                                                                                                                                                                     |
| (ii) Squareness of side                          | $90^\circ \pm 1^\circ$                                                                                                                                                                                                                                       |
| (iii) Radii of corners                           | Outside - between the limits of $0.5t$ and $2.0t$<br>Inside - between the limits of $0.5t$ and $1.5t$ where $t$ equals the specified thickness of section                                                                                                    |
| (iv) Concavity/Convexity (x)                     | $\pm 1\%$ of the length of the side, (this tolerance measured independently of the tolerance on outside dimension.)                                                                                                                                          |
| (v) Angular twist (v)                            | 2 mm plus 0.5 mm per metre maximum. Twist is measured by laying the section, as produced, on a horizontal surface with the face at one end pressed flat. The difference in height above the surface between the two corners at the opposite end is measured. |

## MASS TOLERANCE

The rolling tolerance in accordance with BS 4848, Part 2 is  $\pm 6\%$  on individual lengths; + 6% - 4% on lots of 10 tonnes and over.

## STRAIGHTNESS TOLERANCE

Unless otherwise arranged, the hollow section shall not deviate from straightness by more than 0.2% of the total length as produced, measured at the centre of the length.

## LENGTH TOLERANCE

Mill and Random lengths are supplied as in the following tables. Exacts may be supplied at our option, with a tolerance of +6 mm - 0. Lengths above 12 m in length are subject to an extra for transportation.

# HOT FORMED HOLLOW SECTIONS

## Rectangular Hollow Sections - Length Ranges & Tolerances

| Size               |                   | Welded                  |                        |              | Seamless                  |                          |
|--------------------|-------------------|-------------------------|------------------------|--------------|---------------------------|--------------------------|
| Squares mm         | Rectangles mm     | Standard Mill Lengths m | Special Mill Lengths m | Tolerance mm | Standard Random Lengths m | Special Random Lengths m |
| 20 x 20            | --                | 6.4                     | 5.4 - 7.5              | + 150 - 0    |                           |                          |
| 25 x 25 & 30 x 30  | --                | 6.4 & 7.5               | 5.4 - 7.5              | + 150 - 0    |                           |                          |
| --                 | 50 x 25           | 7.5                     | 5.4 - 7.5              | + 150 - 0    |                           |                          |
| 40 x 40 uti        | 50 x 30 uti       | 7.5, 10 & 12            | 5.4 - 13.7             | + 150 - 0    |                           |                          |
| 100 x 100 x 8      | 120 x 80 x 8      |                         |                        |              |                           |                          |
| 100 x 100 x 10 uti | 120 x 80 x 10 uti | 7.5, 10 & 12            | 6.1 - 14.6             | + 150 - 0    |                           |                          |
| 150 x 150 x 12.5   | 200 x 100 x 12.5  |                         |                        |              |                           |                          |
| 150 x 150 x 16     | 200 x 100 x 16    |                         |                        |              | 10 - 11.2                 | 5.6 - 11.2               |
| 180 x 180 to       | 250 x 150 to      | 10 & 12                 | 9 - 14.8               | + 300 - 0    |                           |                          |
| 400 x 400          | 500 x 300         |                         |                        |              |                           |                          |
| 400 x 400 x 20     | --                | 6 - 9 randoms           | --                     | --           |                           |                          |

Minimum quantities supplied from our works in one size, thickness, length and quality.

Welded - Standard Mill lengths uti 150 x 150 x 12.5 and 200 x 100 x 12.5, 2 bundles, above these sizes 10 tonnes or less at works discretion.

Special Mill lengths - 10 tonnes.

Seamless - Standard Random lengths, 2.5 tonnes.

Special Random lengths, 10 tonnes.

## Circular Hollow Sections - Length Ranges & Tolerances

| Outside Diameter mm | Welded       |                         |                        |              | Seamless     |                         |               |
|---------------------|--------------|-------------------------|------------------------|--------------|--------------|-------------------------|---------------|
|                     | Thickness mm | Standard Mill Lengths m | Special Mill Lengths m | Tolerance mm | Thickness mm | Standard Mill Lengths m | Tolerances mm |
| 21.3 & 26.9         | all          | 6.0 & 6.4               | 5.4 - 7.5              | + 150 - 0    | --           |                         |               |
| 33.7 - 48.3         | all          | 6.0, 6.4 & 7.5          | 5.4 - 7.5              | + 150 - 0    | --           |                         |               |
| 60.3 - 114.3        | all          | 6.0, 6.4, 7.5 & 10      | 5.4 - 12               | + 150 - 0    | --           |                         |               |
| 139.7 - 168.3       | all          | 7.5, 10 & 12            | 6.1 - 14.6             | + 150 - 0    | --           |                         |               |
| 193.7               | uti 12.5     | 7.5, 10 & 12            | 6.1 - 14.6             | + 150 - 0    | 16           | 8, 10 & 12              | + 300 - 0     |
| 219.1               | uti 12.5     | 10 & 12                 | 9 - 14.8               | + 300 - 0    | 16           | 8, 10 & 12              | + 300 - 0     |
|                     |              |                         |                        |              | 20           | 6, 8 & 10               | + 300 - 0     |
| 244.5               | 6.3 - 16     | 10 & 12                 | 9 - 14.8               | + 300 - 0    | 20           | 6, 8 & 10               | + 300 - 0     |
| 273.0               | 6.3 - 16     | 10 & 12                 | 9 - 14.8               | + 300 - 0    | 20           | 6, 8 & 10               | + 300 - 0     |
|                     |              |                         |                        |              | 25           | 4, 6 & 8                | + 300 - 0     |
| 323.9               | 6.3 - 16     | 10 & 12                 | 9 - 14.8               | + 300 - 0    | 20           | 6, 8 & 10               | + 300 - 0     |
|                     |              |                         |                        |              | 25           | 4, 6 & 8                | + 300 - 0     |
| 355.6               | 8.0 - 16     | 10 & 12                 | 9 - 14.8               | + 300 - 0    | 20           | 6, 8 & 10               | + 300 - 0     |
|                     |              |                         |                        |              | 25           | 4, 6 & 8                | + 300 - 0     |
| 406.4               | 10 - 16      | 10 & 12                 | 9 - 14.8               | + 300 - 0    | 20           | 8, 10 & 12              | + 300 - 0     |
|                     |              |                         |                        |              | 25           | 4, 6 & 8                | + 300 - 0     |
|                     |              |                         |                        |              | 32           | 2, 4 & 6                | + 300 - 0     |
| 457.0               | 10 - 16      | 10 & 12                 | 9 - 14.8               | + 300 - 0    | 20           | 8, 10 & 12              | + 300 - 0     |
|                     |              |                         |                        |              | 25           | 6, 8 & 10               | + 300 - 0     |
|                     |              |                         |                        |              | 32           | 4, 6 & 8                | + 300 - 0     |
|                     |              |                         |                        |              | 40           | 2, 4 & 6                | + 300 - 0     |
| 508.0               | 10 - 16      | 10 & 12                 | 9 - 14.8               | + 300 - 0    | 20           | 6, 8 & 10               | + 300 - 0     |
|                     |              |                         |                        |              | 25           | 6, 8 & 10               | + 300 - 0     |
|                     |              |                         |                        |              | 32           | 4, 6 & 8                | + 300 - 0     |
|                     |              |                         |                        |              | 40           | 2, 4 & 6                | + 300 - 0     |
|                     |              |                         |                        |              | 50           | 3, 4 & 5                | + 300 - 0     |

Minimum quantities supplied from our works in one size, thickness, length and quality.

Welded - Standard Mill lengths uti 193.7 - 4/5 tonnes. Over 193.7 - 10 tonnes or less at works discretion.

Special Mill lengths - 10 tonnes.

Seamless - Standard Mill lengths - 10 tonnes, or less at works discretion.

# HOT FORMED HOLLOW SECTIONS

## COMPARABLE SPECIFICATIONS

Compared by tensile and Charpy V-notch impacts

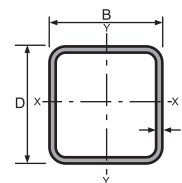
| Specification | Grade     | Tensile strength  | Yield strength<br>minimum | Charpy V-Notch<br>Impact Values* |      |
|---------------|-----------|-------------------|---------------------------|----------------------------------|------|
|               |           | N/mm <sup>2</sup> | N/mm <sup>2</sup>         | Joules                           | °C   |
| BS 4360       | 43C       | 430 - 580         | 275                       | 27                               | 0    |
| ISO 630       | Fe430C    | 430 - 530         | 275                       | 27                               | 0    |
| DIN 17100     | St44 - 2  | 410 - 540         | 275                       | 27                               | + 20 |
| DIN 17100     | St44 - 3  | 410 - 540         | 275                       | 27                               | 0    |
| EURONORM 25   | Fe430C    | 430 - 510         | 275                       | 28                               | 0    |
| SIS 14 14 12  | --        | 430 - 540         | 270                       | --                               | --   |
| UNI 7806      | Fe430C    | 430 min.          | 275                       | 28                               | 0    |
| NBN A21-101   | AE255C    | 410 - 490         | 255                       | 27                               | 0    |
| ONORM M3116   | St430C    | 430 - 540         | 275                       | 27                               | 0    |
| ASTM A36      | --        | 400 - 550         | 250                       | --                               | --   |
| ASTM A501     | --        | 400 min.          | 250                       | --                               | --   |
| NF A49-501    | E-26/3    | 370 - 510         | 255                       | 27                               | 0    |
| JIS G3466     | STKR41    | 402 min.          | 245 min.                  | --                               | --   |
| BS 4360       | 43D       | 430 - 580         | 275                       | 27                               | -20  |
| ISO 630       | Fe430D    | 430 - 530         | 275                       | 27                               | -20  |
| DIN 17100     | St44 - 3N | 410 - 540         | 275                       | 27                               | -20  |
| EURONORM 25   | Fe430D    | 430 - 510         | 275                       | 28                               | -20  |
| UNI 7806      | Fe430D    | 430 min.          | 275                       | 28                               | -20  |
| NBN A21-101   | AE255D    | 410 - 490         | 255                       | 27                               | -20  |
| ONORM M3116   | St430D    | 430 - 540         | 275                       | 27                               | -20  |
| NF A49-501    | E-26/4    | 370 - 510         | 255                       | 27                               | -20  |
| BS 4360       | 43EE      | 430 - 580         | 275                       | 27                               | -50  |
| BS 4360       | 50C       | 490 - 640         | 355                       | 27                               | 0    |
| ISO 630       | Fe 510C   | 490 - 630         | 355                       | 27                               | 0    |
| DIN 17100     | St52 - 3  | 490 - 630         | 355                       | 27                               | 0    |
| EURONORM 25   | Fe510C    | 510 - 610         | 355                       | 28                               | 0    |
| SIS 14 21 32  | --        | 510 min.          | 350                       | --                               | --   |
| UNI 7806      | Fe510C    | 510 min.          | 355                       | 27                               | 0    |
| NBN A21-101   | AE355C    | 510 - 610         | 355                       | 27                               | 0    |
| ONORM M3116   | St510C    | 510 - 630         | 355                       | 27                               | 0    |
| NF A49-501    | E-36/3    | 470 - 610         | 355                       | 28                               | 0    |
| ASTM A618     | --        | 450 min.          | 345                       | --                               | --   |
| JIS G 3466    | STKR50    | 490 min.          | 324 min.                  | --                               | --   |
| BS 4360       | 50D       | 490 - 640         | 355                       | 27                               | -20  |
| ISO 630       | Fe510D    | 490 - 630         | 355                       | 27                               | -20  |
| DIN 17100     | St52 - 3N | 490 - 630         | 355                       | 27                               | -20  |
| EURONORM 25   | Fe510D    | 510 - 610         | 355                       | 28                               | -20  |
| SIS 14 21 34  | --        | 470 - 630         | 350                       | 27                               | -20  |
| UNI 7806      | Fe510D    | 510 min.          | 355                       | 27                               | -20  |
| ONORM M3116   | St510D    | 510 - 630         | 355                       | 27                               | -20  |
| BS 4360       | 50EE      | 490 - 640         | 355                       | 27                               | -50  |
| NBN A21-101   | AE355D    | 510 - 610         | 355                       | 40                               | -20  |
| NF A49-501    | E-36/4    | 470 - 610         | 355                       | 40                               | -20  |
| NORSEC 360    | --        | 490 - 630         | 360                       | 100                              | -20  |
|               |           |                   |                           | 50                               | -60  |
| BS 4360       | 55C       | 550 - 700         | 450                       | 27                               | 0    |

Note:

\* Average value on full size specimens, 10 mm x 10 mm



# HOT FORMED HOLLOW SECTIONS

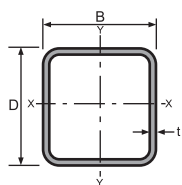


## SQUARE

| Section size<br>D x B | Thickness<br>t | Unit weight<br>M | Section area<br>A | Moment of inertia<br>I | Radius of gyration<br>r | Elastic modulus<br>Z | Plastic modulus<br>S | Torsional<br>J  | Constants<br>C  | Superficial area per m | Approx. length per tonne |
|-----------------------|----------------|------------------|-------------------|------------------------|-------------------------|----------------------|----------------------|-----------------|-----------------|------------------------|--------------------------|
| mm                    | mm             | kg/m             | cm <sup>2</sup>   | cm <sup>4</sup>        | cm                      | cm <sup>3</sup>      | cm <sup>3</sup>      | cm <sup>4</sup> | cm <sup>3</sup> | m <sup>2</sup>         | m                        |
| 20 x 20               | 2.0            | 1.12             | 1.42              | 0.76                   | 0.73                    | 0.76                 | 0.95                 | 1.22            | 1.07            | 0.076                  | 893                      |
|                       | 2.5*           | 1.35             | 1.72              | 0.87                   | 0.71                    | 0.88                 | 1.12                 | 1.41            | 1.21            | 0.075                  | 741                      |
| 25 x 25               | 2.0*           | 1.43             | 1.82              | 1.59                   | 0.94                    | 1.27                 | 1.56                 | 2.52            | 1.81            | 0.096                  | 700                      |
|                       | 2.5*           | 1.74             | 2.22              | 1.85                   | 0.91                    | 1.48                 | 1.86                 | 2.97            | 2.09            | 0.095                  | 574                      |
|                       | 3.0*           | 2.04             | 2.60              | 2.06                   | 0.89                    | 1.65                 | 2.12                 | 3.36            | 2.31            | 0.094                  | 490                      |
|                       | 3.2*           | 2.15             | 2.74              | 2.14                   | 0.88                    | 1.71                 | 2.21                 | 3.49            | 2.38            | 0.093                  | 466                      |
| 30 x 30               | 2.5*           | 2.14             | 2.72              | 3.40                   | 1.12                    | 2.27                 | 2.79                 | 5.40            | 3.22            | 0.115                  | 469                      |
|                       | 3.0*           | 2.51             | 3.20              | 3.84                   | 1.10                    | 2.56                 | 3.21                 | 6.17            | 3.61            | 0.114                  | 399                      |
|                       | 3.2            | 2.65             | 3.38              | 4.00                   | 1.09                    | 2.67                 | 3.37                 | 6.45            | 3.75            | 0.113                  | 378                      |
| 40 x 40               | 2.5*           | 2.92             | 3.72              | 8.67                   | 1.53                    | 4.33                 | 5.21                 | 13.6            | 6.23            | 0.155                  | 343                      |
|                       | 3.0*           | 3.45             | 4.40              | 9.96                   | 1.51                    | 4.98                 | 6.07                 | 15.7            | 7.11            | 0.154                  | 290                      |
|                       | 3.2            | 3.66             | 4.66              | 10.4                   | 1.50                    | 5.22                 | 6.40                 | 16.5            | 7.43            | 0.153                  | 274                      |
|                       | 4.0            | 4.46             | 5.68              | 12.1                   | 1.46                    | 6.07                 | 7.61                 | 19.5            | 8.56            | 0.151                  | 225                      |
|                       | 5.0*           | 5.40             | 6.88              | 13.8                   | 1.42                    | 6.92                 | 8.92                 | 22.6            | 9.65            | 0.149                  | 185                      |
| 50 x 50               | 2.5*           | 3.71             | 4.72              | 17.7                   | 1.94                    | 7.07                 | 8.38                 | 27.4            | 10.2            | 0.195                  | 270                      |
|                       | 3.0*           | 4.39             | 5.60              | 20.5                   | 1.91                    | 8.20                 | 9.83                 | 32.0            | 11.8            | 0.194                  | 228                      |
|                       | 3.2            | 4.66             | 5.94              | 21.6                   | 1.91                    | 8.62                 | 10.4                 | 33.8            | 12.4            | 0.193                  | 215                      |
|                       | 4.0            | 5.72             | 7.28              | 25.5                   | 1.87                    | 10.2                 | 12.5                 | 40.4            | 14.5            | 0.191                  | 175                      |
|                       | 5.0            | 6.97             | 8.88              | 29.6                   | 1.83                    | 11.9                 | 14.9                 | 47.6            | 16.7            | 0.189                  | 144                      |
|                       | 6.0            | 8.15             | 10.4              | 33.0                   | 1.78                    | 13.2                 | 17.0                 | 53.7            | 18.5            | 0.187                  | 123                      |
|                       | 6.3*           | 8.49             | 10.8              | 33.9                   | 1.77                    | 13.6                 | 17.5                 | 55.3            | 18.9            | 0.186                  | 118                      |
| 60 x 60               | 3.0*           | 5.34             | 6.80              | 36.6                   | 2.32                    | 12.2                 | 14.5                 | 56.9            | 17.7            | 0.234                  | 188                      |
|                       | 3.2            | 5.67             | 7.22              | 38.7                   | 2.31                    | 12.9                 | 15.3                 | 60.1            | 18.6            | 0.233                  | 177                      |
|                       | 4.0            | 6.97             | 8.88              | 46.1                   | 2.28                    | 15.4                 | 18.6                 | 72.4            | 22.1            | 0.231                  | 144                      |
|                       | 5.0            | 8.54             | 10.9              | 54.4                   | 2.24                    | 18.1                 | 22.3                 | 86.3            | 25.8            | 0.229                  | 118                      |
|                       | 6.0            | 10.0             | 12.8              | 61.5                   | 2.19                    | 20.5                 | 25.7                 | 98.6            | 28.9            | 0.227                  | 100                      |
|                       | 6.3*           | 10.5             | 13.3              | 63.4                   | 2.18                    | 21.1                 | 26.6                 | 102             | 29.7            | 0.226                  | 95.3                     |
|                       | 8.0*           | 12.8             | 16.3              | 72.4                   | 2.11                    | 24.1                 | 31.4                 | 119             | 33.5            | 0.223                  | 78.1                     |
| 70 x 70               | 3.0*           | 6.28             | 8.0               | 59.6                   | 2.73                    | 17.0                 | 20.0                 | 92.1            | 24.8            | 0.274                  | 160                      |
|                       | 3.6            | 7.46             | 9.5               | 69.5                   | 2.70                    | 19.9                 | 23.6                 | 108             | 28.7            | 0.272                  | 135                      |
|                       | 5.0            | 10.1             | 12.9              | 90.1                   | 2.64                    | 25.7                 | 31.2                 | 142             | 36.8            | 0.269                  | 99.1                     |
|                       | 6.0            | 11.9             | 15.2              | 103                    | 2.60                    | 29.4                 | 36.2                 | 163             | 41.7            | 0.267                  | 84.0                     |
|                       | 6.3*           | 12.5             | 15.9              | 106                    | 2.59                    | 30.4                 | 37.6                 | 169             | 43.0            | 0.266                  | 80.0                     |
|                       | 8.0*           | 15.3             | 19.5              | 123                    | 2.51                    | 35.3                 | 45.0                 | 200             | 49.4            | 0.263                  | 65.4                     |
| 75 x 75               | 3.0            | 6.75             | 8.6               | 74.0                   | 2.93                    | 19.7                 | 23.2                 | 114             | 28.8            | 0.294                  | 148                      |
|                       | 3.6            | 8.02             | 10.2              | 86.5                   | 2.91                    | 23.1                 | 27.3                 | 134             | 33.4            | 0.292                  | 125                      |
|                       | 5.0            | 10.9             | 13.9              | 113                    | 2.85                    | 30.0                 | 36.3                 | 177             | 43.1            | 0.289                  | 91.7                     |
|                       | 6.0            | 12.9             | 16.4              | 129                    | 2.81                    | 34.4                 | 42.2                 | 204             | 49.0            | 0.287                  | 77.5                     |
|                       | 6.3            | 13.4             | 17.1              | 134                    | 2.79                    | 35.6                 | 43.8                 | 212             | 50.6            | 0.286                  | 74.6                     |
|                       | 8.0            | 16.6             | 21.1              | 156                    | 2.72                    | 41.7                 | 52.6                 | 252             | 58.6            | 0.283                  | 60.2                     |
| 80 x 80               | 3.0*           | 7.22             | 9.2               | 90.6                   | 3.14                    | 22.7                 | 26.5                 | 139             | 33.1            | 0.314                  | 139                      |
|                       | 3.6            | 8.59             | 10.9              | 106                    | 3.11                    | 26.5                 | 31.3                 | 164             | 38.5            | 0.312                  | 117                      |
|                       | 5.0            | 11.7             | 14.9              | 139                    | 3.05                    | 34.7                 | 41.7                 | 217             | 49.8            | 0.309                  | 85.5                     |
|                       | 6.0            | 13.8             | 17.6              | 159                    | 3.01                    | 39.8                 | 48.6                 | 251             | 56.9            | 0.307                  | 72.5                     |
|                       | 6.3            | 14.4             | 18.4              | 165                    | 3.00                    | 41.3                 | 50.5                 | 261             | 58.8            | 0.306                  | 69.5                     |
|                       | 8.0*           | 17.8             | 22.7              | 194                    | 2.92                    | 48.6                 | 60.9                 | 312             | 68.5            | 0.303                  | 56.2                     |

Notes:  
 Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.  
 \* Size not included in BS 4848 Part 2  
 † 43C only  
 Length per tonne rounded up

# HOT FORMED HOLLOW SECTIONS



## SQUARE

| Section size<br>D x B | Thickness<br>t | Unit weight<br>M | Section area<br>A | Moment of inertia<br>I | Radius of gyration<br>r | Elastic modulus<br>Z | Plastic modulus<br>S | Torsional<br>J  | Constants<br>C  | Superficial area<br>per m | Approx. length<br>per tonne |
|-----------------------|----------------|------------------|-------------------|------------------------|-------------------------|----------------------|----------------------|-----------------|-----------------|---------------------------|-----------------------------|
| mm                    | mm             | kg/m             | cm <sup>2</sup>   | cm <sup>4</sup>        | cm                      | cm <sup>3</sup>      | cm <sup>3</sup>      | cm <sup>4</sup> | cm <sup>3</sup> | m <sup>2</sup>            | m                           |
| 90 x 90               | 3.6            | 9.72             | 12.4              | 154                    | 3.52                    | 34.1                 | 40.0                 | 237             | 49.7            | 0.352                     | 103                         |
|                       | 5.0            | 13.3             | 16.9              | 202                    | 3.46                    | 45.0                 | 53.6                 | 315             | 64.9            | 0.349                     | 75.2                        |
|                       | 6.0            | 15.7             | 20.0              | 234                    | 3.42                    | 51.9                 | 62.7                 | 367             | 74.4            | 0.347                     | 63.7                        |
|                       | 6.3            | 16.4             | 20.9              | 242                    | 3.41                    | 53.9                 | 65.3                 | 381             | 77.1            | 0.346                     | 61.0                        |
|                       | 8.0*           | 20.4             | 25.9              | 288                    | 3.33                    | 64.0                 | 79.2                 | 459             | 90.7            | 0.343                     | 49.0                        |
| 100 x 100             | 4.0            | 12.0             | 15.3              | 234                    | 3.91                    | 46.8                 | 54.9                 | 361             | 68.2            | 0.391                     | 83.4                        |
|                       | 5.0            | 14.8             | 18.9              | 283                    | 3.87                    | 56.6                 | 67.1                 | 439             | 81.9            | 0.389                     | 67.6                        |
|                       | 6.0            | 17.6             | 22.4              | 328                    | 3.83                    | 65.6                 | 78.6                 | 512             | 94.4            | 0.387                     | 56.8                        |
|                       | 6.3            | 18.4             | 23.4              | 341                    | 3.81                    | 68.2                 | 82.0                 | 533             | 97.9            | 0.386                     | 54.4                        |
|                       | 8.0            | 22.9             | 29.1              | 408                    | 3.74                    | 81.5                 | 99.9                 | 646             | 116             | 0.383                     | 43.7                        |
|                       | 10.0           | 27.9             | 35.5              | 474                    | 3.65                    | 94.9                 | 119                  | 761             | 134             | 0.379                     | 35.9                        |
| 120 x 120             | 5.0            | 18.0             | 22.9              | 503                    | 4.69                    | 83.8                 | 98.4                 | 775             | 122             | 0.469                     | 55.6                        |
|                       | 6.0            | 21.3             | 27.2              | 586                    | 4.64                    | 97.7                 | 116                  | 910             | 141             | 0.467                     | 46.9                        |
|                       | 6.3            | 22.3             | 28.5              | 610                    | 4.63                    | 102                  | 121                  | 949             | 147             | 0.466                     | 44.9                        |
|                       | 8.0            | 27.9             | 35.5              | 738                    | 4.56                    | 123                  | 149                  | 1,159           | 176             | 0.463                     | 35.9                        |
|                       | 10.0           | 34.2             | 43.5              | 870                    | 4.47                    | 145                  | 178                  | 1,381           | 206             | 0.459                     | 29.3                        |
|                       | 12.5*†         | 41.6             | 53.0              | 1,009                  | 4.36                    | 168                  | 212                  | 1,624           | 237             | 0.453                     | 24.0                        |
| 140 x 140             | 5.0*           | 21.1             | 26.9              | 814                    | 5.50                    | 116                  | 136                  | 1,251           | 170             | 0.549                     | 47.4                        |
|                       | 5.6*           | 23.5             | 30.0              | 899                    | 5.48                    | 128                  | 151                  | 1,385           | 187             | 0.548                     | 42.6                        |
|                       | 6.0            | 25.1             | 32.0              | 954                    | 5.46                    | 136                  | 160                  | 1,473           | 198             | 0.547                     | 39.8                        |
|                       | 6.3*           | 26.3             | 33.5              | 994                    | 5.45                    | 142                  | 168                  | 1,538           | 206             | 0.546                     | 38.0                        |
|                       | 7.1*           | 29.4             | 37.5              | 1,099                  | 5.41                    | 157                  | 186                  | 1,706           | 227             | 0.545                     | 34.0                        |
|                       | 8.0*           | 32.9             | 41.9              | 1,212                  | 5.38                    | 173                  | 207                  | 1,889           | 249             | 0.543                     | 30.4                        |
|                       | 10.0*          | 40.4             | 51.5              | 1,441                  | 5.29                    | 206                  | 250                  | 2,269           | 294             | 0.539                     | 24.8                        |
|                       | 12.5*          | 49.5             | 63.0              | 1,691                  | 5.18                    | 242                  | 299                  | 2,695           | 342             | 0.533                     | 20.2                        |
| 150 x 150             | 5.0            | 22.7             | 28.9              | 1,009                  | 5.91                    | 135                  | 157                  | 1,548           | 197             | 0.589                     | 44.1                        |
|                       | 6.0            | 27.0             | 34.4              | 1,185                  | 5.87                    | 158                  | 185                  | 1,826           | 230             | 0.587                     | 37.0                        |
|                       | 6.3            | 28.3             | 36.0              | 1,236                  | 5.86                    | 165                  | 194                  | 1,907           | 240             | 0.586                     | 35.4                        |
|                       | 8.0            | 35.4             | 45.1              | 1,510                  | 5.78                    | 201                  | 240                  | 2,348           | 291             | 0.583                     | 28.3                        |
|                       | 10.0           | 43.6             | 55.5              | 1,803                  | 5.70                    | 240                  | 290                  | 2,829           | 345             | 0.579                     | 23.0                        |
|                       | 12.5           | 53.4             | 68.0              | 2,125                  | 5.59                    | 283                  | 348                  | 3,372           | 403             | 0.573                     | 18.8                        |
|                       | 16.0           | 66.4             | 84.5              | 2,500                  | 5.44                    | 333                  | 421                  | 4,029           | 468             | 0.566                     | 15.1                        |

### Notes:

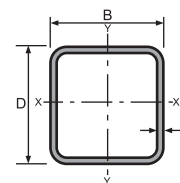
Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.

\* Size not included in BS 4848 Part 2

† 43C only

Length per tonne rounded up

# HOT FORMED HOLLOW SECTIONS

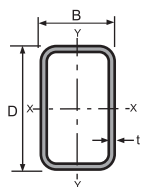


## SQUARE

| Section size<br>D x B | Thickness<br>t | Unit weight<br>M | Section area<br>A | Moment of inertia<br>I | Radius of gyration<br>r | Elastic modulus<br>Z | Plastic modulus<br>S | Torsional<br>J  | Constants<br>C  | Superficial area<br>per m | Approx. length<br>per tonne |
|-----------------------|----------------|------------------|-------------------|------------------------|-------------------------|----------------------|----------------------|-----------------|-----------------|---------------------------|-----------------------------|
| mm                    | mm             | kg/m             | cm <sup>2</sup>   | cm <sup>4</sup>        | cm                      | cm <sup>3</sup>      | cm <sup>3</sup>      | cm <sup>4</sup> | cm <sup>3</sup> | m <sup>2</sup>            | m                           |
| 180 x 180             | 6.0            | 32.6             | 41.6              | 2,093                  | 7.09                    | 233                  | 271                  | 3,211           | 340             | 0.707                     | 30.7                        |
|                       | 6.3            | 34.2             | 43.6              | 2,186                  | 7.08                    | 243                  | 283                  | 3,357           | 355             | 0.706                     | 29.3                        |
|                       | 8.0            | 43.0             | 54.7              | 2,689                  | 7.01                    | 299                  | 352                  | 4,156           | 434             | 0.703                     | 23.3                        |
|                       | 10.0           | 53.0             | 67.5              | 3,237                  | 6.92                    | 360                  | 429                  | 5,041           | 519             | 0.699                     | 18.9                        |
|                       | 12.5           | 65.2             | 83.0              | 3,856                  | 6.82                    | 428                  | 519                  | 6,062           | 613             | 0.693                     | 15.4                        |
|                       | 16.0           | 81.4             | 104               | 4,607                  | 6.66                    | 512                  | 634                  | 7,339           | 725             | 0.686                     | 12.3                        |
| 200 x 200             | 6.0            | 36.4             | 46.4              | 2,903                  | 7.91                    | 290                  | 337                  | 4,444           | 426             | 0.787                     | 27.5                        |
|                       | 6.3            | 38.2             | 48.6              | 3,033                  | 7.90                    | 303                  | 353                  | 4,647           | 444             | 0.786                     | 26.2                        |
|                       | 8.0            | 48.0             | 61.1              | 3,744                  | 7.83                    | 374                  | 439                  | 5,770           | 545             | 0.783                     | 20.9                        |
|                       | 10.0           | 59.3             | 75.5              | 4,525                  | 7.74                    | 452                  | 536                  | 7,020           | 655             | 0.779                     | 16.9                        |
|                       | 12.5           | 73.0             | 93.0              | 5,419                  | 7.63                    | 542                  | 651                  | 8,479           | 779             | 0.773                     | 13.7                        |
|                       | 16.0           | 91.5             | 117               | 6,524                  | 7.48                    | 652                  | 799                  | 10,330          | 929             | 0.766                     | 11.0                        |
| 250 x 250             | 6.0            | 45.8             | 58.4              | 5,784                  | 9.95                    | 463                  | 534                  | 8,816           | 681             | 0.987                     | 21.8                        |
|                       | 6.3            | 48.1             | 61.2              | 6,049                  | 9.94                    | 484                  | 559                  | 9,288           | 712             | 0.986                     | 20.8                        |
|                       | 8.0            | 60.5             | 77.1              | 7,510                  | 9.87                    | 601                  | 699                  | 11,511          | 880             | 0.983                     | 16.6                        |
|                       | 10.0           | 75.0             | 95.5              | 9,141                  | 9.78                    | 731                  | 858                  | 14,086          | 1,065           | 0.979                     | 13.4                        |
|                       | 12.5           | 92.6             | 118               | 11,050                 | 9.68                    | 884                  | 1,048                | 17,139          | 1,279           | 0.973                     | 10.8                        |
|                       | 16.0           | 117              | 149               | 13,480                 | 9.53                    | 1,078                | 1,298                | 21,109          | 1,548           | 0.966                     | 8.55                        |
| 300 x 300             | 8.0*           | 73.1             | 93.1              | 13,209                 | 11.90                   | 881                  | 1,018                | 20,171          | 1,294           | 1.18                      | 13.7                        |
|                       | 10.0           | 90.7             | 116               | 16,150                 | 11.80                   | 1,077                | 1,254                | 24,776          | 1,575           | 1.18                      | 11.1                        |
|                       | 12.5           | 112              | 143               | 19,630                 | 11.70                   | 1,309                | 1,538                | 30,290          | 1,905           | 1.17                      | 8.93                        |
|                       | 16.0           | 142              | 181               | 24,160                 | 11.60                   | 1,610                | 1,916                | 37,566          | 2,327           | 1.17                      | 7.05                        |
| 350 x 350             | 8.0*           | 85.7             | 109               | 21,240                 | 14.00                   | 1,214                | 1,398                | 32,352          | 1,789           | 1.38                      | 11.7                        |
|                       | 10.0           | 106              | 136               | 26,050                 | 13.90                   | 1,489                | 1,725                | 39,840          | 2,186           | 1.38                      | 9.44                        |
|                       | 12.5           | 132              | 168               | 31,810                 | 13.80                   | 1,817                | 2,122                | 48,869          | 2,655           | 1.37                      | 7.58                        |
|                       | 16.0           | 167              | 213               | 39,370                 | 13.60                   | 2,250                | 2,655                | 60,901          | 3,265           | 1.37                      | 5.99                        |
| 400 x 400             | 10.0           | 122              | 156               | 39,350                 | 15.90                   | 1,968                | 2,272                | 60,028          | 2,896           | 1.58                      | 8.20                        |
|                       | 12.5           | 152              | 193               | 48,190                 | 15.80                   | 2,409                | 2,800                | 73,815          | 3,530           | 1.57                      | 6.58                        |
|                       | 16.0           | 192              | 245               | 59,910                 | 15.70                   | 2,995                | 3,514                | 92,310          | 4,363           | 1.57                      | 5.21                        |
|                       | 20.0           | 237              | 302               | 72,396                 | 15.50                   | 3,620                | 4,292                | 112,324         | 5,240           | 1.56                      | 4.22                        |

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 \* Size not included in BS 4848 Part 2  
 † 43C only  
 Length per tonne rounded up

# HOT FORMED HOLLOW SECTIONS



## RECTANGULAR

| Section size<br>D x B | Thickness<br>t | Unit weight<br>M | Section area<br>A | Moment of inertia |                 | Radius of gyration<br>r |      | Elastic modulus<br>Z |                 | Plastic modulus<br>S |                 | Torsional constants<br>J C |                 | Superficial area<br>per m | Approx. length<br>per tonne |
|-----------------------|----------------|------------------|-------------------|-------------------|-----------------|-------------------------|------|----------------------|-----------------|----------------------|-----------------|----------------------------|-----------------|---------------------------|-----------------------------|
|                       |                |                  |                   | x-x               | y-y             | x-x                     | y-y  | x-x                  | y-y             | x-x                  | y-y             | x-x                        | y-y             |                           |                             |
| mm                    | mm             | kg/m             | cm <sup>2</sup>   | cm <sup>4</sup>   | cm <sup>4</sup> | cm                      | cm   | cm <sup>3</sup>      | cm <sup>3</sup> | cm <sup>3</sup>      | cm <sup>3</sup> | cm <sup>4</sup>            | cm <sup>3</sup> | m <sup>2</sup>            | m                           |
| 50 x 25               | 2.5*           | 2.72             | 3.47              | 10.5              | 3.44            | 1.75                    | 1.00 | 4.25                 | 2.75            | 5.41                 | 3.25            | 8.41                       | 4.62            | 0.145                     | 368                         |
|                       | 3.0*           | 3.22             | 4.10              | 12.2              | 3.89            | 1.73                    | 0.98 | 4.88                 | 3.11            | 6.30                 | 3.77            | 9.64                       | 5.21            | 0.144                     | 311                         |
|                       | 3.2*           | 3.41             | 4.34              | 12.8              | 4.05            | 1.72                    | 0.97 | 5.11                 | 3.24            | 6.64                 | 3.96            | 10.1                       | 5.42            | 0.143                     | 294                         |
| 50 x 30               | 2.5*           | 2.92             | 3.72              | 12.0              | 5.30            | 1.80                    | 1.19 | 4.81                 | 3.53            | 6.01                 | 4.16            | 11.7                       | 5.74            | 0.155                     | 343                         |
|                       | 3.0*           | 3.45             | 4.40              | 13.9              | 6.04            | 1.78                    | 1.17 | 5.54                 | 4.03            | 7.01                 | 4.83            | 13.5                       | 6.52            | 0.154                     | 290                         |
|                       | 3.2            | 3.66             | 4.66              | 14.5              | 6.31            | 1.77                    | 1.16 | 5.82                 | 4.21            | 7.39                 | 5.08            | 14.2                       | 6.81            | 0.153                     | 274                         |
|                       | 4.0*           | 4.46             | 5.68              | 17.0              | 7.25            | 1.73                    | 1.13 | 6.80                 | 4.83            | 8.81                 | 6.01            | 16.6                       | 7.79            | 0.151                     | 224                         |
|                       | 5.0*           | 5.40             | 6.88              | 19.5              | 8.13            | 1.68                    | 1.09 | 7.79                 | 5.42            | 10.4                 | 6.98            | 19.0                       | 8.71            | 0.149                     | 185                         |
| 60 x 40               | 2.5*           | 3.71             | 4.72              | 23.1              | 12.2            | 2.21                    | 1.61 | 7.71                 | 6.10            | 9.43                 | 7.09            | 25.0                       | 9.74            | 0.195                     | 270                         |
|                       | 3.0*           | 4.39             | 5.60              | 26.9              | 14.1            | 2.19                    | 1.59 | 8.96                 | 7.04            | 11.1                 | 8.29            | 29.2                       | 11.2            | 0.194                     | 228                         |
|                       | 3.2            | 4.66             | 5.94              | 28.3              | 14.8            | 2.18                    | 1.58 | 9.44                 | 7.39            | 11.7                 | 8.75            | 30.8                       | 11.8            | 0.193                     | 215                         |
|                       | 4.0            | 5.72             | 7.28              | 33.6              | 17.3            | 2.15                    | 1.54 | 11.2                 | 8.67            | 14.1                 | 10.5            | 36.6                       | 13.7            | 0.191                     | 175                         |
|                       | 5.0*           | 6.97             | 8.88              | 39.2              | 20.0            | 2.10                    | 1.50 | 13.1                 | 10.0            | 16.8                 | 12.4            | 43.0                       | 15.8            | 0.189                     | 143                         |
|                       | 6.0*           | 8.15             | 10.4              | 43.9              | 22.1            | 2.06                    | 1.46 | 14.6                 | 11.0            | 19.2                 | 14.1            | 48.3                       | 17.3            | 0.187                     | 129                         |
|                       | 6.3*           | 8.49             | 10.8              | 45.1              | 22.6            | 2.04                    | 1.45 | 15.0                 | 11.3            | 19.9                 | 14.6            | 49.7                       | 17.7            | 0.186                     | 118                         |
| 80 x 40               | 3.0*           | 5.34             | 6.80              | 55.0              | 18.2            | 2.85                    | 1.64 | 13.8                 | 9.10            | 17.3                 | 10.5            | 43.7                       | 15.3            | 0.234                     | 188                         |
|                       | 3.2            | 5.67             | 7.22              | 58.1              | 19.1            | 2.84                    | 1.63 | 14.5                 | 9.56            | 18.3                 | 11.1            | 46.1                       | 16.1            | 0.233                     | 177                         |
|                       | 4.0            | 6.97             | 8.88              | 69.6              | 22.6            | 2.80                    | 1.59 | 17.4                 | 11.3            | 22.2                 | 13.4            | 55.1                       | 18.9            | 0.231                     | 144                         |
|                       | 5.0*           | 8.54             | 10.9              | 82.4              | 26.2            | 2.75                    | 1.55 | 20.6                 | 13.1            | 26.7                 | 15.9            | 65.0                       | 21.9            | 0.229                     | 117                         |
|                       | 6.0            | 10.0             | 12.8              | 93.5              | 29.1            | 2.70                    | 1.51 | 23.4                 | 14.5            | 30.8                 | 18.2            | 73.5                       | 24.3            | 0.227                     | 100                         |
|                       | 6.3*           | 10.5             | 13.3              | 96.5              | 29.8            | 2.69                    | 1.50 | 24.1                 | 14.9            | 31.9                 | 18.8            | 75.8                       | 24.9            | 0.226                     | 95.2                        |
|                       | 8.0*           | 12.8             | 16.3              | 111               | 33.1            | 2.61                    | 1.42 | 27.7                 | 16.6            | 37.8                 | 21.8            | 86.3                       | 27.6            | 0.223                     | 78.1                        |
| 90 x 50               | 3.0*           | 6.28             | 8.0               | 85.4              | 33.8            | 3.27                    | 2.05 | 19.0                 | 13.5            | 23.4                 | 15.5            | 76.4                       | 22.4            | 0.274                     | 160                         |
|                       | 3.6            | 7.46             | 9.5               | 99.8              | 39.1            | 3.24                    | 2.03 | 22.2                 | 15.6            | 27.6                 | 18.1            | 89.3                       | 25.9            | 0.272                     | 135                         |
|                       | 5.0            | 10.1             | 12.9              | 130               | 50.0            | 3.18                    | 1.97 | 28.9                 | 20.0            | 36.6                 | 23.9            | 116                        | 32.9            | 0.269                     | 99.1                        |
|                       | 6.0            | 11.9             | 15.2              | 149               | 56.4            | 3.13                    | 1.93 | 33.1                 | 22.6            | 42.5                 | 27.5            | 133                        | 37.0            | 0.267                     | 84                          |
|                       | 6.3*           | 12.5             | 15.9              | 154               | 58.1            | 3.12                    | 1.91 | 34.2                 | 23.3            | 44.2                 | 28.5            | 138                        | 38.2            | 0.266                     | 80                          |
|                       | 8.0*           | 15.3             | 19.5              | 180               | 66.3            | 3.04                    | 1.84 | 40.0                 | 26.5            | 53.0                 | 33.7            | 161                        | 43.4            | 0.263                     | 65.4                        |
| 100 x 50              | 3.0*           | 6.75             | 8.60              | 111               | 37.1            | 3.59                    | 2.08 | 22.2                 | 14.8            | 27.6                 | 16.9            | 88.3                       | 25.0            | 0.294                     | 149                         |
|                       | 3.2            | 7.18             | 9.14              | 117               | 39.1            | 3.58                    | 2.07 | 23.5                 | 15.6            | 29.2                 | 17.9            | 93.3                       | 26.4            | 0.293                     | 140                         |
|                       | 4.0            | 8.86             | 11.3              | 142               | 46.7            | 3.55                    | 2.03 | 28.4                 | 18.7            | 35.7                 | 21.7            | 113                        | 31.4            | 0.291                     | 113                         |
|                       | 5.0            | 10.9             | 13.9              | 170               | 55.1            | 3.50                    | 1.99 | 34.0                 | 22.0            | 43.3                 | 26.1            | 135                        | 37.0            | 0.289                     | 91.8                        |
|                       | 6.0            | 12.9             | 16.4              | 195               | 62.3            | 3.45                    | 1.95 | 39.0                 | 24.9            | 50.4                 | 30.2            | 154                        | 41.7            | 0.287                     | 77.5                        |
|                       | 6.3*           | 13.4             | 17.1              | 202               | 64.2            | 3.44                    | 1.94 | 40.5                 | 25.7            | 52.5                 | 31.3            | 160                        | 43.0            | 0.286                     | 74.7                        |
|                       | 8.0*           | 16.6             | 21.1              | 238               | 73.5            | 3.36                    | 1.86 | 47.6                 | 29.4            | 63.1                 | 37.1            | 187                        | 49.1            | 0.283                     | 60.2                        |
|                       | 8.0*           | 16.6             | 21.1              | 238               | 73.5            | 3.36                    | 1.86 | 47.6                 | 29.4            | 63.1                 | 37.1            | 187                        | 49.1            | 0.283                     | 60.2                        |
| 100 x 60              | 3.0*           | 7.22             | 9.2               | 125               | 56.2            | 3.69                    | 2.47 | 25.0                 | 18.7            | 30.5                 | 21.3            | 121                        | 30.7            | 0.314                     | 139                         |
|                       | 3.6*           | 8.59             | 10.9              | 147               | 65.4            | 3.66                    | 2.45 | 29.3                 | 21.8            | 36.0                 | 25.1            | 142                        | 35.6            | 0.312                     | 117                         |
|                       | 5.0            | 11.7             | 14.9              | 192               | 84.7            | 3.60                    | 2.39 | 38.5                 | 28.2            | 48.1                 | 33.3            | 187                        | 45.9            | 0.309                     | 85.5                        |
|                       | 6.0            | 13.8             | 17.6              | 222               | 96.6            | 3.55                    | 2.34 | 44.4                 | 32.2            | 56.1                 | 38.6            | 216                        | 52.2            | 0.307                     | 72.5                        |
|                       | 6.3            | 14.4             | 18.4              | 230               | 99.9            | 3.54                    | 2.33 | 46.0                 | 33.3            | 58.4                 | 40.2            | 224                        | 53.9            | 0.306                     | 69.5                        |
|                       | 8.0*           | 17.8             | 22.7              | 272               | 116             | 3.46                    | 2.26 | 54.4                 | 38.7            | 70.5                 | 48.1            | 266                        | 62.4            | 0.303                     | 56.2                        |
| 120 x 60              | 3.6            | 9.72             | 12.4              | 230               | 76.9            | 4.31                    | 2.49 | 38.3                 | 25.6            | 47.6                 | 29.2            | 183                        | 43.3            | 0.352                     | 103                         |
|                       | 5.0            | 13.3             | 16.9              | 304               | 99.9            | 4.24                    | 2.43 | 50.7                 | 33.3            | 63.9                 | 38.8            | 242                        | 56.0            | 0.349                     | 75.2                        |
|                       | 6.0            | 15.7             | 20.0              | 352               | 114             | 4.20                    | 2.39 | 58.7                 | 38.1            | 74.8                 | 45.1            | 279                        | 63.9            | 0.347                     | 63.7                        |
|                       | 6.3            | 16.4             | 20.9              | 366               | 118             | 4.18                    | 2.38 | 61.0                 | 39.4            | 78.0                 | 46.9            | 290                        | 66.0            | 0.346                     | 61                          |
|                       | 8.0*           | 20.4             | 25.9              | 437               | 138             | 4.10                    | 2.31 | 72.8                 | 45.9            | 94.8                 | 56.4            | 344                        | 76.8            | 0.343                     | 49                          |

### Notes:

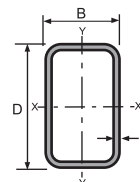
Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.

\* Size not included in BS 4848 Part 2

† 43C only

Length per tonne rounded up

# HOT FORMED HOLLOW SECTIONS



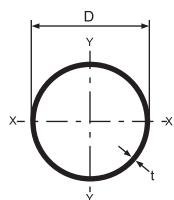
## RECTANGULAR

| Section size | Thickness | Unit weight | Section area    | Moment of inertia |                 | Radius of gyration |      | Elastic modulus |                 | Plastic modulus |                 | Torsional constants |                 | Superficial area per m | Approx. length per t |
|--------------|-----------|-------------|-----------------|-------------------|-----------------|--------------------|------|-----------------|-----------------|-----------------|-----------------|---------------------|-----------------|------------------------|----------------------|
| D x B        | t         | M           | A               | x-x               | y-y             | x-x                | y-y  | x-x             | y-y             | x-x             | y-y             | J                   | C               | m <sup>2</sup>         | m                    |
| mm           | mm        | kg/m        | cm <sup>2</sup> | cm <sup>4</sup>   | cm <sup>4</sup> | cm                 | cm   | cm <sup>3</sup> | cm <sup>3</sup> | cm <sup>3</sup> | cm <sup>3</sup> | cm <sup>4</sup>     | cm <sup>3</sup> |                        |                      |
| 120 x 80     | 5.0       | 14.8        | 18.9            | 370               | 195             | 4.43               | 3.21 | 61.7            | 48.8            | 75.4            | 56.7            | 401                 | 77.9            | 0.389                  | 67.6                 |
|              | 6.0       | 17.6        | 22.4            | 430               | 225             | 4.38               | 3.17 | 71.7            | 56.3            | 88.5            | 66.3            | 467                 | 89.7            | 0.387                  | 56.8                 |
|              | 6.3       | 18.4        | 23.4            | 447               | 234             | 4.37               | 3.16 | 74.6            | 58.4            | 92.3            | 69.1            | 486                 | 93.0            | 0.386                  | 54.4                 |
|              | 8.0       | 22.9        | 29.1            | 537               | 278             | 4.29               | 3.09 | 89.5            | 69.4            | 113             | 83.9            | 586                 | 110             | 0.383                  | 43.7                 |
|              | 10.0      | 27.9        | 35.5            | 628               | 320             | 4.20               | 3.00 | 105             | 80.0            | 134             | 99.4            | 688                 | 126             | 0.379                  | 35.9                 |
| 150 x 100    | 5.0       | 18.7        | 23.9            | 747               | 396             | 5.59               | 4.07 | 99.5            | 79.1            | 121             | 90.8            | 806                 | 127             | 0.489                  | 53.5                 |
|              | 6.0       | 22.3        | 28.4            | 873               | 461             | 5.55               | 4.03 | 116             | 92.1            | 142             | 107             | 945                 | 147             | 0.487                  | 44.8                 |
|              | 6.3       | 23.3        | 29.7            | 910               | 479             | 5.53               | 4.02 | 121             | 95.9            | 148             | 111             | 985                 | 153             | 0.486                  | 43.0                 |
|              | 8.0       | 29.1        | 37.1            | 1,106             | 577             | 5.46               | 3.94 | 147             | 115             | 183             | 137             | 1,202               | 184             | 0.483                  | 34.4                 |
|              | 10.0      | 35.7        | 45.5            | 1,312             | 678             | 5.37               | 3.86 | 175             | 136             | 220             | 164             | 1,431               | 215             | 0.479                  | 28.1                 |
|              | 12.5*†    | 43.6        | 55.5            | 1,532             | 781             | 5.25               | 3.75 | 204             | 156             | 263             | 194             | 1,680               | 246             | 0.473                  | 22.9                 |
| 160 x 80     | 5.0       | 18.0        | 22.9            | 753               | 251             | 5.74               | 3.31 | 94.1            | 62.8            | 117             | 71.7            | 599                 | 106             | 0.469                  | 55.6                 |
|              | 6.0       | 21.3        | 27.2            | 880               | 291             | 5.69               | 3.27 | 110             | 72.8            | 138             | 84.1            | 700                 | 123             | 0.467                  | 46.9                 |
|              | 6.3       | 22.3        | 28.5            | 917               | 302             | 5.68               | 3.26 | 115             | 75.6            | 144             | 87.7            | 729                 | 127             | 0.466                  | 44.9                 |
|              | 8.0       | 27.9        | 35.5            | 1,113             | 361             | 5.60               | 3.19 | 139             | 90.2            | 177             | 107             | 882                 | 151             | 0.463                  | 35.9                 |
|              | 10.0      | 34.2        | 43.5            | 1,318             | 419             | 5.50               | 3.10 | 165             | 105             | 213             | 127             | 1,041               | 175             | 0.459                  | 29.3                 |
|              | 12.5*†    | 41.6        | 53.0            | 1,536             | 476             | 5.38               | 3.00 | 192             | 119             | 254             | 150             | 1,206               | 199             | 0.453                  | 24.0                 |
| 200 x 100    | 5.0       | 22.7        | 28.9            | 1,509             | 509             | 7.23               | 4.20 | 151             | 102             | 186             | 115             | 1,202               | 172             | 0.589                  | 44.1                 |
|              | 6.0       | 27.0        | 34.4            | 1,774             | 593             | 7.18               | 4.15 | 177             | 119             | 221             | 135             | 1,412               | 200             | 0.587                  | 37.0                 |
|              | 6.3       | 28.3        | 36.0            | 1,851             | 618             | 7.17               | 4.14 | 185             | 124             | 231             | 141             | 1,473               | 208             | 0.586                  | 35.4                 |
|              | 8.0       | 35.4        | 45.1            | 2,269             | 747             | 7.09               | 4.07 | 227             | 149             | 286             | 147             | 1,802               | 251             | 0.583                  | 28.3                 |
|              | 10.0      | 43.6        | 55.5            | 2,718             | 881             | 7.00               | 3.98 | 272             | 176             | 346             | 209             | 2,154               | 296             | 0.579                  | 23.0                 |
|              | 12.5      | 53.4        | 68.0            | 3,218             | 1,022           | 6.88               | 3.88 | 322             | 204             | 417             | 249             | 2,541               | 342             | 0.573                  | 18.8                 |
|              | 16.0      | 66.4        | 84.5            | 3,808             | 1,175           | 6.71               | 3.73 | 381             | 235             | 505             | 297             | 2,988               | 393             | 0.566                  | 15.1                 |
| 250 x 150    | 6.0       | 36.4        | 46.4            | 3,997             | 1,807           | 9.28               | 6.24 | 320             | 241             | 387             | 272             | 3,873               | 396             | 0.787                  | 27.5                 |
|              | 6.3       | 38.2        | 48.6            | 4,178             | 1,886           | 9.27               | 6.23 | 334             | 252             | 405             | 284             | 4,049               | 413             | 0.786                  | 26.2                 |
|              | 8.0       | 48.0        | 61.1            | 5,167             | 2,317           | 9.19               | 6.16 | 413             | 309             | 505             | 353             | 5,014               | 506             | 0.783                  | 20.9                 |
|              | 10.0      | 59.3        | 75.5            | 6,259             | 2,784           | 9.10               | 6.07 | 501             | 371             | 618             | 430             | 6,082               | 606             | 0.779                  | 16.9                 |
|              | 12.5      | 73.0        | 93.0            | 7,518             | 3,310           | 8.99               | 5.97 | 601             | 441             | 751             | 520             | 7,317               | 717             | 0.773                  | 13.7                 |
|              | 16.0      | 91.5        | 117             | 9,089             | 3,943           | 8.83               | 5.82 | 727             | 526             | 924             | 635             | 8,863               | 851             | 0.766                  | 11.0                 |
| 300 x 200    | 6.0       | 45.8        | 58.4            | 7,532             | 4,033           | 11.4               | 8.31 | 502             | 403             | 599             | 453             | 8,091               | 651             | 0.987                  | 21.8                 |
|              | 6.3       | 48.1        | 61.2            | 7,880             | 4,216           | 11.3               | 8.30 | 525             | 422             | 627             | 475             | 8,468               | 681             | 0.986                  | 20.8                 |
|              | 8.0       | 60.5        | 77.1            | 9,798             | 5,219           | 11.3               | 8.23 | 653             | 522             | 785             | 593             | 10,549              | 840             | 0.983                  | 16.6                 |
|              | 10.0      | 75.0        | 95.5            | 11,940            | 6,331           | 11.2               | 8.14 | 796             | 633             | 964             | 726             | 12,890              | 1,016           | 0.979                  | 13.4                 |
|              | 12.5      | 92.6        | 118             | 14,460            | 7,619           | 11.1               | 8.04 | 964             | 762             | 1,179           | 866             | 15,654              | 1,217           | 0.973                  | 10.8                 |
|              | 16.0      | 117         | 149             | 17,700            | 9,239           | 10.9               | 7.89 | 1,180           | 924             | 1,462           | 1,094           | 19,227              | 1,469           | 0.966                  | 8.55                 |
| 400 x 200    | 10.0      | 90.7        | 116             | 24,140            | 8,138           | 14.5               | 8.39 | 1,207           | 814             | 1,492           | 916             | 19,236              | 1,377           | 1.18                   | 11.1                 |
|              | 12.5      | 112         | 143             | 29,410            | 9,820           | 14.3               | 8.29 | 1,471           | 982             | 1,831           | 1,120           | 23,408              | 1,657           | 1.17                   | 8.93                 |
|              | 16.0      | 142         | 181             | 36,300            | 11,950          | 14.2               | 8.14 | 1,815           | 1,195           | 2,285           | 1,388           | 28,835              | 2,011           | 1.17                   | 7.05                 |
| 450 x 250    | 10.0      | 106         | 136             | 37,180            | 14,900          | 16.6               | 10.5 | 1,653           | 1,192           | 2,013           | 1,338           | 33,247              | 1,986           | 1.38                   | 9.44                 |
|              | 12.5      | 132         | 168             | 45,470            | 18,100          | 16.5               | 10.4 | 2,021           | 1,448           | 2,478           | 1,642           | 40,668              | 2,407           | 1.37                   | 7.58                 |
|              | 16.0      | 167         | 213             | 56,420            | 22,250          | 16.3               | 10.2 | 2,508           | 1,780           | 3,103           | 2,047           | 50,478              | 2,948           | 1.37                   | 5.99                 |
| 500 x 300    | 10.0*     | 122         | 156             | 54,118            | 24,564          | 18.7               | 12.6 | 2,165           | 1,638           | 2,609           | 1,834           | 52,396              | 2,696           | 1.58                   | 8.20                 |
|              | 12.5*     | 152         | 193             | 66,363            | 29,972          | 18.5               | 12.5 | 2,655           | 1,998           | 3,218           | 2,257           | 64,313              | 3,282           | 1.57                   | 6.58                 |
|              | 16.0*     | 192         | 245             | 82,673            | 37,076          | 18.4               | 12.3 | 3,307           | 2,472           | 4,042           | 2,825           | 80,223              | 4,046           | 1.57                   | 5.21                 |

Notes:  
 Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.

\* Size not included in BS 4848 Part 2

† 43C only  
 Length per tonne rounded up



# HOT FORMED HOLLOW SECTIONS

## CIRCULAR

| Outside diameter | Thickness | Unit weight | Section area    | Moment of inertia | Radius of gyration | Elastic modulus | Plastic modulus | Torsional       | Constants       | Superficial area per m | Approx. length per tonne |
|------------------|-----------|-------------|-----------------|-------------------|--------------------|-----------------|-----------------|-----------------|-----------------|------------------------|--------------------------|
| mm               | t         | kg/m        | cm <sup>2</sup> | cm <sup>4</sup>   | cm                 | cm <sup>3</sup> | cm <sup>3</sup> | cm <sup>4</sup> | cm <sup>3</sup> | m <sup>2</sup>         | m                        |
| 21.3             | 3.2       | 1.43        | 1.82            | 0.77              | 0.65               | 0.72            | 1.06            | 1.54            | 1.44            | 0.067                  | 700                      |
| 26.9             | 3.2       | 1.87        | 2.38            | 1.70              | 0.846              | 1.27            | 1.81            | 3.41            | 2.53            | 0.085                  | 535                      |
| 33.7             | 2.6       | 1.99        | 2.54            | 3.09              | 1.10               | 1.84            | 2.52            | 6.19            | 3.67            | 0.106                  | 503                      |
|                  | 3.2       | 2.41        | 3.07            | 3.60              | 1.08               | 2.14            | 2.99            | 7.21            | 4.28            | 0.106                  | 415                      |
|                  | 4.0       | 2.93        | 3.73            | 4.19              | 1.06               | 2.49            | 3.55            | 8.38            | 4.97            | 0.106                  | 342                      |
| 42.4             | 2.6       | 2.55        | 3.25            | 6.46              | 1.41               | 3.05            | 4.12            | 12.9            | 6.10            | 0.133                  | 393                      |
|                  | 3.2       | 3.09        | 3.94            | 7.62              | 1.39               | 3.59            | 4.93            | 15.2            | 7.19            | 0.133                  | 324                      |
|                  | 4.0       | 3.79        | 4.83            | 8.99              | 1.36               | 4.24            | 5.92            | 18.0            | 8.48            | 0.133                  | 264                      |
| 48.3             | 3.2       | 3.56        | 4.53            | 11.6              | 1.60               | 4.80            | 6.52            | 23.2            | 9.59            | 0.152                  | 281                      |
|                  | 4.0       | 4.37        | 5.57            | 13.8              | 1.57               | 5.70            | 7.87            | 27.5            | 11.4            | 0.152                  | 229                      |
|                  | 5.0       | 5.34        | 6.80            | 16.2              | 1.54               | 6.69            | 9.42            | 32.3            | 13.4            | 0.152                  | 188                      |
| 60.3             | 3.2       | 4.51        | 5.74            | 23.5              | 2.02               | 7.78            | 10.4            | 46.9            | 15.6            | 0.189                  | 222                      |
|                  | 4.0       | 5.55        | 7.07            | 28.2              | 2.00               | 9.34            | 12.7            | 56.3            | 18.7            | 0.189                  | 181                      |
|                  | 5.0       | 6.82        | 8.69            | 33.5              | 1.96               | 11.1            | 15.3            | 67.0            | 22.2            | 0.189                  | 147                      |
| 76.1             | 3.2       | 5.75        | 7.33            | 48.8              | 2.58               | 12.8            | 17.0            | 97.6            | 25.6            | 0.239                  | 174                      |
|                  | 4.0       | 7.11        | 9.06            | 59.1              | 2.55               | 15.5            | 20.8            | 118             | 31.0            | 0.239                  | 141                      |
|                  | 5.0       | 8.77        | 11.2            | 70.9              | 2.52               | 18.6            | 25.3            | 142             | 37.3            | 0.239                  | 115                      |
| 88.9             | 3.2       | 6.76        | 8.62            | 79.2              | 3.03               | 17.8            | 23.5            | 158             | 35.6            | 0.279                  | 148                      |
|                  | 4.0       | 8.38        | 10.7            | 96.3              | 3.00               | 21.7            | 28.9            | 193             | 43.3            | 0.279                  | 120                      |
|                  | 5.0       | 10.3        | 13.2            | 116               | 2.97               | 26.2            | 35.2            | 233             | 52.4            | 0.279                  | 97.1                     |
| 114.3            | 3.6       | 9.83        | 12.5            | 192               | 3.92               | 33.6            | 44.1            | 384             | 67.2            | 0.359                  | 102                      |
|                  | 5.0       | 13.5        | 17.2            | 257               | 3.87               | 45.0            | 59.8            | 514             | 89.9            | 0.359                  | 74.1                     |
|                  | 6.3       | 16.8        | 21.4            | 313               | 3.82               | 54.7            | 73.6            | 625             | 109             | 0.359                  | 59.6                     |
| 139.7            | 5.0       | 16.6        | 21.2            | 481               | 4.77               | 68.8            | 90.8            | 961             | 138             | 0.439                  | 60.3                     |
|                  | 6.3       | 20.7        | 26.4            | 589               | 4.72               | 84.3            | 112             | 1,177           | 169             | 0.439                  | 48.4                     |
|                  | 8.0       | 26.0        | 33.1            | 720               | 4.66               | 103             | 139             | 1,441           | 206             | 0.439                  | 38.5                     |
|                  | 10.0      | 32.0        | 40.7            | 862               | 4.60               | 123             | 169             | 1,724           | 247             | 0.439                  | 31.3                     |
| 168.3            | 5.0       | 20.1        | 25.7            | 856               | 5.78               | 102             | 133             | 1,712           | 203             | 0.529                  | 49.8                     |
|                  | 6.3       | 25.2        | 32.1            | 1,053             | 5.73               | 125             | 165             | 2,107           | 250             | 0.529                  | 39.7                     |
|                  | 8.0       | 31.6        | 40.3            | 1,297             | 5.67               | 154             | 206             | 2,595           | 308             | 0.529                  | 31.7                     |
|                  | 10.0      | 39.0        | 49.7            | 1,564             | 5.61               | 186             | 251             | 3,128           | 372             | 0.529                  | 25.7                     |
| 193.7            | 5.0 *     | 23.3        | 29.6            | 1,320             | 6.67               | 136             | 178             | 2,640           | 273             | 0.609                  | 43.0                     |
|                  | 6.3       | 29.1        | 37.1            | 1,630             | 6.63               | 168             | 221             | 3,260           | 337             | 0.609                  | 34.4                     |
|                  | 8.0       | 36.6        | 46.7            | 2,016             | 6.57               | 208             | 276             | 4,031           | 416             | 0.609                  | 27.4                     |
|                  | 10.0      | 45.3        | 57.7            | 2,442             | 6.50               | 252             | 338             | 4,883           | 504             | 0.609                  | 22.1                     |
|                  | 12.5      | 55.9        | 71.2            | 2,934             | 6.42               | 303             | 411             | 5,869           | 606             | 0.609                  | 17.9                     |
|                  | 16.0 †    | 70.1        | 89.3            | 3,554             | 6.31               | 367             | 507             | 7,109           | 734             | 0.609                  | 14.3                     |
| 219.1            | 5.0 *     | 26.4        | 33.6            | 1,928             | 7.57               | 176             | 229             | 3,856           | 352             | 0.688                  | 37.9                     |
|                  | 6.3       | 33.1        | 42.1            | 2,386             | 7.53               | 218             | 285             | 4,772           | 436             | 0.688                  | 30.3                     |
|                  | 8.0       | 41.6        | 53.1            | 2,960             | 7.47               | 270             | 357             | 5,919           | 540             | 0.688                  | 24.1                     |
|                  | 10.0      | 51.6        | 65.7            | 3,598             | 7.40               | 328             | 438             | 7,197           | 657             | 0.688                  | 19.4                     |
|                  | 12.5      | 63.7        | 81.1            | 4,345             | 7.32               | 397             | 534             | 8,689           | 793             | 0.688                  | 15.7                     |
|                  | 16.0 †    | 80.1        | 102             | 5,297             | 7.20               | 483             | 661             | 10,590          | 967             | 0.688                  | 12.5                     |
|                  | 20.0 †    | 98.2        | 125             | 6,261             | 7.07               | 572             | 795             | 12,520          | 1143            | 0.688                  | 10.2                     |

### Notes:

Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.

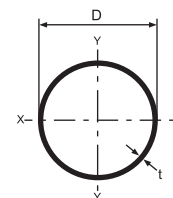
Sizes printed in italics may be supplied Hot Finished, Seamless at our option.

\* Size not included in BS 4848 Part 2

† Grade 50C only

Details of larger sizes of CHS available on application. Length per tonne rounded up

# HOT FORMED HOLLOW SECTIONS



## CIRCULAR

| Outside diameter | Thickness<br>t | Unit weight<br>MA | Section area<br>I | Moment of inertia<br>r | Radius of gyration<br>Z | Elastic modulus<br>S | Plastic modulus<br>J | Torsional<br>C  | Constant<br>per m | Superficial area<br>per tonne | Approx. length |
|------------------|----------------|-------------------|-------------------|------------------------|-------------------------|----------------------|----------------------|-----------------|-------------------|-------------------------------|----------------|
| mm               | mm             | kg/m              | cm <sup>2</sup>   | cm <sup>4</sup>        | cm                      | cm <sup>3</sup>      | cm <sup>3</sup>      | cm <sup>4</sup> | cm <sup>3</sup>   | m <sup>2</sup>                | m              |
| 244.5            | 6.3            | 37.0              | 47.1              | 3,346                  | 8.42                    | 274                  | 358                  | 6,692           | 547               | 0.768                         | 27.1           |
|                  | 8.0            | 46.7              | 59.4              | 4,160                  | 8.37                    | 340                  | 448                  | 8,321           | 681               | 0.768                         | 21.5           |
|                  | 10.0           | 57.8              | 73.7              | 5,073                  | 8.30                    | 415                  | 550                  | 10,150          | 830               | 0.768                         | 17.4           |
|                  | 12.5           | 71.5              | 91.1              | 6,147                  | 8.21                    | 503                  | 673                  | 12,290          | 1,006             | 0.768                         | 14.0           |
|                  | 16.0           | 90.2              | 115               | 7,533                  | 8.10                    | 616                  | 837                  | 15,070          | 1,232             | 0.768                         | 11.1           |
|                  | 20.0 †         | 111               | 141               | 8,957                  | 7.97                    | 733                  | 1,011                | 17,910          | 1,465             | 0.768                         | 9.01           |
| 273              | 6.3            | 41.4              | 52.8              | 4,696                  | 9.43                    | 344                  | 448                  | 9,392           | 688               | 0.858                         | 24.2           |
|                  | 8.0            | 52.3              | 66.6              | 5,852                  | 9.37                    | 429                  | 562                  | 11,700          | 857               | 0.858                         | 19.2           |
|                  | 10.0           | 64.9              | 82.6              | 7,154                  | 9.31                    | 524                  | 692                  | 14,310          | 1,048             | 0.858                         | 15.5           |
|                  | 12.5           | 80.3              | 102               | 8,697                  | 9.22                    | 637                  | 849                  | 17,390          | 1,274             | 0.858                         | 12.5           |
|                  | 16.0           | 101               | 129               | 10,710                 | 9.10                    | 784                  | 1,058                | 21,410          | 1,569             | 0.858                         | 9.91           |
|                  | 20.0 †         | 125               | 159               | 12,800                 | 8.97                    | 938                  | 1,283                | 25,600          | 1,875             | 0.858                         | 8.00           |
|                  | 25.0 †         | 153               | 195               | 15,130                 | 8.81                    | 1,108                | 1,543                | 30,250          | 2,216             | 0.858                         | 6.54           |
| 323.9            | 6.3 *          | 49.3              | 62.9              | 7,929                  | 11.2                    | 490                  | 636                  | 15,850          | 980               | 1.02                          | 20.3           |
|                  | 8.0            | 62.3              | 79.4              | 9,910                  | 11.2                    | 612                  | 799                  | 19,820          | 1,224             | 1.02                          | 16.1           |
|                  | 10.0           | 77.4              | 98.6              | 12,160                 | 11.1                    | 751                  | 986                  | 24,320          | 1,501             | 1.02                          | 13.0           |
|                  | 12.5           | 96.0              | 122               | 14,850                 | 11.0                    | 917                  | 1,213                | 29,690          | 1,833             | 1.02                          | 10.5           |
|                  | 16.0           | 121               | 155               | 18,390                 | 10.9                    | 1,136                | 1,518                | 36,780          | 2,271             | 1.02                          | 8.27           |
|                  | 20.0 †         | 150               | 191               | 22,140                 | 10.8                    | 1,367                | 1,850                | 44,280          | 2,734             | 1.02                          | 6.67           |
|                  | 25.0 †         | 184               | 235               | 26,400                 | 10.6                    | 1,630                | 2,239                | 52,800          | 3,260             | 1.02                          | 5.44           |
|                  |                |                   |                   |                        |                         |                      |                      |                 |                   |                               |                |
| 355.6            | 8.0            | 68.6              | 87.4              | 13,200                 | 12.3                    | 742                  | 967                  | 26,400          | 1,485             | 1.12                          | 14.6           |
|                  | 10.0           | 85.2              | 109               | 16,220                 | 12.2                    | 912                  | 1,195                | 32,450          | 1,825             | 1.12                          | 11.8           |
|                  | 12.5           | 106               | 135               | 19,850                 | 12.1                    | 1,117                | 1,472                | 39,700          | 2,233             | 1.12                          | 9.44           |
|                  | 16.0           | 134               | 171               | 24,660                 | 12.0                    | 1,387                | 1,847                | 49,330          | 2,774             | 1.12                          | 7.47           |
|                  | 20.0 †         | 166               | 211               | 29,790                 | 11.9                    | 1,676                | 2,255                | 59,580          | 3,351             | 1.12                          | 6.03           |
|                  | 25.0 †         | 204               | 260               | 35,680                 | 11.7                    | 2,007                | 2,738                | 71,350          | 4,013             | 1.12                          | 4.91           |
|                  |                |                   |                   |                        |                         |                      |                      |                 |                   |                               |                |
| 406.4            | 10.0           | 97.8              | 125               | 24,480                 | 14.0                    | 1,205                | 1,572                | 48,950          | 2,409             | 1.28                          | 10.3           |
|                  | 12.5           | 121               | 155               | 30,030                 | 13.9                    | 1,478                | 1,940                | 60,060          | 2,956             | 1.28                          | 8.27           |
|                  | 16.0           | 154               | 196               | 37,450                 | 13.8                    | 1,843                | 2,440                | 74,900          | 3,686             | 1.28                          | 6.50           |
|                  | 20.0 †         | 191               | 243               | 45,430                 | 13.7                    | 2,236                | 2,989                | 90,860          | 4,472             | 1.28                          | 5.24           |
|                  | 25.0 †         | 235               | 300               | 54,700                 | 13.5                    | 2,692                | 3,642                | 109,400         | 5,384             | 1.28                          | 4.26           |
|                  | 32.0 †         | 295               | 376               | 66,430                 | 13.3                    | 3,269                | 4,497                | 132,900         | 6,539             | 1.28                          | 3.39           |
|                  |                |                   |                   |                        |                         |                      |                      |                 |                   |                               |                |
| 457              | 10.0           | 110               | 140               | 35,090                 | 15.8                    | 1,536                | 1,998                | 70,180          | 3,071             | 1.44                          | 9.10           |
|                  | 12.5           | 137               | 175               | 43,140                 | 15.7                    | 1,888                | 2,470                | 86,290          | 3,776             | 1.44                          | 7.30           |
|                  | 16.0           | 174               | 222               | 53,960                 | 15.6                    | 2,361                | 3,113                | 107,900         | 4,723             | 1.44                          | 5.75           |
|                  | 20.0 †         | 216               | 275               | 65,680                 | 15.5                    | 2,874                | 3,822                | 131,400         | 5,749             | 1.44                          | 4.63           |
|                  | 25.0 †         | 266               | 339               | 79,420                 | 15.3                    | 3,476                | 4,671                | 158,800         | 6,951             | 1.44                          | 3.76           |
|                  | 32.0 †         | 335               | 427               | 97,010                 | 15.1                    | 4,246                | 5,791                | 194,000         | 8,491             | 1.44                          | 2.99           |
|                  | 40.0 †         | 411               | 524               | 114,900                | 14.8                    | 5,031                | 6,977                | 229,900         | 10,060            | 1.44                          | 2.44           |
|                  |                |                   |                   |                        |                         |                      |                      |                 |                   |                               |                |
| 508              | 10.0 *         | 123               | 156               | 48,520                 | 17.6                    | 1,910                | 2,480                | 97,040          | 3,821             | 1.60                          | 8.14           |
|                  | 12.5 *         | 153               | 195               | 59,760                 | 17.5                    | 2,353                | 3,070                | 119,500         | 4,705             | 1.60                          | 6.54           |
|                  | 16.0 *         | 194               | 247               | 74,910                 | 17.4                    | 2,949                | 3,874                | 149,800         | 5,898             | 1.60                          | 5.16           |
|                  | 20.0 †         | 241               | 307               | 91,430                 | 17.3                    | 3,600                | 4,766                | 182,900         | 7,200             | 1.60                          | 4.15           |
|                  | 25.0 †         | 298               | 379               | 110,900                | 17.1                    | 4,367                | 5,837                | 221,800         | 8,734             | 1.60                          | 3.36           |
|                  | 32.0 †         | 376               | 479               | 136,100                | 16.9                    | 5,360                | 7,261                | 272,300         | 10,720            | 1.60                          | 2.66           |
|                  | 40.0 †         | 462               | 588               | 162,200                | 16.6                    | 6,385                | 8,782                | 324,400         | 12,770            | 1.60                          | 2.17           |
|                  | 50.0 †         | 565               | 719               | 190,900                | 16.3                    | 7,515                | 10,530               | 381,800         | 15,030            | 1.60                          | 1.77           |
|                  |                |                   |                   |                        |                         |                      |                      |                 |                   |                               |                |
|                  |                |                   |                   |                        |                         |                      |                      |                 |                   |                               |                |

Notes:  
 Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.  
 Sizes printed in italics may be supplied Hot Finished, Seamless at our option.  
 \* Size not included in BS 4848 Part 2  
 † Grade 50C only  
 Details of larger sizes of CHS available on application. Length per tonne rounded up

## PIPES

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# PIPES

## TECHNICAL SPECIFICATION REFERENCE -STEEL PIPES

Excerpts from JIS G 3452 & ASTM A-53 Specifications

| Classification                                     | Specification | Grade | Chemical Composition |            |            |           |           | Tensile Test        |                   |                                                      | Bend Test *1           |                   | Flattening<br>Test<br>H |
|----------------------------------------------------|---------------|-------|----------------------|------------|------------|-----------|-----------|---------------------|-------------------|------------------------------------------------------|------------------------|-------------------|-------------------------|
|                                                    |               |       | C<br>max.            | Mn<br>max. | Si<br>max. | P<br>max. | S<br>max. | Tensile<br>Strength | Yield<br>Strength | Elongation                                           | Bending<br>Angle       | Bending<br>Radius |                         |
|                                                    |               |       | %                    | %          | %          | %         | %         | N/mm <sup>2</sup>   | N/mm <sup>2</sup> | %                                                    |                        |                   |                         |
| Carbon<br>Steel Pipes<br>for<br>Ordinary<br>Piping | JIS G 3452    | SGP   | --                   | --         | --         | 0.04      | 0.04      | 290 min.            | --                | * Test Piece<br>No. 11 & 12<br>* Test Piece<br>No. 5 | 30 min.<br><br>25 min. | 90°<br><br>6D     | $\frac{2}{3} D$         |
| ASTM<br>Standard                                   | ASTM A53      | A     | 0.25                 | 0.95       | --         | 0.05      | 0.06      | 330 min.            | 205 min.          | As specified in A-53<br>specification                | 90°                    | 12D               | $\frac{1}{3} D$         |
| Welded Steel<br>Pipes                              | ASTM A53      | B     | 0.30                 | 1.20       | --         | 0.05      | 0.06      | 415 min.            | 240 min.          |                                                      |                        |                   |                         |

### Notes:

- \* - When the tensile test is carried out on No. 12 or No. 5 test piece for the pipe under 8 mm in wall thickness, the minimum value of elongation shall be obtained by subtracting 1.5 % from the thickness values of elongation given in Table above for each 1 mm decrease in wall thickness, and rounding off to an integer in accordance with JIS Z 8401.
- \* - The values of elongation given in Table above shall not applied to the pipe whose nominal size is 32 mm or smaller.
- \*1 - Bend test in table above only applied to pipes of nominal size 2" (50 mm) or smaller.
- H - Distance between the plates
- D - Outside diameter of the pipe

YICK HOE

## TECHNICAL SPECIFICATION REFERENCE - STEEL TUBES

Excerpts from JIS G 3444, JIS G 3445, BS 31 &amp; BS 1139 Specifications

| Classification                                                 | Specification                                                  | Grade                | Chemical Composition |                 |                 |                |                | Tensile Test                 |                            |                             |                             | Bend Test        |                   | Flattening<br>Test<br>H |      |
|----------------------------------------------------------------|----------------------------------------------------------------|----------------------|----------------------|-----------------|-----------------|----------------|----------------|------------------------------|----------------------------|-----------------------------|-----------------------------|------------------|-------------------|-------------------------|------|
|                                                                |                                                                |                      | C<br>max.<br>%       | Mn<br>max.<br>% | Si<br>max.<br>% | P<br>max.<br>% | S<br>max.<br>% | Tensile<br>Strength<br>N/mm² | Yield<br>Strength<br>N/mm² | Elongation<br><br>%         |                             | Bending<br>Angle | Bending<br>Radius |                         |      |
| Carbon Steel<br>Tubes for<br>General<br>Structural<br>Purposes | JIS G 3444                                                     | STK290               | --                   | --              | --              | 0.05           | 0.05           | 290 min.                     | --                         | * Test Piece<br>No.11 & 12  | 30<br>min.                  | 90°              | 6D                | 2/3D                    |      |
|                                                                |                                                                |                      |                      |                 |                 |                |                |                              |                            | * Test Piece<br>No.5        | 25<br>min.                  |                  |                   |                         |      |
|                                                                | JIS G 3444                                                     | STK400               | 0.25                 | --              | -               | 0.04           | 0.04           | 400 min.                     | 235 min.                   | * Test Piece<br>No. 11 &12  | 23<br>min.                  | 90°              | 6D                | 2/3D                    |      |
|                                                                |                                                                |                      |                      |                 |                 |                |                |                              |                            | * Test Piece<br>No. 5       | 18<br>min.                  |                  |                   |                         |      |
|                                                                | JIS G 3444                                                     | STK500               | 0.30                 | 0.35            | 0.30-<br>1.00   | 0.04           | 0.04           | 500 min.                     | 355 min.                   | * Test Piece<br>No. 11 &12  | 15<br>min.                  | 90°              | 8D                | 7/8D                    |      |
|                                                                |                                                                |                      |                      |                 |                 |                |                |                              |                            | * Test Piece<br>No.5        | 10<br>min                   |                  |                   |                         |      |
|                                                                | JIS G 3444                                                     | STK540               | 0.23                 | 0.55            | 1.50            | 0.04           | 0.04           | 540 min.                     | 390 min.                   | * Test Piece<br>No. 11 & 12 | 20<br>min.                  | 90°              | 6D                | 7/8D                    |      |
|                                                                |                                                                |                      |                      |                 |                 |                |                |                              |                            | * Test Piece<br>No.5        | 16<br>min.                  |                  |                   |                         |      |
|                                                                | Carbon Steel<br>Tubes For<br>Machine<br>Structural<br>Purposes | JIS G 3445           | STKM<br>11A          | 0.12            | 0.35            | 0.60           | 0.04           | 0.04                         | 290 min.                   | --                          | * Test Piece<br>No. 11 & 12 | 35<br>min.       | 180°              | 4D                      | 1/2D |
|                                                                |                                                                |                      |                      |                 |                 |                |                |                              |                            |                             | * Test Piece<br>No.5        | 30<br>min.       |                   |                         |      |
| British<br>Standard<br>Galvanized<br>Steel Conduit             | BS31                                                           | Class B<br>(screwed) | --                   | --              | --              | --             | --             | 273<br>to 365                | --                         | Gauge Length<br>(8 in.)     | 15<br>min                   | 90°              | 6D                | --                      |      |
| British<br>Standard<br>Steel Tubes<br>for<br>Scaffolding       | BS1139                                                         |                      | 0.20                 | --              | 0.30            | 0.05           | 0.05           | 340<br>to 480                | 235<br>min.                | Gauge Length<br>Lo=5.65√So  | 24<br>min.                  | --               | --                | --                      |      |
| Window<br>Pipe                                                 | Manufacturer<br>Standard                                       | --                   | --                   | --              | --              | --             | --             | --                           | --                         | --                          | --                          | --               | -                 | -                       |      |
| Earth<br>Pipe                                                  | Manufacturer<br>Standard                                       | --                   | --                   | --              | --              | --             | --             | --                           | --                         | --                          | --                          | --               | --                | --                      |      |

Notes:

- \* - Elongation values in Table above only applied to tubes over 40 mm in outside diameter
- \* - When the tensile test is carried out on No. 12 or No. 5 test piece for the tube under 8 mm in wall thickness the minimum elongation shall be determined by reducing 1.5% per 1 mm of decrease in wall thickness from the values given in the Table above and rounding off the value obtained to integer in accordance with JIS Z 8401.
- H - Distance between flat plates
- D - Outside diameter of the tube

# PIPES

## TECHNICAL SPECIFICATION REFERENCE -STEEL TUBES AND LIPPED CHANNEL

| Classification                        | Standard Specification   |         | Application                                                                | Chemical Requirements (%) |             |           |           |           | Physical Requirements (%) |                   |                   |            |
|---------------------------------------|--------------------------|---------|----------------------------------------------------------------------------|---------------------------|-------------|-----------|-----------|-----------|---------------------------|-------------------|-------------------|------------|
|                                       |                          |         |                                                                            | C                         | Mn          | Si        | P         | S         | Others                    | Tensile Strength  | Yield Strength    | Elongation |
|                                       |                          |         |                                                                            | max.<br>%                 | max.<br>%   | max.<br>% | max.<br>% | max.<br>% | %                         | N/mm <sup>2</sup> | N/mm <sup>2</sup> | %          |
| Circular Hollow Section               | BS1387                   | L       | Carbon Steel Tubes for Ordinary Uses, Water Pipes and Structural Purposes  | 0.20                      | 1.20        | --        | 0.045     | 0.045     | --                        | 320 - 460         | 195 min.          | 20 min.    |
|                                       |                          | M       |                                                                            |                           |             |           |           |           |                           |                   |                   |            |
|                                       |                          | H       |                                                                            |                           |             |           |           |           |                           |                   |                   |            |
|                                       | JIS G 3444 /Mfg Standard | STK 290 | Carbon Steel Tubes for Structural Purposes (Scaffolding)                   | --                        | --          | --        | 0.05      | 0.05      | --                        | 290 min.          | --                | 25 min.    |
|                                       |                          | STK 400 |                                                                            | 0.25                      | --          | --        | 0.04      | 0.04      | --                        | 400 min.          | 235 min.          | 18 min.    |
|                                       |                          | STK 500 |                                                                            | 0.30                      | 0.30 - 1.00 | 0.35      | 0.04      | 0.04      | --                        | 500 min.          | 355 min.          | 10 min.    |
|                                       | JIS G 3445 /Mfg Standard | STKM11A | Carbon Steel Tubes for Machine Structural Purposes                         | 0.12                      | 0.60 max.   | 0.35      | 0.04      | 0.04      | --                        | 290 min.          | --                | 30 min.    |
|                                       | JIS G 3452 /Mfg Standard | SGP     | Carbon Steel Tubes for Ordinary Uses, Water Pipes and Structural Purposes  | --                        | --          | --        | 0.04      | 0.04      | --                        | 290 min.          | --                | 25 min.    |
| Square and Rectangular Hollow Section | ASTM A500/ Mfg Standard  | A       | Carbon Steel Tubes for Structural Purposes and Machine Structural Purposes | 0.30                      | --          | --        | 0.05      | 0.063     | Cu (max) = 0.18           | 310 min.          | 269 min.          | 25 min.    |
|                                       |                          | B       |                                                                            | 0.30                      | --          | --        | 0.05      | 0.063     | Cu (max) = 0.18           | 400 min.          | 317 min.          | 23 min.    |
| Circular Hollow Section               | Mfg Standard             | AA      | Carbon Steel Tubes for Ordinary Uses                                       | 0.20                      | 1.20        | --        | 0.045     | 0.045     | --                        | 250 min.          | 190 min.          | 20 min.    |
| Light Gauge Lipped Channel            | Mfg Standard             |         | For Civil Engineering                                                      | 0.25                      | --          | --        | 0.05      | 0.05      | --                        | 295 min.          | --                | 25 min.    |
| Conduit Pipe                          | BS 31/ Mfg Standard      |         | For Electrical Wiring                                                      | --                        | --          | --        | --        | --        | --                        | 273 - 365         | --                | 15 min.    |
| Window Pipe                           | Mfg Standard             |         | For Window Usage                                                           | --                        | --          | --        | --        | --        | --                        | --                | --                | --         |

## BS 1387:1985 A1, LIGHT, MEDIUM AND HEAVY

### Pertinent Excerpts from BS 1387:1985 Specification

|                                   |                  |   |                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tolerances on Dimensions and Mass | Outside Diameter | : | As shown in Table on opposite page.                                                                                                                                                                                                                                                                   |
|                                   | Wall thickness   | : | Light tubes -8%                                                                                                                                                                                                                                                                                       |
|                                   |                  | : | Medium and Heavy tubes -10%                                                                                                                                                                                                                                                                           |
|                                   | Mass             | : | The mean consignment mass for quantities of 150 m and over of one size shall not deviate by more than $\pm 4\%$ from the mass of consignment calculated from the mass given in Table as appropriate. No single tube shall deviate by more than +10%, -8% from the mass given in Table as appropriate. |

**Bend Test** Black tubes up to and including DN 50 shall be bent cold without any signs of fracture or failure, through 180°C round a former having a radius at the bottom of the groove equal to six times the outside diameter of the tube as given in Table. Hot-dip-zinc coated tubes shall be bent cold without cracking of the steel, through 90° round a former having a radius at the bottom of the groove equal to eight times the outside diameter of tube.

|                     |                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Leak Tightness Test | The test shall be either a hydraulic test at a pressure of 50 bar ( $50 \times 10^5 \text{ N/M}^2$ ), or alternatively, an eddy current test. |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

| Rod Diameters     |                      |
|-------------------|----------------------|
| Nominal Size (DN) | Diameter of rod (mm) |
| 15.00             | 9.50                 |
| 20.00             | 14.30                |
| 25.00             | 20.60                |

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# PIPES

## BRITISH STANDARD WELDED STEEL PIPES

BS 1387:1985

| Thickness<br>Class | Nominal<br>Size |    | Outside Diameter |         | Wall<br>Thickness | Calculated Weight |       |                         |       | No. of<br>Threads<br>per inch | Socket            |                |
|--------------------|-----------------|----|------------------|---------|-------------------|-------------------|-------|-------------------------|-------|-------------------------------|-------------------|----------------|
|                    |                 |    | Maximum          | Minimum |                   | Plain Ends        |       | Threads<br>and Coupling |       |                               | Outer<br>Diameter | Min.<br>Length |
|                    | mm              | in |                  |         | mm                | mm                | mm    | kg/m                    | kg/ft | kg/m                          |                   |                |
| A1                 | 15              | ½  | 21.4             | 21.0    | 1.9               | 0.904             | 0.276 | 0.914                   | 0.279 | 14                            | 27.8              | 38.1           |
|                    | 20              | ¾  | 26.9             | 26.4    | 2.1               | 1.27              | 0.387 | 1.284                   | 0.392 | 14                            | 34.1              | 41.3           |
|                    | 25              | 1  | 33.8             | 33.2    | 2.3               | 1.77              | 0.54  | 1.787                   | 0.545 | 11                            | 42.1              | 47.6           |
|                    | 32              | 1¼ | 42.5             | 41.9    | 2.3               | 2.26              | 0.689 | 2.28                    | 0.695 | 11                            | 51.6              | 54.0           |
|                    | 40              | 1½ | 48.4             | 47.8    | 2.5               | 2.81              | 0.857 | 2.83                    | 0.863 | 11                            | 57.9              | 57.2           |
|                    | 50              | 2  | 60.2             | 59.6    | 2.6               | 3.67              | 1.119 | 3.693                   | 1.126 | 11                            | 70.6              | 63.5           |
|                    | 65              | 2½ | 76.0             | 75.2    | 2.9               | 5.20              | 1.585 | 5.228                   | 1.594 | 11                            | 87.3              | 69.9           |
|                    | 80              | 3  | 88.7             | 87.9    | 2.9               | 6.11              | 1.863 | 6.136                   | 1.871 | 11                            | 101.6             | 76.2           |
|                    | 100             | 4  | 113.9            | 113.0   | 3.2               | 8.70              | 2.652 | 8.736                   | 2.663 | 11                            | 128.6             | 88.9           |
| Light              | 15              | ½  | 21.4             | 21.0    | 2.0               | 0.947             | 0.289 | 0.956                   | 0.291 | 14                            | 27.8              | 38.1           |
|                    | 20              | ¾  | 26.9             | 26.4    | 2.3               | 1.38              | 0.421 | 1.39                    | 0.424 | 14                            | 34.1              | 41.3           |
|                    | 25              | 1  | 33.8             | 33.2    | 2.6               | 1.98              | 0.604 | 2.00                    | 0.61  | 11                            | 42.1              | 47.6           |
|                    | 32              | 1¼ | 42.5             | 41.9    | 2.6               | 2.54              | 0.774 | 2.57                    | 0.783 | 11                            | 51.6              | 54.0           |
|                    | 40              | 1½ | 48.4             | 47.8    | 2.9               | 3.23              | 0.985 | 3.27                    | 0.997 | 11                            | 57.9              | 57.2           |
|                    | 50              | 2  | 60.2             | 59.6    | 2.9               | 4.08              | 1.24  | 4.15                    | 1.26  | 11                            | 70.6              | 63.5           |
|                    | 65              | 2½ | 76.0             | 75.2    | 3.2               | 5.71              | 1.74  | 5.83                    | 1.78  | 11                            | 87.3              | 69.9           |
|                    | 80              | 3  | 88.7             | 87.9    | 3.2               | 6.72              | 2.05  | 6.89                    | 2.10  | 11                            | 101.6             | 76.2           |
|                    | 100             | 4  | 113.9            | 113.0   | 3.6               | 9.75              | 2.97  | 10.00                   | 3.05  | 11                            | 128.6             | 88.9           |

Test Pressure: 50 Bar / 700Psi

## BRITISH STANDARD WELDED STEEL PIPES

BS 1387:1985

| Thickness<br>Class | Nominal<br>Size |    | Outside Diameter |         | Wall<br>Thickness | Calculated Weight |       |                         |       | No. of<br>Threads<br>per inch | Socket            |                |
|--------------------|-----------------|----|------------------|---------|-------------------|-------------------|-------|-------------------------|-------|-------------------------------|-------------------|----------------|
|                    |                 |    | Maximum          | Minimum |                   | Plain Ends        |       | Threads<br>and Coupling |       |                               | Outer<br>Diameter | Min.<br>Length |
|                    | mm              | in |                  |         | mm                | mm                | mm    | kg/m                    | kg/ft | kg/m                          |                   |                |
| Medium             | 15              | ½  | 21.7             | 21.1    | 2.6               | 1.21              | 0.369 | 1.22                    | 0.372 | 14                            | 27.8              | 38.1           |
|                    | 20              | ¾  | 27.2             | 26.6    | 2.6               | 1.56              | 0.475 | 1.57                    | 0.479 | 14                            | 34.1              | 41.3           |
|                    | 25              | 1  | 34.2             | 33.4    | 3.2               | 2.41              | 0.735 | 2.43                    | 0.741 | 11                            | 42.1              | 47.6           |
|                    | 32              | 1¼ | 42.9             | 42.1    | 3.2               | 3.10              | 0.945 | 3.13                    | 0.954 | 11                            | 51.6              | 54.0           |
|                    | 40              | 1½ | 48.8             | 48.0    | 3.2               | 3.57              | 1.09  | 3.61                    | 1.10  | 11                            | 57.9              | 57.2           |
|                    | 50              | 2  | 60.8             | 59.8    | 3.6               | 5.03              | 1.53  | 5.10                    | 1.55  | 11                            | 70.6              | 63.5           |
|                    | 65              | 2½ | 76.6             | 75.4    | 3.6               | 6.43              | 1.96  | 6.55                    | 2.00  | 11                            | 87.3              | 69.9           |
|                    | 80              | 3  | 89.5             | 88.1    | 4.0               | 8.37              | 2.55  | 8.54                    | 2.60  | 11                            | 101.6             | 76.2           |
|                    | 100             | 4  | 114.9            | 113.3   | 4.5               | 12.2              | 3.72  | 12.5                    | 3.81  | 11                            | 128.6             | 88.9           |
|                    | 125             | 5  | 140.6            | 138.7   | 5.0               | 16.6              | 5.06  | 17.1                    | 5.21  | 11                            | 155.6             | 95.3           |
|                    | 150             | 6  | 166.1            | 164.1   | 5.0               | 19.7              | 6.00  | 20.3                    | 6.19  | 11                            | 184.2             | 95.3           |
| Heavy              | 15              | ½  | 21.7             | 21.1    | 3.2               | 1.44              | 0.439 | 1.45                    | 0.442 | 14                            | 27.8              | 38.1           |
|                    | 20              | ¾  | 27.2             | 26.6    | 3.2               | 1.87              | 0.57  | 1.88                    | 0.573 | 14                            | 34.1              | 41.3           |
|                    | 25              | 1  | 34.2             | 33.4    | 4.0               | 2.94              | 0.896 | 2.96                    | 0.902 | 11                            | 42.1              | 47.0           |
|                    | 32              | 1¼ | 42.9             | 42.1    | 4.0               | 3.80              | 1.16  | 3.83                    | 1.17  | 11                            | 51.6              | 54.0           |
|                    | 40              | 1½ | 48.8             | 48.0    | 4.0               | 4.38              | 1.34  | 4.42                    | 1.35  | 11                            | 57.9              | 57.2           |
|                    | 50              | 2  | 60.8             | 59.8    | 4.5               | 6.19              | 1.89  | 6.26                    | 1.91  | 11                            | 70.6              | 63.5           |
|                    | 65              | 2½ | 76.6             | 75.4    | 4.5               | 7.93              | 2.42  | 8.05                    | 2.45  | 11                            | 87.3              | 69.9           |
|                    | 80              | 3  | 89.5             | 88.1    | 5.0               | 10.3              | 3.14  | 10.5                    | 3.20  | 11                            | 101.6             | 76.2           |
|                    | 100             | 4  | 114.9            | 113.3   | 5.4               | 14.5              | 4.42  | 14.8                    | 4.51  | 11                            | 128.6             | 88.9           |
|                    | 125             | 5  | 140.6            | 138.7   | 5.4               | 17.9              | 5.46  | 18.4                    | 5.61  | 11                            | 155.6             | 95.3           |
|                    | 150             | 6  | 166.1            | 164.1   | 5.4               | 21.3              | 6.49  | 21.9                    | 6.68  | 11                            | 184.2             | 95.3           |

Test Pressure: 50 Bar / 700Psi

# PIPES

## BRITISH STANDARD WELDED STEEL PIPES

### BRITISH STANDARD STEEL TUBES FOR SCAFFOLDING (BS 1139: SECTION 1.1:1990)

| Outside Diameter | Wall Thickness | Calculated Weight | Cross -Sectional Area | Second Moment of Inertia | Section Modulus | Radius of Gyration of Area | Plastic Modulus |
|------------------|----------------|-------------------|-----------------------|--------------------------|-----------------|----------------------------|-----------------|
| mm               | mm             | kg/m              | cm <sup>2</sup>       | cm <sup>4</sup>          | cm <sup>3</sup> | cm                         | cm <sup>3</sup> |
| 48.30            | 4.00           | 4.37              | 5.57                  | 13.80                    | 5.70            | 1.57                       | 7.87            |

Applicable Tolerances : Thickness  $\pm 10\%$   
Outside Diameter  $\pm 0.50$  mm

## BRITISH STANDARD GALVANIZED STEEL CONDUIT (BS 31-1940 CLASS B SCREWED)

| Nominal<br>Size |    | Outside Diameter |       |         |       | Thickness |      |         |       | Calculated Weight<br>with Coupler |       |       | No. of<br>Threads | Length of Threads |      |         |      |
|-----------------|----|------------------|-------|---------|-------|-----------|------|---------|-------|-----------------------------------|-------|-------|-------------------|-------------------|------|---------|------|
|                 |    | Nominal          |       | Minimum |       | Nominal   |      | Minimum |       |                                   |       |       |                   | Maximum           |      | Minimum |      |
| in              | mm | in               | mm    | in      | mm    | in        | mm   | in      | mm    | kg/m                              | kg/ft | lb/ft | per inch          | in                | mm   | in      | mm   |
| ¾               | 19 | 0.75             | 19.05 | 0.7387  | 18.76 | 0.064     | 1.63 | 1.52    | 0.060 | 0.713                             | 0.217 | 0.479 | 16                | 0.5625            | 14.3 | 0.500   | 12.7 |
| 1               | 25 | 1.00             | 25.40 | 0.9887  | 25.11 | 0.064     | 1.63 | 1.52    | 0.060 | 0.972                             | 0.296 | 0.653 | 16                | 0.6875            | 17.5 | 0.625   | 15.9 |
| 1¼              | 32 | 1.25             | 31.75 | 1.2387  | 31.46 | 0.064     | 1.63 | 1.52    | 0.060 | 1.240                             | 0.376 | 0.830 | 16                | 0.7500            | 19.1 | 0.6875  | 17.5 |
| 1½              | 38 | 1.50             | 38.10 | 1.4880  | 37.80 | 0.072     | 1.83 | 1.73    | 0.068 | 1.680                             | 0.511 | 1.130 | 14                | 0.8125            | 20.6 | 0.750   | 19.1 |
| 2               | 50 | 2.00             | 50.80 | 1.9880  | 50.50 | 0.080     | 2.03 | 1.93    | 0.076 | 2.510                             | 0.765 | 1.690 | 14                | 0.9375            | 23.8 | 0.875   | 22.2 |

Standard Length 3.81 m (12 ft. 6 in) without coupler

#### Features:

- \* Made of hot-dip galvanized steel strip with extra-smooth surface and highly-adherent zinc coating by the unique tube-making process. The weld zone coating restored in-line
- \* Inside weld bead controlled to a minimum for easier wire pulling
- \* Screwed on both ends to BS 31 and fitted with a zinc-coated coupler on one end
- \* Packed in bare bundles, but the unsocketed ends protected with plastic caps
- \* Easier to cut, thread, bend and pull. Dimensionally accurate. Uniform quality in every respect.

## EARTH PIPE - MANUFACTURER STANDARD

| Nominal Size |    | Outside Diameter | Wall Thickness | Weight |       |       |
|--------------|----|------------------|----------------|--------|-------|-------|
| in           | mm | mm               | mm             | kg/m   | kg/ft | lb/ft |
| 2            | 50 | 60.33            | 1.60           | 2.318  | 0.707 | 1.559 |

Standard Length: 4.78 m (16 ft.)

## WINDOW PIPE - MANUFACTURER STANDARD

| Nominal Size |    | Outside Diameter | Wall Thickness | Weight |       |       |
|--------------|----|------------------|----------------|--------|-------|-------|
| in           | mm | mm               | mm             | kg/m   | kg/ft | lb/ft |
| 0.625        | 16 | 15.90            | 1.20           | 0.435  | 0.133 | 0.293 |

## CARBON STEEL PIPES FOR GENERAL STRUCTURAL PURPOSES

### LIGHT (AA) (MANUFACTURER STANDARD)

| Nominal Size |     | Outside Diameter |         | Nominal Wall Thickness |     | Calculated Weight |       |        |
|--------------|-----|------------------|---------|------------------------|-----|-------------------|-------|--------|
|              |     | Minimum          | Maximum |                        |     | Plain Ends        |       |        |
| in           | mm  | mm               | mm      | in                     | mm  | kg/m              | kg/ft | lb/ft  |
| ½            | 15  | 21.0             | 21.4    | 0.063                  | 1.6 | 0.739             | 0.225 | 0.496  |
| ¾            | 20  | 26.4             | 26.9    | 0.063                  | 1.6 | 0.982             | 0.229 | 0.659  |
| 1            | 25  | 33.2             | 33.8    | 0.069                  | 1.8 | 1.364             | 0.416 | 0.916  |
| 1¼           | 32  | 41.9             | 42.3    | 0.069                  | 1.8 | 1.739             | 0.530 | 1.167  |
| 1½           | 40  | 47.8             | 48.4    | 0.069                  | 1.8 | 1.994             | 0.608 | 1.339  |
| 2            | 50  | 59.6             | 60.2    | 0.069                  | 1.8 | 2.503             | 0.763 | 1.680  |
| 2½           | 65  | 75.2             | 76.0    | 0.072                  | 1.9 | 3.368             | 1.026 | 2.261  |
| 3            | 80  | 87.9             | 88.7    | 0.080                  | 2.1 | 4.330             | 1.320 | 2.907  |
| 3½           | 90  | 101.1            | 102.0   | 0.080                  | 2.1 | 5.150             | 1.570 | 3.460  |
| 4            | 100 | 112.2            | 113.2   | 0.085                  | 2.2 | 5.917             | 1.803 | 3.972  |
| 5            | 125 | 138.7            | 140.6   | 0.150                  | 4.0 | 12.364            | 3.768 | 8.308  |
| 6            | 150 | 164.1            | 166.1   | 0.150                  | 4.0 | 15.069            | 4.593 | 10.117 |

Wall Thickness Tolerance = -10%

### CARBON STEEL CEMENT LINING PIPES (MANUFACTURER STANDARD)

| Outside Diameter | Minimum Nominal Wall Thickness | Outside Diameter |         | Calculated Weight |
|------------------|--------------------------------|------------------|---------|-------------------|
|                  |                                | Maximum          | Minimum |                   |
| mm               | mm                             | mm               | mm      | kg/m              |
| 177.3            | 3.7                            | 179.1            | 175.5   | 17.51             |
|                  | 4.1                            | 179.1            | 175.5   | 19.22             |
| 232.2            | 4.1                            | 234.5            | 229.9   | 25.32             |



# PIPES

## CARBON STEEL PIPES FOR SCAFFOLDING

JIS G 3444/88 (MANUFACTURER STANDARD))

| Nominal Size | Standard Specification | Outside Diameter |       | Wall Thickness | Weight | Reference            |                               |                    |                            |
|--------------|------------------------|------------------|-------|----------------|--------|----------------------|-------------------------------|--------------------|----------------------------|
|              |                        |                  |       |                |        | Cross Sectional Area | Geometrical Moment of Inertia | Modulus of Section | Radius of Gyration of Area |
|              |                        | Min              | Max   |                |        | cm <sup>2</sup>      | cm <sup>4</sup>               | cm <sup>3</sup>    | cm                         |
| mm           |                        | mm               | mm    | mm             | kg/m   |                      |                               |                    |                            |
| 21.7 O/D     | STK 290                | 21.45            | 21.95 | 1.20           | 0.607  | 0.773                | 0.407                         | 0.375              | 0.726                      |
|              |                        |                  |       | 1.50           | 0.747  | 0.952                | 0.488                         | 0.45               | 0.716                      |
|              |                        |                  |       | 1.60           | 0.793  | 1.010                | 0.513                         | 0.473              | 0.713                      |
|              |                        |                  |       | 2.00           | 0.972  | 1.238                | 0.607                         | 0.56               | 0.700                      |
| 42.7 O/D     | STK 400                | 42.45            | 42.95 | 2.30           | 2.29   | 2.919                | 5.97                          | 2.80               | 1.43                       |
|              |                        |                  |       | 2.50           | 2.49   | 3.157                | 6.40                          | 3.00               | 1.42                       |
|              |                        |                  |       | 2.80           | 2.76   | 3.510                | 7.02                          | 3.29               | 1.41                       |
| 48.6 O/D     | STK 500                | 48.35            | 48.85 | 2.30           | 2.63   | 3.345                | 8.99                          | 3.70               | 1.64                       |
|              |                        |                  |       | 2.50           | 2.84   | 3.621                | 9.65                          | 3.97               | 1.63                       |
|              |                        |                  |       | 2.80           | 3.16   | 4.029                | 10.60                         | 4.36               | 1.62                       |
|              |                        |                  |       | 3.20           | 3.58   | 4.564                | 11.80                         | 4.86               | 1.61                       |

Notes:

- 1) When Wall Thickness < 3.0 mm, tolerance =  $\pm 0.3$  mm
- 2) When  $3.0 \text{ mm} \leq \text{wall thickness} < 12 \text{ mm}$ , tolerance =  $\pm 10\%$

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## CARBON STEEL PIPES FOR ORDINARY PIPING

JIS G 3452-1968 (SGP)

| Nominal Size |     | Outside Diameter |        | Thickness |       | Unit weight (plain end) |       | Test Pressure      |     |
|--------------|-----|------------------|--------|-----------|-------|-------------------------|-------|--------------------|-----|
| A            | B   | mm               | in     | mm        | in    | lb/ft                   | kg/m  | kg/cm <sup>2</sup> | Psi |
| 6            | 1/8 | 10.5             | 0.413  | 2.0       | 0.079 | 0.282                   | 0.419 | 25                 | 360 |
| 8            | ¼   | 13.8             | 0.543  | 2.3       | 0.090 | 0.438                   | 0.652 | 25                 | 360 |
| 10           | 3/8 | 17.3             | 0.681  | 2.3       | 0.090 | 0.572                   | 0.851 | 25                 | 360 |
| 15           | ½   | 21.7             | 0.854  | 2.8       | 0.110 | 0.88                    | 1.31  | 25                 | 360 |
| 20           | ¾   | 27.2             | 1.071  | 2.8       | 0.110 | 1.13                    | 1.68  | 25                 | 360 |
| 25           | 1   | 34.0             | 1.338  | 3.2       | 0.126 | 1.63                    | 2.43  | 25                 | 360 |
| 32           | 1¼  | 42.7             | 1.681  | 3.5       | 0.138 | 2.27                    | 3.38  | 25                 | 360 |
| 40           | 1½  | 48.6             | 1.913  | 3.5       | 0.138 | 2.61                    | 3.89  | 25                 | 360 |
| 50           | 2   | 60.5             | 2.382  | 3.8       | 0.150 | 3.57                    | 5.31  | 25                 | 360 |
| 65           | 2½  | 76.3             | 3.004  | 4.2       | 0.165 | 5.02                    | 7.47  | 25                 | 360 |
| 80           | 3   | 89.1             | 3.508  | 4.2       | 0.165 | 5.91                    | 8.79  | 25                 | 360 |
| 90           | 3½  | 101.6            | 4.000  | 4.2       | 0.165 | 6.79                    | 10.1  | 25                 | 360 |
| 100          | 4   | 114.3            | 4.500  | 4.5       | 0.177 | 8.2                     | 12.2  | 25                 | 360 |
| 125          | 5   | 139.8            | 5.504  | 4.5       | 0.177 | 10.1                    | 15.0  | 25                 | 360 |
| 150          | 6   | 165.2            | 6.504  | 5.0       | 0.197 | 13.3                    | 19.8  | 25                 | 360 |
| 175          | 7   | 190.7            | 7.508  | 5.3       | 0.209 | 16.3                    | 24.2  | 25                 | 360 |
| 200          | 8   | 216.3            | 8.516  | 5.8       | 0.228 | 20.2                    | 30.1  | 25                 | 360 |
| 225          | 9   | 241.8            | 9.520  | 6.2       | 0.244 | 24.2                    | 36.0  | 25                 | 360 |
| 250          | 10  | 267.4            | 10.528 | 6.6       | 0.260 | 28.5                    | 42.4  | 25                 | 360 |
| 300          | 12  | 318.5            | 12.539 | 6.9       | 0.272 | 35.6                    | 53.0  | 25                 | 360 |
| 350          | 14  | 355.6            | 14.00  | 7.9       | 0.311 | 45.5                    | 67.7  | 25                 | 360 |
| 400          | 16  | 406.4            | 16.00  | 7.9       | 0.311 | 52.1                    | 77.6  | 25                 | 360 |
| 450          | 18  | 457.2            | 18.00  | 7.9       | 0.311 | 58.7                    | 87.5  | 25                 | 360 |
| 500          | 20  | 508.0            | 20.00  | 7.9       | 0.311 | 65.4                    | 97.4  | 25                 | 360 |

# PIPES

## CARBON STEEL PIPES FOR MACHINE STRUCTURAL PURPOSES

AVAILABLE IN COLD ROLLED, HOT ROLLED MATERIAL (JIS G 3445-1983-STKM 11A)

| Nominal<br>Size |       |      | Outside<br>Diameter |       | Thickness |       |       |       |       |       |       |       |        |       |       |  |  |
|-----------------|-------|------|---------------------|-------|-----------|-------|-------|-------|-------|-------|-------|-------|--------|-------|-------|--|--|
|                 |       |      |                     |       | SWG       | 19    |       |       |       | 18    |       |       |        | 17    |       |  |  |
|                 |       |      |                     |       | mm        | 1.0   |       |       |       | 1.2   |       |       |        | 1.4   |       |  |  |
|                 |       |      |                     |       | in        | 0.039 |       |       |       | 0.047 |       |       |        | 0.065 |       |  |  |
| mm              | in    | mm   |                     | kg/m  | kg/6m     | kg/ft | lb/ft | kg/m  | kg/6m | kg/ft | lb/ft | kg/m  | kg/6m  | kg/ft | lb/ft |  |  |
| 12              | 1/2   | 12.7 |                     | 0.289 | 1.734     | 0.088 | 0.194 | 0.340 | 2.040 | 0.104 | 0.229 | 0.390 | 2.34   | 0.119 | 0.262 |  |  |
| 16              | 5/8   | 15.9 |                     | 0.368 | 2.208     | 0.112 | 0.247 | 0.435 | 2.610 | 0.133 | 0.293 | 0.501 | 3.006  | 0.153 | 0.337 |  |  |
| 19              | 3/4   | 19.1 |                     | 0.447 | 2.682     | 0.136 | 0.300 | 0.530 | 3.180 | 0.162 | 0.357 | 0.611 | 3.666  | 0.186 | 0.410 |  |  |
| 22              | 7/8   | 22.2 |                     | 0.523 | 3.138     | 0.159 | 0.351 | 0.622 | 3.732 | 0.189 | 0.417 | 0.718 | 4.308  | 0.219 | 0.483 |  |  |
| 25              | 1     | 25.4 |                     | 0.602 | 3.612     | 0.183 | 0.404 | 0.716 | 4.296 | 0.218 | 0.481 | 0.829 | 4.974  | 0.253 | 0.558 |  |  |
| 28              | 1-1/8 | 28.6 |                     | 0.681 | 4.086     | 0.208 | 0.459 | 0.811 | 4.866 | 0.247 | 0.545 | 0.939 | 5.634  | 0.286 | 0.631 |  |  |
| 32              | 1-1/4 | 31.8 |                     | 0.760 | 4.560     | 0.232 | 0.512 | 0.906 | 5.436 | 0.276 | 0.609 | 1.050 | 6.30   | 0.320 | 0.706 |  |  |
| 35              | 1-3/8 | 34.9 |                     |       |           |       |       | 1.000 | 6.000 | 0.304 | 0.670 | 1.160 | 6.96   | 0.354 | 0.781 |  |  |
| 38              | 1-1/2 | 38.1 |                     |       |           |       |       | 1.092 | 6.552 | 0.333 | 0.734 | 1.267 | 7.602  | 0.386 | 0.851 |  |  |
| 41              | 1-5/8 | 41.3 |                     |       |           |       |       | 1.187 | 7.122 | 0.362 | 0.798 | 1.378 | 8.268  | 0.420 | 0.926 |  |  |
| 44              | 1-3/4 | 44.5 |                     |       |           |       |       | 1.281 | 7.686 | 0.391 | 0.862 | 1.488 | 8.928  | 0.454 | 1.001 |  |  |
| 47              | 1-7/8 | 47.6 |                     |       |           |       |       | 1.373 | 8.238 | 0.419 | 0.924 | 1.595 | 9.570  | 0.486 | 1.072 |  |  |
| 50              | 2     | 50.8 |                     |       |           |       |       | 1.468 | 8.808 | 0.447 | 0.986 | 1.705 | 10.230 | 0.520 | 1.146 |  |  |
| 54              | 2-1/8 | 54.0 |                     |       |           |       |       | 1.563 | 9.378 | 0.476 | 1.049 | 1.816 | 10.896 | 0.554 | 1.221 |  |  |
| 57              | 2-1/4 | 57.2 |                     |       |           |       |       |       |       |       |       | 1.926 | 11.556 | 0.587 | 1.294 |  |  |
| 60              | 2-3/8 | 60.3 |                     |       |           |       |       |       |       |       |       | 2.033 | 12.198 | 0.620 | 1.367 |  |  |
| 65              | 2-1/2 | 63.5 |                     |       |           |       |       |       |       |       |       | 2.144 | 12.864 | 0.653 | 1.440 |  |  |
| 80              | 3     | 76.2 |                     |       |           |       |       |       |       |       |       |       |        |       |       |  |  |

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## CARBON STEEL PIPES FOR MACHINE STRUCTURAL PURPOSES

AVAILABLE IN COLD ROLLED, HOT ROLLED MATERIAL (JIS G 3445-1983-STKM 11A)

| Nominal<br>Size | Outside<br>Diameter |      | Thickness |       |        |       |       |       |        |       |       |       |        |       |       |
|-----------------|---------------------|------|-----------|-------|--------|-------|-------|-------|--------|-------|-------|-------|--------|-------|-------|
|                 |                     |      | SWG       | 16    |        |       |       | 15    |        |       |       | 14    |        |       |       |
|                 |                     |      | mm        | 1.6   |        |       |       | 1.8   |        |       |       | 2.0   |        |       |       |
|                 |                     |      | in        | 0.063 |        |       |       | 0.071 |        |       |       | 0.079 |        |       |       |
| mm              | in                  | mm   |           | kg/m  | kg/6m  | kg/ft | lb/ft | kg/m  | kg/6m  | kg/ft | lb/ft | kg/m  | kg/6m  | kg/ft | lb/ft |
| 12              | 1/2                 | 12.7 |           | 0.438 | 2.628  | 0.134 | 0.296 | 0.484 | 2.904  | 0.148 | 0.326 | 0.528 | 3.168  | 0.161 | 0.355 |
| 16              | 5/8                 | 15.9 |           | 0.564 | 3.384  | 0.172 | 0.379 | 0.626 | 3.756  | 0.191 | 0.421 | 0.686 | 4.116  | 0.209 | 0.461 |
| 19              | 3/4                 | 19.1 |           | 0.691 | 4.146  | 0.211 | 0.465 | 0.768 | 4.608  | 0.234 | 0.516 | 0.844 | 5.064  | 0.257 | 0.567 |
| 22              | 7/8                 | 22.2 |           | 0.813 | 4.878  | 0.248 | 0.547 | 0.906 | 5.436  | 0.276 | 0.609 | 0.996 | 5.976  | 0.304 | 0.670 |
| 25              | 1                   | 25.4 |           | 0.939 | 5.634  | 0.286 | 0.631 | 1.048 | 6.288  | 0.319 | 0.703 | 1.154 | 6.924  | 0.352 | 0.776 |
| 28              | 1-1/8               | 28.6 |           | 1.066 | 6.396  | 0.325 | 0.717 | 1.190 | 7.140  | 0.363 | 0.80  | 1.312 | 7.872  | 0.400 | 0.882 |
| 32              | 1-1/4               | 31.8 |           | 1.192 | 7.152  | 0.363 | 0.800 | 1.332 | 7.992  | 0.406 | 0.895 | 1.470 | 8.820  | 0.448 | 0.988 |
| 35              | 1-3/8               | 34.9 |           | 1.318 | 7.908  | 0.402 | 0.886 | 1.469 | 8.814  | 0.448 | 0.988 | 1.623 | 9.738  | 0.495 | 1.091 |
| 38              | 1-1/2               | 38.1 |           | 1.440 | 8.640  | 0.439 | 0.968 | 1.612 | 9.672  | 0.491 | 1.083 | 1.781 | 10.686 | 0.543 | 1.197 |
| 41              | 1-5/8               | 41.3 |           | 1.567 | 9.402  | 0.478 | 1.054 | 1.754 | 10.524 | 0.534 | 1.177 | 1.938 | 11.628 | 0.591 | 1.303 |
| 44              | 1-3/4               | 44.5 |           | 1.693 | 10.158 | 0.516 | 1.138 | 1.896 | 11.376 | 0.578 | 1.274 | 2.096 | 12.576 | 0.639 | 1.409 |
| 47              | 1-7/8               | 47.6 |           | 1.815 | 10.890 | 0.553 | 1.219 | 2.033 | 12.198 | 0.620 | 1.367 | 2.249 | 13.494 | 0.686 | 1.512 |
| 50              | 2                   | 50.8 |           | 1.942 | 11.652 | 0.592 | 1.305 | 2.175 | 13.050 | 0.663 | 1.462 | 2.407 | 14.442 | 0.734 | 1.618 |
| 54              | 2-1/8               | 54.0 |           | 2.068 | 12.408 | 0.630 | 1.389 | 2.317 | 13.902 | 0.706 | 1.557 | 2.565 | 15.390 | 0.782 | 1.724 |
| 57              | 2-1/4               | 57.2 |           | 2.194 | 13.164 | 0.669 | 1.475 | 2.459 | 14.754 | 0.750 | 1.654 | 2.723 | 16.338 | 0.830 | 1.830 |
| 60              | 2-3/8               | 60.3 |           | 2.316 | 13.896 | 0.706 | 1.557 | 2.597 | 15.582 | 0.792 | 1.746 | 2.876 | 17.256 | 0.877 | 1.934 |
| 65              | 2-1/2               | 63.5 |           | 2.443 | 14.658 | 0.745 | 1.642 | 2.739 | 16.434 | 0.835 | 1.841 | 3.033 | 18.198 | 0.925 | 2.039 |
| 80              | 3                   | 76.2 |           | 2.944 | 17.644 | 0.897 | 1.978 | 3.303 | 19.818 | 1.007 | 2.220 | 3.660 | 21.960 | 1.116 | 2.460 |

The following size is also available upon request:

| Nominal<br>Size | Outside<br>Diameter |    | Thickness |       |       |       |       |
|-----------------|---------------------|----|-----------|-------|-------|-------|-------|
|                 |                     |    | mm        | 1.60  |       |       |       |
|                 |                     |    | in        | 0.063 |       |       |       |
| mm              | in                  | mm |           | kg/m  | kg/6m | kg/ft | lb/ft |
| 36              | 1.417               | 36 |           | 1.357 | 8.144 | 0.414 | 0.912 |

# PIPES

## CARBON STEEL PIPES FOR GENERAL STRUCTURAL PURPOSES

JIS G 3444-1988-STK 290

JIS G 3444-1988-STK 500

JIS G 3444-1988-STK 400

JIS G 3444-1988-STK 540

| Outside Diameter | Wall Thickness | Calculated Weight | Cross-Sectional Area | Geometrical Moment of Inertia | Modulus of Section | Radius of Gyration of Area |
|------------------|----------------|-------------------|----------------------|-------------------------------|--------------------|----------------------------|
| mm               | mm             | kg/mm             | cm <sup>2</sup>      | cm <sup>4</sup>               | cm <sup>3</sup>    | cm                         |
| 21.7             | 2.0            | 0.972             | 1.238                | 0.607                         | 0.56               | 0.70                       |
| 27.2             | 2.0            | 1.24              | 1.583                | 1.26                          | 0.93               | 0.89                       |
|                  | 2.3            | 1.41              | 1.799                | 1.41                          | 1.03               | 0.88                       |
| 34.0             | 2.3            | 1.80              | 2.291                | 2.89                          | 1.70               | 1.12                       |
| 42.7             | 2.3            | 2.29              | 2.919                | 5.97                          | 2.80               | 1.43                       |
|                  | 2.5            | 2.49              | 3.157                | 6.40                          | 3.00               | 1.42                       |
|                  | 2.8            | 2.76              | 3.510                | 7.02                          | 3.29               | 1.41                       |
| 48.6             | 2.3            | 2.63              | 3.345                | 8.99                          | 3.70               | 1.64                       |
|                  | 2.5            | 2.84              | 3.621                | 9.65                          | 3.97               | 1.63                       |
|                  | 2.8            | 3.16              | 4.029                | 10.6                          | 4.36               | 1.62                       |
|                  | 3.2            | 3.58              | 4.564                | 11.8                          | 4.86               | 1.61                       |
| 60.5             | 2.3            | 3.30              | 4.205                | 17.8                          | 5.90               | 2.06                       |
|                  | 3.2            | 4.52              | 5.760                | 23.7                          | 7.84               | 2.03                       |
|                  | 4.0            | 5.57              | 7.100                | 28.5                          | 9.41               | 2.00                       |
| 76.3             | 2.8            | 5.08              | 6.465                | 43.7                          | 11.5               | 2.60                       |
|                  | 3.2            | 5.77              | 7.349                | 49.2                          | 12.9               | 2.59                       |
|                  | 4.0            | 7.13              | 9.085                | 59.5                          | 15.6               | 2.56                       |
| 89.1             | 2.8            | 5.96              | 7.591                | 70.7                          | 15.9               | 3.05                       |
|                  | 3.2            | 6.78              | 8.636                | 79.8                          | 17.9               | 3.04                       |
|                  | 4.0            | 8.39              | 10.69                | 97.0                          | 21.8               | 3.01                       |
| 101.6            | 3.2            | 7.76              | 9.892                | 120                           | 23.6               | 3.48                       |
|                  | 4.0            | 9.63              | 12.26                | 146                           | 28.8               | 3.45                       |
|                  | 5.0            | 11.90             | 15.17                | 177                           | 34.9               | 3.42                       |
| 114.3            | 3.2            | 8.77              | 11.17                | 172                           | 30.2               | 3.93                       |
|                  | 3.6            | 9.83              | 12.52                | 192                           | 33.6               | 3.92                       |
|                  | 4.5            | 12.20             | 15.52                | 234                           | 41.0               | 3.89                       |
|                  | 5.6            | 15.00             | 19.12                | 283                           | 49.6               | 3.85                       |
| 139.8            | 3.6            | 12.10             | 15.40                | 357                           | 51.1               | 4.82                       |
|                  | 4.0            | 13.40             | 17.07                | 394                           | 56.3               | 4.80                       |
|                  | 4.5            | 15.00             | 19.13                | 438                           | 62.7               | 4.79                       |
|                  | 6.0            | 19.80             | 25.22                | 566                           | 80.9               | 4.74                       |
| 165.2            | 4.5            | 17.80             | 22.72                | 734                           | 88.9               | 5.68                       |
|                  | 5.0            | 19.80             | 25.16                | 808                           | 97.8               | 5.67                       |
|                  | 6.0            | 23.60             | 30.01                | 952                           | 115                | 5.63                       |
|                  | 7.0            | 27.30             | 34.79                | 1,090                         | 132                | 5.60                       |
| 190.7            | 4.5            | 20.70             | 26.32                | 1,140                         | 120                | 6.59                       |
|                  | 5.0            | 22.90             | 29.17                | 1,260                         | 132                | 6.57                       |
|                  | 6.0            | 27.30             | 34.82                | 1,490                         | 156                | 6.53                       |
|                  | 7.0            | 31.70             | 40.40                | 1,710                         | 179                | 6.50                       |
| 216.3            | 4.5            | 23.50             | 29.94                | 1,680                         | 155                | 7.49                       |
|                  | 6.0            | 31.10             | 39.64                | 2,190                         | 203                | 7.44                       |
|                  | 7.0            | 36.10             | 46.03                | 2,520                         | 233                | 7.40                       |

Applicable Tolerances:

|                  |                             |
|------------------|-----------------------------|
| Thickness        | Under 3 mm ± 0.3mm          |
|                  | 3 mm and over to 8 mm ± 10% |
| Outside Diameter | Under 50 mm ± 0.25 mm       |
|                  | 50 mm or over ± 0.5%        |

|                                   |     |
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YICK HOE

# CHANNELS

## SAFE LOADS FOR GRADE 43 STEEL

| Section size | Mass per metre | Safe distributed loads in kilonewtons for spans in metres and deflection coefficients |               |               |               |               |               |               |               |               |               |               |               |               | Critical span L <sub>c</sub> |
|--------------|----------------|---------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------------------------------|
| mm           | kg             | 1.50<br>199.1                                                                         | 2.00<br>112.0 | 2.50<br>71.68 | 3.00<br>49.78 | 3.50<br>36.57 | 4.00<br>28.00 | 4.50<br>22.12 | 5.00<br>17.92 | 5.50<br>14.81 | 6.00<br>12.44 | 7.00<br>9.143 | 8.00<br>7.000 | 9.00<br>5.531 | m                            |
| 76 x 38      | 6.70           | 17                                                                                    | 13            | 10            |               |               |               |               |               |               |               |               |               |               | 1.724                        |
| 102 x 51     | 10.42          | 36                                                                                    | 27            | 22            | 18            |               |               |               |               |               |               |               |               |               | 1.993                        |
| 127 x 64     | 14.90          | 67                                                                                    | 50            | 40            | 33            | 29            | 25            |               |               |               |               |               |               |               | 2.477                        |
| 152 x 76     | 17.88          | 98                                                                                    | 74            | 59            | 49            | 42            | 37            | 33            | 30            |               |               |               |               |               | 2.591                        |
| 152 x 89     | 23.84          | 135                                                                                   | 101           | 81            | 67            | 58            | 50            | 45            | 40            |               |               |               |               |               | 3.629                        |
| 178 x 76     | 20.84          | 132                                                                                   | 99            | 79            | 66            | 57            | 50            | 44            | 40            | 36            | 33            |               |               |               | 2.577                        |
| 178 x 89     | 26.81          | 174                                                                                   | 130           | 104           | 87            | 74            | 65            | 58            | 52            | 47            | 43            |               |               |               | 3.388                        |
| 203 x 76     | 23.82          | 169                                                                                   | 127           | 101           | 84            | 72            | 63            | 56            | 51            | 46            | 42            |               |               |               | 2.497                        |
| 203 x 89     | 29.78          | 216                                                                                   | 162           | 129           | 108           | 92            | 81            | 72            | 65            | 59            | 54            |               |               |               | 3.190                        |
| 229 x 76     | 26.06          | 201                                                                                   | 151           | 121           | 100           | 86            | 75            | 67            | 60            | 55            | 50            | 43            |               |               | 2.317                        |
| 229 x 89     | 32.76          | 261                                                                                   | 196           | 156           | 130           | 112           | 98            | 87            | 78            | 71            | 65            | 56            |               |               | 3.005                        |
| 254 x 76     | 28.29          | 233                                                                                   | 175           | 140           | 117           | 100           | 87            | 78            | 70            | 64            | 58            | 50            | 44            |               | 2.142                        |
| 254 x 89     | 35.74          | 308                                                                                   | 231           | 185           | 154           | 132           | 116           | 103           | 92            | 84            | 77            | 66            | 58            |               | 2.842                        |
| 304 x 89     | 41.69          | 408                                                                                   | 306           | 245           | 204           | 175           | 153           | 136           | 122           | 111           | 102           | 87            | 76            | 68            | 2.537                        |
| 305 x 102    | 46.18          | 474                                                                                   | 356           | 285           | 237           | 203           | 178           | 158           | 142           | 129           | 119           | 102           | 89            | 79            | 3.077                        |
| 381 x 102    | 55.10          | 688                                                                                   | 516           | 413           | 344           | 295           | 258           | 229           | 206           | 188           | 172           | 147           | 129           | 115           | 2.896                        |
| 432 x 102    | 65.54          | 872                                                                                   | 654           | 523           | 436           | 374           | 327           | 291           | 262           | 238           | 218           | 187           | 164           | 145           | 2.707                        |

For explanation of table see page 25.

The loads listed are based on a bending stress of 165 N/mm<sup>2</sup>, and assume adequate lateral support. Without such support the span must not exceed L<sub>c</sub> unless the compressive stress is reduced in accordance with clause 19a (ii) of BS 449.

Loads printed in bold type may cause overloading of the unstiffened web, the capacity of which should be checked. See page 30.

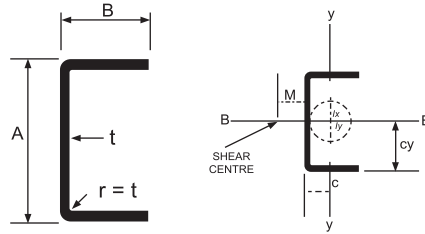
Loads printed in italic type do not cause overloading of the unstiffened web, and do not cause deflection exceeding span/360.

Loads printed in ordinary type should be checked for deflection. See pages 28 and 29.

† Load is based on allowable shear of web and is less than allowable load in bending.

# PLAIN CHANNELS

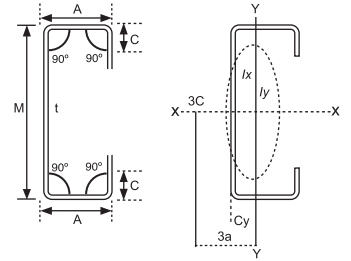
## G 3350-1987



| Section size<br>A x B<br>mm | Unit Weight<br>M<br>kg/m | Thick-<br>ness<br>t<br>mm | Area<br>A<br>cm <sup>2</sup> | Position of Centroid<br>Cx cm Cy cm |       | Moment of Inertia<br>Ix cm <sup>4</sup> Iy cm <sup>4</sup> |       | Radius of gyration<br>Rx cm Ry cm |      | Modulus of Section<br>Zx cm <sup>3</sup> Zy cm <sup>3</sup> |       | Moment Capacity<br>Mx kg/m | Shear Centre<br>M cm | Column Factor<br>Q |
|-----------------------------|--------------------------|---------------------------|------------------------------|-------------------------------------|-------|------------------------------------------------------------|-------|-----------------------------------|------|-------------------------------------------------------------|-------|----------------------------|----------------------|--------------------|
| 60 x 30                     | 1.44                     | 1.6                       | 1.83                         | 0.82                                | 3.00  | 10.33                                                      | 1.63  | 2.37                              | 0.94 | 3.44                                                        | 0.75  | 46.08                      | 1.12                 | 0.77               |
|                             | 2.03                     | 2.3                       | 2.58                         | 0.85                                | 3.00  | 15.13                                                      | 2.25  | 2.33                              | 0.93 | 4.71                                                        | 1.05  | 76.97                      | 1.12                 | 0.96               |
| 75 x 45                     | 2.00                     | 1.6                       | 2.55                         | 1.30                                | 3.75  | 23.86                                                      | 5.38  | 3.05                              | 1.45 | 6.36                                                        | 1.68  | 53.04                      | 1.76                 | 0.46               |
|                             | 2.84                     | 2.3                       | 3.62                         | 1.33                                | 3.75  | 33.08                                                      | 7.51  | 3.02                              | 1.44 | 8.82                                                        | 2.37  | 114.32                     | 1.76                 | 0.76               |
|                             | 3.65                     | 3.0                       | 4.75                         | 1.37                                | 3.75  | 41.58                                                      | 9.51  | 2.98                              | 1.42 | 11.08                                                       | 3.04  | 169.91                     | 1.76                 | 0.90               |
| 100 x 50                    | 2.44                     | 1.6                       | 3.11                         | 1.32                                | 5.00  | 50.00                                                      | 7.88  | 4.00                              | 1.59 | 10.00                                                       | 2.14  | 66.48                      | 1.87                 | 0.33               |
|                             | 3.47                     | 2.3                       | 4.42                         | 1.35                                | 5.00  | 69.83                                                      | 11.05 | 3.97                              | 1.58 | 13.96                                                       | 3.03  | 165.23                     | 1.87                 | 0.66               |
|                             | 4.47                     | 3.0                       | 5.70                         | 1.38                                | 5.00  | 88.45                                                      | 14.05 | 3.93                              | 1.56 | 17.69                                                       | 3.89  | 255.77                     | 1.87                 | 0.85               |
| 125 x 50                    | 3.92                     | 2.3                       | 5.00                         | 1.21                                | 6.25  | 117.36                                                     | 11.84 | 4.84                              | 1.53 | 18.77                                                       | 3.12  | 222.17                     | 1.76                 | 0.61               |
|                             | 5.06                     | 3.0                       | 6.45                         | 1.24                                | 6.25  | 149.25                                                     | 15.07 | 4.80                              | 1.52 | 23.88                                                       | 5.01  | 345.27                     | 1.75                 | 0.81               |
|                             | 7.42                     | 4.5                       | 9.45                         | 1.31                                | 6.25  | 211.76                                                     | 31.43 | 4.73                              | 1.50 | 33.88                                                       | 5.81  | 571.75                     | 1.75                 | 1.00               |
| 150 x 65                    | 4.91                     | 2.3                       | 6.26                         | 1.61                                | 7.50  | 218.16                                                     | 25.86 | 5.90                              | 2.03 | 29.08                                                       | 5.29  | 239.92                     | 2.34                 | 0.40               |
|                             | 6.36                     | 3.0                       | 8.10                         | 1.64                                | 7.50  | 278.84                                                     | 33.12 | 5.86                              | 2.02 | 37.17                                                       | 6.82  | 441.26                     | 2.34                 | 0.63               |
|                             | 9.36                     | 4.5                       | 11.93                        | 1.71                                | 7.50  | 400.15                                                     | 47.69 | 5.79                              | 1.99 | 53.35                                                       | 9.96  | 832.81                     | 2.34                 | 0.91               |
| 175 x 75                    | 5.73                     | 2.3                       | 7.30                         | 1.83                                | 8.75  | 346.87                                                     | 40.19 | 6.89                              | 2.34 | 39.64                                                       | 7.09  | 240.68                     | 2.69                 | 0.28               |
|                             | 7.42                     | 3.0                       | 9.45                         | 1.86                                | 8.75  | 444.65                                                     | 51.59 | 6.85                              | 2.33 | 50.81                                                       | 9.15  | 515.38                     | 2.69                 | 0.51               |
|                             | 10.95                    | 4.5                       | 13.95                        | 1.93                                | 8.75  | 642.23                                                     | 74.74 | 6.78                              | 2.31 | 73.39                                                       | 13.42 | 1061.21                    | 2.69                 | 0.82               |
| 200 x 75                    | 6.18                     | 2.3                       | 7.87                         | 1.70                                | 10.00 | 473.31                                                     | 41.76 | 7.75                              | 2.30 | 47.33                                                       | 7.21  | 287.35                     | 2.48                 | 0.26               |
|                             | 8.01                     | 3.0                       | 10.20                        | 1.73                                | 10.00 | 607.58                                                     | 53.63 | 7.71                              | 2.29 | 60.75                                                       | 9.31  | 616.21                     | 2.58                 | 0.48               |
|                             | 11.84                    | 4.5                       | 15.08                        | 1.80                                | 10.00 | 880.30                                                     | 77.78 | 7.63                              | 2.27 | 88.03                                                       | 13.65 | 1272.76                    | 2.58                 | 0.79               |
| 225 x 75                    | 6.63                     | 2.3                       | 8.45                         | 1.59                                | 11.25 | 624.36                                                     | 43.12 | 8.59                              | 2.25 | 55.49                                                       | 7.30  | 336.94                     | 2.49                 | 0.24               |
|                             | 8.59                     | 3.0                       | 10.95                        | 1.63                                | 11.25 | 802.40                                                     | 55.40 | 8.55                              | 2.24 | 71.32                                                       | 9.43  | 723.37                     | 2.48                 | 0.45               |
|                             | 12.72                    | 4.5                       | 16.20                        | 1.69                                | 11.25 | 1145.51                                                    | 80.39 | 8.47                              | 2.22 | 103.60                                                      | 13.85 | 1497.89                    | 2.48                 | 0.76               |
| 250 x 75                    | 7.08                     | 2.3                       | 9.02                         | 1.50                                | 12.50 | 801.81                                                     | 44.31 | 9.42                              | 2.21 | 64.14                                                       | 7.39  | 389.44                     | 2.39                 | 0.23               |
|                             | 9.18                     | 3.0                       | 11.70                        | 1.53                                | 12.50 | 1031.46                                                    | 56.94 | 9.38                              | 2.20 | 82.51                                                       | 9.54  | 836.88                     | 2.39                 | 0.43               |
|                             | 13.60                    | 4.5                       | 17.33                        | 1.60                                | 12.50 | 1501.38                                                    | 82.67 | 9.30                              | 2.18 | 120.11                                                      | 14.01 | 1736.58                    | 2.39                 | 0.72               |



# CHANNELS



## LIPPED CHANNELS (C-CHANNEL)

JIS G 3350-1987

| Section size<br>A x B x C<br>mm | Unit Weight<br>M<br>kg/m | Thick-ness<br>t<br>mm | Area<br>A<br>cm <sup>2</sup> | Position of Centroid<br>Cx      Cy<br>cm      cm |       | Moment of Inertia<br>Ix      Iy<br>cm <sup>4</sup> cm <sup>4</sup> |        | Radius of gyration<br>Rx      Ry<br>cm      cm |      | Modulus of Section<br>Zx      Zy<br>cm <sup>3</sup> cm <sup>3</sup> |       | Reduced Modulus<br>Zx1<br>cm | Moment Capacity<br>Mx<br>kg/m | Shear Centre<br>M<br>cm | Column Factor<br>Q |
|---------------------------------|--------------------------|-----------------------|------------------------------|--------------------------------------------------|-------|--------------------------------------------------------------------|--------|------------------------------------------------|------|---------------------------------------------------------------------|-------|------------------------------|-------------------------------|-------------------------|--------------------|
| 60 x 30 x 10                    | 1.62                     | 1.6                   | 2.07                         | 1.06                                             | 3.00  | 11.60                                                              | 2.54   | 2.36                                           | 1.10 | 3.87                                                                | 1.31  | 3.87                         | 65.40                         | 1.54                    | 0.97               |
|                                 | 2.25                     | 2.3                   | 2.87                         | 1.05                                             | 3.00  | 15.50                                                              | 3.28   | 2.32                                           | 1.07 | 5.18                                                                | 1.69  | 5.13                         | 87.50                         | 1.53                    | 1.00               |
| 75 x 45 x 15                    | 2.31                     | 1.6                   | 2.95                         | 1.72                                             | 3.75  | 27.10                                                              | 8.69   | 3.03                                           | 1.71 | 7.23                                                                | 3.13  | 7.23                         | 122.00                        | 2.43                    | 0.94               |
|                                 | 3.24                     | 2.3                   | 4.13                         | 1.71                                             | 3.75  | 37.00                                                              | 11.67  | 2.99                                           | 1.67 | 9.88                                                                | 4.19  | 9.88                         | 166.80                        | 2.42                    | 0.99               |
|                                 | 4.12                     | 3.0                   | 5.25                         | 1.71                                             | 3.75  | 45.90                                                              | 14.16  | 2.95                                           | 1.64 | 12.24                                                               | 5.07  | 12.24                        | 206.50                        | 2.42                    | 1.00               |
| 100 x 50 x 20                   | 2.88                     | 1.6                   | 3.67                         | 1.86                                             | 5.00  | 58.30                                                              | 13.97  | 3.98                                           | 1.95 | 11.67                                                               | 4.46  | 11.66                        | 196.80                        | 2.73                    | 0.85               |
|                                 | 4.06                     | 2.3                   | 5.17                         | 1.86                                             | 5.00  | 80.60                                                              | 18.96  | 3.94                                           | 1.91 | 16.12                                                               | 6.04  | 16.12                        | 272.10                        | 2.72                    | 0.95               |
|                                 | 5.18                     | 3.0                   | 6.60                         | 1.85                                             | 5.00  | 100.90                                                             | 23.30  | 3.90                                           | 1.87 | 20.18                                                               | 7.42  | 20.18                        | 340.60                        | 2.71                    | 0.99               |
| 125 x 50 x 20                   | 4.51                     | 2.3                   | 5.74                         | 1.68                                             | 6.25  | 136.40                                                             | 20.54  | 4.87                                           | 1.89 | 21.82                                                               | 6.20  | 21.82                        | 368.20                        | 2.55                    | 0.89               |
|                                 | 5.77                     | 3.0                   | 7.35                         | 1.63                                             | 6.25  | 171.50                                                             | 25.27  | 4.82                                           | 1.85 | 27.44                                                               | 7.62  | 27.44                        | 463.10                        | 2.54                    | 0.95               |
|                                 | 8.31                     | 4.5                   | 10.59                        | 1.67                                             | 6.25  | 237.10                                                             | 33.21  | 4.73                                           | 1.77 | 37.94                                                               | 10.00 | 37.94                        | 640.20                        | 2.51                    | 1.00               |
| 150 x 65 x 20                   | 5.50                     | 2.3                   | 7.01                         | 2.11                                             | 7.50  | 247.70                                                             | 40.99  | 5.94                                           | 2.41 | 33.03                                                               | 9.35  | 33.03                        | 557.40                        | 3.16                    | 0.84               |
|                                 | 7.07                     | 3.0                   | 9.00                         | 2.11                                             | 7.50  | 313.70                                                             | 50.99  | 5.90                                           | 2.37 | 41.83                                                               | 11.62 | 41.83                        | 705.80                        | 3.15                    | 0.91               |
|                                 | 10.25                    | 4.5                   | 13.06                        | 2.10                                             | 7.50  | 440.70                                                             | 68.79  | 5.80                                           | 2.29 | 58.77                                                               | 15.64 | 58.77                        | 991.70                        | 3.12                    | 0.99               |
| 175 x 75 x 20                   | 6.31                     | 2.3                   | 8.04                         | 2.34                                             | 8.75  | 389.30                                                             | 60.97  | 6.95                                           | 2.75 | 44.49                                                               | 11.83 | 44.31                        | 747.70                        | 3.52                    | 0.80               |
|                                 | 8.13                     | 3.0                   | 10.35                        | 2.34                                             | 8.75  | 494.90                                                             | 76.25  | 6.91                                           | 2.71 | 56.56                                                               | 14.78 | 56.56                        | 954.50                        | 3.50                    | 0.87               |
|                                 | 11.84                    | 4.5                   | 15.09                        | 2.33                                             | 8.75  | 701.60                                                             | 104.12 | 6.81                                           | 2.62 | 80.18                                                               | 20.14 | 80.18                        | 1353.10                       | 3.47                    | 0.96               |
| 200 x 75 x 20                   | 6.76                     | 2.3                   | 8.62                         | 2.19                                             | 10.00 | 531.00                                                             | 63.65  | 7.84                                           | 2.71 | 53.10                                                               | 12.00 | 52.89                        | 892.50                        | 3.38                    | 0.75               |
|                                 | 8.71                     | 3.0                   | 11.10                        | 2.19                                             | 10.00 | 676.10                                                             | 79.62  | 7.80                                           | 2.67 | 67.61                                                               | 15.00 | 67.61                        | 1140.90                       | 3.36                    | 0.82               |
|                                 | 12.73                    | 4.5                   | 16.12                        | 2.18                                             | 10.00 | 962.00                                                             | 108.77 | 7.70                                           | 2.58 | 96.20                                                               | 20.46 | 96.20                        | 1623.30                       | 3.33                    | 0.94               |
| 225 x 75 x 20                   | 7.21                     | 2.3                   | 9.19                         | 2.06                                             | 11.25 | 699.60                                                             | 65.99  | 8.72                                           | 2.67 | 62.18                                                               | 12.14 | 61.95                        | 1045.40                       | 3.26                    | 0.71               |
|                                 | 9.30                     | 3.0                   | 11.85                        | 2.06                                             | 11.25 | 892.00                                                             | 82.55  | 8.67                                           | 2.63 | 79.29                                                               | 15.19 | 79.29                        | 1338.00                       | 3.24                    | 0.78               |
|                                 | 13.61                    | 4.5                   | 17.34                        | 2.05                                             | 11.25 | 1273.10                                                            | 112.81 | 8.56                                           | 2.55 | 113.16                                                              | 20.73 | 113.16                       | 1909.60                       | 3.20                    | 0.89               |
| 250 x 75 x 25                   | 7.85                     | 2.3                   | 10.00                        | 2.07                                             | 12.50 | 921.11                                                             | 74.68  | 9.59                                           | 2.73 | 73.68                                                               | 13.77 | 73.43                        | 1239.14                       | 3.34                    | 0.68               |
|                                 | 10.13                    | 3.0                   | 12.90                        | 2.07                                             | 12.50 | 1176.57                                                            | 93.68  | 9.54                                           | 2.69 | 94.12                                                               | 17.27 | 94.12                        | 1588.31                       | 3.32                    | 0.74               |
|                                 | 14.85                    | 4.5                   | 18.91                        | 2.07                                             | 12.50 | 1685.70                                                            | 128.83 | 9.43                                           | 2.60 | 134.85                                                              | 23.74 | 134.85                       | 2275.61                       | 3.27                    | 0.86               |

## DIN 1026 CHANNELS

| Symbol<br>U | Dimensions for |              |      |              |      |              |      |                |                |                | Cross<br>Section<br><br>F<br>cm <sup>2</sup> | Weight<br><br>G<br>kg/m | Surface<br>Area<br><br>U<br>m <sup>2</sup> /m | Relative to bending axis*2        |                                   |                      |                                   |                                   |                      |                                                 |                                    |      | Distance<br>of axis<br>y - y<br><br>e <sub>y</sub> cm<br>X <sub>M</sub> <sup>15</sup> cm |      |
|-------------|----------------|--------------|------|--------------|------|--------------|------|----------------|----------------|----------------|----------------------------------------------|-------------------------|-----------------------------------------------|-----------------------------------|-----------------------------------|----------------------|-----------------------------------|-----------------------------------|----------------------|-------------------------------------------------|------------------------------------|------|------------------------------------------------------------------------------------------|------|
|             | h              | perm<br>var. | b    | perm<br>var. | s    | perm<br>var. | t    | perm<br>var.*1 | r <sub>1</sub> | r <sub>2</sub> |                                              |                         |                                               | x - x                             |                                   |                      | y - y                             |                                   |                      | S <sub>x</sub> <sup>+3</sup><br>cm <sup>3</sup> | s <sub>x</sub> <sup>+4</sup><br>cm |      |                                                                                          |      |
|             |                |              |      |              |      |              |      |                |                |                |                                              |                         |                                               | J <sub>x</sub><br>cm <sup>4</sup> | W <sub>x</sub><br>cm <sup>3</sup> | i <sub>x</sub><br>cm | J <sub>y</sub><br>cm <sup>4</sup> | W <sub>y</sub><br>cm <sup>3</sup> | i <sub>y</sub><br>cm |                                                 |                                    |      |                                                                                          |      |
|             |                |              |      |              |      |              |      |                |                |                |                                              |                         |                                               |                                   |                                   |                      |                                   |                                   |                      |                                                 |                                    |      |                                                                                          |      |
| 30 x 15     | 30             | ±1,5         | 15   | ±1,5         | 4    | ±0,5         | 4,5  | -0,5           | 4,5            | 2              | 2,21                                         | 1,74                    | 0,103                                         | 2,53                              | 1,69                              | 1,07                 | 0,38                              | 0,39                              | 0,42                 | --                                              | --                                 | 0,52 | 0,74                                                                                     |      |
| 30          | 30             |              | 33   |              | 5    |              | 7    |                | 7              | 3,5            | 5,44                                         | 4,27                    | 0,174                                         | 6,39                              | 4,26                              | 1,08                 | 5,33                              | 2,68                              | 0,99                 | --                                              | --                                 | 1,31 | 2,22                                                                                     |      |
| 40 x 20     | 40             |              | 20   |              | 5    |              | 5,5  |                | 5              | 2,5            | 3,66                                         | 2,87                    | 0,142                                         | 7,58                              | 3,79                              | 1,44                 | 1,14                              | 0,86                              | 0,56                 | --                                              | --                                 | 0,67 | 1,01                                                                                     |      |
| 40          | 40             |              | 35   |              | 5    |              | 7    |                | 7              | 3,5            | 6,21                                         | 4,87                    | 0,199                                         | 14,1                              | 7,05                              | 1,50                 | 6,68                              | 3,08                              | 1,04                 | --                                              | --                                 | 1,33 | 2,32                                                                                     |      |
| 50 x 25     | 50             |              | 25   |              | 5    |              | 6    |                | 6              | 3              | 4,92                                         | 3,86                    | 0,181                                         | 16,8                              | 6,73                              | 1,85                 | 2,49                              | 1,48                              | 0,71                 | --                                              | --                                 | 0,81 | 1,34                                                                                     |      |
| 50          | 50             |              | 38   |              | 5    |              | 7    |                | 7              | 3,5            | 7,12                                         | 5,59                    | 0,232                                         | 26,4                              | 10,6                              | 1,92                 | 9,12                              | 3,75                              | 1,13                 | --                                              | --                                 | 1,37 | 2,47                                                                                     |      |
| 60          | 60             |              | 30   |              | 6    |              | 6    |                | 6              | 3              | 6,46                                         | 5,07                    | 0,215                                         | 31,6                              | 10,5                              | 2,21                 | 4,51                              | 2,16                              | 0,84                 | --                                              | --                                 | 0,91 | 1,50                                                                                     |      |
| 65          | 65             |              | 42   |              | ±1,5 |              | 5,5  |                | 7,5            | 7,5            | 4                                            | 9,03                    | 7,09                                          | 0,273                             | 57,5                              | 17,7                 | 2,52                              | 14,1                              | 5,07                 | 1,25                                            | --                                 | --   | 1,42                                                                                     | 2,60 |
| 80          | 80             | ±2,0         | 45   | ±2,0         | 6    | ±0,5         | 8    | -1,0           | 8              | 4              | 11,0                                         | 8,64                    | 0,312                                         | 106                               | 26,5                              | 3,10                 | 19,4                              | 6,36                              | 1,33                 | 15,9                                            | 6,65                               | 1,45 | 2,67                                                                                     |      |
| 100         | 100            |              | 50   |              | 6    |              | ±0,5 |                | 8,5            | 8,5            | 4,5                                          | 13,5                    | 10,6                                          | 0,372                             | 206                               | 41,2                 | 3,91                              | 29,3                              | 8,49                 | 1,47                                            | 24,5                               | 8,42 | 1,55                                                                                     | 2,93 |
| 120         | 120            |              | 55   |              | 7    |              | 9    |                | 9              | 4,5            | 17,0                                         | 13,4                    | 0,434                                         | 364                               | 60,7                              | 4,62                 | 43,2                              | 11,1                              | 1,59                 | 36,3                                            | 10,0                               | 1,60 | 3,03                                                                                     |      |
| 140         | 140            |              | 60   |              | 7    |              | 10   |                | 10             | 5              | 20,4                                         | 16,0                    | 0,489                                         | 605                               | 86,4                              | 5,45                 | 62,7                              | 14,8                              | 1,75                 | 51,4                                            | 11,8                               | 1,75 | 3,37                                                                                     |      |
| 160         | 160            |              | 65   |              | 7,5  |              | 10,5 |                | 10,5           | 5,5            | 24,0                                         | 18,8                    | 0,546                                         | 925                               | 116                               | 6,21                 | 85,3                              | 18,3                              | 1,89                 | 68,8                                            | 13,3                               | 1,84 | 3,56                                                                                     |      |
| 180         | 180            |              | 70   |              | 8    |              | 11   |                | 11             | 5,5            | 28,0                                         | 22,0                    | 0,611                                         | 1350                              | 150                               | 6,95                 | 114                               | 22,4                              | 2,02                 | 89,6                                            | 15,1                               | 1,92 | 3,75                                                                                     |      |
| 200         | 200            |              | 75   |              | 8,5  |              | 11,5 |                | 11,5           | 6              | 32,2                                         | 25,3                    | 0,661                                         | 1910                              | 191                               | 7,70                 | 148                               | 27,0                              | 2,14                 | 114                                             | 16,8                               | 2,01 | 3,94                                                                                     |      |
| 220         | 220            |              | 80   |              | ±2,0 |              | 9    |                | 12,5           | -1,0           | 12,5                                         | 6,5                     | 37,4                                          | 29,4                              | 0,718                             | 2690                 | 245                               | 8,48                              | 197                  | 33,6                                            | 2,30                               | 146  | 18,5                                                                                     | 2,14 |
| 240         | 240            | 85           | 9,5  | 13           |      | 13           | 6,5  | 42,3           | 33,2           | 0,775          | 3600                                         | 300                     | 9,22                                          | 248                               | 39,6                              | 2,42                 | 179                               | 20,1                              | 2,23                 | 4,39                                            |                                    |      |                                                                                          |      |
| 260         | 260            | 90           | 10   | 14           |      | 14           | 7    | 48,3           | 37,9           | 0,834          | 4820                                         | 371                     | 9,99                                          | 317                               | 47,7                              | 2,56                 | 221                               | 21,8                              | 2,36                 | 4,66                                            |                                    |      |                                                                                          |      |
| 280         | 280            | 95           | 10   | 15           |      | 15           | 7,5  | 53,3           | 41,8           | 0,890          | 6280                                         | 448                     | 10,9                                          | 399                               | 57,2                              | 2,74                 | 266                               | 23,6                              | 2,53                 | 5,02                                            |                                    |      |                                                                                          |      |
| 300         | 300            | 100          | ±3,0 | 10           |      | 16           | 16   | 8              | 58,8           | 46,2           | 0,950                                        | 8030                    | 535                                           | 11,7                              | 495                               | 67,8                 | 2,90                              | 316                               | 25,4                 | 2,70                                            | 5,41                               |      |                                                                                          |      |
| 320         | 320            | 100          |      | 14           |      | 17,5         | 17,5 | 8,75           | 75,8           | 59,5           | 0,982                                        | 10870                   | 679                                           | 12,1                              | 597                               | 80,6                 | 2,81                              | 413                               | 26,3                 | 2,60                                            | 4,82                               |      |                                                                                          |      |
| 350         | 350            | 100          |      | ±2,5         |      | 14           | ±0,7 | 16             | -1,5           | 16             | 8                                            | 77,3                    | 60,6                                          | 1,05                              | 12840                             | 734                  | 12,9                              | 570                               | 75,0                 | 2,72                                            | 459                                | 28,6 | 2,40                                                                                     | 4,45 |
| 380         | 380            | 102          |      |              |      | 13,5         | 16   | 16             | 8              | 80,4           | 63,1                                         | 1,11                    | 15760                                         | 829                               | 14,0                              | 615                  | 78,7                              | 2,77                              | 507                  | 31,1                                            | 2,38                               | 4,58 |                                                                                          |      |
| 400         | 400            | 110          |      |              | 14   | 18           | 18   | 9              | 91,5           | 71,8           | 1,18                                         | 20350                   | 1020                                          | 14,9                              | 846                               | 102                  | 3,04                              | 618                               | 32,9                 | 2,65                                            | 5,11                               |      |                                                                                          |      |

\*1 The permissible plus variation is limited by the permissible exceeding of weight.

\*2 J = moment of inertia; W = section modulus; i = radius of gyration, referenced in each case to the bending axis concerned.

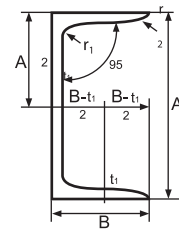
\*3 S<sub>x</sub> = static moment of half cross-section

\*4 s<sub>x</sub> = J<sub>x</sub> : S<sub>x</sub> = distance between compression and tension centres

\*5 x<sub>M</sub> = distance of shear centre M from the y - y axis

The cross-sections, weights, surface areas and static values have been calculated from the dimensions detailed in the Table.

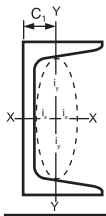
# CHANNELS



## U-CHANNELS (Inside slope 5°)

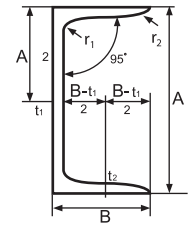
| Section size<br>A x B | Unit weight<br>M |       | Section depth<br>A |      | Flange width<br>B |     | Flange thickness<br>t1 |       | Web thickness<br>t2 |       | Corner radius |       |     |       |
|-----------------------|------------------|-------|--------------------|------|-------------------|-----|------------------------|-------|---------------------|-------|---------------|-------|-----|-------|
|                       |                  |       |                    |      |                   |     |                        |       |                     |       | r1            |       | r2  |       |
| mm (in)               | kg/m             | lb/ft | mm                 | in   | mm                | in  | mm                     | in    | mm                  | in    | mm            | in    | mm  | in    |
| 127 x 64 (5 x 2 1/2)  | 14.90            | 10.0  | 127.0              | 5.0  | 63.5              | 2.5 | 9.2                    | 0.362 | 6.4                 | 0.252 | 10.7          | 0.421 | 2.4 | 0.094 |
| 152 x 76 (6 x 3)      | 17.88            | 12.0  | 152.4              | 6.0  | 76.2              | 3.0 | 9.0                    | 0.354 | 6.4                 | 0.252 | 12.2          | 0.480 | 2.4 | 0.094 |
| 152 x 89 (6 x 3 1/2)  | 23.84            | 16.0  | 152.4              | 6.0  | 88.9              | 3.5 | 11.6                   | 0.457 | 7.1                 | 0.280 | 13.7          | 0.539 | 3.2 | 0.126 |
| 178 x 76 (7 x 3)      | 20.84            | 14.0  | 177.8              | 7.0  | 76.2              | 3.0 | 10.3                   | 0.405 | 6.6                 | 0.260 | 12.2          | 0.480 | 3.2 | 0.126 |
| 178 x 89 (7 x 3 1/2)  | 26.81            | 18.0  | 177.8              | 7.0  | 88.9              | 3.5 | 12.3                   | 0.484 | 7.6                 | 0.299 | 13.7          | 0.539 | 3.2 | 0.126 |
| 203 x 76 (8 x 3)      | 23.82            | 16.0  | 203.2              | 8.0  | 76.2              | 3.0 | 11.2                   | 0.441 | 7.1                 | 0.280 | 12.2          | 0.480 | 3.2 | 0.126 |
| 203 x 89 (8 x 3 1/2)  | 29.78            | 20.0  | 203.2              | 8.0  | 88.9              | 3.5 | 12.9                   | 0.508 | 8.1                 | 0.319 | 13.7          | 0.539 | 3.2 | 0.126 |
| 229 x 76 (9 x 3)      | 26.06            | 17.5  | 228.6              | 9.0  | 76.2              | 3.0 | 11.2                   | 0.441 | 7.6                 | 0.299 | 12.2          | 0.480 | 3.2 | 0.126 |
| 229 x 89 (9 x 3 1/2)  | 32.76            | 22.0  | 228.6              | 9.0  | 88.9              | 3.5 | 13.3                   | 0.524 | 8.6                 | 0.339 | 13.7          | 0.539 | 3.2 | 0.126 |
| 254 x 76 (10 x 3)     | 28.29            | 19.0  | 254.0              | 10.0 | 76.2              | 3.0 | 10.9                   | 0.429 | 8.1                 | 0.319 | 12.2          | 0.480 | 3.2 | 0.126 |
| 254 x 89 (10 x 3 1/2) | 35.74            | 24.0  | 254.0              | 10.0 | 88.9              | 3.5 | 13.6                   | 0.535 | 9.1                 | 0.358 | 13.7          | 0.539 | 3.2 | 0.126 |
| 305 x 89 (12 x 3 1/2) | 41.69            | 28.0  | 304.8              | 12.0 | 88.9              | 3.5 | 13.7                   | 0.539 | 10.2                | 0.402 | 13.7          | 0.539 | 3.2 | 0.126 |
| 305 x 102 (12 x 4)    | 46.18            | 31.0  | 304.8              | 12.0 | 101.6             | 4.0 | 14.8                   | 0.583 | 10.2                | 0.402 | 15.2          | 0.598 | 4.8 | 0.189 |
| 381 x 102 (15 x 4)    | 55.10            | 37.0  | 381.0              | 15.0 | 101.6             | 4.0 | 16.3                   | 0.642 | 10.4                | 0.409 | 15.2          | 0.598 | 4.8 | 0.189 |
| 432 x 102 (17 x 4)    | 65.54            | 44.0  | 431.8              | 17.0 | 101.6             | 4.0 | 16.8                   | 0.661 | 12.2                | 0.480 | 15.2          | 0.598 | 4.8 | 0.189 |

# CHANNELS



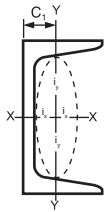
| Section size          | Section area    |                 | Centre of gravity |       | Moment of inertia |                 |                 |                 | Radius of gyration |       |      |       | Modulus of section |                 |                 |                 |
|-----------------------|-----------------|-----------------|-------------------|-------|-------------------|-----------------|-----------------|-----------------|--------------------|-------|------|-------|--------------------|-----------------|-----------------|-----------------|
|                       | A               |                 | y                 |       | Ix                |                 | Iy              |                 | ix                 |       | iy   |       | Zx                 |                 | Zy              |                 |
| mm (in)               | cm <sup>2</sup> | in <sup>2</sup> | cm                | in    | cm <sup>4</sup>   | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm                 | in    | cm   | in    | cm <sup>3</sup>    | in <sup>3</sup> | cm <sup>3</sup> | in <sup>3</sup> |
| 127 x 64 (5 x 2 1/2)  | 18.98           | 2.94            | 1.94              | 0.764 | 482.5             | 11.59           | 67.23           | 1.62            | 5.04               | 1.984 | 1.88 | 0.74  | 75.99              | 4.637           | 15.25           | 0.931           |
| 152 x 76 (6 x 3)      | 22.77           | 3.53            | 2.21              | 0.87  | 851.5             | 20.46           | 113.8           | 2.73            | 6.12               | 2.409 | 2.24 | 0.882 | 111.8              | 6.822           | 21.05           | 1.285           |
| 152 x 89 (6 x 3 1/2)  | 30.36           | 4.71            | 2.86              | 1.126 | 1166              | 28.01           | 215.1           | 5.17            | 6.20               | 2.441 | 2.66 | 1.047 | 153.0              | 9.337           | 35.70           | 2.178           |
| 178 x 76 (7 x 3)      | 26.54           | 4.11            | 2.20              | 0.866 | 1337              | 32.12           | 134.0           | 3.22            | 7.10               | 2.795 | 2.25 | 0.886 | 150.4              | 9.178           | 24.72           | 1.509           |
| 178 x 89 (7 x 3 1/2)  | 34.15           | 5.29            | 2.76              | 1.087 | 1753              | 42.12           | 241.0           | 5.79            | 7.16               | 2.819 | 2.66 | 1.047 | 197.2              | 12.034          | 39.29           | 2.398           |
| 203 x 76 (8 x 3)      | 30.34           | 4.70            | 2.13              | 0.839 | 1950              | 46.85           | 151.3           | 3.63            | 8.02               | 3.157 | 2.23 | 0.878 | 192.0              | 11.717          | 27.59           | 1.684           |
| 203 x 89 (8 x 3 1/2)  | 37.94           | 5.88            | 2.65              | 1.043 | 2491              | 59.85           | 264.4           | 6.35            | 8.10               | 3.189 | 2.64 | 1.039 | 245.2              | 14.963          | 42.34           | 2.584           |
| 229 x 76 (9 x 3)      | 33.20           | 5.15            | 2.00              | 0.787 | 2610              | 62.70           | 158.7           | 3.81            | 8.87               | 3.492 | 2.19 | 0.862 | 228.3              | 13.932          | 28.22           | 1.722           |
| 229 x 89 (9 x 3 1/2)  | 41.73           | 6.47            | 2.53              | 0.996 | 3387              | 81.37           | 285.0           | 6.85            | 9.01               | 3.547 | 2.61 | 1.028 | 296.4              | 18.087          | 44.82           | 2.735           |
| 254 x 76 (10 x 3)     | 36.03           | 5.58            | 1.86              | 0.732 | 3367              | 80.89           | 162.6           | 3.91            | 9.67               | 3.807 | 2.12 | 0.835 | 265.1              | 16.177          | 28.21           | 1.721           |
| 254 x 89 (10 x 3 1/2) | 45.52           | 7.06            | 2.42              | 0.953 | 4448              | 106.86          | 302.4           | 7.27            | 9.88               | 3.89  | 2.58 | 1.016 | 350.2              | 21.370          | 46.70           | 2.850           |
| 305 x 89 (12 x 3 1/2) | 53.11           | 8.23            | 2.18              | 0.858 | 7061              | 169.64          | 325.4           | 7.82            | 11.5               | 4.528 | 2.48 | 0.976 | 463.3              | 28.272          | 48.49           | 2.959           |
| 305 x 102 (12 x 4)    | 58.83           | 9.12            | 2.66              | 1.047 | 8214              | 197.34          | 499.5           | 12.00           | 11.8               | 4.646 | 2.91 | 1.146 | 539.0              | 32.892          | 66.59           | 4.064           |
| 381 x 102 (15 x 4)    | 70.19           | 10.88           | 2.52              | 0.992 | 14894             | 357.82          | 579.7           | 13.93           | 14.6               | 5.748 | 2.87 | 1.130 | 781.8              | 47.708          | 75.86           | 4.629           |
| 432 x 102 (17 x 4)    | 83.49           | 12.94           | 2.32              | 0.913 | 21399             | 514.11          | 628.6           | 15.10           | 16.0               | 6.299 | 2.74 | 1.079 | 991.1              | 60.480          | 80.14           | 4.890           |

# CHANNELS



## U-CHANNELS (Inside slope 5°)

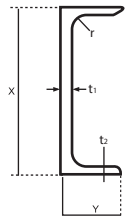
| Section size<br>A x B x t1 | Unit weight<br>M |       | Section depth<br>A |        | Flange width<br>B |       | Web thickness<br>t1 |       | Flange thickness<br>t2 |       | Corner radius |       |       |       |
|----------------------------|------------------|-------|--------------------|--------|-------------------|-------|---------------------|-------|------------------------|-------|---------------|-------|-------|-------|
|                            | kg/m             | lb/ft | mm                 | in     | mm                | in    | mm                  | in    | mm                     | in    | r1            |       | r2    |       |
| 75 x 40 x 5                | 6.92             | 4.65  | 75                 | 2.953  | 40                | 1.575 | 5.0                 | 0.197 | 7.0                    | 0.276 | 8             | 0.315 | 4.0   | 0.157 |
| 100 x 50 x 5               | 9.36             | 6.29  | 100                | 3.937  | 50                | 1.969 | 5.0                 | 0.197 | 7.5                    | 0.295 | 8             | 0.315 | 4.0   | 0.157 |
| 125 x 65 x 6               | 13.4             | 9.004 | 125                | 4.921  | 65                | 2.559 | 6.0                 | 0.236 | 8.0                    | 0.315 | 8             | 0.315 | 4.0   | 0.157 |
| 150 x 75 x 6.5 18.6        | 12.50            | 150   | 5.906              | 75     | 2.953             | 6.5   | 0.256               | 10.0  | 0.394                  | 10    | 0.394         | 5.0   | 0.197 |       |
| 150 x 75 x 9               | 24.0             | 16.13 | 150                | 5.906  | 75                | 2.953 | 9.0                 | 0.354 | 12.5                   | 0.492 | 15            | 0.591 | 7.5   | 0.295 |
| 180 x 75 x 7               | 21.4             | 14.35 | 180                | 7.087  | 75                | 2.953 | 7.0                 | 0.276 | 10.5                   | 0.413 | 11            | 0.433 | 5.5   | 0.217 |
| 180 x 90 x 7.5 27.1        | 18.21            | 180   | 7.087              | 90     | 3.543             | 7.5   | 0.295               | 12.5  | 0.492                  | 13    | 0.512         | 6.5   | 0.256 |       |
| 200 x 80 x 7.5 24.6        | 16.53            | 200   | 7.874              | 80     | 3.15              | 7.5   | 0.295               | 11.0  | 0.433                  | 12    | 0.472         | 6.0   | 0.236 |       |
| 200 x 90 x 8               | 30.3             | 20.40 | 200                | 7.874  | 90                | 3.543 | 8.0                 | 0.315 | 13.5                   | 0.531 | 14            | 0.551 | 7.0   | 0.276 |
| 230 x 80 x 8               | 28.4             | 19.08 | 230                | 9.055  | 80                | 3.15  | 8.0                 | 0.315 | 12.0                   | 0.472 | 13            | 0.512 | 6.5   | 0.256 |
| 230 x 90 x 8.5 33.1        | 22.24            | 230   | 9.055              | 90     | 3.543             | 8.5   | 0.335               | 13.5  | 0.531                  | 15    | 0.591         | 7.5   | 0.295 |       |
| 250 x 80 x 8               | 30.2             | 20.29 | 250                | 9.843  | 80                | 3.15  | 8.0                 | 0.315 | 12.5                   | 0.492 | 14            | 0.551 | 7.0   | 0.276 |
| 250 x 90 x 9               | 34.6             | 23.15 | 250                | 9.843  | 90                | 3.543 | 9.0                 | 0.354 | 13.0                   | 0.512 | 14            | 0.551 | 7.0   | 0.276 |
| 250 x 90 x 11 40.2         | 26.90            | 250   | 9.843              | 90     | 3.543             | 11.0  | 0.433               | 14.5  | 0.571                  | 17    | 0.669         | 8.5   | 0.335 |       |
| 280 x 100 x 9 38.8         | 26.07            | 280   | 11.024             | 100    | 3.937             | 9.0   | 0.354               | 13.0  | 0.512                  | 14    | 0.551         | 7.0   | 0.276 |       |
| 280 x 100 x 11.5           | 48.2             | 32.39 | 280                | 11.024 | 100               | 3.937 | 11.5                | 0.453 | 16.0                   | 0.630 | 18            | 0.709 | 9.0   | 0.354 |
| 300 x 90 x 9               | 38.1             | 25.57 | 300                | 11.811 | 90                | 3.543 | 9.0                 | 0.354 | 12.0                   | 0.512 | 14            | 0.551 | 7.0   | 0.276 |
| 300 x 90 x 10 43.8         | 29.32            | 300   | 11.811             | 90     | 3.543             | 10.0  | 0.394               | 15.5  | 0.611                  | 19    | 0.748         | 9.5   | 0.374 |       |
| 380 x 100 x 10.5           | 54.5             | 36.60 | 380                | 14.961 | 100               | 3.937 | 10.5                | 0.413 | 16.0                   | 0.630 | 18            | 0.709 | 9.0   | 0.354 |
| 380 x 100 x 13             | 62.0             | 41.66 | 380                | 14.961 | 100               | 3.937 | 13.0                | 0.512 | 16.5                   | 0.650 | 18            | 0.709 | 9.0   | 0.354 |



# CHANNELS

| Section<br>Size<br>A x B x t1 | Section<br>area<br>A |                 | Centre<br>of gravity<br>Cy |       | Moment<br>of inertia |                 |                 |                 | Radius<br>of gyration |       |      |       | Modulus<br>of section |                 |                 |                 |
|-------------------------------|----------------------|-----------------|----------------------------|-------|----------------------|-----------------|-----------------|-----------------|-----------------------|-------|------|-------|-----------------------|-----------------|-----------------|-----------------|
|                               | cm <sup>2</sup>      | in <sup>2</sup> | cm                         | in    | Ix                   |                 | Iy              |                 | ix                    |       | iy   |       | Zx                    |                 | Zy              |                 |
|                               | cm <sup>2</sup>      | in <sup>2</sup> | cm                         | in    | cm <sup>4</sup>      | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm                    | in    | cm   | in    | cm <sup>3</sup>       | in <sup>3</sup> | cm <sup>3</sup> | in <sup>3</sup> |
| 75 x 40 x 5                   | 8.818                | 1.367           | 1.27                       | 0.500 | 75.9                 | 1.823           | 12.4            | 0.298           | 2.93                  | 1.154 | 1.19 | 0.469 | 20.2                  | 1.233           | 4.54            | 0.277           |
| 100 x 50 x 5                  | 11.92                | 1.848           | 1.55                       | 0.610 | 189                  | 4.541           | 26.9            | 0.646           | 3.98                  | 1.567 | 1.50 | 0.591 | 37.8                  | 2.307           | 7.82            | 0.477           |
| 125 x 65 x 6                  | 17.11                | 2.652           | 1.94                       | 0.764 | 425                  | 10.21           | 65.5            | 1.571           | 4.99                  | 1.965 | 1.96 | 0.772 | 68.0                  | 4.150           | 14.4            | 0.879           |
| 150 x 75 x 6.5                | 23.71                | 3.675           | 2.31                       | 0.909 | 864                  | 20.76           | 122             | 2.931           | 6.04                  | 2.378 | 2.27 | 0.894 | 115                   | 7.018           | 23.6            | 1.440           |
| 150 x 75 x 9                  | 30.49                | 4.741           | 2.31                       | 0.009 | 1060                 | 25.47           | 151             | 3.628           | 5.87                  | 2.311 | 2.22 | 0.874 | 141                   | 8.604           | 29.1            | 1.776           |
| 180 x 75 x 7                  | 27.20                | 4.316           | 2.15                       | 0.846 | 1380                 | 33.15           | 137             | 3.291           | 7.13                  | 2.807 | 2.24 | 0.882 | 154                   | 9.398           | 25.5            | 1.556           |
| 180 x 90 x 7.5                | 34.57                | 5.358           | 2.85                       | 0.858 | 1840                 | 44.21           | 258             | 6.198           | 7.29                  | 2.870 | 2.73 | 1.075 | 204                   | 12.45           | 42.0            | 2.563           |
| 200 x 80 x 7.5                | 31.33                | 4.856           | 2.24                       | 0.882 | 1950                 | 46.85           | 177             | 4.252           | 7.89                  | 3.106 | 2.38 | 0.937 | 195                   | 11.90           | 30.8            | 1.88            |
| 200 x 90 x 8                  | 38.65                | 5.991           | 2.77                       | 1.091 | 2490                 | 59.82           | 286             | 6.871           | 8.03                  | 3.161 | 2.72 | 1.071 | 249                   | 15.19           | 45.9            | 2.801           |
| 230 x 80 x 8                  | 36.12                | 5.499           | 2.15                       | 0.846 | 2900                 | 69.67           | 200             | 4.805           | 8.96                  | 3.528 | 2.35 | 0.925 | 252                   | 15.38           | 34.2            | 2.087           |
| 230 x 90 x 8.5                | 42.14                | 6.531           | 2.58                       | 1.016 | 3490                 | 83.85           | 303             | 7.280           | 9.10                  | 3.583 | 2.68 | 1.055 | 304                   | 18.55           | 47.3            | 2.886           |
| 250 x 80 x 8                  | 38.51                | 5.969           | 2.11                       | 0.831 | 3630                 | 87.21           | 210             | 5.045           | 9.71                  | 3.823 | 2.33 | 0.921 | 291                   | 17.76           | 35.7            | 2.179           |
| 250 x 90 x 9                  | 44.07                | 6.831           | 2.42                       | 0.953 | 4180                 | 100.4           | 306             | 7.352           | 9.74                  | 3.835 | 2.64 | 1.039 | 335                   | 20.44           | 46.5            | 2.838           |
| 250 x 90 x 11                 | 51.17                | 7.931           | 2.39                       | 0.941 | 4690                 | 112.7           | 342             | 8.217           | 9.47                  | 3.768 | 2.58 | 1.016 | 375                   | 22.88           | 51.7            | 3.155           |
| 280 x 100 x 9                 | 49.37                | 7.652           | 2.64                       | 1.040 | 5930                 | 142.5           | 428             | 10.28           | 11.0                  | 4.331 | 2.95 | 1.161 | 423                   | 25.81           | 58.2            | 3.552           |
| 280 x 100 x 11.5              | 61.37                | 9.512           | 2.68                       | 1.055 | 7150                 | 171.8           | 515             | 12.37           | 10.8                  | 4.252 | 2.90 | 1.142 | 510                   | 31.12           | 70.4            | 4.296           |
| 300 x 90 x 9                  | 48.57                | 7.531           | 2.23                       | 0.878 | 6440                 | 154.7           | 325             | 7.808           | 11.5                  | 4.528 | 2.59 | 1.020 | 429                   | 26.18           | 48.0            | 2.929           |
| 300 x 90 x 10                 | 55.74                | 8.64            | 2.33                       | 0.917 | 7400                 | 177.8           | 373             | 8.961           | 11.5                  | 4.528 | 2.59 | 1.020 | 494                   | 30.15           | 56.0            | 3.417           |
| 380 x 100 x 10.5              | 69.38                | 10.76           | 2.41                       | 0.949 | 14500                | 348.4           | 557             | 13.38           | 14.5                  | 5.709 | 2.83 | 1.114 | 762                   | 46.50           | 73.3            | 4.473           |
| 380 x 100 x 13                | 78.96                | 12.54           | 2.29                       | 0.902 | 15600                | 374.8           | 584             | 14.03           | 14.1                  | 5.551 | 2.72 | 1.071 | 822                   | 50.16           | 75.8            | 4.625           |

# CHANNELS



## INCH SERIES

| Designation | Depth | Flange | Thickness |        | Radius of Fillet | Sectional Area  | Unit Weight |       |       | Weight per |       |       |
|-------------|-------|--------|-----------|--------|------------------|-----------------|-------------|-------|-------|------------|-------|-------|
|             |       |        | Web       | Flange |                  |                 |             |       |       | 20 ft      | 30 ft | 40 ft |
|             | H     | B      | t1        | t2     | r                |                 |             |       |       | kg         | kg    | kg    |
|             | in.   | in.    | in.       | in.    | in.              | in <sup>2</sup> | kg/m        | kg/ft | lb/ft |            |       |       |
| C3 x 4.1    | 3     | 1.41   | 0.17      | 0.273  | 0.27             | 1.21            | 6.10        | 1.86  | 4.1   | 37.2       | 55.8  | 74.4  |
| C4 x 5.4    | 4     | 1.584  | 0.184     | 0.296  | 0.28             | 1.59            | 8.04        | 2.45  | 5.4   | 49.0       | 73.5  | 98    |
| C5 x 6.7    | 5     | 1.75   | 0.19      | 0.32   | 0.29             | 1.97            | 9.97        | 3.04  | 6.7   | 60.8       | 91.2  | 122   |
| C6 x 8.2    | 6     | 1.92   | 0.20      | 0.343  | 0.30             | 2.40            | 12.2        | 3.72  | 8.2   | 74.4       | 112   | 149   |
| C8 x 11.5   | 8     | 2.26   | 0.22      | 0.39   | 0.32             | 3.30            | 17.1        | 5.22  | 11.5  | 104        | 157   | 209   |
| C10 x 15.3  | 10    | 2.60   | 0.24      | 0.436  | 0.34             | 4.49            | 22.8        | 6.94  | 15.3  | 139        | 208   | 278   |
| C12 x 20.7  | 12    | 2.942  | 0.282     | 0.501  | 0.38             | 6.09            | 30.8        | 9.39  | 20.7  | 188        | 282   | 376   |



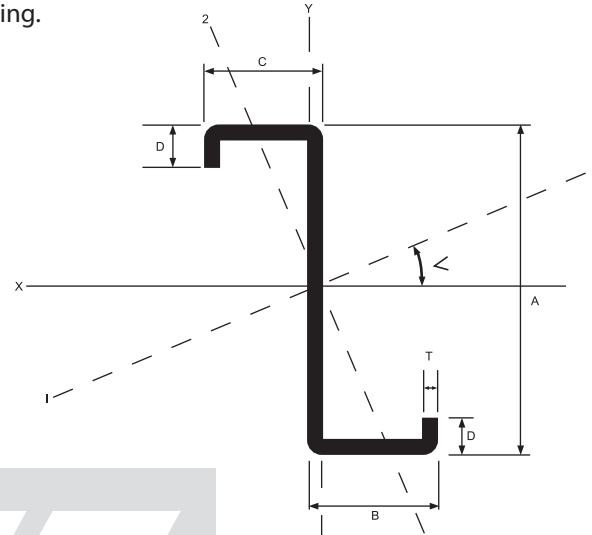
## HIGH-TENSILE GALVANISED Z-PURLINS

High-Tensile Galvanised Z-Purlins are roll formed from high tensile Zinc coated steel with base steel Thickness of 1.60 mm, 2.00 mm and 2.50 mm.

They are suitable for roofing and wall cladding supports and for structural frames of buildings. Due to the light weight and high strength of the steel and the Zinc-coated surface, High-Tensile Galvanised Z-Purlins are versatile and economical and require minimal maintenance throughout the life span of the building.

### MATERIAL SPECIFICATION

|                      |                                                                                |
|----------------------|--------------------------------------------------------------------------------|
| Base Steel Thickness | : 1.60 mm, 2.00 mm and 2.50 mm                                                 |
| Steel Grade          | : ASTM A446 Grade E Modified<br>[AS 1397-G450]                                 |
| Yield Stress         | : 450 Mpa (minimum)                                                            |
| Zinc Coating         | : 275g/m <sup>2</sup> minimum coating mass                                     |
| Mechanical Property  | : Y/P Min. 450 Mpa<br>T/S Min. 510 Mpa<br>E/LMin. 10%                          |
| Tolerances           | : Depth : 1 mm<br>Flange width : 2 mm<br>Length : 3 mm<br>Hole Centres : 1.5mm |



| Dimensions             |                    |    |    |    |     |                        |                    |    |      |    |     |
|------------------------|--------------------|----|----|----|-----|------------------------|--------------------|----|------|----|-----|
| Section Identification | Section Dimensions |    |    |    |     | Section Identification | Section Dimensions |    |      |    |     |
|                        | A                  | B  | C  | D  | t   |                        | A                  | B  | C    | D  | t   |
|                        | mm                 | mm | mm | mm | mm  |                        | mm                 | mm | mm   | mm | mm  |
| BSZ 1610               | 102                | 57 | 51 | 16 | 1.6 | BSZ 1615               | 153                | 74 | 67.5 | 16 | 1.6 |
| BSZ 2010               | 102                | 57 | 51 | 16 | 2.0 | BSZ 2015               | 153                | 74 | 67.5 | 16 | 2.0 |
| BSZ 2510               | 102                | 57 | 51 | 16 | 2.5 | BSZ 2515               | 153                | 74 | 67.5 | 16 | 2.5 |
| BSZ 1612               | 127                | 57 | 51 | 16 | 1.6 | BSZ 1620               | 203                | 74 | 67.5 | 16 | 1.6 |
| BSZ 2012               | 127                | 57 | 51 | 16 | 2.0 | BSZ 2020               | 203                | 74 | 67.5 | 16 | 2.0 |
| BSZ 2512               | 127                | 57 | 51 | 16 | 2.5 | BSZ 2520               | 203                | 74 | 67.5 | 16 | 2.5 |

### Properties

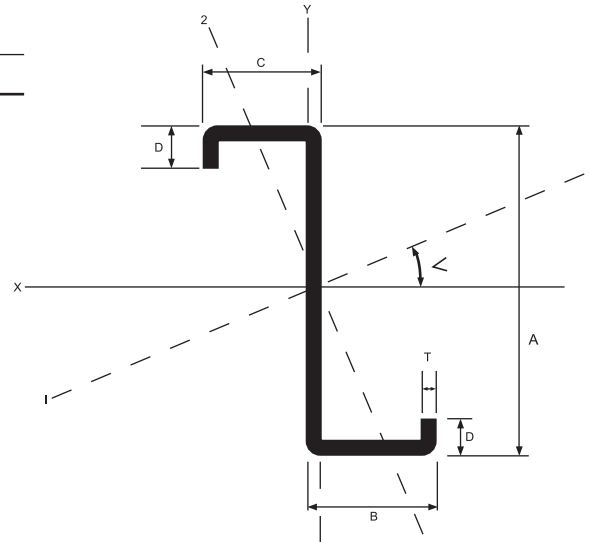
| Section Identification | Mass | Area            | XX-AXIS                         |                                 |      | YY-AXIS                         |                                 |      | 11 -AXIS                        | 22-AXIS                         | ANGLE |
|------------------------|------|-----------------|---------------------------------|---------------------------------|------|---------------------------------|---------------------------------|------|---------------------------------|---------------------------------|-------|
|                        |      |                 | IX                              | ZX                              | RX   | IY                              | ZY                              | RY   | I1                              | I2                              |       |
|                        | kg/m | mm <sup>2</sup> | 10 <sup>6</sup> mm <sup>4</sup> | 10 <sup>3</sup> mm <sup>3</sup> | mm   | 10 <sup>6</sup> mm <sup>4</sup> | 10 <sup>3</sup> mm <sup>3</sup> | mm   | 10 <sup>6</sup> mm <sup>4</sup> | 10 <sup>6</sup> mm <sup>4</sup> |       |
| BSZ 1610               | 3.05 | 373             | 0.62                            | 12.16                           | 40.6 | 0.250                           | 4.94                            | 25.8 | 0.70                            | 0.160                           | 24.5  |
| BSZ 2010               | 3.72 | 442             | 0.77                            | 15.14                           | 41.8 | 0.310                           | 6.22                            | 26.5 | 0.98                            | 0.190                           | 24.4  |
| BSZ 2510               | 4.60 | 534             | 0.96                            | 18.84                           | 42.4 | 0.380                           | 7.60                            | 26.6 | 1.11                            | 0.230                           | 24.3  |
| BSZ 1612               | 3.20 | 408             | 1.03                            | 16.18                           | 50.1 | 0.261                           | 5.20                            | 25.3 | 1.04                            | 0.243                           | 22.9  |
| BSZ 2012               | 3.94 | 510             | 1.28                            | 20.16                           | 51.1 | 0.380                           | 7.60                            | 27.3 | 1.30                            | 0.351                           | 22.7  |
| BSZ 2512               | 4.89 | 638             | 1.60                            | 25.20                           | 52.2 | 0.470                           | 9.45                            | 27.5 | 1.63                            | 0.434                           | 22.6  |
| BSZ 1615               | 4.01 | 512             | 2.01                            | 26.30                           | 62.7 | 0.646                           | 9.69                            | 35.5 | 2.32                            | 0.335                           | 23.3  |
| BSZ 2015               | 4.92 | 598             | 2.50                            | 32.70                           | 65.4 | 0.811                           | 12.2                            | 37.2 | 2.86                            | 0.451                           | 22.7  |
| BSZ 2515               | 6.01 | 731             | 3.12                            | 40.70                           | 66.2 | 1.031                           | 15.6                            | 38.1 | 3.53                            | 0.619                           | 22.1  |
| BSZ 1620               | 4.70 | 572             | 3.86                            | 38.10                           | 82.6 | 0.635                           | 9.52                            | 33.5 | 4.15                            | 0.350                           | 15.9  |
| BSZ 2020               | 5.74 | 698             | 4.85                            | 47.80                           | 84.1 | 0.811                           | 12.2                            | 34.4 | 5.16                            | 0.501                           | 15.0  |
| BSZ 2520               | 7.03 | 855             | 6.05                            | 59.60                           | 85.1 | 1.021                           | 15.4                            | 34.9 | 6.40                            | 0.675                           | 14.2  |



# Z-PURLINS

## HIGH-TENSILE GALVANISED Z-PURLINS (OTHER AVAILABLE SIZES)

| Section Identification | Section Dimensions |    |    |    |     | Area<br>mm <sup>2</sup> | Mass Per Unit length<br>kg/m |
|------------------------|--------------------|----|----|----|-----|-------------------------|------------------------------|
|                        | A                  | B  | C  | D  | t   |                         |                              |
|                        | mm                 | mm | mm | mm | mm  |                         |                              |
| SZ 100-16              | 102                | 53 | 49 | 14 | 1.6 | 344                     | 2.78                         |
| SZ 100-20              | 102                | 53 | 48 | 14 | 2.0 | 430                     | 3.45                         |
| SZ 150-16              | 152                | 65 | 61 | 16 | 1.6 | 472                     | 3.84                         |
| SZ 150-20              | 152                | 65 | 61 | 16 | 2.0 | 590                     | 4.77                         |
| SZ 150-25              | 152                | 66 | 60 | 16 | 2.5 | 738                     | 5.93                         |
| SZ 200-16              | 203                | 79 | 74 | 20 | 1.6 | 592                     | 4.82                         |
| SZ 200-20              | 203                | 79 | 74 | 20 | 2.0 | 750                     | 6.00                         |
| SZ 200-25              | 203                | 79 | 73 | 20 | 2.5 | 938                     | 7.46                         |
| SZ 250-20              | 250                | 78 | 72 | 20 | 2.0 | 840                     | 6.76                         |
| SZ 250-25              | 250                | 78 | 72 | 20 | 2.5 | 1050                    | 8.24                         |



| Section Identification | Section Dimensions |    |    |    |     | Area<br>mm <sup>2</sup> | Mass/<br>unit<br>length<br>kg/m | Second<br>Moment of Area        |                                 | Section<br>Modulus              |                                 | Radius of<br>Gyration |      | Form<br>Factor<br>Q |
|------------------------|--------------------|----|----|----|-----|-------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|-----------------------|------|---------------------|
|                        | A                  | B  | C  | D  | t   |                         |                                 | IX                              | IY                              | ZX                              | ZY                              | RX                    | RY   |                     |
|                        | mm                 | mm | mm | mm | mm  |                         |                                 | 10 <sup>6</sup> mm <sup>4</sup> | 10 <sup>6</sup> mm <sup>4</sup> | 10 <sup>3</sup> mm <sup>3</sup> | 10 <sup>3</sup> mm <sup>3</sup> | mm                    | mm   |                     |
| SZ 100-16              | 102                | 53 | 49 | 14 | 1.6 | 344                     | 2.78                            | 0.570                           | 0.210                           | 11.18                           | 4.02                            | 40.7                  | 24.7 | 0.840               |
| SZ 100-20              | 102                | 53 | 48 | 14 | 2.0 | 430                     | 3.45                            | 0.702                           | 0.262                           | 13.76                           | 5.03                            | 40.4                  | 24.7 | 0.891               |
| SZ 150-16              | 152                | 65 | 61 | 16 | 1.6 | 472                     | 3.84                            | 1.701                           | 0.409                           | 22.39                           | 6.37                            | 60.0                  | 29.4 | 0.703               |
| SZ 150-20              | 152                | 65 | 61 | 16 | 2.0 | 590                     | 4.77                            | 2.111                           | 0.513                           | 27.77                           | 8.02                            | 59.8                  | 29.5 | 0.771               |
| SZ 150-25              | 152                | 66 | 60 | 16 | 2.5 | 738                     | 5.93                            | 2.608                           | 0.648                           | 34.31                           | 10.00                           | 59.5                  | 29.6 | 0.823               |
| SZ 200-16              | 203                | 79 | 74 | 20 | 1.6 | 592                     | 4.82                            | 3.756                           | 0.663                           | 37.00                           | 8.47                            | 79.7                  | 33.5 | 0.578               |
| SZ 200-20              | 203                | 79 | 74 | 20 | 2.0 | 750                     | 6.00                            | 4.742                           | 0.888                           | 46.72                           | 11.38                           | 79.5                  | 34.4 | 0.667               |
| SZ 200-25              | 203                | 79 | 73 | 20 | 2.5 | 938                     | 7.46                            | 5.872                           | 1.114                           | 57.85                           | 14.33                           | 79.1                  | 34.5 | 0.737               |
| SZ 250-20              | 250                | 78 | 72 | 20 | 2.0 | 840                     | 6.76                            | 7.665                           | 0.852                           | 61.32                           | 11.07                           | 95.5                  | 31.9 | 0.599               |
| SZ 250-25              | 250                | 78 | 72 | 20 | 2.5 | 1050                    | 8.24                            | 9.522                           | 1.075                           | 76.18                           | 14.01                           | 95.2                  | 32.0 | 0.658               |

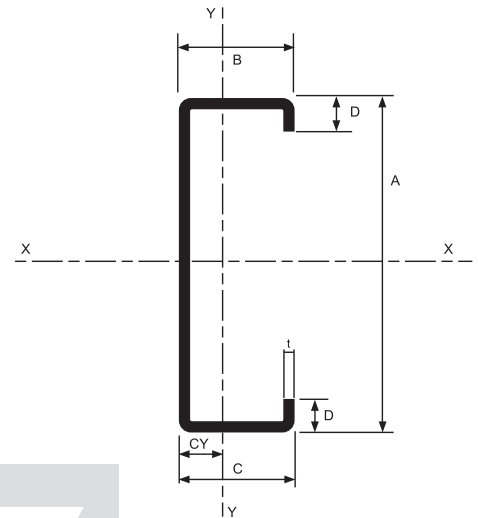
## HIGH-TENSILE GALVANISED C-PURLINS

High-Tensile Galvanised C-Purlins (Sections) are roll formed from high-tensile Zinc coated steel with base steel Thickness of 1.60 mm, 2.00 mm and 2.50 mm.

They are suitable for roofing and wall cladding supports and for structural frames of buildings. Due to the light weight and high strength of the steel and the Zinc-coated surface. High-Tensile Galvanised C-Purlins are versatile and economical and require minimal maintenance throughout the life span of the building.

### MATERIAL SPECIFICATION

|                      |   |                                                                                      |
|----------------------|---|--------------------------------------------------------------------------------------|
| Base Steel Thickness | : | 1.60 mm, 2.00 mm and 2.50 mm                                                         |
| Steel Grade          | : | ASTM A446 Grade E Modified<br>[AS 1397-G450]                                         |
| Yield Stress         | : | 450 Mpa (minimum)                                                                    |
| Zinc Coating         | : | 275g/m <sup>2</sup> minimum coating mass                                             |
| Mechanical Property  | : | Y/P Min. 450 Mpa<br>T/S Min. 510 Mpa<br>E/L Min. 10%                                 |
| Tolerances           | : | Depth : ± 1 mm<br>Flange width : ± 2 mm<br>Length : ± 3 mm<br>Hole Centres : ± 1.5mm |



| Dimensions             |                    |    |    |    |     |                        |                    |    |    |    |     |
|------------------------|--------------------|----|----|----|-----|------------------------|--------------------|----|----|----|-----|
| Section Identification | Section Dimensions |    |    |    |     | Section Identification | Section Dimensions |    |    |    |     |
|                        | A                  | B  | C  | D  | t   |                        | A                  | B  | C  | D  | t   |
|                        | mm                 | mm | mm | mm | mm  |                        | mm                 | mm | mm | mm | mm  |
| BSC 1610               | 102                | 51 | 51 | 16 | 1.6 | BSC 1615               | 153                | 71 | 71 | 16 | 1.6 |
| BSC 2010               | 102                | 51 | 51 | 16 | 2.0 | BSC 2015               | 153                | 71 | 71 | 16 | 2.0 |
| BSC 2510               | 102                | 51 | 51 | 16 | 2.5 | BSC 2515               | 153                | 71 | 71 | 16 | 2.5 |
| BSC 1612               | 127                | 51 | 51 | 16 | 1.6 | BSC 1620               | 203                | 71 | 71 | 16 | 1.6 |
| BSC 2012               | 127                | 51 | 51 | 16 | 2.0 | BSC 2020               | 203                | 71 | 71 | 16 | 2.0 |
| BSC 2512               | 127                | 51 | 51 | 16 | 2.5 | BSC 2520               | 203                | 71 | 71 | 16 | 2.5 |

### Properties

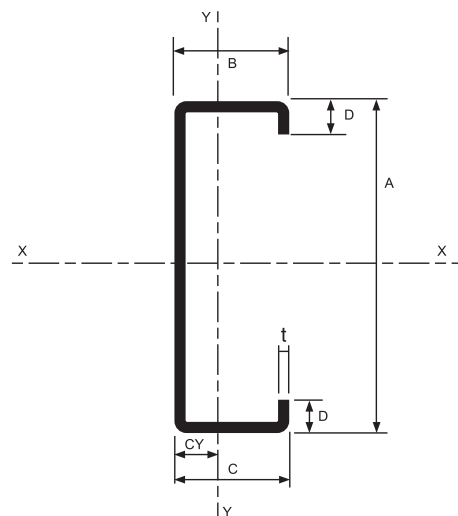
| Section Identification | Mass | Area            | XX-AXIS                         |                                 |      | YY-AXIS                         |                                 |      | CY    |
|------------------------|------|-----------------|---------------------------------|---------------------------------|------|---------------------------------|---------------------------------|------|-------|
|                        |      |                 | IX                              | ZX                              | RX   | IY                              | ZY                              | RY   |       |
|                        | kg/m | mm <sup>2</sup> | 10 <sup>6</sup> mm <sup>4</sup> | 10 <sup>3</sup> mm <sup>3</sup> | mm   | 10 <sup>6</sup> mm <sup>4</sup> | 10 <sup>3</sup> mm <sup>3</sup> | mm   | mm    |
| BSC 1610               | 3.05 | 373             | 0.60                            | 11.77                           | 40.6 | 0.14                            | 4.11                            | 19.5 | 18.24 |
| BSC 2010               | 3.72 | 442             | 0.76                            | 14.84                           | 41.9 | 0.18                            | 5.46                            | 20.3 | 18.60 |
| BSC 2510               | 4.60 | 534             | 0.95                            | 18.56                           | 42.7 | 0.22                            | 7.01                            | 20.9 | 19.05 |
| BSC 1612               | 3.20 | 408             | 1.00                            | 15.72                           | 50.0 | 0.15                            | 4.29                            | 19.4 | 11.08 |
| BSC 2012               | 3.94 | 510             | 1.25                            | 19.67                           | 50.1 | 0.19                            | 5.49                            | 19.5 | 16.42 |
| BSC 2512               | 4.89 | 638             | 1.56                            | 24.56                           | 50.1 | 0.24                            | 7.13                            | 19.6 | 17.35 |
| BSC 1615               | 4.01 | 512             | 1.99                            | 26.30                           | 62.3 | 0.34                            | 6.90                            | 25.8 | 21.69 |
| BSC 2015               | 4.92 | 598             | 2.48                            | 32.42                           | 64.4 | 0.41                            | 8.61                            | 26.2 | 23.36 |
| BSC 2515               | 6.01 | 731             | 3.11                            | 40.65                           | 65.2 | 0.50                            | 10.24                           | 26.3 | 24.11 |
| BSC 1620               | 4.70 | 572             | 3.80                            | 37.44                           | 80.8 | 0.37                            | 7.09                            | 25.1 | 19.18 |
| BSC 2020               | 5.74 | 698             | 4.75                            | 46.80                           | 82.5 | 0.45                            | 9.39                            | 25.4 | 20.16 |
| BSC 2520               | 7.03 | 855             | 5.94                            | 58.52                           | 83.4 | 0.61                            | 12.23                           | 26.8 | 20.80 |

# C-PURLINS

## HIGH-TENSILE GALVANISED C-PURLINS (OTHER AVAILABLE SIZES)

The following purlin sizes provide an easy reference.

| Section Identification | Dimensions |    |    |    |     | Mass<br>kg/m |
|------------------------|------------|----|----|----|-----|--------------|
|                        | A          | B  | C  | D  | t   |              |
|                        | mm         | mm | mm | mm | mm  |              |
| SC100-16               | 100        | 50 | 50 | 14 | 1.6 | 2.78         |
| SC100-20               | 100        | 50 | 50 | 14 | 2.0 | 3.45         |
| SC150-16               | 150        | 65 | 65 | 16 | 1.6 | 3.84         |
| SC150-20               | 150        | 65 | 65 | 16 | 2.0 | 4.77         |
| SC150-25               | 150        | 65 | 65 | 16 | 2.5 | 5.93         |
| SC200-16               | 200        | 75 | 75 | 20 | 1.6 | 4.82         |
| SC200-20               | 200        | 75 | 75 | 20 | 2.0 | 6.00         |
| SC200-25               | 200        | 75 | 75 | 20 | 2.5 | 7.46         |



## DESIGN PRINCIPLE FOR PURLIN SELECTION TABLES

The purlin selection tables are derived basing on BS 5950 : Part 5 : 1987 "Code of Practice for design of cold formed sections".

- Purlin is simply supported with pinned joints over supports.
- Roof Pitch  
The purlin selection tables are only applicable for roof pitch less than or equal to 30°.
- Loading  
Besides dead load arising from roof self weight, the purlins are designed based on a live load of 0.25KN/m<sup>2</sup> (Malaysian Building Bylaw Clause 63) and wind load (uplift) of 0.50KN/m<sup>2</sup> respectively.
  - 1.4 D.L. + 1.6 L.L. (The most severe inward loading combination)
  - 1.0 D.L. + 1.4 W.L. (The most severe outward loading combination)
- Deflection  
Under serviceability limit state the limiting deflection under total load is L/150 whereas under live load only is L/180 (where L is the purlin span).
- Design Strength  $P_y$   
The design strength  $P_y$  for purlins is to be taken as the minimum yield strength of 450N/mm<sup>2</sup>

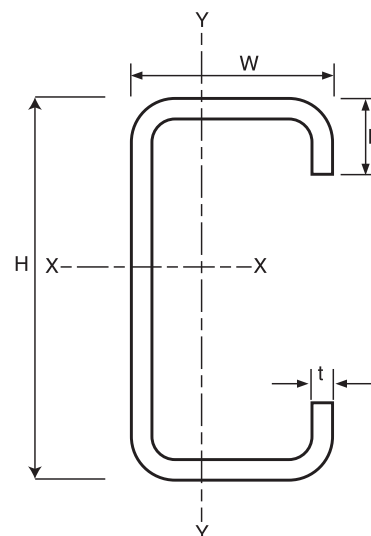


TABLE NO. 1 - PURLIN DIMENSIONS AND PROPERTIES

| Section Identification | Dimensions |    |    |     | Mass per unit<br>kg/m | Area<br>mm <sup>2</sup> | Effective section modulus       |                                 | Full section modulus            |                                 | Second moment of Area           |                                 | Radius of Gyration |      |
|------------------------|------------|----|----|-----|-----------------------|-------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|--------------------|------|
|                        | H          | W  | L  | t   |                       |                         | Zey                             | Zxx                             | Zyy                             | Ixx                             | lyy                             | rxx                             | ryy                |      |
|                        | mm         | mm | mm | mm  |                       |                         | 10 <sup>3</sup> mm <sup>3</sup> | 10 <sup>3</sup> mm <sup>3</sup> | 10 <sup>3</sup> mm <sup>3</sup> | 10 <sup>3</sup> mm <sup>3</sup> | 10 <sup>6</sup> mm <sup>4</sup> | 10 <sup>6</sup> mm <sup>4</sup> | mm                 | mm   |
| SC100-16               | 100        | 50 | 14 | 1.6 | 2.78                  | 355                     | 11.36                           | 3.60                            | 11.60                           | 3.82                            | 0.579                           | 0.125                           | 40.4               | 18.8 |
| SC100-20               | 100        | 50 | 14 | 2.0 | 3.45                  | 440                     | 14.22                           | 4.52                            | 14.30                           | 4.65                            | 0.713                           | 0.153                           | 40.3               | 18.6 |
| SC150-16               | 150        | 65 | 16 | 1.6 | 3.84                  | 489                     | 21.78                           | 4.51                            | 23.60                           | 6.31                            | 1.770                           | 0.282                           | 60.2               | 24.0 |
| SC150-20               | 150        | 65 | 16 | 2.0 | 4.77                  | 608                     | 28.44                           | 5.67                            | 29.20                           | 7.74                            | 2.187                           | 0.346                           | 60.0               | 23.9 |
| SC150-25               | 150        | 65 | 16 | 2.5 | 5.93                  | 755                     | 35.80                           | 7.08                            | 35.90                           | 9.44                            | 2.696                           | 0.422                           | 59.8               | 23.6 |
| SC200-16               | 200        | 75 | 20 | 1.6 | 4.82                  | 614                     | 34.11                           | 6.28                            | 38.50                           | 8.94                            | 3.848                           | 0.472                           | 79.2               | 27.7 |
| SC200-20               | 200        | 75 | 20 | 2.0 | 6.00                  | 764                     | 44.89                           | 7.92                            | 47.70                           | 11.00                           | 4.768                           | 0.580                           | 79.0               | 27.6 |
| SC200-25               | 200        | 75 | 20 | 2.5 | 7.46                  | 950                     | 58.07                           | 9.93                            | 58.90                           | 13.50                           | 5.893                           | 0.710                           | 78.8               | 27.3 |

Note : Effective section modulus are calculated based on the effective cross section which taken into consideration of local buckling effect (Section Four BS5950 Part 5 : 1987).

## HIGH-TENSILE GALVANISED C-PURLINS

### PURLIN SELECTION TABLES FOR ROOFING APPLICATION

The following purlin selection tables provide an easy reference for selection of appropriate purlin size in accordance with the bay width or column spacing of the building.

These tables can be generally used in Singapore and Malaysia. However, for optimum designs where specific roofing and insulation loads are to be used or for any other structure or for application outside Singapore and Malaysia, contact CHYE HIN HARDWARE PTE LTD for free technical advice.

TABLE NO. 2 - PURLIN SELECTION TABLE

Roof decking (0.61 mm TCT at 6.55kg/m<sup>2</sup> distributed load) on C-Purlin

| Span mm | Purlin spacing 1200 mm c/c | Purlin spacing 1500 mm c/c | Purlin spacing 1800 mm c/c | Purlin spacing 2100 mm c/c |
|---------|----------------------------|----------------------------|----------------------------|----------------------------|
| 5000    | SC100-16                   | SC100-20                   | SC150-16                   | SC150-16                   |
| 6000    | SC150-16                   | SC150-16                   | SC150-16                   | SC150-20                   |
| 7000    | SC150-20                   | SC200-16                   | SC200-16                   | SC200-16                   |
| 7500    | SC200-16                   | SC200-16                   | SC200-20                   | SC200-16                   |
| 8000    | SC200-16                   | SC200-16                   | SC200-16                   | SC200-16                   |
| 9000    | SC200-16                   | SC200-16                   | SC200-20                   | SC200-25                   |
| 10000   | SC200-20                   | SC200-25                   | --                         | --                         |
| 11000   | SC200-25                   | --                         | --                         | --                         |

TABLE NO. 3 - PURLIN SELECTION TABLE

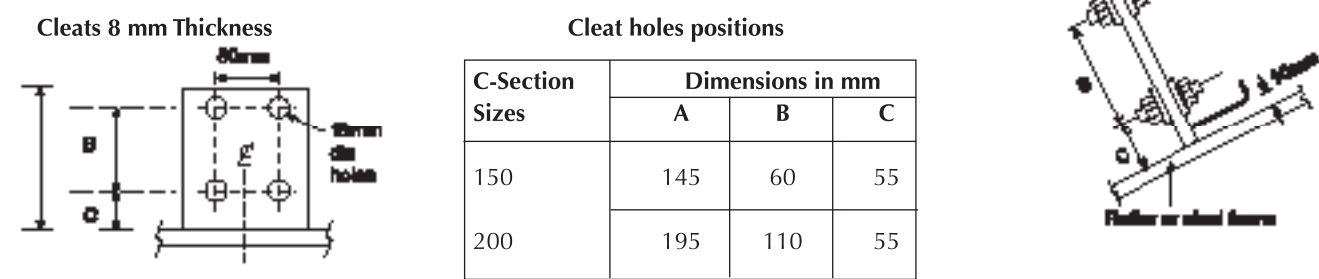
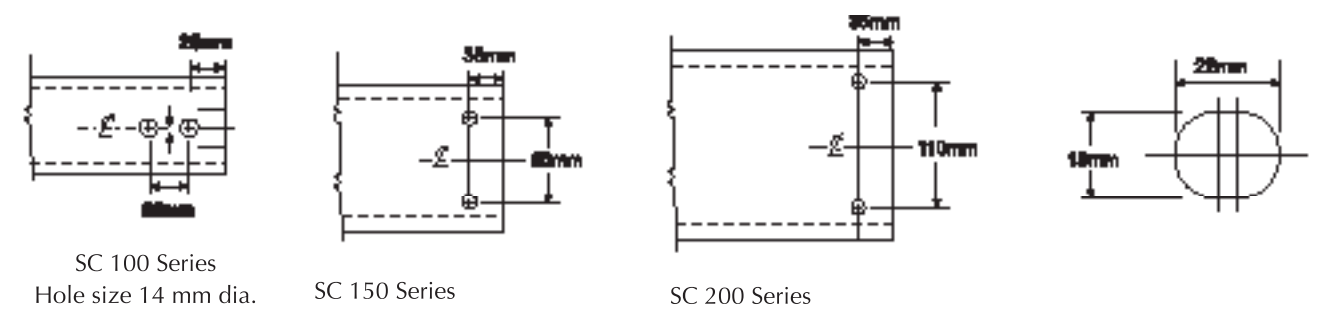
Roof decking (0.61 mm TCT at 6.55kg/m<sup>2</sup> distributed load) on  
50mm thk insulation wool (32 kg/m<sup>3</sup>) on  
1 layer Aluminium Foil on  
1 layer Chicken mesh on C- Purlin

| Span mm | Purlin spacing 1200 mm c/c | Purlin spacing 1500 mm c/c | Purlin spacing 1800 mm c/c | Purlin spacing 2100 mm c/c |
|---------|----------------------------|----------------------------|----------------------------|----------------------------|
| 5000    | SC100-20                   | SC100-20                   | SC150-16                   | SC150-16                   |
| 6000    | SC150-16                   | SC150-16                   | SC150-16                   | SC150-20                   |
| 7000    | SC150-20                   | SC200-16                   | SC200-16                   | SC200-16                   |
| 7500    | SC200-16                   | SC200-16                   | SC200-20                   | SC200-16                   |
| 8000    | SC200-16                   | SC200-16                   | SC200-16                   | SC200-20                   |
| 9000    | SC200-16                   | SC200-20                   | SC200-25                   | SC200-25                   |
| 10000   | SC200-20                   | SC200-25                   | --                         | --                         |

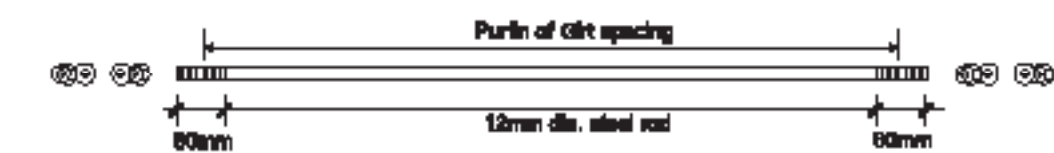
Note: All purlins shall have one bridging at midspan except for purlins in shaded area where 2 bridging are provided at L/3 and 2L/3 between supports.

# C-PURLINS

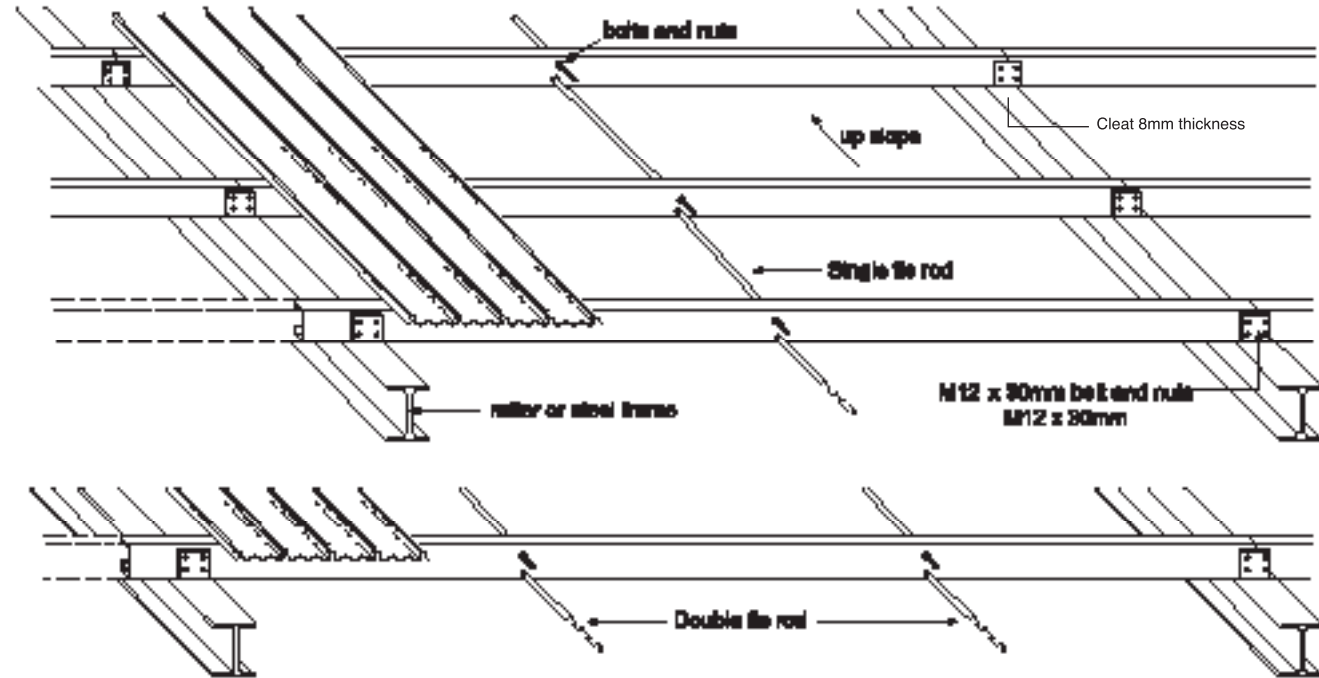
## HOLING, CLEAT AND TIE ROD FOR C-PURLIN AND C-GIRTS



## DETAIL OF STANDARD TIE ROD



## TYPICAL ARRANGEMENT OF SIMPLE SPAN C-PURLINS



## ANGLES

EQUAL ANGLES

142

UNEQUAL ANGLES

144

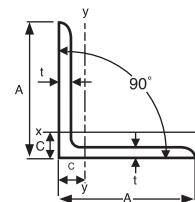
INVERTED ANGLES

150



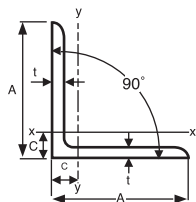
YICK HOE

# ANGLES



## EQUAL

| Section size<br>A x B<br>mm | Thickness<br>t<br>mm | Unit weight<br>M<br>kg/m | Reduce of fillet<br>r<br>mm | Section area<br>A<br>cm <sup>2</sup> | Centre of gravity<br>Cx = Cy<br>cm | Axis                       |               |          |                            |
|-----------------------------|----------------------|--------------------------|-----------------------------|--------------------------------------|------------------------------------|----------------------------|---------------|----------|----------------------------|
|                             |                      |                          |                             |                                      |                                    | Ix = Iy<br>cm <sup>4</sup> | ix = iy<br>cm | iv<br>cm | Zx = Zy<br>cm <sup>3</sup> |
| 20 x 20                     | 3.0                  | 0.885                    | 4.0                         | 1.127                                | 0.60                               | 0.61                       | 0.59          | 0.38     | 0.28                       |
| 25 x 25                     | 3.0                  | 1.12                     | 4.0                         | 1.427                                | 0.719                              | 0.797                      | 0.747         | 0.483    | 0.448                      |
|                             | 5.0                  | 1.76                     | 4.0                         | 2.246                                | 0.79                               | 1.18                       | 0.72          | 0.48     | 0.69                       |
| 30 x 30                     | 3.0                  | 1.36                     | 4.0                         | 1.727                                | 0.844                              | 1.42                       | 0.908         | 0.585    | 0.661                      |
|                             | 5.0                  | 2.16                     | 4.0                         | 2.746                                | 0.92                               | 2.14                       | 0.88          | 0.57     | 1.03                       |
| 40 x 40                     | 3.0                  | 1.83                     | 4.5                         | 2.336                                | 1.09                               | 3.53                       | 1.23          | 0.79     | 1.21                       |
|                             | 4.0                  | 2.42                     | 6.0                         | 3.079                                | 1.12                               | 4.48                       | 1.21          | 0.78     | 1.56                       |
|                             | 5.0                  | 2.95                     | 4.5                         | 3.755                                | 1.17                               | 5.42                       | 1.20          | 0.774    | 1.91                       |
|                             | 6.0                  | 3.52                     | 6.0                         | 4.48                                 | 1.20                               | 6.33                       | 1.19          | 0.77     | 2.26                       |
| 45 x 45                     | 4.0                  | 2.74                     | 6.5                         | 3.492                                | 1.24                               | 6.50                       | 1.36          | 0.880    | 2.00                       |
|                             | 5.0                  | 3.38                     | 6.5                         | 4.302                                | 1.28                               | 7.91                       | 1.36          | 0.874    | 2.46                       |
|                             | 6.0                  | 3.96                     | 6.5                         | 5.044                                | 1.32                               | 9.00                       | 1.34          | 0.86     | 2.83                       |
| 50 x 50                     | 3.0                  | 2.33                     | 7.0                         | 2.96                                 | 1.31                               | 6.86                       | 1.52          | 0.99     | 1.86                       |
|                             | 4.0                  | 3.06                     | 6.5                         | 3.892                                | 1.37                               | 9.06                       | 1.53          | 0.983    | 2.49                       |
|                             | 5.0                  | 3.77                     | 6.5                         | 4.802                                | 1.41                               | 11.1                       | 1.52          | 0.976    | 3.08                       |
|                             | 6.0                  | 4.43                     | 6.5                         | 5.644                                | 1.44                               | 12.6                       | 1.50          | 0.963    | 3.55                       |
|                             | 8.0                  | 5.78                     | 6.5                         | 7.364                                | 1.52                               | 16.1                       | 1.48          | 0.96     | 4.62                       |
| 60 x 60                     | 5.0                  | 4.55                     | 6.5                         | 5.802                                | 1.66                               | 19.6                       | 1.84          | 1.18     | 4.52                       |
|                             | 6.0                  | 5.42                     | 6.5                         | 6.91                                 | 1.69                               | 22.8                       | 1.82          | 1.17     | 5.29                       |
|                             | 7.0                  | 6.21                     | 6.5                         | 7.914                                | 1.73                               | 25.9                       | 1.81          | 1.16     | 6.06                       |
| 65 x 65                     | 5.0                  | 5.00                     | 8.5                         | 6.367                                | 1.77                               | 25.3                       | 1.99          | 1.28     | 5.35                       |
|                             | 6.0                  | 5.91                     | 8.5                         | 7.527                                | 1.81                               | 29.4                       | 1.98          | 1.27     | 6.26                       |
|                             | 8.0                  | 7.66                     | 8.5                         | 9.761                                | 1.88                               | 36.8                       | 1.94          | 1.25     | 7.96                       |
|                             | 10.0                 | 9.42                     | 8.5                         | 12.0                                 | 1.96                               | 44.4                       | 1.92          | 1.25     | 9.79                       |
| 70 x 70                     | 6.0                  | 6.38                     | 8.5                         | 8.127                                | 1.93                               | 37.1                       | 2.14          | 1.37     | 7.33                       |
|                             | 7.0                  | 7.38                     | 8.5                         | 9.40                                 | 1.97                               | 42.4                       | 2.12          | 1.37     | 8.43                       |
|                             | 8.0                  | 8.29                     | 8.5                         | 10.56                                | 2.01                               | 46.6                       | 2.10          | 1.35     | 9.34                       |
|                             | 10.0                 | 10.20                    | 8.5                         | 13.0                                 | 2.09                               | 56.4                       | 2.08          | 1.35     | 11.5                       |
| 75 x 75                     | 6.0                  | 6.85                     | 8.5                         | 8.727                                | 2.06                               | 46.1                       | 2.30          | 1.48     | 8.47                       |
|                             | 8.0                  | 9.03                     | 10.0                        | 11.5                                 | 2.13                               | 58.9                       | 2.26          | 1.46     | 11.0                       |
|                             | 9.0                  | 9.96                     | 8.5                         | 12.69                                | 2.17                               | 64.4                       | 2.25          | 1.45     | 12.1                       |
|                             | 12.0                 | 13.00                    | 8.5                         | 16.56                                | 2.29                               | 81.9                       | 2.22          | 1.44     | 15.7                       |
| 80 x 80                     | 6.0                  | 7.32                     | 8.5                         | 9.327                                | 2.18                               | 56.4                       | 2.46          | 1.58     | 9.7                        |
|                             | 7.0                  | 8.49                     | 10.0                        | 10.8                                 | 2.21                               | 64.2                       | 2.44          | 1.57     | 11.1                       |
|                             | 8.0                  | 9.66                     | 10.0                        | 12.3                                 | 2.26                               | 72.3                       | 2.42          | 1.55     | 12.6                       |
|                             | 9.0                  | 10.7                     | 10.0                        | 13.59                                | 2.30                               | 79.2                       | 2.41          | 1.55     | 13.9                       |
|                             | 12.0                 | 13.9                     | 10.0                        | 17.76                                | 2.41                               | 101.0                      | 2.38          | 1.54     | 18.1                       |
| 90 x 90                     | 6.0                  | 8.28                     | 10.0                        | 10.55                                | 2.42                               | 80.7                       | 2.77          | 1.78     | 12.3                       |
|                             | 7.0                  | 9.59                     | 10.0                        | 12.22                                | 2.46                               | 93.0                       | 2.76          | 1.77     | 14.2                       |
|                             | 8.0                  | 10.9                     | 11.0                        | 13.9                                 | 2.50                               | 104.0                      | 2.74          | 1.76     | 16.1                       |
|                             | 9.0                  | 12.2                     | 11.0                        | 15.5                                 | 2.54                               | 116.0                      | 2.74          | 1.76     | 18.0                       |
|                             | 10.0                 | 13.3                     | 10.0                        | 17.0                                 | 2.57                               | 125.0                      | 2.71          | 1.74     | 19.5                       |
|                             | 12.0                 | 15.9                     | 11.0                        | 20.3                                 | 2.70                               | 148.0                      | 2.70          | 1.75     | 23.3                       |
|                             | 13.0                 | 17.0                     | 10.0                        | 21.71                                | 2.69                               | 156.0                      | 2.68          | 1.73     | 24.8                       |



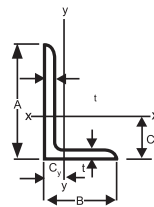
# ANGLES

## EQUAL

| Section size<br>A x B<br>mm | Thickness<br>t<br>mm | Unit weight<br>M<br>kg/m | Reduce of fillet<br>r<br>mm | Section area<br>A<br>cm <sup>2</sup> | Centre of gravity<br>Cx = Cy<br>cm | Axis                       |               |          |                            |
|-----------------------------|----------------------|--------------------------|-----------------------------|--------------------------------------|------------------------------------|----------------------------|---------------|----------|----------------------------|
|                             |                      |                          |                             |                                      |                                    | Ix = Iy<br>cm <sup>4</sup> | ix = iy<br>cm | iv<br>cm | Zx = Zy<br>cm <sup>3</sup> |
| 100 x 100                   | 6.0                  | 9.2                      | 8.5                         | 11.67                                | 2.67                               | 111.3                      | 3.09          | 1.95     | 15.2                       |
|                             | 7.0                  | 10.7                     | 10.0                        | 13.62                                | 2.71                               | 129.0                      | 3.08          | 1.98     | 17.7                       |
|                             | 8.0                  | 12.2                     | 12.0                        | 15.5                                 | 2.74                               | 145.0                      | 3.06          | 1.96     | 19.9                       |
|                             | 10.0                 | 15.0                     | 10.0                        | 19.0                                 | 2.82                               | 175.0                      | 3.04          | 1.95     | 24.4                       |
|                             | 12.0                 | 17.8                     | 12.0                        | 22.7                                 | 3.02                               | 207.0                      | 3.10          | 1.94     | 29.1                       |
|                             | 13.0                 | 19.1                     | 10.0                        | 24.31                                | 2.94                               | 220.0                      | 3.00          | 1.94     | 31.1                       |
|                             | 15.0                 | 21.9                     | 12.0                        | 27.9                                 | 2.98                               | 249.0                      | 3.05          | 1.93     | 35.6                       |
| 120 x 120                   | 8.0                  | 14.7                     | 13.0                        | 18.7                                 | 3.69                               | 255.0                      | 3.96          | 2.37     | 29.1                       |
|                             | 10.0                 | 18.2                     | 13.0                        | 23.2                                 | 3.67                               | 313.0                      | 3.94          | 2.36     | 36.0                       |
|                             | 12.0                 | 21.6                     | 13.0                        | 27.5                                 | 3.65                               | 368.0                      | 3.91          | 2.35     | 42.7                       |
|                             | 15.0                 | 26.6                     | 13.0                        | 33.9                                 | 3.62                               | 445.0                      | 3.86          | 2.33     | 52.4                       |
| 130 x 130                   | 8.0                  | 15.9                     | 13.0                        | 20.42                                | 3.49                               | 331.75                     | 4.03          | 2.59     | 34.88                      |
|                             | 9.0                  | 17.9                     | 12.0                        | 22.74                                | 3.53                               | 366.0                      | 4.01          | 2.57     | 38.7                       |
|                             | 10.0                 | 19.7                     | 13.0                        | 25.26                                | 3.57                               | 406.04                     | 4.01          | 2.57     | 43.08                      |
|                             | 12.0                 | 23.5                     | 12.0                        | 29.76                                | 3.64                               | 467.0                      | 3.96          | 2.54     | 49.9                       |
|                             | 15.0                 | 28.8                     | 12.0                        | 36.75                                | 3.76                               | 568.0                      | 3.93          | 2.53     | 61.5                       |
|                             | 16.0                 | 30.7                     | 13.0                        | 39.38                                | 3.81                               | 669.86                     | 3.90          | 2.51     | 66.31                      |
| 150 x 150                   | 10.0                 | 23.0                     | 16.0                        | 29.3                                 | 4.62                               | 624.0                      | 4.76          | 2.97     | 56.90                      |
|                             | 12.0                 | 27.3                     | 14.0                        | 34.77                                | 4.14                               | 740.0                      | 4.61          | 2.96     | 68.1                       |
|                             | 15.0                 | 33.6                     | 14.0                        | 42.74                                | 4.24                               | 888.0                      | 4.56          | 2.92     | 82.6                       |
|                             | 18.0                 | 40.1                     | 14.0                        | 51.0                                 | 4.54                               | 1,050.0                    | 4.65          | 2.92     | 88.7                       |
|                             | 19.0                 | 41.9                     | 14.0                        | 53.38                                | 4.40                               | 1,090.0                    | 4.52          | 2.91     | 103.0                      |
| 175 x 175                   | 12.0                 | 31.8                     | 15.0                        | 40.52                                | 4.73                               | 1,170.0                    | 5.37          | 3.44     | 91.6                       |
|                             | 15.0                 | 39.4                     | 15.0                        | 50.21                                | 4.85                               | 1,440.0                    | 5.35          | 3.42     | 114.0                      |
| 200 x 200                   | 15.0                 | 45.3                     | 17.0                        | 57.75                                | 5.47                               | 2,180.0                    | 6.14          | 3.93     | 150.0                      |
|                             | 16.0                 | 48.5                     | 18.0                        | 61.8                                 | 6.16                               | 2,340.0                    | 6.30          | 3.94     | 162.0                      |
|                             | 18.0                 | 54.2                     | 18.0                        | 69.1                                 | 6.13                               | 2,600.0                    | 6.27          | 3.93     | 181.0                      |
|                             | 20.0                 | 59.9                     | 17.0                        | 76.3                                 | 5.67                               | 2,820.0                    | 6.09          | 3.90     | 197.0                      |
|                             | 24.0                 | 71.1                     | 18.0                        | 90.6                                 | 6.06                               | 3,333.0                    | 6.22          | 3.90     | 235.0                      |
|                             | 25.0                 | 73.6                     | 17.0                        | 93.75                                | 5.87                               | 3,420.0                    | 6.04          | 3.88     | 242.0                      |
| 250 x 250                   | 25.0                 | 93.7                     | 24.0                        | 119.4                                | 7.10                               | 6,950.0                    | 7.63          | 4.89     | 388.0                      |
|                             | 28.0                 | 104.0                    | 24.0                        | 133.0                                | 7.23                               | 7,690.0                    | 7.61          | 4.89     | 433.0                      |
|                             | 32.0                 | 118.0                    | 24.0                        | 150.0                                | 7.38                               | 8,600.0                    | 7.57          | 4.87     | 488.0                      |
|                             | 35.0                 | 128.0                    | 24.0                        | 162.6                                | 7.45                               | 9,110.0                    | 7.48          | 4.83     | 519.0                      |

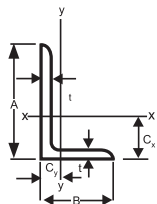


# ANGLES



## UNEQUAL

| Section<br>Size<br>A x B x t | Thickness |       | Unit<br>Weight |       | Leg<br>Length |       |    |       | Corner radius |       |     |       | Section<br>Area |                 | Centre of Gravity |       |      |       |
|------------------------------|-----------|-------|----------------|-------|---------------|-------|----|-------|---------------|-------|-----|-------|-----------------|-----------------|-------------------|-------|------|-------|
|                              | t         |       | M              |       | A             |       | B  |       | r1            |       | r2  |       | A               |                 | Cx                |       | Cy   |       |
|                              | mm        | in    | kg/m           | lb/ft | mm            | in    | mm | in    | mm            | in    | mm  | in    | cm <sup>2</sup> | in <sup>2</sup> | cm                | in    | cm   | in    |
| 40 x 20 x 3                  | 3         | 0.118 | 1.37           | 0.921 | 40            | 1.575 | 20 | 0.787 | 5             | 0.197 | 2   | 0.079 | 1.747           | 0.271           | 1.41              | 0.555 | 0.44 | 0.173 |
| 40 x 20 x 5                  | 5         | 0.197 | 2.16           | 1.451 | 40            | 1.575 | 20 | 0.787 | 5             | 0.197 | 3.5 | 0.138 | 2.751           | 0.426           | 1.50              | 0.591 | 0.51 | 0.201 |
| 50 x 35 x 4                  | 4         | 0.157 | 2.58           | 1.734 | 50            | 1.969 | 35 | 1.378 | 6.5           | 0.256 | 3   | 0.118 | 3.292           | 0.510           | 1.58              | 0.622 | 0.85 | 0.335 |
| 50 x 35 x 6                  | 6         | 0.236 | 3.72           | 2.500 | 50            | 1.969 | 35 | 1.378 | 6.5           | 0.256 | 4.5 | 0.177 | 4.744           | 0.735           | 1.66              | 0.654 | 0.92 | 0.362 |
| 60 x 50 x 5                  | 5         | 0.197 | 4.16           | 2.795 | 60            | 2.362 | 50 | 1.969 | 6.5           | 0.256 | 3   | 0.118 | 5.302           | 0.822           | 1.79              | 0.705 | 1.30 | 0.512 |
| 60 x 50 x 7                  | 7         | 0.276 | 5.66           | 3.803 | 60            | 2.362 | 50 | 1.969 | 6.5           | 0.256 | 4.5 | 0.177 | 7.214           | 1.118           | 1.87              | 0.736 | 1.37 | 0.539 |
| 65 x 50 x 5                  | 5         | 0.197 | 4.36           | 2.930 | 65            | 2.559 | 50 | 1.969 | 6.5           | 0.256 | 3   | 0.118 | 5.552           | 0.861           | 1.99              | 0.783 | 1.25 | 0.492 |
| 65 x 50 x 7                  | 7         | 0.276 | 5.94           | 3.992 | 65            | 2.559 | 50 | 1.969 | 6.5           | 0.256 | 4.5 | 0.177 | 7.564           | 1.172           | 2.07              | 0.815 | 1.33 | 0.524 |
| 65 x 50 x 9                  | 9         | 0.354 | 7.49           | 5.033 | 65            | 2.559 | 50 | 1.969 | 6.5           | 0.256 | 4.5 | 0.177 | 9.544           | 1.479           | 2.14              | 0.843 | 1.40 | 0.551 |
| 70 x 60 x 6                  | 6         | 0.236 | 5.91           | 3.971 | 70            | 2.756 | 60 | 2.362 | 8.5           | 0.335 | 4   | 0.157 | 7.527           | 1.167           | 2.07              | 0.815 | 1.58 | 0.622 |
| 70 x 60 x 8                  | 8         | 0.315 | 7.66           | 5.147 | 70            | 2.756 | 60 | 2.362 | 8.5           | 0.335 | 6   | 0.236 | 9.761           | 1.513           | 2.14              | 0.843 | 1.65 | 0.661 |
| 70 x 60 x 10                 | 10        | 0.394 | 9.42           | 6.33  | 70            | 2.756 | 60 | 2.362 | 8.5           | 0.335 | 6   | 0.236 | 12.00           | 1.86            | 2.24              | 0.874 | 1.73 | 0.681 |
| 75 x 50 x 6                  | 6         | 0.236 | 5.67           | 3.810 | 75            | 2.953 | 50 | 1.969 | 8.5           | 0.335 | 4   | 0.157 | 7.227           | 1.120           | 2.43              | 0.957 | 1.20 | 0.472 |
| 75 x 50 x 8                  | 8         | 0.315 | 7.35           | 4.939 | 75            | 2.953 | 50 | 1.969 | 8.5           | 0.335 | 6   | 0.236 | 9.261           | 1.451           | 2.50              | 0.984 | 1.27 | 0.500 |
| 75 x 50 x 10                 | 10        | 0.394 | 9.03           | 6.068 | 75            | 2.953 | 50 | 1.969 | 8.5           | 0.335 | 6   | 0.236 | 11.50           | 1.782           | 2.58              | 1.016 | 1.35 | 0.531 |
| 75 x 65 x 6                  | 6         | 0.236 | 6.38           | 4.287 | 75            | 2.953 | 65 | 2.559 | 8.5           | 0.335 | 4   | 0.157 | 8.127           | 1.260           | 2.19              | 0.862 | 1.70 | 0.669 |
| 75 x 65 x 8                  | 8         | 0.315 | 8.29           | 5.571 | 75            | 2.953 | 65 | 2.559 | 8.5           | 0.335 | 6   | 0.236 | 10.56           | 1.637           | 2.26              | 0.890 | 1.77 | 0.697 |
| 75 x 65 x 10                 | 10        | 0.394 | 10.2           | 6.854 | 75            | 2.953 | 65 | 2.559 | 8.5           | 0.335 | 6   | 0.236 | 13.00           | 2.015           | 2.34              | 0.921 | 1.85 | 0.728 |
| 80 x 60 x 6                  | 6         | 0.236 | 6.38           | 4.287 | 80            | 3.15  | 60 | 2.362 | 8.5           | 0.335 | 4   | 0.157 | 8.127           | 1.260           | 2.46              | 0.969 | 1.48 | 0.583 |
| 80 x 60 x 8                  | 8         | 0.315 | 8.29           | 5.571 | 80            | 3.15  | 60 | 2.362 | 8.5           | 0.335 | 6   | 0.236 | 10.56           | 1.637           | 2.54              | 1.000 | 1.55 | 0.610 |
| 80 x 60 x 10                 | 10        | 0.394 | 10.2           | 6.854 | 80            | 3.15  | 60 | 2.362 | 8.5           | 0.335 | 6   | 0.236 | 13.00           | 2.015           | 2.62              | 1.031 | 1.63 | 0.642 |
| 80 x 70 x 6                  | 6         | 0.236 | 6.85           | 4.603 | 80            | 3.15  | 70 | 2.756 | 8.5           | 0.335 | 4   | 0.157 | 8.727           | 1.353           | 2.32              | 0.913 | 1.83 | 0.72  |
| 80 x 70 x 9                  | 9         | 0.354 | 9.96           | 6.693 | 80            | 3.15  | 70 | 2.756 | 8.5           | 0.335 | 6   | 0.236 | 12.69           | 1.967           | 2.43              | 0.957 | 1.93 | 0.76  |
| 80 x 70 x 12                 | 12        | 0.472 | 13.0           | 8.736 | 80            | 3.15  | 70 | 2.756 | 8.5           | 0.335 | 6   | 0.236 | 16.56           | 2.567           | 2.54              | 1.00  | 2.05 | 0.807 |
| 90 x 60 x 6                  | 6         | 0.236 | 6.85           | 4.603 | 90            | 3.543 | 60 | 2.362 | 8.5           | 0.335 | 4   | 0.157 | 8.727           | 1.353           | 2.88              | 1.134 | 1.40 | 0.551 |
| 90 x 60 x 9                  | 9         | 0.354 | 9.96           | 6.693 | 90            | 3.543 | 60 | 2.362 | 8.5           | 0.335 | 6   | 0.236 | 12.69           | 1.967           | 2.99              | 1.177 | 1.51 | 0.594 |
| 90 x 60 x 12                 | 12        | 0.472 | 13.0           | 8.736 | 90            | 3.543 | 60 | 2.362 | 8.5           | 0.335 | 6   | 0.236 | 16.56           | 2.567           | 3.11              | 1.244 | 1.63 | 0.642 |
| 90 x 75 x 6                  | 6         | 0.236 | 7.56           | 5.08  | 90            | 3.543 | 75 | 2.953 | 8.5           | 0.335 | 4   | 0.157 | 9.626           | 1.492           | 2.64              | 1.039 | 1.90 | 0.748 |
| 90 x 75 x 9                  | 9         | 0.354 | 11.0           | 7.392 | 90            | 3.543 | 75 | 2.953 | 8.5           | 0.335 | 6   | 0.236 | 14.04           | 2.176           | 2.75              | 1.083 | 2.01 | 0.791 |
| 90 x 75 x 12                 | 12        | 0.472 | 14.4           | 9.676 | 90            | 3.543 | 75 | 2.953 | 8.5           | 0.335 | 6   | 0.236 | 18.36           | 2.846           | 2.87              | 1.13  | 2.12 | 0.835 |

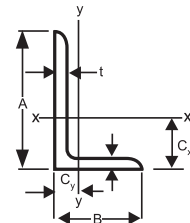


# ANGLES

## UNEQUAL

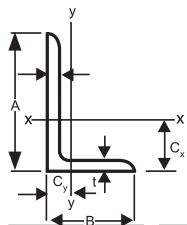
| Section Size<br>A x B x t | Moment of Inertia |                 |                 |                 | Max Iu          |                 |                 |                 | Radius of Gyration |       |      |       |        |       |        |       | tan   | Modulus of Section |                 |                 |                 |
|---------------------------|-------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|--------------------|-------|------|-------|--------|-------|--------|-------|-------|--------------------|-----------------|-----------------|-----------------|
|                           | Ix                |                 | Iy              |                 |                 |                 | Min Iv          |                 | ix                 |       | iy   |       | Max Iu |       | Min Iv |       | a     | Zx                 |                 | Zy              |                 |
|                           | cm <sup>4</sup>   | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm                 | in    | cm   | in    | cm     | in    | cm     | in    |       | cm <sup>3</sup>    | in <sup>3</sup> | cm <sup>3</sup> | in <sup>3</sup> |
| 40 x 20 x 3               | 2.81              | 0.068           | 0.47            | 0.011           | 2.98            | 0.072           | 0.31            | 0.007           | 1.27               | 0.50  | 0.52 | 0.205 | 1.31   | 0.516 | 0.42   | 0.165 | 0.257 | 1.09               | 0.067           | 0.30            | 0.018           |
| 40 x 20 x 5               | 4.21              | 0.11            | 0.68            | 0.016           | 4.43            | 0.106           | 0.41            | 0.01            | 1.24               | 0.488 | 0.50 | 0.197 | 1.27   | 0.50  | 0.41   | 0.161 | 0.243 | 1.68               | 0.103           | 0.46            | 0.208           |
| 50 x 35 x 4               | 8.08              | 0.194           | 3.25            | 0.078           | 9.49            | 0.228           | 1.84            | 0.044           | 1.57               | 0.618 | 0.99 | 0.39  | 1.70   | 0.669 | 0.75   | 0.295 | 0.475 | 2.36               | 0.144           | 1.23            | 0.075           |
| 50 x 35 x 6               | 12.2              | 0.269           | 4.43            | 0.106           | 13.1            | 0.315           | 2.55            | 0.061           | 1.54               | 0.606 | 0.97 | 0.382 | 1.66   | 0.654 | 0.73   | 0.287 | 0.466 | 3.35               | 0.204           | 1.72            | 0.105           |
| 60 x 50 x 5               | 18.5              | 0.445           | 11.7            | 0.281           | 24.4            | 0.586           | 5.82            | 0.14            | 1.87               | 0.736 | 1.48 | 0.583 | 2.14   | 0.843 | 1.05   | 0.413 | 0.679 | 4.40               | 0.269           | 3.16            | 0.193           |
| 60 x 50 x 7               | 24.4              | 0.586           | 15.2            | 0.365           | 31.9            | 0.766           | 7.68            | 0.185           | 1.84               | 0.724 | 1.45 | 0.571 | 2.10   | 0.827 | 1.03   | 0.406 | 0.673 | 5.89               | 0.359           | 4.2             | 0.256           |
| 65 x 50 x 5               | 23.2              | 0.557           | 11.9            | 0.286           | 28.8            | 0.692           | 6.33            | 0.152           | 2.04               | 0.803 | 1.47 | 0.579 | 2.28   | 0.898 | 1.07   | 0.421 | 0.577 | 5.14               | 0.314           | 3.19            | 0.195           |
| 65 x 50 x 7               | 30.6              | 0.735           | 15.6            | 0.375           | 37.9            | 0.911           | 8.35            | 0.201           | 2.01               | 0.791 | 1.44 | 0.567 | 2.24   | 0.882 | 1.05   | 0.413 | 0.571 | 6.90               | 0.421           | 4.25            | 0.259           |
| 65 x 50 x 9               | 37.8              | 0.908           | 19.2            | 0.461           | 46.6            | 1.12            | 10.4            | 0.25            | 1.99               | 0.783 | 1.42 | 0.559 | 2.21   | 0.87  | 1.05   | 0.413 | 0.565 | 8.69               | 0.53            | 5.33            | 0.325           |
| 70 x 60 x 6               | 35.3              | 0.848           | 23.9            | 0.574           | 47.6            | 1.144           | 11.7            | 0.281           | 2.17               | 0.854 | 1.78 | 0.701 | 2.15   | 0.988 | 1.24   | 0.488 | 0.719 | 7.17               | 0.438           | 5.4             | 0.33            |
| 70 x 60 x 8               | 44.4              | 1.067           | 29.8            | 0.716           | 59.5            | 1.429           | 14.7            | 0.353           | 2.13               | 0.839 | 1.75 | 0.689 | 2.47   | 0.872 | 1.23   | 0.484 | 0.714 | 9.13               | 0.457           | 6.85            | 0.418           |
| 70 x 60 x 10              | 53.6              | 1.288           | 36.0            | 0.865           | 71.7            | 1.723           | 17.9            | 0.43            | 2.11               | 0.831 | 1.73 | 0.681 | 2.44   | 0.961 | 1.22   | 0.48  | 0.711 | 11.2               | 0.683           | 8.41            | 0.513           |
| 75 x 50 x 6               | 40.4              | 0.971           | 14.4            | 0.346           | 46.4            | 1.115           | 8.33            | 0.20            | 2.36               | 0.929 | 1.41 | 0.555 | 2.53   | 0.996 | 1.07   | 0.421 | 0.434 | 7.97               | 0.486           | 3.78            | 0.231           |
| 75 x 50 x 8               | 50.8              | 1.221           | 17.8            | 0.428           | 58.1            | 1.396           | 10.4            | 0.25            | 2.33               | 0.917 | 1.38 | 0.543 | 2.49   | 0.98  | 1.05   | 0.413 | 0.427 | 10.2               | 0.622           | 4.77            | 0.291           |
| 75 x 50 x 10              | 61.4              | 1.475           | 21.4            | 0.514           | 70.0            | 1.682           | 12.7            | 0.257           | 2.31               | 0.909 | 1.36 | 0.535 | 2.47   | 0.972 | 1.05   | 0.413 | 0.421 | 12.5               | 0.763           | 5.85            | 0.357           |
| 75 x 65 x 6               | 44.1              | 1.06            | 30.7            | 0.738           | 49.9            | 1.439           | 14.8            | 0.356           | 2.33               | 0.917 | 1.94 | 0.764 | 2.72   | 1.071 | 1.35   | 0.531 | 0.737 | 8.3                | 0.506           | 6.39            | 0.523           |
| 75 x 65 x 8               | 55.5              | 1.333           | 38.4            | 0.923           | 75.3            | 1.809           | 18.6            | 0.447           | 2.29               | 0.902 | 1.91 | 0.752 | 2.67   | 1.051 | 1.33   | 0.524 | 0.733 | 10.6               | 0.647           | 8.13            | 0.496           |
| 75 x 65 x 10              | 67.3              | 1.617           | 46.5            | 1.117           | 91.0            | 2.186           | 22.8            | 0.548           | 2.27               | 0.894 | 1.89 | 0.744 | 2.65   | 1.043 | 1.32   | 0.52  | 0.73  | 13.0               | 0.793           | 9.99            | 0.61            |
| 80 x 60 x 6               | 51.5              | 1.237           | 24.8            | 0.596           | 62.9            | 1.511           | 13.4            | 0.322           | 2.52               | 0.992 | 1.75 | 0.689 | 2.78   | 1.094 | 1.28   | 0.504 | 0.548 | 9.30               | 0.568           | 5.5             | 0.336           |
| 80 x 60 x 8               | 65.0              | 1.562           | 31.0            | 0.745           | 79.1            | 1.90            | 16.9            | 0.406           | 2.48               | 0.976 | 1.71 | 0.673 | 2.74   | 1.079 | 1.26   | 0.496 | 0.542 | 11.9               | 0.726           | 6.97            | 0.425           |
| 80 x 60 x 10              | 78.7              | 1.891           | 37.4            | 0.899           | 95.6            | 2.297           | 20.5            | 0.493           | 2.46               | 0.969 | 1.7  | 0.669 | 2.52   | 0.992 | 1.26   | 0.496 | 0.538 | 14.6               | 0.891           | 8.55            | 0.522           |
| 80 x 70 x 6               | 54.1              | 1.30            | 38.6            | 0.927           | 75.3            | 1.785           | 18.4            | 0.442           | 2.49               | 0.98  | 2.1  | 0.827 | 2.92   | 1.15  | 1.45   | 0.571 | 0.752 | 9.52               | 0.581           | 7.46            | 0.455           |
| 80 x 70 x 9               | 75.8              | 1.821           | 53.8            | 1.293           | 104             | 2.499           | 25.9            | 0.622           | 2.44               | 0.961 | 2.06 | 0.811 | 2.86   | 1.126 | 1.43   | 0.563 | 0.748 | 13.6               | 0.83            | 10.6            | 0.647           |
| 80 x 70 x 12              | 96.6              | 1.321           | 68.3            | 1.641           | 131             | 3.147           | 33.4            | 0.802           | 2.41               | 0.949 | 2.03 | 0.799 | 2.82   | 1.11  | 1.42   | 0.559 | 0.744 | 17.7               | 1.08            | 13.8            | 0.842           |
| 90 x 60 x 6               | 71.5              | 1.718           | 23.6            | 0.615           | 82.3            | 1.977           | 14.8            | 0.356           | 2.86               | 1.126 | 1.71 | 0.674 | 3.07   | 1.209 | 1.30   | 0.512 | 0.437 | 11.7               | 0.714           | 5.47            | 0.34            |
| 90 x 60 x 9               | 101               | 2.427           | 35.4            | 0.85            | 115             | 2.763           | 20.7            | 0.497           | 2.82               | 1.11  | 1.67 | 0.657 | 3.01   | 1.185 | 1.28   | 0.504 | 0.43  | 16.7               | 1.091           | 7.9             | 0.482           |
| 90 x 60 x 12              | 128               | 3.075           | 44.8            | 1.076           | 146             | 3.508           | 26.7            | 0.641           | 2.78               | 1.094 | 1.65 | 0.65  | 2.97   | 1.169 | 1.27   | 0.50  | 0.423 | 21.8               | 1.33            | 10.2            | 0.622           |
| 90 x 75 x 6               | 76.9              | 1.848           | 48.6            | 1.168           | 101             | 2.427           | 24.2            | 0.581           | 2.83               | 1.114 | 2.25 | 0.886 | 3.25   | 1.28  | 1.58   | 0.622 | 0.681 | 12.1               | 0.738           | 8.68            | 0.53            |
| 90 x 75 x 9               | 109               | 2.619           | 68.1            | 1.636           | 143             | 3.436           | 34.1            | 0.819           | 2.78               | 1.094 | 2.2  | 0.866 | 3.19   | 1.257 | 1.56   | 0.614 | 0.676 | 17.4               | 1.062           | 12.4            | 0.757           |
| 90 x 75 x 12              | 139               | 3.34            | 86.8            | 2.085           | 182             | 4.373           | 44.0            | 1.057           | 2.75               | 1.083 | 2.17 | 0.854 | 3.14   | 1.236 | 1.55   | 0.61  | 0.672 | 22.6               | 1.379           | 16.1            | 0.982           |

# ANGLES



## UNEQUAL

| Section Size<br>A x B x t | Thickness |       | Unit Weight |        | Leg Length |       |     |       | Corner radius |       |     |       | Section Area    |                 | Centre of Gravity |       |      |       |
|---------------------------|-----------|-------|-------------|--------|------------|-------|-----|-------|---------------|-------|-----|-------|-----------------|-----------------|-------------------|-------|------|-------|
|                           | t         |       | M           |        | A          |       | B   |       | r1            |       | r2  |       | A               |                 | Cx                |       | Cy   |       |
|                           | mm        | in    | kg/m        | lb/ft  | mm         | in    | mm  | in    | mm            | in    | mm  | in    | cm <sup>2</sup> | in <sup>2</sup> | cm                | in    | cm   | in    |
| 90 x 80 x 7               | 7         | 0.276 | 9.04        | 6.075  | 90         | 3.543 | 80  | 3.15  | 10            | 0.394 | 5   | 0.197 | 11.52           | 1.786           | 2.59              | 1.02  | 2.1  | 0.827 |
| 90 x 80 x 10              | 10        | 0.394 | 12.6        | 8.467  | 90         | 3.543 | 80  | 3.15  | 10            | 0.394 | 7   | 0.276 | 16.0            | 2.48            | 2.71              | 1.067 | 2.21 | 0.87  |
| 90 x 80 x 13              | 13        | 0.512 | 16.0        | 10.751 | 90         | 3.543 | 80  | 3.15  | 10            | 0.394 | 7   | 0.276 | 20.41           | 3.164           | 2.82              | 1.11  | 2.33 | 0.917 |
| 100 x 65 x 7              | 7         | 0.276 | 8.77        | 5.893  | 100        | 3.937 | 65  | 2.559 | 10            | 0.394 | 5   | 0.197 | 11.17           | 1.731           | 3.23              | 1.272 | 1.52 | 0.598 |
| 100 x 65 x 8              | 8         | 0.315 | 9.94        | 6.679  | 100        | 3.937 | 65  | 2.559 | 10            | 0.394 | 4.8 | 0.189 | 12.7            | 1.984           | 3.27              | 1.287 | 1.55 | 0.61  |
| 100 x 65 x 9              | 9         | 0.354 | 11.0        | 7.392  | 100        | 3.937 | 65  | 2.559 | 10            | 0.394 | 7   | 0.276 | 14.04           | 2.176           | 3.31              | 1.303 | 1.59 | 0.626 |
| 100 x 65 x 10             | 10        | 0.394 | 12.3        | 8.265  | 100        | 3.937 | 65  | 2.559 | 10            | 0.394 | 4.8 | 0.189 | 15.6            | 2.418           | 3.36              | 1.323 | 1.63 | 0.642 |
| 100 x 65 x 12             | 12        | 0.472 | 14.4        | 9.676  | 100        | 3.937 | 65  | 2.559 | 10            | 0.394 | 7   | 0.276 | 18.36           | 2.846           | 3.43              | 1.35  | 1.7  | 0.669 |
| 100 x 75 x 7              | 7         | 0.276 | 9.32        | 6.263  | 100        | 3.937 | 75  | 2.953 | 10            | 0.394 | 5   | 0.197 | 11.87           | 1.84            | 3.06              | 1.205 | 1.84 | 0.724 |
| 100 x 75 x 8              | 8         | 0.315 | 10.6        | 7.123  | 100        | 3.937 | 75  | 2.953 | 10            | 0.394 | 4.8 | 0.189 | 13.5            | 2.093           | 3.1               | 1.22  | 1.87 | 0.736 |
| 100 x 75 x 10             | 10        | 0.394 | 13.0        | 8.736  | 100        | 3.937 | 75  | 2.953 | 10            | 0.394 | 7   | 0.276 | 16.5            | 2.557           | 3.18              | 1.252 | 1.95 | 0.764 |
| 100 x 75 x 12             | 12        | 0.472 | 15.4        | 10.348 | 100        | 3.937 | 75  | 2.953 | 10            | 0.394 | 4.8 | 0.189 | 19.7            | 3.054           | 3.27              | 1.287 | 2.03 | 0.799 |
| 100 x 75 x 13             | 13        | 0.512 | 16.5        | 11.087 | 100        | 3.937 | 75  | 2.953 | 10            | 0.394 | 7   | 0.276 | 21.06           | 3.264           | 3.3               | 1.3   | 2.06 | 0.811 |
| 100 x 80 x 7              | 7         | 0.276 | 9.59        | 6.444  | 100        | 3.937 | 80  | 3.15  | 10            | 0.394 | 5   | 0.196 | 12.22           | 1.894           | 2.98              | 1.173 | 2.00 | 0.787 |
| 100 x 80 x 10             | 10        | 0.394 | 13.3        | 8.937  | 100        | 3.937 | 80  | 3.15  | 10            | 0.394 | 7   | 0.276 | 17.0            | 2.635           | 3.1               | 1.22  | 2.11 | 0.831 |
| 100 x 80 x 13             | 13        | 0.512 | 17.0        | 11.423 | 100        | 3.937 | 80  | 3.15  | 10            | 0.394 | 7   | 0.276 | 21.71           | 3.365           | 3.22              | 1.268 | 2.23 | 0.878 |
| 100 x 90 x 7              | 7         | 0.276 | 10.1        | 6.787  | 100        | 3.937 | 90  | 3.543 | 10            | 0.394 | 5   | 0.197 | 12.92           | 2.003           | 2.84              | 1.118 | 2.35 | 0.925 |
| 100 x 90 x 10             | 10        | 0.394 | 14.1        | 9.475  | 100        | 3.937 | 90  | 3.543 | 10            | 0.394 | 7   | 0.276 | 18.00           | 2.79            | 2.96              | 1.165 | 2.46 | 0.969 |
| 100 x 90 x 13             | 13        | 0.512 | 18.1        | 12.162 | 100        | 3.937 | 90  | 3.543 | 10            | 0.394 | 7   | 0.276 | 23.01           | 3.567           | 3.07              | 1.209 | 2.58 | 1.016 |
| 125 x 75 x 6.5            | 6.5       | 0.256 | 9.98        | 6.706  | 125        | 4.921 | 75  | 2.953 | 11            | 0.433 | 4.8 | 0.189 | 12.7            | 1.968           | 4.06              | 1.599 | 1.61 | 0.634 |
| 125 x 75 x 7              | 7         | 0.276 | 10.7        | 7.19   | 125        | 4.921 | 75  | 2.953 | 10            | 0.394 | 5   | 0.197 | 13.62           | 2.111           | 4.1               | 1.615 | 1.64 | 0.646 |
| 125 x 75 x 8              | 8         | 0.315 | 12.2        | 8.198  | 125        | 4.921 | 75  | 2.953 | 11            | 0.433 | 4.8 | 0.189 | 15.5            | 2.403           | 4.14              | 1.63  | 1.68 | 0.661 |
| 125 x 75 x 10             | 10        | 0.394 | 15.0        | 10.012 | 125        | 4.921 | 75  | 2.953 | 10            | 0.394 | 7   | 0.276 | 19.0            | 2.945           | 4.23              | 1.665 | 1.75 | 0.689 |
| 125 x 75 x 12             | 12        | 0.472 | 17.8        | 11.961 | 125        | 4.921 | 75  | 2.953 | 11            | 0.433 | 4.8 | 0.189 | 22.7            | 3.519           | 4.31              | 1.697 | 1.84 | 0.724 |
| 125 x 75 x 13             | 13        | 0.512 | 19.1        | 12.835 | 125        | 4.921 | 75  | 2.953 | 10            | 0.394 | 7   | 0.276 | 24.31           | 3.768           | 4.35              | 1.713 | 1.87 | 0.736 |
| 125 x 90 x 7              | 7         | 0.276 | 11.5        | 7.728  | 125        | 4.921 | 90  | 3.543 | 10            | 0.394 | 5   | 0.197 | 14.67           | 2.274           | 3.84              | 1.512 | 2.11 | 0.831 |
| 125 x 90 x 10             | 10        | 0.394 | 16.1        | 10.819 | 125        | 4.921 | 90  | 3.543 | 10            | 0.394 | 7   | 0.276 | 20.5            | 3.177           | 3.95              | 1.555 | 2.22 | 0.874 |
| 125 x 90 x 13             | 13        | 0.512 | 20.6        | 13.843 | 125        | 4.921 | 90  | 3.543 | 10            | 0.394 | 7   | 0.276 | 26.26           | 4.07            | 4.08              | 1.606 | 2.34 | 0.921 |
| 137 x 102 x 6.4           | 6.4       | 0.252 | 11.7        | 7.862  | 137        | 5.394 | 102 | 4.016 | 11            | 0.433 | 4.8 | 0.189 | 14.8            | 2.294           | 4.09              | 1.61  | 2.36 | 0.929 |
| 137 x 102 x 7.9           | 7.9       | 0.311 | 14.5        | 9.744  | 137        | 5.394 | 102 | 4.016 | 11            | 0.433 | 4.8 | 0.189 | 18.4            | 2.852           | 4.17              | 1.642 | 2.44 | 0.961 |
| 137 x 102 x 9.5           | 9.5       | 0.374 | 17.3        | 11.625 | 137        | 5.394 | 102 | 4.016 | 11            | 0.433 | 4.8 | 0.189 | 21.9            | 3.395           | 4.22              | 1.661 | 2.49 | 0.98  |

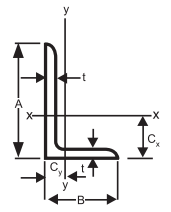


# ANGLES

## UNEQUAL

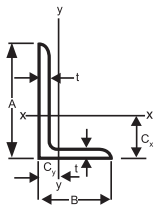
| Section<br>Size<br>A x B x t | Moment of<br>Inertia |                 |                 |                 | Radius of<br>Gyratation |                 |                 |                 |      |       |      |       | tan    |       |        |       | Modulus of<br>Section |                 |                 |                 |                 |
|------------------------------|----------------------|-----------------|-----------------|-----------------|-------------------------|-----------------|-----------------|-----------------|------|-------|------|-------|--------|-------|--------|-------|-----------------------|-----------------|-----------------|-----------------|-----------------|
|                              | Ix                   |                 | Iy              |                 | Max Iu                  |                 | Min Iv          |                 | ix   |       | iy   |       | Max Iu |       | Min Iv |       | a                     | Zx              |                 | Zy              |                 |
|                              | cm <sup>4</sup>      | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup>         | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm   | in    | cm   | in    | cm     | in    | cm     | in    |                       | cm <sup>3</sup> | in <sup>3</sup> | cm <sup>3</sup> | in <sup>3</sup> |
| 90 x 80 x 7                  | 89.6                 | 2.153           | 66.5            | 1.598           | 125                     | 3.003           | 31.3            | 0.752           | 2.79 | 1.098 | 2.4  | 0.945 | 3.29   | 1.295 | 1.65   | 0.65  | 0.777                 | 14.0            | 0.854           | 11.3            | 0.69            |
| 90 x 80 x 10                 | 121                  | 2.907           | 89.1            | 2.141           | 167                     | 4.018           | 42.1            | 1.011           | 2.74 | 1.078 | 2.36 | 0.929 | 3.24   | 1.276 | 1.62   | 0.638 | 0.774                 | 19.1            | 1.166           | 15.4            | 0.94            |
| 90 x 80 x 13                 | 151                  | 3.628           | 111             | 2.668           | 208                     | 4.997           | 53.3            | 1.281           | 2.72 | 1.071 | 2.33 | 0.917 | 3.19   | 1.256 | 1.62   | 0.638 | 0.77                  | 24.4            | 1.489           | 19.6            | 1.196           |
| 100 x 65 x 7                 | 112                  | 2.091           | 37.6            | 0.903           | 128                     | 3.075           | 22.0            | 0.529           | 3.17 | 1.248 | 1.84 | 0.724 | 3.39   | 1.335 | 1.4    | 0.551 | 0.415                 | 16.6            | 1.013           | 7.55            | 0.461           |
| 100 x 65 x 8                 | 127                  | 3.051           | 42.2            | 1.014           | 144                     | 3.46            | 24.8            | 0.596           | 3.16 | 1.244 | 1.83 | 0.72  | 3.37   | 1.327 | 1.4    | 0.551 | 0.414                 | 18.9            | 1.153           | 8.54            | 0.521           |
| 100 x 65 x 9                 | 138                  | 3.315           | 45.6            | 1.096           | 157                     | 3.772           | 26.8            | 0.644           | 3.14 | 1.236 | 1.8  | 0.709 | 3.34   | 1.335 | 1.38   | 0.543 | 0.411                 | 20.6            | 1.257           | 9.28            | 0.566           |
| 100 x 65 x 10                | 154                  | 3.7             | 51.0            | 1.225           | 175                     | 4.204           | 30.1            | 0.723           | 3.14 | 1.236 | 1.81 | 0.713 | 3.35   | 1.319 | 1.39   | 0.547 | 0.41                  | 23.2            | 1.416           | 10.5            | 0.641           |
| 100 x 65 x 12                | 177                  | 4.252           | 58.1            | 1.396           | 200                     | 4.882           | 34.7            | 0.834           | 3.11 | 1.224 | 1.78 | 0.701 | 3.3    | 1.3   | 1.37   | 0.539 | 0.405                 | 27.0            | 1.648           | 12.1            | 0.738           |
| 100 x 75 x 7                 | 118                  | 2.835           | 57.0            | 1.369           | 144                     | 3.46            | 30.7            | 0.738           | 3.15 | 1.24  | 2.19 | 0.862 | 3.49   | 1.374 | 1.61   | 0.634 | 0.548                 | 17.0            | 1.037           | 10.1            | 0.616           |
| 100 x 75 x 8                 | 133                  | 3.195           | 64.1            | 1.54            | 163                     | 3.916           | 34.6            | 0.831           | 3.14 | 1.236 | 2.18 | 0.858 | 3.47   | 1.366 | 1.6    | 0.63  | 0.547                 | 19.3            | 1.178           | 11.4            | 0.696           |
| 100 x 75 x 10                | 159                  | 3.82            | 76.1            | 1.828           | 194                     | 4.661           | 41.3            | 0.992           | 3.11 | 1.224 | 2.15 | 0.846 | 3.43   | 1.35  | 1.58   | 0.622 | 0.543                 | 23.3            | 1.422           | 13.7            | 0.836           |
| 100 x 75 x 12                | 189                  | 4.541           | 90.2            | 2.167           | 230                     | 5.526           | 49.5            | 1.189           | 3.1  | 1.22  | 2.14 | 0.843 | 3.42   | 1.346 | 1.59   | 0.626 | 0.54                  | 28.0            | 1.709           | 16.5            | 1.007           |
| 100 x 75 x 13                | 199                  | 4.781           | 94.8            | 2.278           | 242                     | 5.814           | 52.2            | 1.254           | 3.08 | 1.213 | 2.12 | 0.835 | 3.39   | 1.335 | 1.57   | 0.618 | 0.538                 | 29.7            | 1.812           | 17.4            | 1.062           |
| 100 x 80 x 7                 | 120                  | 2.883           | 68.6            | 1.643           | 154                     | 3.7             | 35.3            | 0.848           | 3.14 | 1.236 | 2.37 | 0.933 | 3.55   | 1.398 | 1.7    | 0.669 | 0.625                 | 17.2            | 1.05            | 11.4            | 0.696           |
| 100 x 80 x 10                | 163                  | 3.916           | 91.9            | 2.208           | 207                     | 4.973           | 47.6            | 1.144           | 3.09 | 1.217 | 2.33 | 0.917 | 3.49   | 1.374 | 1.67   | 0.659 | 0.621                 | 23.6            | 1.44            | 15.6            | 0.952           |
| 100 x 80 x 13                | 204                  | 4.901           | 115             | 2.763           | 258                     | 6.198           | 60.1            | 1.444           | 3.06 | 1.205 | 2.3  | 0.906 | 3.45   | 1.358 | 1.66   | 0.654 | 0.616                 | 30.1            | 1.837           | 19.9            | 1.215           |
| 100 x 90 x 7                 | 125                  | 3.003           | 95.9            | 2.304           | 177                     | 4.252           | 44.5            | 1.069           | 3.11 | 1.224 | 2.73 | 1.075 | 3.7    | 1.457 | 1.85   | 0.728 | 0.799                 | 17.5            | 1.068           | 14.5            | 0.879           |
| 100 x 90 x 10                | 169                  | 4.06            | 129             | 3.099           | 238                     | 5.718           | 60.1            | 1.444           | 3.07 | 1.209 | 2.68 | 1.055 | 3.64   | 1.433 | 1.83   | 0.72  | 0.797                 | 24.0            | 1.465           | 19.8            | 1.208           |
| 100 x 90 x 13                | 212                  | 5.093           | 162             | 3.892           | 298                     | 7.159           | 76.0            | 1.826           | 3.06 | 1.205 | 2.65 | 1.043 | 3.6    | 1.417 | 1.82   | 0.717 | 0.794                 | 30.6            | 1.867           | 25.2            | 1.538           |
| 125 x 75 x 6.5               | 204                  | 4.901           | 56.1            | 1.348           | 228                     | 4.821           | 34.3            | 0.825           | 4.01 | 1.579 | 2.1  | 0.827 | 4.23   | 1.665 | 1.64   | 0.646 | 0.36                  | 24.2            | 1.477           | 9.5             | 0.58            |
| 125 x 75 x 7                 | 219                  | 5.261           | 60.4            | 1.451           | 243                     | 5.138           | 36.4            | 0.875           | 4.01 | 1.579 | 2.11 | 0.831 | 4.23   | 1.665 | 1.63   | 0.642 | 0.362                 | 26.1            | 1.593           | 10.3            | 0.629           |
| 125 x 75 x 8                 | 247                  | 5.934           | 67.6            | 1.624           | 274                     | 6.583           | 40.9            | 0.983           | 4.0  | 1.575 | 2.09 | 0.823 | 4.2    | 1.654 | 1.63   | 0.642 | 0.359                 | 29.6            | 1.806           | 11.6            | 0.708           |
| 125 x 75 x 10                | 298                  | 7.159           | 80.9            | 1.944           | 330                     | 7.928           | 49.0            | 1.177           | 3.86 | 1.52  | 2.06 | 0.811 | 4.17   | 1.642 | 1.61   | 0.634 | 0.357                 | 36.1            | 2.203           | 14.1            | 0.86            |
| 125 x 75 x 12                | 354                  | 8.505           | 95.5            | 2.294           | 391                     | 9.394           | 58.5            | 1.405           | 3.95 | 1.555 | 2.05 | 0.807 | 4.15   | 1.634 | 1.61   | 0.634 | 0.353                 | 43.2            | 2.636           | 16.9            | 1.031           |
| 125 x 75 x 13                | 376                  | 9.033           | 101             | 2.427           | 414                     | 9.946           | 61.9            | 1.487           | 3.92 | 1.543 | 2.04 | 0.803 | 4.13   | 1.626 | 1.6    | 0.63  | 0.352                 | 46.1            | 2.813           | 17.9            | 1.092           |
| 125 x 90 x 7                 | 233                  | 5.598           | 102             | 2.451           | 279                     | 6.703           | 56.2            | 1.35            | 3.99 | 1.571 | 2.64 | 1.039 | 4.36   | 1.717 | 1.96   | 0.772 | 0.51                  | 26.9            | 1.642           | 14.8            | 0.903           |
| 125 x 90 x 10                | 318                  | 7.64            | 138             | 3.315           | 380                     | 9.129           | 76.1            | 1.828           | 3.94 | 1.551 | 2.59 | 1.02  | 4.3    | 1.693 | 1.93   | 0.76  | 0.506                 | 37.2            | 2.27            | 20.4            | 1.245           |
| 125 x 90 x 13                | 401                  | 9.634           | 165             | 3.964           | 479                     | 11.51           | 87.2            | 2.095           | 3.91 | 1.539 | 2.51 | 0.988 | 4.27   | 1.681 | 1.82   | 0.717 | 0.499                 | 47.5            | 2.899           | 24.8            | 1.513           |
| 137 x 102 x 6.4              | 287                  | 6.895           | 137             | 3.291           | 352                     | 8.457           | 74.7            | 1.795           | 4.37 | 1.72  | 3.02 | 1.189 | 4.84   | 1.91  | 2.23   | 0.878 | 0.544                 | 29.9            | 1.825           | 17.6            | 1.074           |
| 137 x 102 x 7.9              | 353                  | 8.481           | 169             | 4.06            | 428                     | 10.283          | 90.4            | 2.172           | 4.34 | 1.709 | 3.0  | 1.181 | 4.82   | 1.9   | 2.22   | 0.874 | 0.544                 | 37.3            | 2.276           | 21.8            | 1.33            |
| 137 x 102 x 9.5              | 417                  | 10.018          | 194             | 4.661           | 506                     | 12.157          | 107             | 2.571           | 4.34 | 1.709 | 3.00 | 1.181 | 4.80   | 1.89  | 2.2    | 0.866 | 0.543                 | 43.8            | 2.673           | 25.8            | 1.574           |

# ANGLES



## UNEQUAL

| Section Size<br>A x B x t | Thickness<br>t |       | Unit Weight<br>M |        | Leg Length |       |     |       | Corner radius |       |     |       | Section Area<br>A |                 | Centre of Gravity |       |      |       |
|---------------------------|----------------|-------|------------------|--------|------------|-------|-----|-------|---------------|-------|-----|-------|-------------------|-----------------|-------------------|-------|------|-------|
|                           |                |       |                  |        | A          |       | B   |       | r1            |       | r2  |       |                   |                 | Cx                |       | Cy   |       |
|                           | mm             | in    | kg/m             | lb/ft  | mm         | in    | mm  | in    | mm            | in    | mm  | in    | cm <sup>2</sup>   | in <sup>2</sup> | cm                | in    | cm   | in    |
| 150 x 75 x 7              | 7              | 0.276 | 12.1             | 8.131  | 150        | 5.906 | 75  | 2.953 | 12            | 0.472 | 5   | 0.197 | 15.46             | 2.396           | 5.17              | 2.035 | 1.5  | 0.591 |
| 150 x 75 x 9              | 9              | 0.354 | 15.3             | 10.281 | 150        | 5.906 | 75  | 2.953 | 12            | 0.472 | 8.5 | 0.335 | 19.44             | 3.013           | 5.25              | 2.067 | 1.56 | 0.614 |
| 150 x 75 x 10             | 10             | 0.394 | 17.0             | 11.423 | 150        | 5.906 | 75  | 2.953 | 11            | 0.433 | 4.8 | 0.189 | 21.6              | 3.348           | 5.32              | 2.094 | 1.61 | 0.634 |
| 150 x 75 x 12             | 12             | 0.472 | 20.1             | 13.574 | 150        | 5.906 | 75  | 2.953 | 12            | 0.472 | 8.5 | 0.335 | 25.56             | 3.962           | 5.39              | 2.122 | 1.68 | 0.661 |
| 150 x 75 x 15             | 15             | 0.591 | 24.8             | 16.665 | 150        | 5.906 | 75  | 2.953 | 11            | 0.433 | 4.8 | 0.189 | 31.6              | 4.898           | 5.53              | 2.177 | 1.81 | 0.713 |
| 150 x 90 x 9              | 9              | 0.354 | 16.4             | 11.02  | 150        | 5.906 | 90  | 3.543 | 12            | 0.472 | 6   | 0.236 | 20.94             | 3.246           | 4.96              | 1.953 | 2.0  | 0.787 |
| 150 x 90 x 10             | 10             | 0.394 | 18.2             | 12.23  | 150        | 5.906 | 90  | 3.543 | 12            | 0.472 | 4.8 | 0.189 | 23.2              | 3.596           | 5.0               | 1.969 | 2.04 | 0.803 |
| 150 x 90 x 12             | 12             | 0.472 | 21.6             | 14.515 | 150        | 5.906 | 90  | 3.543 | 12            | 0.472 | 8.5 | 0.335 | 27.36             | 4.241           | 5.07              | 1.996 | 2.1  | 0.828 |
| 150 x 90 x 15             | 15             | 0.591 | 26.6             | 17.874 | 150        | 5.906 | 90  | 3.543 | 12            | 0.472 | 8.5 | 0.335 | 33.75             | 5.231           | 5.2               | 2.047 | 2.22 | 0.874 |
| 150 x 100 x 9             | 9              | 0.354 | 17.1             | 11.49  | 150        | 5.906 | 100 | 3.937 | 12            | 0.472 | 6   | 0.236 | 21.84             | 3.385           | 4.77              | 1.878 | 2.32 | 0.913 |
| 150 x 100 x 12            | 12             | 0.472 | 22.4             | 15.052 | 150        | 5.906 | 100 | 3.937 | 12            | 0.472 | 8.5 | 0.335 | 28.56             | 4.427           | 4.88              | 1.921 | 2.41 | 0.949 |
| 150 x 100 x 15            | 15             | 0.591 | 27.7             | 18.614 | 150        | 5.906 | 100 | 3.937 | 12            | 0.472 | 8.5 | 0.335 | 35.25             | 5.464           | 5.1               | 1.972 | 2.53 | 0.996 |
| 175 x 90 x 9              | 9              | 0.354 | 18.2             | 12.23  | 175        | 6.89  | 90  | 3.543 | 12            | 0.472 | 6   | 0.236 | 23.19             | 3.594           | 6.04              | 2.373 | 1.85 | 0.728 |
| 175 x 90 x 12             | 12             | 0.472 | 23.8             | 15.993 | 175        | 6.89  | 90  | 3.543 | 12            | 0.472 | 8.5 | 0.335 | 30.36             | 4.706           | 6.16              | 2.425 | 1.96 | 0.772 |
| 175 x 90 x 15             | 15             | 0.591 | 29.4             | 19.756 | 175        | 6.89  | 90  | 3.543 | 12            | 0.472 | 8.5 | 0.335 | 37.5              | 5.812           | 6.29              | 2.476 | 2.08 | 0.819 |
| 200 x 100 x 10            | 10             | 0.394 | 23.0             | 15.455 | 200        | 7.874 | 100 | 3.937 | 15            | 0.591 | 4.8 | 0.189 | 29.2              | 4.526           | 6.93              | 2.728 | 2.01 | 0.791 |
| 200 x 100 x 12            | 12             | 0.472 | 27.3             | 18.345 | 200        | 7.874 | 100 | 3.937 | 15            | 0.591 | 4.8 | 0.189 | 34.8              | 5.394           | 7.03              | 2.768 | 2.1  | 0.827 |
| 200 x 100 x 15            | 15             | 0.591 | 33.7             | 22.645 | 200        | 7.874 | 100 | 3.937 | 15            | 0.591 | 4.8 | 0.189 | 43.0              | 6.665           | 7.16              | 2.819 | 2.22 | 0.874 |
| 200 x 150 x 12            | 12             | 0.472 | 32.0             | 21.503 | 200        | 7.874 | 150 | 5.906 | 15            | 0.591 | 4.8 | 0.189 | 40.8              | 6.324           | 6.08              | 2.394 | 3.61 | 1.421 |
| 200 x 150 x 15            | 15             | 0.591 | 39.6             | 26.61  | 200        | 7.874 | 150 | 5.906 | 15            | 0.591 | 4.8 | 0.189 | 50.5              | 7.828           | 6.21              | 2.445 | 3.73 | 1.469 |
| 200 x 150 x 18            | 18             | 0.709 | 47.1             | 31.65  | 200        | 7.874 | 150 | 5.906 | 15            | 0.591 | 4.8 | 0.189 | 60.0              | 9.3             | 6.33              | 2.492 | 3.85 | 1.516 |



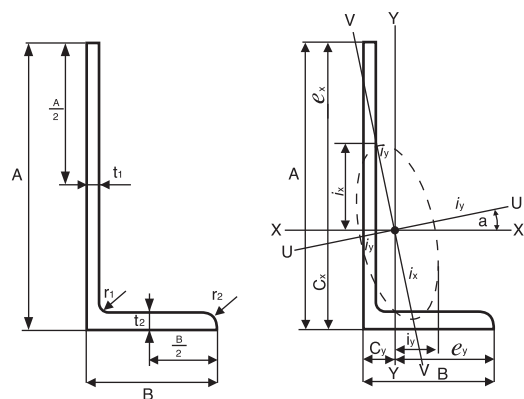
## UNEQUAL

| Section<br>Size<br>A x B x t | Moment of<br>Inertia |                 |                 |                 | Radius of       |                 |                 |                 |          |       |      |       | tan    |       |        |       | Modulus of |                 |                 |                 |                 |
|------------------------------|----------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------|-------|------|-------|--------|-------|--------|-------|------------|-----------------|-----------------|-----------------|-----------------|
|                              | Ix                   |                 | Iy              |                 | Max Iu          |                 | Min Iv          |                 | Gyration |       |      |       | Max Iu |       | Min Iv |       | a          | Section         |                 |                 |                 |
|                              |                      |                 |                 |                 |                 |                 |                 |                 | ix       |       | iy   |       |        |       |        |       |            | Zx              |                 | Zy              |                 |
|                              | cm <sup>4</sup>      | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm       | in    | cm   | in    | cm     | in    | cm     | in    |            | cm <sup>3</sup> | in <sup>3</sup> | cm <sup>3</sup> | in <sup>3</sup> |
| 150 x 75 x 7                 | 364                  | 8.745           | 63.1            | 1.516           | 386             | 9.274           | 40.5            | 0.973           | 4.85     | 1.909 | 2.02 | 0.795 | 5.0    | 1.969 | 1.62   | 0.638 | 0.265      | 37.0            | 2.258           | 10.5            | 0.641           |
| 150 x 75 x 9                 | 446                  | 7.151           | 75.6            | 1.816           | 473             | 11.36           | 48.5            | 1.17            | 4.79     | 1.886 | 1.97 | 0.776 | 4.93   | 1.941 | 1.58   | 0.622 | 0.26       | 45.8            | 2.795           | 12.7            | 0.775           |
| 150 x 75 x 10                | 501                  | 12.036          | 85.8            | 2.061           | 532             | 12.78           | 55.3            | 1.329           | 4.81     | 1.894 | 1.99 | 0.783 | 4.96   | 1.953 | 1.6    | 0.63  | 0.261      | 51.8            | 3.161           | 14.6            | 0.891           |
| 150 x 75 x 12                | 580                  | 13.935          | 97.4            | 2.34            | 614             | 14.75           | 63.3            | 1.521           | 4.76     | 1.874 | 1.95 | 0.768 | 4.9    | 1.929 | 1.57   | 0.618 | 0.257      | 60.3            | 3.68            | 16.7            | 1.019           |
| 150 x 75 x 15                | 713                  | 17.13           | 120             | 2.883           | 754             | 18.11           | 78.8            | 1.893           | 4.75     | 1.87  | 1.94 | 0.764 | 4.88   | 1.921 | 1.58   | 0.622 | 0.254      | 75.3            | 4.595           | 21.0            | 1.281           |
| 150 x 90 x 9                 | 484                  | 16.63           | 133             | 3.195           | 537             | 12.9            | 80.2            | 1.927           | 4.81     | 1.894 | 2.52 | 0.992 | 5.06   | 1.992 | 1.96   | 0.772 | 0.362      | 48.2            | 2.941           | 19.0            | 1.159           |
| 150 x 90 x 10                | 533                  | 12.8            | 146             | 3.508           | 591             | 14.2            | 88.3            | 2.121           | 4.8      | 1.89  | 2.51 | 0.988 | 5.05   | 1.988 | 1.95   | 0.768 | 0.36       | 53.3            | 3.253           | 21.0            | 1.281           |
| 150 x 90 x 12                | 619                  | 14.87           | 168             | 4.036           | 684             | 16.43           | 102             | 2.451           | 4.75     | 1.87  | 2.47 | 0.972 | 5.0    | 1.969 | 1.93   | 0.76  | 0.357      | 62.3            | 3.802           | 24.3            | 1.483           |
| 150 x 90 x 15                | 753                  | 18.09           | 202             | 4.853           | 831             | 19.96           | 124             | 2.979           | 4.72     | 1.858 | 2.45 | 0.965 | 4.99   | 1.965 | 1.92   | 0.756 | 0.353      | 76.8            | 4.687           | 29.9            | 1.825           |
| 150 x 100 x 9                | 502                  | 12.06           | 179             | 4.3             | 580             | 13.93           | 101             | 2.427           | 4.79     | 1.886 | 2.86 | 1.126 | 5.15   | 2.028 | 2.15   | 0.846 | 0.441      | 49.0            | 2.99            | 23.3            | 1.422           |
| 150 x 100 x 12               | 642                  | 15.42           | 229             | 5.502           | 738             | 17.73           | 133             | 3.195           | 4.74     | 1.866 | 2.83 | 1.114 | 5.08   | 2.0   | 2.15   | 0.846 | 0.435      | 63.4            | 3.869           | 30.2            | 1.843           |
| 150 x 100 x 15               | 781                  | 18.76           | 276             | 6.631           | 896             | 21.55           | 161             | 3.868           | 4.71     | 1.854 | 2.8  | 1.102 | 5.04   | 1.984 | 2.14   | 0.843 | 0.432      | 78.2            | 4.772           | 37.0            | 2.258           |
| 175 x 90 x 9                 | 741                  | 17.8            | 138             | 3.315           | 790             | 18.93           | 78.4            | 1.884           | 5.65     | 2.224 | 2.44 | 0.961 | 5.84   | 2.299 | 1.84   | 0.724 | 0.277      | 64.6            | 3.942           | 19.3            | 1.178           |
| 175 x 90 x 12                | 950                  | 22.82           | 174             | 4.18            | 1010            | 24.37           | 111             | 2.667           | 5.6      | 2.205 | 2.39 | 0.941 | 5.78   | 2.276 | 1.91   | 0.752 | 0.274      | 83.8            | 5.114           | 24.7            | 1.507           |
| 175 x 90 x 15                | 1160                 | 27.87           | 210             | 5.045           | 1230            | 29.55           | 136             | 3.267           | 5.56     | 2.189 | 2.37 | 0.933 | 5.74   | 2.26  | 1.9    | 0.748 | 0.27       | 103             | 6.285           | 30.4            | 1.855           |
| 200 x 100 x 10               | 1220                 | 29.31           | 210             | 5.045           | 1290            | 30.99           | 135             | 3.243           | 6.46     | 2.543 | 2.68 | 1.055 | 6.65   | 2.618 | 2.15   | 0.846 | 0.263      | 93.2            | 5.687           | 26.3            | 1.605           |
| 200 x 100 x 12               | 1440                 | 34.6            | 247             | 5.934           | 1530            | 36.76           | 159             | 3.82            | 6.43     | 2.531 | 2.67 | 1.051 | 6.63   | 2.61  | 2.14   | 0.843 | 0.262      | 111             | 6.774           | 31.3            | 1.91            |
| 200 x 100 x 15               | 1758                 | 42.24           | 299             | 7.183           | 1863            | 44.76           | 194             | 4.661           | 6.4      | 2.52  | 2.64 | 1.039 | 6.58   | 2.591 | 2.13   | 0.839 | 0.259      | 137             | 8.36            | 38.4            | 2.343           |
| 200 x 150 x 12               | 1652                 | 39.69           | 803             | 19.29           | 2024            | 48.63           | 431             | 10.35           | 6.36     | 2.504 | 4.44 | 1.748 | 7.04   | 2.772 | 3.25   | 1.28  | 0.552      | 119             | 7.262           | 70.5            | 4.302           |
| 200 x 150 x 15               | 2022                 | 48.58           | 979             | 23.52           | 2475            | 59.46           | 527             | 12.66           | 6.33     | 2.492 | 4.4  | 1.732 | 7.0    | 2.756 | 3.23   | 1.272 | 0.55       | 147             | 8.97            | 86.9            | 5.303           |
| 200 x 150 x 18               | 2376                 | 57.08           | 1146            | 27.53           | 2902            | 69.72           | 618             | 14.85           | 6.29     | 2.476 | 4.37 | 1.72  | 6.95   | 2.736 | 3.21   | 1.264 | 0.548      | 174             | 10.62           | 103             | 6.285           |

# ANGLES

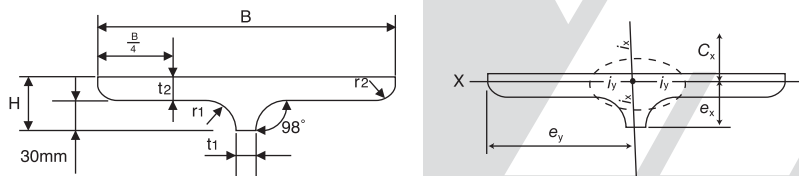
## INVERTED ANGLES (ANGLES WITH UNEQUAL LEGS AND THICKNESS)

Geometrical moment of inertia :  $I = ai^2$   
 Radius of gyration of area :  $i = \sqrt{I/a}$   
 Modulus of section :  $Z = I/e$   
 (a - sectional area)



Standard Sectional Dimensions of Unequal Leg and Unequal Thickness Angles and Their Sectional Area, Unit Mass and Sectional Characteristics

| Standard sectional dimension |    |    |    |     | Sectional area  | Unit mass | Informative reference              |      |                                                  |     |       |     |                                 |      |      |      |        |                                       |      |
|------------------------------|----|----|----|-----|-----------------|-----------|------------------------------------|------|--------------------------------------------------|-----|-------|-----|---------------------------------|------|------|------|--------|---------------------------------------|------|
|                              |    |    |    |     |                 |           | Position of centre of gravity (cm) |      | Geometrical moment of inertia (cm <sup>4</sup> ) |     |       |     | Radius of gyration of area (cm) |      |      |      | tan a  | Modulus of section (cm <sup>3</sup> ) |      |
| A x B                        | t1 | t2 | r1 | r2  | Cx              | Cy        |                                    |      |                                                  |     |       |     |                                 |      |      |      |        |                                       |      |
| mm                           | mm | mm | mm | mm  | cm <sup>2</sup> | kg/m      |                                    |      |                                                  |     |       |     |                                 |      |      |      |        |                                       |      |
| 200 x 90                     | 9  | 14 | 14 | 7   | 29.66           | 23.3      | 6.36                               | 2.15 | 1210                                             | 200 | 1290  | 125 | 6.39                            | 2.60 | 6.58 | 2.05 | 0.263  | 88.7                                  | 29.2 |
| 250 x 90                     | 10 | 15 | 17 | 8.5 | 37.47           | 29.4      | 8.61                               | 1.92 | 2440                                             | 223 | 2520  | 147 | 8.08                            | 2.44 | 8.20 | 1.98 | 0.182  | 149                                   | 31.5 |
|                              | 12 | 16 | 17 | 8.5 | 42.95           | 33.7      | 8.99                               | 1.89 | 2790                                             | 238 | 2870  | 160 | 8.07                            | 2.35 | 8.18 | 1.93 | 0.173  | 174                                   | 33.5 |
| 300 x 90                     | 11 | 16 | 19 | 9.5 | 46.22           | 36.3      | 11.0                               | 1.76 | 4370                                             | 245 | 4440  | 168 | 9.72                            | 2.30 | 9.80 | 1.90 | 0.136  | 229                                   | 33.8 |
|                              | 13 | 17 | 19 | 9.5 | 52.67           | 41.3      | 11.3                               | 1.75 | 4940                                             | 259 | 5020  | 181 | 9.68                            | 2.22 | 9.76 | 1.85 | 0.128  | 265                                   | 35.8 |
| 350 x 100                    | 12 | 17 | 22 | 11  | 57.74           | 45.3      | 13.0                               | 1.87 | 7440                                             | 362 | 7550  | 251 | 11.3                            | 2.50 | 11.4 | 2.08 | 0.124  | 338                                   | 44.5 |
| 400 x 100                    | 13 | 18 | 24 | 12  | 68.59           | 53.8      | 15.4                               | 1.77 | 11500                                            | 388 | 11600 | 277 | 12.9                            | 2.38 | 13.0 | 2.01 | 0.0996 | 467                                   | 47.1 |



Standard Sectional Dimensions of T Section and Their Sectional Area, Unit Mass and Sectional Characteristics

| Standard sectional dimension |     |    |    |    |    |    | Unit tional area | mass | Informative reference              |    |                                                  |      |                                 |      |                                       |      |
|------------------------------|-----|----|----|----|----|----|------------------|------|------------------------------------|----|--------------------------------------------------|------|---------------------------------|------|---------------------------------------|------|
| Nominal section              | B   | H  | t1 | t2 | r1 | r2 |                  |      | Position of centre of gravity (cm) |    | Geometrical moment of inertia (cm <sup>4</sup> ) |      | Radius of gyration of area (cm) |      | Modulus of section (cm <sup>3</sup> ) |      |
| B x t2                       | mm  | mm | mm | mm | mm | mm | cm <sup>2</sup>  | kg/m | Cx                                 | Cy | Ix                                               | Iy   | ix                              | iy   | Zx                                    | Zy   |
| 150 x 9                      | 150 | 39 | 12 | 9  | 8  | 3  | 18.52            | 14.5 | 0.934                              | 0  | 16.5                                             | 254  | 0.942                           | 3.70 | 5.55                                  | 33.8 |
| 150 x 12                     | 150 | 42 | 12 | 12 | 8  | 3  | 23.02            | 18.1 | 1.02                               | 0  | 20.7                                             | 338  | 0.949                           | 3.83 | 6.52                                  | 45.1 |
| 150 x 15                     | 150 | 45 | 12 | 15 | 8  | 3  | 27.52            | 21.6 | 1.13                               | 0  | 25.9                                             | 423  | 0.971                           | 3.92 | 7.70                                  | 56.4 |
| 200 x 12                     | 200 | 42 | 12 | 12 | 8  | 3  | 29.02            | 22.8 | 0.935                              | 0  | 22.3                                             | 799  | 0.877                           | 5.25 | 6.83                                  | 79.9 |
| 200 x 16                     | 200 | 46 | 12 | 16 | 8  | 3  | 37.02            | 29.1 | 1.09                               | 0  | 30.5                                             | 1070 | 0.907                           | 5.37 | 8.68                                  | 107  |
| 200 x 19                     | 200 | 49 | 12 | 19 | 8  | 3  | 43.02            | 33.8 | 1.22                               | 0  | 38.5                                             | 1270 | 0.946                           | 5.43 | 10.4                                  | 127  |
| 200 x 22                     | 200 | 52 | 12 | 22 | 8  | 3  | 49.02            | 38.5 | 1.35                               | 0  | 48.3                                             | 1470 | 0.993                           | 5.47 | 12.6                                  | 147  |
| 250 x 16                     | 250 | 46 | 12 | 16 | 20 | 3  | 46.05            | 36.2 | 1.06                               | 0  | 33.6                                             | 2080 | 0.854                           | 6.72 | 9.49                                  | 167  |
| 250 x 19                     | 250 | 49 | 12 | 19 | 20 | 3  | 53.55            | 42.0 | 1.19                               | 0  | 43.1                                             | 2470 | 0.897                           | 6.80 | 11.6                                  | 198  |
| 250 x 22                     | 250 | 52 | 12 | 22 | 20 | 3  | 61.05            | 47.9 | 1.33                               | 0  | 55.0                                             | 2870 | 0.949                           | 6.85 | 14.2                                  | 229  |
| 250 x 25                     | 250 | 55 | 12 | 25 | 20 | 3  | 68.55            | 53.8 | 1.46                               | 0  | 69.6                                             | 3260 | 1.01                            | 6.90 | 17.2                                  | 261  |

## MILD STEEL TEE BARS - WEIGHT TABLE

|           |             |               |             |             |               |             |
|-----------|-------------|---------------|-------------|-------------|---------------|-------------|
| Size (mm) | 30 x 30 x 4 | 35 x 35 x 4.5 | 40 x 40 x 4 | 40 x 40 x 5 | 45 x 45 x 5.5 | 50 x 50 x 6 |
| kg/m      | 1.77        | 2.33          | 2.38        | 2.96        | 3.67          | 4.44        |
| 6m        | 10.62       | 13.98         | 14.28       | 17.76       | 22.02         | 26.64       |

|           |             |             |             |                |                |  |
|-----------|-------------|-------------|-------------|----------------|----------------|--|
| Size (mm) | 60 x 60 x 7 | 70 x 70 x 8 | 80 x 80 x 9 | 100 x 100 x 11 | 120 x 120 x 13 |  |
| kg/m      | 6.23        | 8.32        | 10.68       | 16.40          | 28.20          |  |
| 6m        | 37.38       | 49.92       | 64.08       | 98.40          | 169.20         |  |

|                         |     |
|-------------------------|-----|
| BARS                    |     |
| FLAT BARS               | 152 |
| BULB FLATS              | 154 |
| SQUARE BARS             | 155 |
| DEFORMED AND ROUND BARS | 156 |



YICK HOE



# BARS

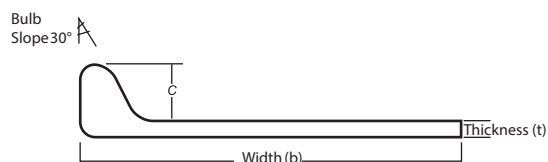
## FLAT

| Section size |       | Unit weight<br>M<br>kg/m | Section area<br>A<br>cm <sup>2</sup> | Section size |       | Unit weight<br>M<br>kg/m | Section area<br>A<br>cm <sup>2</sup> |
|--------------|-------|--------------------------|--------------------------------------|--------------|-------|--------------------------|--------------------------------------|
| thickness    | width |                          |                                      | thickness    | width |                          |                                      |
| mm           | mm    |                          |                                      | mm           | mm    |                          |                                      |
| 3.0          | 12.0  | 0.283                    | 0.36                                 | 9.0          | 19.0  | 1.34                     | 1.71                                 |
| 3.0          | 16.0  | 0.377                    | 0.48                                 | 9.0          | 25.0  | 1.77                     | 2.25                                 |
| 3.0          | 19.0  | 0.447                    | 0.57                                 | 9.0          | 32.0  | 2.26                     | 2.88                                 |
| 3.0          | 25.0  | 0.589                    | 0.75                                 | 9.0          | 35.0  | 2.47                     | 3.15                                 |
| 3.0          | 32.0  | 0.754                    | 0.96                                 | 9.0          | 38.0  | 2.68                     | 3.42                                 |
| 3.0          | 38.0  | 0.895                    | 0.14                                 | 9.0          | 44.0  | 3.11                     | 3.96                                 |
| 3.0          | 50.0  | 1.18                     | 1.50                                 | 9.0          | 50.0  | 3.53                     | 4.50                                 |
| 4.5          | 12.0  | 0.424                    | 0.54                                 | 9.0          | 65.0  | 4.59                     | 5.85                                 |
| 4.5          | 16.0  | 0.565                    | 0.72                                 | 9.0          | 75.0  | 5.30                     | 6.75                                 |
| 4.5          | 19.0  | 0.67                     | 0.85                                 | 9.0          | 90.0  | 6.36                     | 8.10                                 |
| 4.5          | 25.0  | 0.88                     | 1.125                                | 9.0          | 100.0 | 7.06                     | 9.999                                |
| 4.5          | 32.0  | 1.13                     | 1.44                                 | 9.0          | 125.0 | 8.83                     | 11.25                                |
| 4.5          | 38.0  | 1.34                     | 1.71                                 | 9.0          | 150.0 | 10.6                     | 13.50                                |
| 4.5          | 44.0  | 1.55                     | 1.98                                 | 9.0          | 180.0 | 12.7                     | 16.20                                |
| 4.5          | 50.0  | 1.77                     | 2.25                                 | 9.0          | 200.0 | 14.1                     | 18.00                                |
| 4.5          | 65.0  | 2.30                     | 2.93                                 | 9.0          | 230.0 | 16.2                     | 20.70                                |
| 4.5          | 75.0  | 2.60                     | 3.37                                 | 9.0          | 250.0 | 17.7                     | 22.50                                |
| 4.5          | 100.0 | 3.55                     | 4.50                                 | 9.0          | 300.0 | 21.2                     | 27.00                                |
| 5.0          | 25.0  | 0.98                     | 1.25                                 | 12.0         | 25.0  | 2.36                     | 3.00                                 |
| 5.0          | 50.0  | 1.96                     | 2.50                                 | 12.0         | 32.0  | 3.01                     | 3.84                                 |
| 6.0          | 12.0  | 0.57                     | 0.72                                 | 12.0         | 35.0  | 3.31                     | 4.20                                 |
| 6.0          | 16.0  | 0.754                    | 0.96                                 | 12.0         | 38.0  | 3.58                     | 4.56                                 |
| 6.0          | 19.0  | 0.895                    | 1.14                                 | 12.0         | 44.0  | 4.14                     | 5.28                                 |
| 6.0          | 25.0  | 1.13                     | 1.50                                 | 12.0         | 50.0  | 4.71                     | 6.00                                 |
| 6.0          | 32.0  | 1.51                     | 1.92                                 | 12.0         | 65.0  | 6.12                     | 7.80                                 |
| 6.0          | 38.0  | 1.79                     | 2.28                                 | 12.0         | 75.0  | 7.06                     | 9.00                                 |
| 6.0          | 44.0  | 2.07                     | 2.64                                 | 12.0         | 90.0  | 8.48                     | 10.80                                |
| 6.0          | 50.0  | 2.36                     | 3.00                                 | 12.0         | 100.0 | 9.42                     | 12.00                                |
| 6.0          | 65.0  | 3.06                     | 3.90                                 | 12.0         | 125.0 | 11.8                     | 15.00                                |
| 6.0          | 75.0  | 3.53                     | 4.50                                 | 12.0         | 150.0 | 14.1                     | 18.00                                |
| 6.0          | 90.0  | 4.24                     | 5.40                                 | 12.0         | 180.0 | 17.0                     | 21.60                                |
| 6.0          | 100.0 | 4.71                     | 6.00                                 | 12.0         | 200.0 | 18.8                     | 24.00                                |
| 6.0          | 125.0 | 5.89                     | 7.50                                 | 12.0         | 230.0 | 21.7                     | 27.60                                |
| 6.0          | 150.0 | 7.05                     | 9.00                                 | 12.0         | 250.0 | 23.6                     | 30.00                                |
| 6.0          | 200.0 | 9.42                     | 12.00                                | 12.0         | 280.0 | 26.4                     | 33.60                                |
| 6.0          | 250.0 | 11.78                    | 15.00                                | 12.0         | 300.0 | 28.3                     | 36.00                                |
| 6.0          | 300.0 | 14.13                    | 18.00                                | 16.0         | 25.0  | 3.14                     | 4.00                                 |
| 8.0          | 25.0  | 1.57                     | 2.00                                 | 16.0         | 32.0  | 4.02                     | 5.12                                 |
| 8.0          | 32.0  | 2.01                     | 2.56                                 | 16.0         | 38.0  | 4.77                     | 6.08                                 |
| 8.0          | 38.0  | 2.39                     | 3.04                                 | 16.0         | 44.0  | 5.53                     | 7.04                                 |
| 8.0          | 44.0  | 2.76                     | 3.52                                 | 16.0         | 50.0  | 6.28                     | 8.00                                 |
| 8.0          | 50.0  | 3.14                     | 4.00                                 | 16.0         | 65.0  | 8.16                     | 10.40                                |
| 8.0          | 65.0  | 5.65                     | 5.20                                 | 16.0         | 75.0  | 9.42                     | 12.00                                |
| 8.0          | 75.0  | 4.71                     | 6.00                                 | 16.0         | 90.0  | 11.3                     | 14.40                                |
| 8.0          | 90.0  | 5.65                     | 7.20                                 | 16.0         | 100.0 | 12.6                     | 16.00                                |
| 8.0          | 100.0 | 6.28                     | 8.00                                 | 16.0         | 125.0 | 15.7                     | 20.00                                |
| 8.0          | 125.0 | 7.85                     | 10.00                                | 16.0         | 150.0 | 18.8                     | 24.00                                |

## FLAT

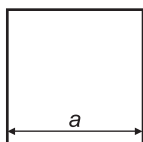
| Section size |       | Unit weight<br>M<br>kg/m | Section area<br>A<br>cm <sup>2</sup> | Section size |       | Unit weight<br>M<br>kg/m | Section area<br>A<br>cm <sup>2</sup> |
|--------------|-------|--------------------------|--------------------------------------|--------------|-------|--------------------------|--------------------------------------|
| thickness    | width |                          |                                      | thickness    | width |                          |                                      |
| mm           | mm    |                          |                                      | mm           | mm    |                          |                                      |
| 16.0         | 180.0 | 22.6                     | 28.0                                 | 25.0         | 90.0  | 17.7                     | 22.5                                 |
| 16.0         | 200.0 | 25.1                     | 32.0                                 | 25.0         | 100.0 | 19.6                     | 25.0                                 |
| 16.0         | 230.0 | 28.9                     | 36.8                                 | 25.0         | 125.0 | 24.5                     | 31.25                                |
| 16.0         | 250.0 | 31.4                     | 50.0                                 | 25.0         | 150.0 | 29.4                     | 37.5                                 |
| 16.0         | 280.0 | 35.2                     | 44.8                                 | 25.0         | 180.0 | 35.3                     | 45.0                                 |
| 16.0         | 300.0 | 37.7                     | 48.0                                 | 25.0         | 200.0 | 39.2                     | 50.0                                 |
| 19.0         | 38.0  | 5.67                     | 7.22                                 | 25.0         | 230.0 | 45.1                     | 57.5                                 |
| 19.0         | 44.0  | 6.56                     | 8.36                                 | 25.0         | 250.0 | 49.1                     | 62.5                                 |
| 19.0         | 50.0  | 7.46                     | 9.50                                 | 25.0         | 280.0 | 55.0                     | 70.0                                 |
| 19.0         | 65.0  | 9.69                     | 12.35                                | 25.0         | 300.0 | 58.9                     | 75.0                                 |
| 19.0         | 75.0  | 11.2                     | 14.25                                | 28.0         | 100.0 | 22.0                     | 28.0                                 |
| 19.0         | 90.0  | 13.4                     | 17.1                                 | 28.0         | 125.0 | 27.5                     | 35.0                                 |
| 19.0         | 100.0 | 14.9                     | 19.0                                 | 28.0         | 150.0 | 33.0                     | 42.0                                 |
| 19.0         | 125.0 | 18.6                     | 23.75                                | 28.0         | 180.0 | 39.6                     | 50.4                                 |
| 19.0         | 150.0 | 22.4                     | 28.5                                 | 28.0         | 200.0 | 44.0                     | 56.0                                 |
| 19.0         | 180.0 | 26.8                     | 34.2                                 | 28.0         | 230.0 | 50.6                     | 64.4                                 |
| 19.0         | 200.0 | 29.8                     | 38.0                                 | 28.0         | 250.0 | 55.0                     | 70.0                                 |
| 19.0         | 230.0 | 34.3                     | 43.7                                 | 28.0         | 280.0 | 61.5                     | 78.4                                 |
| 19.0         | 250.0 | 37.3                     | 47.5                                 | 28.0         | 300.0 | 65.9                     | 84.0                                 |
| 19.0         | 280.0 | 41.8                     | 53.2                                 | 32.0         | 100.0 | 25.1                     | 32.0                                 |
| 19.0         | 300.0 | 44.7                     | 57.0                                 | 32.0         | 125.0 | 31.4                     | 40.0                                 |
| 22.0         | 50.0  | 8.64                     | 11.0                                 | 32.0         | 150.0 | 37.7                     | 48.0                                 |
| 22.0         | 65.0  | 11.2                     | 14.3                                 | 32.0         | 180.0 | 45.2                     | 57.6                                 |
| 22.0         | 75.0  | 13.0                     | 16.5                                 | 32.0         | 200.0 | 50.2                     | 64.0                                 |
| 22.0         | 90.0  | 15.5                     | 19.8                                 | 32.0         | 230.0 | 57.8                     | 73.6                                 |
| 22.0         | 100.0 | 17.3                     | 22.0                                 | 32.0         | 250.0 | 62.8                     | 80.0                                 |
| 22.0         | 125.0 | 21.6                     | 27.5                                 | 32.0         | 280.0 | 70.3                     | 89.6                                 |
| 22.0         | 150.0 | 25.9                     | 33.0                                 | 32.0         | 300.0 | 75.4                     | 96.0                                 |
| 22.0         | 180.0 | 31.1                     | 39.6                                 | 36.0         | 100.0 | 28.3                     | 36.0                                 |
| 22.0         | 200.0 | 34.5                     | 44.0                                 | 36.0         | 125.0 | 35.3                     | 45.0                                 |
| 22.0         | 230.0 | 39.7                     | 50.6                                 | 36.0         | 150.0 | 42.4                     | 54.0                                 |
| 22.0         | 250.0 | 43.2                     | 55.0                                 | 36.0         | 180.0 | 50.9                     | 64.8                                 |
| 22.0         | 280.0 | 48.4                     | 61.0                                 | 36.0         | 200.0 | 56.5                     | 72.0                                 |
| 22.0         | 300.0 | 51.8                     | 66.0                                 | 36.0         | 230.0 | 65.0                     | 82.8                                 |
| 25.0         | 50.0  | 9.81                     | 12.5                                 | 36.0         | 250.0 | 70.6                     | 90.0                                 |
| 25.0         | 65.0  | 12.8                     | 16.25                                | 36.0         | 280.0 | 79.1                     | 100.8                                |
| 25.0         | 75.0  | 14.7                     | 18.75                                | 36.0         | 300.0 | 84.8                     | 108.0                                |

# BARS



## BULB FLATS

| Width<br>b<br>mm | Thickness<br>t<br>mm | Bulb Height<br>C<br>mm | Area of Section<br>A<br>mm <sup>2</sup> | Mass Per Metre<br>kg/m |
|------------------|----------------------|------------------------|-----------------------------------------|------------------------|
| 120              | 6.0                  | 17.0                   | 9.31                                    | 7.31                   |
|                  | 7.0                  | 17.0                   | 10.5                                    | 8.25                   |
|                  | 8.0                  | 17.0                   | 11.7                                    | 9.19                   |
| 140              | 6.5                  | 19.0                   | 11.7                                    | 9.21                   |
|                  | 7.0                  | 19.0                   | 12.4                                    | 9.74                   |
|                  | 8.0                  | 19.0                   | 13.8                                    | 10.83                  |
|                  | 10.0                 | 19.0                   | 16.6                                    | 13.03                  |
| 160              | 7.0                  | 22.0                   | 14.6                                    | 11.4                   |
|                  | 8.0                  | 22.0                   | 16.2                                    | 12.7                   |
|                  | 9.0                  | 22.0                   | 17.8                                    | 14.0                   |
|                  | 11.5                 | 22.0                   | 21.8                                    | 17.3                   |
| 180              | 8.0                  | 25.0                   | 18.9                                    | 14.8                   |
|                  | 9.0                  | 25.0                   | 20.7                                    | 16.2                   |
|                  | 10.0                 | 25.0                   | 22.5                                    | 17.6                   |
|                  | 11.5                 | 25.0                   | 25.2                                    | 19.7                   |
| 200              | 8.5                  | 28.0                   | 22.6                                    | 17.8                   |
|                  | 9.0                  | 28.0                   | 23.6                                    | 18.5                   |
|                  | 10.0                 | 28.0                   | 25.6                                    | 20.1                   |
|                  | 11.0                 | 28.0                   | 27.6                                    | 21.7                   |
|                  | 12.0                 | 28.0                   | 29.6                                    | 23.2                   |
| 220              | 9.0                  | 31.0                   | 26.8                                    | 21.0                   |
|                  | 10.0                 | 31.0                   | 29.0                                    | 22.8                   |
|                  | 11.0                 | 31.0                   | 31.2                                    | 24.5                   |
|                  | 12.0                 | 31.0                   | 33.4                                    | 26.2                   |
| 240              | 9.5                  | 34.0                   | 31.2                                    | 24.4                   |
|                  | 10.0                 | 34.0                   | 32.4                                    | 25.4                   |
|                  | 11.0                 | 34.0                   | 34.9                                    | 27.4                   |
|                  | 12.0                 | 34.0                   | 37.3                                    | 29.3                   |
| 260              | 10.0                 | 37.0                   | 36.1                                    | 28.3                   |
|                  | 11.0                 | 37.0                   | 38.7                                    | 30.3                   |
|                  | 12.0                 | 37.0                   | 41.3                                    | 32.4                   |
| 280              | 10.5                 | 40.0                   | 41.2                                    | 32.4                   |
|                  | 11.0                 | 40.0                   | 42.6                                    | 33.5                   |
|                  | 12.0                 | 40.0                   | 45.5                                    | 35.7                   |
|                  | 13.0                 | 40.0                   | 48.4                                    | 37.9                   |
| 300              | 11.0                 | 43.0                   | 46.7                                    | 36.7                   |
|                  | 12.0                 | 43.0                   | 49.7                                    | 39.0                   |
|                  | 13.0                 | 43.0                   | 52.8                                    | 41.5                   |
| 320              | 11.5                 | 46.0                   | 52.6                                    | 41.2                   |
|                  | 12.0                 | 46.0                   | 54.2                                    | 42.5                   |
|                  | 13.0                 | 46.0                   | 57.4                                    | 45.0                   |
|                  | 14.0                 | 46.0                   | 60.6                                    | 47.5                   |
| 340              | 12.0                 | 49.0                   | 58.8                                    | 46.1                   |
|                  | 13.0                 | 49.0                   | 62.2                                    | 48.8                   |
|                  | 14.0                 | 49.0                   | 65.5                                    | 51.5                   |
|                  | 15.0                 | 49.0                   | 69.0                                    | 54.2                   |
| 370              | 12.5                 | 53.5                   | 67.8                                    | 53.1                   |
|                  | 13.0                 | 53.5                   | 69.6                                    | 54.6                   |
|                  | 14.0                 | 53.5                   | 73.3                                    | 57.5                   |
|                  | 15.0                 | 53.5                   | 77.0                                    | 60.5                   |
|                  | 16.0                 | 53.5                   | 80.7                                    | 63.5                   |
| 400              | 13.0                 | 58.0                   | 77.4                                    | 60.8                   |
|                  | 14.0                 | 58.0                   | 81.4                                    | 63.9                   |
|                  | 15.0                 | 58.0                   | 85.4                                    | 67.0                   |
|                  | 16.0                 | 58.0                   | 89.4                                    | 70.2                   |
| 430              | 14.0                 | 62.5                   | 89.7                                    | 70.6                   |
|                  | 15.0                 | 62.5                   | 94.1                                    | 73.9                   |
|                  | 17.0                 | 62.5                   | 103.0                                   | 80.6                   |
|                  | 20.0                 | 62.5                   | 115.0                                   | 90.8                   |



## SQUARE

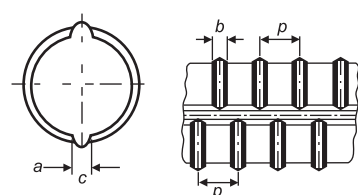
| Section size | Unit weight<br>M |       | Side length<br>a |       | Section Area<br>A |                 | Moment of inertia | Radius of gyration | Modulus of section |
|--------------|------------------|-------|------------------|-------|-------------------|-----------------|-------------------|--------------------|--------------------|
|              | kg/m             | lb/ft | mm               | in    | cm <sup>2</sup>   | in <sup>2</sup> |                   |                    |                    |
| 9            | 0.64             | 0.43  | 9                | 0.354 | 0.81              | 0.126           | --                | --                 | --                 |
| 10           | 0.79             | 0.531 | 10               | 0.393 | 1.0               | 0.1549          | --                | --                 | --                 |
| 12           | 1.13             | 0.759 | 12               | 0.47  | 1.44              | 0.2231          | --                | --                 | --                 |
| 16           | 2.01             | 1.351 | 16               | 0.63  | 2.56              | 0.3968          | 0.0132            | 0.181              | 0.0415             |
| 18           | 2.54             | 1.707 | 18               | 0.709 | 3.24              | 0.5022          | 0.0209            | 0.205              | 0.0592             |
| 19           | 2.83             | 1.902 | 19               | 0.748 | 3.61              | 0.5596          | 0.0262            | 0.217              | 0.0696             |
| 20           | 3.14             | 2.11  | 20               | 0.787 | 4.0               | 0.612           | --                | --                 | --                 |
| 22           | 3.80             | 2.553 | 22               | 0.866 | 4.84              | 0.7502          | 0.0468            | 0.252              | 0.108              |
| 25           | 4.91             | 3.299 | 25               | 0.984 | 6.25              | 0.9688          | 0.0783            | 0.283              | 0.159              |
| 28           | 6.15             | 4.133 | 28               | 1.102 | 7.84              | 1.215           | 0.123             | 0.319              | 0.223              |
| 30           | 7.07             | 4.751 | 30               | 1.181 | 9.00              | 1.395           | 0.162             | 0.343              | 0.275              |
| 32           | 8.04             | 5.403 | 32               | 1.260 | 10.24             | 1.587           | 0.210             | 0.362              | 0.333              |
| 35           | 9.62             | 6.464 | 35               | 1.378 | 12.25             | 1.899           | 0.300             | 0.398              | 0.436              |
| 36           | 10.2             | 6.854 | 36               | 1.417 | 12.96             | 2.009           | 0.336             | 0.409              | 0.475              |
| 38           | 11.3             | 7.593 | 38               | 1.496 | 14.44             | 2.238           | 0.418             | 0.433              | 0.558              |
| 44           | 15.2             | 10.21 | 44               | 1.732 | 19.36             | 3.001           | --                | --                 | --                 |
| 50           | 19.6             | 13.17 | 50               | 1.969 | 25.00             | 3.875           | 1.252             | 0.567              | 1.269              |
| 55           | 23.7             | 15.93 | 55               | 2.165 | 30.25             | 4.689           | 1.833             | 0.626              | 1.69               |
| 60           | 28.3             | 19.02 | 60               | 2.362 | 36.00             | 5.580           | 2.595             | 0.681              | 2.197              |
| 65           | 33.2             | 22.31 | 65               | 2.559 | 42.25             | 6.549           | 3.580             | 0.740              | 2.795              |
| 70           | 38.5             | 25.87 | 70               | 2.756 | 49.00             | 7.595           | 4.805             | 0.795              | 3.491              |
| 75           | 44.2             | 29.70 | 75               | 2.953 | 56.25             | 8.719           | 6.342             | 0.854              | 4.29               |
| 80           | 50.2             | 33.73 | 80               | 3.150 | 64.00             | 9.92            | 8.192             | 0.909              | 5.205              |
| 85           | 56.7             | 38.10 | 85               | 3.346 | 72.25             | 11.20           | 10.450            | 0.965              | 6.224              |
| 90           | 63.6             | 42.74 | 90               | 3.543 | 81.00             | 12.56           | 13.141            | 1.024              | 7.384              |
| 95           | 70.8             | 47.58 | 95               | 3.740 | 90.25             | 13.99           | 16.312            | 1.079              | 8.726              |
| 100          | 78.5             | 52.75 | 100              | 3.937 | 100.0             | 15.50           | 20.012            | 1.138              | 10.191             |
| 110          | 95.0             | 63.84 | 110              | 4.331 | 121.0             | 18.76           | 29.309            | 1.252              | 13.547             |
| 120          | 113.0            | 75.93 | 120              | 4.724 | 144.0             | 22.32           | 41.562            | 1.362              | 17.575             |
| 130          | 133.0            | 89.37 | 130              | 5.118 | 169.0             | 26.20           | 57.177            | 1.476              | 22.334             |
| 140          | 154.0            | 103.5 | 140              | 5.512 | 196.0             | 30.38           | 76.877            | 1.591              | 27.888             |
| 150          | 177.0            | 118.9 | 150              | 5.906 | 225.0             | 34.88           | 101.381           | 1.705              | 34.295             |
| 160          | 201.0            | 135.1 | 160              | 6.299 | 256.0             | 39.68           | 131.195           | 1.819              | 41.679             |

# BARS

## DEFORMED AND ROUND

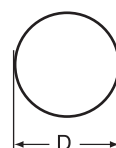
### DEFORMED

| Section size | Unit weight<br>M | Cross section area<br>A |
|--------------|------------------|-------------------------|
| mm           | kg/m             | cm <sup>2</sup>         |
| 6.0          | 0.222            | 0.283                   |
| 10.0         | 0.617            | 0.785                   |
| 12.0         | 0.888            | 1.131                   |
| 13.0         | 1.04             | 1.327                   |
| 14.0         | 1.21             | 1.539                   |
| 16.0         | 1.58             | 2.011                   |
| 18.0         | 2.00             | 2.545                   |
| 19.0         | 2.23             | 2.835                   |
| 20.0         | 2.47             | 3.142                   |
| 22.0         | 2.98             | 3.801                   |
| 24.0         | 3.55             | 4.524                   |
| 25.0         | 3.85             | 4.909                   |
| 26.0         | 4.17             | 5.309                   |
| 28.0         | 4.83             | 6.158                   |
| 29.0         | 5.19             | 6.605                   |
| 30.0         | 5.55             | 7.069                   |
| 32.0         | 6.31             | 8.042                   |
| 35.0         | 7.55             | 9.621                   |
| 38.0         | 8.90             | 11.34                   |



### ROUND

| Section size | Unit weight<br>M | Cross section area<br>A |
|--------------|------------------|-------------------------|
| mm           | kg/m             | cm <sup>2</sup>         |
| 6.0          | 0.222            | 0.283                   |
| 10.0         | 0.617            | 0.785                   |
| 12.0         | 0.888            | 1.131                   |
| 13.0         | 1.04             | 1.327                   |
| 14.0         | 1.21             | 1.539                   |
| 16.0         | 1.58             | 2.011                   |
| 18.0         | 2.00             | 2.545                   |
| 19.0         | 2.23             | 2.835                   |
| 20.0         | 2.47             | 3.142                   |
| 22.0         | 2.98             | 3.801                   |
| 24.0         | 3.55             | 4.524                   |
| 25.0         | 3.85             | 4.909                   |
| 26.0         | 4.17             | 5.309                   |
| 28.0         | 4.83             | 6.158                   |
| 29.0         | 5.19             | 6.605                   |
| 30.0         | 5.55             | 7.069                   |
| 32.0         | 6.31             | 8.042                   |
| 35.0         | 7.55             | 9.621                   |
| 38.0         | 8.90             | 11.34                   |
| 44.0         | 11.9             | 15.21                   |
| 50.0         | 15.4             | 19.63                   |
| 65.0         | 26.0             | 33.18                   |
| 75.0         | 34.7             | 44.18                   |
| 90.0         | 49.9             | 63.62                   |
| 100.0        | 61.7             | 78.54                   |
| 125.00       | 96.3             | 122.7                   |
| 150.00       | 139.0            | 176.7                   |
| 200.00       | 247.0            | 314.2                   |



|                                          |     |
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# PLATES

## PLATES-SPECIFICATIONS

| Application                                              | Standard                                                                                                |                                                                                                                     |                                                                                                                                    |                                                                                                                                      |                                                                                                                                                 |                                                                                                                                            |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | KS                                                                                                      | JIS                                                                                                                 | ASTM                                                                                                                               | DIN                                                                                                                                  | BS                                                                                                                                              | Others                                                                                                                                     |
| Structural Steel<br>Plates                               | KS D3503<br>SS34,41,50<br>KS D3514<br>SWS 41A,B,C<br>SWS 50A,B,C<br>SWS 50YA, YB<br>SWS 53B,C<br>SWS 58 | JIS G3101<br>SS 34,41,50<br>JIS G3106<br>SM 41A,B,C<br>SM 50A,B,C<br>SM 50YA,B<br>SM 58<br>JIS G3128<br>SHY 70N, NS | A283<br>Gr A,B,C,D<br>A36<br>A284<br>Gr A,B,C,D<br>A572<br>Gr 42,50,60,65<br>A633<br>Gr A,C,D,E<br>A678 Gr B,C<br>A656<br>A514 B,F | DIN 17100<br>U, RST 34-1,2<br>ST 37-1,2,3<br>U, RST 42-1<br>RST 42-2,3<br>RST 46-2,3<br>ST 50-1,2<br>ST 52-3<br>ST 60-1,2<br>ST 70-2 | BS 4360<br>Gr 40A,B,C,D,E<br>Gr 43A,B,C,D,E<br>Gr 50A,B,C,D,E<br>Gr 55C, E, F                                                                   | ISO R630<br>Fe 33, 37<br>CAS G4021<br>IS 226<br>ST 42 (S)<br>ST 42 (S,C)<br>IS 2062<br>ST 42 (W)<br>ST 42 (W,C)<br>ISO R630<br>Fe 42,44,52 |
| Weathering &<br>Corrosion<br>Resistant<br>Steel Plates   |                                                                                                         | JIS G3114<br>SMA 41A,B,C<br>SMA 50A,B,C<br>SMA 58                                                                   | A441<br>A588<br>Gr A,B,C,D,E,F                                                                                                     |                                                                                                                                      |                                                                                                                                                 |                                                                                                                                            |
| Steel Plates<br>for<br>Shipbuilding                      | KR<br>RA,RB,RD,RE<br>RA32,RD32,RE32<br>RA36,RD36,RE36                                                   | NK<br>KA,KB,KD,KE<br>KA32, KD32<br>KA36, KD36                                                                       | A131<br>A,B,D,E<br>ABS<br>A,B,D,E<br>AH32, DH32,<br>EH32                                                                           | GL<br>A32, D32, E32<br>A36, D36, E36                                                                                                 | LR<br>AH32, DH32,<br>EH32<br>AHS 34,<br>DHS 34,<br>EHS 34<br>AH36, DH36,<br>EH36                                                                | NV, CR, BV, RINA<br>A,B,C,D<br>NV, RINA<br>A27,D27,E27<br>A32,D32,E32<br>CR, BV<br>A32,D32,E32<br>A36,D36,E36                              |
| Carbon Steel<br>Plates for Boilers<br>& Pressure Vessels | KS D3560<br>SBB 35,42,<br>46,49<br>KS D3540<br>SG 11,42,46,49<br>KS D3521<br>SPPV 24,32,36,<br>46,50    | JIS G3103<br>SB 35,42,49<br>JIS G3118<br>SGV 42,46,49<br>JIS G3115<br>SPV 24,32,46<br>50                            | A285<br>Gr A,B,C<br>A515<br>Gr 55,60,65,70<br>A516<br>Gr 55,60,65,70<br>A442<br>Gr 55, 60<br>A537<br>Class 1,2<br>A612             | DIN 17155<br>HII, HII<br>HIII, HIV<br>17Mn4<br>19Mn5<br>VDEhw680<br>TTST 41<br>TTST45                                                | BS 1501-151<br>Gr 360A,B<br>Gr 400A,B<br>Gr 430A,B<br>BS 1501-221<br>Gr 460A,B<br>BS 1501-224<br>Gr 400A,B<br>Gr 430A,B<br>Gr 460A<br>Gr 490A,B |                                                                                                                                            |

## PLATES-SPECIFICATIONS

| Application                                                 | Standard                                             |                                                                                                 |                                                                                                |                                                  |                                              |        |
|-------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------|--------|
|                                                             | KS                                                   | JIS                                                                                             | ASTM                                                                                           | DIN                                              | BS                                           | Others |
| Low Alloy Steel<br>Plates for Boilers &<br>Pressure Vessels | KS D3560<br>SBB46M,49M                               | JIS G3103<br>SB46M,49M<br>JIS G3119<br>SBV 1A, 1B,<br>2,3<br>JIS G3120<br>SQV 1A,B,<br>SQV 3A,B | A202<br>Gr A,B<br>A204<br>Gr A,B,C<br>A533 Type A<br>Class 1,2,3<br>A533 Type C<br>Class 1,2,3 | WSN 5423<br>16Mo5<br>DIN 17155<br>15Mo3<br>16Mo5 | BS 1501-240<br>1/4Mo<br>BS 1501-261<br>1/4Mo |        |
| Steel Plates for<br>Low Temperature                         |                                                      | JIS G3126<br>SLA 24<br>SLA 33A,B<br>SLA 37<br>JIS G3127<br>SL3N26<br>SL3N28<br>SL3N45           | A662<br>Gr A,B,C<br>A203<br>Gr A                                                               | WDEhw680<br>TTST 41<br>TTST 45                   | BS 1501-224<br>Gr 430 A,B<br>Gr 460 A,B      |        |
| Carbon Steel Pipes<br>for Pressure Service                  | KS D3562<br>SPPS 38,42                               | JIS G3454<br>STPG 38,42                                                                         |                                                                                                |                                                  |                                              |        |
| Steel Plates for<br>Line Pipes                              |                                                      |                                                                                                 | A53<br>API 5L- A,B<br>API 5LX - X42,<br>X46,X52,X56,<br>X60,X65,X70                            |                                                  |                                              |        |
| Automobile &<br>Machine Steel<br>Plates                     | KS D3519<br>SAPH 32<br>SAPH 38<br>SAPH 41<br>SAPH 45 | JIS G3113<br>SAPH 32<br>SAPH 38<br>SAPH 41<br>SAPH 45                                           |                                                                                                |                                                  |                                              |        |
| Machine Structural<br>Steel Plates                          | KS D3752<br>SM10C~<br>SM58C                          | JIS G4051<br>S10C~<br>S58C                                                                      | SAE 1010 ~<br>SAE 1060                                                                         |                                                  |                                              |        |



# PLATES

## SPECIFICATIONS

### (1) ABSTRACT FROM MAIN SPECIFICATION (Classified by Standards Association)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                          |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------|--------------------------|---------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------|
| <b>1. JIS Specifications</b><br>(Japanese Industrial Standards) <table> <tr> <td>G3101</td> <td>Steel plates for general structures</td> </tr> <tr> <td>G3103</td> <td>Steel plates for boilers</td> </tr> <tr> <td>G3106</td> <td>Steel plates for welded structures</td> </tr> <tr> <td>G3114</td> <td>Hot-rolled atmospheric corrosion resisting steel plates for welded structures</td> </tr> <tr> <td>G3115</td> <td>Steel plates for pressure vessels</td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | G3101                                                                                    | Steel plates for general structures                                        | G3103                                                                                                                                                            | Steel plates for boilers                                 | G3106                                                                                                                                                                                                                               | Steel plates for welded structures              | G3114                                                                     | Hot-rolled atmospheric corrosion resisting steel plates for welded structures   | G3115                                            | Steel plates for pressure vessels                           | <b>6. BS Specifications</b><br>(British Standards Institution) <table> <tr> <td>BS15</td> <td>Steel plates for general structures</td> </tr> <tr> <td>BS1501-151</td> <td>Carbon steel plates for pressure vessels</td> </tr> <tr> <td>BS2762</td> <td>Notch ductile steel plates for bridges and general building construction</td> </tr> <tr> <td>BS4360</td> <td>Weldable structural steel plates</td> </tr> </table> | BS15                                               | Steel plates for general structures | BS1501-151                                                                               | Carbon steel plates for pressure vessels | BS2762                                                                              | Notch ductile steel plates for bridges and general building construction                                                                                                                   | BS4360                               | Weldable structural steel plates                                     |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| G3101                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Steel plates for general structures                                                      |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| G3103                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Steel plates for boilers                                                                 |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| G3106                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Steel plates for welded structures                                                       |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| G3114                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Hot-rolled atmospheric corrosion resisting steel plates for welded structures            |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| G3115                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Steel plates for pressure vessels                                                        |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| BS15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Steel plates for general structures                                                      |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| BS1501-151                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Carbon steel plates for pressure vessels                                                 |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| BS2762                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Notch ductile steel plates for bridges and general building construction                 |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| BS4360                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Weldable structural steel plates                                                         |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| <b>2. Kawasaki Specifications</b> <table> <tr> <td>VP</td> <td>Steel plates for gas vessels</td> </tr> <tr> <td>AG</td> <td>Steel plates for galvanizing pots</td> </tr> <tr> <td>PV</td> <td>Steel plates for pressure vessels</td> </tr> <tr> <td>RIVER ACE</td> <td>Quenched and tempered high tensile strength steel plates</td> </tr> <tr> <td>KH, HTP</td> <td>High tensile strength steel plates</td> </tr> <tr> <td>RIVER TEN</td> <td>Atmospheric corrosion resisting steel plates</td> </tr> <tr> <td>RIVER TEN</td> <td>Atmospheric corrosion resisting steel plates for welded structures</td> </tr> <tr> <td>RIVER TEN-R</td> <td>Atmospheric corrosion resisting steel plates for rolling stock</td> </tr> <tr> <td>RIVER TEN</td> <td>Sulfuric-acid resisting steel plates</td> </tr> <tr> <td>MARINER</td> <td>Anti-marine steel plates</td> </tr> <tr> <td>KL, RIVER ACE</td> <td>Steel plates for low temperature service</td> </tr> </table> | VP                                                                                       | Steel plates for gas vessels                                               | AG                                                                                                                                                               | Steel plates for galvanizing pots                        | PV                                                                                                                                                                                                                                  | Steel plates for pressure vessels               | RIVER ACE                                                                 | Quenched and tempered high tensile strength steel plates                        | KH, HTP                                          | High tensile strength steel plates                          | RIVER TEN                                                                                                                                                                                                                                                                                                                                                                                                                | Atmospheric corrosion resisting steel plates       | RIVER TEN                           | Atmospheric corrosion resisting steel plates for welded structures                       | RIVER TEN-R                              | Atmospheric corrosion resisting steel plates for rolling stock                      | RIVER TEN                                                                                                                                                                                  | Sulfuric-acid resisting steel plates | MARINER                                                              | Anti-marine steel plates | KL, RIVER ACE | Steel plates for low temperature service | <b>7. BV Specifications</b><br>(Bureau Veritas) <table> <tr> <td>BV</td> <td>Steel for construction of hulls and engine parts</td> </tr> </table> | BV | Steel for construction of hulls and engine parts |
| VP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Steel plates for gas vessels                                                             |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| AG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Steel plates for galvanizing pots                                                        |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| PV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Steel plates for pressure vessels                                                        |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| RIVER ACE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Quenched and tempered high tensile strength steel plates                                 |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| KH, HTP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | High tensile strength steel plates                                                       |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| RIVER TEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Atmospheric corrosion resisting steel plates                                             |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| RIVER TEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Atmospheric corrosion resisting steel plates for welded structures                       |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| RIVER TEN-R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Atmospheric corrosion resisting steel plates for rolling stock                           |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| RIVER TEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sulfuric-acid resisting steel plates                                                     |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| MARINER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Anti-marine steel plates                                                                 |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| KL, RIVER ACE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Steel plates for low temperature service                                                 |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| BV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Steel for construction of hulls and engine parts                                         |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| <b>3. AB Specifications</b><br>(American Bureau of Shipping Rules) <table> <tr> <td>AB</td> <td>Ordinary strength hull structural steel plates</td> </tr> <tr> <td>AB</td> <td>High strength hull structural steel plates</td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AB                                                                                       | Ordinary strength hull structural steel plates                             | AB                                                                                                                                                               | High strength hull structural steel plates               | <b>8. CR Specifications</b><br>(China Corporation Register of Shipping) <table> <tr> <td>CR</td> <td>Quality and testing of materials of hull, equipment, boiler and machinery</td> </tr> </table>                                  | CR                                              | Quality and testing of materials of hull, equipment, boiler and machinery |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| AB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Ordinary strength hull structural steel plates                                           |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| AB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | High strength hull structural steel plates                                               |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| CR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Quality and testing of materials of hull, equipment, boiler and machinery                |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| <b>4. AAR Specifications</b><br>(Association of American Railroads) <table> <tr> <td>M128</td> <td>High tensile strength plates for tank car tanks and other pressure vessels</td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M128                                                                                     | High tensile strength plates for tank car tanks and other pressure vessels | <b>9. DIN Specifications</b><br>(Deutscher Normenausschuss) <table> <tr> <td>DIN 17100</td> <td>Steel plates for general structural purposes</td> </tr> </table> | DIN 17100                                                | Steel plates for general structural purposes                                                                                                                                                                                        |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| M128                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | High tensile strength plates for tank car tanks and other pressure vessels               |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| DIN 17100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Steel plates for general structural purposes                                             |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| <b>5. ASTM Specifications</b><br>(American Society for Testing of Materials) <table> <tr> <td>A113</td> <td>Structural steel plates for locomotives and cars</td> </tr> <tr> <td>A131</td> <td>Structural carbon steel plates for ships</td> </tr> <tr> <td>A242</td> <td>High strength low alloy structural steel plates</td> </tr> <tr> <td>A283</td> <td>Low and intermediate tensile strength carbon-steel plates of structural quality</td> </tr> <tr> <td>A387</td> <td>Chromium-molybdenum alloy steel plates for pressure vessels</td> </tr> <tr> <td>A455</td> <td>Carbon-manganese steel plates for pressure vessels</td> </tr> <tr> <td>A515</td> <td>Carbon steel plates for pressure vessels for intermediate and higher temperature service</td> </tr> <tr> <td>A516</td> <td>Carbon steel plates for pressure vessels for moderate and lower temperature service</td> </tr> </table>                                                             | A113                                                                                     | Structural steel plates for locomotives and cars                           | A131                                                                                                                                                             | Structural carbon steel plates for ships                 | A242                                                                                                                                                                                                                                | High strength low alloy structural steel plates | A283                                                                      | Low and intermediate tensile strength carbon-steel plates of structural quality | A387                                             | Chromium-molybdenum alloy steel plates for pressure vessels | A455                                                                                                                                                                                                                                                                                                                                                                                                                     | Carbon-manganese steel plates for pressure vessels | A515                                | Carbon steel plates for pressure vessels for intermediate and higher temperature service | A516                                     | Carbon steel plates for pressure vessels for moderate and lower temperature service | <b>10. CSA Specifications</b><br>(Canadian Standards Association) <table> <tr> <td>G40.8</td> <td>Structural steel plates with improved resistance to brittle fracture</td> </tr> </table> | G40.8                                | Structural steel plates with improved resistance to brittle fracture |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| A113                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Structural steel plates for locomotives and cars                                         |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| A131                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Structural carbon steel plates for ships                                                 |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| A242                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | High strength low alloy structural steel plates                                          |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| A283                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Low and intermediate tensile strength carbon-steel plates of structural quality          |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| A387                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Chromium-molybdenum alloy steel plates for pressure vessels                              |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| A455                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Carbon-manganese steel plates for pressure vessels                                       |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| A515                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Carbon steel plates for pressure vessels for intermediate and higher temperature service |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| A516                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Carbon steel plates for pressure vessels for moderate and lower temperature service      |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| G40.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Structural steel plates with improved resistance to brittle fracture                     |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| <b>11. LR Specifications</b><br>(Lloyd's Register of Shipping Rules) <table> <tr> <td>LR</td> <td>Mild steel plates for hull structures</td> </tr> <tr> <td>LR</td> <td>Higher tensile steel plates for hull structures</td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | LR                                                                                       | Mild steel plates for hull structures                                      | LR                                                                                                                                                               | Higher tensile steel plates for hull structures          | <b>12. NK Specifications</b><br>(Nippon Kaiji Kyokai) <table> <tr> <td>NK</td> <td>Ordinary steel plates for hull construction</td> </tr> <tr> <td>NK</td> <td>High strength steel plates for hull construction</td> </tr> </table> | NK                                              | Ordinary steel plates for hull construction                               | NK                                                                              | High strength steel plates for hull construction |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| LR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mild steel plates for hull structures                                                    |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| LR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Higher tensile steel plates for hull structures                                          |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| NK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Ordinary steel plates for hull construction                                              |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| NK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | High strength steel plates for hull construction                                         |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| <b>13. NV Specifications</b><br>(Norske Veritas) <table> <tr> <td>NV</td> <td>Hull structural steel plates</td> </tr> <tr> <td>NV</td> <td>Shipbuilding steel plates with controlled yield strength</td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | NV                                                                                       | Hull structural steel plates                                               | NV                                                                                                                                                               | Shipbuilding steel plates with controlled yield strength | <b>14. GL Specifications</b><br>(Germanischer Lloyd) <table> <tr> <td>GL</td> <td>Hull structural steel plates</td> </tr> <tr> <td>GL</td> <td>High tensile hull structural steel plates</td> </tr> </table>                        | GL                                              | Hull structural steel plates                                              | GL                                                                              | High tensile hull structural steel plates        |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| NV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Hull structural steel plates                                                             |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| NV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Shipbuilding steel plates with controlled yield strength                                 |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| GL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Hull structural steel plates                                                             |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |
| GL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | High tensile hull structural steel plates                                                |                                                                            |                                                                                                                                                                  |                                                          |                                                                                                                                                                                                                                     |                                                 |                                                                           |                                                                                 |                                                  |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                     |                                                                                          |                                          |                                                                                     |                                                                                                                                                                                            |                                      |                                                                      |                          |               |                                          |                                                                                                                                                   |    |                                                  |

## (2) LIST OF SPECIFICATION (Classified by Tensile Strength)

| Tensile strength class | Uses                                                  | Specifications                | Tensile strength kg/mm <sup>2</sup> | Yield point kg/mm <sup>2</sup>                |
|------------------------|-------------------------------------------------------|-------------------------------|-------------------------------------|-----------------------------------------------|
| 41                     | General structures                                    | JIS G3101 SS41                | 41                                  | t≤16 25 16<t≤40 24 40<t 22                    |
|                        | Welded structures                                     | JIS G3106 SM41 A,B,C          | 41                                  | t≤16 25 16<t≤40 24 40<t 22                    |
|                        |                                                       | BS4360-40A.B.C.D.E            | 41                                  | t≤16 23 16<t≤38 22<br>38<t≤64 22 38<t≤ 102 21 |
|                        | Structures                                            | ASTM A36                      | 41                                  | 25                                            |
|                        | Bridges, general structures                           | BS 2762 ND IA, IIA, IIIA      | 41                                  | 10<t≤19 24 19≤t 22                            |
|                        | Atmospheric corrosion resist steel                    | RIVER TEN 41                  | 41                                  | t≤16 25 16<t≤40 24 40<t 22                    |
|                        | Atmospheric corrosion resist steel, welded structures | RIVER TEN 41S                 | 41                                  | t≤16 25                                       |
|                        | Boilers                                               | JIS G31.14 SMA41 A,B,C        | 41                                  | t≤16 19 16<t 22 40<t 24                       |
|                        |                                                       | RIVER TEN 41 A,B,C            | 41                                  | t≤16 25 16≤t 40 24 40<t 22                    |
|                        |                                                       | NV2-0, 1,2, 3                 | 41                                  | --                                            |
|                        |                                                       | BV BP41                       | 41                                  | --                                            |
|                        |                                                       | CR Grade CRI                  | 41                                  | --                                            |
|                        |                                                       | DIN 17155 HII                 | 41                                  | t≤16 26 16<t≤ 40 25<br>40<t≤ 60 24            |
|                        | Pressure vessels                                      | JIS G3115 SPV24               | 41                                  | t≤40 24 40<t 22                               |
|                        |                                                       | BS 1501-151-26 A,B            | 41                                  | t≤16 23 16<t≤ 32 23<br>32<t≤ 64 22            |
|                        |                                                       | BS 1501-211-26A, 221-26A.B    | 41                                  | t≤16 24.1 16<t≤ 32 23.5<br>32<t≤ 64 22.7      |
|                        |                                                       | BS 1501-224-26                | 41                                  | t≤16 26.8 16<t≤ 32 25.2<br>32<t≤ 64 23.6      |
|                        | Gas vessels                                           | RIVER TEN VP41                | 41                                  | 28                                            |
|                        | Low temperature service                               | NK KT35                       | 41                                  | --                                            |
|                        |                                                       | RIVER TEN KL33A,B             | 41                                  | 33                                            |
|                        | Automobiles                                           | RIVER TEN KFR41               | 41                                  | t≤6 26 t=8 24                                 |
|                        | Wheels                                                | RIVER TEN WH41                | 41                                  | --                                            |
|                        | Hull structures                                       | NK Grade A, B, C, D, E        | 41                                  | --                                            |
|                        |                                                       | AB Grade A, R, B, C, CS, D, E | 41                                  | --                                            |
|                        |                                                       | LR Grade A, B, C, D, E        | 41                                  | --                                            |
|                        |                                                       | NV Grade A, W, C, D, E        | 41                                  | --                                            |
|                        |                                                       | BV Grade A, B, C, D           | 41                                  | --                                            |
|                        |                                                       | CR Grade A, B, C, D, E        | 41                                  | --                                            |
|                        |                                                       | NDS XXG3101                   | 41                                  | 23                                            |
|                        |                                                       | GL                            | 41                                  | --                                            |
|                        |                                                       | ASTM- A131A                   | 41                                  | 22                                            |
| 42                     | General structures                                    | DIN 17100 Ust 42-1, 2 RSt     | 42                                  | t≤16 26 16<t ≤ 40 25<br>40<t≤ 100 24          |
|                        |                                                       | 42-1, 2 St 42-3               |                                     |                                               |
|                        | Locomotives, cars                                     | ASTM A113A                    | 42                                  | 23                                            |
|                        | Structures                                            | ASTM A283D                    | 42                                  | 23                                            |
|                        | Machine structures                                    | ASTM A284C                    | 42                                  | 21                                            |
|                        |                                                       | ASTM A284D                    | 42                                  | 23                                            |
|                        | Structures                                            | ASTM A441 (101.6 < t ≤ 203)   | 42                                  | 28                                            |
|                        | Structures                                            | IS 226, IS 2062               | 42                                  | t<6-6<t≤ 20 26<br>20<t ≤40 24 40<t 23         |
|                        | Structures                                            | LR LM42                       | 42                                  | t≤16 23.5 16<t≤40 22.5<br>40t≤63 21.5         |
|                        |                                                       |                               |                                     |                                               |
|                        |                                                       |                               |                                     |                                               |

# PLATES

## SPECIFICATION

| Tensile strength class | Uses                                                  | Specifications               | Tensile strength kg/mm <sup>2</sup> | Yield point kg/mm <sup>2</sup>     |
|------------------------|-------------------------------------------------------|------------------------------|-------------------------------------|------------------------------------|
| 42                     | Boilers                                               | JIS G3103-SB42               | 42                                  | 23                                 |
|                        |                                                       | LR LP421, 422                | 42                                  | --                                 |
|                        |                                                       | NK KP42A, B                  | 42                                  | 0.5T                               |
|                        |                                                       | AB APE, APL                  | 42                                  | 22.5                               |
|                        | Pressure vessels                                      | ASTM A387C                   | 42                                  | 25                                 |
|                        |                                                       | ASTM A387D                   | 42                                  | 21                                 |
|                        | Low temperature vessels                               | ASTM A410LT5                 | 42                                  | 21                                 |
|                        | Pressure vessels                                      | ASTM A442-60                 | 42                                  | 22                                 |
|                        | Pressure vessels, intermediate and higher temperature | ASTM A515-60                 | 42                                  | 22                                 |
|                        | Pressure vessels, moderate and lower temperature      | ASTM A516-60                 | 42                                  | 22                                 |
|                        | Pressure vessels                                      | RIVER TEN PV42A              | 42                                  | TSX0.5                             |
| 44                     | Structures                                            | ASTM A242 Type               | 44                                  | 29.5                               |
|                        | Structures                                            | ASTM A440 (38.1<t≤101.6)     | 44                                  | 30                                 |
|                        |                                                       | ASTM A441 (38.1<t≤101.6)     | 44                                  | 30                                 |
|                        | General structures                                    | BS 15-2                      | 44                                  | 9.53<t≤19.05 25<br>19.05<t≤38.1 24 |
|                        |                                                       | BS 15-3                      | 44                                  | 38.1≤t 23                          |
|                        |                                                       | BS 1449 En2D/A, 2D/B, 2D     | 44                                  | 25                                 |
|                        | Railroads                                             | BS 24-621, 631               | 44                                  | --                                 |
|                        | Bridges, general structures                           | BS 2762 NDIB, IIB, IIIB, IVB | 44                                  | 10<t≤19 25 19<t 24                 |
|                        | Welded structures                                     | BS4360-43A.B.C.D.E           | 44                                  | t≤16 25 16<t≤38 24                 |
|                        |                                                       |                              | 44                                  | 38<t≤64 23 64<t≤102 22             |
|                        | Structures                                            | DIN 17100 RSt46-2, St46-3    | 44                                  | t≤16 29 16<t≤40 28                 |
|                        |                                                       |                              | 44                                  | 40≤t≤100 27                        |
|                        | Structures                                            | IRS-M5                       | 44                                  | --                                 |
|                        | Boilers                                               | NV NV3-0, 1,2                | 44                                  | t≤16 28 16<t≤40 26                 |
|                        |                                                       |                              | 44                                  | 46<t≤100 25                        |
|                        |                                                       | NV NV7-1                     | 44                                  | t≤16 28 16<t≤40 27                 |
|                        | Pressure vessels                                      | NV NV7-2                     | 44                                  | 46<t≤100 27                        |
|                        |                                                       | BS 1501-151-28A              | 44                                  | t≤16 31 16<t≤40 30                 |
|                        |                                                       |                              | 44                                  | 40<t≤100 30                        |
|                        |                                                       | BS 1501-151-28B              | 44                                  | t≤16 25 16<t≤32 24                 |
|                        |                                                       |                              | 44                                  | 32<t≤64 23                         |
|                        |                                                       | BS 1501-161-28A              | 44                                  | t≤16 25 16<t≤32 25                 |
|                        |                                                       |                              | 44                                  | 32<t≤64 24                         |
|                        |                                                       | BS 1501-161-28B              | 44                                  | t≤16 25.2 16<t≤32 23.6             |
|                        |                                                       |                              | 44                                  | 32<t≤64 23.3                       |
|                        |                                                       | BS 1501-211-28A, 221-28A     | 44                                  | t≤16 25.2 16<t≤32 25.2             |
|                        |                                                       |                              | 44                                  | 32<t≤64 23.9                       |
|                        |                                                       | BS 1501-211-28B, 221-28B     | 44                                  | t≤16 26 16<t≤32 25.2               |
|                        |                                                       |                              | 44                                  | 32<t≤64 24.4                       |
|                        | Boilers                                               | BS 1501-224-28               | 44                                  | t≤16 26 16<t≤32 26                 |
|                        |                                                       |                              | 44                                  | 32<t≤64 24.7                       |
|                        |                                                       | DIN 17155 HIII               | 44                                  | t≤16 29.1 16<t≤32 27.6             |
|                        |                                                       |                              | 44                                  | 32<t≤64 26                         |
|                        |                                                       | DIN 17155 15Mo3              | 44                                  | t≤16 28 16<t≤40 27                 |

## SPECIFICATIONS

| Tensile strength class | Uses                                                           | Specifications           | Tensile strength kg/mm <sup>2</sup> | Yield point kg/mm <sup>2</sup>             |
|------------------------|----------------------------------------------------------------|--------------------------|-------------------------------------|--------------------------------------------|
| 44                     | Boilers                                                        | DIN 17155 13CrMo44       | 44                                  | t≤16 31 16<t≤40 30<br>40<t≤60 30           |
| 45                     | Gas vessels                                                    | RIVER TEN VP45           | 45                                  | 30                                         |
|                        | Wheels                                                         | RIVER TEN WH45           | 45                                  | 31                                         |
|                        | Hull structures                                                | GL-F32, G32              | 45                                  | t≤35 32 35<t≤50 31                         |
|                        | Hull structures                                                | NDS XXG3102B NS30C       | 45                                  | 30                                         |
|                        | Boilers                                                        | NV7-3                    | 45                                  | 29                                         |
|                        | Low temperature service                                        | RIVER TEN KL36B          | 45                                  | 36                                         |
|                        | Structures                                                     | CSA G40.8                | 45                                  | t≤15.9 28 15.9<t≤25.4 26<br>25.4<t≤38.1 25 |
|                        | Vessels, intermediate and higher temperature                   | ASTM A515-65             | 45                                  | 24                                         |
| 46                     | Hull structures                                                | NDS XXG3102B-NS30B       | 46                                  | 31                                         |
|                        | Boilers                                                        | JIS G3103 SB46           | 46                                  | 25                                         |
|                        | Boilers                                                        | JIS G3103 SB46M          | 46                                  | 25                                         |
|                        |                                                                | NK KP46                  | 46                                  | 0.5T                                       |
|                        |                                                                | AB APF, APM              | 46                                  | 24.5                                       |
|                        |                                                                | AB APH                   | 46                                  | 26                                         |
|                        | Boilers, pressure vessels                                      | ASTM A203A, A203D        | 46                                  | 26                                         |
|                        |                                                                | ASTM A204A               | 46                                  | 26                                         |
|                        | Pressure vessels                                               | ASTM A387B               | 46                                  | 28                                         |
|                        | Boilers, pressure vessels, intermediate and higher temperature | ASTM A515-65             | 46                                  | 25                                         |
|                        | Pressure vessels, moderate and lower temperature               | ASTM A516-65             | 46                                  | 25                                         |
|                        | Low temperature service                                        | NK KT50N, 50Q            | 46                                  | --                                         |
|                        | Low temperature pressure vessels                               | ASTM A203ALT, DLT        | 46                                  | 26                                         |
|                        | Rolling stock (Atmospheric corrosion resisting)                | RIVER TEN-R(C)           | 46                                  | 32                                         |
| 47                     | Structures                                                     | ASTM A440 (19<t≤38)      | 47                                  | 32                                         |
|                        |                                                                | ASTM A441 (19<t≤38)      | 47                                  | 32                                         |
|                        | Pressure vessels                                               | BS 1501-211-30A, 221-30A | 47                                  | t≤16 27.9 16<t≤32 26.9<br>32<t≤64 26.5     |
|                        |                                                                | BS 1501-211-30B, 221-30B | 47                                  | t≤16 27.9 16<t≤32 27.9<br>32<t≤26.5        |
|                        |                                                                | BS 1501-213-30           | 47                                  | t≤16 32.3 16<t≤32 31.5<br>32<t≤64 30.7     |
|                        |                                                                | BS 1501-224-30           | 47                                  | t≤16 31.5 16<t≤32 29.9<br>32<t≤64 28.3     |
|                        | Boilers                                                        | LP LP471,472             | 47                                  | --                                         |
|                        |                                                                | NV Grade NV5-0, 1,2      | 47                                  | t≤16 29 16<t≤40 28<br>40<t≤100 26          |
|                        |                                                                | DIN 17155 HIV            | 47                                  | t≤16 29 16<t≤40 28<br>40<t≤6 27            |
|                        |                                                                | DIN 17155 17Mn4          | 47                                  | t≤16 29 16<t≤40 28<br>40<t≤60 28           |

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## SPECIFICATIONS

| Tensile strength class                                    | Uses                                                           | Specifications               | Tensile strength kg/mm <sup>2</sup> | Yield point kg/mm <sup>2</sup> |
|-----------------------------------------------------------|----------------------------------------------------------------|------------------------------|-------------------------------------|--------------------------------|
| 49                                                        | Structures                                                     | ASTM A242 Type 1             | 49                                  | 35                             |
|                                                           | Structures                                                     | ASTM A440 (t≤19)             | 49                                  | 35                             |
|                                                           |                                                                | ASTM A441 (t≤19)             | 49                                  | 35                             |
|                                                           |                                                                | Boilers                      | JIS G3103 SB49, 49M                 | 49                             |
|                                                           | Boilers, pressure vessels                                      | NK KP49                      | 49                                  | 27                             |
|                                                           |                                                                | ASTM A203B, A203E            | 49                                  | 28                             |
|                                                           |                                                                | ASTM A204B                   | 49                                  | 28                             |
|                                                           |                                                                | ASTM A225A                   | 49                                  | 28                             |
|                                                           |                                                                | AB APG, API                  | 49.5                                | 27                             |
|                                                           | Boilers, pressure vessels, intermediate and higher temperature | ASTM A515-70                 | 49                                  | 26                             |
| Boilers, pressure vessels, moderate and lower temperature | ASTM A516-70FR, F                                              | 49                           | 27                                  |                                |
| Low temperature pressure vessels                          | ASTM A203ELT                                                   | 49                           | 28                                  |                                |
| Rolling stock (Atmospheric corrosion resisting)           | RIVER TEN-R(H,P)                                               | 49                           | 35                                  |                                |
| 50                                                        | General structures                                             | JIS G3101 SS50               | 50                                  | t≤16 29 16<t≤40 28 40<t 26     |
|                                                           | Welded structures                                              | JIS G3106 SM50A,B,C          | 50                                  | t≤16 33 16<t≤40 32 40<t 30     |
|                                                           |                                                                | JIS G3106 SM50YA, YB         | 50                                  | t≤16 37 16<t≤40 36 40<t 34     |
|                                                           |                                                                | BS4360-50A.B.C.D.E           | 50                                  | t≤16 36 16<t≤38 35             |
|                                                           |                                                                |                              |                                     | 39<t≤64 34 64<t≤102 33         |
|                                                           | Gas vessels                                                    | RIVER TEN VP50               | 50                                  | 36                             |
|                                                           | High yield point structures                                    | BS 968                       | 50                                  | t≤9.5 36 9.5<t≤16 36           |
|                                                           |                                                                |                              |                                     | 16<t≤31.7 35 31.7<t≤50.8 36    |
|                                                           |                                                                |                              |                                     | 50.8 ≤ t-                      |
|                                                           | Structures                                                     | DIN 17100 St50-1, 50-2       | 50                                  | t≤16 30 16<t≤ 40 29            |
|                                                           |                                                                |                              |                                     | 40<t ≤100 28                   |
|                                                           | Hull structures                                                | NK Kind of No. 1 Grade A,D,E | 50                                  | 32                             |
|                                                           |                                                                | AB Grade AH,BH,CH,DHS,EH     | 50                                  | 32                             |
|                                                           |                                                                | GL-F36, G36                  | 50                                  | t≤35 36 35<t≤50 35             |
|                                                           | High tensile steel                                             | RIVER TEN KH36A,B            | 50                                  | t≤16 37 16<t≤40 36 40<t 34     |
|                                                           | Atmospheric corrosion resist, steel                            | RIVER TEN 50M                | 50                                  | t≤16 33 16<t≤40 32 40<t 30     |
|                                                           | Atmospheric corrosion resist, welded structures                | JIS G3114 SMA50A,B,C         | 50                                  | t≤16 19 16<t 22 40<t 24        |
|                                                           |                                                                | RIVER TEN 50A,B,C            | 50                                  | t≤16 37 16<t≤40 36 40<t 34     |
|                                                           | Sulfuric-acid corrosion resisting steel                        | RIVER TEN 50K                | 50                                  | 33                             |
|                                                           | Anti-marine steel                                              | RIVER TEN MARINER            | 50                                  | 36                             |
|                                                           | Boilers, pressure vessels                                      | JIS G3115 SPV32              | 50                                  | t≤40 32 40<t 30                |
|                                                           |                                                                | NV Grade NV4-0,1,2,3,4       | 50                                  | t≤16 34 16<t≤40 33             |
|                                                           |                                                                |                              |                                     | 40<t≤100 32                    |
|                                                           |                                                                |                              |                                     |                                |
|                                                           | Pressure vessels                                               | NV Grade NV25-1,2,3,4        | 50                                  | 22                             |
|                                                           |                                                                | BS 1501-211-32A, 221-32A     | 50                                  | t≤16 29.6 16<t≤32 29           |
|                                                           |                                                                |                              | 32<t≤64 27.9                        |                                |
|                                                           |                                                                | BS 1501-211-32B, 221-32B     | 50                                  | t≤16 29.6 16<t≤32 29.6         |
|                                                           |                                                                |                              |                                     | 32<t≤64 28.8                   |

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| Tensile strength class | Uses                                            | Specifications              | Tensile strength kg/mm <sup>2</sup> | Yield point kg/mm <sup>2</sup>                                             |
|------------------------|-------------------------------------------------|-----------------------------|-------------------------------------|----------------------------------------------------------------------------|
| 50                     | Pressure vessels                                | BS 1501-213-32              | 50                                  | t≤16 34.6 16<t≤32 33.9<br>32<t≤64 33.1                                     |
|                        |                                                 | BS 1501-224-32              | 50                                  | t≤16 33.1 16<t≤32 31.5<br>32<t≤64 29.9                                     |
|                        | Low temperature service Structures              | NK Kind of No.2 K5T-50Q     | 50                                  | --                                                                         |
|                        |                                                 | DIN 17100 St50, 50-2        | 50                                  | t≤16 30 16<t≤40 29<br>40<t≤100 28                                          |
| 52                     | Structures                                      | DIN 17100 St52-3            | 52                                  | t<16 36 16<t≤30 35<br>30<t≤50 34                                           |
|                        | Boilers                                         | LR LP521,522                | 52                                  | --                                                                         |
|                        |                                                 | IS 2002-2B                  | 52                                  | T.S. x 0.50                                                                |
|                        |                                                 | NV Grade NV6                | 52                                  | t≤16 33 16<t≤40 32<br>40<t≤100 31                                          |
|                        |                                                 | DIN 17155 19Mn5             | 52                                  | t≤16 33 16<t≤40 32<br>46<t≤60 32                                           |
|                        | Pressure vessels                                | ASTM A302A                  | 52                                  | 31                                                                         |
|                        |                                                 |                             |                                     |                                                                            |
|                        | Pressure vessels                                | ASTM A302A                  | 52                                  | 31                                                                         |
| 53                     | Welded structures                               | JIS G3106 SM53B,C           | 53                                  | t≤16 37 16<t≤40 36 40<t 34                                                 |
|                        | High tensile steel, welded structures           | WES (HPI) Kind of No.1 HW36 | 53                                  | 36                                                                         |
|                        | Atmospheric corrosion resist, welded structures | RIVER TEN 53B,C             | 53                                  | t≤16 37 16<t≤40 36 40<t 34                                                 |
|                        | Boilers, pressure vessels                       | AB APJ                      | 53                                  | 30.5                                                                       |
|                        |                                                 | ASTM A204C                  | 53                                  | 30                                                                         |
|                        |                                                 | ASTM A299                   | 53                                  | 30                                                                         |
|                        |                                                 | ASTM A302A                  | 53                                  | 32                                                                         |
|                        |                                                 | JIS G3115 SPV36             | 53                                  | t≤40 36 40<t 34                                                            |
|                        | Pressure vessels                                | ASTM A387D, A387E           | 53                                  | 32                                                                         |
|                        | Unfired pressure vessel                         | ASTM A455A                  | 53                                  | 26                                                                         |
| 55                     | General structures                              | JIS G3101 SS55              | 55                                  | t<16 41 16<t≤40 40 40<t -                                                  |
|                        | Structures                                      | IS 961 St58-HT, HTC         | 55                                  | 6<t≤28 36 28<t≤45 35<br>45<t≤63 52 63<t 50                                 |
|                        | Welded structures                               | BS4360-55C.E                | 55                                  | t<16 45 16<t≤38 44<br>38<t≤64 42                                           |
|                        | Railroads                                       | BS24-632A, B                | 55                                  | t≤25.4 33 t>25.4 31                                                        |
|                        | Hull structures                                 | GL-G42                      | 55                                  | t≤35 42 35<t≤50 41                                                         |
|                        | Gas vessels                                     | RIVER TEN VP55              | 55                                  | 38                                                                         |
|                        | High tensile steel                              | RIVER TEN KH40A, B          | 55                                  | t≤16 41 16<t≤40 40 40<t 38                                                 |
|                        |                                                 | RIVER TEN HP55W,WA,WB       | 55                                  | 36                                                                         |
|                        | Abrasion resist steel                           | RIVER TEN CrMn55            | 55                                  | 34                                                                         |
|                        |                                                 |                             |                                     |                                                                            |
|                        | Abrasion resist steel                           | RIVER TEN CrMn55            | 55                                  | 34                                                                         |
| 56                     | Boilers                                         | JIS G3103 SB56M             | 56                                  | 35                                                                         |
|                        | Boilers, pressure vessels                       | ASTM A302B,C,D              | 56                                  | 35                                                                         |
| 57                     | Tank car tanks, pressure vessels                | AAR M128                    | 57                                  | 35                                                                         |
|                        | High tensile steel, welded structures           | RIVER TEN HTP57W,WA,WB      | 57                                  | 40                                                                         |
|                        |                                                 | WES (HPI) Kind of No.2 HW40 | 57                                  | 40                                                                         |
| 58                     | Welded structures                               | JIS G3106 SM58              | 58                                  | t≤16 47 16<t≤40 46 40<t 44                                                 |
|                        | Bridges, general structures                     | BS548                       | 58                                  | 6.4<t≤31.8 36 31.8<t≤44.5 35<br>44.5<t≤57.1 33 57.1<t≤69.9 32<br>69.9≤t 30 |

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## SPECIFICATIONS

| Tensile strength class | Uses                                                          | Specifications                  | Tensile strength kg/mm <sup>2</sup> | Yield point kg/mm <sup>2</sup>                          |
|------------------------|---------------------------------------------------------------|---------------------------------|-------------------------------------|---------------------------------------------------------|
| 58                     | Structures                                                    | IS 961 St55-HTW. HTWC           | 58                                  | 6<t≤16 36 16<t≤32 35<br>32<t≤63 34 63<t 29              |
|                        | Hull structures                                               | NDS XXG3103 NS46B, 46C          | 58                                  | 46                                                      |
|                        | Pressure vessels                                              | IIS G3115 SPV 46                | 58                                  | t≤40 46 40<t 46                                         |
|                        | Atmospheric corrosion resist, welded structures               | JIS G3114 SMA58<br>RIVER TEN 58 | 58<br>58                            | t≤16 19 16<t 26 40 < t 20<br>t≤16 47 16<t≤40 46 40<t 44 |
| 60                     | Structures                                                    | DIN 17100 St60-1, 2             | 60                                  | t≤16 34 16<t≤40 33<br>40<t≤100 32                       |
|                        | Hull structures                                               | NK Kind of No.2 Grade A,D,E     | 60                                  | 46                                                      |
|                        | Quenched and tempered high tensile steel                      | RIVER ACE 60                    | 60                                  | 50                                                      |
|                        | Non-tempered high tensile steel                               | RIVER TEN HTP60W                | 60                                  | 45                                                      |
|                        | High tensile steel, welded structures                         | WES (HPI) Kind of No.3 HW45     | 60                                  | 45                                                      |
|                        | Abrasion resist steel                                         | RIVER TEN Cr Mn60               | 60                                  | 40                                                      |
| 62                     | Quenched and tempered high tensile steel                      | RIVER ACE 60H                   | 62                                  | 50                                                      |
|                        | Atmospheric corrosion resist. high tensile, welded structures | WES (HPI) Kind of No.4 HW50     | 62                                  | 50                                                      |
|                        | Atmospheric corrosion resist, welded structures               | RIVER TEN 62                    | 62                                  | t≤16 50 16<t≤40 50 40<t 50                              |
|                        | Low temperature pressure vessels                              | JIS G3115 SPV50                 | 62                                  | t≤40 50 40<t -                                          |
|                        | Low temperature service                                       | RIVER ACE 60L                   | 62                                  | 50                                                      |
| 68                     | High tensile steel, welded structures                         | WES (HPI) Kind of No.5 HW56     | 68                                  | 56                                                      |
| 70                     | Structures                                                    | DIN 17100 St70-2                | 70                                  | t≤16 37 16<t≤40 36<br>40<t≤100 35                       |
|                        | Quenched and tempered high tensile steel                      | RIVER ACE 70                    | 70                                  | 60                                                      |
|                        | Low temperature pressure vessels                              | ASTM A353LT4                    | 70                                  | 53                                                      |
| 74                     | Quenched and tempered high tensile steel                      | RIVER ACE 70H                   | 74                                  | 63                                                      |
|                        | High tensile steel, welded structures                         | WES (HPI) Kind of No.6 HW63     | 74                                  | 63                                                      |
|                        | Low temperature service                                       | RIVER ACE 70L                   | 74                                  | 63                                                      |
| 80                     | Quenched and tempered high tensile steel                      | RIVER ACE K-O                   | 80                                  | 70                                                      |
|                        | High tensile steel, welded structures                         | WES (HPI) Kind of No.7 HW70     | 80                                  | 70                                                      |
| 88                     | High tensile steel, welded structures                         | WES (HPI) Kind of No. 8 HW80    | 88                                  | 80                                                      |
| 97                     | High tensile steel, welded structures                         | WES (HPI) Kind of No.9 HW90     | 97                                  | 90                                                      |

## Imperial Units JIS (18.524 kg/in ft<sup>2</sup>)

| Thickness |        | Unit weight        | Width x Length, ft |        |        |        |        |        |        |        |
|-----------|--------|--------------------|--------------------|--------|--------|--------|--------|--------|--------|--------|
| t         |        | M                  | 4 x 8              | 5 x 10 | 5 x 20 | 5 x 30 | 6 x 20 | 6 x 24 | 6 x 30 | 8 x 30 |
| in        | mm     | kg/ft <sup>2</sup> | 32                 | 50     | 100    | 150    | 120    | 144    | 180    | 240    |
| 1/16      | 1.59   | 1.158              | 37.1               | 57.9   | 116    | 174    | 139    | 167    | 208    | 278    |
| 1/8       | 3.18   | 2.316              | 74.1               | 116    | 232    | 347    | 278    | 334    | 417    | 556    |
| 3/16      | 4.76   | 3.473              | 111                | 174    | 347    | 521    | 417    | 500    | 625    | 834    |
| 1/4       | 6.35   | 4.631              | 148                | 232    | 463    | 695    | 556    | 667    | 834    | 1,111  |
| 5/16      | 7.94   | 5.789              | 185                | 289    | 579    | 868    | 695    | 834    | 1,042  | 1,389  |
| 3/8       | 9.53   | 6.946              | 222                | 347    | 695    | 1,042  | 834    | 1,000  | 1,250  | 1,667  |
| 7/16      | 11.11  | 8.104              | 259                | 405    | 810    | 1,256  | 972    | 1,167  | 1,459  | 1,945  |
| 1/2       | 12.70  | 9.262              | 296                | 463    | 926    | 1,389  | 1,111  | 1,334  | 1,667  | 2,223  |
| 9/16      | 14.24  | 10.42              | 333                | 521    | 1,042  | 1,563  | 1,250  | 1,500  | 1,876  | 2,501  |
| 5/8       | 15.88  | 11.58              | 371                | 579    | 1,158  | 1,737  | 1,390  | 1,668  | 2,084  | 2,779  |
| 11/16     | 17.46  | 12.74              | 408                | 637    | 1,274  | 1,911  | 1,529  | 1,835  | 2,293  | 3,058  |
| 3/4       | 19.05  | 13.89              | 444                | 694    | 1,389  | 2,084  | 1,667  | 2,000  | 2,500  | 3,334  |
| 13/16     | 20.64  | 15.05              | 482                | 752    | 1,505  | 2,254  | 1,806  | 2,167  | 2,709  | 3,612  |
| 7/8       | 22.23  | 16.21              | 519                | 810    | 1,621  | 2,432  | 1,945  | 2,334  | 2,918  | 3,890  |
| 15/16     | 23.81  | 17.37              | 556                | 868    | 1,737  | 2,606  | 2,084  | 2,501  | 3,127  | 4,169  |
| 1         | 25.40  | 18.52              | 593                | 926    | 1,852  | 2,778  | 2,222  | 2,667  | 3,334  | 4,445  |
| 1 1/8     | 28.58  | 20.84              | 667                | 1,042  | 2,084  | 3,126  | 2,501  | 3,001  | 3,751  | 5,002  |
| 1 1/4     | 31.75  | 23.16              | 741                | 1,158  | 2,316  | 3,474  | 2,779  | 3,335  | 4,169  | 5,558  |
| 1 3/8     | 34.93  | 25.47              | 815                | 1,274  | 2,547  | 3,820  | 3,056  | 3,668  | 4,585  | 6,113  |
| 1 1/2     | 38.10  | 27.79              | 889                | 1,390  | 2,779  | 4,168  | 3,335  | 4,002  | 5,002  | 6,670  |
| 1 5/8     | 41.28  | 30.10              | 963                | 1,505  | 3,010  | 4,515  | 3,612  | 4,334  | 5,418  | 7,224  |
| 1 3/4     | 44.45  | 32.42              | 1,037              | 1,621  | 3,242  | 4,863  | 3,890  | 4,668  | 5,836  | 7,781  |
| 1 7/8     | 47.63  | 34.73              | 1,111              | 1,736  | 3,473  | 5,210  | 4,168  | 5,001  | 6,251  | 8,335  |
| 2         | 50.80  | 37.05              | 1,186              | 1,852  | 3,705  | 5,558  | 4,446  | 5,335  | 6,669  | 8,892  |
| 2 1/2     | 63.50  | 46.31              | 1,482              | 2,316  | 4,631  | 6,946  | 5,557  | 6,669  | 8,336  | 11,114 |
| 3         | 76.20  | 55.57              | 1,778              | 2,778  | 5,557  | 8,336  | 6,668  | 8,002  | 10,003 | 13,337 |
| 4         | 101.60 | 74.10              | 2,371              | 3,705  | 7,410  | 11,115 | 8,892  | 10,670 | 13,338 | 17,784 |
| 5         | 127.00 | 92.62              | 2,964              | 4,631  | 9,262  | 13,893 | 11,114 | 13,337 | 16,672 | 22,229 |
| 6         | 152.40 | 111.1              | 3,557              | 5,558  | 11,114 | 16,672 | 13,338 | 16,005 | 20,006 | 26,675 |



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Metric Units (7.85 kg/mm m<sup>2</sup>-0.7293 kg/mm ft<sup>2</sup>)

| Thickness | Unit weight | Width x Length, ft |       |        |        |        |        |        |        |        |        |        |
|-----------|-------------|--------------------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| t         | M           | 3 x 6              | 4 x 8 | 4 x 10 | 4 x 16 | 4 x 20 | 5 x 10 | 5 x 20 | 5 x 30 | 5 x 40 | 6 x 30 | 6 x 40 |
| mm        | kg          | 18                 | 32    | 40     | 64     | 80     | 50     | 100    | 150    | 200    | 180    | 240    |
| 1.20.8752 | 15.8        | 28.0               | 35.0  | 56.0   | 70.0   | 43.8   | 87.5   | 131    | 175    | 158    | 210    |        |
| 1.61.167  | 21.0        | 37.3               | 46.7  | 74.7   | 93.4   | 58.4   | 117    | 175    | 233    | 210    | 280    |        |
| 1.91.386  | 24.9        | 44.4               | 55.4  | 88.7   | 111    | 69.3   | 139    | 208    | 277    | 249    | 333    |        |
| 2.31.677  | 30.2        | 53.7               | 67.1  | 107    | 134    | 83.9   | 168    | 252    | 335    | 302    | 402    |        |
| 2.61.896  | 34.1        | 60.7               | 75.8  | 121    | 152    | 94.8   | 190    | 284    | 379    | 341    | 455    |        |
| 3.02.188  | 39.4        | 70.0               | 87.5  | 140    | 175    | 109    | 219    | 328    | 438    | 394    | 525    |        |
| 3.22.334  | 42.0        | 74.7               | 93.4  | 148    | 187    | 117    | 233    | 350    | 467    | 420    | 560    |        |
| 4.53.282  | 59.1        | 105                | 131   | 210    | 263    | 164    | 328    | 492    | 656    | 591    | 788    |        |
| 5.03.646  | 65.6        | 117                | 146   | 233    | 292    | 182    | 365    | 547    | 729    | 656    | 875    |        |
| 6         | 4.376       | 78.8               | 140   | 175    | 280    | 350    | 219    | 438    | 656    | 875    | 788    | 1,050  |
| 7         | 5.105       | 91.9               | 163   | 204    | 327    | 408    | 255    | 510    | 766    | 1,021  | 919    | 1,225  |
| 8         | 5.834       | 105                | 187   | 233    | 373    | 467    | 292    | 583    | 875    | 1,167  | 1,050  | 1,400  |
| 9         | 6.564       | 118                | 210   | 263    | 420    | 525    | 328    | 656    | 985    | 1,313  | 1,182  | 1,575  |
| 10        | 7.293       | 131                | 233   | 292    | 467    | 583    | 365    | 729    | 1,094  | 1,459  | 1,313  | 1,750  |
| 11        | 8.022       | 144                | 257   | 321    | 513    | 642    | 401    | 802    | 1,203  | 1,604  | 1,444  | 1,925  |
| 12        | 8.752       | 158                | 280   | 350    | 560    | 700    | 438    | 875    | 1,313  | 1,750  | 1,575  | 2,100  |
| 12.7      | 9.262       | 167                | 296   | 370    | 593    | 741    | 463    | 926    | 1,389  | 1,852  | 1,667  | 2,223  |
| 13        | 9.481       | 171                | 303   | 379    | 607    | 758    | 474    | 948    | 1,422  | 1,896  | 1,707  | 2,275  |
| 14        | 10.21       | 184                | 327   | 408    | 653    | 817    | 510    | 1,021  | 1,532  | 2,042  | 1,838  | 2,450  |
| 15        | 10.94       | 197                | 350   | 438    | 700    | 875    | 547    | 1,094  | 1,641  | 2,188  | 1,969  | 2,626  |
| 16        | 11.67       | 210                | 373   | 467    | 747    | 934    | 584    | 1,167  | 1,750  | 2,334  | 2,101  | 2,801  |
| 17        | 12.40       | 223                | 397   | 496    | 794    | 992    | 620    | 1,240  | 1,860  | 2,480  | 2,232  | 2,976  |
| 18        | 13.13       | 236                | 420   | 525    | 840    | 1,050  | 656    | 1,313  | 1,970  | 2,625  | 2,363  | 3,151  |
| 19        | 13.86       | 249                | 444   | 554    | 887    | 1,109  | 693    | 1,386  | 2,079  | 2,772  | 2,495  | 3,326  |
| 20        | 14.59       | 263                | 467   | 584    | 934    | 1,167  | 730    | 1,459  | 2,188  | 2,918  | 2,626  | 3,502  |
| 21        | 15.32       | 276                | 490   | 613    | 980    | 1,226  | 766    | 1,532  | 2,298  | 3,064  | 2,758  | 3,677  |
| 22        | 16.04       | 289                | 513   | 642    | 1,027  | 1,283  | 802    | 1,604  | 2,406  | 3,208  | 2,887  | 3,850  |
| 23        | 16.77       | 302                | 537   | 671    | 1,073  | 1,342  | 838    | 1,677  | 2,516  | 3,354  | 3,019  | 4,025  |
| 24        | 17.50       | 315                | 560   | 700    | 1,120  | 1,400  | 875    | 1,750  | 2,625  | 3,500  | 3,150  | 4,200  |
| 25        | 18.23       | 328                | 583   | 729    | 1,167  | 1,458  | 912    | 1,823  | 2,734  | 3,646  | 3,281  | 4,375  |
| 25.4      | 18.52       | 333                | 593   | 741    | 1,185  | 1,482  | 926    | 1,852  | 2,778  | 3,704  | 3,334  | 4,445  |
| 26        | 18.96       | 341                | 607   | 758    | 1,213  | 1,517  | 948    | 1,896  | 2,844  | 3,792  | 3,413  | 4,550  |
| 27        | 19.69       | 354                | 630   | 788    | 1,260  | 1,575  | 984    | 1,969  | 2,954  | 3,938  | 3,544  | 4,726  |
| 28        | 20.42       | 368                | 653   | 817    | 1,307  | 1,634  | 1,021  | 2,042  | 3,063  | 4,084  | 3,676  | 4,901  |
| 29        | 21.15       | 381                | 677   | 846    | 1,354  | 1,692  | 1,058  | 2,115  | 3,172  | 4,230  | 3,807  | 5,076  |
| 30        | 21.88       | 394                | 700   | 875    | 1,400  | 1,750  | 1,094  | 2,188  | 3,282  | 4,376  | 3,938  | 5,251  |
| 32        | 23.34       | 420                | 747   | 934    | 1,494  | 1,867  | 1,167  | 2,334  | 3,501  | 4,668  | 4,201  | 5,602  |
| 34        | 24.80       | 446                | 794   | 992    | 1,587  | 1,984  | 1,240  | 2,480  | 3,720  | 4,960  | 4,464  | 5,952  |
| 36        | 26.25       | 472                | 840   | 1,050  | 1,680  | 2,100  | 1,312  | 2,625  | 3,938  | 5,250  | 4,725  | 6,300  |
| 38        | 27.71       | 499                | 887   | 1,108  | 1,773  | 2,217  | 1,386  | 2,771  | 4,156  | 5,542  | 4,988  | 6,650  |
| 40        | 29.17       | 525                | 933   | 1,167  | 1,867  | 2,334  | 1,458  | 2,917  | 4,376  | 5,834  | 5,251  | 7,001  |
| 45        | 32.82       | 591                | 1,050 | 1,313  | 2,100  | 2,626  | 1,641  | 3,282  | 4,923  | 6,564  | 5,908  | 7,877  |
| 50        | 36.46       | 656                | 1,167 | 1,458  | 2,334  | 2,917  | 1,823  | 3,646  | 5,469  | 7,292  | 6,563  | 8,750  |
| 55        | 40.11       | 722                | 1,284 | 1,604  | 2,567  | 3,209  | 2,006  | 4,011  | 6,016  | 8,022  | 7,220  | 9,626  |
| 60        | 43.76       | 788                | 1,400 | 1,750  | 2,801  | 3,501  | 2,188  | 4,376  | 6,564  | 8,752  | 7,877  | 10,502 |
| 65        | 47.40       | 853                | 1,517 | 1,896  | 3,034  | 3,792  | 2,370  | 4,740  | 7,110  | 9,480  | 8,532  | 11,376 |
| 70        | 51.05       | 919                | 1,634 | 2,042  | 3,267  | 4,084  | 2,552  | 5,105  | 7,658  | 10,210 | 9,189  | 12,252 |
| 80        | 58.34       | 1,050              | 1,867 | 2,334  | 3,734  | 4,667  | 2,917  | 5,834  | 8,751  | 11,670 | 10,500 | 14,002 |
| 90        | 65.64       | 1,182              | 2,100 | 2,626  | 4,201  | 5,251  | 3,282  | 6,564  | 9,846  | 13,128 | 11,815 | 15,754 |
| 100       | 72.93       | 1,313              | 2,334 | 2,917  | 4,668  | 5,834  | 3,647  | 7,293  | 10,940 | 14,586 | 13,127 | 17,503 |
| 110       | 80.22       | 1,444              | 2,567 | 3,209  | 5,134  | 6,418  | 4,011  | 8,022  | 12,033 | 16,044 | 14,440 | 19,253 |
| 120       | 87.52       | 1,575              | 2,801 | 3,501  | 5,601  | 7,002  | 4,376  | 8,752  | 13,128 | 17,504 | 15,754 | 21,005 |
| 130       | 94.81       | 1,707              | 3,034 | 3,792  | 6,080  | 7,585  | 4,741  | 9,481  | 14,222 | 18,962 | 17,066 | 22,754 |
| 140       | 102.11,838  | 3,267              | 4,084 | 6,534  | 8,168  | 5,105  | 10,210 | 15,315 | 20,420 | 18,378 | 24,504 |        |
| 150       | 109.4       | 1,969              | 3,501 | 4,376  | 7,002  | 8,752  | 5,470  | 10,940 | 16,410 | 21,880 | 19,692 | 26,256 |

## TECHNICAL SPECIFICATION REFERENCE - STEEL PLATES, SHEETS

Excerpts from JIS G 3131, JIS G 3101, JIS G 3302 & JIS G 3141 Specifications

| Classification                                                       | Specification | Grade | Chemical Composition |                 |                 |                |                | Tensile Test                             |                                        |            |                                |            | Bend Test <sup>*2</sup> |                                                            |
|----------------------------------------------------------------------|---------------|-------|----------------------|-----------------|-----------------|----------------|----------------|------------------------------------------|----------------------------------------|------------|--------------------------------|------------|-------------------------|------------------------------------------------------------|
|                                                                      |               |       | C<br>max.<br>%       | Mn<br>max.<br>% | Si<br>max.<br>% | P<br>max.<br>% | S<br>max.<br>% | Tensile<br>Strength<br>N/mm <sup>2</sup> | Yield<br>Strength<br>N/mm <sup>2</sup> | Elongation |                                |            | Bending<br>Angle        | Bending<br>Radius                                          |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        | Test PC    | Thickness                      | %          |                         |                                                            |
| Hot Rolled<br>Steel Plates<br>and Sheets<br>&<br>Chequered<br>Plates | JIS G 3131    | SPHC  | 0.15                 | 0.60            | --              | 0.05           | 0.05           | 270<br>min.                              | --                                     | No.5       | 1.2mm & over,<br>up to 1.6 mm  | 27<br>min. | 180°                    | Flat on<br>itself<br><br>0.5t                              |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        |            | 1.6mm & over,<br>up to 3.2 mm  | 29<br>min. |                         |                                                            |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        |            | 3.2mm & over                   | 31<br>min. |                         |                                                            |
|                                                                      | JIS G 3101    | SS330 | --                   | --              | --              | 0.05           | 0.05           | 330<br>- 430                             | 205<br>min.                            | No.5       | 5 mm & under                   | 26<br>min. | 180°                    | 0.5t                                                       |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        | No.1A      | over 5mm-<br>16mm              | 21<br>min. |                         |                                                            |
|                                                                      | JIS G 3101    | SS400 | --                   | --              | --              | 0.05           | 0.05           | 400<br>- 510                             | 245<br>min.                            | No.5       | 5 mm & under                   | 21<br>min. | 180°                    | 1.5t                                                       |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        | No.1A      | over 5mm-<br>16mm              | 17<br>min. |                         |                                                            |
| Galvanised<br>Steel Sheets                                           | JIS G 3302    | SGCC  | --                   | --              | --              | --             | --             | 270<br>min.                              | 205<br>min.                            | --         | Up to 1.6 mm                   | --         | 180°                    | t <sup>*1</sup><br>t <sup>*1</sup><br><br>2t <sup>*1</sup> |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        |            | 1.6 mm & over,<br>up to 3.0 mm |            |                         |                                                            |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        |            | 3.0 mm & over                  |            |                         |                                                            |
|                                                                      | JIS G 3302    | SGHC  | --                   | --              | --              | --             | --             | 270<br>min.                              | 205<br>min.                            | --         | Up to 1.6 mm                   | --         | 180°                    | t <sup>*1</sup><br>t <sup>*1</sup><br><br>2t <sup>*1</sup> |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        |            | 1.6 mm & over,<br>Up to 3.0 mm |            |                         |                                                            |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        |            | 3.0 mm & over                  |            |                         |                                                            |
| Cold Rolled<br>Steel Sheets                                          | JIS G 3141    | SPCC  | 0.12                 | 0.50            | --              | 0.04           | 0.045          | 270<br>min.                              | --                                     | No. 5      | 0.6 mm & over,<br>up to 1.0 mm | 36<br>min. | 180°                    | Flat on<br>itself                                          |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        |            | 1.0 mm & over,<br>up to 1.6 mm | 37<br>min. |                         |                                                            |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        |            | 1.6mm & over,<br>up to 2.5 mm  | 38<br>min. |                         |                                                            |
|                                                                      |               |       |                      |                 |                 |                |                |                                          |                                        |            | 2.5 mm & over                  | 39<br>min. |                         |                                                            |

Notes:

\*1 - Up to grade Z27 maximum

\*2 - Bend Test test piece No. 3

t - Thickness

# PLATES

## CHEQUERED (FLOOR) PLATES

### WEIGHT TABLE

| Thickness | Unit weight       | Width x Length |             |               |               |               |               |               |               |
|-----------|-------------------|----------------|-------------|---------------|---------------|---------------|---------------|---------------|---------------|
|           |                   | (mm)           |             |               |               |               |               |               |               |
| mm        | kg/m <sup>2</sup> | 914 x 1,829    | 914 x 3,658 | 1,219 x 2,438 | 1,219 x 3,048 | 1,219 x 4,877 | 1,219 x 6,096 | 1,524 x 3,048 | 1,524 x 6,096 |
| 2.3       | 19.73             | 33.0           | 66.0        | 58.6          | 73.3          | 117           | 147           | 91.6          | 183           |
| 3.0       | 25.26             | 42.2           | 84.5        | 75.1          | 93.9          | 150.2         | 187.7         | 117.3         | 235           |
| 3.2       | 26.79             | 44.8           | 89.6        | 79.6          | 99.6          | 159           | 199           | 124           | 249           |
| 4.5       | 36.99             | 61.8           | 124         | 110           | 137           | 220           | 275           | 172           | 344           |
| 6.0       | 48.77             | 81.5           | 163         | 145           | 181           | 290           | 362           | 227           | 453           |
| 8.0       | 64.47             | 108            | 216         | 192           | 240           | 383           | 479           | 299           | 599           |
| 9.0       | 72.32             | 121            | 242         | 215           | 269           | 430           | 537           | 336           | 672           |
| 10        | 80.17             | 134            | 268         | 238           | 298           | 477           | 596           | 372           | 745           |
| 12        | 95.87             | 167            | 320         | 285           | 356           | 570           | 712           | 445           | 891           |

### WEIGHT TABLE

| Thickness | Unit weight        | Width x Length |        |       |        |        |        |        |        |
|-----------|--------------------|----------------|--------|-------|--------|--------|--------|--------|--------|
|           |                    | (inch)         |        |       |        |        |        |        |        |
| in        | lb/ft <sup>2</sup> | 3 x 6          | 3 x 12 | 4 x 8 | 4 x 10 | 4 x 16 | 4 x 20 | 5 x 10 | 5 x 20 |
| 0.091     | 4.04               | 72.75          | 145.5  | 129.2 | 161.6  | 257.9  | 324.1  | 201.9  | 403.4  |
| 0.118     | 5.17               | 93.09          | 186.2  | 165.5 | 206.9  | 331.0  | 413.7  | 258.6  | 517.2  |
| 0.126     | 5.49               | 98.77          | 197.5  | 175.5 | 219.6  | 350.5  | 438.7  | 273.4  | 549    |
| 0.177     | 7.58               | 136.2          | 273.4  | 242.5 | 302    | 485    | 606.3  | 379.2  | 758.4  |
| 0.236     | 9.99               | 179.7          | 359.4  | 319.7 | 399    | 639.3  | 798.1  | 500.4  | 998.7  |
| 0.315     | 13.2               | 238.1          | 476.2  | 423.3 | 529.1  | 844.4  | 1,056  | 659.2  | 1,321  |
| 0.354     | 14.8               | 266.8          | 533.5  | 474   | 593.9  | 948    | 1,184  | 740.8  | 1,482  |
| 0.394     | 16.4               | 295.4          | 590.8  | 524.7 | 657    | 1,052  | 1,314  | 820.1  | 1,642  |
| 0.472     | 19.6               | 368.2          | 705.5  | 628.3 | 748.8  | 1,257  | 1,570  | 981.1  | 1,964  |

## COLD ROLLED COILS AND SHEETS - SPECIFICATIONS

| Application                              |            | Standard        |               |      |                      |       |              |                 |
|------------------------------------------|------------|-----------------|---------------|------|----------------------|-------|--------------|-----------------|
|                                          |            | KS              | JIS           | ASTM | BS                   | DIN   |              | POSCO           |
|                                          |            |                 |               |      |                      | 1623  | 1624         |                 |
| Commercial Quality                       |            | SCP 1           | SPCC          | A366 | CR 4<br>CS 4         | St 12 | St 0<br>St 1 | CSP 1<br>CSP 1D |
| Drawing Quality                          |            | SCP 2           | SPCD          | A619 | CR 3<br>CS 3         | St 13 | St 2         | CSP 2           |
| Deep Drawing Quality                     |            | SCP 3<br>SCP 3N | SPCE<br>SPCEN | A620 | CR 2<br>CS 2         | St 14 | St 3<br>St 4 | CSP 3<br>CSP 3N |
| Extra Deep Drawing Quality               |            | --              | --            | --   | CR 1<br>CS 1         | --    | --           | CSP 3E          |
| Structural Quality                       |            | --              | --            | A611 | CR 34/20<br>CR 37/23 | --    | --           | --              |
| Porcelain Enameling                      | Commercial | --              | SPP           | A424 | --                   | --    | --           | CESP 1          |
|                                          | Drawing    |                 |               |      |                      |       |              | CESP 2          |
| Superior Atmospheric Corrosion Resisting |            | --              | SPA-C         | --   | --                   | --    | --           | CASP            |
| High Strength Steel                      |            | --              | --            | --   | --                   | --    | --           | CHSP            |

## GENERAL COLD ROLLED STEELS

| Classification | Standard |       |                 | Characteristics                                                                                                                       | End use                                                                |
|----------------|----------|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
|                | KS       | JIS   | POSCO           |                                                                                                                                       |                                                                        |
| Class 1        | SCP 1    | SPCC  | CSP 1<br>CSP 1D | General steel used for flatting, bending process (CSP 1) and light drawing (CSP 1D)                                                   | Refrigerator, Laundry machine, Other electrical appliances, Drum, etc. |
| Class 2        | SCP 2    | SPCD  | CSP 2           | This quality has a better characteristics of processing for drawing quality                                                           | Drawing parts of automobile door, etc.                                 |
| Class 3        | SCP 3    | SPCE  | CSP 3           | This quality improved characteristics of processing of deep drawing material through metallurgical adjustment of metal grain boundary | Automobile fender, Electrical cable case                               |
|                | SCP 3N   | SPCEN | CSP 3N          | (Ç: Guarantee non-aging characteristics when designated by the buyer)                                                                 |                                                                        |
|                | --       | --    | CSP 3E          | Extra deep drawing steel which has superior characteristics of .drawability than SCP 3N                                               | Automobile rear floor panel, etc.                                      |

# PLATES

## COLD ROLLED STEEL SHEETS

(JIS G3141)

### DIMENSIONS AND WEIGHTS

| For Reference only |        | Thickness | Sizes  | lb/pce   | kg/pce  | Pcs/M. Ton |
|--------------------|--------|-----------|--------|----------|---------|------------|
| Gauge No.          | mm     | mm        | ft     |          |         |            |
| 30 SWG             | 0.315  | 0.3       | 3 x 6  | 8.6821   | 3.9382  | 253.9231   |
| 30 BG              | 0.3124 |           | 3 x 8  | 11.5762  | 5.2509  | 190.4435   |
| 30 USG             | 0.318  |           | 4 x 8  | 15.4349  | 7.0012  | 142.8327   |
| 30 SWG             | 0.300  |           |        |          |         |            |
| 28 BWG             | 0.360  | 0.35      | 3 x 6  | 10.1292  | 4.5946  | 217.6468   |
| 29 SWG             | 0.3454 |           | 3 x 8  | 13.5056  | 6.1261  | 163.2360   |
| 29 USG             | 0.357  |           | 4 x 8  | 18.0075  | 8.1682  | 122.4260   |
| 29 BG              | 0.3531 |           |        |          |         |            |
| 26 BWG             | 0.460  | 0.45      | 3 x 6  | 13.0233  | 5.9074  | 169.2792   |
| 26 SWG             | 0.457  |           | 3 x 8  | 17.3644  | 7.8765  | 126.9600   |
| 27 BG              | 0.4432 |           | 4 x 8  | 23.1525  | 10.5019 | 95.2209    |
|                    |        |           |        |          |         |            |
| 25 BWG             | 0.510  | 0.5       | 3 x 6  | 14.4703  | 6.5637  | 152.3531   |
| 25 SWG             | 0.508  |           | 3 x 8  | 19.2938  | 8.7517  | 114.2636   |
| 26 BG              | 0.4981 |           | 4 x 8  | 25.7250  | 11.6688 | 85.6986    |
|                    |        |           |        |          |         |            |
| 24 BWG             | 0.560  | 0.55      | 3 x 6  | 15.9174  | 7.2201  | 138.5023   |
| 24 SWG             | 0.559  |           | 3 x 8  | 21.2231  | 9.6268  | 103.8767   |
| 25 BG              | 0.5598 |           | 4 x 8  | 28.2975  | 12.8357 | 77.9078    |
|                    |        |           |        |          |         |            |
| 23 SWG             | 0.610  | 0.6       | 3 x 6  | 17.3644  | 7.8765  | 126.9600   |
| 23 BWG             | 0.640  |           | 3 x 8  | 23.1525  | 10.5019 | 95.2209    |
| 24 USG             | 0.635  |           | 4 x 8  | 30.8700  | 14.0026 | 72.5624    |
| 24 BG              | 0.6298 |           |        |          |         |            |
| 22 SWG             | 0.711  | 0.7       | 3 x 6  | 20.2584  | 9.1892  | 108.8234   |
| 22 BWG             | 0.710  |           | 3 x 8  | 27.0112  | 12.2522 | 81.6180    |
| 23 USG             | 0.714  |           | 4 x 8  | 36.0150  | 16.3363 | 61.2134    |
| 23 BG              | 0.7066 |           |        |          |         |            |
| 21 BWG             | 0.810  | 0.8       | 3 x 6  | 23.1525  | 10.5019 | 95.2209    |
| 21 SWG             | 0.813  |           | 3 x 8  | 30.8700  | 14.0026 | 71.4153    |
| 22 USG             | 0.794  |           | 4 x 8  | 41.1600  | 18.6701 | 53.5616    |
|                    |        |           |        |          |         |            |
| 20 BWG             | 0.890  | 0.89      | 3 x 6  | 25.7572  | 11.6834 | 85.5916    |
| 21 BG              | 0.8865 |           | 3 x 8  | 34.3429  | 15.5779 | 64.1935    |
|                    |        |           | 4 x 8  | 45.7905  | 20.7705 | 48.1452    |
|                    |        |           | 4 x 10 | 57.2400  | 25.9639 | 38.5046    |
| 20 SWG             | 0.914  | 0.914     | 3 x 6  | 26.4517  | 11.9985 | 83.3438    |
|                    |        |           | 3 x 8  | 35.2690  | 15.9980 | 62.5079    |
|                    |        |           | 4 x 8  | 47.0253  | 21.3306 | 46.8810    |
|                    |        |           |        |          |         |            |
| 19 BWG             | 1.070  | 1.0       | 3 x 6  | 28.9406  | 13.1274 | 76.1765    |
| 19 SWG             | 1.016  |           | 3 x 8  | 38.5875  | 17.5032 | 57.1325    |
| 20 USG             | 0.952  |           | 4 x 8  | 51.4500  | 23.3376 | 42.8493    |
| 20 BG              | 0.9957 |           |        |          |         |            |
| 18 SWG             | 1.219  | 1.2       | 3 x 6  | 34.7287  | 15.7529 | 63.4804    |
| 18 BG              | 1.257  |           | 3 x 8  | 46.3050  | 21.0039 | 47.6102    |
| 18 BWG             | 1.240  |           | 4 x 8  | 61.7399  | 28.0051 | 35.7078    |
| 18 USG             | 1.270  |           |        |          |         |            |
| 16 USG             | 1.588  | 1.5       | 3 x 6  | 43.4109  | 19.6911 | 50.7844    |
| 17 BWG             | 1.470  |           | 3 x 8  | 57.8812  | 26.2548 | 38.0883    |
|                    |        |           | 4 x 8  | 77.1749  | 35.0063 | 28.5663    |
|                    |        |           |        |          |         |            |
| 14 SWG             | 2.032  | 2.0       | 3 x 6  | 57.8812  | 26.2548 | 38.0883    |
| 14 USG             | 1.984  |           | 3 x 8  | 77.1749  | 35.0063 | 28.5663    |
| 14 BG              | 1.994  |           | 4 x 8  | 102.8999 | 46.6751 | 21.4247    |
|                    |        |           |        |          |         |            |
| 12 SWG             | 2.642  | 2.6       | 3 x 6  | 75.2454  | 34.1311 | 29.2908    |
| 10 BG              | 2.590  |           | 3 x 8  | 100.3272 | 45.5081 | 21.9681    |
|                    |        |           | 4 x 8  | 133.7696 | 60.6775 | 16.4761    |
|                    |        |           |        |          |         |            |

## ELECTROLYTIC GALVANISED COILS AND SHEETS -SPECIFICATIONS

| Application               | Standard |     |               |                 |                   |                 |       |            |       |      |               |
|---------------------------|----------|-----|---------------|-----------------|-------------------|-----------------|-------|------------|-------|------|---------------|
|                           | Pure Zn  |     |               |                 | Fe-Zn             | Ni-Zn           | Pb-Sn | Base Metal |       |      |               |
|                           | KS       | JIS | ASTM          | POSCO           | POSCO Designation |                 |       | KS         | JIS   | ASTM | POSCO         |
| Commercial                | SECC     |     | A591-CQ       | EGSC            | EFSC              | ENSC            | ETSC  | --         | SPCC  | A366 | CSP 1         |
| Drawing                   | SECD     |     | A591-DQ       | EGSD            | EFSD              | ENSD            | ETSD  | SCP 2      | SPCD  | A619 | CSP 2         |
| Deep Drawing              | SECE     |     | A591-DQ<br>SK | EGSE            | EFSP              | ENSP            | ETSP  | SCP 3      | SPCE  | A620 | CSP 3         |
| Non-Aging<br>Deep Drawing | SECEN    |     |               | EGSN            | EFSN              | ENSN            | --    | SCP 3N     | SPCEN | A620 | CSP 3N        |
| Extra Deep Drawing        | --       |     | --            | EGSE            | EFSE              | ENSE            | --    | --         | --    | --   | CSP 3E        |
| High Strength Steel       | --       |     | --            | EGCHSP<br>35/60 | EFCHSP<br>35/60   | ENCHSP<br>35/60 | --    | --         | --    | --   | CHSP<br>35/60 |

## WEIGHT TABLE (Standard Width & Length - 4' X 8')

| Thickness | Weight |
|-----------|--------|
| mm        | kg/sht |
| 0.36      | 8.46   |
| 0.40      | 9.44   |
| 0.45      | 10.62  |
| 0.50      | 11.77  |
| 0.55      | 12.94  |
| 0.60      | 14.11  |
| 0.70      | 16.44  |
| 0.80      | 18.77  |
| 0.85      | 19.94  |
| 0.90      | 21.11  |
| 1.00      | 23.44  |
| 1.20      | 28.11  |
| 1.40      | 32.79  |

| Thickness | Weight |
|-----------|--------|
| mm        | kg/sht |
| 1.50      | 36.127 |
| 1.55      | 36.22  |
| 1.60      | 37.44  |
| 1.80      | 42.12  |
| 1.90      | 44.43  |
| 2.00      | 46.77  |
| 2.30      | 53.77  |
| 2.50      | 58.43  |
| 2.60      | 60.75  |
| 2.80      | 65.43  |
| 2.90      | 67.72  |
| 3.00      | 70.12  |
| 3.20      | 74.72  |

# PLATES

## HOT DIP GALVANISED COILS AND SHEETS - SPECIFICATIONS

| Application                  | Standard |              |      |                      |           |                                          |
|------------------------------|----------|--------------|------|----------------------|-----------|------------------------------------------|
|                              | KS       | JIS          | ASTM | BS                   | DIN 17162 |                                          |
|                              | D3506    | G3302        | A525 | 2989                 | Part 1    | Part 2                                   |
| Commercial Quality           | SBHG 1   | SPGC         | A526 | Z1                   | St 01Z    | --                                       |
| Lock Forming Quality         | SBHG 2   | --           | A527 | Z1                   | St 02Z    | --                                       |
| Drawing Quality              | SBHG 3   | SPGD         | A587 | Z3                   | St 03Z    | --                                       |
| Deep Drawing Quality         | --       | SPGDD        | --   | Z4                   | St 04Z    | --                                       |
| Deep Drawing Special Killed  | --       | --           | A642 | Z5                   | St 05Z    | --                                       |
| Extra Deep Drawing Quality   | --       | --           | --   | --                   | --        | --                                       |
| Roofing & Siding Quality     | --       | SPGR<br>SPGA | A361 | --                   | --        | --                                       |
| Structural Quality           | SBHG4    | SPGS         | A446 | Z25, Z28<br>Z32, Z55 | --        | St E250<br>St E280<br>St E320<br>St E350 |
| Corrugating General Quality  | SBHG 1-W | SPGH         | A361 | BS 3083              | DIN 59231 | --                                       |
| Corrugating Building Quality | --       | SPGW         | --   | --                   | --        | --                                       |
| High Strength Steel          | --       | --           | --   | --                   | --        | --                                       |

## GALVANISED STEEL SHEETS

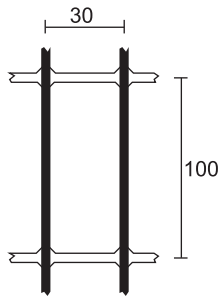
JIS G3302 - 1987 SGCC      JIS G3302 - 1987 SGHC

| Thickness         | Width x Length    |         |        |             |         |        |        |         |        |
|-------------------|-------------------|---------|--------|-------------|---------|--------|--------|---------|--------|
|                   | 4'x 8'            |         |        |             |         |        |        |         |        |
|                   | 1219 mm x 2438 mm |         |        |             |         |        |        |         |        |
|                   | Z 18              |         |        | Z 22 / Z 25 |         |        | Z 27   |         |        |
| Calculated Weight |                   |         |        |             |         |        |        |         |        |
| mm                | kg/pc             | lb/pc   | pcs/mt | kg/pc       | lb/pc   | pcs/mt | kg/pc  | lb/pc   | pcs/mt |
| 0.30              | 7.724             | 17.028  | 129.5  | 7.905       | 17.428  | 126.5  | 8.131  | 17.926  | 123.0  |
| 0.35              | 8.891             | 19.600  | 112.5  | 9.072       | 20.000  | 110.2  | 9.298  | 20.498  | 107.6  |
| 0.36              | 9.124             | 20.114  | 109.6  | 9.305       | 20.514  | 107.5  | 9.531  | 21.012  | 104.9  |
| 0.40              | 10.057            | 22.172  | 99.4   | 10.238      | 22.571  | 97.7   | 10.464 | 23.069  | 95.6   |
| 0.45              | 11.223            | 24.743  | 89.1   | 11.405      | 25.143  | 87.7   | 11.631 | 25.641  | 86.0   |
| 0.50              | 12.390            | 27.315  | 80.7   | 12.571      | 27.715  | 79.5   | 12.797 | 28.212  | 78.1   |
| 0.55              | 13.556            | 29.886  | 73.8   | 13.738      | 30.286  | 72.8   | 13.964 | 30.784  | 71.6   |
| 0.60              | 14.723            | 32.458  | 67.9   | 14.904      | 32.858  | 67.1   | 15.130 | 33.356  | 66.1   |
| 0.65              | 15.889            | 35.030  | 62.9   | 16.071      | 35.429  | 62.2   | 16.297 | 35.927  | 61.4   |
| 0.70              | 17.056            | 37.601  | 58.6   | 17.237      | 38.001  | 58.0   | 17.463 | 38.499  | 57.3   |
| 0.75              | 18.222            | 40.173  | 54.9   | 18.404      | 40.573  | 54.3   | 18.629 | 41.071  | 53.7   |
| 0.80              | 19.389            | 42.745  | 51.6   | 19.570      | 43.144  | 51.1   | 19.796 | 43.642  | 50.5   |
| 0.85              | 20.555            | 45.316  | 48.6   | 20.737      | 45.716  | 48.2   | 20.962 | 46.214  | 47.7   |
| 0.90              | 21.722            | 47.888  | 46.0   | 21.903      | 48.287  | 45.7   | 22.129 | 48.785  | 45.2   |
| 1.00              | 24.055            | 53.031  | 41.6   | 24.236      | 53.431  | 41.3   | 24.462 | 53.929  | 40.9   |
| 1.10              | 26.388            | 58.174  | 37.9   | 26.569      | 58.574  | 37.6   | 26.795 | 59.072  | 37.3   |
| 1.20              | 28.721            | 63.318  | 34.8   | 28.902      | 63.717  | 34.6   | 29.128 | 64.215  | 34.3   |
| 1.40              | 33.387            | 73.604  | 30.0   | 33.568      | 74.004  | 29.8   | 33.794 | 74.502  | 29.6   |
| 1.45              | 34.553            | 76.176  | 28.9   | 34.734      | 76.575  | 28.8   | 34.960 | 77.073  | 28.6   |
| 1.50              | 35.720            | 78.747  | 28.0   | 35.901      | 79.147  | 27.9   | 36.127 | 79.645  | 27.7   |
| 1.55              | 36.886            | 81.319  | 27.1   | 37.067      | 81.719  | 27.0   | 37.293 | 82.217  | 26.8   |
| 1.60              | 38.052            | 83.891  | 26.3   | 38.234      | 84.290  | 26.2   | 38.460 | 84.788  | 26.0   |
| 1.80              | 42.718            | 94.177  | 23.4   | 42.900      | 94.577  | 23.3   | 43.126 | 95.075  | 23.2   |
| 1.90              | 45.051            | 99.320  | 22.2   | 45.233      | 99.720  | 22.1   | 45.459 | 100.218 | 22.0   |
| 2.00              | 47.384            | 104.463 | 21.1   | 47.566      | 104.863 | 21.0   | 47.791 | 105.361 | 20.9   |
| 2.20              | 52.050            | 114.750 | 19.2   | 52.232      | 115.150 | 19.1   | 52.457 | 115.648 | 19.1   |
| 2.30              | 54.383            | 119.893 | 18.4   | 54.564      | 120.293 | 18.3   | 54.790 | 120.791 | 18.3   |
| 2.50              | 59.049            | 130.180 | 16.9   | 59.230      | 130.579 | 16.9   | 59.456 | 131.077 | 16.8   |
| 2.60              | 61.382            | 135.323 | 16.3   | 61.563      | 135.723 | 16.2   | 61.789 | 136.221 | 16.2   |
| 2.80              | 66.048            | 145.609 | 15.1   | 66.229      | 146.009 | 15.1   | 66.455 | 146.507 | 15.0   |
| 2.90              | 68.381            | 150.753 | 14.6   | 68.562      | 151.152 | 14.6   | 68.788 | 151.650 | 14.5   |
| 3.00              | 70.714            | 155.896 | 14.1   | 70.895      | 156.296 | 14.1   | 71.121 | 156.794 | 14.1   |



# GALVANISED SERRATED GRATINGS

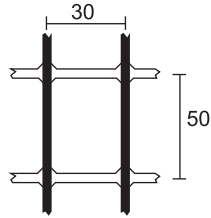
## STANDARD TYPES



Series 3/1

Load factor 1.26

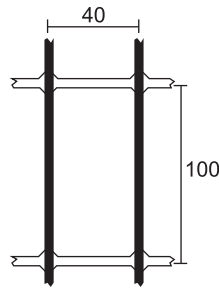
B/bars/M 34



Series 3/5

Load factor 1.26

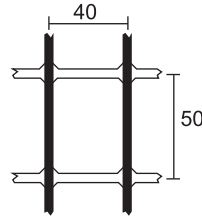
B/bars/M 34



Series 4/1

Load factor 1.0

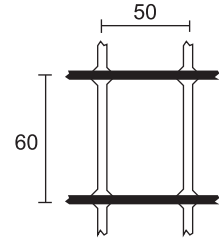
B/bars/M 26



Series 4/5

Load factor 1.0

B/bars/M 26



Series 6/5

Load factor 0.63

B/bars/M 17

## GRATINGS LOADING TABLE "4/1" SERIES - based on type "4/1" plan (factor 1)

| B/bar    | B/bar<br>type | Serrate<br>factor<br>"Z" | Banded<br>mass/<br>kg/m <sup>2</sup> | Load | Span in mm |       |       |        |       |      |      |      |      |      |        | Max.<br>recomm<br>Span |
|----------|---------------|--------------------------|--------------------------------------|------|------------|-------|-------|--------|-------|------|------|------|------|------|--------|------------------------|
|          |               |                          |                                      |      |            |       | Heavy | Medium | Light |      |      |      |      |      |        |                        |
|          |               |                          |                                      |      | 500        | 750   | 900   | 1000   | 1250  | 1500 | 1800 | 2100 | 2400 | 2750 | Class  |                        |
| 25 x 3   | 1             | 0.84                     | 20                                   | U    | 4224       | 1770  | 1024  | 746    | 382   | 221  |      |      |      |      | Light  | 1250                   |
|          |               |                          |                                      | d    | 1.76       | 3.75  | 4.5   | 5      | 6.25  | 7.5  |      |      |      |      | Medium | 1100                   |
|          |               |                          |                                      | C    | 1055       | 703   | 575   | 466    | 296   | 207  |      |      |      |      | Heavy  | 1000                   |
|          |               |                          |                                      | d    | 1.42       | 3.19  | 4.5   | 5      | 6.25  | 7.5  |      |      |      |      |        |                        |
| 25 x 4.5 | 2             | 0.84                     | 28                                   | U    | 6336       | 2656  | 1536  | 1119   | 572   | 332  |      |      |      |      | Light  | 1500                   |
|          |               |                          |                                      | d    | 1.76       | 3.75  | 4.5   | 5      | 6.25  | 7.5  |      |      |      |      | Medium | 1250                   |
|          |               |                          |                                      | C    | 1583       | 1055  | 863   | 699    | 447   | 310  |      |      |      |      | Heavy  | 1100                   |
|          |               |                          |                                      | d    | 1.42       | 3.19  | 4.5   | 5      | 6.25  | 7.5  |      |      |      |      |        |                        |
| 30 x 3   | 4             | 0.87                     | 23.1                                 | U    | 6082       | 2703  | 1751  | 1276   | 653   | 378  | 218  | 131  |      |      | Light  | 1500                   |
|          |               |                          |                                      | d    | 1.49       | 3.35  | 4.5   | 5      | 6.25  | 7.5  | 9    | 10   |      |      | Medium | 1300                   |
|          |               |                          |                                      | C    | 1520       | 1014  | 844   | 760    | 510   | 354  | 245  | 172  |      |      | Heavy  | 1200                   |
|          |               |                          |                                      | d    | 1.19       | 2.68  | 3.86  | 4.77   | 6.25  | 7.5  | 9    | 10   |      |      |        |                        |
| 30 x 4.5 | 5             | 0.87                     | 32.7                                 | U    | 9124       | 4055  | 2627  | 1915   | 980   | 567  | 327  | 197  |      |      | Light  | 1800                   |
|          |               |                          |                                      | d    | 1.49       | 3.35  | 4.5   | 5      | 6.25  | 7.5  | 9    | 10   |      |      | Medium | 1500                   |
|          |               |                          |                                      | C    | 2281       | 1521  | 1267  | 1140   | 765   | 531  | 368  | 258  |      |      | Heavy  | 1300                   |
|          |               |                          |                                      | d    | 1.19       | 2.68  | 3.86  | 4.77   | 6.25  | 7.5  | 9    | 10   |      |      |        |                        |
| 40 x 3   | 7             | 0.9                      | 29.5                                 | U    | 10813      | 4806  | 3337  | 2703   | 1544  | 893  | 517  | 310  | 181  | 113  | Light  | 2100                   |
|          |               |                          |                                      | d    | 1.12       | 2.52  | 3.63  | 4.48   | 6.25  | 7.5  | 9    | 10   | 10   | 10   | Medium | 1800                   |
|          |               |                          |                                      | C    | 2703       | 1802  | 1502  | 1351   | 1081  | 837  | 581  | 407  | 272  | 191  | Heavy  | 1500                   |
|          |               |                          |                                      | d    | 0.9        | 2.02  | 2.91  | 3.59   | 5.61  | 7.5  | 9    | 10   | 10   | 10   |        |                        |
| 40 x 4.5 | 8             | 0.9                      | 42.2                                 | U    | 16222      | 7209  | 5006  | 4055   | 2316  | 1340 | 775  | 464  | 272  | 170  | Light  | 2350                   |
|          |               |                          |                                      | d    | 1.12       | 2.52  | 3.63  | 4.48   | 6.25  | 7.5  | 9    | 10   | 10   | 10   | Medium | 2000                   |
|          |               |                          |                                      | C    | 4055       | 2703  | 2253  | 2027   | 1633  | 1255 | 872  | 610  | 408  | 286  | Heavy  | 1800                   |
|          |               |                          |                                      | d    | 0.9        | 2.02  | 2.91  | 3.59   | 5.61  | 7.5  | 9    | 10   | 10   | 10   |        |                        |
| 50 x 5   | 11            | 0.92                     | 57.2                                 | U    | 28162      | 12516 | 8692  | 7040   | 4506  | 2925 | 1691 | 1014 | 594  | 370  | Light  | 2750                   |
|          |               |                          |                                      | d    | 0.89       | 2     | 2.89  | 3.56   | 5.57  | 7.5  | 9    | 10   | 10   | 10   | Medium | 2400                   |
|          |               |                          |                                      | C    | 7040       | 4694  | 3911  | 3520   | 2816  | 2347 | 1903 | 1332 | 892  | 626  | Heavy  | 2200                   |
|          |               |                          |                                      | d    | 0.71       | 1.61  | 2.31  | 2.85   | 4.46  | 6.42 | 9    | 10   | 10   | 10   |        |                        |

U=kg/m<sup>2</sup> UDL      C=kg/m width line load      d= Deflection at centre

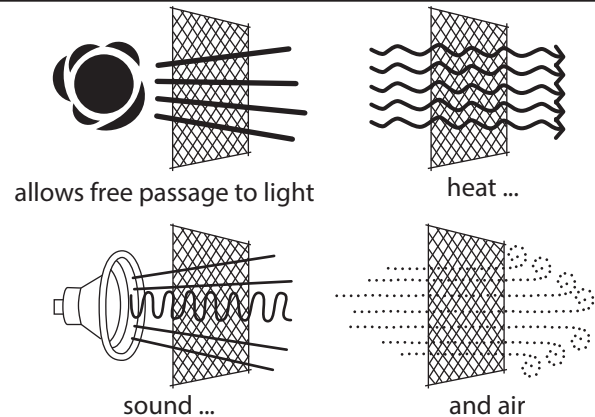
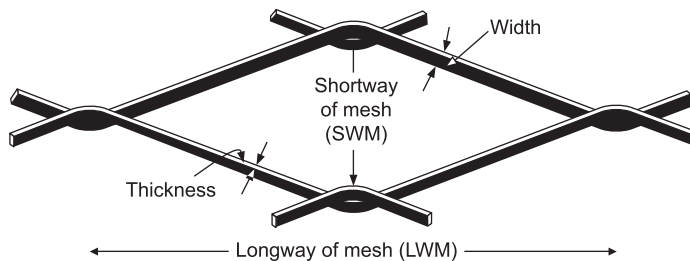
Note: Galvanizing will increase masses/m<sup>2</sup> by 9% on average

Deflection criteria : 1/200 of span or 10mm whichever is the lesser

Bending stress criteria : 165 MPa

# EXPANDED METAL

## PROPERTIES

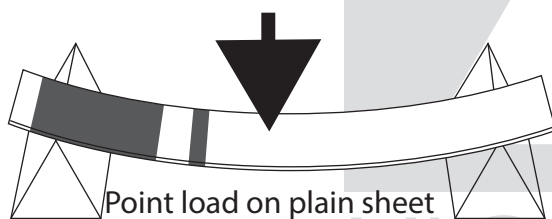


The main feature of Expanded Metal is that the mesh is formed from a single piece of metal. A metal sheet, plate or foil is passed through an expanding machine having a serrated knife which shears and stretches the metal into strands but leaves them joined at the knuckles.

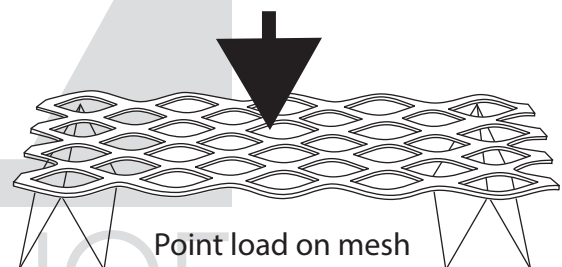
This high strength-to-weight ratio material is technically functional in many protective and structural applications. The jointless expanded metal ensures that every strand will perform as a structural member, distributing the load over a wide area.

### Advantages

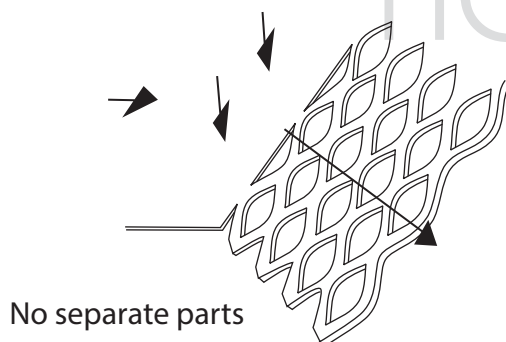
1. No separate parts. Nothing to work loose, no fretting strands. Stays together under stress, even when cut into irregular shapes.
2. The strands are angled to the plane of the complete sheet of mesh in a truss-like formation which ensures excellent load bearing properties.
3. Protection. The range includes meshes small enough to exclude even the smallest fingers where moving parts are within finger length of the guards.
4. Screening effect - especially in anti-dazzle fences.
5. Reinforcement. The meshes anchor immovably and the angle strand formation further strengthens the reinforced material.
6. Provides anti-slip surface e.g. catwalk, stairtreads and ramps.



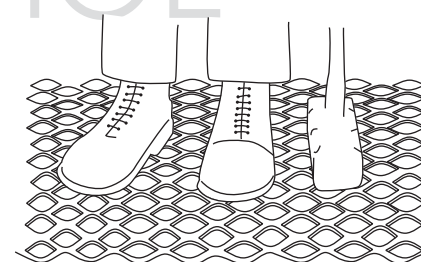
Point load on plain sheet



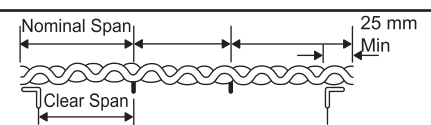
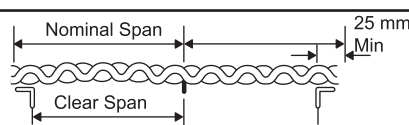
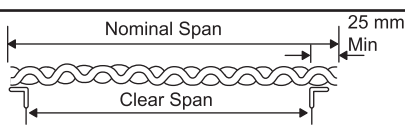
Point load on mesh



No separate parts



Anti-slip



### QUICK SELECTION TABLE FOR WALKWAYS

| LOAD                               | NOMINAL SPAN OF MESH - SIMPLY SUPPORTED |                      |                   |
|------------------------------------|-----------------------------------------|----------------------|-------------------|
|                                    | 600 mm                                  | 750 mm               | 900 mm            |
| Distributed - 2.5 kPa              | 1000 - G1201 - 1502                     | 1000 - G 1201        | 1000 (3.8) - 1501 |
| Distributed - 5.0 kPa              | 1000 - G 1201 - 1502                    | 1000 (3.8)           | -                 |
| Concentrated - 1.1 kN @ 1.2 m CTRS | 1000, 1502, 1000 (3.8)                  | 1000 - G 1201 - 1501 | 1000 (3.8) - 1501 |

\* These meshes whilst safe for the spans shown, may be springy, underfoot.

# EXPANDED METAL

## SPECIFICATIONS

| B.S No. | Code No. | Material | Nominal Size of Mesh |     | Thickness | Strand Width | Approx. Nominal Wt. kg/m <sup>2</sup> ± 10% |
|---------|----------|----------|----------------------|-----|-----------|--------------|---------------------------------------------|
|         |          |          | SWM                  | LWM |           |              |                                             |
|         |          |          | mm                   | mm  |           | mm           | mm                                          |
| XM 21   | 3001     | STEEL    | 76                   | 200 | 6.0       | 12.5         | 16.5                                        |
| XM 22   | 3002     | STEEL    |                      |     | 6.0       | 11.0         | 14.5                                        |
| XM 23   | 3003     | STEEL    |                      |     | 4.5       | 12.0         | 12.4                                        |
| XM 24   | 3004     | STEEL    |                      |     | 6.0       | 7.5          | 10.4                                        |
| XM 25   | 3005     | STEEL    |                      |     | 4.5       | 9.0          | 9.3                                         |
| XM 26   | 3006     | STEEL    |                      |     | 6.0       | 6.0          | 8.3                                         |
| XM 27   | 3007     | STEEL    |                      |     | 4.5       | 7.0          | 7.0                                         |
| XM 28   | 3008     | STEEL    |                      |     | 4.5       | 6.0          | 6.2                                         |
| XM 29   | 3009     | STEEL    |                      |     | 3.0       | 7.5          | 5.1                                         |
| XM 30   | 3010     | STEEL    |                      |     | 4.5       | 4.5          | 4.6                                         |
| XM 31   | 3011     | STEEL    |                      |     | 3.0       | 6.0          | 4.1                                         |
| XM 33   | 3013     | STEEL    |                      |     | 3.0       | 3.0          | 2.0                                         |
| XM 41   | 1501     | STEEL    | 38                   | 114 | 4.5       | 9.5          | 18.0                                        |
| XM 42   | 1502     | STEEL    |                      |     | 4.5       | 7.5          | 15.5                                        |
| XM 43   | 1503     | STEEL    |                      |     | 4.5       | 6.0          | 12.4                                        |
| XM 44   | 1504     | STEEL    |                      |     | 3.0       | 6.0          | 8.3                                         |
| XM 45   | 1505     | STEEL    |                      |     | 3.0       | 4.5          | 6.2                                         |
| XM 46   | 1506     | STEEL    |                      |     | 3.0       | 3.0          | 4.1                                         |
| XM 47   | 1507     | STEEL    |                      |     | 1.5       | 6.0          | 4.1                                         |
| XM 48   | 1508     | STEEL    |                      |     | 1.5       | 4.5          | 3.1                                         |
| XM 49   | 1509     | STEEL    |                      |     | 1.5       | 3.0          | 2.0                                         |
| XM 61   | 1001     | STEEL    | 25                   | 76  | 3.0       | 4.5          | 8.1                                         |
| XM 63   | 1003     | STEEL    |                      |     | 3.0       | 3.0          | 5.4                                         |
| XM 65   | 1005     | STEEL    |                      |     | 1.5       | 4.5          | 4.0                                         |
| XM 66   | 1006     | STEEL    |                      |     | 1.5       | 3.0          | 2.7                                         |
| XM 71   | 7502     | STEEL    | 19                   | 50  | 3.0       | 4.5          | 10.4                                        |
| XM 72   | 7503     | STEEL    |                      |     | 3.0       | 3.0          | 7.1                                         |
| XM 73   | 7504     | STEEL    |                      |     | 1.5       | 6.0          | 6.1                                         |
| XM 75   | 7506     | STEEL    |                      |     | 1.5       | 3.0          | 3.5                                         |
| XM 76   | 7507     | STEEL    |                      |     | 1.5       | 2.3          | 2.7                                         |

| Code       | Nominal size of mesh (mm) |        | Dimensions of strands (mm) |       | Approx. nominal weight kg/m <sup>2</sup> ± 1 0% |
|------------|---------------------------|--------|----------------------------|-------|-------------------------------------------------|
|            | S.W.M.                    | L.W.M. | Thickness                  | Width |                                                 |
| G 1020     | 16                        | 38     | 1.2                        | 2.0   | 1.9                                             |
| G 1032     | 25                        | 40     | 3.0                        | 3.0   | 5.4                                             |
| G 1201     | 32                        | 114    | 4.5                        | 7.5   | 13.1                                            |
| G 1232     | 33                        | 102    | 3.0                        | 3.0   | 4.5                                             |
| G 2263     | 65                        | 185    | 3.0                        | 6.0   | 3.2                                             |
| G 3092     | 38                        | 76     | 3.0                        | 4.5   | 4.5                                             |
| G 5011     | 45                        | 135    | 4.5                        | 11.0  | 15.0                                            |
| 1000       | 30                        | 76     | 4.5                        | 6.0   | 15.7                                            |
| 1000 (3.8) | 30                        | 76     | 4.5                        | 10.0  | 22.1                                            |
| ½ X18G     | 13                        | 28     | 1.2                        | 1.5   | 2.3                                             |

Tolerances :

Reference to sheet size, strand width, thickness and weights are approximate only. Whilst every effort is made for these figures to be accurate, it must be understood that the specification of goods manufactured and supplied by the company is subject to normal trade tolerances.

The company policy is one of continuous development : we therefore reserve the right to alter specifications, etc without notice.



YICK HOE

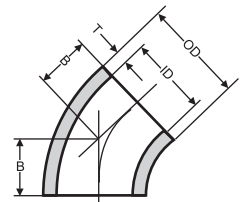
# WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)

## DIMENSIONS OF FITTING ENDS

| Nominal<br>Pipe<br>Size | Outside<br>Diameter<br>OD | Inside Diameter<br>ID |       |       |       | Wall Thickness<br>T |      |      |      |
|-------------------------|---------------------------|-----------------------|-------|-------|-------|---------------------|------|------|------|
|                         |                           | STD                   | S-40  | XS    | S-80  | STD                 | S-40 | XS   | S-80 |
| ½                       | 21.3                      | 15.7                  | 15.7  | 13.9  | 13.9  | 2.8                 | 2.8  | 3.7  | 3.7  |
| ¾                       | 26.7                      | 20.9                  | 20.9  | 18.9  | 18.9  | 2.9                 | 2.9  | 3.9  | 3.9  |
| 1                       | 33.4                      | 26.6                  | 26.6  | 24.4  | 24.4  | 3.4                 | 3.4  | 4.5  | 4.5  |
| 1¼                      | 42.2                      | 35.0                  | 35.0  | 32.4  | 32.4  | 3.6                 | 3.6  | 4.9  | 4.9  |
| 1½                      | 48.3                      | 40.9                  | 40.9  | 38.1  | 38.1  | 3.7                 | 3.7  | 5.1  | 5.1  |
| 2                       | 60.3                      | 52.5                  | 52.5  | 49.3  | 49.3  | 3.9                 | 3.9  | 5.5  | 5.5  |
| 2½                      | 73.0                      | 62.6                  | 62.6  | 59.0  | 59.0  | 5.2                 | 5.2  | 7.0  | 7.0  |
| 3                       | 88.9                      | 77.9                  | 77.9  | 73.7  | 73.7  | 5.5                 | 5.5  | 7.6  | 7.6  |
| 3½                      | 101.6                     | 90.2                  | 90.2  | 85.4  | 85.4  | 5.7                 | 5.7  | 8.1  | 8.1  |
| 4                       | 114.3                     | 102.3                 | 102.3 | 97.1  | 97.1  | 6.0                 | 6.0  | 8.6  | 8.6  |
| 5                       | 141.3                     | 128.1                 | 128.1 | 122.3 | 122.3 | 6.6                 | 6.6  | 9.5  | 9.5  |
| 6                       | 168.3                     | 154.1                 | 154.1 | 146.3 | 146.3 | 7.1                 | 7.1  | 11.0 | 11.0 |
| 8                       | 219.1                     | 202.7                 | 202.7 | 193.7 | 193.7 | 8.2                 | 8.2  | 12.7 | 12.7 |
| 10                      | 273.1                     | 254.5                 | 254.5 | 247.7 | 242.9 | 9.3                 | 9.3  | 12.7 | 15.1 |
| 12                      | 323.9                     | 304.9                 | 303.3 | 298.5 | 289.1 | 9.5                 | 10.3 | 12.7 | 17.4 |
| 14                      | 355.6                     | 336.6                 | 333.4 | 330.2 | 317.6 | 9.5                 | 11.1 | 12.7 | 19.0 |
| 16                      | 406.4                     | 387.4                 | 381.0 | 381.0 | 363.6 | 9.5                 | 12.7 | 12.7 | 21.4 |
| 18                      | 457.2                     | 438.2                 | 428.6 | 431.8 | 409.6 | 9.5                 | 14.3 | 12.7 | 23.8 |

Note: Dimensions of outside diameter and wall thickness conform to ANSI B 36.10



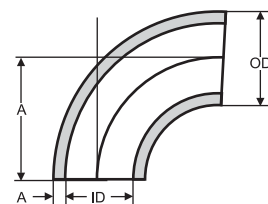
## 45° ELBOWS

ANSI B 16.9

| Nominal<br>Pipe<br>Size | Outside<br>Diameter<br>OD | STD   |     | XS    |      | Centre to Face<br>B | Approx. Weight<br>kg |        |
|-------------------------|---------------------------|-------|-----|-------|------|---------------------|----------------------|--------|
|                         |                           | ID    | T   | ID    | T    |                     | STD                  | XS     |
| ½                       | 21.3                      | 15.7  | 2.8 | 13.9  | 3.7  | 15.9                | 0.0379               | 0.0485 |
| ¾                       | 26.7                      | 20.9  | 2.9 | 18.9  | 3.9  | 11.1                | 0.0378               | 0.0494 |
| 1                       | 33.4                      | 26.6  | 3.4 | 24.4  | 4.5  | 22.2                | 0.0748               | 0.0968 |
| 1¼                      | 42.2                      | 35.0  | 3.6 | 32.4  | 4.9  | 25.4                | 0.127                | 0.167  |
| 1½                      | 48.3                      | 40.9  | 3.7 | 38.1  | 5.1  | 28.6                | 0.182                | 0.243  |
| 2                       | 60.3                      | 52.5  | 3.9 | 49.3  | 5.5  | 34.9                | 0.326                | 0.448  |
| 2½                      | 73.0                      | 62.6  | 5.2 | 59.0  | 7.0  | 44.5                | 0.646                | 0.854  |
| 3                       | 88.9                      | 77.9  | 5.5 | 73.7  | 7.6  | 50.8                | 1.01                 | 1.37   |
| 3½                      | 101.6                     | 90.2  | 5.7 | 85.4  | 8.1  | 57.2                | 1.42                 | 1.95   |
| 4                       | 114.3                     | 102.3 | 6.0 | 97.1  | 8.6  | 63.5                | 1.92                 | 2.67   |
| 5                       | 141.3                     | 128.1 | 6.6 | 122.3 | 9.5  | 79.4                | 3.26                 | 4.63   |
| 6                       | 168.3                     | 154.1 | 7.1 | 146.3 | 11.0 | 95.3                | 5.08                 | 7.64   |
| 8                       | 219.1                     | 202.7 | 8.2 | 193.7 | 12.7 | 127.0               | 10.2                 | 15.5   |
| 10                      | 273.1                     | 254.5 | 9.3 | 247.7 | 12.7 | 158.8               | 18.1                 | 24.4   |
| 12                      | 323.9                     | 304.9 | 9.5 | 298.5 | 12.7 | 190.5               | 26.5                 | 35.0   |
| 14                      | 355.6                     | 336.6 | 9.5 | 330.2 | 12.7 | 222.3               | 34.1                 | 45.0   |
| 16                      | 406.4                     | 387.4 | 9.5 | 381.0 | 12.7 | 254.0               | 44.6                 | 59.0   |
| 18                      | 457.2                     | 438.2 | 9.5 | 431.8 | 12.7 | 285.8               | 56.6                 | 74.9   |

# WROUGHT STEEL BUTT-WELDING FITTINGS

## Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)

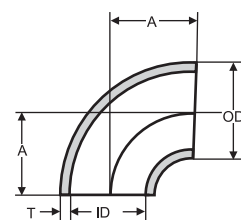


### 90° ELBOWS Long Radius

#### ANSI B 16.9

| Nominal<br>Pipe<br>Size | Outside<br>Diameter<br>OD | STD   |     | XS    |      | Centre to Face<br>A | Approx. Weight<br>kg |        |
|-------------------------|---------------------------|-------|-----|-------|------|---------------------|----------------------|--------|
|                         |                           | ID    | T   | ID    | T    |                     | STD                  | XS     |
| 1/2                     | 21.3                      | 15.7  | 2.8 | 13.9  | 3.7  | 38.1                | 0.0759               | 0.0971 |
| 3/4                     | 26.7                      | 20.9  | 2.9 | 18.9  | 3.9  | 28.6                | 0.0756               | 0.0987 |
| 1                       | 33.4                      | 26.6  | 3.4 | 24.4  | 4.5  | 38.1                | 0.15                 | 0.194  |
| 1 1/4                   | 42.2                      | 35.0  | 3.6 | 32.4  | 4.9  | 47.6                | 0.253                | 0.334  |
| 1 1/2                   | 48.3                      | 40.9  | 3.7 | 38.1  | 5.1  | 57.2                | 0.364                | 0.486  |
| 2                       | 60.3                      | 52.5  | 3.9 | 49.3  | 5.5  | 76.2                | 0.652                | 0.895  |
| 2 1/2                   | 73.0                      | 62.6  | 5.2 | 59.0  | 7.0  | 95.3                | 1.29                 | 1.71   |
| 3                       | 88.9                      | 77.9  | 5.5 | 73.7  | 7.6  | 114.3               | 2.03                 | 2.74   |
| 3 1/2                   | 101.6                     | 90.2  | 5.7 | 85.4  | 8.1  | 133.4               | 2.84                 | 3.90   |
| 4                       | 114.3                     | 102.3 | 6.0 | 97.1  | 8.6  | 152.4               | 3.85                 | 5.34   |
| 5                       | 141.3                     | 128.1 | 6.6 | 122.3 | 9.5  | 190.5               | 6.52                 | 9.26   |
| 6                       | 168.3                     | 154.1 | 7.1 | 146.3 | 11.0 | 228.6               | 10.2                 | 15.3   |
| 8                       | 219.1                     | 202.7 | 8.2 | 193.7 | 12.7 | 304.8               | 20.4                 | 30.9   |
| 10                      | 273.1                     | 254.5 | 9.3 | 247.7 | 12.7 | 381.0               | 36.1                 | 48.8   |
| 12                      | 323.9                     | 304.9 | 9.5 | 298.5 | 12.7 | 457.2               | 53.0                 | 69.9   |
| 14                      | 355.6                     | 336.6 | 9.5 | 330.2 | 12.7 | 533.4               | 68.1                 | 89.9   |
| 16                      | 406.4                     | 387.4 | 9.5 | 381.0 | 12.7 | 609.6               | 89.3                 | 118.0  |
| 18                      | 457.2                     | 438.2 | 9.5 | 431.8 | 12.7 | 685.8               | 113.0                | 150.0  |

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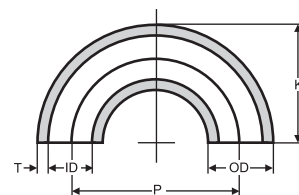
### 90° ELBOWS Short Radius

#### ANSI B 16.9

| Nominal<br>Pipe<br>Size | Outside<br>Diameter<br>OD | STD   |     | XS    |      | Centre to Face<br>A | Approx. Weight |       |
|-------------------------|---------------------------|-------|-----|-------|------|---------------------|----------------|-------|
|                         |                           | ID    | T   | ID    | T    |                     | STD            | XS    |
| 1                       | 33.4                      | 26.6  | 3.4 | 24.4  | 4.5  | 25.4                | 0.0998         | 0.129 |
| 1 1/4                   | 42.2                      | 35.0  | 3.6 | 32.4  | 4.9  | 31.8                | 0.169          | 0.223 |
| 1 1/2                   | 48.3                      | 40.9  | 3.7 | 38.1  | 5.1  | 38.1                | 0.243          | 0.324 |
| 2                       | 60.3                      | 52.5  | 3.9 | 49.3  | 5.5  | 50.8                | 0.435          | 0.597 |
| 2 1/2                   | 73.0                      | 62.6  | 5.2 | 59.0  | 7.0  | 63.5                | 0.861          | 1.14  |
| 3                       | 88.9                      | 77.9  | 5.5 | 73.7  | 7.6  | 76.2                | 1.35           | 1.83  |
| 3 1/2                   | 101.6                     | 90.2  | 5.7 | 85.4  | 8.1  | 88.9                | 1.89           | 2.60  |
| 4                       | 114.3                     | 102.3 | 6.0 | 97.1  | 8.6  | 101.6               | 2.56           | 3.56  |
| 5                       | 141.3                     | 128.1 | 6.6 | 122.3 | 9.5  | 127.0               | 4.35           | 6.17  |
| 6                       | 168.3                     | 154.1 | 7.1 | 146.3 | 11.0 | 152.4               | 6.77           | 10.2  |
| 8                       | 219.1                     | 202.7 | 8.2 | 193.7 | 12.7 | 203.2               | 13.6           | 20.6  |
| 10                      | 273.1                     | 254.5 | 9.3 | 247.7 | 12.7 | 254.0               | 24.1           | 32.5  |
| 12                      | 323.9                     | 304.9 | 9.5 | 298.5 | 12.7 | 304.8               | 35.3           | 46.6  |
| 14                      | 355.6                     | 336.6 | 9.5 | 330.2 | 12.7 | 355.6               | 45.4           | 60.0  |
| 16                      | 406.4                     | 387.4 | 9.5 | 381.0 | 12.7 | 406.4               | 59.5           | 78.7  |
| 18                      | 457.2                     | 438.2 | 9.5 | 431.8 | 12.7 | 457.2               | 75.5           | 99.9  |

# WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)

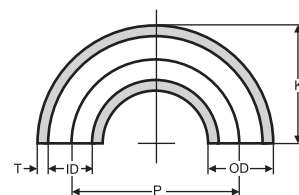


180° RETURNS Long Radius

ANSI B 16.9

| Nominal<br>Pipe<br>Size | Outside<br>Diameter<br>OD | STD   |     | XS    |      | Centre to<br>centre<br>P | Centre to<br>Face<br>K | Approx. Weight<br>kg |       |
|-------------------------|---------------------------|-------|-----|-------|------|--------------------------|------------------------|----------------------|-------|
|                         |                           | ID    | T   | ID    | T    |                          |                        | STD                  | XS    |
| ½                       | 21.3                      | 15.7  | 2.8 | --    | --   | 76.2                     | 47.6                   | 0.152                | --    |
| ¾                       | 26.7                      | 20.9  | 2.9 | 18.9  | 3.9  | 57.2                     | 42.9                   | 0.151                | 0.197 |
| 1                       | 33.4                      | 26.6  | 3.4 | 24.4  | 4.5  | 76.2                     | 55.6                   | 0.299                | 0.387 |
| 1¼                      | 42.2                      | 35.0  | 3.6 | 32.4  | 4.9  | 95.3                     | 69.9                   | 0.506                | 0.668 |
| 1½                      | 48.3                      | 40.9  | 3.7 | 38.1  | 5.1  | 114.3                    | 82.6                   | 0.728                | 0.972 |
| 2                       | 60.3                      | 52.5  | 3.9 | 49.3  | 5.5  | 152.4                    | 106.4                  | 1.30                 | 1.79  |
| 2½                      | 73.0                      | 62.6  | 5.2 | 59.0  | 7.0  | 190.5                    | 131.8                  | 2.58                 | 3.42  |
| 3                       | 88.9                      | 77.9  | 5.5 | 73.7  | 7.6  | 228.6                    | 158.8                  | 4.05                 | 5.48  |
| 3½                      | 101.6                     | 90.2  | 5.7 | 85.4  | 8.1  | 266.7                    | 184.2                  | 5.68                 | 7.81  |
| 4                       | 114.3                     | 102.3 | 6.0 | 97.1  | 8.6  | 304.8                    | 209.6                  | 7.69                 | 10.7  |
| 5                       | 141.3                     | 128.1 | 6.6 | 122.3 | 9.5  | 381.0                    | 261.9                  | 13.0                 | 18.5  |
| 6                       | 168.3                     | 154.1 | 7.1 | 146.3 | 11.0 | 457.2                    | 312.7                  | 20.3                 | 30.6  |
| 8                       | 219.1                     | 202.7 | 8.2 | 193.7 | 12.7 | 609.6                    | 414.3                  | 40.7                 | 61.9  |
| 10                      | 273.1                     | 254.5 | 9.3 | 247.7 | 12.7 | 762.0                    | 517.5                  | 72.2                 | 97.5  |
| 12                      | 323.9                     | 304.9 | 9.5 | 298.5 | 12.7 | 914.4                    | 619.1                  | 106.0                | 140.0 |

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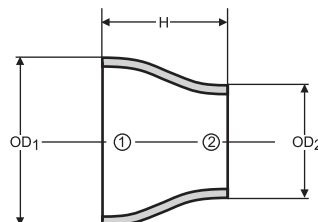
180° RETURNS Short Radius

ANSI B 16.9

| Nominal<br>Pipe<br>Size | Outside<br>Diameter<br>OD | STD   |     | XS    |      | Centre to<br>centre<br>P | Centre to<br>Face<br>K | Approx. Weight<br>kg |       |
|-------------------------|---------------------------|-------|-----|-------|------|--------------------------|------------------------|----------------------|-------|
|                         |                           | ID    | T   | ID    | T    |                          |                        | STD                  | XS    |
| 1                       | 33.4                      | 26.6  | 3.4 | 24.4  | 4.5  | 50.8                     | 41.3                   | 0.200                | 0.258 |
| 1¼                      | 42.2                      | 35.0  | 3.6 | 32.4  | 4.9  | 63.5                     | 52.4                   | 0.338                | 0.446 |
| 1½                      | 48.3                      | 40.9  | 3.7 | 38.1  | 5.1  | 76.2                     | 61.9                   | 0.485                | 0.647 |
| 2                       | 60.3                      | 52.5  | 3.9 | 49.3  | 5.5  | 101.6                    | 81.0                   | 0.87                 | 1.19  |
| 2½                      | 73.0                      | 62.6  | 5.2 | 59.0  | 7.0  | 127.0                    | 100.0                  | 1.72                 | 2.28  |
| 3                       | 88.9                      | 77.9  | 5.5 | 73.7  | 7.6  | 152.4                    | 120.7                  | 2.70                 | 3.65  |
| 3½                      | 101.6                     | 90.2  | 5.7 | 85.4  | 8.1  | 177.8                    | 139.7                  | 3.79                 | 5.20  |
| 4                       | 114.3                     | 102.3 | 6.0 | 97.1  | 8.6  | 203.2                    | 158.8                  | 5.13                 | 7.12  |
| 5                       | 141.3                     | 128.1 | 6.6 | 122.3 | 9.5  | 254.0                    | 196.9                  | 8.69                 | 12.3  |
| 6                       | 168.3                     | 154.1 | 7.1 | 146.3 | 11.0 | 304.8                    | 236.5                  | 13.5                 | 20.4  |
| 8                       | 219.1                     | 202.7 | 8.2 | 193.7 | 12.7 | 406.4                    | 312.7                  | 27.1                 | 41.2  |
| 10                      | 273.1                     | 254.5 | 9.3 | 247.7 | 12.7 | 508.0                    | 390.5                  | 48.1                 | 65.0  |
| 12                      | 323.9                     | 304.9 | 9.5 | 298.5 | 12.7 | 609.6                    | 466.7                  | 70.7                 | 93.3  |

# WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)



## REDUCERS

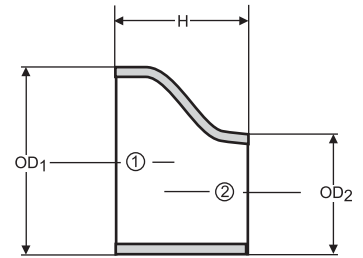
## ANSI B 16.9

| Nominal Pipe Size<br>① x ② | Large End       | Small End       | Length | Approx. Weight<br>kg |        |
|----------------------------|-----------------|-----------------|--------|----------------------|--------|
|                            | OD <sub>1</sub> | OD <sub>2</sub> | H      | STD                  | XS     |
| 1/4 x 1/2                  | 26.7            | 21.3            | 38.1   | 0.0571               | 0.0741 |
| 1 x 3/4                    | 33.4            | 26.7            | 50.8   | 0.113                | 0.145  |
| 1 x 1/2                    |                 | 21.3            |        | 0.102                | 0.131  |
| 1 1/4 x 1                  |                 | 33.4            |        | 0.153                | 0.201  |
| 1 1/4 x 3/4                | 42.2            | 26.7            | 50.8   | 0.139                | 0.182  |
| 1 1/4 x 1/2                |                 | 21.3            |        | 0.128                | 0.167  |
| 1 1/2 x 1 1/4              |                 | 42.2            |        | 0.240                | 0.319  |
| 1 1/2 x 1                  | 48.3            | 33.4            | 63.5   | 0.216                | 0.286  |
| 1 1/2 x 3/4                |                 | 26.7            |        | 0.198                | 0.261  |
| 1 1/2 x 1/2                |                 | 21.3            |        | 0.184                | 0.241  |
| 2 x 1 1/2                  |                 | 48.3            |        | 0.372                | 0.509  |
| 2 x 1 1/4                  | 60.3            | 42.2            | 76.2   | 0.351                | 0.479  |
| 2 x 1                      |                 | 33.4            |        | 0.321                | 0.437  |
| 2 x 3/4                    |                 | 26.7            |        | 0.298                | 0.404  |
| 2 1/2 x 2                  |                 | 60.3            |        | 0.697                | 0.919  |
| 2 1/2 x 1 1/2              | 73.0            | 48.3            | 88.9   | 0.633                | 0.832  |
| 2 1/2 x 1 1/4              |                 | 42.2            |        | 0.602                | 0.788  |
| 2 1/2 x 1                  |                 | 33.4            |        | 0.556                | 0.727  |
| 3 x 2 1/2                  |                 | 73.0            |        | 0.911                | 1.23   |
| 3 x 2                      | 88.9            | 60.3            | 88.9   | 0.842                | 1.13   |
| 3 x 1 1/2                  |                 | 48.3            |        | 0.778                | 1.04   |
| 3 x 1 1/4                  |                 | 42.2            |        | 0.747                | 0.999  |
| 3 1/2 x 3                  |                 | 88.9            |        | 1.29                 | 1.77   |
| 3 1/2 x 2 1/2              |                 | 73.0            |        | 1.18                 | 1.62   |
| 3 1/2 x 2                  | 101.6           | 60.3            | 101.6  | 1.10                 | 1.50   |
| 3 1/2 x 1 1/2              |                 | 48.3            |        | 1.03                 | 1.40   |
| 3 1/2 x 1 1/4              |                 | 42.2            |        | 0.984                | 1.34   |
| 4 x 3 1/2                  |                 | 101.6           |        | 1.54                 | 2.13   |
| 4 x 3                      |                 | 88.9            |        | 1.45                 | 2.01   |
| 4 x 2 1/2                  | 114.3           | 73.0            | 101.6  | 1.35                 | 1.86   |
| 4 x 2                      |                 | 60.3            |        | 1.27                 | 1.75   |
| 4 x 1 1/2                  |                 | 48.3            |        | 1.19                 | 1.64   |



# WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)



## REDUCERS

ANSI B 16.9

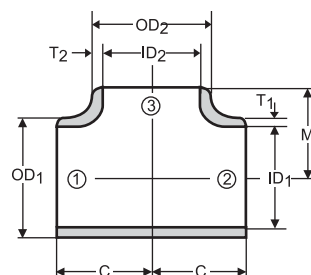
| Nominal Pipe Size<br>①x ② | Large End       | Small End       | Length | Approx. Weight<br>kg |      |
|---------------------------|-----------------|-----------------|--------|----------------------|------|
|                           | OD <sub>1</sub> | OD <sub>2</sub> | H      | STD                  | XS   |
| 5 x 4                     | 141.3           | 114.3           | 127.0  | 2.50                 | 3.55 |
| 5 x 3½                    |                 | 101.6           |        | 2.39                 | 3.38 |
| 5 x 3                     |                 | 88.9            |        | 2.28                 | 3.21 |
| 5 x 2½                    |                 | 73.0            |        | 2.14                 | 3.02 |
| 5 x 2                     |                 | 60.3            |        | 2.03                 | 2.86 |
| 6 x 5                     | 168.3           | 141.3           | 139.7  | 3.64                 | 5.46 |
| 6 x 4                     |                 | 114.3           |        | 3.35                 | 5.02 |
| 6 x 3½                    |                 | 101.6           |        | 3.22                 | 4.82 |
| 6 x 3                     |                 | 88.3            |        | 3.09                 | 4.62 |
| 6 x 2½                    |                 | 73.0            |        | 2.94                 | 4.38 |
| 8 x 6                     | 219.1           | 168.3           | 152.4  | 5.78                 | 8.75 |
| 8 x 5                     |                 | 141.3           |        | 5.45                 | 8.24 |
| 8 x 4                     |                 | 114.3           |        | 5.15                 | 7.77 |
| 8 x 3½                    |                 | 101.6           |        | 5.01                 | 7.55 |
| 10 x 8                    |                 | 219.1           |        | 9.74                 | 13.1 |
| 10 x 6                    | 273.1           | 168.3           | 177.8  | 8.96                 | 12.1 |
| 10 x 5                    |                 | 141.3           |        | 8.58                 | 11.3 |
| 10 x 4                    |                 | 114.3           |        | 8.21                 | 11.0 |
| 12 x 10                   |                 | 273.1           |        | 13.9                 | 18.3 |
| 12 x 8                    | 323.9           | 219.1           | 203.2  | 12.9                 | 17.0 |
| 12 x 6                    |                 | 168.3           |        | 12.1                 | 15.9 |
| 12 x 5                    |                 | 141.3           |        | 11.6                 | 15.3 |
| 14 x 12                   |                 | 323.9           |        | 25.5                 | 33.8 |
| 14 x 10                   | 355.6           | 273.1           | 330.2  | 23.7                 | 31.4 |
| 14 x 8                    |                 | 219.1           |        | 21.9                 | 27.7 |
| 16 x 14                   |                 | 355.6           |        | 31.0                 | 41.1 |
| 16 x 12                   | 406.4           | 323.9           | 355.6  | 29.8                 | 39.5 |
| 16 x 10                   |                 | 273.1           |        | 28.0                 | 37.0 |
| 18 x 16                   |                 | 406.4           |        | 37.7                 | 50.1 |
| 18 x 14                   | 457.2           | 355.6           | 381.0  | 35.7                 | 47.4 |
| 18 x 12                   |                 | 323.9           |        | 34.5                 | 45.7 |

Note: (1) Wall thickness and inside diameter conform to ANSI B 36.10.

(2) Shows available sizes of eccentric reducer.

# WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)



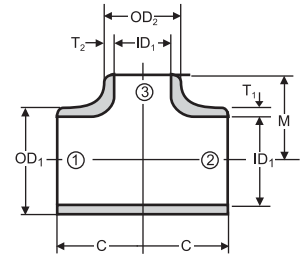
TEES

ANSI B 16.9

| Nominal Pipe<br>Size<br>①x ②x ③ | Outside<br>Diameter |                 | Center<br>to End |      | STD             |                |                 |                | XS              |                |                 |                | Approx.<br>Weight (kg) |      |
|---------------------------------|---------------------|-----------------|------------------|------|-----------------|----------------|-----------------|----------------|-----------------|----------------|-----------------|----------------|------------------------|------|
|                                 | OD <sub>1</sub>     | OD <sub>2</sub> | C                | M    | ID <sub>1</sub> | T <sub>1</sub> | ID <sub>2</sub> | T <sub>2</sub> | ID <sub>1</sub> | T <sub>1</sub> | ID <sub>2</sub> | T <sub>2</sub> | STD                    | XS   |
| ½ x ½ x ½                       | 21.3                | 21.3            | 25.4             | 25.4 | 15.7            | 2.8            | 15.7            | 2.8            | 13.9            | 3.7            | 13.9            | 3.7            | 0.11                   | 0.16 |
| ¾ x ¾ x ¾                       | 26.7                | 26.7            | 28.6             | 28.6 | 20.9            | 2.9            | 20.9            | 2.9            | 18.9            | 3.9            | 18.9            | 3.9            | 0.17                   | 0.21 |
| ¾ x ½                           | 26.7                | 21.3            | 28.6             | 28.6 | 20.9            | 2.9            | 15.7            | 2.8            | 18.9            | 3.9            | 13.9            | 3.7            | 0.15                   | 0.19 |
| 1x1x1                           | 33.4                | 33.4            | 38.1             | 38.1 | 26.6            | 3.4            | 26.6            | 3.4            | 24.4            | 4.5            | 24.4            | 4.5            | 0.35                   | 0.42 |
| 1x ¾                            | 33.4                | 26.7            | 38.1             | 38.1 | 26.6            | 3.4            | 20.9            | 2.9            | 24.4            | 4.5            | 18.9            | 3.9            | 0.30                   | 0.36 |
| 1x ½                            | 33.4                | 21.3            | 38.1             | 38.1 | 26.6            | 3.4            | 15.7            | 2.8            | 24.4            | 4.5            | 13.9            | 3.7            | 0.29                   | 0.36 |
| 1¼ x 1¼ x 1¼                    | 42.2                | 42.2            | 47.6             | 47.6 | 35.0            | 3.6            | 35.0            | 3.6            | 32.4            | 4.9            | 32.4            | 4.9            | 0.61                   | 0.75 |
| 1¼ x 1                          | 42.2                | 33.4            | 47.6             | 47.6 | 35.0            | 3.6            | 26.6            | 3.4            | 32.4            | 4.9            | 24.4            | 4.5            | 0.53                   | 0.66 |
| 1¼ x ¾                          | 42.2                | 26.7            | 47.6             | 47.6 | 35.0            | 3.6            | 20.9            | 2.9            | 32.4            | 4.9            | 18.9            | 3.9            | 0.50                   | 0.63 |
| 1¼ x ½                          | 42.2                | 21.3            | 47.6             | 47.6 | 35.0            | 3.6            | 15.7            | 2.8            | 32.4            | 4.9            | 13.9            | 3.7            | 0.49                   | 0.61 |
| 1½ x 1½ x 1½                    | 48.3                | 48.3            | 57.2             | 57.2 | 40.9            | 3.7            | 40.9            | 3.7            | 38.1            | 5.1            | 38.1            | 5.1            | 0.92                   | 1.13 |
| 1½ x 1¼                         | 48.3                | 42.2            | 57.2             | 57.2 | 40.9            | 3.7            | 35.0            | 3.6            | 38.1            | 5.1            | 32.4            | 4.9            | 0.80                   | 1.00 |
| 1½ x 1                          | 48.3                | 33.4            | 57.2             | 57.2 | 40.9            | 3.7            | 26.6            | 3.4            | 38.1            | 5.1            | 24.4            | 4.5            | 0.76                   | 0.95 |
| 1½ x ¾                          | 48.3                | 26.7            | 57.2             | 57.2 | 40.9            | 3.7            | 20.9            | 2.9            | 38.1            | 5.1            | 18.9            | 3.9            | 0.73                   | 0.93 |
| 1½ x ½                          | 48.3                | 21.3            | 57.2             | 57.2 | 40.9            | 3.7            | 15.7            | 2.8            | 38.1            | 5.1            | 13.9            | 3.7            | 0.72                   | 0.90 |
| 2 x 2 x 2                       | 60.3                | 60.3            | 63.5             | 63.5 | 52.5            | 3.9            | 52.5            | 3.9            | 49.3            | 5.5            | 49.3            | 5.5            | 1.34                   | 1.70 |
| 2 x 1½                          | 60.3                | 48.3            | 63.5             | 60.3 | 52.5            | 3.9            | 40.9            | 3.7            | 49.3            | 5.5            | 38.1            | 5.1            | 1.21                   | 1.54 |
| 2 x 1¼                          | 60.3                | 42.2            | 63.5             | 57.2 | 52.5            | 3.9            | 35.0            | 3.6            | 49.3            | 5.5            | 32.4            | 4.9            | 1.12                   | 1.48 |
| 2 x 1                           | 60.3                | 33.4            | 63.5             | 50.8 | 52.5            | 3.9            | 26.6            | 3.4            | 49.3            | 5.5            | 24.4            | 4.5            | 1.11                   | 1.44 |
| 2 x ¾                           | 60.3                | 26.7            | 63.5             | 44.5 | 52.5            | 3.9            | 20.9            | 2.9            | 49.3            | 5.5            | 18.9            | 3.9            | 1.10                   | 1.12 |
| 2½ x 2½ x 2½                    | 73.0                | 73.0            | 76.2             | 76.2 | 62.6            | 5.2            | 62.6            | 5.2            | 59.0            | 7.0            | 59.0            | 7.0            | 2.37                   | 2.93 |
| 2½ x 2                          | 73.0                | 60.3            | 76.2             | 69.9 | 62.6            | 5.2            | 52.5            | 3.9            | 59.0            | 7.0            | 49.3            | 5.5            | 2.08                   | 2.62 |
| 2½ x 1½                         | 73.0                | 48.3            | 76.2             | 66.7 | 62.6            | 5.2            | 40.9            | 3.7            | 59.0            | 7.0            | 38.1            | 5.1            | 2.05                   | 2.57 |
| 2½ x 1¼                         | 73.0                | 42.2            | 76.2             | 63.5 | 62.6            | 5.2            | 35.0            | 3.6            | 59.0            | 7.0            | 32.4            | 4.9            | 2.00                   | 2.52 |
| 3 x 3 x 3                       | 88.9                | 88.9            | 85.7             | 85.7 | 77.9            | 5.5            | 77.9            | 5.5            | 73.7            | 7.6            | 73.7            | 7.6            | 3.38                   | 4.29 |
| 3 x 2½                          | 88.9                | 73.0            | 85.7             | 82.6 | 77.9            | 5.5            | 62.6            | 5.2            | 73.7            | 7.6            | 59.0            | 7.0            | 3.09                   | 3.91 |
| 3 x 2                           | 88.9                | 60.3            | 85.7             | 76.2 | 77.9            | 5.5            | 52.5            | 3.9            | 73.7            | 7.6            | 49.3            | 5.5            | 2.96                   | 3.79 |
| 3 x 1½                          | 88.9                | 48.3            | 85.7             | 73.0 | 77.9            | 5.5            | 40.9            | 3.7            | 73.7            | 7.6            | 38.1            | 5.1            | 2.93                   | 3.77 |
| 3 x 1¼                          | 88.9                | 42.2            | 85.7             | 69.9 | 77.9            | 5.5            | 35.0            | 3.6            | 73.7            | 7.6            | 32.4            | 4.9            | 2.88                   | 3.71 |
| 3½ x 3½ x 3½                    | 101.6               | 101.6           | 95.3             | 95.3 | 90.2            | 5.7            | 90.2            | 5.7            | 85.4            | 8.1            | 85.4            | 8.1            | 4.48                   | 5.73 |
| 3½ x 3                          | 101.6               | 88.9            | 95.3             | 92.1 | 90.2            | 5.7            | 77.9            | 5.5            | 85.4            | 8.1            | 73.7            | 7.6            | 4.10                   | 5.32 |
| 3½ x 2½                         | 101.6               | 73.0            | 95.3             | 88.9 | 90.2            | 5.7            | 62.6            | 5.2            | 85.4            | 8.1            | 59.0            | 7.0            | 4.00                   | 5.23 |
| 3½ x 2                          | 101.6               | 60.3            | 95.3             | 82.6 | 90.2            | 5.7            | 52.5            | 3.9            | 85.4            | 8.1            | 49.3            | 5.5            | 3.86                   | 5.10 |
| 3½ x 1½                         | 101.6               | 48.3            | 95.3             | 79.4 | 90.2            | 5.7            | 40.9            | 3.7            | 85.4            | 8.1            | 38.1            | 5.1            | 3.86                   | 5.04 |

# WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)

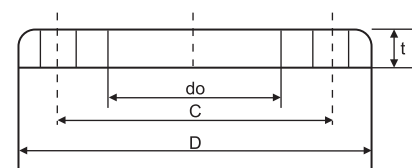


## REDUCING

## ANSI B 16.9

| Nominal Pipe<br>Size<br>①x ②x ③ | Outside<br>Diameter |                 | Center<br>to End |       | STD             |                |                 |                | XS              |                |                 |                | Approx.     |      |
|---------------------------------|---------------------|-----------------|------------------|-------|-----------------|----------------|-----------------|----------------|-----------------|----------------|-----------------|----------------|-------------|------|
|                                 |                     |                 |                  |       | Run             |                | Branch          |                | Run             |                | Branch          |                | Weight (kg) |      |
|                                 | OD <sub>1</sub>     | OD <sub>2</sub> | C                | M     | ID <sub>1</sub> | T <sub>1</sub> | ID <sub>2</sub> | T <sub>2</sub> | ID <sub>1</sub> | T <sub>1</sub> | ID <sub>2</sub> | T <sub>2</sub> | STD         | XS   |
| 4 x 4 x 4                       | 114.3               | 114.3           | 104.8            | 104.8 | 102.3           | 6.0            | 102.3           | 6.0            | 97.1            | 8.6            | 97.1            | 8.6            | 5.73        | 7.47 |
| x 3½                            | 114.3               | 101.6           | 104.8            | 101.6 | 102.3           | 6.0            | 90.2            | 5.7            | 97.1            | 8.6            | 85.4            | 8.1            | 5.28        | 6.96 |
| x 3                             | 114.3               | 88.9            | 104.8            | 98.4  | 102.3           | 6.0            | 77.9            | 5.5            | 97.1            | 8.6            | 73.7            | 7.6            | 5.18        | 6.82 |
| x 2½                            | 114.3               | 73.0            | 104.8            | 95.3  | 102.3           | 6.0            | 62.6            | 5.2            | 97.1            | 8.6            | 59.0            | 7.0            | 5.13        | 6.68 |
| x 2                             | 114.3               | 60.3            | 104.8            | 88.9  | 102.3           | 6.0            | 52.5            | 3.9            | 97.1            | 8.6            | 49.3            | 5.5            | 5.00        | 6.59 |
| x 1½                            | 114.3               | 48.3            | 104.8            | 85.7  | 102.3           | 6.0            | 40.9            | 3.7            | 97.1            | 8.6            | 38.1            | 5.1            | 4.96        | 6.54 |
| 5 x 5 x 5                       | 141.3               | 141.3           | 123.8            | 123.8 | 128.1           | 6.6            | 128.1           | 6.6            | 122.3           | 9.5            | 122.3           | 9.5            | 9.00        | 12.0 |
| x 4                             | 141.3               | 114.3           | 123.8            | 117.5 | 128.1           | 6.6            | 102.3           | 6.0            | 122.3           | 9.5            | 97.1            | 8.6            | 8.23        | 11.0 |
| x 3½                            | 141.3               | 101.6           | 123.8            | 114.3 | 128.1           | 6.6            | 90.2            | 5.7            | 122.3           | 9.5            | 85.4            | 8.1            | 8.10        | 10.9 |
| x 3                             | 141.3               | 88.9            | 123.8            | 111.1 | 128.1           | 6.6            | 77.9            | 5.5            | 122.3           | 9.5            | 73.7            | 7.6            | 8.00        | 10.8 |
| x 2½                            | 141.3               | 73.0            | 123.8            | 108.0 | 128.1           | 6.6            | 62.6            | 5.2            | 122.3           | 9.5            | 59.0            | 7.0            | 7.90        | 10.7 |
| x 2                             | 141.3               | 60.3            | 123.8            | 104.8 | 128.1           | 6.6            | 52.5            | 3.9            | 122.3           | 9.5            | 49.3            | 5.5            | 7.80        | 10.6 |
| 6 x 6 x 6                       | 168.3               | 168.3           | 142.9            | 142.9 | 154.1           | 7.1            | 154.1           | 7.1            | 146.3           | 11.0           | 146.3           | 11.0           | 13.3        | 19.1 |
| x 5                             | 168.3               | 141.3           | 142.9            | 136.5 | 154.1           | 7.1            | 128.1           | 6.6            | 146.3           | 11.0           | 122.3           | 9.5            | 12.1        | 17.2 |
| x 4                             | 168.3               | 114.3           | 142.9            | 130.2 | 154.1           | 7.1            | 102.3           | 6.0            | 146.3           | 11.0           | 97.1            | 8.6            | 11.9        | 16.9 |
| x 3½                            | 168.3               | 101.6           | 142.9            | 127.0 | 154.1           | 7.1            | 90.2            | 5.7            | 146.3           | 11.0           | 85.4            | 8.1            | 11.8        | 16.9 |
| x 3                             | 168.3               | 88.9            | 142.9            | 123.0 | 154.1           | 7.1            | 77.9            | 5.5            | 146.3           | 11.0           | 73.7            | 7.6            | 11.8        | 16.8 |
| x 2½                            | 168.3               | 73.0            | 142.9            | 120.7 | 154.1           | 7.1            | 62.6            | 5.2            | 146.3           | 11.0           | 59.0            | 7.0            | 11.7        | 16.8 |
| 8 x 8 x 8                       | 219.1               | 219.1           | 177.8            | 177.8 | 202.7           | 8.2            | 202.7           | 8.2            | 193.7           | 12.7           | 193.7           | 12.7           | 24.4        | 34.8 |
| x 6                             | 219.1               | 168.3           | 177.8            | 168.3 | 202.7           | 8.2            | 154.1           | 7.1            | 193.7           | 12.7           | 122.3           | 9.5            | 22.1        | 32.1 |
| x 5                             | 219.1               | 141.3           | 177.8            | 161.9 | 202.7           | 8.2            | 128.1           | 6.6            | 193.7           | 12.7           | 97.1            | 8.6            | 21.8        | 31.6 |
| x 4                             | 219.1               | 114.3           | 177.8            | 155.6 | 202.7           | 8.2            | 102.3           | 6.0            | 193.7           | 12.7           | 85.4            | 8.1            | 21.8        | 31.5 |
| x 3½                            | 219.1               | 101.6           | 177.8            | 152.4 | 202.7           | 8.2            | 90.2            | 5.7            | 193.7           | 12.7           | 73.7            | 7.6            | 21.7        | 31.4 |
| 10 x 10 x 10                    | 273.1               | 273.1           | 215.9            | 215.9 | 254.5           | 9.3            | 254.5           | 9.3            | 247.7           | 12.7           | 247.7           | 12.7           | 41.4        | 55.9 |
| x 8                             | 273.1               | 219.1           | 215.9            | 203.2 | 254.5           | 9.3            | 202.7           | 8.2            | 247.7           | 12.7           | 193.7           | 12.7           | 37.7        | 52.9 |
| x 6                             | 273.1               | 168.3           | 215.9            | 193.7 | 254.5           | 9.3            | 154.1           | 7.1            | 247.7           | 12.7           | 122.3           | 9.5            | 37.1        | 50.4 |
| x 5                             | 273.1               | 141.3           | 215.9            | 190.5 | 254.5           | 9.3            | 128.1           | 6.6            | 247.7           | 12.7           | 97.1            | 8.6            | 37.0        | 49.3 |
| 12 x 12 x 12                    | 323.9               | 323.9           | 254.0            | 254.0 | 304.9           | 9.5            | 304.9           | 9.5            | 298.5           | 12.7           | 298.5           | 12.7           | 60.0        | 78.7 |
| x 10                            | 323.9               | 273.1           | 254.0            | 241.3 | 304.9           | 9.5            | 254.5           | 9.3            | 298.5           | 12.7           | 247.7           | 12.7           | 55.0        | 75.2 |

## JIS FLANGES (JIS B2220)



### JIS FLANGES (5K)

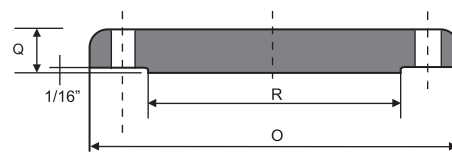
| Nominal Dia.<br>of Flange |     | Outer<br>Dia. of<br>Pipe | Inner<br>Dia. of<br>Flange<br>do | Outer<br>Dia. of<br>Flange<br>D | Thickness<br>of<br>Flange<br>t | Hole of Bolt        |                 |           | Size<br>of<br>Bolt | Weight<br>kg |
|---------------------------|-----|--------------------------|----------------------------------|---------------------------------|--------------------------------|---------------------|-----------------|-----------|--------------------|--------------|
|                           |     |                          |                                  |                                 |                                | Distance<br>of Hole | No. of<br>Holes | Bore<br>C |                    |              |
| in                        | mm  |                          |                                  |                                 |                                |                     |                 |           |                    |              |
| 1/2                       | 15  | 21.7                     | 22.2                             | 80                              | 9                              | 60                  | 4               | 12        | M10                | 0.30         |
| 3/4                       | 20  | 27.2                     | 27.7                             | 85                              | 10                             | 65                  | 4               | 12        | M10                | 0.37         |
| 1                         | 25  | 34.0                     | 34.5                             | 95                              | 10                             | 75                  | 4               | 12        | M10                | 0.45         |
| 1 1/4                     | 32  | 42.7                     | 43.2                             | 115                             | 12                             | 90                  | 4               | 15        | M12                | 0.78         |
| 1 1/2                     | 40  | 48.6                     | 49.1                             | 120                             | 12                             | 95                  | 4               | 15        | M12                | 0.83         |
| 2                         | 50  | 60.5                     | 61.1                             | 130                             | 14                             | 105                 | 4               | 15        | M12                | 1.07         |
| 2 1/2                     | 65  | 76.3                     | 77.1                             | 155                             | 14                             | 130                 | 4               | 15        | M12                | 1.49         |
| 3                         | 80  | 89.1                     | 90.0                             | 180                             | 14                             | 145                 | 4               | 19        | M16                | 1.99         |
| 3 1/2                     | 90  | 101.6                    | 102.6                            | 190                             | 14                             | 155                 | 4               | 19        | M16                | 2.08         |
| 4                         | 100 | 114.3                    | 115.4                            | 200                             | 16                             | 165                 | 8               | 19        | M16                | 2.39         |
| 5                         | 125 | 139.8                    | 141.2                            | 235                             | 16                             | 200                 | 8               | 19        | M16                | 3.23         |
| 6                         | 150 | 165.2                    | 166.6                            | 265                             | 18                             | 230                 | 8               | 19        | M16                | 4.41         |
| 8                         | 200 | 216.3                    | 218.0                            | 320                             | 20                             | 280                 | 8               | 23        | M20                | 6.33         |
| 10                        | 250 | 267.4                    | 269.5                            | 385                             | 22                             | 345                 | 12              | 23        | M20                | 9.45         |
| 12                        | 300 | 318.5                    | 321.0                            | 430                             | 22                             | 390                 | 12              | 23        | M20                | 10.30        |
| 14                        | 350 | 355.6                    | 358.1                            | 480                             | 24                             | 435                 | 12              | 25        | M22                | 14.00        |
| 16                        | 400 | 406.4                    | 409.0                            | 540                             | 24                             | 495                 | 16              | 25        | M22                | 16.90        |
| 18                        | 450 | 457.2                    | 460.0                            | 605                             | 24                             | 555                 | 16              | 25        | M22                | 21.36        |
| 20                        | 500 | 508.0                    | 511.0                            | 665                             | 24                             | 605                 | 20              | 25        | M22                | 23.50        |
| 22                        | 550 | 558.8                    | 562.0                            | 720                             | 26                             | 665                 | 20              | 27        | M24                | 33.40        |
| 24                        | 600 | 609.6                    | 613.0                            | 770                             | 26                             | 715                 | 20              | 27        | M24                | 36.90        |

### JIS FLANGES (10K)

| Nominal Dia.<br>of Flange |     | Outer<br>Dia. of<br>Pipe | Inner<br>Dia. of<br>Flange<br>do | Outer<br>Dia. of<br>Flange<br>D | Thickness<br>of<br>Flange<br>t | Hole of Bolt        |                 |           | Size<br>of<br>Bolt | Weight<br>kg |
|---------------------------|-----|--------------------------|----------------------------------|---------------------------------|--------------------------------|---------------------|-----------------|-----------|--------------------|--------------|
|                           |     |                          |                                  |                                 |                                | Distance<br>of Hole | No. of<br>Holes | Bore<br>C |                    |              |
| in                        | mm  |                          |                                  |                                 |                                |                     |                 |           |                    |              |
| 1/2                       | 15  | 21.7                     | 22.2                             | 95                              | 12                             | 70                  | 4               | 15        | M12                | 0.57         |
| 3/4                       | 20  | 27.2                     | 27.7                             | 100                             | 14                             | 75                  | 4               | 15        | M12                | 0.73         |
| 1                         | 25  | 34.0                     | 34.5                             | 125                             | 14                             | 90                  | 4               | 19        | M16                | 1.13         |
| 1 1/4                     | 32  | 42.7                     | 43.2                             | 135                             | 16                             | 100                 | 4               | 19        | M16                | 1.48         |
| 1 1/2                     | 40  | 48.6                     | 49.1                             | 140                             | 16                             | 105                 | 4               | 19        | M16                | 1.56         |
| 2                         | 50  | 60.5                     | 61.1                             | 155                             | 16                             | 120                 | 4               | 19        | M16                | 1.88         |
| 2 1/2                     | 65  | 76.3                     | 77.1                             | 175                             | 18                             | 140                 | 4               | 19        | M16                | 2.60         |
| 3                         | 80  | 89.1                     | 90.0                             | 185                             | 18                             | 150                 | 8               | 19        | M16                | 2.61         |
| 3 1/2                     | 90  | 101.6                    | 102.6                            | 195                             | 18                             | 160                 | 8               | 19        | M16                | 2.76         |
| 4                         | 100 | 114.3                    | 115.4                            | 210                             | 18                             | 175                 | 8               | 19        | M16                | 3.14         |
| 5                         | 125 | 139.8                    | 141.2                            | 250                             | 20                             | 210                 | 8               | 23        | M20                | 4.77         |
| 6                         | 150 | 165.2                    | 166.6                            | 280                             | 22                             | 240                 | 8               | 23        | M20                | 6.34         |
| 8                         | 200 | 216.3                    | 218.0                            | 330                             | 22                             | 290                 | 12              | 23        | M20                | 7.53         |
| 10                        | 250 | 267.4                    | 269.5                            | 400                             | 24                             | 355                 | 12              | 25        | M22                | 11.80        |
| 12                        | 300 | 318.5                    | 321.0                            | 445                             | 26                             | 400                 | 16              | 25        | M22                | 13.60        |
| 14                        | 350 | 355.6                    | 358.1                            | 490                             | 26                             | 445                 | 16              | 25        | M22                | 17.20        |
| 16                        | 400 | 406.4                    | 409.0                            | 560                             | 28                             | 510                 | 16              | 27        | M24                | 24.00        |
| 18                        | 450 | 457.2                    | 460.0                            | 620                             | 30                             | 565                 | 20              | 27        | M24                | 32.00        |
| 20                        | 500 | 508.0                    | 511.0                            | 675                             | 30                             | 620                 | 20              | 27        | M24                | 35.00        |
| 22                        | 550 | 558.8                    | 562.0                            | 745                             | 32                             | 680                 | 20              | 33        | M30                | 47.00        |
| 24                        | 600 | 609.6                    | 613.0                            | 795                             | 32                             | 730                 | 24              | 33        | M30                | 49.00        |

# FLANGES

## ANSI FLANGES (ANSI B16.5)

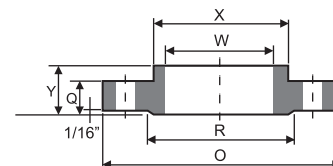


### 150LB. BLIND FLANGES

| Nominal<br>Pipe<br>Size | Outside<br>Dia. of<br>Flange | Thickness<br>of<br>Flange | Dia. of<br>Raised<br>Face | Drilling Template |                  |                        | Approx.<br>Weight |
|-------------------------|------------------------------|---------------------------|---------------------------|-------------------|------------------|------------------------|-------------------|
|                         |                              |                           |                           | No. of<br>Holes   | Dia. of<br>Bolts | Dia. of<br>Bolt Circle |                   |
|                         | O                            | Q                         | R                         |                   |                  |                        | kg                |
| 1/2                     | 3 1/2                        | 7/16                      | 1 3/8                     | 4                 | 1/2              | 2 3/8                  | 0.5               |
| 3/4                     | 3 7/8                        | 1/2                       | 1 11/16                   | 4                 | 1/2              | 2 3/4                  | 0.9               |
| 1                       | 4 1/4                        | 9/16                      | 2                         | 4                 | 1/2              | 3 1/8                  | 0.9               |
| 1 1/4                   | 4 5/8                        | 5/8                       | 2 1/2                     | 4                 | 1/2              | 3 1/2                  | 1.4               |
| 1 1/2                   | 5                            | 11/16                     | 2 7/8                     | 4                 | 1/2              | 3 7/8                  | 1.8               |
| 2                       | 6                            | 3/4                       | 3 5/8                     | 4                 | 5/8              | 4 3/4                  | 2.3               |
| 2 1/2                   | 7                            | 7/8                       | 4 1/8                     | 4                 | 5/8              | 5 1/2                  | 3.2               |
| 3                       | 7 1/2                        | 15/16                     | 5                         | 4                 | 5/8              | 6                      | 4.1               |
| 3 1/2                   | 8 1/2                        | 15/16                     | 5 1/2                     | 8                 | 5/8              | 7                      | 5.9               |
| 4                       | 9                            | 15/16                     | 6 3/16                    | 8                 | 5/8              | 7 1/2                  | 7.7               |
| 5                       | 10                           | 15/16                     | 7 5/16                    | 8                 | 3/4              | 8 1/2                  | 9.1               |
| 6                       | 11                           | 1                         | 8 1/2                     | 8                 | 3/4              | 9 1/2                  | 11.8              |
| 8                       | 13 1/2                       | 1 1/8                     | 10 5/8                    | 8                 | 3/4              | 11 3/4                 | 20.4              |
| 10                      | 16                           | 1 3/16                    | 12 3/4                    | 12                | 7/8              | 14 1/4                 | 31.8              |
| 12                      | 19                           | 1 1/4                     | 15                        | 12                | 7/8              | 17                     | 49.9              |
| 14                      | 21                           | 1 3/8                     | 16 1/4                    | 12                | 1                | 18 3/4                 | 63.5              |
| 16                      | 23 1/2                       | 1 7/16                    | 18 1/2                    | 16                | 1                | 21 1/4                 | 81.6              |
| 18                      | 25                           | 1 9/16                    | 21                        | 16                | 1 1/8            | 22 3/4                 | 99.8              |
| 20                      | 27 1/2                       | 1 11/16                   | 23                        | 20                | 1 1/8            | 25                     | 129.3             |
| 24                      | 32                           | 1 7/8                     | 27 1/2                    | 20                | 1 1/4            | 29 1/2                 | 195.0             |

### 300LB. BLIND FLANGES

| Nominal<br>Pipe<br>Size | Outside<br>Dia. of<br>Flange | Thickness<br>of<br>Flange | Dia. of<br>Raised<br>Face | Drilling Template |                  |                        | Approx.<br>Weight |
|-------------------------|------------------------------|---------------------------|---------------------------|-------------------|------------------|------------------------|-------------------|
|                         |                              |                           |                           | No. of<br>Holes   | Dia. of<br>Bolts | Dia. of<br>Bolt Circle |                   |
|                         | O                            | Q                         | R                         |                   |                  |                        | kg                |
| 1/2                     | 3 3/4                        | 9/16                      | 1 3/8                     | 4                 | 1/2              | 2 5/8                  | 0.9               |
| 3/4                     | 4 5/8                        | 5/8                       | 1 11/16                   | 4                 | 5/8              | 3 1/4                  | 1.4               |
| 1                       | 4 7/8                        | 11/16                     | 2                         | 4                 | 5/8              | 3 1/2                  | 1.4               |
| 1 1/4                   | 5 1/4                        | 3/4                       | 2 1/2                     | 4                 | 5/8              | 3 7/8                  | 2.0               |
| 1 1/2                   | 6 1/8                        | 13/16                     | 2 7/8                     | 4                 | 3/4              | 4 1/2                  | 2.7               |
| 2                       | 6 1/2                        | 7/8                       | 3 5/8                     | 8                 | 5/8              | 5                      | 3.6               |
| 2 1/2                   | 7 1/2                        | 1                         | 4 1/8                     | 8                 | 3/4              | 5 7/8                  | 5.4               |
| 3                       | 8 1/4                        | 1 1/8                     | 5                         | 8                 | 3/4              | 6 5/8                  | 7.3               |
| 3 1/2                   | 9                            | 1 3/16                    | 5 1/2                     | 8                 | 3/4              | 7 1/4                  | 10.0              |
| 4                       | 10                           | 1 1/4                     | 6 3/16                    | 8                 | 3/4              | 7 7/8                  | 12.3              |
| 5                       | 11                           | 1 3/8                     | 7 5/16                    | 8                 | 3/4              | 9 1/4                  | 17.1              |
| 6                       | 12 1/2                       | 1 7/16                    | 8 1/2                     | 12                | 3/4              | 10 5/8                 | 22.2              |
| 8                       | 15                           | 1 5/8                     | 10 5/8                    | 12                | 7/8              | 13                     | 36.7              |
| 10                      | 17 1/2                       | 1 7/8                     | 12 3/4                    | 16                | 1                | 15 1/4                 | 56.7              |
| 12                      | 20 1/2                       | 2                         | 15                        | 16                | 1 1/8            | 17 3/4                 | 83.9              |
| 14                      | 23                           | 2 1/8                     | 16 1/4                    | 20                | 1 1/8            | 20 1/4                 | 113.4             |
| 16                      | 25 1/2                       | 2 1/4                     | 18 1/2                    | 20                | 1 1/4            | 22 1/2                 | 133.8             |
| 18                      | 28                           | 2 3/8                     | 21                        | 24                | 1 1/4            | 24 3/4                 | 179.2             |
| 20                      | 30 1/2                       | 2 1/2                     | 23                        | 24                | 1 1/4            | 27                     | 229.1             |
| 24                      | 36                           | 2 3/4                     | 27 1/4                    | 24                | 1 1/2            | 32                     | 358.3             |



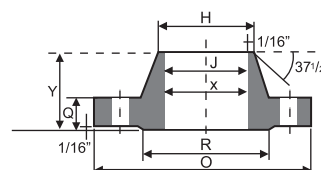
## 150LB. SLIP-ON FLANGES

| Nominal Pipe Size | Outside Dia. of Flange | Thickness of Flange | Dia. of Raised Face | Dia. of Hub at Base | Length thru Hub | Dia. of Bore | Drilling Template |               |                     | Approx. Weight |
|-------------------|------------------------|---------------------|---------------------|---------------------|-----------------|--------------|-------------------|---------------|---------------------|----------------|
|                   |                        |                     |                     |                     |                 |              | No. of Holes      | Dia. of Bolts | Dia. of Bolt Circle |                |
|                   | O                      | Q                   | R                   | X                   | Y               | W            |                   |               |                     | kg             |
| 1/2               | 3 1/2                  | 7/16                | 1 3/8               | 1 3/16              | 5/8             | 0.88         | 4                 | 1/2           | 2 3/8               | 0.4            |
| 3/4               | 3 7/8                  | 1/2                 | 1 11/16             | 1 1/2               | 5/8             | 1.09         | 4                 | 1/2           | 2 3/4               | 0.6            |
| 1                 | 4 1/4                  | 9/16                | 2                   | 1 15/16             | 1 1/16          | 1.36         | 4                 | 1/2           | 3 1/8               | 0.8            |
| 1 1/4             | 4 5/8                  | 5/8                 | 2 1/2               | 2 5/16              | 1 3/16          | 1.70         | 4                 | 1/2           | 3 1/2               | 1.1            |
| 1 1/2             | 5                      | 11/16               | 2 7/8               | 2 9/16              | 7/8             | 1.95         | 4                 | 1/2           | 3 7/8               | 1.4            |
| 2                 | 6                      | 3/4                 | 3 5/8               | 3 1/16              | 1               | 2.44         | 4                 | 5/8           | 4 3/4               | 2.3            |
| 2 1/2             | 7                      | 7/8                 | 4 1/8               | 3 9/16              | 1 1/8           | 2.94         | 4                 | 5/8           | 5 1/2               | 3.5            |
| 3                 | 7 1/2                  | 15/16               | 5                   | 4 1/4               | 1 3/16          | 3.57         | 4                 | 5/8           | 6                   | 4.0            |
| 3 1/2             | 8 1/2                  | 15/16               | 5 1/2               | 4 13/16             | 1 1/4           | 4.07         | 8                 | 5/8           | 7                   | 4.9            |
| 4                 | 9                      | 15/16               | 6 3/16              | 5 5/16              | 1 5/16          | 4.57         | 8                 | 5/8           | 7 1/2               | 5.6            |
| 5                 | 10                     | 15/16               | 7 5/16              | 6 7/16              | 1 7/16          | 5.66         | 8                 | 3/4           | 8 1/2               | 6.8            |
| 6                 | 11                     | 1                   | 8 1/2               | 7 9/16              | 1 9/16          | 6.72         | 8                 | 3/4           | 9 1/2               | 8.5            |
| 8                 | 13 1/2                 | 1 1/8               | 10 5/8              | 9 11/16             | 1 3/4           | 8.72         | 8                 | 3/4           | 11 3/4              | 13.4           |
| 10                | 16                     | 1 3/16              | 12 3/4              | 12                  | 1 15/16         | 10.88        | 12                | 7/8           | 14 1/4              | 19.5           |
| 12                | 19                     | 1 1/4               | 15                  | 14 3/8              | 2 3/16          | 12.88        | 12                | 7/8           | 17                  | 29.0           |
| 14                | 21                     | 1 3/8               | 16 1/4              | 15 3/4              | 2 1/4           | 14.14        | 12                | 1             | 18 3/4              | 38.6           |
| 16                | 23 1/2                 | 1 7/16              | 18 1/2              | 18                  | 2 1/2           | 16.16        | 16                | 1             | 21 1/4              | 45.0           |
| 18                | 25                     | 1 9/16              | 21                  | 19 7/8              | 2 11/16         | 18.18        | 16                | 1 1/8         | 22 3/4              | 54.4           |
| 20                | 27 1/2                 | 1 11/16             | 23                  | 22                  | 2 7/8           | 20.20        | 20                | 1 1/8         | 25                  | 70.3           |
| 24                | 32                     | 1 7/8               | 27 1/4              | 26 1/8              | 3 1/4           | 24.25        | 20                | 1 1/4         | 29 1/2              | 90.0           |

## 300LB. SLIP-ON FLANGES

| Nominal Pipe Size | Outside Dia. of Flange | Thickness of Flange | Dia. of Raised Face | Dia. of Hub at Base | Length thru Hub | Dia. of Bore | Drilling Template |               |                     | Approx. Weight |
|-------------------|------------------------|---------------------|---------------------|---------------------|-----------------|--------------|-------------------|---------------|---------------------|----------------|
|                   |                        |                     |                     |                     |                 |              | No. of Holes      | Dia. of Bolts | Dia. of Bolt Circle |                |
|                   | O                      | Q                   | R                   | X                   | Y               | W            |                   |               |                     | kg             |
| 1/2               | 3 3/4                  | 9/16                | 1 3/8               | 1 1/2               | 7/8             | 0.88         | 4                 | 1/2           | 2 5/8               | 0.7            |
| 3/4               | 4 5/8                  | 5/8                 | 1 11/16             | 1 7/8               | 1               | 1.09         | 4                 | 5/8           | 3 1/4               | 1.2            |
| 1                 | 4 7/8                  | 11/16               | 2                   | 2 1/8               | 1 1/16          | 1.36         | 4                 | 5/8           | 3 1/2               | 1.5            |
| 1 1/4             | 5 1/4                  | 3/4                 | 2 1/2               | 2 1/2               | 1 1/16          | 1.70         | 4                 | 5/8           | 3 7/8               | 1.8            |
| 1 1/2             | 6 1/8                  | 13/16               | 2 7/8               | 2 3/4               | 1 3/16          | 1.95         | 4                 | 3/4           | 4 1/2               | 2.7            |
| 2                 | 6 1/2                  | 7/8                 | 3 5/8               | 3 5/16              | 1 5/16          | 2.44         | 8                 | 5/8           | 5                   | 3.1            |
| 2 1/2             | 7 1/2                  | 1                   | 4 1/8               | 3 15/16             | 1 1/2           | 2.94         | 8                 | 3/4           | 5 7/8               | 4.5            |
| 3                 | 8 1/4                  | 1 1/8               | 5                   | 4 5/8               | 1 11/16         | 3.57         | 8                 | 3/4           | 6 5/8               | 6.2            |
| 3 1/2             | 9                      | 1 3/16              | 5 1/2               | 5 1/4               | 1 3/4           | 4.07         | 8                 | 3/4           | 7 1/4               | 7.6            |
| 4                 | 10                     | 1 1/4               | 6 3/16              | 5 3/4               | 1 7/8           | 4.57         | 8                 | 3/4           | 7 7/8               | 10.0           |
| 5                 | 11                     | 1 3/8               | 7 5/16              | 7                   | 2               | 5.66         | 8                 | 3/4           | 9 1/4               | 12.7           |
| 6                 | 12 1/2                 | 1 7/16              | 8 1/2               | 8 1/8               | 2 1/16          | 6.72         | 12                | 3/4           | 10 5/8              | 16.6           |
| 8                 | 15                     | 1 5/8               | 10 5/8              | 10 1/4              | 2 7/16          | 8.72         | 12                | 7/8           | 13                  | 26.5           |
| 10                | 17 1/2                 | 1 7/8               | 12 3/4              | 12 5/8              | 2 5/8           | 10.88        | 16                | 1             | 15 1/4              | 36.7           |
| 12                | 20 1/2                 | 2                   | 15                  | 14 3/4              | 2 7/8           | 12.88        | 16                | 1 1/8         | 17 3/4              | 52.2           |
| 14                | 23                     | 2 1/8               | 16 1/4              | 16 3/4              | 3               | 14.14        | 20                | 1 1/8         | 20 1/4              | 74.4           |
| 16                | 25 1/2                 | 2 1/4               | 18 1/2              | 19                  | 3 1/4           | 16.16        | 20                | 1 1/4         | 22 1/2              | 99.8           |
| 18                | 28                     | 2 3/8               | 21                  | 21                  | 3 1/2           | 18.18        | 24                | 1 1/4         | 24 3/4              | 127.0          |
| 20                | 30 1/2                 | 2 1/2               | 23                  | 23 1/8              | 3 3/4           | 20.20        | 24                | 1 1/4         | 27                  | 147.0          |
| 24                | 36                     | 2 3/4               | 27 1/4              | 27 5/8              | 4 3/16          | 24.25        | 24                | 1 1/2         | 32                  | 222.0          |

# FLANGES



## 150LB. WELDING NECK FLANGES

| Nominal Pipe Size | Outside Dia. of Flange | Thickness of Flange | Dia. of Raised Face | Dia. of Hub at Base | Min. Dia. of Hub at Pt. of Welding | Length thru Hub | Dia. of Bore | Drilling Template |               |                     |
|-------------------|------------------------|---------------------|---------------------|---------------------|------------------------------------|-----------------|--------------|-------------------|---------------|---------------------|
|                   |                        |                     |                     |                     |                                    |                 |              | No. of Holes      | Dia. of Bolts | Dia. of Bolt Circle |
|                   | O                      | Q                   | R                   | X                   | H                                  | Y               | J            |                   |               |                     |
| 1/2               | 3 1/2                  | 7/16                | 1 3/8               | 1 3/16              | 0.84                               | 1 7/8           | 0.62         | 4                 | 1/2           | 2 3/8               |
| 3/4               | 3 7/8                  | 1/2                 | 1 11/16             | 1 1/2               | 1.05                               | 2 1/16          | 0.82         | 4                 | 1/2           | 2 3/4               |
| 1                 | 4 1/4                  | 9/16                | 2                   | 1 15/16             | 1.32                               | 2 3/16          | 1.05         | 4                 | 1/2           | 3 1/8               |
| 1 1/4             | 4 5/8                  | 5/8                 | 2 1/2               | 2 15/16             | 1.66                               | 2 1/4           | 1.38         | 4                 | 1/2           | 3 1/2               |
| 1 1/2             | 5                      | 11/16               | 2 7/8               | 2 9/16              | 1.90                               | 2 7/16          | 1.61         | 4                 | 1/2           | 3 7/8               |
| 2                 | 6                      | 3/4                 | 3 5/8               | 3 1/16              | 2.38                               | 2 1/2           | 2.07         | 4                 | 5/8           | 4 3/4               |
| 2 1/2             | 7                      | 7/8                 | 4 1/8               | 3 9/16              | 2.88                               | 2 3/4           | 2.47         | 4                 | 5/8           | 5 1/2               |
| 3                 | 7 1/2                  | 15/16               | 5                   | 4 1/4               | 3.50                               | 2 3/4           | 3.07         | 4                 | 5/8           | 6                   |
| 3 1/2             | 8 1/2                  | 15/16               | 5 1/2               | 4 13/16             | 4.00                               | 2 13/16         | 3.55         | 8                 | 5/8           | 7                   |
| 4                 | 9                      | 15/16               | 6 3/16              | 5 5/16              | 4.50                               | 3               | 4.03         | 8                 | 5/8           | 7 1/2               |
| 5                 | 10                     | 15/16               | 7 9/16              | 6 7/16              | 5.56                               | 3 1/2           | 5.05         | 8                 | 3/4           | 8 1/2               |
| 6                 | 11                     | 1                   | 8 1/2               | 7 5/16              | 6.63                               | 3 1/2           | 6.07         | 8                 | 3/4           | 9 1/2               |
| 8                 | 13 1/2                 | 1 1/8               | 10 5/8              | 9 11/16             | 8.63                               | 4               | 7.98         | 8                 | 3/4           | 11 3/4              |
| 10                | 16                     | 1 3/16              | 12 3/4              | 12                  | 10.75                              | 4               | 10.02        | 12                | 7/8           | 14 1/4              |
| 12                | 19                     | 1 1/4               | 15                  | 14 3/8              | 12.75                              | 4 1/2           | 12.00        | 12                | 7/8           | 17                  |
| 14                | 21                     | 1 3/8               | 16 1/4              | 15 3/4              | 14.00                              | 5               | 13.25        | 12                | 1             | 18 3/4              |
| 16                | 23 1/2                 | 1 7/16              | 18 1/2              | 18                  | 16.00                              | 5               | 15.25        | 16                | 1             | 21 1/4              |
| 18                | 25                     | 1 9/16              | 21                  | 19 7/8              | 18.00                              | 5 1/2           | 17.25        | 16                | 1 1/8         | 22 3/4              |
| 20                | 27 1/2                 | 1 11/16             | 23                  | 22                  | 20.00                              | 5 11/16         | 19.25        | 20                | 1 1/8         | 25                  |
| 24                | 32                     | 1 7/8               | 27 1/4              | 26 1/8              | 24.00                              | 6               | 23.25        | 20                | 1 1/4         | 29 1/2              |

## 300LB. WELDING NECK FLANGES

| Nominal Pipe Size | Outside Dia. of Flange | Thickness of Flange | Dia. of Raised Face | Dia. of Hub at Base | Min. Dia. of Hub at Pt. of Welding | Length thru Hub | Dia. of Bore | Drilling Template |               |                     | Approx. Weight |
|-------------------|------------------------|---------------------|---------------------|---------------------|------------------------------------|-----------------|--------------|-------------------|---------------|---------------------|----------------|
|                   |                        |                     |                     |                     |                                    |                 |              | No. of Holes      | Dia. of Bolts | Dia. of Bolt Circle |                |
|                   | O                      | Q                   | R                   | X                   | H                                  | Y               | J            |                   |               |                     | kg             |
| 1/2               | 3 3/4                  | 9/16                | 1 3/8               | 1 1/2               | 0.84                               | 2 1/16          | 0.62         | 4                 | 1/2           | 2 5/8               | 0.9            |
| 3/4               | 4 5/8                  | 5/8                 | 1 11/16             | 1 7/8               | 1.05                               | 2 1/4           | 0.82         | 4                 | 5/8           | 3 1/4               | 1.5            |
| 1                 | 4 7/8                  | 11/16               | 2                   | 2 1/8               | 1.32                               | 2 7/16          | 1.05         | 4                 | 5/8           | 3 1/2               | 1.7            |
| 1 1/4             | 5 1/4                  | 3/4                 | 2 1/2               | 2 1/2               | 1.66                               | 2 9/16          | 1.38         | 4                 | 5/8           | 3 7/8               | 2.1            |
| 1 1/2             | 6 1/8                  | 13/16               | 2 7/8               | 2 3/4               | 1.90                               | 2 11/16         | 1.61         | 4                 | 3/4           | 4 1/2               | 3.2            |
| 2                 | 6 1/2                  | 7/8                 | 3 5/8               | 3 5/16              | 2.38                               | 2 3/4           | 2.07         | 8                 | 5/8           | 5                   | 3.6            |
| 2 1/2             | 7 1/2                  | 1                   | 4 1/8               | 3 15/16             | 2.88                               | 3               | 2.47         | 8                 | 3/4           | 5 7/8               | 5.4            |
| 3                 | 8 1/4                  | 1 1/8               | 5                   | 4 5/8               | 3.50                               | 3 1/8           | 3.07         | 8                 | 3/4           | 6 5/8               | 7.5            |
| 3 1/2             | 9                      | 1 3/16              | 5 1/2               | 5 1/4               | 4.00                               | 3 3/16          | 3.55         | 8                 | 3/4           | 7 1/4               | 9.2            |
| 4                 | 10                     | 1 1/4               | 6 3/16              | 5 3/4               | 4.50                               | 3 3/8           | 4.03         | 8                 | 3/4           | 7 7/8               | 12.7           |
| 5                 | 11                     | 1 3/8               | 7 5/16              | 7                   | 5.56                               | 3 7/8           | 5.05         | 8                 | 3/4           | 9 1/4               | 16.0           |
| 6                 | 12 1/2                 | 1 7/16              | 8 1/2               | 8 1/8               | 6.63                               | 3 7/8           | 6.07         | 12                | 3/4           | 10 5/8              | 21.0           |
| 8                 | 15                     | 1 5/8               | 10 5/8              | 10 1/4              | 8.63                               | 4 3/8           | 7.98         | 12                | 7/8           | 13                  | 33.0           |
| 10                | 17 1/2                 | 1 7/8               | 12 3/4              | 12 5/8              | 10.75                              | 4 5/8           | 10.02        | 16                | 1             | 15 1/4              | 46.6           |
| 12                | 20 1/2                 | 2                   | 15                  | 14 3/4              | 12.75                              | 5 1/8           | 12.00        | 16                | 1 1/8         | 17 3/4              | 69.0           |
| 14                | 23                     | 2 1/8               | 16 1/4              | 16 3/4              | 14.00                              | 5 5/8           | 13.25        | 20                | 1 1/8         | 20 1/4              | 93.4           |
| 16                | 25 1/2                 | 2 1/4               | 18 1/2              | 19                  | 16.00                              | 5 3/4           | 15.25        | 20                | 1 1/4         | 22 1/2              | 119.6          |
| 18                | 28                     | 2 3/8               | 21                  | 21                  | 18.00                              | 6 1/4           | 17.25        | 24                | 1 1/4         | 24 3/4              | 150.5          |
| 20                | 30 1/2                 | 2 1/2               | 23                  | 23 1/8              | 20.00                              | 6 3/8           | 19.25        | 24                | 1 1/4         | 27                  | 184.4          |
| 24                | 36                     | 2 3/4               | 27 1/4              | 27 5/8              | 24.00                              | 6 5/8           | 23.25        | 24                | 1 1/2         | 32                  | 274.7          |

## CLASS 600

| Nominal Size in | Dia. of Flange                       | Bolt Circle Dia.                     | No. of Bolts | Dia. of Bolts                      | Dia. of Holes                      | Dia. of Raised Face                  | Height of Raised Face | Thickness of Flange |              |                                      |
|-----------------|--------------------------------------|--------------------------------------|--------------|------------------------------------|------------------------------------|--------------------------------------|-----------------------|---------------------|--------------|--------------------------------------|
|                 |                                      |                                      |              |                                    |                                    |                                      |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel                |
| 1/2             | 3 <sup>3</sup> / <sub>4</sub> (95)   | 2 <sup>5</sup> / <sub>8</sub> (67)   | 4            | 1/2 (13)                           | 5/8 (16)                           | 1 <sup>3</sup> / <sub>8</sub> (35)   | 1/4 (6)               | -                   | -            | 9/16 (14)                            |
| 3/4             | 4 <sup>5</sup> / <sub>8</sub> (117)  | 3 <sup>1</sup> / <sub>4</sub> (83)   | 4            | 5/8 (16)                           | 3/4 (19)                           | 1 <sup>11</sup> / <sub>16</sub> (43) | 1/4 (6)               | -                   | -            | 5/8 (16)                             |
| 1               | 4 <sup>7</sup> / <sub>8</sub> (124)  | 3 <sup>1</sup> / <sub>2</sub> (89)   | 4            | 5/8 (16)                           | 3/4 (19)                           | 2 (51)                               | 1/4 (6)               | -                   | -            | 1 <sup>11</sup> / <sub>16</sub> (17) |
| 1 1/4           | 5 <sup>1</sup> / <sub>4</sub> (133)  | 3 <sup>7</sup> / <sub>8</sub> (98)   | 4            | 5/8 (16)                           | 3/4 (19)                           | 2 1/2 (64)                           | 1/4 (6)               | -                   | -            | 1 <sup>3</sup> / <sub>16</sub> (21)  |
| 1 1/2           | 6 <sup>1</sup> / <sub>8</sub> (156)  | 4 <sup>1</sup> / <sub>2</sub> (114)  | 4            | 3/4 (19)                           | 7/8 (22)                           | 2 <sup>7</sup> / <sub>8</sub> (73)   | 1/4 (6)               | -                   | -            | 7/8 (22)                             |
| 2               | 6 <sup>1</sup> / <sub>2</sub> (165)  | 5 (127)                              | 8            | 5/8 (16)                           | 3/4 (19)                           | 3 <sup>5</sup> / <sub>8</sub> (92)   | 1/4 (6)               | -                   | -            | 1 (25)                               |
| 2 1/2           | 7 <sup>1</sup> / <sub>2</sub> (191)  | 5 <sup>7</sup> / <sub>8</sub> (149)  | 8            | 3/4 (19)                           | 7/8 (22)                           | 4 <sup>1</sup> / <sub>8</sub> (105)  | 1/4 (6)               | -                   | -            | 1 <sup>1</sup> / <sub>8</sub> (29)   |
| 3               | 8 <sup>1</sup> / <sub>4</sub> (210)  | 6 <sup>5</sup> / <sub>8</sub> (168)  | 8            | 3/4 (19)                           | 7/8 (22)                           | 5 (127)                              | 1/4 (6)               | -                   | -            | 1 <sup>1</sup> / <sub>4</sub> (32)   |
| 4               | 10 <sup>3</sup> / <sub>4</sub> (273) | 8 <sup>1</sup> / <sub>2</sub> (216)  | 8            | 7/8 (22)                           | 1 (25)                             | 6 <sup>3</sup> / <sub>16</sub> (157) | 1/4 (6)               | -                   | -            | 1 <sup>1</sup> / <sub>2</sub> (38)   |
| 5               | 13 (330)                             | 10 <sup>1</sup> / <sub>2</sub> (267) | 8            | 1 (25)                             | 1 <sup>1</sup> / <sub>8</sub> (29) | 7 <sup>5</sup> / <sub>16</sub> (186) | 1/4 (6)               | -                   | -            | 1 <sup>3</sup> / <sub>4</sub> (44)   |
| 6               | 14 (356)                             | 11 <sup>1</sup> / <sub>2</sub> (292) | 12           | 1 (25)                             | 1 <sup>1</sup> / <sub>8</sub> (29) | 8 <sup>1</sup> / <sub>2</sub> (216)  | 1/4 (6)               | -                   | -            | 1 <sup>7</sup> / <sub>8</sub> (48)   |
| 8               | 16 <sup>1</sup> / <sub>2</sub> (419) | 13 <sup>3</sup> / <sub>4</sub> (349) | 12           | 1 <sup>1</sup> / <sub>8</sub> (29) | 1 <sup>1</sup> / <sub>4</sub> (32) | 10 <sup>5</sup> / <sub>8</sub> (270) | 1/4 (6)               | -                   | -            | 2 <sup>3</sup> / <sub>16</sub> (56)  |
| 10              | 20 (508)                             | 17 (432)                             | 16           | 1 <sup>1</sup> / <sub>4</sub> (32) | 1 <sup>3</sup> / <sub>8</sub> (35) | 12 <sup>3</sup> / <sub>4</sub> (324) | 1/4 (6)               | -                   | -            | 2 <sup>1</sup> / <sub>2</sub> (64)   |
| 12              | 22 (559)                             | 19 <sup>1</sup> / <sub>4</sub> (489) | 20           | 1 <sup>1</sup> / <sub>4</sub> (32) | 1 <sup>3</sup> / <sub>8</sub> (35) | 15 (381)                             | 1/4 (6)               | -                   | -            | 2 <sup>5</sup> / <sub>8</sub> (67)   |
| 14              | 23 <sup>3</sup> / <sub>4</sub> (603) | 20 <sup>3</sup> / <sub>4</sub> (527) | 20           | 1 <sup>3</sup> / <sub>8</sub> (35) | 1 <sup>1</sup> / <sub>2</sub> (38) | 16 <sup>1</sup> / <sub>4</sub> (413) | 1/4 (6)               | -                   | -            | 2 <sup>3</sup> / <sub>4</sub> (70)   |
| 16              | 27 (686)                             | 23 <sup>3</sup> / <sub>4</sub> (603) | 20           | 1 <sup>1</sup> / <sub>2</sub> (38) | 1 <sup>5</sup> / <sub>8</sub> (41) | 18 <sup>1</sup> / <sub>2</sub> (470) | 1/4 (6)               | -                   | -            | 3 (76)                               |
| 18              | 29 <sup>1</sup> / <sub>4</sub> (743) | 25 <sup>3</sup> / <sub>4</sub> (654) | 20           | 1 <sup>5</sup> / <sub>8</sub> (41) | 1 <sup>3</sup> / <sub>4</sub> (44) | 21 (533)                             | 1/4 (6)               | -                   | -            | 3 <sup>1</sup> / <sub>4</sub> (83)   |
| 20              | 32 (813)                             | 28 <sup>1</sup> / <sub>2</sub> (724) | 24           | 1 <sup>5</sup> / <sub>8</sub> (41) | 1 <sup>3</sup> / <sub>4</sub> (44) | 23 (584)                             | 1/4 (6)               | -                   | -            | 3 <sup>1</sup> / <sub>2</sub> (89)   |
| 24              | 37 (940)                             | 33 (838)                             | 24           | 1 <sup>7</sup> / <sub>8</sub> (48) | 2 (51)                             | 27 <sup>1</sup> / <sub>4</sub> (692) | 1/4 (6)               | -                   | -            | 4 (102)                              |

## CLASS 900

| Nominal Size in | Dia. of Flange                       | Bolt Circle Dia.                     | No. of Bolts | Dia. of Bolts                      | Dia. of Holes                      | Dia. of Raised Face                  | Height of Raised Face | Thickness of Flange |              |                                     |
|-----------------|--------------------------------------|--------------------------------------|--------------|------------------------------------|------------------------------------|--------------------------------------|-----------------------|---------------------|--------------|-------------------------------------|
|                 |                                      |                                      |              |                                    |                                    |                                      |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel               |
| 1/2             | 4 <sup>3</sup> / <sub>4</sub> (121)  | 3 <sup>1</sup> / <sub>4</sub> (83)   | 4            | 3/4 (19)                           | 7/8 (22)                           | 1 <sup>3</sup> / <sub>8</sub> (35)   | 1/4 (6)               | -                   | -            | 7/8 (22)                            |
| 3/4             | 5 <sup>1</sup> / <sub>8</sub> (130)  | 3 <sup>1</sup> / <sub>2</sub> (89)   | 4            | 3/4 (19)                           | 7/8 (22)                           | 1 <sup>11</sup> / <sub>16</sub> (43) | 1/4 (6)               | -                   | -            | 1 (25)                              |
| 1               | 5 <sup>7</sup> / <sub>8</sub> (149)  | 4 (102)                              | 4            | 7/8 (22)                           | 1 (25)                             | 2 (51)                               | 1/4 (6)               | -                   | -            | 1 <sup>1</sup> / <sub>8</sub> (29)  |
| 1 1/4           | 6 <sup>1</sup> / <sub>4</sub> (159)  | 4 <sup>3</sup> / <sub>8</sub> (111)  | 4            | 7/8 (22)                           | 1 (25)                             | 2 1/2 (64)                           | 1/4 (6)               | -                   | -            | 1 <sup>1</sup> / <sub>8</sub> (29)  |
| 1 1/2           | 7 (178)                              | 4 <sup>7</sup> / <sub>8</sub> (124)  | 4            | 1 (25)                             | 1 <sup>1</sup> / <sub>8</sub> (29) | 2 <sup>7</sup> / <sub>8</sub> (73)   | 1/4 (6)               | -                   | -            | 1 <sup>1</sup> / <sub>4</sub> (32)  |
| 2               | 8 <sup>1</sup> / <sub>2</sub> (216)  | 6 <sup>1</sup> / <sub>2</sub> (165)  | 8            | 7/8 (22)                           | 1 (25)                             | 3 <sup>5</sup> / <sub>8</sub> (92)   | 1/4 (6)               | -                   | -            | 1 <sup>1</sup> / <sub>2</sub> (38)  |
| 2 1/2           | 9 <sup>5</sup> / <sub>8</sub> (244)  | 7 <sup>1</sup> / <sub>2</sub> (191)  | 8            | 1 (25)                             | 1 <sup>1</sup> / <sub>8</sub> (29) | 4 <sup>1</sup> / <sub>8</sub> (105)  | 1/4 (6)               | -                   | -            | 1 <sup>5</sup> / <sub>8</sub> (41)  |
| 3               | 9 <sup>1</sup> / <sub>2</sub> (241)  | 7 <sup>1</sup> / <sub>2</sub> (192)  | 8            | 7/8 (22)                           | 1 (25)                             | 5 (127)                              | 1/4 (6)               | -                   | -            | 1 <sup>1</sup> / <sub>2</sub> (38)  |
| 4               | 11 <sup>1</sup> / <sub>2</sub> (292) | 9 <sup>1</sup> / <sub>4</sub> (235)  | 8            | 1 <sup>1</sup> / <sub>8</sub> (29) | 1 <sup>1</sup> / <sub>4</sub> (32) | 6 <sup>3</sup> / <sub>16</sub> (157) | 1/4 (6)               | -                   | -            | 1 <sup>3</sup> / <sub>4</sub> (44)  |
| 5               | 13 <sup>3</sup> / <sub>4</sub> (349) | 11 (279)                             | 8            | 1 <sup>1</sup> / <sub>4</sub> (32) | 1 <sup>3</sup> / <sub>8</sub> (35) | 7 <sup>5</sup> / <sub>16</sub> (186) | 1/4 (6)               | -                   | -            | 2 (51)                              |
| 6               | 15 (381)                             | 12 <sup>1</sup> / <sub>2</sub> (318) | 12           | 1 <sup>1</sup> / <sub>8</sub> (29) | 1 <sup>1</sup> / <sub>4</sub> (32) | 8 <sup>1</sup> / <sub>2</sub> (216)  | 1/4 (6)               | -                   | -            | 2 <sup>3</sup> / <sub>16</sub> (56) |
| 8               | 18 <sup>1</sup> / <sub>2</sub> (470) | 15 <sup>1</sup> / <sub>2</sub> (394) | 12           | 1 <sup>3</sup> / <sub>8</sub> (35) | 1 <sup>1</sup> / <sub>2</sub> (38) | 10 <sup>5</sup> / <sub>8</sub> (270) | 1/4 (6)               | -                   | -            | 2 <sup>1</sup> / <sub>2</sub> (64)  |
| 10              | 21 <sup>1</sup> / <sub>2</sub> (546) | 18 <sup>1</sup> / <sub>2</sub> (470) | 16           | 1 <sup>3</sup> / <sub>8</sub> (35) | 1 <sup>1</sup> / <sub>2</sub> (38) | 12 <sup>3</sup> / <sub>4</sub> (324) | 1/4 (6)               | -                   | -            | 2 <sup>3</sup> / <sub>4</sub> (70)  |
| 12              | 24 (610)                             | 21 (533)                             | 20           | 1 <sup>3</sup> / <sub>8</sub> (35) | 1 <sup>1</sup> / <sub>2</sub> (38) | 15 (381)                             | 1/4 (6)               | -                   | -            | 3 <sup>1</sup> / <sub>8</sub> (80)  |
| 14              | 25 <sup>1</sup> / <sub>4</sub> (641) | 22 (559)                             | 20           | 1 <sup>1</sup> / <sub>2</sub> (38) | 1 <sup>5</sup> / <sub>8</sub> (41) | 16 <sup>1</sup> / <sub>4</sub> (413) | 1/4 (6)               | -                   | -            | 3 <sup>3</sup> / <sub>8</sub> (86)  |
| 16              | 27 <sup>3</sup> / <sub>4</sub> (705) | 24 <sup>1</sup> / <sub>4</sub> (616) | 20           | 1 <sup>5</sup> / <sub>8</sub> (41) | 1 <sup>3</sup> / <sub>4</sub> (44) | 18 <sup>1</sup> / <sub>2</sub> (470) | 1/4 (6)               | -                   | -            | 3 <sup>1</sup> / <sub>2</sub> (89)  |
| 18              | 31 (787)                             | 27 (686)                             | 20           | 1 <sup>7</sup> / <sub>8</sub> (48) | 2 (51)                             | 21 (533)                             | 1/4 (6)               | -                   | -            | 4 (102)                             |
| 20              | 33 <sup>3</sup> / <sub>4</sub> (857) | 29 <sup>1</sup> / <sub>2</sub> (749) | 20           | 2 (51)                             | 2 <sup>1</sup> / <sub>8</sub> (54) | 23 (584)                             | 1/4 (6)               | -                   | -            | 4 <sup>1</sup> / <sub>4</sub> (108) |
| 24              | 41 (1041)                            | 35 <sup>1</sup> / <sub>2</sub> (902) | 20           | 2 <sup>1</sup> / <sub>2</sub> (64) | 2 <sup>5</sup> / <sub>8</sub> (67) | 27 <sup>1</sup> / <sub>4</sub> (692) | 1/4 (6)               | -                   | -            | 5 <sup>1</sup> / <sub>2</sub> (140) |



# FLANGES

## CLASS 1500

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 4 3/4(121)     | 3 1/4(83)        | 4            | 3/4(19)       | 7/8(22)       | 1 3/8(35)           | 1/4(6)                | -                   | -            | 7/8(22)               |
| 3/4             | 5 1/8(130)     | 3 1/2(89)        | 4            | 3/4(19)       | 7/8(22)       | 1 11/16(43)         | 1/4(6)                | -                   | -            | 1(25)                 |
| 1               | 5 7/8(149)     | 4(102)           | 4            | 7/8(22)       | 1(25)         | 2(51)               | 1/4(6)                | -                   | -            | 1 1/8(29)             |
| 1 1/4           | 6 1/4(159)     | 4 3/8(111)       | 4            | 7/8(22)       | 1(25)         | 2 1/2(64)           | 1/4(6)                | -                   | -            | 1 1/8(29)             |
| 1 1/2           | 7(178)         | 4 7/8(124)       | 4            | 1(25)         | 1 1/8(29)     | 2 7/8(73)           | 1/4(6)                | -                   | -            | 1 1/4(32)             |
| 2               | 8 1/2(216)     | 6 1/2(165)       | 8            | 7/8(22)       | 1(25)         | 3 5/8(92)           | 1/4(6)                | -                   | -            | 1 1/2(38)             |
| 2 1/2           | 9 5/8(244)     | 7 1/2(191)       | 8            | 1(25)         | 1 1/8(29)     | 4 1/8(105)          | 1/4(6)                | -                   | -            | 1 5/8(41)             |
| 3               | 10 1/2(267)    | 8(203)           | 8            | 1 1/8(29)     | 1 1/4(32)     | 5(127)              | 1/4(6)                | -                   | -            | 1 7/8(48)             |
| 4               | 12 1/4(311)    | 9 1/2(241)       | 8            | 1 1/4(32)     | 1 3/8(35)     | 6 3/16(157)         | 1/4(6)                | -                   | -            | 2 1/8(54)             |
| 5               | 14 3/4(375)    | 11 1/2(292)      | 8            | 1 1/2(38)     | 1 5/8(41)     | 7 5/16(186)         | 1/4(6)                | -                   | -            | 2 7/8(73)             |
| 6               | 15 1/2(394)    | 12 1/2(318)      | 12           | 1 3/8(35)     | 1 1/2(38)     | 8 1/2(216)          | 1/4(6)                | -                   | -            | 3 1/4(83)             |
| 8               | 19(483)        | 15 1/2(394)      | 12           | 1 5/8(41)     | 1 3/4(44)     | 10 5/8(270)         | 1/4(6)                | -                   | -            | 3 5/8(92)             |
| 10              | 23(584)        | 19(483)          | 12           | 1 7/8(48)     | 2(51)         | 12 3/4(324)         | 1/4(6)                | -                   | -            | 4 1/4(108)            |
| 12              | 26 1/2(673)    | 22 1/2(571)      | 16           | 2(51)         | 2 1/8(54)     | 15(381)             | 1/4(6)                | -                   | -            | 4 7/8(124)            |
| 14              | 29 1/2(749)    | 25(635)          | 16           | 2 1/4(57)     | 2 3/8(60)     | 16 1/4(413)         | 1/4(6)                | -                   | -            | 5 1/4(133)            |
| 16              | 32 1/2(826)    | 27 3/4(705)      | 16           | 2 1/2(64)     | 2 5/8(67)     | 18 1/2(470)         | 1/4(6)                | -                   | -            | 5 3/4(146)            |
| 18              | 36(914)        | 30 1/2(775)      | 16           | 2 3/4(70)     | 2 7/8(73)     | 21(533)             | 1/4(6)                | -                   | -            | 6 3/8(162)            |
| 20              | 38 3/4(984)    | 32 3/4(832)      | 16           | 3(76)         | 3 1/8(79)     | 23(584)             | 1/4(6)                | -                   | -            | 7(178)                |
| 24              | 46(1168)       | 39(991)          | 16           | 3 1/2(89)     | 3 5/8(92)     | 27 1/4(692)         | 1/4(6)                | -                   | -            | 8(203)                |

YICK HOE

## SLIP-ON FLANGES(BS 4504)

### PN 6

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 80             | 55               | 4            | M10           | 11            | 40                  | 2                     | 12                  | -            | -                     |
| 3/4             | 90             | 65               | 4            | M10           | 11            | 50                  | 2                     | 14                  | -            | -                     |
| 1               | 100            | 75               | 4            | M10           | 11            | 60                  | 2                     | 14                  | -            | -                     |
| 1 1/4           | 120            | 90               | 4            | M12           | 14            | 70                  | 2                     | 16                  | -            | -                     |
| 1 1/2           | 130            | 100              | 4            | M12           | 14            | 80                  | 3                     | 16                  | -            | -                     |
| 2               | 140            | 110              | 4            | M12           | 14            | 90                  | 3                     | 16                  | -            | -                     |
| 2 1/2           | 160            | 130              | 4            | M12           | 14            | 110                 | 3                     | 16                  | -            | -                     |
| 3               | 190            | 150              | 4            | M16           | 18            | 128                 | 3                     | 18                  | -            | -                     |
| 4               | 210            | 170              | 4            | M16           | 18            | 148                 | 3                     | 18                  | -            | -                     |
| 5               | 240            | 200              | 8            | M16           | 18            | 178                 | 3                     | 20                  | -            | -                     |
| 6               | 265            | 225              | 8            | M16           | 18            | 202                 | 3                     | 20                  | -            | -                     |
| 8               | 320            | 280              | 8            | M16           | 18            | 258                 | 3                     | 22                  | -            | -                     |
| 10              | 375            | 335              | 12           | M16           | 18            | 312                 | 3                     | 24                  | -            | -                     |
| 12              | 440            | 395              | 12           | M20           | 22            | 365                 | 4                     | 24                  | -            | -                     |
| 14              | 490            | 445              | 12           | M20           | 22            | 415                 | 4                     | 26                  | -            | -                     |
| 16              | 540            | 495              | 16           | M20           | 22            | 465                 | 4                     | 28                  | -            | -                     |
| 18              | 595            | 550              | 16           | M20           | 22            | 520                 | 4                     | 30                  | -            | -                     |
| 20              | 645            | 600              | 20           | M20           | 22            | 570                 | 4                     | 30                  | -            | -                     |
| 24              | 755            | 705              | 20           | M24           | 26            | 670                 | 5                     | 30                  | -            | -                     |

### PN 10

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 95             | 65               | 4            | M12           | 14            | 45                  | 2                     | 14                  | -            | -                     |
| 3/4             | 105            | 75               | 4            | M12           | 14            | 58                  | 2                     | 16                  | -            | -                     |
| 1               | 115            | 85               | 4            | M12           | 14            | 68                  | 2                     | 16                  | -            | -                     |
| 1 1/4           | 140            | 100              | 4            | M16           | 18            | 78                  | 2                     | 18                  | -            | -                     |
| 1 1/2           | 150            | 110              | 4            | M16           | 18            | 88                  | 3                     | 18                  | -            | -                     |
| 2               | 165            | 125              | 4            | M16           | 18            | 102                 | 3                     | 20                  | -            | -                     |
| 2 1/2           | 185            | 145              | 4            | M16           | 18            | 122                 | 3                     | 20                  | -            | -                     |
| 3               | 200            | 160              | 8            | M16           | 18            | 138                 | 3                     | 22                  | -            | -                     |
| 4               | 220            | 180              | 8            | M16           | 18            | 158                 | 3                     | 24                  | -            | -                     |
| 5               | 250            | 210              | 8            | M16           | 18            | 188                 | 3                     | 26                  | -            | -                     |
| 6               | 285            | 240              | 8            | M20           | 22            | 212                 | 3                     | 26                  | -            | -                     |
| 8               | 340            | 295              | 8            | M20           | 22            | 268                 | 3                     | 26                  | -            | -                     |
| 10              | 395            | 350              | 12           | M20           | 22            | 320                 | 3                     | 28                  | -            | -                     |
| 12              | 445            | 400              | 12           | M20           | 22            | 370                 | 4                     | 28                  | -            | -                     |
| 14              | 505            | 460              | 16           | M20           | 22            | 430                 | 4                     | 30                  | -            | -                     |
| 16              | 565            | 515              | 16           | M24           | 23            | 482                 | 4                     | 32                  | -            | -                     |
| 18              | 615            | 565              | 20           | M24           | 26            | 532                 | 4                     | 32                  | -            | -                     |
| 20              | 670            | 620              | 20           | M24           | 26            | 585                 | 4                     | 34                  | -            | -                     |
| 24              | 780            | 725              | 20           | M27           | 30            | 685                 | 5                     | 36                  | -            | -                     |

# FLANGES

## PN 16

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 95             | 65               | 4            | M12           | 14            | 45                  | 2                     | 14                  | 6            | 16                    |
| 3/4             | 105            | 75               | 4            | M12           | 14            | 58                  | 2                     | 16                  | 6            | 18                    |
| 1               | 115            | 85               | 4            | M12           | 14            | 68                  | 2                     | 16                  | 8            | 18                    |
| 1 1/4           | 140            | 100              | 4            | M16           | 18            | 78                  | 2                     | 18                  | 8            | 18                    |
| 1 1/2           | 150            | 110              | 4            | M16           | 18            | 88                  | 3                     | 18                  | 9            | 18                    |
| 2               | 165            | 125              | 4            | M16           | 18            | 102                 | 3                     | 20                  | 11           | 20                    |
| 2 1/2           | 185            | 145              | 4            | M16           | 18            | 122                 | 3                     | 20                  | 13           | 18                    |
| 3               | 200            | 160              | 8            | M16           | 18            | 138                 | 3                     | 22                  | 13           | 20                    |
| 4               | 220            | 180              | 8            | M16           | 18            | 158                 | 3                     | 24                  | 16           | 20                    |
| 5               | 250            | 210              | 8            | M16           | 18            | 188                 | 3                     | 26                  | -            | 22                    |
| 6               | 285            | 240              | 8            | M20           | 22            | 212                 | 3                     | 26                  | -            | 22                    |
| 8               | 340            | 295              | 12           | M20           | 22            | 268                 | 3                     | 30                  | -            | 24                    |
| 10              | 405            | 355              | 12           | M24           | 26            | 320                 | 3                     | 32                  | -            | 26                    |
| 12              | 460            | 410              | 12           | M24           | 26            | 378                 | 4                     | 32                  | -            | 28                    |
| 14              | 520            | 470              | 16           | M24           | 26            | 438                 | 4                     | 36                  | -            | 30                    |
| 16              | 580            | 525              | 16           | M27           | 30            | 490                 | 4                     | 38                  | -            | 32                    |
| 18              | 640            | 585              | 20           | M27           | 30            | 550                 | 4                     | 40                  | -            | -                     |
| 20              | 715            | 650              | 20           | M30           | 33            | 610                 | 4                     | 42                  | -            | 36                    |
| 24              | 840            | 770              | 20           | M33           | 36            | 725                 | 5                     | 48                  | -            | 40                    |

## PN 25

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 95             | 65               | 4            | M12           | 14            | 45                  | 2                     | 16                  | 8            | 16                    |
| 3/4             | 105            | 75               | 4            | M12           | 14            | 58                  | 2                     | 18                  | 8            | 18                    |
| 1               | 115            | 85               | 4            | M12           | 14            | 68                  | 2                     | 18                  | 9            | 18                    |
| 1 1/4           | 140            | 100              | 4            | M16           | 18            | 78                  | 2                     | 20                  | 9            | 18                    |
| 1 1/2           | 150            | 110              | 4            | M16           | 18            | 88                  | 3                     | 20                  | 11           | 18                    |
| 2               | 165            | 125              | 4            | M16           | 18            | 102                 | 3                     | 22                  | 11           | 20                    |
| 2 1/2           | 185            | 145              | 8            | M16           | 18            | 122                 | 3                     | 24                  | 13           | 22                    |
| 3               | 200            | 160              | 8            | M16           | 18            | 138                 | 3                     | 26                  | 14           | 24                    |
| 4               | 235            | 190              | 8            | M20           | 22            | 162                 | 3                     | 28                  | 17           | 24                    |
| 5               | 270            | 220              | 8            | M24           | 26            | 188                 | 3                     | 30                  | -            | 26                    |
| 6               | 300            | 250              | 8            | M24           | 26            | 218                 | 3                     | 34                  | -            | 28                    |
| 8               | 360            | 310              | 12           | M24           | 26            | 278                 | 3                     | 34                  | -            | 30                    |
| 10              | 425            | 370              | 12           | M27           | 30            | 335                 | 3                     | 36                  | -            | 32                    |
| 12              | 485            | 430              | 16           | M27           | 30            | 395                 | 4                     | 40                  | -            | 34                    |
| 14              | 555            | 490              | 16           | M30           | 33            | 450                 | 4                     | 44                  | -            | 38                    |
| 16              | 620            | 550              | 16           | M33           | 36            | 505                 | 4                     | 48                  | -            | 40                    |
| 20              | 730            | 660              | 20           | M33           | 36            | 615                 | 4                     | 52                  | -            | 44                    |
| 24              | 845            | 770              | 20           | M36           | 39            | 720                 | 5                     | -                   | -            | 46                    |

## PN 40

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 95             | 65               | 4            | M12           | 14            | 45                  | 2                     | -                   | 9            | 16                    |
| 3/4             | 105            | 75               | 4            | M12           | 14            | 58                  | 2                     | -                   | 9            | 18                    |
| 1               | 115            | 85               | 4            | M12           | 14            | 68                  | 2                     | -                   | 11           | 18                    |
| 1 1/4           | 140            | 100              | 4            | M16           | 18            | 78                  | 2                     | -                   | 11           | 18                    |
| 1 1/2           | 150            | 110              | 4            | M16           | 18            | 88                  | 3                     | -                   | 13           | 18                    |
| 2               | 165            | 125              | 4            | M16           | 18            | 102                 | 3                     | -                   | 13           | 20                    |
| 2 1/2           | 185            | 145              | 8            | M16           | 18            | 122                 | 3                     | -                   | 14           | 22                    |
| 3               | 200            | 160              | 8            | M16           | 18            | 138                 | 3                     | -                   | 16           | 24                    |
| 4               | 235            | 190              | 8            | M20           | 22            | 162                 | 3                     | -                   | 19           | 24                    |
| 5               | 270            | 220              | 8            | M24           | 26            | 188                 | 3                     | -                   | -            | 26                    |
| 6               | 300            | 250              | 8            | M24           | 26            | 218                 | 3                     | -                   | -            | 28                    |
| 8               | 375            | 320              | 12           | M27           | 30            | 285                 | 3                     | -                   | -            | 34                    |
| 10              | 450            | 385              | 12           | M30           | 33            | 345                 | 3                     | -                   | -            | 38                    |
| 12              | 515            | 450              | 16           | M30           | 33            | 410                 | 4                     | -                   | -            | 42                    |
| 14              | 580            | 510              | 16           | M33           | 36            | 465                 | 4                     | -                   | -            | 46                    |
| 16              | 660            | 585              | 16           | M36           | 39            | 535                 | 4                     | -                   | -            | 50                    |
| 20              | 755            | 670              | 20           | M39           | 42            | 615                 | 4                     | -                   | -            | 52                    |
| 24              | 890            | 795              | 20           | M45           | 48            | 735                 | 5                     | -                   | -            | 60                    |

## PN 64

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 105            | 75               | 4            | M12           | 14            | 45                  | 2                     | -                   | -            | 20                    |
| 3/4             | 130            | 90               | 4            | M16           | 18            | 58                  | 2                     | -                   | -            | 22                    |
| 1               | 140            | 100              | 4            | M16           | 18            | 68                  | 2                     | -                   | -            | 24                    |
| 1 1/4           | 155            | 110              | 4            | M20           | 22            | 78                  | 2                     | -                   | -            | 26                    |
| 1 1/2           | 170            | 125              | 4            | M20           | 22            | 88                  | 3                     | -                   | -            | 28                    |
| 2               | 180            | 135              | 4            | M20           | 22            | 102                 | 3                     | -                   | -            | 26                    |
| 2 1/2           | 205            | 160              | 8            | M20           | 22            | 122                 | 3                     | -                   | -            | 26                    |
| 3               | 215            | 170              | 8            | M20           | 22            | 138                 | 3                     | -                   | -            | 28                    |
| 4               | 250            | 200              | 8            | M24           | 26            | 162                 | 3                     | -                   | -            | 30                    |
| 5               | 295            | 240              | 8            | M27           | 30            | 188                 | 3                     | -                   | -            | 34                    |
| 6               | 345            | 280              | 8            | M30           | 33            | 218                 | 3                     | -                   | -            | 36                    |
| 8               | 415            | 345              | 12           | M33           | 36            | 285                 | 3                     | -                   | -            | 42                    |
| 10              | 470            | 400              | 12           | M33           | 36            | 345                 | 3                     | -                   | -            | 46                    |
| 12              | 530            | 460              | 16           | M33           | 36            | 410                 | 4                     | -                   | -            | 52                    |
| 14              | 600            | 525              | 16           | M36           | 39            | 465                 | 4                     | -                   | -            | 56                    |
| 16              | 670            | 585              | 16           | M39           | 42            | 535                 | 4                     | -                   | -            | 60                    |
| 20              | 800            | 705              | 20           | M45           | 48            | 615                 | 4                     | -                   | -            | 68                    |
| 24              | 930            | 820              | 20           | M52           | 56            | 735                 | 4                     | -                   | -            | 76                    |

# FLANGES

## PN 100

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 105            | 75               | 4            | M12           | 14            | 45                  | 2                     | -                   | -            | 20                    |
| 3/4             | 130            | 90               | 4            | M16           | 18            | 58                  | 2                     | -                   | -            | 22                    |
| 1               | 140            | 100              | 4            | M16           | 18            | 68                  | 2                     | -                   | -            | 24                    |
| 1 1/4           | 155            | 110              | 4            | M20           | 22            | 78                  | 2                     | -                   | -            | 26                    |
| 1 1/2           | 170            | 125              | 4            | M20           | 22            | 88                  | 3                     | -                   | -            | 28                    |
| 2               | 195            | 145              | 4            | M24           | 26            | 102                 | 3                     | -                   | -            | 30                    |
| 2 1/2           | 220            | 170              | 8            | M24           | 26            | 122                 | 3                     | -                   | -            | 34                    |
| 3               | 230            | 180              | 8            | M24           | 26            | 138                 | 3                     | -                   | -            | 36                    |
| 4               | 265            | 210              | 8            | M27           | 30            | 162                 | 3                     | -                   | -            | 40                    |
| 5               | 315            | 250              | 8            | M30           | 33            | 188                 | 3                     | -                   | -            | 40                    |
| 6               | 355            | 290              | 12           | M30           | 33            | 218                 | 3                     | -                   | -            | 44                    |
| 8               | 430            | 360              | 12           | M33           | 36            | 285                 | 3                     | -                   | -            | 52                    |
| 10              | 505            | 430              | 12           | M36           | 39            | 345                 | 3                     | -                   | -            | 60                    |
| 12              | 585            | 500              | 16           | M39           | 42            | 410                 | 4                     | -                   | -            | 68                    |
| 14              | 655            | 560              | 16           | M45           | 48            | 465                 | 4                     | -                   | -            | 74                    |
| 16              | 715            | 620              | 16           | M45           | 48            | 535                 | 4                     | -                   | -            | 78                    |
| 20              | 870            | 760              | 20           | M52           | 56            | 615                 | 4                     | -                   | -            | 94                    |
| 24              | 990            | 875              | 20           | M56           | 62            | 735                 | 4                     | -                   | -            | 104                   |

## PN 160

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 105            | 75               | 4            | M12           | 14            | 45                  | 2                     | -                   | -            | 20                    |
| 3/4             | 130            | 90               | 4            | M16           | 18            | 58                  | 2                     | -                   | -            | 22                    |
| 1               | 140            | 100              | 4            | M16           | 18            | 68                  | 2                     | -                   | -            | 24                    |
| 1 1/4           | 155            | 110              | 4            | M20           | 22            | 78                  | 2                     | -                   | -            | 26                    |
| 1 1/2           | 170            | 125              | 4            | M20           | 22            | 88                  | 3                     | -                   | -            | 28                    |
| 2               | 195            | 145              | 4            | M24           | 26            | 102                 | 3                     | -                   | -            | 30                    |
| 2 1/2           | 220            | 170              | 8            | M24           | 26            | 122                 | 3                     | -                   | -            | 34                    |
| 3               | 230            | 180              | 8            | M24           | 26            | 138                 | 3                     | -                   | -            | 36                    |
| 4               | 265            | 210              | 8            | M27           | 30            | 162                 | 3                     | -                   | -            | 40                    |
| 5               | 315            | 250              | 8            | M30           | 33            | 188                 | 3                     | -                   | -            | 44                    |
| 6               | 355            | 290              | 12           | M30           | 33            | 218                 | 3                     | -                   | -            | 50                    |
| 8               | 430            | 360              | 12           | M33           | 36            | 285                 | 3                     | -                   | -            | 60                    |
| 10              | 515            | 430              | 12           | M39           | 42            | 345                 | 3                     | -                   | -            | 68                    |
| 12              | 585            | 500              | 16           | M39           | 42            | 410                 | 4                     | -                   | -            | 78                    |

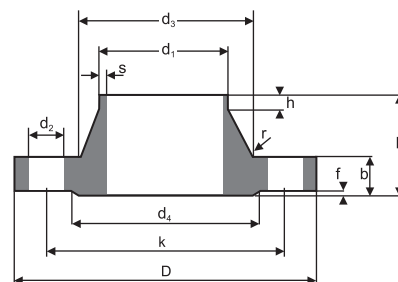
## PN 250

| Nominal<br>Size<br>in | Dia. of<br>Flange | Bolt<br>Circle<br>Dia. | No.<br>of<br>Bolts | Dia.<br>of<br>Bolts | Dia.<br>of<br>Holes | Dia. of<br>Raised<br>Face | Height<br>of Raised<br>Face | Thickness of Flange |                 |                          |
|-----------------------|-------------------|------------------------|--------------------|---------------------|---------------------|---------------------------|-----------------------------|---------------------|-----------------|--------------------------|
|                       |                   |                        |                    |                     |                     |                           |                             | Grey<br>Cast Iron   | Copper<br>Alloy | Cast and<br>Forged Steel |
| 1/2                   | 130               | 90                     | 4                  | M16                 | 18                  | 45                        | 2                           | -                   | -               | 26                       |
| 3/4                   | 135               | 95                     | 4                  | M16                 | 18                  | 58                        | 2                           | -                   | -               | 28                       |
| 1                     | 150               | 105                    | 4                  | M20                 | 22                  | 68                        | 2                           | -                   | -               | 28                       |
| 1 1/4                 | 165               | 120                    | 4                  | M20                 | 22                  | 78                        | 2                           | -                   | -               | 32                       |
| 1 1/2                 | 185               | 135                    | 4                  | M24                 | 26                  | 88                        | 3                           | -                   | -               | 34                       |
| 2                     | 200               | 150                    | 8                  | M24                 | 26                  | 102                       | 3                           | -                   | -               | 38                       |
| 2 1/2                 | 230               | 180                    | 8                  | M24                 | 26                  | 122                       | 3                           | -                   | -               | 42                       |
| 3                     | 255               | 200                    | 8                  | M27                 | 30                  | 138                       | 3                           | -                   | -               | 46                       |
| 4                     | 300               | 230                    | 8                  | M30                 | 33                  | 162                       | 3                           | -                   | -               | 54                       |
| 5                     | 340               | 275                    | 12                 | M30                 | 33                  | 188                       | 3                           | -                   | -               | 60                       |
| 6                     | 390               | 320                    | 12                 | M33                 | 36                  | 218                       | 3                           | -                   | -               | 68                       |
| 8                     | 485               | 400                    | 12                 | M39                 | 42                  | 285                       | 3                           | -                   | -               | 82                       |
| 10                    | 585               | 490                    | 12                 | M45                 | 48                  | 345                       | 3                           | -                   | -               | 100                      |
| 12                    | 690               | 590                    | 16                 | M48                 | 52                  | 410                       | 4                           | -                   | -               | 120                      |

YICK HOE

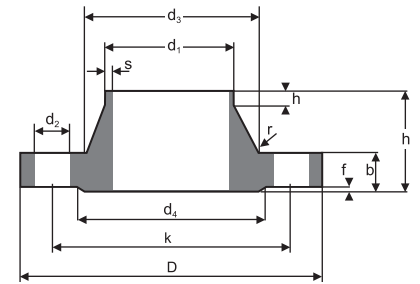
# FLANGES

## WELDING NECK FLANGES



DIN 2633 PN16 Raised face to DIN 2526, form C

| Tube<br>DN | Flange               |         |         |         | Hub                  |                      |         |         |                           | Raised face          |         | Drilling Template |        |                      |              |
|------------|----------------------|---------|---------|---------|----------------------|----------------------|---------|---------|---------------------------|----------------------|---------|-------------------|--------|----------------------|--------------|
|            | d <sub>1</sub><br>mm | D<br>mm | b<br>mm | k<br>mm | h <sub>1</sub><br>mm | d <sub>3</sub><br>mm | s<br>mm | r<br>mm | h <sub>2</sub><br>mm<br>≈ | d <sub>4</sub><br>mm | f<br>mm | Number            | Thread | d <sub>2</sub><br>mm | Weight<br>kg |
| 15         | 20<br>21.3           | 95      | 14      | 65      | 35                   | 30<br>32             | 2       | 4       | 6                         | 45                   | 2       | 4                 | M 12   | 14                   | 0.648        |
| 20         | 25<br>26.9           | 105     | 16      | 75      | 38                   | 38<br>40             | 2.3     | 4       | 6                         | 58                   | 2       | 4                 | M 12   | 14                   | 0.952        |
| 25         | 30<br>33.7           | 115     | 16      | 85      | 38<br>45             | 42<br>45             | 2.6     | 4       | 6                         | 68                   | 2       | 4                 | M 12   | 14                   | 1.140        |
| 32         | 38<br>42.4           | 140     | 16      | 100     | 40<br>56             | 52<br>56             | 2.6     | 6       | 6                         | 78                   | 2       | 4                 | M 16   | 18                   | 1.690        |
| 40         | 44.5<br>48.3         | 150     | 16      | 110     | 42<br>64             | 60<br>64             | 2.6     | 6       | 7                         | 88                   | 3       | 4                 | M 16   | 18                   | 1.860        |
| 50         | 57<br>60.3           | 165     | 18      | 125     | 45<br>75             | 72<br>75             | 2.9     | 6       | 8                         | 102                  | 3       | 4                 | M 16   | 18                   | 2.530        |
| 65         | 76.1                 | 185     | 18      | 145     | 45                   | 90                   | 2.9     | 6       | 10                        | 122                  | 3       | 4                 | M 16   | 18                   | 3.060        |
| 80         | 88.9                 | 200     | 20      | 160     | 50                   | 105                  | 3.2     | 8       | 10                        | 138                  | 3       | 8                 | M 16   | 18                   | 3.700        |
| 100        | 108<br>114.3         | 220     | 20      | 180     | 52<br>131            | 125<br>131           | 3.6     | 8       | 12                        | 158                  | 3       | 8                 | M 16   | 18                   | 4.620        |
| 125        | 133<br>139.7         | 250     | 22      | 210     | 55                   | 150<br>156           | 4       | 8       | 12                        | 188                  | 3       | 8                 | M 16   | 18                   | 6.300        |
| 150        | 159<br>168.3         | 285     | 22      | 240     | 55                   | 175<br>184           | 4.5     | 10      | 12                        | 212                  | 3       | 8                 | M 20   | 22                   | 7.750        |
| 200        | 219.1                | 340     | 24      | 295     | 62                   | 235                  | 5.9     | 10      | 16                        | 268                  | 3       | 12                | M 20   | 22                   | 11.000       |



DIN 2635 PN 40 Raised face to DIN 2526, form C

| Tube<br>DN | Flange               |         |         |         | Hub                  |                      |         |         |                      | Raised face               |         | Drilling Template |        |                      |              |
|------------|----------------------|---------|---------|---------|----------------------|----------------------|---------|---------|----------------------|---------------------------|---------|-------------------|--------|----------------------|--------------|
|            | d <sub>1</sub><br>mm | D<br>mm | b<br>mm | k<br>mm | h <sub>1</sub><br>mm | d <sub>3</sub><br>mm | s<br>mm | r<br>mm | h <sub>2</sub><br>mm | d <sub>4</sub><br>mm<br>≈ | f<br>mm | Number            | Thread | d <sub>2</sub><br>mm | Weight<br>kg |
| 10         | 14<br>17.2           | 90      | 16      | 60      | 35                   | 25<br>28             | 1.8     | 4       | 6                    | 40                        | 2       | 4                 | M 12   | 14                   | 0.661        |
| 15         | 20<br>21.3           | 95      | 16      | 65      | 38                   | 30<br>32             | 2       | 4       | 6                    | 45                        | 2       | 4                 | M 12   | 14                   | 0.746        |
| 20         | 25<br>26.9           | 105     | 18      | 75      | 40                   | 38<br>40             | 2.3     | 4       | 6                    | 58                        | 2       | 4                 | M 12   | 14                   | 1.060        |
| 25         | 30<br>33.7           | 115     | 18      | 85      | 40                   | 42<br>46             | 2.6     | 4       | 6                    | 68                        | 2       | 4                 | M 12   | 14                   | 1.290        |
| 32         | 38<br>42.4           | 140     | 18      | 100     | 42                   | 52<br>56             | 2.6     | 6       | 6                    | 78                        | 2       | 4                 | M 16   | 18                   | 1.880        |
| 40         | 44.5<br>48.3         | 150     | 18      | 110     | 45                   | 60<br>64             | 2.6     | 6       | 7                    | 88                        | 3       | 4                 | M 16   | 18                   | 2.330        |
| 50         | 57<br>60.3           | 165     | 20      | 125     | 48                   | 72<br>75             | 2.9     | 6       | 8                    | 102                       | 3       | 4                 | M 16   | 18                   | 2.820        |
| 65         | 76.1                 | 185     | 22      | 145     | 52                   | 90                   | 2.9     | 6       | 10                   | 122                       | 3       | 8                 | M 16   | 18                   | 3.740        |
| 80         | 88.9                 | 200     | 24      | 160     | 58                   | 105                  | 3.2     | 8       | 12                   | 138                       | 3       | 8                 | M 16   | 18                   | 4.750        |
| 100        | 108<br>114.3         | 235     | 24      | 190     | 65                   | 128<br>134           | 3.6     | 8       | 12                   | 162                       | 3       | 8                 | M 20   | 22                   | 6.520        |
| 125        | 133<br>139.7         | 270     | 26      | 220     | 68                   | 155<br>162           | 4       | 8       | 12                   | 188                       | 3       | 8                 | M 24   | 26                   | 9.070        |
| 150        | 159<br>168.3         | 300     | 28      | 250     | 75                   | 182<br>192           | 4.5     | 10      | 12                   | 218                       | 3       | 8                 | M 24   | 26                   | 11.800       |
| 200        | 219.1                | 375     | 34      | 320     | 88                   | 244                  | 6.3     | 10      | 16                   | 285                       | 3       | 12                | M 27   | 30                   | 21.500       |



# FLANGES

## SLIP-ON FLANGES (BS10)

TABLE A

| Nominal Size in               | Dia. of Flange                       | Bolt Circle Dia.                     | No. of Bolts | Dia. of Bolts | Dia. of Holes                       | Dia. of Raised Face | Height of Raised Face | Thickness of Flange                 |                                     |                       |
|-------------------------------|--------------------------------------|--------------------------------------|--------------|---------------|-------------------------------------|---------------------|-----------------------|-------------------------------------|-------------------------------------|-----------------------|
|                               |                                      |                                      |              |               |                                     |                     |                       | Grey Cast Iron                      | Copper Alloy                        | Cast and Forged Steel |
| 1/2                           | 3 <sup>3</sup> / <sub>4</sub> (95)   | 2 <sup>5</sup> / <sub>8</sub> (67)   | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 1/2(13)                             | 1/4(6)                              | -                     |
| 3/4                           | 4(102)                               | 2 <sup>7</sup> / <sub>8</sub> (73)   | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 1/2(13)                             | 1/4(6)                              | -                     |
| 1                             | 4 <sup>1</sup> / <sub>2</sub> (114)  | 3 <sup>1</sup> / <sub>4</sub> (83)   | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 1/2(13)                             | 5/16(8)                             | -                     |
| 1 <sup>1</sup> / <sub>4</sub> | 4 <sup>3</sup> / <sub>4</sub> (121)  | 3 <sup>7</sup> / <sub>16</sub> (87)  | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 5/8(16)                             | 5/16(8)                             | -                     |
| 1 <sup>1</sup> / <sub>2</sub> | 5 <sup>1</sup> / <sub>4</sub> (133)  | 3 <sup>7</sup> / <sub>8</sub> (98)   | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 5/8(16)                             | 3/8(10)                             | -                     |
| 2                             | 6(152)                               | 4 <sup>1</sup> / <sub>2</sub> (114)  | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 5/8(16)                             | 3/8(10)                             | -                     |
| 2 <sup>1</sup> / <sub>2</sub> | 6 <sup>1</sup> / <sub>2</sub> (165)  | 5(127)                               | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 1 <sup>1</sup> / <sub>16</sub> (17) | 7/16(11)                            | -                     |
| 3                             | 7 <sup>1</sup> / <sub>4</sub> (184)  | 5 <sup>3</sup> / <sub>4</sub> (146)  | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 1 <sup>1</sup> / <sub>16</sub> (17) | 1/2(13)                             | -                     |
| 4                             | 8 <sup>1</sup> / <sub>2</sub> (216)  | 7(178)                               | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 3/4(19)                             | 5/8(16)                             | -                     |
| 5                             | 10(254)                              | 8 <sup>1</sup> / <sub>4</sub> (210)  | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 3/4(19)                             | 1 <sup>1</sup> / <sub>16</sub> (17) | -                     |
| 6                             | 11(279)                              | 9 <sup>1</sup> / <sub>4</sub> (235)  | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 1 <sup>3</sup> / <sub>16</sub> (21) | 1 <sup>1</sup> / <sub>16</sub> (17) | -                     |
| 8                             | 13 <sup>1</sup> / <sub>4</sub> (337) | 11 <sup>1</sup> / <sub>2</sub> (292) | 8            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 7/8(22)                             | 3/4(19)                             | -                     |
| 10                            | 16(406)                              | 14(356)                              | 8            | 3/4(19)       | 7/8(22)                             | -                   | -                     | 1 <sup>5</sup> / <sub>16</sub> (24) | 3/4(19)                             | -                     |
| 12                            | 18(457)                              | 16(406)                              | 8            | 3/4(19)       | 7/8(22)                             | -                   | -                     | 1 <sup>5</sup> / <sub>16</sub> (24) | 7/8(22)                             | -                     |
| 14                            | 20 <sup>3</sup> / <sub>4</sub> (527) | 18 <sup>1</sup> / <sub>2</sub> (470) | 8            | 7/8(22)       | 1(25)                               | -                   | -                     | 1(25)                               | 1(25)                               | -                     |
| 16                            | 22 <sup>3</sup> / <sub>4</sub> (578) | 20 <sup>1</sup> / <sub>2</sub> (521) | 12           | 7/8(22)       | 1(25)                               | -                   | -                     | 1 <sup>1</sup> / <sub>16</sub> (27) | 1(25)                               | -                     |
| 18                            | 25 <sup>1</sup> / <sub>4</sub> (641) | 23(584)                              | 12           | 7/8(22)       | 1(25)                               | -                   | -                     | 1 <sup>1</sup> / <sub>16</sub> (27) | 1 <sup>1</sup> / <sub>16</sub> (27) | -                     |
| 20                            | 27 <sup>3</sup> / <sub>4</sub> (705) | 25 <sup>1</sup> / <sub>4</sub> (641) | 12           | 7/8(22)       | 1(25)                               | -                   | -                     | 1 <sup>1</sup> / <sub>8</sub> (29)  | 1 <sup>1</sup> / <sub>8</sub> (29)  | -                     |
| 24                            | 32 <sup>1</sup> / <sub>2</sub> (826) | 29 <sup>3</sup> / <sub>4</sub> (756) | 12           | 1(25)         | 1 <sup>1</sup> / <sub>8</sub> (29)  | -                   | -                     | 1 <sup>3</sup> / <sub>16</sub> (30) | 1 <sup>3</sup> / <sub>16</sub> (30) | -                     |

TABLE D

| Nominal Size in               | Dia. of Flange                       | Bolt Circle Dia.                     | No. of Bolts | Dia. of Bolts | Dia. of Holes                       | Dia. of Raised Face | Height of Raised Face | Thickness of Flange                 |                                     |                                     |
|-------------------------------|--------------------------------------|--------------------------------------|--------------|---------------|-------------------------------------|---------------------|-----------------------|-------------------------------------|-------------------------------------|-------------------------------------|
|                               |                                      |                                      |              |               |                                     |                     |                       | Grey Cast Iron                      | Copper Alloy                        | Cast and Forged Steel               |
| 1/2                           | 3 <sup>3</sup> / <sub>4</sub> (95)   | 2 <sup>5</sup> / <sub>8</sub> (67)   | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 1/2(13)                             | 1/4(6)                              | 3/8(10)                             |
| 3/4                           | 4(102)                               | 2 <sup>7</sup> / <sub>8</sub> (73)   | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 1/2(13)                             | 1/4(6)                              | 3/8(10)                             |
| 1                             | 4 <sup>1</sup> / <sub>2</sub> (114)  | 3 <sup>1</sup> / <sub>4</sub> (83)   | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 1/2(13)                             | 5/16(8)                             | 3/8(10)                             |
| 1 <sup>1</sup> / <sub>4</sub> | 4 <sup>3</sup> / <sub>4</sub> (121)  | 3 <sup>7</sup> / <sub>16</sub> (87)  | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 5/8(16)                             | 5/16(8)                             | 1/2(13)                             |
| 1 <sup>1</sup> / <sub>2</sub> | 5 <sup>1</sup> / <sub>4</sub> (133)  | 3 <sup>7</sup> / <sub>8</sub> (98)   | 4            | 1/2(13)       | 9/16(14)                            | -                   | -                     | 5/8(16)                             | 3/8(10)                             | 1/2(13)                             |
| 2                             | 6(152)                               | 4 <sup>1</sup> / <sub>2</sub> (114)  | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 1 <sup>1</sup> / <sub>16</sub> (17) | 3/8(10)                             | 9/16(14)                            |
| 2 <sup>1</sup> / <sub>2</sub> | 6 <sup>1</sup> / <sub>2</sub> (165)  | 5(127)                               | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 1 <sup>1</sup> / <sub>16</sub> (17) | 7/16(11)                            | 9/16(14)                            |
| 3                             | 7 <sup>1</sup> / <sub>4</sub> (184)  | 5 <sup>3</sup> / <sub>4</sub> (146)  | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 3/4(19)                             | 1/2(13)                             | 9/16(14)                            |
| 4                             | 8 <sup>1</sup> / <sub>2</sub> (216)  | 7(178)                               | 4            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 3/4(19)                             | 5/8(16)                             | 1 <sup>1</sup> / <sub>16</sub> (17) |
| 5                             | 10(254)                              | 8 <sup>1</sup> / <sub>4</sub> (210)  | 8            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 1 <sup>3</sup> / <sub>16</sub> (21) | 1 <sup>1</sup> / <sub>16</sub> (17) | 1 <sup>1</sup> / <sub>16</sub> (17) |
| 6                             | 11(279)                              | 9 <sup>1</sup> / <sub>4</sub> (235)  | 8            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 1 <sup>3</sup> / <sub>16</sub> (21) | 1 <sup>1</sup> / <sub>16</sub> (17) | 1 <sup>1</sup> / <sub>16</sub> (17) |
| 8                             | 13 <sup>1</sup> / <sub>4</sub> (337) | 11 <sup>1</sup> / <sub>2</sub> (292) | 8            | 5/8(16)       | 1 <sup>1</sup> / <sub>16</sub> (17) | -                   | -                     | 7/8(22)                             | 3/4(19)                             | 3/4(19)                             |
| 10                            | 16(406)                              | 14(356)                              | 8            | 3/4(19)       | 7/8(22)                             | -                   | -                     | 1(25)                               | 3/4(19)                             | 3/4(19)                             |
| 12                            | 18(457)                              | 16(406)                              | 12           | 3/4(19)       | 7/8(22)                             | -                   | -                     | 1(25)                               | 7/8(22)                             | 7/8(22)                             |
| 14                            | 20 <sup>3</sup> / <sub>4</sub> (527) | 18 <sup>1</sup> / <sub>2</sub> (470) | 12           | 7/8(22)       | 1(25)                               | -                   | -                     | 1 <sup>1</sup> / <sub>8</sub> (29)  | 1(25)                               | 1(25)                               |
| 16                            | 22 <sup>3</sup> / <sub>4</sub> (578) | 20 <sup>1</sup> / <sub>2</sub> (521) | 12           | 7/8(22)       | 1(25)                               | -                   | -                     | 1 <sup>1</sup> / <sub>8</sub> (29)  | 1(25)                               | 1(25)                               |
| 18                            | 25 <sup>1</sup> / <sub>4</sub> (641) | 23(584)                              | 12           | 7/8(22)       | 1(25)                               | -                   | -                     | 1 <sup>1</sup> / <sub>4</sub> (32)  | 1 <sup>1</sup> / <sub>8</sub> (29)  | 1 <sup>1</sup> / <sub>8</sub> (29)  |
| 20                            | 27 <sup>3</sup> / <sub>4</sub> (705) | 25 <sup>1</sup> / <sub>4</sub> (641) | 16           | 7/8(22)       | 1(25)                               | -                   | -                     | 1 <sup>1</sup> / <sub>4</sub> (32)  | 1 <sup>1</sup> / <sub>4</sub> (32)  | 1 <sup>1</sup> / <sub>4</sub> (32)  |
| 24                            | 32 <sup>1</sup> / <sub>2</sub> (826) | 29 <sup>3</sup> / <sub>4</sub> (756) | 16           | 1(25)         | 1 <sup>1</sup> / <sub>8</sub> (29)  | -                   | -                     | 1 <sup>3</sup> / <sub>8</sub> (35)  | 1 <sup>3</sup> / <sub>8</sub> (35)  | 1 <sup>3</sup> / <sub>8</sub> (35)  |

TABLE E

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 3 3/4(95)      | 2 5/8(67)        | 4            | 1/2(13)       | 9/16(14)      | -                   | -                     | 1/2(13)             | 1/4(6)       | 3/8(10)               |
| 3/4             | 4(102)         | 2 7/8(73)        | 4            | 1/2(13)       | 9/16(14)      | -                   | -                     | 1/2(13)             | 1/4(6)       | 3/8(10)               |
| 1               | 4 1/2(114)     | 3 1/4(83)        | 4            | 1/2(13)       | 9/16(14)      | -                   | -                     | 1/2(13)             | 5/16(8)      | 3/8(10)               |
| 1 1/4           | 4 3/4(121)     | 3 7/16(87)       | 4            | 1/2(13)       | 9/16(14)      | -                   | -                     | 5/8(16)             | 5/16(8)      | 1/2(13)               |
| 1 1/2           | 5 1/4(133)     | 3 7/8(98)        | 4            | 1/2(13)       | 9/16(14)      | -                   | -                     | 5/8(16)             | 3/8(10)      | 1/2(13)               |
| 2               | 6(152)         | 4 1/2(114)       | 4            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 3/4(19)             | 3/8(10)      | 9/16(14)              |
| 2 1/2           | 6 1/2(165)     | 5(127)           | 4            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 3/4(19)             | 7/16(11)     | 9/16(14)              |
| 3               | 7 1/4(184)     | 5 3/4(146)       | 4            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 3/4(19)             | 1/2(13)      | 9/16(14)              |
| 4               | 8 1/2(216)     | 7(178)           | 8            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 7/8(22)             | 5/8(16)      | 1 1/16(17)            |
| 5               | 10(254)        | 8 1/4(210)       | 8            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 7/8(22)             | 1 1/16(17)   | 1 1/16(17)            |
| 6               | 11(279)        | 9 1/4(235)       | 8            | 3/4(19)       | 7/8(22)       | -                   | -                     | 7/8(22)             | 1 1/16(17)   | 1 1/16(17)            |
| 8               | 13 1/4(337)    | 11 1/2(292)      | 8            | 3/4(19)       | 7/8(22)       | -                   | -                     | 1(25)               | 3/4(19)      | 3/4(19)               |
| 10              | 16(406)        | 14(356)          | 12           | 3/4(19)       | 7/8(22)       | -                   | -                     | 1(25)               | 7/8(22)      | 7/8(22)               |
| 12              | 18(457)        | 16(406)          | 12           | 7/8(22)       | 1(25)         | -                   | -                     | 1 1/8(29)           | 1(25)        | 1(25)                 |
| 14              | 20 3/4(527)    | 18 1/2(470)      | 12           | 7/8(22)       | 1(25)         | -                   | -                     | 1 1/4(32)           | 1(25)        | 1(25)                 |
| 16              | 22 3/4(578)    | 20 1/2(521)      | 12           | 7/8(22)       | 1(25)         | -                   | -                     | 1 1/4(32)           | 1(25)        | 1(25)                 |
| 18              | 25 1/4(641)    | 23(584)          | 16           | 7/8(22)       | 1(25)         | -                   | -                     | 1 3/8(35)           | 1 1/8(29)    | 1 1/8(29)             |
| 20              | 27 3/4(705)    | 25 1/4(641)      | 16           | 7/8(22)       | 1(25)         | -                   | -                     | 1 1/2(38)           | 1 1/4(32)    | 1 1/4(32)             |
| 24              | 32 1/2(826)    | 29 3/4(756)      | 16           | 1 1/8(29)     | 1 1/4(32)     | -                   | -                     | 1 5/8(41)           | 1 1/2(38)    | 1 1/2(38)             |

TABLE F

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 3 3/4(95)      | 2 5/8(67)        | 4            | 1/2(13)       | 9/16(14)      | -                   | -                     | 1/2(13)             | 5/16(8)      | 3/8(10)               |
| 3/4             | 4(102)         | 2 7/8(73)        | 4            | 1/2(13)       | 9/16(14)      | -                   | -                     | 1/2(13)             | 5/16(8)      | 3/8(10)               |
| 1               | 4 3/4(121)     | 3 7/16(87)       | 4            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 1/2(13)             | 3/8(10)      | 3/8(10)               |
| 1 1/4           | 5 1/4(133)     | 3 7/8(98)        | 4            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 5/8(16)             | 3/8(10)      | 1/2(13)               |
| 1 1/2           | 5 1/2(140)     | 4 1/8(105)       | 4            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 5/8(16)             | 7/16(11)     | 1/2(13)               |
| 2               | 6 1/2(165)     | 5(127)           | 4            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 3/4(19)             | 7/16(11)     | 5/8(16)               |
| 2 1/2           | 7 1/4(184)     | 5 3/4(146)       | 8            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 3/4(19)             | 1/2(13)      | 5/8(16)               |
| 3               | 8(203)         | 6 1/2(165)       | 8            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 3/4(19)             | 9/16(14)     | 5/8(16)               |
| 4               | 9(229)         | 7 1/2(191)       | 8            | 5/8(16)       | 1 1/16(17)    | -                   | -                     | 7/8(22)             | 1 1/16(17)   | 3/4(19)               |
| 5               | 11(279)        | 9 1/4(235)       | 8            | 3/4(19)       | 7/8(22)       | -                   | -                     | 1(25)               | 3/4(19)      | 7/8(22)               |
| 6               | 12(305)        | 10 1/4(260)      | 12           | 3/4(19)       | 7/8(22)       | -                   | -                     | 1(25)               | 7/8(22)      | 7/8(22)               |
| 8               | 14 1/2(368)    | 12 3/4(324)      | 12           | 3/4(19)       | 7/8(22)       | -                   | -                     | 1 1/8(29)           | 1(25)        | 1(25)                 |
| 10              | 17(432)        | 15(381)          | 12           | 7/8(22)       | 1(25)         | -                   | -                     | 1 1/8(29)           | 1(25)        | 1(25)                 |
| 12              | 19 1/4(489)    | 17 1/4(438)      | 16           | 7/8(22)       | 1(25)         | -                   | -                     | 1 1/4(32)           | 1 1/8(29)    | 1 1/8(29)             |
| 14              | 21 3/4(552)    | 19 1/2(495)      | 16           | 1(25)         | 1 1/8(29)     | -                   | -                     | 1 3/8(35)           | 1 1/4(32)    | 1 1/4(32)             |
| 16              | 24(610)        | 21 3/4(552)      | 20           | 1(25)         | 1 1/8(29)     | -                   | -                     | 1 3/8(35)           | 1 1/4(32)    | 1 1/4(32)             |
| 18              | 26 1/2(673)    | 24(610)          | 20           | 1 1/8(29)     | 1 1/4(32)     | -                   | -                     | 1 1/2(38)           | 1 3/8(35)    | 1 3/8(35)             |
| 20              | 29(737)        | 26 1/2(673)      | 24           | 1 1/8(29)     | 1 1/4(32)     | -                   | -                     | 1 5/8(41)           | 1 1/2(38)    | 1 1/2(38)             |
| 24              | 33 1/2(851)    | 30 3/4(781)      | 24           | 1 1/4(32)     | 1 3/8(35)     | -                   | -                     | 1 3/4(44)           | 1 5/8(41)    | 1 5/8(41)             |

# FLANGES

TABLE H

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 4 1/2(114)     | 3 1/4(83)        | 4            | 5/8(16)       | 11/16(17)     | 2 1/4(57)           | 1/16(2)               | 5/8(16)             | 3/8(10)      | 1/2(13)               |
| 3/4             | 4 1/2(114)     | 3 1/4(83)        | 4            | 5/8(16)       | 11/16(17)     | 2 1/4(57)           | 1/16(2)               | 5/8(16)             | 3/8(10)      | 1/2(13)               |
| 1               | 4 3/4(121)     | 3 7/16(87)       | 4            | 5/8(16)       | 11/16(17)     | 2 1/2(64)           | 1/16(2)               | 3/4(19)             | 7/16(11)     | 9/16(14)              |
| 1 1/4           | 5 1/4(133)     | 3 7/8(98)        | 4            | 5/8(16)       | 11/16(17)     | 3(76)               | 1/16(2)               | 7/8(22)             | 7/16(11)     | 11/16(17)             |
| 1 1/2           | 5 1/2(140)     | 4 1/8(105)       | 4            | 5/8(16)       | 11/16(17)     | 3 1/4(83)           | 1/16(2)               | 7/8(22)             | 1/2(13)      | 11/16(17)             |
| 2               | 6 1/2(165)     | 5(127)           | 4            | 5/8(16)       | 11/16(17)     | 4(102)              | 1/16(2)               | 1(25)               | 1/2(13)      | 3/4(19)               |
| 2 1/2           | 7 1/4(184)     | 5 3/4(146)       | 8            | 5/8(16)       | 11/16(17)     | 4 1/2(114)          | 1/16(2)               | 1(25)               | 9/16(14)     | 3/4(19)               |
| 3               | 8(203)         | 6 1/2(165)       | 8            | 5/8(16)       | 11/16(17)     | 5(127)              | 1/16(2)               | 1 1/8(29)           | 5/8(16)      | 7/8(22)               |
| 4               | 9(229)         | 7 1/2(191)       | 8            | 5/8(16)       | 11/16(17)     | 6(152)              | 1/16(2)               | 1 1/4(32)           | 3/4(19)      | 1(25)                 |
| 5               | 11(279)        | 9 1/4(235)       | 8            | 3/4(19)       | 7/8(22)       | 7(178)              | 1/16(2)               | 1 3/8(35)           | 7/8(22)      | 1 1/8(29)             |
| 6               | 12(305)        | 10 1/4(260)      | 12           | 3/4(19)       | 7/8(22)       | 8 1/4(210)          | 1/16(2)               | 1 3/8(35)           | 1(25)        | 1 1/8(29)             |
| 8               | 14 1/2(368)    | 12 3/4(324)      | 12           | 3/4(19)       | 7/8(22)       | 10 1/4(260)         | 1/16(2)               | 1 1/2(38)           | 1 1/4(32)    | 1 1/4(32)             |
| 10              | 17(432)        | 15(381)          | 12           | 7/8(22)       | 1(25)         | 12 1/4(311)         | 1/16(2)               | 1 5/8(41)           | 1 3/8(35)    | 1 3/8(35)             |
| 12              | 19 1/4(489)    | 17 1/4(438)      | 16           | 7/8(22)       | 1(25)         | 14 1/4(362)         | 1/16(2)               | 1 3/4(44)           | 1 1/2(38)    | 1 1/2(38)             |
| 14              | 21 3/4(552)    | 19 1/2(495)      | 16           | 1(25)         | 1 1/8(29)     | 16 1/2(419)         | 1/16(2)               | 1 7/8(48)           | 1 5/8(41)    | 1 5/8(41)             |
| 16              | 24(610)        | 21 3/4(552)      | 20           | 1(25)         | 1 1/8(29)     | 19(483)             | 1/16(2)               | 2(51)               | 1 3/4(44)    | 1 3/4(44)             |
| 18              | 26 1/2(673)    | 24(610)          | 20           | 1 1/8(29)     | 1 1/4(32)     | 21(533)             | 1/16(2)               | 2 1/8(54)           | 1 7/8(48)    | 1 7/8(48)             |
| 20              | 29(737)        | 26 1/2(673)      | 24           | 1 1/8(29)     | 1 1/4(32)     | 23 1/2(597)         | 1/16(2)               | 2 1/4(57)           | 2(51)        | 2(51)                 |
| 24              | 33 1/2(851)    | 30 3/4(781)      | 24           | 1 1/4(32)     | 1 3/8(35)     | 27 1/2(699)         | 1/16(2)               | 2 1/2(64)           | 2 1/4(57)    | 2 1/4(57)             |

TABLE J

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 4 1/2(114)     | 3 1/4(83)        | 4            | 5/8(16)       | 11/16(17)     | 2 1/4(57)           | 1/16(2)               | -                   | 5/8(16)      | 5/8(16)               |
| 3/4             | 4 1/2(114)     | 3 1/4(83)        | 4            | 5/8(16)       | 11/16(17)     | 2 1/4(57)           | 1/16(2)               | -                   | 5/8(16)      | 5/8(16)               |
| 1               | 4 3/4(121)     | 3 7/16(87)       | 4            | 5/8(16)       | 11/16(17)     | 2 1/2(64)           | 1/16(2)               | -                   | 3/4(19)      | 3/4(19)               |
| 1 1/4           | 5 1/4(133)     | 3 7/8(98)        | 4            | 5/8(16)       | 11/16(17)     | 3(76)               | 1/16(2)               | -                   | 3/4(19)      | 3/4(19)               |
| 1 1/2           | 5 1/2(140)     | 4 1/8(105)       | 4            | 5/8(16)       | 11/16(17)     | 3 1/4(83)           | 1/16(2)               | -                   | 7/8(22)      | 7/8(22)               |
| 2               | 6 1/2(165)     | 5(127)           | 4            | 3/4(19)       | 7/8(22)       | 4(102)              | 1/16(2)               | -                   | 1(25)        | 1(25)                 |
| 2 1/2           | 7 1/4(184)     | 5 3/4(146)       | 8            | 3/4(19)       | 7/8(22)       | 4 1/2(114)          | 1/16(2)               | -                   | 1(25)        | 1(25)                 |
| 3               | 8(203)         | 6 1/2(165)       | 8            | 3/4(19)       | 7/8(22)       | 5(127)              | 1/16(2)               | -                   | 1 1/4(32)    | 1 1/4(32)             |
| 4               | 9(229)         | 7 1/2(191)       | 8            | 3/4(19)       | 7/8(22)       | 6(152)              | 1/16(2)               | -                   | 1 3/8(35)    | 1 3/8(35)             |
| 5               | 11(279)        | 9 1/4(235)       | 8            | 7/8(22)       | 1(25)         | 7(178)              | 1/16(2)               | -                   | 1 1/2(38)    | 1 1/2(38)             |
| 6               | 12(305)        | 10 1/4(260)      | 12           | 7/8(22)       | 1(25)         | 8 1/4(210)          | 1/16(2)               | -                   | 1 1/2(38)    | 1 1/2(38)             |
| 8               | 14 1/2(368)    | 12 3/4(324)      | 12           | 7/8(22)       | 1(25)         | 10 1/4(260)         | 1/16(2)               | -                   | 1 5/8(41)    | 1 5/8(41)             |
| 10              | 17(432)        | 15(381)          | 12           | 1(25)         | 1 1/8(29)     | 12 1/4(311)         | 1/16(2)               | -                   | -            | 1 7/8(48)             |
| 12              | 19 1/4(489)    | 17 1/4(438)      | 16           | 1(25)         | 1 1/8(29)     | 14 1/4(362)         | 1/16(2)               | -                   | -            | 2(51)                 |
| 14              | 21 3/4(552)    | 19 1/2(495)      | 16           | 1 1/8(29)     | 1 1/4(32)     | 16 1/2(419)         | 1/16(2)               | -                   | -            | 2 1/8(54)             |
| 16              | 24(610)        | 21 3/4(552)      | 20           | 1 1/8(29)     | 1 1/4(32)     | 19(483)             | 1/16(2)               | -                   | -            | 2 1/4(57)             |
| 18              | 26 1/2(673)    | 24(610)          | 20           | 1 1/4(32)     | 1 3/8(35)     | 21(533)             | 1/16(2)               | -                   | -            | 2 3/8(60)             |
| 20              | 29(737)        | 26 1/2(673)      | 24           | 1 1/4(32)     | 1 3/8(35)     | 23 1/2(597)         | 1/16(2)               | -                   | -            | 2 1/2(64)             |
| 24              | 33 1/2(851)    | 30 3/4(781)      | 24           | 1 3/8(35)     | 1 1/2(38)     | 27 1/2(699)         | 1/16(2)               | -                   | -            | 2 3/4(70)             |

TABLE K

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 4 1/2(114)     | 3 1/4(83)        | 4            | 5/8(16)       | 1 1/16(17)    | 2 1/2(64)           | 1/16(2)               | -                   | 3/4(19)      | 3/4(19)               |
| 3/4             | 4 1/2(114)     | 3 1/4(83)        | 4            | 5/8(16)       | 1 1/16(17)    | 2 1/2(64)           | 1/16(2)               | -                   | 3/4(19)      | 3/4(19)               |
| 1               | 5(127)         | 3 3/4(95)        | 4            | 5/8(16)       | 1 1/16(17)    | 3(76)               | 1/16(2)               | -                   | 7/8(22)      | 7/8(22)               |
| 1 1/4           | 5 1/4(133)     | 3 7/8(98)        | 4            | 5/8(16)       | 1 1/16(17)    | 3(76)               | 1/16(2)               | -                   | 7/8(22)      | 7/8(22)               |
| 1 1/2           | 6(152)         | 4 1/2(114)       | 4            | 3/4(19)       | 7/8(22)       | 3 1/2(89)           | 1/16(2)               | -                   | 1(25)        | 1(25)                 |
| 2               | 6 1/2(165)     | 5(127)           | 8            | 5/8(16)       | 1 1/16(17)    | 4(102)              | 1/16(2)               | -                   | 1(25)        | 1(25)                 |
| 2 1/2           | 7 1/4(184)     | 5 3/4(146)       | 8            | 3/4(19)       | 7/8(22)       | 4 1/2(114)          | 1/16(2)               | -                   | 1 1/8(29)    | 1 1/8(29)             |
| 3               | 8(203)         | 6 1/2(165)       | 8            | 3/4(19)       | 7/8(22)       | 5(127)              | 1/16(2)               | -                   | 1 1/4(32)    | 1 1/4(32)             |
| 4               | 9 1/2(241)     | 7 3/4(197)       | 8            | 7/8(22)       | 1(25)         | 6(152)              | 1/16(2)               | -                   | 1 3/8(35)    | 1 3/8(35)             |
| 5               | 11(279)        | 9 1/4(235)       | 12           | 7/8(22)       | 1(25)         | 7(178)              | 1/16(2)               | -                   | 1 5/8(41)    | 1 5/8(41)             |
| 6               | 12(305)        | 10 1/4(260)      | 12           | 7/8(22)       | 1(25)         | 8 1/4(210)          | 1/16(2)               | -                   | 1 5/8(41)    | 1 5/8(41)             |
| 8               | 14 1/2(368)    | 12 1/2(318)      | 12           | 1(25)         | 1 1/8(29)     | 10 1/4(260)         | 1/16(2)               | -                   | 1 7/8(48)    | 1 7/8(48)             |
| 10              | 17(432)        | 15(381)          | 16           | 1(25)         | 1 1/8(29)     | 12 1/4(311)         | 1/16(2)               | -                   | -            | 2(51)                 |
| 12              | 19 1/4(489)    | 17(432)          | 16           | 1 1/8(29)     | 1 1/4(32)     | 14 1/4(362)         | 1/16(2)               | -                   | -            | 2 1/4(57)             |
| 14              | 22 1/2(572)    | 20(508)          | 16           | 1 1/4(32)     | 1 3/8(35)     | 16 1/2(419)         | 1/16(2)               | -                   | -            | 2 3/8(60)             |
| 16              | 24 3/4(629)    | 22 1/4(565)      | 20           | 1 1/4(32)     | 1 3/8(35)     | 19(483)             | 1/16(2)               | -                   | -            | 2 5/8(67)             |
| 18              | 28 1/4(718)    | 25 3/4(654)      | 20           | 1 3/8(35)     | 1 1/2(38)     | 22 1/2(572)         | 1/16(2)               | -                   | -            | 3(76)                 |
| 20              | 31(787)        | 28(711)          | 20           | 1 1/2(38)     | 1 5/8(41)     | 24 1/2(622)         | 1/16(2)               | -                   | -            | 3 1/4(83)             |

TABLE R

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 4 1/2(114)     | 3 1/4(83)        | 4            | 5/8(16)       | 1 1/16(17)    | 2 1/2(64)           | 1/16(2)               | -                   | -            | 3/4(19)               |
| 3/4             | 4 1/2(114)     | 3 1/4(83)        | 4            | 5/8(16)       | 1 1/16(17)    | 2 1/2(64)           | 1/16(2)               | -                   | -            | 3/4(19)               |
| 1               | 5(127)         | 3 3/4(95)        | 4            | 5/8(16)       | 1 1/16(17)    | 3(76)               | 1/16(2)               | -                   | -            | 7/8(22)               |
| 1 1/4           | 5 1/4(133)     | 3 7/8(98)        | 4            | 5/8(16)       | 1 1/16(17)    | 3(76)               | 1/16(2)               | -                   | -            | 7/8(22)               |
| 1 1/2           | 6(152)         | 4 1/2(114)       | 4            | 3/4(19)       | 7/8(22)       | 3 1/2(89)           | 1/16(2)               | -                   | -            | 1(25)                 |
| 2               | 6 1/2(165)     | 5(127)           | 8            | 5/8(16)       | 1 1/16(17)    | 4(102)              | 1/16(2)               | -                   | -            | 1(25)                 |
| 2 1/2           | 7 1/4(184)     | 5 3/4(146)       | 8            | 3/4(19)       | 7/8(22)       | 4 1/2(114)          | 1/16(2)               | -                   | -            | 1 1/8(29)             |
| 3               | 8(203)         | 6 1/2(165)       | 8            | 3/4(19)       | 7/8(22)       | 5(127)              | 1/16(2)               | -                   | -            | 1 1/4(32)             |
| 4               | 9 1/2(241)     | 7 3/4(197)       | 8            | 7/8(22)       | 1(25)         | 6(152)              | 1/16(2)               | -                   | -            | 1 3/8(35)             |
| 5               | 11(279)        | 9 1/4(235)       | 12           | 7/8(22)       | 1(25)         | 7(178)              | 1/16(2)               | -                   | -            | 1 5/8(41)             |
| 6               | 12(305)        | 10 1/4(260)      | 12           | 7/8(22)       | 1(25)         | 8 1/4(210)          | 1/16(2)               | -                   | -            | 1 5/8(41)             |
| 8               | 14 1/2(368)    | 12 3/4(324)      | 12           | 1(25)         | 1 1/8(29)     | 10 1/4(260)         | 1/16(2)               | -                   | -            | 2(51)                 |
| 10              | 17(432)        | 15 1/4(387)      | 16           | 1(25)         | 1 1/8(29)     | 12 1/4(311)         | 1/16(2)               | -                   | -            | 2 1/4(57)             |
| 12              | 20(508)        | 18(457)          | 16           | 1 1/8(29)     | 1 1/4(32)     | 14 1/4(362)         | 1/16(2)               | -                   | -            | 2 1/2(64)             |
| 14              | 23(584)        | 20 3/4(527)      | 16           | 1 1/4(32)     | 1 3/8(35)     | 16 1/2(419)         | 1/16(2)               | -                   | -            | 2 3/4(70)             |
| 16              | 25 1/4(641)    | 23(584)          | 20           | 1 1/4(32)     | 1 3/8(35)     | 19(483)             | 1/16(2)               | -                   | -            | 3(76)                 |
| 18              | 29(737)        | 26 1/2(673)      | 20           | 1 3/8(35)     | 1 1/2(38)     | 22 1/2(572)         | 1/16(2)               | -                   | -            | 3 1/4(83)             |
| 20              | 31 3/4(806)    | 28 3/4(730)      | 20           | 1 1/2(38)     | 1 5/8(41)     | 24 1/2(622)         | 1/16(2)               | -                   | -            | 3 1/2(89)             |

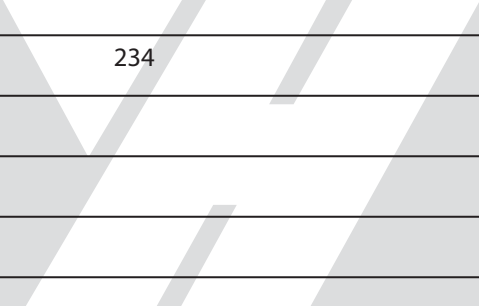
# FLANGES

TABLE S

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 5(127)         | 3 1/2(89)        | 4            | 3/4(19)       | 7/8(22)       | 2 1/2(64)           | 1/16(2)               | -                   | -            | 7/8(22)               |
| 3/4             | 5(127)         | 3 1/2(89)        | 4            | 3/4(19)       | 7/8(22)       | 2 1/2(64)           | 1/16(2)               | -                   | -            | 7/8(22)               |
| 1               | 5 1/2(140)     | 4(102)           | 4            | 3/4(19)       | 7/8(22)       | 3(76)               | 1/16(2)               | -                   | -            | 1(25)                 |
| 1 1/4           | 5 3/4(146)     | 4 1/4(114)       | 4            | 3/4(19)       | 7/8(22)       | 3 1/4(83)           | 1/16(2)               | -                   | -            | 1 1/8(29)             |
| 1 1/2           | 6 1/4(159)     | 4 3/4(121)       | 4            | 3/4(19)       | 7/8(22)       | 3 1/2(89)           | 1/16(2)               | -                   | -            | 1 1/8(29)             |
| 2               | 6 3/4(171)     | 5 1/4(133)       | 8            | 3/4(19)       | 7/8(22)       | 3 1/2(89)           | 1/16(2)               | -                   | -            | 1 1/4(32)             |
| 2 1/2           | 7 1/4(184)     | 5 3/4(146)       | 8            | 3/4(19)       | 7/8(22)       | 4(102)              | 1/16(2)               | -                   | -            | 1 1/4(32)             |
| 3               | 8(203)         | 6 1/2(165)       | 8            | 7/8(22)       | 1(25)         | 4 1/2(114)          | 1/16(2)               | -                   | -            | 1 3/8(35)             |
| 4               | 9 3/4(248)     | 8(203)           | 8            | 1(25)         | 1 1/8(29)     | 6 1/4(159)          | 1/16(2)               | -                   | -            | 1 5/8(41)             |
| 5               | 11 1/4(286)    | 9 1/4(235)       | 12           | 7/8(22)       | 1(25)         | 7 1/2(191)          | 1/16(2)               | -                   | -            | 1 3/4(44)             |
| 6               | 12 3/4(324)    | 10 3/4(273)      | 12           | 1(25)         | 1 1/8(29)     | 8 1/4(210)          | 1/16(2)               | -                   | -            | 2(51)                 |
| 8               | 16 1/4(413)    | 14(356)          | 12           | 1 1/4(32)     | 1 3/8(35)     | 10 3/4(273)         | 1/16(2)               | -                   | -            | 2 1/2(64)             |
| 10              | 19(483)        | 16 3/4(425)      | 16           | 1 1/4(32)     | 1 3/8(35)     | 13(330)             | 1/16(2)               | -                   | -            | 2 7/8(73)             |
| 12              | 22 3/4(578)    | 20(508)          | 16           | 1 1/2(38)     | 1 5/8(41)     | 15(381)             | 1/16(2)               | -                   | -            | 3 1/4(83)             |
| 14              | 25 1/2(648)    | 22 3/4(578)      | 20           | 1 1/2(38)     | 1 5/8(41)     | 17 1/4(438)         | 1/16(2)               | -                   | -            | 3 3/4(95)             |
| 16              | 29 1/4(743)    | 26(660)          | 20           | 1 3/4(44)     | 1 7/8(48)     | 19 1/2(495)         | 1/16(2)               | -                   | -            | 4 1/4(108)            |

TABLE T

| Nominal Size in | Dia. of Flange | Bolt Circle Dia. | No. of Bolts | Dia. of Bolts | Dia. of Holes | Dia. of Raised Face | Height of Raised Face | Thickness of Flange |              |                       |
|-----------------|----------------|------------------|--------------|---------------|---------------|---------------------|-----------------------|---------------------|--------------|-----------------------|
|                 |                |                  |              |               |               |                     |                       | Grey Cast Iron      | Copper Alloy | Cast and Forged Steel |
| 1/2             | 5 1/2(140)     | 4(102)           | 4            | 3/4(19)       | 7/8(22)       | 2 1/2(64)           | 1/16(2)               | -                   | -            | 1(25)                 |
| 3/4             | 5 1/2(140)     | 4(102)           | 4            | 3/4(19)       | 7/8(22)       | 2 1/2(64)           | 1/16(2)               | -                   | -            | 1(25)                 |
| 1               | 5 3/4(146)     | 4 1/4(114)       | 4            | 3/4(19)       | 7/8(22)       | 3(76)               | 1/16(2)               | -                   | -            | 1 1/8(29)             |
| 1 1/4           | 6 1/4(159)     | 4 3/4(121)       | 4            | 7/8(22)       | 1(25)         | 3 1/4(83)           | 1/16(2)               | -                   | -            | 1 1/4(32)             |
| 1 1/2           | 6 3/4(171)     | 5 1/4(133)       | 8            | 3/4(19)       | 7/8(22)       | 3 1/2(89)           | 1/16(2)               | -                   | -            | 1 3/8(35)             |
| 2               | 7 1/4(184)     | 5 3/4(146)       | 8            | 3/4(19)       | 7/8(22)       | 4(102)              | 1/16(2)               | -                   | -            | 1 3/8(35)             |
| 2 1/2           | 8(203)         | 6 1/2(165)       | 8            | 7/8(22)       | 1(25)         | 4 1/2(114)          | 1/16(2)               | -                   | -            | 1 5/8(41)             |
| 3               | 9 1/4(235)     | 7 1/2(191)       | 8            | 1(25)         | 1 1/8(29)     | 5(127)              | 1/16(2)               | -                   | -            | 1 7/8(48)             |
| 4               | 11 1/4(286)    | 9 1/4(235)       | 8            | 1 1/8(29)     | 1 1/4(32)     | 6 1/4(159)          | 1/16(2)               | -                   | -            | 2 1/4(57)             |
| 5               | 12 3/4(324)    | 10 3/4(273)      | 12           | 1 1/8(29)     | 1 1/4(32)     | 8 1/4(210)          | 1/16(2)               | -                   | -            | 2 5/8(67)             |
| 6               | 14 3/4(375)    | 12 1/2(318)      | 12           | 1 1/4(32)     | 1 3/8(35)     | 9(229)              | 1/16(2)               | -                   | -            | 2 7/8(73)             |
| 8               | 18 3/4(476)    | 16(406)          | 12           | 1 1/2(38)     | 1 5/8(41)     | 11 3/4(298)         | 1/16(2)               | -                   | -            | 3 1/2(89)             |
| 10              | 22(559)        | 19 1/4(489)      | 16           | 1 1/2(38)     | 1 5/8(41)     | 14(356)             | 1/16(2)               | -                   | -            | 4 1/4(108)            |
| 12              | 25 3/4(654)    | 22 1/2(572)      | 16           | 1 3/4(44)     | 1 7/8(48)     | 16 1/4(413)         | 1/16(2)               | -                   | -            | 4 3/4(121)            |



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# STAINLESS STEEL PRODUCTS

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## GENERAL INFORMATION OF STAINLESS STEELS

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### RESISTANT TO CORROSION

Addition of the element chromium to steel enhances its resistance to corrosion and staining. Steels that contain a comparatively large amount of chromium are therefore called stainless steels.

Stainless steels have many areas of application, including household articles, building materials, rolling stock as well as chemical and petrochemical equipment. They are produced in a wide variety of types to suit specific end-uses.

### ATTRACTIVENESS EXPANDS APPLICABILITY

Needless to say, the bulk of stainless steels are used where corrosion resistance is a basic requirement. Recently, however, their unique, attractive appearance has come to be widely appreciated, causing a sharp increase in their architectural applications.

### HEAT AND LOW-TEMPERATURE RESISTANCE

Stainless steels retain high strength in elevated temperature service; the austenitic grades, in particular, have superior ductility at low temperatures.

### SHEETS IN GREATEST DEMAND

While stainless steels are produced in various product forms such as sheets, shapes, rods, bars and tubes, those in sheet form presently register by far the fastest gain in market demand. This is because they have come to be used in large quantities as exterior and interior building materials, thus expanding a new market.

### THREE BASIC CLASSES

There are many types of stainless steel. Although different in both mechanical properties and chemical composition, they may be generally grouped into three basic classes.

#### 13 Cr CLASS (MARTENSITIC)

Generally, a group of steels with the carbon content of up to 0.2% and the chromium content ranging between 11.5% and 15.0%. These steels become martensitic in structure when heat treated. Since they are hardenable by heat treatment, steels of this class can be produced in many different degrees of hardness by varying heating conditions. Martensitic stainless steels are hardly corroded in air and are outstanding in hardness, although in weldability and workability they are inferior to the other two classes. Steels of this class are primarily used for the manufacture of cutlery.

#### 18 Cr CLASS(FERRITIC)

Generally, steels having carbon up to 0.2% and chromium between 16.0% and 20.0%. Not hardenable by heat treatment, they are basically used in the annealed condition. These steels are more workable, more weldable, and more corrosion-resistant than martensitic steels. Ferritic steels have good mechanical and physical properties and effectively resist both atmospheric corrosion and strong oxidizing solutions. They are widely used in nitric acid-base chemical processing equipment as well as in buildings, kitchen sinks, and appliances.

#### 18 Cr-8 Ni CLASS (AUSTENITIC)

Generally, steels containing a maximum of 0.15% carbon, a minimum of 16.0% chromium and a minimum of 6.0% nickel. In plain carbon steel, an austenitic structure is formed only at high temperatures. However, stainless steels of this class, because they retain a stable austenite form even at room temperature, are called austenitic stainless steels.

Of the three classes, this has the highest corrosion resistance. Steels of this class combine low yield point with high tensile strength and good elongation, thus offering the best cold-working properties. While it is not possible to change their mechanical properties by heat treatment, their tensile strength and hardness can be greatly enhanced by work hardening. Because of these characteristics, these steels are most extensively used in chemical plants and equipment



# STAINLESS STEEL PRODUCTS

In using these data it should be understood that they are results of laboratory tests and are indicative only of the conditions under which they are run, and may be considered as a basis for recommendation, but not for guarantee. As actual operating conditions can rarely be duplicated in the laboratory it is desirable to test further under actual service. The letters A, B, C, etc., represent the approximate corrosion penetration per month as follows:

## CORROSION RESISTANCE DATA

| substance<br>and condition | temp F  | 304 | 316 | substance<br>and condition          | temp F  | 304 | 316 | substance<br>and condition | temp F  | 304  | 316 |
|----------------------------|---------|-----|-----|-------------------------------------|---------|-----|-----|----------------------------|---------|------|-----|
| Acetic acid                |         |     |     | beer                                |         | A   | A   | 5% agitated                | 70°     | A    | A   |
| 5% agitated                | 70°     | A   | A   | benzene                             | 70°     | A   | A   | 5% aerated                 | 70°     | A    | A   |
| 5% aerated                 | 70°     | A   | A   | benzol                              | hot     | A   | A   | 50% aqueous solution       | hot     | A    | A   |
| 10% agitated               | 70°     | A   | A   | blood (meat juices)                 | cold    | A1  | A   | copper sulphate            |         |      |     |
| 10% aerated                | 70°     | A   | A   | boric acid                          |         |     |     | 5% agitated still          | 70°     | A    | A   |
| 20% agitated               | 70°     | A   | A   | 5% solution                         | 70°     | A   | A   | 5% aerated                 | 70°     | A    | A   |
| 20% aerated                | 70°     | A   | A   | 5% solution hot                     |         |     |     | saturated solution         | boiling | A    | A   |
| 50%                        | 70°     | A   | A   | 5% solution boiling                 |         |     | A1  | creosote (coal tar)        | hot     | A    | A   |
| 50%                        | boiling | C   | B   | saturated solution                  | 70°     | A1  | A1  | creosote oil               | hot     | A    | A   |
| 80%                        | 70°     | A   | A   | boiling                             |         |     | A1  | dyewood liquor             | 70°     | A3   | A   |
| 80%                        | boiling | D   | B   | buttermilk                          | 70°     | A   | A   | ether                      | 70°     | A    | A   |
| 100%                       | 70°     | A   | A   | Calcium carbonate                   | 70°     | A   | A   | ethyl acetate              |         |      |     |
| 100%                       | boiling | C   | B   | calcium chlorate                    |         |     |     | concentrated solution      | 70°     | A    |     |
| 100%-150 lb. pressure      | 400°    | E   | C   | dilute solution                     | 70°     | A   | A   | ethyl chloride             | 70°     | A    | A   |
| acetone                    | boiling | A   | A   | dilute solution                     | hot     | A   | A   | ethylene chloride          | 70°     | A    | A   |
| acetylene                  | 70°     |     |     | calcium chloride                    |         |     |     | ethylene glycol            |         |      |     |
| concentrated               | 70°     | A   |     | dilute solution                     | 70°     | B2  | A1  | concentrated               | 70°     | A    |     |
| commercially pure          | 70°     |     |     | concentrated solution               | 70°     | B2  | A1  | Ferric chloride            |         |      |     |
| alcohol, ethyl             | 70°     | A   | A   | calcium hydroxide                   |         |     |     | 1% solution still          | 70°     | B41  | A1  |
| alcohol, methyl            | boiling | A   | A   | 10%                                 | boiling | A   | A   | boiling                    | D41     | C1   |     |
|                            | boiling | A   | A   | 20%                                 | boiling | A   | A   | 5% solution still          | 70°     | D431 | C1  |
| alum (chrome) 5%           | 70°     | A   | A   | 50%                                 | boiling | C   | B   | 5% agitated                | 70°     | C431 | C1  |
| aluminium chloride         | 150°    | C1  | B   | calcium sulphate                    |         |     |     | 5% aerated                 | 70°     | C431 | C1  |
| aluminium                  | 70°     | A   | A   | saturated                           | 70°     | A   | A   | ferric hydroxide           |         |      |     |
| aluminium sulphate         | 70°     | D   | C   | carbonated water                    |         | A   | A   | (hydrated iron oxide)      | 70°     | A    | A   |
| 10%                        | molten  | E   | E   | carbonic acid                       |         |     |     | ferric nitrate             |         |      |     |
| saturated                  |         |     |     | saturated solution                  | 70°     | A   |     | 1% still                   | 70°     | A    | A   |
| ammonia                    | 70°     | A   | A   | carbon monoxide gas                 | 870°C.  | A   | A   | 5% still                   | 70°     | A    | A   |
| all concentrations         | boiling | B   | A   | carbon tetrachloride                | 760°C.  | A   | A   | 1% agitated                | 70°     | A    | A   |
| gas                        | 70°     | A   | A   | pure                                |         |     |     | 5% agitated                | 70°     | A    | A   |
| ammonia liquor             | boiling | B   | A   | aqueous solution 5-10%              | 70°     | A   | A   | 1% aerated                 | 70°     | A    | A   |
| ammonium chloride          |         |     |     | chloracetic acid                    | 70°     | C1  |     | 5% aerated                 | 70°     | A    | A   |
| 1% still                   | 70°     | A   | A   | chloric acid                        | 70°     | D   | C   | ferric sulphate            |         |      |     |
| aerated                    | hot     | D   |     | chlorinated water                   | 70°     | E   | D   | 1% and 5% still            | 70°     | A1   | A   |
| agitated                   | boiling | A   | A   | saturated                           |         |     |     | aerated or agitated        | 70°     | A1   | A   |
| 10% solution               | 70°     | A   | A   | chlorine gas - dry                  | 70°     | C1  | B1  | ferrous chloride           |         |      |     |
| 28% solution               | 70°     | A   | A   | gas - moist                         | 70°     | C   | B   | saturated solution         | 70°     |      | A   |
| 50% solution               | boiling | A1  | A1  | gas                                 | 70°     | D   | C   | ferrous sulphate           |         |      |     |
| ammonium nitrate           | boiling | B1  | A1  | chloroform                          | 100°C.  | E   | D   | dilute solution            | 70°     | A    | A   |
| all concentrations         | boiling | B1  | A1  | chromic acid                        | 70°     | A   | A   | fluorine                   | 70°     | E    | E   |
| agitated                   |         |     |     | 5%                                  |         |     |     | formaldehyde               |         |      |     |
| aerated                    | 70°     | A   | A   | 10% CP                              | 70°     | A   | A   | 40% solution               |         | A1   | A1  |
| saturated                  | boiling | B1  | A1  | 50% commercial (cont. SO2)          | boiling | C   | B   | formic acid                |         |      |     |
| ammonium phosphate 5%      |         |     |     | cider                               | 70°     | D1  | C   | 5% still                   | 70°     | B    | A   |
| ammonium sulphate          | 70°     | A   | A   | citric acid                         | 70°     | A   | A   | fruit juices               | 150°    | B    | A   |
| 1% & 5% agitated           | 70°     | A   | A   | 5% still                            |         |     |     | fuel oil                   | 70°     | A    | A   |
| aerated                    | boiling | A   | A   | 5% still                            | 70°     | A   | A   | containing sulphuric       | hot     | A    | A   |
| 10%                        | 70°     | A   | A   | 15%                                 | 150°    | A   | A   | gasoline                   |         | C    | B   |
| saturated                  | 70°     | A   | A   | concentrated                        | 70°     | A   | A   | gelatine                   | 70°     | A    | A   |
| ammonium sulphite          | boiling | B1  | A1  | coca cola syrup (pure)              | boiling | B   | A   | gule - dry                 | 70°     | A    | A   |
| aniline 3%                 | boiling | B1  | A1  | coffee                              | 70°     | A   | A   | solution - acid            | 70°     | B1   | A   |
| concentrated crude         | cold    | A   | A   | copper carbonate                    | boiling | A   | A   | glycerine                  | 140°    | B1   | A   |
| barium chloride 5%         | boiling | A   | A   | sat. sol. in 50% NH <sub>3</sub> OH |         |     |     | Hydrachloric acid          | 70°     | A    | A   |
| saturated                  | boiling | A   | A   | copper chloride                     |         | A   | A   | all concentrations         |         | E    | E   |
| aqueous solution           | 70°     | A   | A   | 1% agitated                         | 70°     | B1  | A1  | hydrocyanic acid           | 70°     | A    | A   |
| barium nitrate             | 70°     | A   | A   | 1% aerated                          | 70°     | B1  | A1  | hydrofluosilicic acid      |         | E    | D   |
| aqueous solution           | hot     | B1  | A1  | 5% agitated                         | 70°     | C1  | B1  | hydrogen sulphide          |         |      |     |
| barium sulphate            | 70°     | A   | A   | 5% aerated                          | 70°     | E1  | D1  | dry                        |         | A    | A   |
| barytes-blanc fixe         |         |     |     | copper cyanide                      |         |     |     | wet                        |         | B3   | A3  |
| barium sulphide            |         |     |     | saturated solution                  | boiling | A   | A   | Ink                        |         | B3   | A   |
| saturated solution         | 70°     | A   |     | copper nitrate                      |         |     |     | iodine                     |         | E    | D   |
| solution                   | 70°     |     | A   | 1% still                            | 70°     | A   | A   | iodoform                   |         | A    | A   |
|                            |         |     |     | 1% agitated                         | 70°     | A   | A   | Kerosene                   | 70°     | A    | A   |
|                            |         |     |     | 1% aerated                          | 70°     | A   | A   | ketchup                    | 70°     | A1   | A   |
|                            |         |     |     | 5% still                            | 70°     | A   | A   |                            |         |      |     |

- 1 Subject to pitting at air line or when allowed to dry.
- 2 Keep solutions alkaline.



# STAINLESS STEEL PRODUCTS

fully resistant : A less than .00035 inches  
satisfactorily resistant : B .00035-.0035 inches  
fairly resistant : C .0035-.010 inches  
slightly resistant : D .010-.035 inches  
not resistant : E over .035 inches

| substance<br>and condition         | temp F    | 304 | 316 | substance<br>and condition            | temp F  | 304 | 316 | substance<br>and condition | temp F  | 304 | 316 |
|------------------------------------|-----------|-----|-----|---------------------------------------|---------|-----|-----|----------------------------|---------|-----|-----|
| Lactic acid                        | 70°       | A   | A   | 5% still                              | 70°     | A1  | A1  | sodium sulphate            | 70°     | A   | A   |
| 5%                                 | 150°      | B   | A   | 5% agitated                           | 70°     | A   | A   | 5% still                   | 70°     | A   | A   |
| 10%                                | boiling   | D   | B   | 5% aerated                            | 70°     | A   | A   | all concentrations         | 70°     | A   | A   |
| 10%                                | 150°      | C   | B   |                                       | boiling | A   | A   | sodium sulphide            |         | B1  | A   |
| lard                               | 70°       | A   | A   | potassium ferricyanide                | 70°     | A   | A   | saturated                  |         |     |     |
| lead                               | molten    | B   | B   | 5%                                    |         |     |     | sodium sulphite            | 70°     | A   | A   |
| linseed oil                        | 70°       | A   | A   | potassium ferrocyanide                | 70°     | A   | A   | 5%                         | 150°    | A   | A   |
| Magnesium chloride                 |           |     |     | 5%                                    |         |     |     | 10%                        |         |     |     |
| 1% & 5% still                      | 70°       | A1  | A   | potassium hydroxide                   | 70°     | A   | A   | sodium thiosulphate        | 70°     | A   | A3  |
|                                    | hot       | C1  | B1  | 5% still                              | 70°     | A   | A   | saturated solution         | 70°     |     | A   |
| magnesium sulphate                 | cold, hot | A   | A   | 5% agitated                           | 70°     | A   | A   | acid fixing bath (hypo)    | 70°     |     | A3  |
| malic acid                         | cold, hot | B   | A   | 5% aerated                            | 70°     | A   | A   | 25% solution               | 70°     |     | A3  |
| mayonnaise                         | 70°       | A1  | A   | 27%                                   | boiling | A   | A   |                            | boiling |     |     |
| mercuric chloride                  |           |     |     | 50%                                   | boiling | B   | A   | stearic acid               |         | A   | A   |
| dilute solutions                   |           | E1  | D1  | potassium nitrate                     |         |     |     | sugar juice                |         | A   | A   |
| mercury                            |           | A   | A   | 1% still                              | 70°     | A   | A   | sulphur chloride           |         | E   | D   |
| methanol                           |           |     |     | 1% agitated                           | 70°     | A   | A   | sulphur dioxide            |         |     |     |
| (methyl alcohol)                   |           | A   | A   | 1% aerated                            | 70°     | A   | A   | gas-moist                  | 70°     | B   | A   |
| milk, fresh or sour                | hot, cold | A   | A   | 5% aerated                            | 70°     | A   | A   | gas                        | 575°    | A   | A   |
| mixed acids                        |           |     |     | 5% still & agitated                   | 70°     | A   | A   | sulphur-dry                | molten  | A   | A   |
| 53% H <sub>2</sub> SO <sub>4</sub> | cold      | A   | A   | potassium nitrate                     | hot     | A   | A   | wet                        |         | B1  | A1  |
| 45% HNO <sub>3</sub>               | cold      | A   | A   | potassium oxalate                     |         | A   | A   | sulphuric acid             |         |     |     |
| mustard                            | 70°       | A1  | A1  | potassium permanganate                |         |     |     | 5%                         | 70°     | C   | B   |
| Naphtha                            | 70°       | A   | A   | 5%                                    | 70°     | A   | A   | 5%                         | boiling | E   | C   |
| naphtha - crude                    | 70°       | A   | A   | potassium sulphate                    | 70°     | A   | A   | 10%                        | 70°     | C   | B   |
| nickel chloride                    |           |     |     | 1% still                              | 70°     | A   | A   | 10%                        | boiling | E   | D   |
| solution                           | 70°       | A1  | A1  | 1% agitated                           | 70°     | A   | A   | 50%                        | 70°     | D   | C   |
| nickel sulphate                    | hot, cold | A   | A   | 1% aerated                            | 70°     | A   | A   | 50%                        | boiling | E   | D   |
| nitric acid                        |           |     |     | 5% still                              | 70°     | A   | A   | concentrated               | 70°     | A   | A   |
| 5% solution                        | 70°       | A   | A   | 5% agitated                           | 70°     | A   | A   | concentrated               | boiling | D   | D   |
| 20% solution                       | 70°       | A   | A   | 5% aerated                            | 70°     | A   | A   | concentrated               | 300°    | E   | E   |
| 50% solution                       | 70°       | A   | A   |                                       | hot     | A   | A   | fuming                     | 70°     | C   | B   |
| 50% solution                       | boiling   | A   | A   | potassium sulphide                    |         |     |     | sulphurous acid            |         |     |     |
| 65% solution                       | boiling   | B   | B   | (salt)                                |         | A   | A   | saturated                  | 70°     | C   | B   |
| concentrated                       | 70°       | A   | A   | pyrogalllic acid                      |         | A   | A   | saturated 60 lb.           |         |     |     |
| concentrated                       | boiling   | D   | D   | Rosin                                 |         | A   | A   | pressure                   | 250°    | C   | B   |
| Oils, crude                        | hot, cold | A3  | A3  | Sea water                             | molten  | A1  | A1  | saturated 70/125 lb.       |         |     |     |
| oils, vegetable, mineral           | hot, cold | A3  | A   | silver bromide                        |         | B1  | A1  | pressure                   | 310°    | C   | B   |
| oleic acid                         | 70°       | A1  | A   | silver chloride                       |         | E   | E   | 150 lb. pressure           | 375°    | C   | B   |
| oxalic acid                        |           |     |     | silver nitrate                        |         | A   | A   | sulphurous spray           | 70°     | D1  | D1  |
| 5%                                 | hot, cold | A   | A   | soap                                  | 70°     | A   | A   | Tannic acid                | 70°     | A   | A   |
| 10%                                | 70°       | A   | A   | sodium acetate-moist                  |         | A1  | A   |                            | 150°    | B   | A   |
| 10%                                | boiling   | D   | C   | sodium bicarbonate                    |         |     |     | tin                        | molten  | C   | C   |
| Paraffin                           | hot, cold | A   | A   | all concentrations                    | 70°     | A   | A   | trichloroacetic acid       | 70°     | E   | E   |
| petroleum ether                    |           | A   | A   | 5% still                              | 150°    | A   | A   | trichlorethylene           | 70°     | A1  |     |
| phenol                             |           | A   | A   | sodium bisulphate                     |         |     |     | Varnish                    | 70°     | A   | A   |
| phosphoric acid                    |           |     |     | solution                              | 70°     | A   | A   |                            | hot     | A   | A   |
| 1%                                 | 70°       | A4  | A4  | saturated solution                    | 70°     | E   |     | Vegetable juices           |         | A   | A   |
| 5% still                           | 70°       | A   | A   | 2g + 1gH <sub>2</sub> SO <sub>4</sub> |         |     |     | vinegar fumes              |         | B   | A   |
| 5% agitated                        | 70°       | A   | A   | per liter                             | 68°     |     | A   | vinegar-still              | 70°     | A   | A   |
| 5% aerated                         | 70°       | A   | A   |                                       | 212°    |     | B   | agitated                   |         | A   | A   |
| 10% still                          | 70°       | C   | A   | sodium carbonate                      |         |     |     | aerated                    | 70°     | A   | A   |
| 10% agitated                       | 70°       | C   | B   | 5%                                    | 70°     | A   | A   | Zinc                       | molten  | E   | E   |
| 10% aerated                        | 70°       | C   | B   | 5%                                    | 150°    | A   | A   | zinc chloride              |         |     |     |
| picric acid                        | 70°       | A   | A   | sodium chloride                       |         |     |     | 5% still                   | 70°     | A1  | A1  |
| potassium bichromate               | 70°       | A   | A   | 5% still                              | 70°     | A1  | A   |                            | boiling | B1  | B1  |
| potassium bromide                  | 70°       | A   | A   | 5% still                              | 150°    | A1  | A   | zinc sulphate              |         |     |     |
| potassium carbonate                | 70°       | B1  | A1  | 20% aerated                           | 70°     | A1  | A   | 5%                         | 70°     | A   | A   |
| 1% still                           | 70°       | A   | A   | saturated                             | 70°     | A1  | A   | saturated                  | 70°     | A   | A   |
| 1% agitated                        | 70°       | A   | A   | saturated                             | boiling | B1  | A   | 25%                        | boiling | A   | A   |
| 1% aerated                         | 70°       | A   | A   | sodium fluoride                       |         |     |     |                            |         |     |     |
| potassium carbonate                | hot       | A   | A   | 5% solution                           |         | B1  | A1  |                            |         |     |     |
| potassium chlorate                 |           | A   | A   | sodium hyroxide                       |         | A   | A   |                            |         |     |     |
| potassium chloride                 |           |     |     | sodium hypochlorite                   |         |     |     |                            |         |     |     |
| 1% still                           | 70°       | A1  | A1  | 5% still                              |         | B1  | A1  |                            |         |     |     |
| 1% agitated                        | 70°       | A   | A   | sodium hyposulphite                   | 70°     | A3  | A   |                            |         |     |     |
| 1% aerated                         | 70°       | A   | A   | sodium nitrate                        | fused   | C   | B   |                            |         |     |     |

3. May attack when sulphuric acid is present.  
4. May attack when hydrochloric acid is present.

# STAINLESS STEEL COILS & SHEETS

## SPECIFICATIONS

| Application          |             | Standard                                             |                                                                           |                                                                                                                              |                                                                                                                     |
|----------------------|-------------|------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|                      |             | AISI                                                 | JIS                                                                       | BS                                                                                                                           | DIN                                                                                                                 |
| Stainless Steel      | Austenitic  | 301,302<br>304, 304L<br>316, 316L<br>321, 347<br>385 | SUS 301, 302<br>SUS 304, 304L<br>SUS 316, 316L<br>SUS 321, 347<br>SUS 385 | 301S21, 302S17<br>302S25<br>304S15, 304S12<br>304S16<br>316S16, 316S12<br>316S49, 316S37<br>321S12, 347S17<br>321S49, 347S49 | X12CrNi177<br>X12CrNi188<br>X5CrNi1810<br>X2CrNi1911<br>X5CrNiMo17122<br>X2CrNiMo17132<br>X6CrNiMoTi1810<br>X6CrNi1 |
|                      | Ferritic    | 405<br>409<br>430                                    | SUS 405<br>SUS 409<br>SUS 430                                             | 405S17<br>409S17<br>430S15                                                                                                   | X6CrAl13<br>X6CrTi12<br>X6Cr17                                                                                      |
|                      | Martensitic | 403,410<br>420, 420J2<br>431,501<br>502              | SUS 403, 410<br>SUS 420J1, 420J2<br>SUS 431                               | 430S17, 410S21<br>420S29, 420S45<br>431S29                                                                                   | X15Cr13, X6Cr13<br>X20Cr13, X46Cr13<br>X22CrNi117                                                                   |
| Heat Resisting Steel |             | 314,446<br>H-46, 661                                 | SUH 446<br>SUH 600, 661                                                   |                                                                                                                              | X15CrNiSi2520                                                                                                       |

YICK HOE

# STAINLESS STEEL SHEETS & PLATES

## SPECIFICATIONS: CHEMICAL COMPOSITION OF STAINLESS STEEL SHEETS

| Type of Structure      | Type of Composition          | Corresponding Designation |      |      | Chemical Composition, % |         |            |        |        |             |             |             | Other Elements   |
|------------------------|------------------------------|---------------------------|------|------|-------------------------|---------|------------|--------|--------|-------------|-------------|-------------|------------------|
|                        |                              | JIS                       | AISI | DIN  | C                       | Si max. | Mn         | P max. | S max. | Ni          | Cr          | Mo          |                  |
| Austenite              | 17Cr-5Ni-7Mn                 | SUS 201                   | 201  | --   | 0.15 max.               | 1.00    | 5.5-7.5    | 0.06   | 0.03   | 3.5 - 5.5   | 16.0 - 18.0 | --          | N: 0.25 max.     |
|                        | 18Cr-6Ni-10Mn                | SUS 202                   | 202  | --   | 0.15 max.               | 1.00    | 7.5 - 10.0 | 0.06   | 0.03   | 4.0-5.5     | 16.0 - 18.0 | --          | N: 0.25 max.     |
|                        | 17Cr-7Ni                     | SUS 301                   | 301  | 4310 | 0.15 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 6.0-8.0     | 16.0 - 18.0 | --          | --               |
|                        | 18Cr-8Ni-high C              | SUS 302                   | 302  | 4300 | 0.15 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 8.0 - 10.0  | 17.0 - 19.0 | --          | --               |
|                        | 18Cr-8Ni                     | SUS 304                   | 304  | 4301 | 0.08 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 8.0 - 10.5  | 18.0 - 20.0 | --          | --               |
|                        | 18Cr-8Ni -extra-low C        | SUS 304L                  | 304L | 4306 | 0.03 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 9.0 - 13.0  | 18.0 - 20.0 | --          | --               |
|                        | 18Cr-12Ni                    | SUS 305                   | 305  | 3955 | 0.12 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 10.5 - 13.0 | 17.0 - 19.0 | --          | --               |
|                        | 23Cr-12Ni                    | SUS 309S                  | 309S | 4845 | 0.03 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 12.0-15.0   | 22.0 - 24.0 | --          | --               |
|                        | 25Cr-20Ni                    | SUS 310S                  | 310S | --   | 0.08 max.               | 1.50    | 2.00       | 0.04   | 0.03   | 19.0-22.0   | 24.0 - 26.0 | --          | --               |
|                        | 18Cr-12Ni-2.5Mo              | SUS 316                   | 316  | 4401 | 0.08 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 10.0-14.0   | 16.0 - 18.0 | 2.0-3.0     | --               |
|                        | 18Cr-12Ni-7.5 Mo-extra-low C | SUS 316L                  | 316L | 4404 | 0.03 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 12.0 - 15.0 | 16.0 - 18.0 | 2.0-3.0     | --               |
|                        | 18Cr-12Ni-2Mo -2Cu           | SUS 316J1                 |      | 4505 | 0.03 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 10.0 - 14.0 | 17.0 - 19.0 | 1.2 - 2.75  | Cu: 1.0-2.5      |
|                        | 18Cr-13Ni-3.5Mo              | SUS 317                   | 317  | 4402 | 0.03 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 18.0-15.0   | 19.0 - 20.0 | 3.0-4.0     | --               |
|                        | 18Cr-13Ni-3.5 Mo-extra-low C | SUS 317L                  | 317L | --   | 0.03 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 11.0 - 15.0 | 18.0 - 20.0 | 3.0 - 4.0   | --               |
|                        | 18Cr-8Ni-Ti                  | SUS 321                   | 321  | 4541 | 0.03 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 9.0 - 13.0  | 17.0 - 19.0 | --          | Ti: 5 x C% min   |
|                        | 18Cr-9Ni-Nb                  | SUS 347                   | 347  | 4550 | 0.03 max.               | 1.00    | 2.00       | 0.04   | 0.03   | 9.0-13.0    | 17.0 - 19.0 | --          | Nb+Ta: 10xC% min |
| Ferrite                | 13Cr-Al                      | SUS 405                   | 405  | 4002 | 0.03 max.               | 1.00    | 1.00       | 0.04   | 0.03   | 0.60 max.   | 11.5 - 16.0 | --          | Al: 0.1-0.3      |
|                        | 16Cr                         | SUS 429                   | 429  | 4009 | 0.12 max.               | 1.00    | 1.00       | 0.04   | 0.03   | --          | 14.0 - 16.0 | --          | --               |
|                        | 18Cr                         | SUS 430                   | 430  | 4016 | 0.12 max.               | 0.75    | 1.00       | 0.04   | 0.03   | 0.60 max.   | 16.0 - 18.0 | --          | --               |
|                        | 18Cr-Mo                      | SUS 434                   | 434  | 4113 | 0.12 max.               | 1.00    | 1.00       | 0.04   | 0.03   | --          | 16.0 - 18.0 | 0.75 - 1.25 | --               |
| Martensite             | 13Cr-low Si                  | SUS 403                   | 403  | 4024 | 0.15 max.               | 0.50    | 1.00       | 0.04   | 0.03   | 0.60 max.   | 11.5 - 13.0 | --          | --               |
|                        | 13Cr                         | SUS 410                   | 410  | 4000 | 0.15 max.               | 1.00    | 1.00       | 0.04   | 0.03   | 0.60 max.   | 11.5 - 13.5 | --          | --               |
|                        | 13Cr-high C                  | SUS 420J2                 | 420  | 4021 | 0.26-0.40               | 1.00    | 1.00       | 0.04   | 0.03   | --          | 12.0 - 14.0 | --          | --               |
|                        | 18Cr-high C                  | SUS 440A                  | 440A | --   | 0.6 - 0.75              | 1.00    | 1.00       | 0.04   | 0.03   | --          | 16.0 - 18.0 | 0.75 max.   | --               |
| Precipitation Hardened | 17Cr-7Ni-1Al                 | SUS 631                   | 631  | --   | 0.09 max.               | 1.00    | 1.00       | 0.04   | 0.03   | 6.5-7.75    | 16.0-18.0   | --          | Al: 0.75-1.5     |

# STAINLESS STEEL SHEETS & PLATES

## SPECIFICATIONS: MECHANICAL PROPERTIES OF STAINLESS STEEL SHEETS

| Type of Structure      | Type of Composition           | Corresponding Designation |      |                     | Mechanical Properties |                      |        |        |                    |                       |                       |                                      |
|------------------------|-------------------------------|---------------------------|------|---------------------|-----------------------|----------------------|--------|--------|--------------------|-----------------------|-----------------------|--------------------------------------|
|                        |                               |                           |      |                     | Tensile Test          |                      |        |        |                    | Hardness Test         |                       | Bend Test<br>180° Radius<br>(t, min) |
|                        |                               | Tensile Strength min.     |      | Yield Strength min. |                       | Elongation %<br>min. |        |        |                    |                       |                       |                                      |
|                        |                               | JIS                       | AISI | DIN                 | kg/mm²                | psi                  | kg/mm² | psi    | JIS No.13 Specimen | Rockwell B Scale max. | Vickers Hardness max. |                                      |
| Austenite              | 17Cr-5Ni-7Mn                  | SUS 201                   | 201  | --                  | 65                    | 93,000               | 25     | 35,800 | 40                 | 100                   | 253                   | --                                   |
|                        | 18Cr-6Ni-10Mn                 | SUS 202                   | 202  | --                  | 60                    | 85,500               | 25     | 35,800 | 40                 | 95                    | 218                   | --                                   |
|                        | 17Cr-7Ni                      | SUS 301                   | 301  | 4310                | 53                    | 75,800               | 21     | 30,000 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-8Ni-high C               | SUS 302                   | 302  | 4300                | 53                    | 75,800               | 21     | 30,000 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-8Ni                      | SUS 304                   | 304  | 4301                | 53                    | 75,800               | 21     | 30,000 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-8Ni<br>-extra -low C     | SUS 304L                  | 304L | 4306                | 49                    | 69,500               | 18     | 25,500 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-12Ni                     | SUS 305                   | 305  | 3955                | 49                    | 69,500               | 18     | 25,500 | 40                 | 90                    | 200                   | --                                   |
|                        | 23Cr-12Ni                     | SUS 309S                  | 309S | 4845                | 53                    | 75,800               | 21     | 30,000 | 40                 | 90                    | 200                   | --                                   |
|                        | 25Cr-20Ni                     | SUS 310S                  | 310S | --                  | 53                    | 75,800               | 21     | 30,000 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-12Ni-2.5Mo               | SUS 316                   | 316  | 4401                | 53                    | 75,800               | 21     | 30,000 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-12Ni-7.5 Mo-extra-low C  | SUS 316L                  | 316L | 4404                | 49                    | 69,500               | 18     | 25,500 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-12Ni-2Mo -2Cu            | SUS 316J1                 |      | 4505                | 53                    | 75,800               | 21     | 30,000 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-13Ni-3.5Mo               | SUS 317                   | 317  | 4402                | 53                    | 75,800               | 21     | 30,000 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-13Ni-3.5 Mo-extra- low C | SUS 317L                  | 317L | --                  | 49                    | 69,500               | 18     | 25,800 | 40                 | 90                    | 200                   | --                                   |
|                        | 18Cr-8Ni-Ti                   | SUS 321                   | 321  | 4541                | 53                    | 75,800               | 21     | 30,000 | 40                 | 90                    | 200                   | --                                   |
| 18Cr-9Ni-Nb            | SUS 347                       | 347                       | 4550 | 53                  | 75,800                | 21                   | 30,000 | 40     | 90                 | 200                   | --                    |                                      |
| Ferrite                | 13Cr-Al                       | SUS 405                   | 405  | 4002                | 42                    | 60,000               | 18     | 25,500 | 20                 | 88                    | 200                   | t < 8.05 t                           |
|                        | 16Cr                          | SUS 429                   | 429  | 4009                | 46                    | 65,800               | 21     | 30,000 | 22                 | 88                    | 200                   | 1.0 t                                |
|                        | 18Cr                          | SUS 430                   | 430  | 4016                | 46                    | 65,800               | 21     | 30,000 | 22                 | 88                    | 200                   | 1.0 t                                |
|                        | 18Cr-Mo                       | SUS 434                   | 434  | 4113                | 46                    | 65,800               | 21     | 30,000 | 22                 | 88                    | 200                   | 1.0 t                                |
| Martensite             | 13Cr-low Si                   | SUS 403                   | 403  | 4024                | 45                    | 64,000               | 21     | 30,000 | 20                 | 88                    | 200                   | 1.0 t                                |
|                        | 13Cr                          | SUS 410                   | 410  | 4000                | 45                    | 64,000               | 21     | 30,000 | 20                 | 88                    | 200                   | 1.0 t                                |
|                        | 13Cr-high C                   | SUS 420J2                 | 420  | 4021                | 55                    | 78,500               | 23     | 32,750 | 18                 | 93                    | 210                   | --                                   |
|                        | 18Cr-high C                   | SUS 440A                  | 440A | --                  | 60                    | 85,500               | 25     | 35,800 | 15                 | 97                    | 230                   | --                                   |
| Precipitation Hardened | 17Cr-7Ni-1Al                  | SUS 631                   | 631  | --                  | 105                   | 148,000              | 39     | 55,500 | 20                 | 92                    | 200                   | --                                   |

# STAINLESS STEEL SHEETS & PLATES

## MAIN USES

| JIS Designation | AISI Designation | Characteristics and Main Uses                                                                                                                                                                                           |
|-----------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SUS 410         | 410              | The most common grade of economical 13 Cr steels. In the annealed condition, this grade may be drawn or formed. Used for machine structures, home appliances, etc.                                                      |
| SUS 403         | 403              | Suited for steam turbine blades, jet engines, compressor blades and other applications involving high stresses.                                                                                                         |
| SUS 405         | 405              | Possesses corrosion resistance equivalent to that of 13 Cr steel. Used in applications involving welding. Machine structures, food-processing machinery, chemical equipment, etc.                                       |
| SUS 420 J2      | 420              | High carbon content results in high abrasion resistance. Greater hardness may be obtained by heat treatment. Used for cutlery, machine parts, etc.                                                                      |
| SUS 434         | 434              | Designed to resist atmospheric corrosion especially for use as automotive trim. Used for wheel covers, hub caps, grilles, mouldings, etc.                                                                               |
| SUS 430         | 430              | The most common and widely used of the ferritic grades. This steel may be drawn or formed and has high corrosion resistance. Used for office and kitchen equipment, buildings, automobiles, etc.                        |
| SUS 301         | 301              | This grade can be cold worked to very high strength levels. Corrosion resistance is somewhat inferior to SUS 304. Used for rolling stock, automobiles, home appliances, etc.                                            |
| SUS 302         | 302              | More resistant to corrosion than SUS 301. Suited for oil, chemical, paper and food-processing machinery, dairy installations, buildings, kitchen equipment and utensils, etc.                                           |
| SUS 304         | 304              | Slightly more resistant to general corrosion than SUS 302. Suited for the same range of applications as SUS 302.                                                                                                        |
| SUS 304L        | 304L             | A special SUS 304 grade for use where carbide precipitation after welding is undesirable. Used for chemical and petrochemical equipment, nuclear reactor equipment, oil-refining equipment, etc.                        |
| SUS 321         | 321              | Suited for use in a 500° to 850°C temperature range in which carbides tend to precipitate and in applications where stress relieving (or annealing) is not feasible. Used in the chemical industry, etc.                |
| SUS 316         | 316              | High resistant to such reducing acids as sulphuric, sulphurous, acetic and hydrochloric acids. Also, high creep strength at elevated temperatures. Used in applications in the paper, textile, and chemical industries. |
| SUS 316L        | 316L             | A special SUS 316 grade intended for use where carbide precipitation after welding is undesirable.                                                                                                                      |
| SUS 347         | 347              | Suited for the same applications as SUS 321. Widely used for the construction of manufacturing equipment involving welding.                                                                                             |

# STAINLESS STEEL SHEETS & PLATES

## FINISHES

| Finish Designation | Finishing Method and Degree of Finish                                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No. 1              | Hot-rolled, annealed and descaled. Generally used in industrial applications, where heat or corrosion resistance is required but surface smoothness is not particularly important. |
| No. 2D             | Dull, smooth finish produced by cold rolling, annealing and pickling. Steel is soft and has a silver white surface, suited for deep drawing as well as general uses.               |
| No. 2B             | Bright, smooth cold-rolled finish obtained by skinpass rolling of No. 2D. Specified for general use.                                                                               |
| No. 3              | Intermediate polished finish obtained with rather coarse abrasives (#100 to #120 mesh). A higher luster can be attained by further polishing after fabrication.                    |
| No. 4              | A standard polished finish produced with finer abrasives (#150 to #180 mesh). Widely specified for restaurant and kitchen equipment and dairy processing equipment.                |
| #240               | Polish finish of about #240. Used when finer finish than No. 4 finish is desired such as equipment for restaurants and household kitchens.                                         |
| #320               | Polishing finish of about #320.                                                                                                                                                    |
| #400               | High luster finish produced by polishing No. 2B first with #400 buffing iron and then with buffing cloth. Suited for general use.                                                  |
| H.L.               | Special polished finish having continuous lines produced with abrasives of an appropriate grain size. Mainly used in architectural applications.                                   |
| BA                 | Highly reflective finish produced by cold rolling, bright annealing and-temper rolling. Used where a lustrous surface is specially required.                                       |

### Note:

Upon request, strip surface protection is provided such as bonding of vinyl sheets or vinyl coating.

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# STAINLESS STEEL SHEETS & PLATES

## Sizes and Unit Weights

| Size      | 1219 x 2438 (4'x 8') |        |                |        | 1524 x 3048 (5'x 10') |        |                |        |
|-----------|----------------------|--------|----------------|--------|-----------------------|--------|----------------|--------|
| AISI      | Type 304, 304L       |        | Type 316, 316L |        | Type 304, 304L        |        | Type 316, 316L |        |
| Th / Wgt. | kg/pc                | pcs/mt | kg/pc          | pcs/mt | kg/pc                 | pcs/mt | kg/pc          | pcs/mt |
| 0.3       | 7.07                 | 141.4  | 7.11           | 140.70 | 11.06                 | 90.42  | 11.13          | 89.85  |
| 0.4       | 9.42                 | 106.2  | 9.48           | 105.50 | 14.75                 | 67.80  | 14.84          | 67.39  |
| 0.45      | 10.60                | 94.34  | 10.67          | 93.72  | 16.59                 | 60.28  | 16.70          | 59.88  |
| 0.5       | 11.78                | 84.89  | 11.85          | 84.39  | 18.44                 | 54.23  | 18.56          | 53.88  |
| 0.55      | 12.95                | 77.22  | 13.04          | 76.69  | 20.28                 | 49.31  | 20.41          | 49.00  |
| 0.6       | 14.13                | 70.77  | 14.22          | 70.32  | 22.12                 | 45.21  | 22.27          | 44.90  |
| 0.7       | 16.49                | 60.64  | 16.59          | 60.28  | 25.81                 | 38.74  | 25.98          | 38.49  |
| 0.8       | 18.84                | 53.08  | 18.96          | 52.74  | 29.50                 | 33.90  | 29.69          | 33.68  |
| 0.9       | 21.20                | 47.17  | 21.33          | 46.88  | 33.18                 | 30.14  | 33.40          | 29.94  |
| 1         | 23.55                | 42.46  | 23.70          | 42.19  | 36.87                 | 27.12  | 37.11          | 26.95  |
| 1.2       | 28.26                | 35.39  | 28.44          | 35.16  | 44.24                 | 22.60  | 44.53          | 22.46  |
| 1.5       | 35.33                | 28.30  | 35.55          | 28.13  | 55.31                 | 18.08  | 55.67          | 17.96  |
| 2         | 47.10                | 21.23  | 47.40          | 21.10  | 73.74                 | 13.56  | 74.22          | 13.47  |
| 2.3       | 54.17                | 18.46  | 54.51          | 18.35  | 84.80                 | 11.79  | 85.35          | 11.72  |
| 2.5       | 58.88                | 16.98  | 59.25          | 16.88  | 92.18                 | 10.85  | 92.78          | 10.78  |
| 3         | 70.65                | 14.15  | 71.10          | 14.06  | 110.6                 | 9.04   | 111.33         | 8.98   |
| 4         | 94.20                | 10.62  | 94.80          | 10.55  | 147.5                 | 6.78   | 148.4          | 6.74   |
| 4.5       | 106.0                | 9.44   | 106.7          | 9.38   | 165.9                 | 6.03   | 167.0          | 5.99   |
| 5         | 117.8                | 8.49   | 118.5          | 8.44   | 184.4                 | 5.42   | 185.6          | 5.39   |
| 6         | 141.3                | 7.08   | 142.2          | 7.03   | 221.2                 | 4.52   | 222.7          | 4.49   |
| 7.5       | 176.6                | 5.66   | 177.8          | 5.63   | 276.5                 | 3.62   | 278.3          | 3.59   |
| 8         | 188.4                | 5.31   | 189.6          | 5.27   | 295.0                 | 3.39   | 296.9          | 3.37   |
| 9         | 212.0                | 4.72   | 213.3          | 4.69   | 331.9                 | 3.01   | 334.0          | 2.99   |
| 10        | 235.5                | 4.25   | 237.0          | 4.22   | 368.7                 | 2.71   | 371.1          | 2.69   |
| 12        | 282.6                | 3.54   | 284.4          | 3.52   | 442.4                 | 2.26   | 445.3          | 2.25   |
| 15        | 353.3                | 2.83   | 355.5          | 2.81   | 553.1                 | 1.81   | 556.7          | 1.80   |
| 18        | 423.9                | 2.36   | 426.6          | 2.34   | 663.7                 | 1.51   | 668.0          | 1.50   |
| 20        | 471.0                | 2.12   | 474.0          | 2.11   | 737.4                 | 1.36   | 742.2          | 1.35   |
| 22        | 518.1                | 1.93   | 521.4          | 1.92   | 811.2                 | 1.23   | 816.4          | 1.22   |
| 25        | 588.8                | 1.70   | 592.5          | 1.69   | 921.8                 | 1.08   | 927.8          | 1.08   |
| 30        | 706.5                | 1.42   | 711.0          | 1.41   | 1,106.0               | 0.90   | 1,113.0        | 0.90   |
| 32        | 753.6                | 1.33   | 758.4          | 1.32   | 1,180.0               | 0.85   | 1,188.0        | 0.84   |
| 38        | 894.9                | 1.12   | 900.6          | 1.11   | 1,401.0               | 0.71   | 1,410.0        | 0.71   |
| 40        | 942.0                | 1.06   | 948.0          | 1.05   | 1,475.0               | 0.68   | 1,484.0        | 0.67   |
| 44        | 1,036.0              | 0.97   | 1,043.0        | 0.96   | 1,622.0               | 0.62   | 1,633.0        | 0.61   |
| 50        | 1,178.0              | 0.85   | 1,185.0        | 0.84   | 1,844.0               | 0.54   | 1,856.0        | 0.54   |
| 65        | 1,531.0              | 0.65   | 1,541.0        | 0.65   | 2,397.0               | 0.42   | 2,412.0        | 0.41   |
| 75        | 1,766.0              | 0.57   | 1,778.0        | 0.56   | 2,765.0               | 0.36   | 2,783.0        | 0.36   |

# STAINLESS STEEL ANGLES & FLATS

Stainless Steel Angles and Flats are produced by hot-rolling.

After fully annealed, they are subjected to special chemical treatment for descaling, which gives the products the following advantages:

1. Clean and silverly coloured surface.
2. Smooth and tight skinned surface by this descaling process, which enables to achieve fine surface finish by polishing.
3. Excellent corrosion resistance and bending quality by full annealing.

Hot rolled stainless steel angles and flats are superior to cold formed products with regard to minimum internal stress, better corrosion resistance and lower cost.

For applications where the corrosion resistance after welding is essential, Extra Low Carbon Stainless Steel Angles and Flats (Ex. AISI 304 L, 316 L) is particularly recommended.

CHEMICAL COMPOSITION AISI, ASTM A276-78

| Type | C max. | Mn max. | P max. | S max. | Si max. | Cr          | Ni          | Mo        | Ti         |
|------|--------|---------|--------|--------|---------|-------------|-------------|-----------|------------|
| 302  | 0.15   | 2.0     | 0.045  | 0.03   | 1.0     | 17.0 - 19.0 | 8.0 - 10.0  | --        |            |
| 304  | 0.08   | 2.0     | 0.045  | 0.03   | 1.0     | 18.0 - 20.0 | 8.0 - 10.5  | --        |            |
| 304L | 0.03   | 2.0     | 0.045  | 0.03   | 1.0     | 18.0 - 20.0 | 8.0 - 12.0  | --        |            |
| 316  | 0.08   | 2.0     | 0.045  | 0.03   | 1.0     | 16.0 - 18.0 | 10.0 - 14.0 | 2.0 - 3.0 |            |
| 316L | 0.03   | 2.0     | 0.045  | 0.03   | 1.0     | 16.0 - 18.0 | 10.0 - 14.0 | 2.0 - 3.0 |            |
| 321  | 0.08   | 2.0     | 0.045  | 0.03   | 1.0     | 17.0 - 19.0 | 9.0 - 12.0  | --        | 5 x C min. |
| 430  | 0.12   | 1.0     | 0.040  | 0.03   | 1.0     | 16.0 - 18.0 | --          | --        |            |



# STAINLESS STEEL ANGLES & FLATS

## TYPICAL CHARACTERISTICS FOR CHEMICAL COMPOSITION

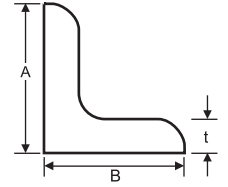
| AISI | Typical Characteristics                                                                                                                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 302  | Basic, general purpose type of non-magnetic, austenite stainless steel with good corrosion resistance and mechanical properties.<br>Retains ductility and impact strength at low temperature.<br>Good cold workability.                                                                                                                       |
| 304  | Low carbon variation of TYPE 302.<br>Superior corrosion resistance and weldability to TYPE 302.<br>Widely used for general purposes as well as parts fabricated by welding.                                                                                                                                                                   |
| 304L | Extra low carbon eliminates the harmful carbide precipitation due to welding.<br>Suitable for applications where post-weld annealing is unpracticable.                                                                                                                                                                                        |
| 316  | Additional of Molybdenum strengthens corrosion resistance to reducing acids such as sulphuric, acetic, phosphoric and chloric acid.<br>Improved pitting resistance and superior creep strength at high temperature to TYPE 302, 304.<br>Used for various chemical industries such as pulp-, photograph-, dye-, textile-, fertilizer industry. |
| 316L | Extra low carbon eliminates the harmful carbide precipitation due to welding.<br>Suitable for applications where post-weld annealing is unpracticable and which are used in severely corrosive atmosphere.                                                                                                                                    |
| 321  | Addition of Titanium prevents intergranular corrosion due to welding.<br>Strong creep strength at high temperature.<br>Can be used at high temperature (800 - 1500°F).                                                                                                                                                                        |
| 430  | Typical ferritic stainless steel, magnetic.<br>Superior corrosion resistance to TYPE 410.<br>Suitable for applications used in weak corrosive atmosphere, kitchen equipment and utensils.                                                                                                                                                     |

## MECHANICAL PROPERTIES ASTM A276-78

| Type | Condition | Size | Tension Test            |               |                     |          |
|------|-----------|------|-------------------------|---------------|---------------------|----------|
|      |           |      | Y.S<br>(0.2%)<br>p.s.i. | T.S<br>p.s.i. | EL.<br>(2 in.)<br>% | R.A<br>% |
|      |           |      | min.                    | min.          | min.                | min.     |
| 302  | annealed  | all  | 30,000                  | 75,000        | 40                  | 50       |
| 304  | annealed  | all  | 30,000                  | 75,000        | 40                  | 40       |
| 304L | annealed  | all  | 25,000                  | 70,000        | 40                  | 50       |
| 316  | annealed  | all  | 30,000                  | 75,000        | 40                  | 50       |
| 316L | annealed  | all  | 25,000                  | 70,000        | 40                  | 50       |
| 321  | annealed  | all  | 30,000                  | 75,000        | 40                  | 50       |
| 430  | annealed  | all  | 40,000                  | 70,000        | 20                  | 45       |

# STAINLESS STEEL ANGLES & FLATS

## ANGLE (Sizes and Unit Weights)



### IMPERIAL UNITS: APPROX. WT. LBS/FT

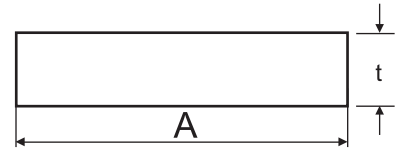
| Legs<br>(A x B) in. | Thickness (t) in. |       |      |      |      |       |
|---------------------|-------------------|-------|------|------|------|-------|
|                     | 1/8               | 3/16  | 1/4  | 5/16 | 3/8  | 1/2   |
| 3/4 x 3/4           | 0.599             | 0.856 |      |      |      |       |
| 1 x 1               | 0.814             | 1.18  | 1.52 |      |      |       |
| 1 1/4 x 1 1/4       | 1.03              | 1.51  | 1.96 |      |      |       |
| 1 1/2 x 1 1/2       | 1.25              | 1.83  | 2.39 |      |      |       |
| 2 x 2               | 1.68              | 2.49  | 3.26 | 4.00 | 4.73 |       |
| 2 1/2 x 2 1/2       |                   | 3.13  | 4.14 | 5.08 | 6.02 |       |
| 3 x 3               |                   |       | 5.01 | 6.19 | 7.34 | 9.57  |
| 3 1/2 x 3 1/2       |                   |       | 5.88 | 7.27 | 8.63 | 11.30 |
| 4 x 4               |                   |       | 6.75 | 8.35 | 9.95 | 13.00 |

### METRIC UNITS: APPROX. WT. KG/M

| Legs<br>(A x B) mm | Thickness (t) mm |      |      |      |      |      |      |      |      |      |      |
|--------------------|------------------|------|------|------|------|------|------|------|------|------|------|
|                    | 3                | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   | 13   |
| 20 x 20            | 0.908            | 1.16 | 1.41 |      |      |      |      |      |      |      |      |
| 25 x 25            | 1.15             | 1.48 | 1.81 | 2.12 |      |      |      |      |      |      |      |
| 30 x 30            | 1.39             | 1.80 | 2.21 | 2.58 |      |      |      |      |      |      |      |
| 35 x 35            | 1.64             | 2.13 | 2.62 | 3.00 |      |      |      |      |      |      |      |
| 40 x 40            | 1.88             | 2.45 | 3.03 | 3.56 |      |      |      |      |      |      |      |
| 50 x 50            | 2.41             | 3.14 | 3.87 | 4.55 |      |      | 6.60 | 7.26 |      |      |      |
| 60 x 60            |                  |      | 4.68 | 5.53 | 6.39 |      |      |      |      |      |      |
| 65 x 65            |                  |      | 5.13 | 6.07 | 6.98 | 7.87 | 8.78 | 9.67 |      |      |      |
| 70 x 70            |                  |      |      | 6.55 | 7.54 | 8.51 | 9.50 | 10.5 |      |      |      |
| 75 x 75            |                  |      |      | 7.03 | 8.11 | 9.16 | 10.2 | 11.3 | 12.3 | 13.3 | 14.3 |
| 80 x 80            |                  |      |      | 7.52 | 8.67 | 9.80 | 11.0 | 12.0 | 13.2 | 14.3 | 15.4 |
| 90 x 90            |                  |      |      | 8.53 | 9.85 | 11.1 | 12.5 | 13.7 | 15.0 | 16.2 | 17.5 |
| 100 x 100          |                  |      |      | 9.50 | 11.0 | 12.4 | 13.9 | 15.3 | 16.7 | 18.2 | 19.5 |

# STAINLESS STEEL ANGLES & FLATS

## FLAT (Sizes and Unit Weights)



IMPERIAL UNIT: APPROX. WT. LBS/FT

| Width<br>(A) in. | Thickness (t) in. |       |       |       |       |      |      |       |       |
|------------------|-------------------|-------|-------|-------|-------|------|------|-------|-------|
|                  | 1/8               | 3/16  | 1/4   | 5/16  | 3/8   | 1/2  | 5/8  | 3/4   | 1     |
| 1/2              | 0.217             | 0.326 |       |       |       |      |      |       |       |
| 5/8              | 0.272             | 0.408 | 0.543 |       |       |      |      |       |       |
| 3/4              | 0.326             | 0.489 | 0.652 | 0.816 | 0.979 | 1.30 |      |       |       |
| 1                | 0.435             | 0.652 | 0.87  | 1.09  | 1.30  | 1.74 | 2.18 | 2.61  |       |
| 1 1/4            | 0.544             | 0.816 | 1.09  | 1.36  | 1.63  | 2.18 | 2.72 | 3.26  |       |
| 1 1/2            | 0.653             | 0.979 | 1.30  | 1.63  | 1.96  | 2.61 | 3.26 | 3.92  | 5.22  |
| 1 3/4            | 0.761             | 1.14  | 1.52  | 1.90  | 2.28  | 3.04 |      |       | 6.09  |
| 2                | 0.87              | 1.30  | 1.74  | 2.18  | 2.61  | 3.48 | 4.35 | 5.22  | 6.96  |
| 2 1/2            | 1.09              | 1.63  | 2.18  | 2.72  | 3.26  | 4.35 | 5.44 | 6.52  | 8.70  |
| 3                |                   | 1.96  | 2.61  | 3.26  | 3.91  | 5.22 | 6.52 | 7.83  | 10.40 |
| 3 1/2            |                   |       | 3.04  | 3.81  | 4.57  | 6.09 | 7.61 | 9.14  | 12.20 |
| 4                |                   |       | 3.48  | 4.35  | 5.22  | 6.96 | 8.70 | 10.40 | 13.90 |

METRIC UNIT: APPROX. WT. KG/M

| Width<br>(A) mm | Thickness (t) mm |       |       |       |      |      |      |      |      |      |      |      |
|-----------------|------------------|-------|-------|-------|------|------|------|------|------|------|------|------|
|                 | 3                | 4     | 5     | 6     | 8    | 9    | 10   | 12   | 13   | 15   | 20   | 25   |
| 12              | 0.29             |       |       | 0.58  |      |      |      |      |      |      |      |      |
| 13              | 0.314            | 0.419 |       |       |      |      |      |      |      |      |      |      |
| 14              | 0.339            | 0.451 | 0.564 | 0.677 |      |      |      |      |      |      |      |      |
| 15              | 0.363            | 0.484 | 0.604 |       |      |      |      |      |      |      |      |      |
| 16              | 0.387            | 0.516 | 0.645 | 0.774 |      |      |      |      |      |      |      |      |
| 20              | 0.484            | 0.645 | 0.806 | 0.967 | 1.29 | 1.45 | 1.61 | 1.93 |      |      |      |      |
| 25              | 0.604            | 0.806 | 1.01  | 1.21  | 1.61 | 1.81 | 2.02 | 2.42 | 2.62 | 3.02 | 4.03 |      |
| 30              | 0.725            | 0.967 | 1.21  | 1.45  | 1.93 | 2.18 | 2.42 | 2.90 | 3.14 | 3.63 | 4.84 |      |
| 35              | 0.846            | 1.13  | 1.41  | 1.69  | 2.26 | 2.54 | 2.82 | 3.39 | 3.67 | 4.23 | 5.64 |      |
| 40              | 0.967            | 1.29  | 1.61  | 1.93  | 2.58 | 2.90 | 3.22 | 3.87 | 4.19 | 4.84 | 6.45 | 8.06 |
| 45              | 1.09             | 1.45  | 1.81  | 2.18  | 2.90 | 3.26 | 3.63 | 4.35 | 4.72 |      |      | 9.07 |
| 50              | 1.21             | 1.61  | 2.02  | 2.42  | 3.22 | 3.63 | 4.03 | 4.84 | 5.24 | 6.04 | 8.06 | 10.1 |
| 60              |                  |       | 2.42  | 2.90  | 3.87 | 4.35 | 4.84 | 5.80 | 6.29 | 7.25 | 9.67 | 12.1 |
| 65              |                  |       | 2.62  | 3.14  | 4.19 | 4.72 | 5.24 | 6.29 | 6.81 | 7.86 | 10.5 | 13.1 |
| 70              |                  |       | 2.82  | 3.39  | 4.51 | 5.08 | 5.64 | 6.77 | 7.33 | 8.46 | 11.3 | 14.1 |
| 75              |                  |       | 3.02  | 3.63  | 4.84 | 5.44 | 6.04 | 7.25 | 7.86 | 9.07 | 12.1 | 15.1 |
| 80              |                  |       |       | 3.87  | 5.16 | 5.80 | 6.45 | 7.74 | 8.38 | 9.67 | 12.9 | 16.1 |
| 90              |                  |       |       | 4.35  | 5.80 | 6.38 | 7.25 | 8.70 | 9.43 | 10.9 | 14.5 | 18.1 |
| 100             |                  |       |       | 4.84  | 6.45 | 7.25 | 8.06 | 9.67 | 10.5 | 12.1 | 16.1 | 20.2 |

# STAINLESS STEEL BARS

## ROUND BARS

| Type<br>Number | UNS<br>Number | Equivalent to |        |      | Chemical Composition, % |                  |       |      |                 |                  |                  |                | Other<br>Elements |
|----------------|---------------|---------------|--------|------|-------------------------|------------------|-------|------|-----------------|------------------|------------------|----------------|-------------------|
|                |               | JIS           | BS     | DIN  | C                       | Mn               | P     | S    | Si              | Cr               | Ni               | Mo             |                   |
| 201            | (S20100)      | SUS201        | --     | --   | 0.15                    | 5.50 ~<br>7.50   | 0.06  | 0.03 | 1.00            | 16.00 ~<br>18.00 | 3.50 ~<br>5.50   | --             | N 0.25            |
| 202            | (S20200)      | SUS202        | 284S16 | --   | 0.15                    | 7.50 ~<br>10.00  | 0.06  | 0.03 | 1.00            | 17.00 ~<br>19.00 | 4.00 ~<br>6.00   | --             | N 0.25            |
| 205            | (S20500)      | --            | --     | --   | 0.12 ~<br>0.25          | 14.00 ~<br>15.50 | 0.06  | 0.03 | 1.00            | 16.50 ~<br>18.00 | 1.00 ~<br>1.75   | --             | N 0.32 / 0.40     |
| 301            | (S30100)      | SUS301        | 301S21 | 4310 | 0.15                    | 2.00             | 0.045 | 0.03 | 1.00            | 16.00 ~<br>18.00 | 6.00 ~<br>8.00   | --             |                   |
| 302            | (S30200)      | SUS302        | 302S25 | 4300 | 0.15                    | 2.00             | 0.045 | 0.03 | 1.00            | 17.00 ~<br>19.00 | 8.00 ~<br>10.00  | --             |                   |
| 302B           | (S30215)      | SUS302B       | --     | --   | 0.15                    | 2.00             | 0.045 | 0.03 | 2.00 ~<br>3.00* | 17.00 ~<br>19.00 | 8.00 ~<br>10.00  | --             |                   |
| 303            | (S30300)      | SUS303        | 303S21 | 4305 | 0.15                    | 2.00             | 0.20  | 0.15 | 1.00            | 17.00 ~<br>19.00 | 8.00 ~<br>10.00  | 0.60           |                   |
| 303Se          | (S30323)      | SUS303Se      | 303S41 | --   | 0.15                    | 2.00             | 0.20  | 0.06 | 1.00            | 17.00 ~<br>19.00 | 8.00 ~<br>10.00  | --             | Se 0.15 min.      |
| 304            | (S30400)      | SUS304        | 304S15 | 4301 | 0.08                    | 2.00             | 0.045 | 0.03 | 1.00            | 18.00 ~<br>20.00 | 8.00 ~<br>10.50  | --             |                   |
| 304L           | (S30403)      | SUS304L       | 304S12 | 4306 | 0.03                    | 2.00             | 0.045 | 0.03 | 1.00            | 18.00 ~<br>20.00 | 8.00 ~<br>12.00  | --             |                   |
|                | (S30430)      | --            | --     | --   | 0.08                    | 2.00             | 0.045 | 0.03 | 1.00            | 17.00 ~<br>19.00 | 8.00 ~<br>10.00  | --             | Cu 3.00/4.00      |
| 304N           | (S30451)      | SUS304N1      | --     | --   | 0.08                    | 2.00             | 0.045 | 0.03 | 1.00            | 18.00 ~<br>20.00 | 8.00 ~<br>10.00  | --             | N 0.10/0.16       |
| 305            | (S30500)      | SUS305        | 305S19 | --   | 0.12                    | 2.00             | 0.045 | 0.03 | 1.00            | 17.00 ~<br>19.00 | 10.50 ~<br>13.00 | --             |                   |
| 308            | (S30800)      | SUS308        | 305S19 | --   | 0.08                    | 2.00             | 0.045 | 0.03 | 1.00            | 19.00 ~<br>21.00 | 10.00 ~<br>12.00 | --             |                   |
| 309            | (S30900)      | SUH309        | 309S24 | --   | 0.20                    | 2.00             | 0.045 | 0.03 | 1.00            | 22.00 ~<br>24.00 | 12.00 ~<br>15.00 | --             |                   |
| 309S           | (S30908)      | SUS309S       | 309S24 | --   | 0.08                    | 2.00             | 0.045 | 0.03 | 1.00            | 22.00 ~<br>24.00 | 12.00 ~<br>15.00 | --             |                   |
| 310            | (S31000)      | SUH310        | 310S24 | --   | 0.25                    | 2.00             | 0.045 | 0.03 | 1.50            | 24.00 ~<br>26.00 | 19.00 ~<br>22.00 | --             |                   |
| 310S           | (S31008)      | SUS310S       | 310S24 | 4845 | 0.08                    | 2.00             | 0.045 | 0.03 | 1.50            | 24.00 ~<br>26.00 | 19.00 ~<br>22.00 | --             |                   |
| 314            | (S31400)      | --            | --     | --   | 0.25                    | 2.00             | 0.045 | 0.03 | 1.50 ~<br>3.00  | 23.00 ~<br>26.00 | 19.00 ~<br>22.00 | --             |                   |
| 316            | (S31600)      | SUS316        | 316S16 | 4401 | 0.08                    | 2.00             | 0.045 | 0.03 | 1.00            | 16.00 ~<br>18.00 | 10.00 ~<br>14.00 | 2.00 ~<br>3.00 |                   |
| 316F           | (S31620)      | --            | --     | --   | 0.08                    | 2.00             | 0.20  | 0.10 | 1.00            | 16.00 ~<br>18.00 | 10.00 ~<br>14.00 | 1.75 ~<br>2.50 |                   |
| 316L           | (S31603)      | SUS316L       | 316S16 | 4402 | 0.03                    | 2.00             | 0.045 | 0.03 | 1.00            | 16.00 ~<br>18.00 | 10.00 ~<br>14.00 | 2.00 ~<br>3.00 |                   |

Note:

- \* May be added at the manufacturer's discretion.
- Products excluded from the above table are also available upon request.

# STAINLESS STEEL BARS

## ROUND BARS

| Type<br>Number | UNS<br>Number | Equivalent to        |                  |      | Chemical Composition, % |      |       |       |                |                  |                  |                | Other<br>Elements                                |
|----------------|---------------|----------------------|------------------|------|-------------------------|------|-------|-------|----------------|------------------|------------------|----------------|--------------------------------------------------|
|                |               | JIS                  | BS               | DIN  | C                       | Mn   | P     | S     | Si             | Cr               | Ni               | Mo             |                                                  |
| 316N           | (S31651)      | SUS316N              | --               | --   | 0.08                    | 2.00 | 0.045 | 0.03  | 1.00           | 16.00 ~<br>18.00 | 10.00 ~<br>14.00 | 2.30 ~<br>3.00 | N 0.10/0.16                                      |
| 317            | (S31700)      | SUS317               | 316S12           | --   | 0.08                    | 2.00 | 0.045 | 0.03  | 1.00           | 18.00 ~<br>20.00 | 11.00 ~<br>15.00 | 3.00 ~<br>4.00 |                                                  |
| 317L           | (S31703)      | SUS317L              | --               | 4438 | 0.03                    | 2.00 | 0.045 | 0.03  | 1.00           | 18.00 ~<br>20.00 | 11.00 ~<br>15.00 | 3.00 ~<br>4.00 |                                                  |
| 321            | (S32100)      | SUS321               | 317S16           | 4541 | 0.08                    | 2.00 | 0.045 | 0.03  | 1.00           | 17.00 ~<br>19.00 | 9.00 ~<br>12.00  | --             | Ti 5 x C min.                                    |
| 329            | (S32900)      | SUS327J1             | 317S12           | --   | 0.10                    | 2.00 | 0.04  | 0.03  | 1.00           | 25.00 ~<br>30.00 | 3.00 ~<br>6.00   | 1.00 ~<br>2.00 |                                                  |
| 330            | (N08330)      | SUH330               | 321S12<br>321S20 | --   | 0.08                    | 2.00 | 0.04  | 0.03  | 0.75 ~<br>1.50 | 17.00 ~<br>20.00 | 34.00 ~<br>37.00 | --             |                                                  |
| 347            | (S34700)      | SUS347               | 347S17           | 4450 | 0.08                    | 2.00 | 0.045 | 0.03  | 1.00           | 17.00 ~<br>19.00 | 9.00 ~<br>13.00  | --             | Cb + Ta 10 x C min.                              |
| 348            | (S34800)      | --                   | --               | --   | 0.08                    | 2.00 | 0.045 | 0.03  | 1.00           | 17.00 ~<br>19.00 | 9.00 ~<br>13.00  | --             | Cb + Ta 10 x C min.<br>Ta 0.10 max. Co 0.20 max. |
| 384            | (S38400)      | SU384                | --               | --   | 0.08                    | 2.00 | 0.045 | 0.03  | 1.00           | 11.00 ~<br>17.00 | 17.00 ~<br>19.00 | --             |                                                  |
| 403            | (S40300)      | SUS403               | 403S17           | --   | 0.15                    | 1.00 | 0.04  | 0.03  | 0.50           | 11.50 ~<br>13.00 |                  |                |                                                  |
| 405            | (S40500)      | SUS405               | 405S17           | 4002 | 0.08                    | 1.00 | 0.04  | 0.03  | 1.00           | 11.50 ~<br>14.50 | --               | --             | Al 0.10/0.30                                     |
| 409            | (S40900)      | SUH409               | --               | --   | 0.08                    | 1.00 | 0.045 | 0.045 | 1.00           | 10.50 ~<br>11.75 |                  |                | Ti 6 x C min / 0.75 max                          |
| 410            | (S41000)      | SUS410               | 410S21           | 4006 | 0.15                    | 1.00 | 0.04  | 0.03  | 1.00           | 11.50 ~<br>13.50 |                  |                |                                                  |
| 414            | (S41400)      | --                   | --               | --   | 0.15                    | 1.00 | 0.04  | 0.03  | 1.00           | 11.50 ~<br>13.50 | 1.25 ~<br>2.50   |                |                                                  |
| 416            | (S41600)      | SUS416               | 416S21           | --   | 0.15                    | 1.25 | 0.06  | 0.15  | 1.00           | 12.00 ~<br>14.00 | --               | 0.60*          |                                                  |
| 416Se          | (S41623)      | --                   | --               | --   | 0.15                    | 1.25 | 0.06  | 0.06  | 1.00           | 12.00 ~<br>14.00 | --               | --             | Se 0.15 min.                                     |
| 420            | (S42000)      | SUS420J1<br>SUS420J2 | 420S37<br>420S45 | 4021 | 0.15                    | 1.00 | 0.04  | 0.03  | 1.00           | 12.00 ~<br>14.00 | --               | --             |                                                  |
| 420F           | (S42020)      | SUS420F              | --               | --   | 0.15                    | 1.25 | 0.06  | 0.15  | 1.00           | 12.00 ~<br>14.00 | --               | 0.60*          |                                                  |
| 422            | (S42200)      | SUH616               | --               | --   | 0.20 ~<br>0.25          | 1.00 | 0.025 | 0.025 | 0.75           | 11.00 ~<br>13.00 | 0.50 ~<br>1.00   | 0.75 ~<br>1.25 | V 0.15/0.30<br>W 0.75/1.25                       |
| 429            | (S42900)      | SUS429               | --               | --   | 0.12                    | 1.00 | 0.04  | 0.03  | 1.00           | 14.00 ~<br>16.00 |                  |                |                                                  |
| 430            | (S43000)      | SUS430               | 430S15           | 4016 | 0.12                    | 1.00 | 0.04  | 0.03  | 1.00           | 16.00 ~<br>18.00 |                  |                |                                                  |

Note:

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- Products excluded from the above table are also available upon request.

# STAINLESS STEEL BARS

## ROUND BARS

| Type<br>Number | UNS<br>Number | Equivalent to |        |      | Chemical Composition, % |      |      |       |      |                  |                  |                | Other<br>Elements                                   |
|----------------|---------------|---------------|--------|------|-------------------------|------|------|-------|------|------------------|------------------|----------------|-----------------------------------------------------|
|                |               | JIS           | BS     | DIN  | C                       | Mn   | P    | S     | Si   | Cr               | Ni               | Mo             |                                                     |
| 430F           | (S43020)      | SUS430F       | --     | 4104 | 0.12                    | 1.25 | 0.06 | 0.15  | 1.00 | 16.00 ~<br>18.00 | --               | 0.60*          |                                                     |
| 430FSe         | (S43023)      | --            | --     | --   | 0.12                    | 1.25 | 0.06 | 0.06  | 1.00 | 16.00 ~<br>18.00 | --               | --             | Se 0.15 max.                                        |
| 431            | (S43100)      | SUS431        | 431S29 | 4037 | 0.20                    | 1.00 | 0.04 | 0.03  | 1.00 | 15.00 ~<br>17.00 | 1.25 ~<br>2.50   | --             |                                                     |
| 434            | (S43400)      | SUS434        | 434S19 | 4113 | 0.12                    | 1.00 | 0.04 | 0.03  | 1.00 | 16.00 ~<br>18.00 |                  | 0.75 ~<br>1.25 |                                                     |
| 436            | (S43600)      | --            | --     | --   | 0.12                    | 1.00 | 0.04 | 0.03  | 1.00 | 16.00 ~<br>18.00 |                  | 0.75 ~<br>1.25 | Cb + Ta 5 x C min.                                  |
| 440A           | (S44002)      | SUS440A       | --     | --   | 0.60~<br>0.75           | 1.00 | 0.04 | 0.03  | 1.00 | 16.00 ~<br>18.00 |                  | 0.75           |                                                     |
| 440B           | (S44003)      | SUS440B       | --     | --   | 0.75~<br>0.95           | 1.00 | 0.04 | 0.03  | 1.00 | 16.00 ~<br>18.00 |                  | 0.75           |                                                     |
| 440C           | (S44004)      | SUS440C       | --     | --   | 0.95~<br>1.20           | 1.00 | 0.04 | 0.03  | 1.00 | 16.00 ~<br>18.00 |                  | 0.75           |                                                     |
| 442            | (S44200)      | --            | --     | --   | 0.20                    | 1.00 | 0.04 | 0.03  | 1.00 | 18.00 ~<br>23.00 |                  |                |                                                     |
| 446            | (S44600)      | SUH446        | --     | --   | 0.20                    | 1.50 | 0.04 | 0.03  | 1.00 | 23.00 ~<br>27.00 |                  |                | N0.25                                               |
| 501            | (S50100)      | --            | --     | --   | 0.10                    | 1.00 | 0.04 | 0.03  | 1.00 | 4.00 ~<br>6.00   |                  | 0.40 ~<br>0.65 |                                                     |
| 502            | (S50200)      | --            | --     | --   | 0.10                    | 1.00 | 0.04 | 0.03  | 1.00 | 4.00 ~<br>6.00   |                  | 0.40 ~<br>0.65 |                                                     |
|                | (S13800)      | --            | --     | --   | 0.05                    | 0.10 | 0.01 | 0.008 | 0.10 | 12.25 ~<br>13.25 | 7.50 ~<br>8.50   | 2.00 ~<br>2.50 | Al 0.09/1.35<br>N 0.01                              |
|                | (S15500)      | --            | --     | --   | 0.07                    | 1.00 | 0.04 | 0.03  | 1.00 | 14.00 ~<br>15.00 | 3.50 ~<br>5.50   | --             | Cu 3.00/5.00<br>Cb + Ta 0.15/0.45                   |
|                | (S17400)      | SUS630        | --     | --   | 0.07                    | 1.00 | 0.04 | 0.03  | 1.00 | 15.50 ~<br>17.50 | 3.00 ~<br>5.00   | --             | Cu 3.00/5.00<br>Cb + Ta 0.15/0.45                   |
|                | (S17700)      | SUS631        | --     | --   | 0.09                    | 1.00 | 0.04 | 0.04  | 1.00 | 16.00 ~<br>18.00 | 6.50 ~<br>7.75   | --             | Al 0.75/1.50                                        |
|                | (K66286)      | SUH660        | --     | --   | 0.08                    | 2.00 | 0.04 | 0.03  | 1.00 | 13.50 ~<br>16.00 | 24.00 ~<br>17.00 | 1.00 ~<br>1.50 | Ti 1.90/2.35, Al 0.35<br>V 0.1/0.5,<br>B 0.001/0.01 |

Note:

- \* May be added at the manufacturer's discretion.
- Products excluded from the above table are also available upon request.

# STAINLESS STEEL BARS

ROUND BARS (Sizes and Unit Weights)

| Size         | lb/ft | kg/ft | Size   | lb/ft  | kg/ft  |
|--------------|-------|-------|--------|--------|--------|
| 3.00 mm      | 0.037 | 0.017 | 2"     | 10.803 | 4.901  |
| 1/8"         | 0.042 | 0.019 | 2 1/4" | 13.673 | 6.202  |
| 4.50 mm      | 0.085 | 0.039 | 2 1/2" | 16.88  | 7.657  |
| 3/16"        | 0.096 | 0.044 | 2 3/4" | 20.425 | 9.265  |
| 6.00 mm      | 0.150 | 0.068 | 3"     | 24.307 | 11.026 |
| 1/4"         | 0.169 | 0.077 | 3 1/4" | 28.526 | 12.94  |
| 7.50 mm      | 0.235 | 0.107 | 3 1/2" | 33.084 | 15.007 |
| 5/16"        | 0.265 | 0.120 | 3 3/4" | 37.978 | 17.227 |
| 9.00 mm      | 0.339 | 0.154 | 4"     | 43.211 | 19.601 |
| 3/8"         | 0.380 | 0.172 | 4 1/2" | 54.689 | 24.807 |
| 12.00 mm     | 0.603 | 0.274 | 5"     | 67.516 | 30.626 |
| 1/2"         | 0.675 | 0.306 | 5 1/2" | 81.695 | 37.057 |
| 5/8"         | 1.055 | 0.479 | 6"     | 97.224 | 44.101 |
| 3/4"         | 1.519 | 0.689 | 6 1/2" | 114.10 | 51.757 |
| 7/8"         | 2.067 | 0.938 | 7"     | 132.33 | 60.026 |
| 1"           | 2.701 | 1.225 | 7 1/2" | 151.91 | 68.907 |
| 1 1/8" 3.418 | 1.550 | 8"    | 172.84 | 78.401 |        |
| 1 1/4" 4.220 | 1.914 | 9"    | 218.75 | 99.226 |        |
| 1 3/8" 5.107 | 2.317 | 10"   | 270.07 | 122.50 |        |
| 1 1/2" 6.076 | 2.756 | 11"   | 326.78 | 148.23 |        |
| 1 5/8" 7.132 | 3.235 | 12"   | 388.89 | 176.40 |        |
| 1 3/4" 8.271 | 3.752 | 13"   | 456.41 | 207.03 |        |
| 1 7/8" 9.495 | 4.307 | 14"   | 529.33 | 240.10 |        |

# STAINLESS STEEL BARS

| HEXAGON BARS (Sizes and Unit Weights) |        |        | SQUARE BARS (Sizes and Unit Weights) |        |        |
|---------------------------------------|--------|--------|--------------------------------------|--------|--------|
| Size                                  | lb/ft  | kg/ft  | Size                                 | lb/ft  | kg/ft  |
| 3.00 mm                               | 0.041  | 0.019  | 3.00 mm                              | 0.048  | 0.022  |
| 1/8"                                  | 0.046  | 0.021  | 1/8"                                 | 0.054  | 0.024  |
| 4.50 mm                               | 0.093  | 0.042  | 4.50 mm                              | 0.108  | 0.049  |
| 3/16"                                 | 0.105  | 0.047  | 3/16"                                | 0.121  | 0.055  |
| 6.00 mm                               | 0.166  | 0.075  | 6.00 mm                              | 0.192  | 0.087  |
| 1/4"                                  | 0.186  | 0.084  | 1/4"                                 | 0.215  | 0.097  |
| 7.50 mm                               | 0.259  | 0.118  | 7.50 mm                              | 0.299  | 0.136  |
| 5/16"                                 | 0.290  | 0.132  | 5/16"                                | 0.335  | 0.152  |
| 9.00 mm                               | 0.373  | 0.169  | 9.00 mm                              | 0.431  | 0.196  |
| 3/8"                                  | 0.418  | 0.190  | 3/8"                                 | 0.483  | 0.219  |
| 12.00 mm                              | 0.664  | 0.301  | 12.00 mm                             | 0.766  | 0.348  |
| 1/2"                                  | 0.744  | 0.337  | 1/2"                                 | 0.859  | 0.390  |
| 5/8"                                  | 1.162  | 0.527  | 5/8"                                 | 1.341  | 0.609  |
| 3/4"                                  | 1.673  | 0.759  | 3/4"                                 | 1.932  | 0.876  |
| 7/8"                                  | 2.277  | 1.033  | 7/8"                                 | 2.629  | 1.193  |
| 1"                                    | 2.974  | 1.350  | 1"                                   | 3.434  | 1.558  |
| 1 1/8" 3.765                          | 1.708  | 1 1/8" | 4.346                                | 1.972  |        |
| 1 1/4" 4.648                          | 2.109  | 1 1/4" | 5.366                                | 2.434  |        |
| 1 3/8" 5.624                          | 2.552  | 1 3/8" | 6.492                                | 2.946  |        |
| 1 1/2" 6.693                          | 3.037  | 1 1/2" | 7.727                                | 3.506  |        |
| 1 5/8" 7.854                          | 3.564  | 1 5/8" | 9.068                                | 4.114  |        |
| 1 3/4" 9.109                          | 4.133  | 1 3/4" | 10.517                               | 4.772  |        |
| 1 7/8" 10.457                         | 4.745  | 1 7/8" | 12.073                               | 5.478  |        |
| 2"                                    | 11.898 | 5.398  | 2"                                   | 13.736 | 6.232  |
| 2 1/4" 15.058                         | 6.832  | 2 1/4" | 17.385                               | 7.888  |        |
| 2 1/2" 18.590                         | 8.435  | 2 1/2" | 21.463                               | 9.738  |        |
| 2 3/4" 22.494                         | 10.206 | 2 3/4" | 25.970                               | 11.783 |        |
| 3"                                    | 26.770 | 12.146 | 3"                                   | 30.906 | 14.023 |
| 3 1/4" 31.418                         | 14.255 | 3 1/4" | 36.272                               | 16.457 |        |
| 3 1/2" 36.437                         | 16.532 | 3 1/2" | 42.067                               | 19.086 |        |
| 3 3/4" 41.828                         | 41.828 | 18.978 | 3 3/4"                               | 48.291 | 21.910 |
| 4"                                    | 47.591 | 21.593 | 4"                                   | 54.944 | 24.929 |

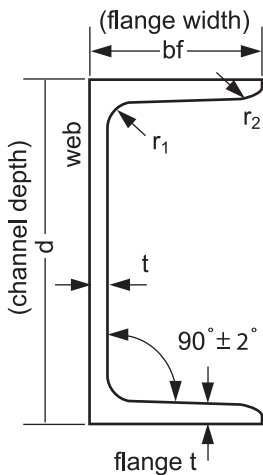


# STAINLESS STEEL WELDED CHANNELS

AISI 304

| Size (in.)                           | Tolerance         |                               | AISI 304. 304L      | AISI 316. 316L      |
|--------------------------------------|-------------------|-------------------------------|---------------------|---------------------|
| d x bf x t                           | d                 | bf                            | Unit weight (lb/ft) | Unit weight (lb/ft) |
| 3 x 1 $\frac{1}{2}$ x $\frac{3}{16}$ | $\pm\frac{1}{8}$  | $\pm\frac{1}{8}$              | 3.70                | 3.70                |
| 4 x 2 x $\frac{3}{16}$               | $\pm\frac{1}{8}$  | $\pm\frac{1}{8}$              | 5.03                | 5.03                |
| 4 x 2 x $\frac{1}{4}$                | $\pm\frac{1}{8}$  | $\pm\frac{1}{8}$              | 6.58                | 6.58                |
| 5 x 2 $\frac{1}{2}$ x $\frac{1}{4}$  | $\pm\frac{3}{16}$ | $\pm\frac{1}{8}$              | 8.40                | 8.40                |
| 6 x 3 x $\frac{1}{4}$                | $\pm\frac{3}{16}$ | $\pm\frac{1}{8}$              | 10.14               | 10.14               |
| 6 x 3 x $\frac{3}{8}$                | $\pm\frac{3}{16}$ | $\pm\frac{1}{8}$              | 14.77               | 14.77               |
| 8 x 4 x $\frac{3}{8}$                | $\pm\frac{3}{16}$ | $+\frac{3}{16} - \frac{1}{8}$ | 20.02               | 20.02               |

| Size (mm)      | Tolerance  |            | AISI 304 . 304L    | AISI 316. 316L     |
|----------------|------------|------------|--------------------|--------------------|
| d x bf x t     | d          | bf         | Unit weight (kg/m) | Unit weight (kg/m) |
| 80 x 40 x 5    | $\pm 2.00$ | $\pm 2.00$ | 6.08               | 6.08               |
| 100 x 50 x 4   | $\pm 2.00$ | $\pm 2.00$ | 6.28               | 6.28               |
| 100 x 50 x 6   | $\pm 2.00$ | $\pm 2.00$ | 9.16               | 9.16               |
| 130 x 65 x 6   | $\pm 3.00$ | $\pm 3.00$ | 12.20              | 12.20              |
| 150 x 75 x 6   | $\pm 3.00$ | $\pm 3.00$ | 14.13              | 14.13              |
| 150 x 75 x 9   | $\pm 3.00$ | $\pm 3.00$ | 20.53              | 20.53              |
| 200 x 100 x 10 | $\pm 4.00$ | $\pm 4.00$ | 30.80              | 30.80              |



## MECHANICAL PROPERTIES OF AISI 304 BUTT WELDED JOINT

| Joint Thickness (inch) | Tension test     |             |        | Bend Test (180°) |      |
|------------------------|------------------|-------------|--------|------------------|------|
|                        | Y.S. 0.2% P.S.I. | T.S. P.S.I. | E.L. % | Face             | Root |
| $\frac{3}{16}$         | 50200            | 95800       | 57.7   | Good             | Good |
| $\frac{1}{4}$          | 48800            | 93100       | 57.2   | Good             | Good |
| $\frac{3}{8}$          | 41200            | 85200       | 43.2   | Good             | Good |

| Joint Thickness (mm) | Tension test                    |                            |        | Bend Test (180°) |      |
|----------------------|---------------------------------|----------------------------|--------|------------------|------|
|                      | Y.S. 0.2% (kg/mm <sup>2</sup> ) | T.S. (kg/mm <sup>2</sup> ) | E.L. % | Face             | Root |
| 5                    | 34                              | 62                         | 57.7   | Good             | Good |
| 6                    | 35                              | 62                         | 57.2   | Good             | Good |
| 10                   | 37                              | 60                         | 43.2   | Good             | Good |

# STAINLESS STEEL WELDED TUBINGS

## WELDED STAINLESS STEEL TUBING

### MECHANICAL PROPERTIES

| Type     | Round and square                     |                                        |                                | Round   |                                 |                                 |         |
|----------|--------------------------------------|----------------------------------------|--------------------------------|---------|---------------------------------|---------------------------------|---------|
|          | Tension test                         |                                        |                                | Bending | Flattering                      |                                 | Flaring |
|          | Yield strength<br>kg/mm <sup>2</sup> | Tensile strength<br>kg/mm <sup>2</sup> | Elongation<br>(longitudinal) % |         | No.1                            | No.2                            |         |
| AISI 430 | > 25                                 | > 42                                   | > 20                           | 180°4D  | < <sup>2</sup> / <sub>3</sub> D | < <sup>1</sup> / <sub>2</sub> D | 1.14D   |
| AISI 304 | > 21                                 | > 53                                   | > 35                           | 180°3D  | < <sup>2</sup> / <sub>3</sub> D | < <sup>1</sup> / <sub>2</sub> D | 1.2D    |
| AISI 316 | > 21                                 | > 53                                   | > 35                           | 180°3D  | < <sup>2</sup> / <sub>3</sub> D | < <sup>1</sup> / <sub>2</sub> D | 1.2D    |

Notes:

- (1) Yield strength test shall not be conducted unless requested. Other mechanical property values are available on request.
- (2) For the tension test of round tubing, test piece No. 11 or 12 specified in JIS Z 2201 (Tension Test Pieces for Materials) shall be used. Test piece No. 12 shall be taken from an unwelded portion.
- (3) For the tension test of square tubing, test tubing, test piece No. 5 specified in JIS Z 2201 shall be used. However, for a side length of less than 25 mm, the test shall be conducted on a test piece with a width equivalent to its side length. The test piece shall be taken from an unwelded portion.
- (4) Unless otherwise specified, the values of No. 1 shall be used.

### TYPES OF MANUFACTURES

| Surface Finish              |            |              |                                                 |
|-----------------------------|------------|--------------|-------------------------------------------------|
|                             | Finish     | Nisshin Code | Description                                     |
| Round tubing                | Unpolished | S2           | As welded                                       |
|                             | 120 GRIT   | S 240        | Polished by #120 belt                           |
|                             | 120 GRIT   | SB 240       | Buffed S 240                                    |
|                             | 180 GRIT   | SB 400       | Buffed after polishing by #180 belt             |
|                             | 180 GRIT   | GL           | Hair-line finished after polishing by #180 belt |
|                             | 320 GRIT   | SB 600       | Buffed after polishing by #320 belt             |
|                             | 400 GRIT   | BL 400       | Buffed after polishing by #400 belt             |
|                             | MIRROR     | MR           | Buffed after polishing by #600 belt             |
| Square & rectangular tubing | Unpolished | S2           | As welded                                       |
|                             | 180 GRIT   | HL180        | Hair-line finished after polishing by #180 belt |

### CHEMICAL COMPOSITION

| Type     | Chemical Composition (%) |       |       |       |       |       |        |          |
|----------|--------------------------|-------|-------|-------|-------|-------|--------|----------|
|          | C                        | Si    | Mn    | P     | S     | Ni    | Cr     | Mo       |
| AISI 430 | <0.12                    | <1.00 | <1.00 | <0.04 | <0.03 | --    | 16 ~18 | --       |
| AISI 304 | <0.08                    | <1.00 | <2.00 | <0.04 | <0.03 | 8~11  | 18~20  | --       |
| AISI 316 | <0.08                    | <1.00 | <2.00 | <0.04 | <0.03 | 11~14 | 16 ~18 | 200 ~300 |

# STAINLESS STEEL WELDED TUBINGS

## ROUND TUBING

TABLE OF UNIT WEIGHT

| Outside diameter |      | Type     | Wall Thickness (mm) |        |       |        |       |        |       |        |       |        |       |        |       |        |       |        |      |        |      |        |
|------------------|------|----------|---------------------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|------|--------|------|--------|
|                  |      |          | 0.5                 |        | 0.6   |        | 0.7   |        | 0.8   |        | 1.0   |        | 1.2   |        | 1.5   |        | 2.0   |        | 2.5  |        | 3.0  |        |
|                  |      |          | kg/m                | kg/20' | kg/m  | kg/20' | kg/m  | kg/20' | kg/m  | kg/20' | kg/m  | kg/20' | kg/m  | kg/20' | kg/m  | kg/20' | kg/m  | kg/20' | kg/m | kg/20' | kg/m | kg/20' |
| 1/2              | 12.7 | AISI 304 | 0.152               | 0.927  | 0.181 | 1.10   | 0.209 | 1.27   | 0.237 | 1.44   | 0.291 | 1.77   | 0.344 | 2.10   | 0.418 | 2.55   |       |        |      |        |      |        |
|                  |      | AISI 430 | 0.148               | 0.902  | 0.176 | 1.07   | 0.203 | 1.24   | 0.203 | 1.40   | 0.283 | 1.73   | 0.334 | 2.04   | 0.406 | 2.47   |       |        |      |        |      |        |
| 5/8              | 15.9 | AISI 304 | 0.192               | 1.17   | 0.229 | 1.40   | 0.265 | 1.62   | 0.301 | 1.83   | 0.371 | 2.26   | 0.439 | 2.68   | 0.538 | 3.28   |       |        |      |        |      |        |
|                  |      | AISI 430 | 0.186               | 1.13   | 0.222 | 1.35   | 0.257 | 1.57   | 0.292 | 1.78   | 0.36  | 2.19   | 0.427 | 2.60   | 0.522 | 3.18   |       |        |      |        |      |        |
|                  | 16   | AISI 304 | 0.193               | 1.18   | 0.23  | 1.40   | 0.267 | 1.63   | 0.303 | 1.85   | 0.374 | 2.28   | 0.442 | 2.69   | 0.542 | 3.30   |       |        |      |        |      |        |
|                  |      | AISI 430 | 0.187               | 1.14   | 0.224 | 1.37   | 0.259 | 1.58   | 0.294 | 1.79   | 0.363 | 2.21   | 0.43  | 2.62   | 0.562 | 3.21   |       |        |      |        |      |        |
| 3/4              | 19.1 | AISI 304 | 0.232               | 1.41   | 0.277 | 1.69   | 0.321 | 1.96   | 0.365 | 2.23   | 0.451 | 2.75   | 0.533 | 3.26   | 0.658 | 4.01   |       |        |      |        |      |        |
|                  |      | AISI 430 | 0.225               | 1.37   | 0.269 | 1.64   | 0.321 | 1.90   | 0.354 | 2.16   | 0.438 | 2.67   | 0.52  | 3.17   | 0.639 | 3.90   |       |        |      |        |      |        |
| 7/8              | 22.2 | AISI 304 |                     |        | 0.323 | 1.97   | 0.375 | 2.29   | 0.426 | 2.60   | 0.528 | 3.22   | 0.628 | 3.83   | 0.773 | 4.71   | 1.01  | 6.16   |      |        |      |        |
|                  |      | AISI 430 |                     |        | 0.314 | 1.91   | 0.364 | 2.33   | 0.414 | 2.52   | 0.513 | 3.13   | 0.61  | 3.72   | 0.751 | 4.58   | 0.977 | 5.96   |      |        |      |        |
|                  | 25   | AISI 304 |                     |        | 0.365 | 2.23   | 0.429 | 2.62   | 0.482 | 2.94   | 0.598 | 3.65   | 0.711 | 4.33   | 0.878 | 5.35   | 1.15  | 7.01   |      |        |      |        |
|                  |      | AISI 430 |                     |        | 0.354 | 2.16   | 0.411 | 2.51   | 0.468 | 2.85   | 0.581 | 3.54   | 0.691 | 4.21   | 0.853 | 5.20   | 1.11  | 6.77   |      |        |      |        |
| 1                | 25.4 | AISI 304 |                     |        | 0.371 | 2.26   | 0.431 | 2.63   | 0.49  | 2.99   | 0.608 | 3.71   | 0.723 | 4.41   | 0.893 | 5.44   | 1.17  | 7.13   |      |        |      |        |
|                  |      | AISI 430 |                     |        | 0.36  | 2.19   | 0.417 | 2.54   | 0.476 | 2.90   | 0.59  | 3.60   | 0.702 | 4.28   | 0.867 | 5.29   | 1.13  | 6.89   |      |        |      |        |
| 1 1/8            | 28.6 | AISI 304 |                     |        | 0.418 | 2.55   | 0.486 | 2.96   | 0.554 | 3.38   | 0.687 | 4.19   | 0.819 | 4.99   | 1.01  | 6.16   | 1.32  | 8.05   |      |        |      |        |
|                  |      | AISI 430 |                     |        | 0.406 | 2.47   | 0.472 | 2.88   | 0.538 | 3.28   | 0.668 | 4.07   | 0.795 | 4.85   | 0.983 | 5.99   | 1.29  | 7.86   |      |        |      |        |
| 1 1/4            | 31.8 | AISI 304 |                     |        |       |        | 0.542 | 3.30   | 0.618 | 3.77   | 0.767 | 4.68   | 0.915 | 5.58   | 1.13  | 6.89   | 1.48  | 9.02   | 1.82 | 11.1   | 2.15 | 13.1   |
|                  |      | AISI 430 |                     |        |       |        | 0.527 | 3.21   | 0.60  | 3.66   | 0.745 | 4.54   | 0.888 | 5.41   | 1.10  | 6.71   | 1.44  | 8.78   | 1.77 | 10.8   | 2.09 | 12.7   |
|                  | 32   | AISI 304 |                     |        |       |        | 0.546 | 3.33   | 0.622 | 3.79   | 0.772 | 4.71   | 0.921 | 5.61   | 1.14  | 6.95   | 1.49  | 9.08   | 1.84 | 11.2   | 2.17 | 13.2   |
|                  |      | AISI 430 |                     |        |       |        | 0.53  | 3.23   | 0.604 | 3.68   | 0.75  | 4.57   | 0.894 | 5.45   | 1.11  | 6.77   | 1.45  | 8.84   | 1.78 | 10.9   | 2.10 | 19.8   |
| 1 1/2            | 38.1 | AISI 304 |                     |        |       |        | 0.652 | 3.97   | 0.743 | 4.53   | 0.924 | 5.63   | 1.10  | 6.71   | 1.37  | 8.35   | 1.80  | 11.0   | 2.22 | 13.5   | 2.62 | 16.0   |
|                  |      | AISI 430 |                     |        |       |        | 0.633 | 3.86   | 0.722 | 4.40   | 0.897 | 5.47   | 1.07  | 6.52   | 1.33  | 8.11   | 1.75  | 10.7   | 2.15 | 13.1   | 2.55 | 15.3   |
| 1 5/8            | 41.3 | AISI 304 |                     |        |       |        |       |        | 0.807 | 4.92   | 1.00  | 6.10   | 1.20  | 7.32   | 1.49  | 9.08   | 1.96  | 11.9   | 2.42 | 14.8   | 2.86 | 17.4   |
|                  |      | AISI 430 |                     |        |       |        |       |        | 0.784 | 4.78   | 0.974 | 5.94   | 1.16  | 7.07   | 1.44  | 8.78   | 1.90  | 11.6   | 2.35 | 14.3   | 2.78 | 16.9   |
| 1 3/4            | 44.5 | AISI 304 |                     |        |       |        |       |        | 0.871 | 5.31   | 1.08  | 6.58   | 1.29  | 7.86   | 1.61  | 9.81   | 2.12  | 12.9   | 2.62 | 16.0   | 3.10 | 18.9   |
|                  |      | AISI 430 |                     |        |       |        |       |        | 0.846 | 5.16   | 1.05  | 6.40   | 1.26  | 7.68   | 1.56  | 9.51   | 2.06  | 12.6   | 2.54 | 15.5   | 3.01 | 18.3   |
|                  | 45   | AISI 304 |                     |        |       |        |       |        | 0.881 | 5.37   | 1.10  | 6.71   | 1.31  | 7.99   | 1.63  | 9.94   | 2.14  | 13.0   | 2.65 | 16.2   | 3.14 | 19.1   |
|                  |      | AISI 430 |                     |        |       |        |       |        | 0.855 | 5.21   | 1.06  | 6.46   | 1.27  | 7.74   | 1.58  | 9.63   | 2.08  | 12.7   | 2.57 | 15.7   | 3.05 | 18.6   |
| 1.9              | 48.3 | AISI 304 |                     |        |       |        |       |        | 0.947 | 5.77   | 1.18  | 7.19   | 1.41  | 8.60   | 1.75  | 10.7   | 2.31  | 14.1   | 2.85 | 17.4   | 3.39 | 20.7   |
|                  |      | AISI 430 |                     |        |       |        |       |        | 0.92  | 5.61   | 1.14  | 6.95   | 1.37  | 8.35   | 1.70  | 10.4   | 2.24  | 13.7   | 2.77 | 16.9   | 3.29 | 20.1   |
|                  | 50   | AISI 304 |                     |        |       |        |       |        |       |        | 1.22  | 7.44   | 1.46  | 8.90   | 1.81  | 11.0   | 2.39  | 14.6   | 2.96 | 18.0   | 3.51 | 21.4   |
|                  |      | AISI 430 |                     |        |       |        |       |        |       |        | 1.19  | 7.25   | 1.42  | 8.66   | 1.76  | 10.7   | 2.32  | 14.1   | 2.87 | 17.5   | 3.41 | 20.8   |
| 2                | 50.8 | AISI 304 |                     |        |       |        |       |        |       |        | 1.24  | 7.56   | 1.48  | 9.02   | 1.84  | 11.2   | 2.43  | 14.8   | 3.01 | 18.3   | 3.57 | 21.8   |
|                  |      | AISI 430 |                     |        |       |        |       |        |       |        | 1.20  | 7.32   | 1.44  | 8.78   | 1.79  | 10.9   | 2.36  | 14.4   | 2.92 | 17.8   | 3.47 | 21.2   |
|                  | 60.8 | AISI 304 |                     |        |       |        |       |        |       |        |       |        | 1.77  | 10.8   | 2.20  | 13.4   | 2.91  | 17.7   | 3.61 | 22.0   | 4.30 | 26.2   |
|                  |      | AISI 430 |                     |        |       |        |       |        |       |        |       |        | 1.72  | 10.5   | 2.14  | 13.0   | 2.83  | 17.3   | 3.51 | 21.4   | 4.17 | 25.4   |
| 2 1/2            | 63.5 | AISI 304 |                     |        |       |        |       |        |       |        |       |        | 1.86  | 11.3   | 2.32  | 14.1   | 3.06  | 18.7   | 3.80 | 23.2   | 4.52 | 27.6   |
|                  |      | AISI 430 |                     |        |       |        |       |        |       |        |       |        | 1.81  | 11.0   | 2.25  | 13.7   | 2.98  | 18.2   | 3.69 | 22.5   | 4.39 | 26.8   |
| 3                | 76.2 | AISI 304 |                     |        |       |        |       |        |       |        |       |        | 2.24  | 13.7   | 2.79  | 17.0   | 3.70  | 22.6   | 4.59 | 28.0   | 5.47 | 33.3   |
|                  |      | AISI 430 |                     |        |       |        |       |        |       |        |       |        | 2.18  | 13.3   | 2.71  | 16.5   | 3.59  | 21.9   | 4.46 | 27.2   | 5.31 | 32.4   |

# STAINLESS STEEL WELDED TUBINGS

## SQUARE TUBING

TABLE OF UNIT WEIGHT

| Side length<br>mm (inch)                           | Type     | Wall thickness mm, BWG (inch) |        |       |        |       |        |      |        |      |        |              |        |              |        |
|----------------------------------------------------|----------|-------------------------------|--------|-------|--------|-------|--------|------|--------|------|--------|--------------|--------|--------------|--------|
|                                                    |          | 1.0                           |        | 1.2   |        | 1.5   |        | 2.0  |        | 3.0  |        | BWG18(0.049) |        | BWG16(0.065) |        |
|                                                    |          | kg/m                          | kg/20' | kg/m  | kg/20' | kg/m  | kg/20' | kg/m | kg/20' | kg/m | kg/20' | kg/ft        | kg/20' | kg/ft        | kg/20' |
| 12 x 12                                            | AISI 304 | 0.347                         | 2.12   | 0.409 | 2.49   |       |        |      |        |      |        | 0.128        | 2.57   |              |        |
|                                                    | AISI 430 | 0.337                         | 2.05   | 0.397 | 2.42   |       |        |      |        |      |        | 0.125        | 2.49   |              |        |
| 12.7 x 12.7<br>( $\frac{1}{2}$ x $\frac{1}{2}$ )   | AISI 304 | 0.369                         | 2.25   | 0.436 | 2.66   |       |        |      |        |      |        | 0.137        | 2.74   |              |        |
|                                                    | AISI 430 | 0.359                         | 2.19   | 0.423 | 2.58   |       |        |      |        |      |        | 0.133        | 2.66   |              |        |
| 14 x 14                                            | AISI 304 | 0.411                         | 2.51   | 0.485 | 2.96   |       |        |      |        |      |        | 0.152        | 3.05   |              |        |
|                                                    | AISI 430 | 0.399                         | 2.43   | 0.471 | 2.87   |       |        |      |        |      |        | 0.148        | 2.96   |              |        |
| 15.9 x 15.9<br>( $\frac{5}{8}$ x $\frac{5}{8}$ )   | AISI 304 | 0.471                         | 2.87   | 0.558 | 3.40   |       |        |      |        |      |        | 0.175        | 3.51   |              |        |
|                                                    | AISI 430 | 0.457                         | 2.79   | 0.542 | 3.30   |       |        |      |        |      |        | 0.17         | 3.40   |              |        |
| 16 x 16                                            | AISI 304 | 0.474                         | 2.89   | 0.562 | 3.43   |       |        |      |        |      |        | 0.176        | 3.53   |              |        |
|                                                    | AISI 430 | 0.46                          | 2.8    | 0.545 | 3.32   |       |        |      |        |      |        | 0.171        | 3.43   |              |        |
| 19 x 19                                            | AISI 304 | 0.569                         | 3.47   | 0.676 | 4.12   | 0.831 | 5.07   |      |        |      |        | 0.212        | 4.25   | 0.276        | 5.52   |
|                                                    | AISI 430 | 0.553                         | 3.37   | 0.656 | 4.00   | 0.807 | 4.92   |      |        |      |        | 0.206        | 4.13   | 0.268        | 5.36   |
| 19.1 x 19.1<br>( $\frac{3}{4}$ x $\frac{3}{4}$ )   | AISI 304 | 0.572                         | 3.49   | 0.68  | 4.15   | 0.836 | 5.10   |      |        |      |        | 0.214        | 4.27   | 0.278        | 5.56   |
|                                                    | AISI 430 | 0.556                         | 3.39   | 0.66  | 4.02   | 0.811 | 4.94   |      |        |      |        | 0.207        | 4.15   | 0.27         | 5.39   |
| 20 x 20                                            | AISI 304 | 0.601                         | 3.66   | 0.714 | 4.35   | 0.878 | 5.35   |      |        |      |        | 0.224        | 4.49   | 0.292        | 5.85   |
|                                                    | AISI 430 | 0.584                         | 3.56   | 0.693 | 4.22   | 0.853 | 5.20   |      |        |      |        | 0.218        | 4.36   | 0.284        | 5.68   |
| 21 x 21                                            | AISI 304 | 0.633                         | 3.86   | 0.752 | 4.58   | 0.926 | 5.64   |      |        |      |        | 0.236        | 4.72   | 0.308        | 6.16   |
|                                                    | AISI 430 | 0.614                         | 3.74   | 0.73  | 4.45   | 0.899 | 5.48   |      |        |      |        | 0.23         | 4.59   | 0.299        | 5.99   |
| 22.2 x 22.2<br>( $\frac{7}{8}$ x $\frac{7}{8}$ )   | AISI 304 | 0.671                         | 4.09   | 0.798 | 4.86   | 0.983 | 5.99   |      |        |      |        | 0.251        | 5.02   | 0.326        | 6.52   |
|                                                    | AISI 430 | 0.651                         | 3.97   | 0.774 | 4.72   | 0.955 | 5.82   |      |        |      |        | 0.244        | 4.88   | 0.317        | 6.34   |
| 25 x 25                                            | AISI 304 | 0.76                          | 4.63   | 0.904 | 5.51   | 1.12  | 6.83   |      |        |      |        | 0.284        | 5.69   | 0.372        | 7.44   |
|                                                    | AISI 430 | 0.738                         | 4.5    | 0.878 | 5.35   | 1.08  | 6.58   |      |        |      |        | 0.276        | 5.52   | 0.36         | 7.19   |
| 25.4 x 25.4<br>(1 x 1)                             | AISI 304 | 0.772                         | 4.71   | 0.919 | 5.60   | 1.14  | 6.95   |      |        |      |        | 0.289        | 5.79   | 0.378        | 7.56   |
|                                                    | AISI 430 | 0.75                          | 4.57   | 0.893 | 5.44   | 1.10  | 6.71   |      |        |      |        | 0.281        | 5.61   | 0.369        | 7.38   |
| 28 x 28                                            | AISI 304 | 0.855                         | 5.21   | 1.02  | 6.22   | 1.26  | 7.68   |      |        |      |        | 0.32         | 6.40   | 0.421        | 8.41   |
|                                                    | AISI 430 | 0.83                          | 5.06   | 0.989 | 6.03   | 1.22  | 7.44   |      |        |      |        | 0.311        | 6.22   | 0.408        | 8.17   |
| 30 x 30                                            | AISI 304 | 0.918                         | 5.60   | 1.09  | 6.64   | 1.35  | 8.23   | 1.77 | 10.8   |      |        | 0.344        | 6.89   | 0.452        | 9.03   |
|                                                    | AISI 430 | 0.892                         | 5.44   | 1.06  | 6.46   | 1.32  | 8.05   | 1.72 | 10.5   |      |        | 0.335        | 6.71   | 0.439        | 8.78   |
| 31.8 x 31.8<br>( $1\frac{1}{4}$ x $1\frac{1}{4}$ ) | AISI 304 | 0.975                         | 5.94   | 1.16  | 7.07   | 1.44  | 8.78   | 1.89 | 11.5   |      |        | 0.366        | 7.32   | 0.482        | 9.63   |
|                                                    | AISI 430 | 0.947                         | 5.77   | 1.13  | 6.89   | 1.40  | 8.53   | 1.83 | 11.2   |      |        | 0.357        | 7.13   | 0.466        | 9.33   |
| 32 x 32                                            | AISI 304 | 0.982                         | 5.99   | 1.17  | 7.13   | 1.45  | 8.84   | 1.90 | 11.6   |      |        | 0.369        | 7.38   | 0.485        | 9.69   |
|                                                    | AISI 430 | 0.953                         | 5.81   | 1.14  | 6.95   | 1.41  | 8.60   | 1.85 | 11.3   |      |        | 0.357        | 7.13   | 0.469        | 9.39   |
| 34 x 34                                            | AISI 304 | 1.05                          | 6.40   | 1.25  | 7.62   | 1.54  | 9.39   | 2.03 | 12.4   |      |        | 0.393        | 7.86   | 0.515        | 10.3   |
|                                                    | AISI 430 | 1.01                          | 6.16   | 1.21  | 7.38   | 1.50  | 9.14   | 1.97 | 12.0   |      |        | 0.381        | 7.62   | 0.50         | 10.0   |
| 35 x 35                                            | AISI 304 | 1.08                          | 6.58   | 1.28  | 7.80   | 1.59  | 9.69   | 2.09 | 12.7   |      |        | 0.405        | 8.11   | 0.531        | 10.6   |
|                                                    | AISI 430 | 1.05                          | 6.40   | 1.25  | 7.62   | 1.55  | 9.45   | 2.03 | 12.4   |      |        | 0.393        | 7.86   | 0.515        | 10.3   |
| 38 x 38                                            | AISI 304 | 1.17                          | 7.13   | 1.40  | 8.53   | 1.73  | 10.5   | 2.28 | 13.9   |      |        | 0.439        | 8.78   | 0.579        | 11.6   |
|                                                    | AISI 430 | 1.14                          | 6.95   | 1.36  | 8.29   | 1.68  | 10.2   | 2.22 | 13.5   |      |        | 0.427        | 8.53   | 0.564        | 11.3   |
| 38.1 x 38.1<br>( $1\frac{1}{2}$ x $1\frac{1}{2}$ ) | AISI 304 | 1.18                          | 7.19   | 1.40  | 8.53   | 1.74  | 10.6   | 2.29 | 14.0   |      |        | 0.442        | 8.84   | 0.582        | 11.6   |
|                                                    | AISI 430 | 1.14                          | 6.95   | 1.36  | 8.29   | 1.69  | 10.3   | 2.22 | 13.5   |      |        | 0.43         | 8.60   | 0.564        | 11.3   |
| 40 x 40                                            | AISI 304 | 1.24                          | 7.56   | 1.48  | 9.02   | 1.83  | 11.2   | 2.41 | 14.7   | 3.52 | 21.5   | 0.463        | 9.27   | 0.613        | 12.3   |
|                                                    | AISI 430 | 1.20                          | 7.32   | 1.43  | 8.72   | 1.78  | 10.9   | 2.34 | 14.3   | 3.42 | 20.8   | 0.451        | 9.02   | 0.594        | 11.9   |
| 46 x 46                                            | AISI 304 | 1.43                          | 8.72   | 1.70  | 10.4   | 2.12  | 12.9   | 2.79 | 17.0   | 4.09 | 24.9   | 0.536        | 10.7   | 0.707        | 14.1   |
|                                                    | AISI 430 | 1.38                          | 8.41   | 1.65  | 10.1   | 2.05  | 12.5   | 2.71 | 16.5   | 3.97 | 24.2   | 0.521        | 10.4   | 0.686        | 13.7   |
| 50 x 50                                            | AISI 304 |                               |        | 1.86  | 11.3   | 2.31  | 14.1   | 3.04 | 18.5   | 4.47 | 27.2   | 0.585        | 11.7   | 0.771        | 15.4   |
|                                                    | AISI 430 |                               |        | 1.80  | 11.0   | 2.24  | 13.7   | 2.96 | 18.0   | 4.34 | 26.5   | 0.567        | 11.3   | 0.75         | 15.0   |
| 50.8 x 50.8<br>(2 x 2)                             | AISI 304 |                               |        | 1.89  | 11.5   | 2.34  | 14.3   | 3.09 | 18.8   | 4.55 | 27.7   | 0.594        | 11.9   | 0.783        | 15.7   |
|                                                    | AISI 430 |                               |        | 1.83  | 11.2   | 2.28  | 13.9   | 3.00 | 18.3   | 4.42 | 26.9   | 0.576        | 11.5   | 0.762        | 15.2   |
| 60 x 60                                            | AISI 304 |                               |        | 2.24  | 13.7   | 2.78  | 16.9   | 3.68 | 22.4   | 5.42 | 33.0   | 0.704        | 14.1   | 0.93         | 18.6   |
|                                                    | AISI 430 |                               |        | 2.17  | 13.2   | 2.70  | 16.5   | 3.57 | 21.8   | 5.27 | 32.1   | 0.683        | 13.7   | 0.902        | 18.0   |

# STAINLESS STEEL WELDED TUBINGS

## RECTANGULAR TUBING

### TABLE OF UNIT WEIGHT

| Side length<br>mm (inch)   | Type     | Wall thickness mm, BWG (inch) |        |       |        |       |        |      |        |                |        |                |        |
|----------------------------|----------|-------------------------------|--------|-------|--------|-------|--------|------|--------|----------------|--------|----------------|--------|
|                            |          | 1.0                           |        | 1.2   |        | 1.5   |        | 2.0  |        | BWG 18 (0.049) |        | BWG 16 (0.065) |        |
|                            |          | kg/m                          | kg/20' | kg/m  | kg/20' | kg/m  | kg/20' | kg/m | kg/20' | kg/ft          | kg/20' | kg/ft          | kg/20' |
| 19 x 10                    | AISI 304 | 0.426                         | 2.60   | 0.505 | 3.08   |       |        |      |        | 0.158          | 3.17   |                |        |
|                            | AISI 430 | 0.414                         | 2.52   | 0.49  | 2.99   |       |        |      |        | 0.154          | 3.08   |                |        |
| 20 x 10                    | AISI 304 | 0.442                         | 2.69   | 0.524 | 3.19   |       |        |      |        | 0.164          | 3.29   |                |        |
|                            | AISI 430 | 0.43                          | 2.62   | 0.508 | 3.10   |       |        |      |        | 0.16           | 3.19   |                |        |
| 20 x 15                    | AISI 304 | 0.522                         | 3.18   | 0.619 | 3.77   |       |        |      |        | 0.194          | 3.89   |                |        |
|                            | AISI 430 | 0.507                         | 3.09   | 0.601 | 3.66   |       |        |      |        | 0.189          | 3.77   |                |        |
| 22 x 10                    | AISI 304 | 0.474                         | 2.89   | 0.562 | 3.43   |       |        |      |        | 0.176          | 3.53   |                |        |
|                            | AISI 430 | 0.46                          | 2.80   | 0.545 | 3.32   |       |        |      |        | 0.171          | 3.43   |                |        |
| 22 x 12                    | AISI 304 | 0.506                         | 3.08   | 0.60  | 3.66   |       |        |      |        | 0.188          | 3.77   |                |        |
|                            | AISI 430 | 0.491                         | 2.99   | 0.582 | 3.55   |       |        |      |        | 0.183          | 3.66   |                |        |
| 24 x 14                    | AISI 304 | 0.569                         | 3.47   | 0.676 | 4.12   | 0.831 | 5.07   |      |        | 0.212          | 4.25   | 0.276          | 5.52   |
|                            | AISI 430 | 0.553                         | 3.37   | 0.656 | 4.00   | 0.807 | 4.92   |      |        | 0.206          | 4.13   | 0.268          | 5.36   |
| 25 x 14                    | AISI 304 | 0.585                         | 3.57   | 0.695 | 4.23   | 0.855 | 5.21   |      |        | 0.218          | 4.36   | 0.284          | 5.68   |
|                            | AISI 430 | 0.568                         | 3.46   | 0.675 | 4.11   | 0.83  | 5.06   |      |        | 0.212          | 4.24   | 0.276          | 5.52   |
| 25.4 x 12.7<br>(1 x 1/2)   | AISI 304 | 0.571                         | 3.48   | 0.678 | 4.13   | 0.833 | 5.08   |      |        | 0.213          | 4.26   | 0.277          | 5.54   |
|                            | AISI 430 | 0.554                         | 3.38   | 0.658 | 4.01   | 0.809 | 4.93   |      |        | 0.207          | 4.14   | 0.269          | 5.38   |
| 30 x 10                    | AISI 304 | 0.601                         | 3.66   | 0.714 | 4.35   | 0.878 | 5.35   |      |        | 0.224          | 4.49   | 0.292          | 5.85   |
|                            | AISI 430 | 0.584                         | 3.56   | 0.693 | 4.22   | 0.853 | 5.20   |      |        | 0.218          | 4.36   | 0.284          | 5.68   |
| 30 x 20                    | AISI 304 | 0.76                          | 4.63   | 0.904 | 5.51   | 1.12  | 6.83   |      |        | 0.284          | 5.69   | 0.372          | 7.44   |
|                            | AISI 430 | 0.738                         | 4.50   | 0.878 | 5.35   | 1.08  | 6.58   |      |        | 0.276          | 5.52   | 0.36           | 7.19   |
| 32 x 16                    | AISI 304 | 0.728                         | 4.44   | 0.866 | 5.28   | 1.07  | 6.52   |      |        | 0.272          | 5.44   | 0.357          | 7.13   |
|                            | AISI 430 | 0.707                         | 4.31   | 0.841 | 5.13   | 1.04  | 6.34   |      |        | 0.265          | 5.29   | 0.344          | 6.89   |
| 32 x 17                    | AISI 304 | 0.744                         | 4.54   | 0.885 | 5.39   | 1.09  | 6.64   |      |        | 0.278          | 5.57   | 0.363          | 7.25   |
|                            | AISI 430 | 0.722                         | 4.40   | 0.859 | 5.24   | 1.06  | 6.46   |      |        | 0.27           | 5.41   | 0.354          | 7.07   |
| 38.1 x 25.4<br>(1 1/2 x 1) | AISI 304 | 0.974                         | 5.94   | 1.16  | 7.07   | 1.44  | 8.78   |      |        | 0.366          | 7.32   | 0.479          | 9.57   |
|                            | AISI 430 | 0.945                         | 5.76   | 1.13  | 6.89   | 1.40  | 8.53   |      |        | 0.354          | 7.07   | 0.466          | 9.33   |
| 40 x 10                    | AISI 304 | 0.76                          | 4.63   | 0.904 | 5.51   | 1.12  | 6.83   |      |        | 0.284          | 5.69   | 0.372          | 7.44   |
|                            | AISI 430 | 0.738                         | 4.50   | 0.878 | 5.35   | 1.08  | 6.58   |      |        | 0.276          | 5.52   | 0.36           | 7.19   |
| 40 x 15                    | AISI 304 | 0.839                         | 5.11   | 0.999 | 6.09   | 1.24  | 7.56   |      |        | 0.314          | 6.28   | 0.411          | 8.23   |
|                            | AISI 430 | 0.815                         | 4.97   | 0.97  | 5.91   | 1.20  | 7.32   |      |        | 0.305          | 6.10   | 0.399          | 7.99   |
| 40 x 18                    | AISI 304 | 0.886                         | 5.40   | 1.06  | 6.46   | 1.31  | 7.99   |      |        | 0.332          | 6.64   | 0.436          | 8.72   |
|                            | AISI 430 | 0.861                         | 5.25   | 1.03  | 6.28   | 1.27  | 7.74   |      |        | 0.323          | 6.46   | 0.424          | 8.47   |
| 40 x 20                    | AISI 304 | 0.918                         | 5.60   | 1.09  | 6.64   | 1.35  | 8.23   | 1.77 | 10.8   | 0.344          | 6.89   | 0.451          | 9.02   |
|                            | AISI 430 | 0.892                         | 5.44   | 1.06  | 6.46   | 1.31  | 7.99   | 1.72 | 10.5   | 0.335          | 6.71   | 0.439          | 8.78   |
| 40 x 25                    | AISI 304 |                               |        | 1.19  | 7.25   | 1.47  | 8.96   | 1.93 | 11.8   | 0.375          | 7.50   | 0.491          | 9.81   |
|                            | AISI 430 |                               |        | 1.16  | 7.07   | 1.43  | 8.72   | 1.88 | 11.5   | 0.363          | 7.25   | 0.479          | 9.57   |
| 40 x 28                    | AISI 304 |                               |        | 1.25  | 7.62   | 1.54  | 9.39   | 2.03 | 12.4   | 0.393          | 7.86   | 0.515          | 10.3   |
|                            | AISI 430 |                               |        | 1.21  | 7.38   | 1.50  | 9.14   | 1.97 | 12.0   | 0.381          | 7.62   | 0.50           | 10.0   |
| 50 x 25                    | AISI 304 |                               |        | 1.38  | 8.41   | 1.71  | 10.4   | 2.25 | 13.7   | 0.433          | 8.66   | 0.57           | 11.4   |
|                            | AISI 430 |                               |        | 1.34  | 8.17   | 1.66  | 10.1   | 2.19 | 13.4   | 0.421          | 8.41   | 0.555          | 11.1   |
| 50.8 x 25.4<br>(2 x 1)     | AISI 304 |                               |        | 1.40  | 8.53   | 1.74  | 10.6   | 2.29 | 14.0   | 0.442          | 8.84   | 0.582          | 11.6   |
|                            | AISI 430 |                               |        | 1.36  | 8.29   | 1.69  | 10.3   | 2.22 | 13.5   | 0.43           | 8.60   | 0.564          | 11.3   |
| 51x 26                     | AISI 304 |                               |        | 1.42  | 8.66   | 1.76  | 10.7   | 2.31 | 14.1   | 0.445          | 8.90   | 0.588          | 11.8   |
|                            | AISI 430 |                               |        | 1.38  | 8.41   | 1.71  | 10.4   | 2.25 | 13.7   | 0.433          | 8.66   | 0.57           | 11.4   |
| 60 x 20                    | AISI 304 |                               |        | 1.48  | 9.02   | 1.83  | 11.2   | 2.41 | 14.7   | 0.463          | 9.27   | 0.613          | 12.3   |
|                            | AISI 430 |                               |        | 1.43  | 8.72   | 1.78  | 10.9   | 2.34 | 14.3   | 0.451          | 9.02   | 0.594          | 11.9   |
| 60 x 40                    | AISI 304 |                               |        | 1.86  | 11.3   | 2.31  | 14.1   | 3.04 | 18.5   | 0.585          | 11.7   | 0.771          | 15.4   |
|                            | AISI 430 |                               |        | 1.80  | 11.0   | 2.24  | 13.7   | 2.96 | 18.0   | 0.567          | 11.3   | 0.75           | 15.0   |
| 75 x 45                    | AISI 304 |                               |        |       |        | 2.78  | 16.9   | 3.68 | 22.4   |                |        | 0.93           | 18.6   |
|                            | AISI 430 |                               |        |       |        | 2.70  | 16.5   | 3.57 | 21.8   |                |        | 0.902          | 18.0   |
| 76.2 x 25.4<br>(3 x 1)     | AISI 304 |                               |        |       |        | 2.34  | 14.3   | 3.09 | 18.8   |                |        | 0.783          | 15.7   |
|                            | AISI 430 |                               |        |       |        | 2.28  | 13.9   | 3.00 | 18.3   |                |        | 0.762          | 15.2   |
| 80x40                      | AISI 304 |                               |        |       |        | 2.78  | 16.9   | 3.68 | 22.4   |                |        | 0.93           | 18.6   |
|                            | AISI 430 |                               |        |       |        | 2.70  | 16.5   | 3.57 | 21.8   |                |        | 0.902          | 18.0   |

# STAINLESS STEEL WELDED TUBINGS

\*SWG = Standard Wire Gauge  
BWG = Birmingham Wire Gauge  
SWG can almost invariably be  
replaced by BWG as the wall of  
BWG is slightly thicker.

## INSTRUMENTATION TUBE

| Outside<br>Diameter | Wall<br>Thickness                            | Nominal*<br>Size                                                | Weight                                             | 3R12<br>SS<br>2352<br>AISI<br>304L | 3R60<br>2353<br>316L       | 8R30<br>2337<br>321   | 8R70<br>2350<br>(316Ti) |
|---------------------|----------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------|------------------------------------|----------------------------|-----------------------|-------------------------|
| mm                  | mm                                           |                                                                 | kg/m                                               |                                    |                            |                       |                         |
| 6                   | 1.0<br>1.5                                   |                                                                 | 0.125<br>0.169                                     | •                                  | •<br>•                     | •<br>•                | •<br>•                  |
| 6.35                | 0.89<br>0.91<br>1.22<br>1.24                 | 1/4" X 20 BWG<br>20 SWG<br>18 SWG<br>18 BWG                     | 0.122<br>0.124<br>0.157<br>0.159                   |                                    | •<br>•<br>•<br>•           |                       |                         |
| 8                   | 1.0<br>1.5<br>2.0                            |                                                                 | 0.175<br>0.244<br>0.300                            | •<br>•<br>•                        | •<br>•<br>•                | •<br>•<br>•           | •<br>•<br>•             |
| 9.53                | 0.89<br>0.91<br>1.22<br>1.24<br>1.63<br>1.65 | 3/8" x 20 BWG<br>20 SWG<br>18 SWG<br>18 BWG<br>16 SWG<br>16 BWG | 0.193<br>0.196<br>0.254<br>0.257<br>0.322<br>0.326 |                                    | •<br>•<br>•<br>•<br>•<br>• |                       |                         |
| 10                  | 1.0<br>1.5<br>2.0                            |                                                                 | 0.225<br>0.319<br>0.400                            | •<br>•<br>•                        | •<br>•<br>•                | •<br>•<br>•           | •<br>•<br>•             |
| 10.2                | 2.0                                          |                                                                 | 0.411                                              | •                                  | •                          |                       |                         |
| 12                  | 1.0<br>1.5<br>2.0                            |                                                                 | 0.275<br>0.394<br>0.500                            | •<br>•<br>•                        | •<br>•<br>•                | •<br>•<br>•           | •<br>•<br>•             |
| 12.7                | 1.22<br>1.24<br>1.63<br>1.65                 | 1/2" X 18 SWG<br>18 BWG<br>16 SWG<br>16 BWG                     | 0.350<br>0.356<br>0.452<br>0.456                   |                                    | •<br>•<br>•<br>•           |                       |                         |
| 14                  | 1.0<br>1.5<br>2.0                            |                                                                 | 0.325<br>0.469<br>0.600                            | •<br>•<br>•                        | •<br>•<br>•                | •<br>•<br>•           | •<br>•<br>•             |
| 15                  | 1.0<br>1.5<br>2.0                            |                                                                 | 0.350<br>0.507<br>0.651                            | •<br>•<br>•                        | •<br>•<br>•                | •<br>•<br>•           | •<br>•<br>•             |
| 15.88               | 1.22<br>1.24                                 | 5/8" X 18 SWG<br>18 BWG                                         | 0.448<br>0.454                                     |                                    | •<br>•                     |                       |                         |
| 16                  | 1.0<br>1.5<br>2.0<br>2.5<br>3.0              |                                                                 | 0.375<br>0.544<br>0.701<br>0.845<br>0.976          | •<br>•<br>•<br>•<br>•              | •<br>•<br>•<br>•<br>•      | •<br>•<br>•<br>•<br>• | •<br>•<br>•<br>•<br>•   |
| 17.2                | 1.6<br>2.9                                   |                                                                 | 0.625<br>1.038                                     | •<br>•                             | •<br>•                     | •<br>•                | •<br>•                  |
| 18                  | 1.0<br>1.5<br>2.0<br>2.5<br>3.0              |                                                                 | 0.425<br>0.619<br>0.801<br>0.970<br>1.126          | •<br>•<br>•<br>•<br>•              | •<br>•<br>•<br>•<br>•      | •<br>•<br>•<br>•<br>• | •<br>•<br>•<br>•<br>•   |
| 19.05               | 1.22<br>1.24                                 | 3/4" X 18 SWG<br>18 BWG                                         | 0.544<br>0.553                                     |                                    | •<br>•                     |                       |                         |
| 20                  | 1.5<br>2.0<br>2.5<br>3.0<br>4.0              |                                                                 | 0.694<br>0.901<br>1.095<br>1.28<br>1.60            | •<br>•<br>•<br>•<br>•              | •<br>•<br>•<br>•<br>•      | •<br>•<br>•<br>•<br>• | •<br>•<br>•<br>•<br>•   |

### TOLERANCES

Metric sizes acc. to ISO 1127

| Outside<br>Diameter               | Wall<br>Thickness              |
|-----------------------------------|--------------------------------|
| ±0.75%, but min.<br>±0.30 mm (D3) | ±10%, but min.<br>±0.2 mm (T3) |

Tube in imperial sizes acc. to ASTM A213/A450

| Size  | Outside<br>Diameter | Wall Thickness<br>average wall* | min wall |
|-------|---------------------|---------------------------------|----------|
| mm    | mm                  | %                               | %        |
| <25.4 | +0.10/-0.10         | + 10/-10                        | + 20/0   |

\* Valid for stocked sizes.

### SPECIFICATIONS

3R12 (AISI 304L) and 3R60 (AISI 316L)  
SS 219711 (condition 22, i.e. 3R12=SS 2352-22)  
DIN 17458, testing level 1  
NFA 49-117  
8R30 (AISI 321) and 8R70 (AISI 316Ti)  
DIN 17458, testing level 1  
Imperial size  
3R60 (AISI 316L) ASTM A213, A269

# STAINLESS STEEL PIPES

| Symbol                   |         | Chemical Composition (%) |       |       |       |       |                 |                  |               |                  |
|--------------------------|---------|--------------------------|-------|-------|-------|-------|-----------------|------------------|---------------|------------------|
| JIS                      | ASTM    | C                        | Si    | Mn    | P     | S     | Ni              | Cr               | Mo            | Others           |
| SUS 304TP<br>SUS 304TB   |         | ≤0.08                    | ≤1.00 | ≤2.00 | ≤0.04 | ≤0.03 | 8.00~<br>11.00  | 18.00~<br>20.00  | --            |                  |
|                          | TP 304  | ≤0.08                    | ≤0.75 | ≤2.00 | ≤0.04 | ≤0.03 | 8.00~<br>11.00  | 18.00~<br>20.00  | --            |                  |
| SUS 304LTP<br>SUS 304LTB |         | ≤0.03                    | ≤1.00 | ≤2.00 | ≤0.04 | ≤0.03 | 9.00~<br>13.00  | 18.00~<br>20.00  | --            |                  |
|                          | TP 304L | ≤0.035*                  | ≤0.75 | ≤2.00 | ≤0.04 | ≤0.03 | 8.00~<br>13.00  | 18.00 ~<br>20.00 | --            |                  |
| SUS 321TP<br>SUS 321TB   |         | ≤0.08                    | ≤1.00 | ≤2.00 | ≤0.04 | ≤0.03 | 9.00~<br>13.00  | 17.00~<br>19.00  | --            | Ti 5 x C% min.   |
|                          | TP 321  | ≤0.08                    | ≤0.75 | ≤2.00 | ≤0.04 | ≤0.03 | 9.00~<br>13.00  | 17.00~<br>19.00  | --            | Ti 5 x C% ~ 0.60 |
| SUS 316TP<br>SUS 316TB   |         | ≤0.08                    | ≤1.00 | ≤2.00 | ≤0.04 | ≤0.03 | 10.00~<br>14.00 | 16.00~<br>18.00  | 2.00~<br>3.00 |                  |
|                          | TP 316  | ≤0.08                    | ≤0.75 | ≤2.00 | ≤0.04 | ≤0.03 | 11.00~<br>14.00 | 16.00~<br>18.00  | 2.00~<br>3.00 |                  |
| SUS 316LTP<br>SUS 316LTB |         | ≤0.03                    | ≤1.00 | ≤2.00 | ≤0.04 | ≤0.03 | 12.00~<br>16.00 | 16.00~<br>18.00  | 2.00~<br>3.00 |                  |
|                          | TP 316L | ≤0.035*                  | ≤0.75 | ≤2.00 | ≤0.04 | ≤0.03 | 10.00~<br>15.00 | 16.00~<br>18.00  | 2.00~<br>3.00 |                  |
| SUS 309STP<br>SUS 309STB |         | ≤0.15                    | ≤1.00 | ≤2.00 | ≤0.04 | ≤0.03 | 12.00~<br>15.00 | 22.00~<br>24.00  | --            |                  |
|                          | TP 309  | ≤0.15                    | ≤0.75 | ≤2.00 | ≤0.04 | ≤0.03 | 12.00~<br>15.00 | 22.00~<br>24.00  | --            |                  |
| SUS 310STP<br>SUS 310STB |         | ≤0.15                    | ≤1.50 | ≤2.00 | ≤0.04 | ≤0.03 | 19.00~<br>22.00 | 24.00~<br>26.00  | --            |                  |
|                          | TP 310  | ≤0.15                    | ≤0.75 | ≤2.00 | ≤0.04 | ≤0.03 | 19.00~<br>22.00 | 24.00~<br>26.00  | --            |                  |
| SUS 321HTP<br>SUS 321HTB |         | 0.04 ~<br>0.10           | ≤0.75 | ≤2.00 | ≤0.03 | ≤0.03 | 9.00~<br>13.00  | 17.00~<br>20.00  | --            | Ti 4 x C% ~ 0.60 |
|                          | TP 321H | 0.04 ~<br>0.10           | ≤0.75 | ≤2.00 | ≤0.04 | ≤0.03 | 9.00~<br>13.00  | 17.00~<br>20.00  | --            | Ti 4 x C% ~ 0.60 |

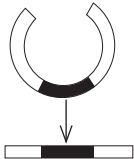
## Note:

- The above chemical composition shall be applied even when product analysis is requested by the purchasers. However, the carbon content of SUS 340LTP/LTB and SUS 316 LTP/LTB shall be 0.035% or less.
- Those with an asterisk (\*) having either a small diameter (i.e., outside diameter less than 0.5 in or 12.70 mm) or a thin wall (i.e., average wall thickness less than 0.049 in or 12.40 mm) or minimum wall thickness (less than 0.044 in or 1.12 mm) and requiring several drawings, shall have carbon content of 0.04% or less.

# STAINLESS STEEL PIPES

All stainless steel pipes and tubes for piping, boilers and heat exchangers subjected to solution heat treatment as follows.

| Symbol      |                                 | Solution Heat Treatment |                |
|-------------|---------------------------------|-------------------------|----------------|
| For piping  | For boilers and heat exchangers | Heat temperature (°C)   | Cooling method |
| SUS 304 TP  | SUS 304 TB                      | 1010°C min.             | Rapid cooling  |
| SUS 304L TP | SUS 304L TB                     | 1010°C min.             | Rapid cooling  |
| SUS 321 TP  | SUS 321 TB                      | 920°C min.              | Rapid cooling  |
| SUS 316 TP  | SUS 316 TB                      | 1010°C min.             | Rapid cooling  |
| SUS 316L TP | SUS 316L TB                     | 1010°C min.             | Rapid cooling  |
| SUS 309S TP | SUS 309S TB                     | 1030°C min.             | Rapid cooling  |
| SUS 310S TP | SUS 310S TB                     | 1030°C min.             | Rapid cooling  |
| SUS 321H TP | SUS 321H TB                     | 1095°C min.             | Rapid cooling  |
| TP 304      |                                 | 1040°C min.             | Rapid cooling  |
| TP 304L     |                                 | 1040°C min.             | Rapid cooling  |
| TP 321      |                                 | 1040°C min.             | Rapid cooling  |
| TP 316      |                                 | 1040°C min.             | Rapid cooling  |
| TP 316L     |                                 | 1040°C min.             | Rapid cooling  |
| TP 309      |                                 | 1040°C min.             | Rapid cooling  |
| TP 310      |                                 | 1040°C min.             | Rapid cooling  |
| TP 321H     |                                 | 1095°C min.             | Rapid cooling  |

| Symbol         | Tension test                           |                                      |                |            | Flattening test                                                                                                                                                                                                                                                                   | Flaring test                         | Reverse flattening test                                                                                                                                                           |
|----------------|----------------------------------------|--------------------------------------|----------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | Tensile strength (kg/mm <sup>2</sup> ) | Yield strength (kg/mm <sup>2</sup> ) | Elongation (%) |            |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
|                |                                        |                                      | Longitudinal   | Transverse |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| SUS 304TP/TB   |                                        |                                      |                |            | $H = \frac{(1 + e) t}{e + \frac{t}{D}}$<br>where,<br>H = Distance between flattening plates (mm)<br>t = Wall thickness (mm)<br>D = Outside diameter (mm)<br>(mm)<br>e = 0.09 (constant)<br>Note: Welded section shall be affixed at an angle 90° to the direction of compression. | 1.2 D<br>(D = Outside diameter (mm)) | <div>When flattened reversely, no cracks or breaks shall occur on its welded section.</div>  |
| TP 304         | 53 min.                                | 21 min.                              | 35 min.        | 25 min.    |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| SUS 304LTP/LTB |                                        |                                      |                |            |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| TP 304L        | 49 min.                                | 18 min.                              | 35 min.        | 25 min.    |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| SUS 321TP/TB   |                                        |                                      |                |            |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| TP 321         | 53 min.                                | 21 min.                              | 35 min.        | 25 min.    |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| SUS 316TP/TB   |                                        |                                      |                |            |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| TP 316         | 53 min.                                | 21 min.                              | 35 min.        | 25 min.    |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| SUS 316LTP/LTB |                                        |                                      |                |            |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| TP 316L        | 49 min.                                | 28 min.                              | 35 min.        | 25 min.    |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| SUS 309STP/STB |                                        |                                      |                |            |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| TP 309         | 53 min.                                | 21 min.                              | 35 min.        | 25 min.    |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| SUS 310STP/STB |                                        |                                      |                |            |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| TP 310S        | 53 min.                                | 21 min.                              | 35 min.        | 25 min.    |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| SUS 321HTP/HTB |                                        |                                      |                |            |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |
| TP 321H        | 53 min.                                | 21 min.                              | 35 min.        | 25 min.    |                                                                                                                                                                                                                                                                                   |                                      |                                                                                                                                                                                   |

Notes:

1. Tensile strength shall not be measured unless specifically requested.
2. Flaring test and reverse flattening test shall be conducted only on tubes for boilers and heat exchanger



# STAINLESS STEEL PIPES

| JIS                  | AISI             | Characteristics                                                                                                                                                                                                                                                                                                                                                 | Uses                                                                                                                                                                                                    |
|----------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SUS 304              | TP 304           | Most widely used stainless steel. Because of its nickel content, it has excellent corrosion and heat resistance, strength under low temperature and improved mechanical properties. It has work hardening characteristic, and hardening by heat treatment is impossible. Has no magnetism.                                                                      | Home appliances and kitchen utensils, architectural ornaments, vehicles, auto parts, medical equipment, and various applications in the food, chemical, and textile industries.                         |
| SUS 304L             | TP 304L          | Ni-Cr steel with extremely low carbon content. Although its degree of corrosion resistance under normal conditions is similar to that of SUS 304, it has greater resistance against intergranular corrosion after welding or stress-relief heat treatment. Has no magnetism.                                                                                    | Parts and structures in the chemical, petroleum, coal and pharmaceutical industries to which heat treatment is difficult to apply. Normally used at temperatures below 400°C.                           |
| SUS 321              | TP 321           | Excellent corrosion resistance. Other properties are similar to those of SUS 304. The resistance to intergranular corrosion is improved by addition of 18-8 type. It has no magnetism, and it is especially suited for use at temperatures between 430°C - 900°C. Though having no magnetism in annealed state, it becomes slightly magnetized by cold working. | Exhaust pipes of aircraft engines, boilers, jet engine parts, heating furnace parts, and other parts used in the chemical industry to which heat treatment after welding is difficult to apply.         |
| SUS 316              | TP 316           | Substantial amounts of nickel and chromium content greatly improve heat and corrosion resistance.<br><br>Work hardening characteristic but not magnetism.                                                                                                                                                                                                       | Various uses in the chemical, food, photographic, textile, paper and pulp industries. Especially suited for exterior of structures located near coastal areas.                                          |
| SUS 316L             | TP 316L          | Nickel-chromium steel containing molybdenum and an extremely small amount of carbon. Physical properties are similar to those of SUS 316. Excellent resistance against intergranular corrosion after welding or stress-relief heat treatment. Has no magnetism.                                                                                                 | Typical applications similar to those of SUS 316. For machine parts and equipment which are not easily subjected to heat treatment after welding. Especially suited for use at temperatures below 420°C |
| SUS 309S<br>SUS 310S | TP 309<br>TP 310 | Substantial amount of nickel and chromium content greatly improve heat and corrosion resistance.                                                                                                                                                                                                                                                                | Jet engine parts, tanks for chemicals, combustion apparatus parts, boilers, and gasturbine parts.                                                                                                       |

# STAINLESS STEEL PIPES

## SEAMLESS/WELDED (Dimension and Weight)

ASTM A312TP 304/304L/316/316L/321/317/347/310, ETC. ANSI-B36.19

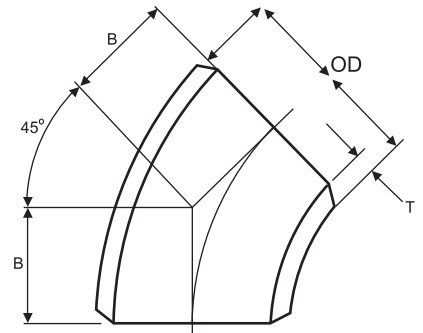
| N.B.   | Outside Diameter |        | SCH5S |       |        | SCH10S |       |        |
|--------|------------------|--------|-------|-------|--------|--------|-------|--------|
|        |                  |        | Thick | Thick | Weight | Thick  | Thick | Weight |
|        | inch             | mm     | mm    | mm    | kg/m   | inch   | mm    | kg/m   |
| 1/8"   | 0.405            | 10.29  | --    | --    | --     | 0.049  | 1.24  | 0.277  |
| 1/4"   | 0.54             | 13.72  | --    | --    | --     | 0.065  | 1.65  | 0.491  |
| 3/8"   | 0.675            | 17.15  | --    | --    | --     | 0.065  | 1.65  | 0.631  |
| 1/2"   | 0.84             | 21.34  | 0.065 | 1.65  | 0.801  | 0.083  | 2.11  | 1.00   |
| 3/4"   | 1.05             | 26.67  | 0.065 | 1.65  | 1.02   | 0.083  | 2.11  | 1.28   |
| 1"     | 1.315            | 33.40  | 0.065 | 1.65  | 1.29   | 0.109  | 2.77  | 2.09   |
| 1 1/4" | 1.66             | 42.16  | 0.065 | 1.65  | 1.65   | 0.109  | 2.77  | 2.69   |
| 1 1/2" | 1.90             | 48.26  | 0.065 | 1.65  | 1.90   | 0.109  | 2.77  | 3.11   |
| 2"     | 2.375            | 60.33  | 0.065 | 1.65  | 2.39   | 0.109  | 2.77  | 3.93   |
| 2 1/2" | 2.875            | 73.03  | 0.083 | 2.11  | 3.69   | 0.12   | 3.05  | 5.26   |
| 3"     | 3.50             | 88.90  | 0.083 | 2.11  | 4.52   | 0.12   | 3.05  | 6.46   |
| 3 1/2" | 4.00             | 101.60 | 0.083 | 2.11  | 5.18   | 0.12   | 3.05  | 7.41   |
| 4"     | 4.50             | 114.30 | 0.083 | 2.11  | 5.84   | 0.12   | 3.05  | 8.37   |
| 5"     | 5.563            | 141.30 | 0.109 | 2.71  | 9.46   | 0.134  | 3.40  | 11.60  |
| 6"     | 6.625            | 168.28 | 0.109 | 2.77  | 11.30  | 0.134  | 3.40  | 13.80  |
| 8"     | 8.625            | 219.08 | 0.109 | 2.77  | 14.80  | 0.148  | 3.76  | 20.00  |
| 10"    | 10.75            | 273.05 | 0.134 | 3.40  | 22.60  | 0.165  | 4.19  | 27.80  |
| 12"    | 12.75            | 323.85 | 0.156 | 3.96  | 31.20  | 0.18   | 4.57  | 36.00  |
| 14"    | 14.00            | 355.60 | 0.156 | 3.96  | 34.30  | 0.188  | 4.78  | 41.40  |
| 16"    | 16.00            | 406.40 | 0.165 | 4.19  | 41.60  | 0.188  | 4.78  | 47.30  |
| 18"    | 18.00            | 457.20 | 0.165 | 4.19  | 46.80  | 0.188  | 4.78  | 53.30  |
| 20"    | 20.00            | 508.00 | 0.188 | 4.78  | 59.30  | 0.218  | 5.54  | 68.60  |
| 22"    | 22.00            | 558.80 | 0.188 | 4.78  | 65.30  | 0.218  | 5.54  | 75.60  |
| 24"    | 24.00            | 609.60 | 0.218 | 5.54  | 82.50  | 0.25   | 6.36  | 94.50  |
| 30"    | 30.00            | 762.00 | 0.25  | 6.36  | 118.30 | 0.312  | 7.92  | 147.30 |

ASTM A312 TP 304/304L/316/316L/321/317/347/310, ETC. ANSI-B36.19

| N.B.   | Outside Diameter |        | SCH40S |       |        | SCH80S |       |        |
|--------|------------------|--------|--------|-------|--------|--------|-------|--------|
|        |                  |        | Thick  | Thick | Weight | Thick  | Thick | Weight |
|        | inch             | mm     | mm     | mm    | kg/m   | inch   | mm    | kg/m   |
| 1/8"   | 0.405            | 10.29  | 0.068  | 1.73  | 0.365  | 0.095  | 2.41  | 0.468  |
| 1/4"   | 0.54             | 13.72  | 0.088  | 2.24  | 0.634  | 0.119  | 3.02  | 0.797  |
| 3/8"   | 0.675            | 17.15  | 0.091  | 2.31  | 0.845  | 0.126  | 3.20  | 1.10   |
| 1/2"   | 0.84             | 21.34  | 0.109  | 2.77  | 1.27   | 0.147  | 3.73  | 1.62   |
| 3/4"   | 1.05             | 26.67  | 0.113  | 2.87  | 1.68   | 0.154  | 3.91  | 2.19   |
| 1"     | 1.315            | 33.40  | 0.133  | 3.38  | 2.50   | 0.179  | 4.55  | 3.42   |
| 1 1/4" | 1.66             | 42.16  | 0.14   | 3.56  | 3.39   | 0.191  | 4.85  | 4.46   |
| 1 1/2" | 1.90             | 48.26  | 0.146  | 3.68  | 4.05   | 0.20   | 5.08  | 5.41   |
| 2"     | 2.375            | 60.33  | 0.154  | 3.91  | 5.44   | 0.218  | 5.54  | 7.49   |
| 2 1/2" | 2.875            | 73.03  | 0.203  | 5.61  | 8.64   | 0.276  | 7.01  | 11.40  |
| 3"     | 3.50             | 88.90  | 0.216  | 5.49  | 11.30  | 0.30   | 7.62  | 15.30  |
| 3 1/2" | 4.00             | 101.60 | 0.226  | 5.74  | 13.60  | 0.318  | 8.08  | 18.60  |
| 4"     | 4.50             | 114.30 | 0.237  | 6.02  | 16.10  | 0.337  | 8.56  | 22.30  |
| 5"     | 5.563            | 141.30 | 0.258  | 6.55  | 21.80  | 0.375  | 9.25  | 30.90  |
| 6"     | 6.625            | 168.28 | 0.28   | 7.11  | 28.30  | 0.432  | 10.97 | 42.60  |
| 8"     | 8.625            | 219.08 | 0.322  | 8.18  | 42.50  | 0.50   | 12.70 | 64.60  |
| 10"    | 10.75            | 273.05 | 0.365  | 9.27  | 60.30  | 0.50   | 12.70 | 81.50  |
| 12"    | 12.75            | 323.85 | 0.375  | 9.52  | 73.80  | 0.50   | 12.70 | 97.40  |
| 14"    | 14.00            | 355.60 | --     | --    | --     | --     | --    | --     |
| 16"    | 16.00            | 406.40 | --     | --    | --     | --     | --    | --     |
| 18"    | 18.00            | 457.20 | --     | --    | --     | --     | --    | --     |
| 20"    | 20.00            | 508.00 | --     | --    | --     | --     | --    | --     |
| 22"    | 22.00            | 558.80 | --     | --    | --     | --     | --    | --     |
| 24"    | 24.00            | 609.60 | --     | --    | --     | --     | --    | --     |
| 30"    | 30.00            | 762.00 | --     | --    | --     | --     | --    | --     |

# STAINLESS STEEL FITTINGS

## FITTINGS

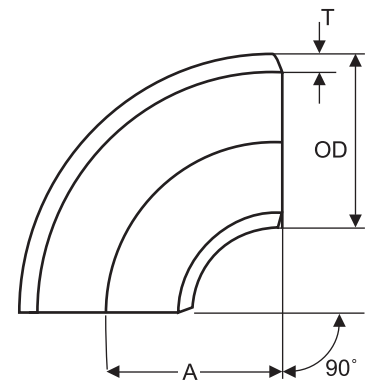


### 45° LONG RADIUS ELBOWS

| Dimensions              |                         |                    |         |         |         | Unit: inch            |
|-------------------------|-------------------------|--------------------|---------|---------|---------|-----------------------|
| Nominal<br>Pipe<br>Size | Outside<br>Dia.<br>(OD) | Wall Thickness (T) |         |         |         | Center<br>to<br>End B |
|                         |                         | Sch.5S             | Sch.10S | Sch.40S | Sch.80S |                       |
| 1/2                     | 0.840                   | 0.065              | 0.083   | 0.109   | 0.147   | 0.625                 |
| 3/4                     | 1.050                   | 0.065              | 0.083   | 0.113   | 0.154   | 0.438                 |
| 1                       | 1.315                   | 0.065              | 0.109   | 0.133   | 0.179   | 0.875                 |
| 1 1/4                   | 1.660                   | 0.065              | 0.109   | 0.140   | 0.191   | 1.00                  |
| 1 1/2                   | 1.900                   | 0.065              | 0.109   | 0.145   | 0.200   | 1.12                  |
| 2                       | 2.375                   | 0.065              | 0.109   | 0.154   | 0.218   | 1.38                  |
| 2 1/2                   | 2.875                   | 0.083              | 0.120   | 0.203   | 0.276   | 1.75                  |
| 3                       | 3.500                   | 0.083              | 0.120   | 0.216   | 0.300   | 2.00                  |
| 3 1/2                   | 4.000                   | 0.083              | 0.120   | 0.226   | 0.318   | 2.25                  |
| 4                       | 4.500                   | 0.083              | 0.120   | 0.237   | 0.337   | 2.50                  |
| 5                       | 5.563                   | 0.109              | 0.134   | 0.258   | 0.375   | 3.12                  |
| 6                       | 6.625                   | 0.109              | 0.134   | 0.280   | 0.432   | 3.75                  |
| 8                       | 8.625                   | 0.109              | 0.148   | 0.322   | 0.500   | 5.00                  |
| 10                      | 10.750                  | 0.134              | 0.165   | 0.365   | 0.500   | 6.25                  |
| 12                      | 12.750                  | 0.156              | 0.180   | 0.375   | 0.500   | 7.50                  |
| 14                      | 14.000                  | 0.250              | 0.312   | 0.375   | 0.500   | 8.75                  |
| 16                      | 16.000                  | 0.250              | 0.312   | 0.375   | 0.500   | 10.00                 |
| 18                      | 18.000                  | 0.250              | 0.312   | 0.375   | 0.500   | 11.25                 |
| 20                      | 20.000                  | 0.250              | 0.375   | 0.375   | 0.500   | 12.50                 |
| 22                      | 22.000                  | 0.250              | 0.375   | 0.375   | 0.500   | 13.50                 |
| 24                      | 24.000                  | 0.250              | 0.375   | 0.375   | 0.500   | 15.00                 |
| 26                      | 26.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 16.00                 |
| 28                      | 28.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 17.25                 |
| 30                      | 30.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 18.50                 |
| 32                      | 32.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 19.75                 |
| 34                      | 34.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 21.00                 |
| 36                      | 36.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 22.25                 |
| 38                      | 38.000                  |                    |         | 0.375   | 0.500   | 23.62                 |
| 40                      | 40.000                  |                    |         | 0.375   | 0.500   | 24.88                 |
| 42                      | 42.000                  |                    |         | 0.375   | 0.500   | 26.00                 |
| 44                      | 44.000                  |                    |         | 0.375   | 0.500   | 27.38                 |
| 46                      | 47.000                  |                    |         | 0.375   | 0.500   | 28.62                 |
| 48                      | 48.000                  |                    |         | 0.375   | 0.500   | 29.88                 |

| Weights                 |        |         |         |         | Unit: kg |
|-------------------------|--------|---------|---------|---------|----------|
| Nominal<br>Pipe<br>Size | Sch.5S | Sch.10S | Sch.40S | Sch.80S |          |
| 1/2                     | 0.024  | 0.030   | 0.039   | 0.050   |          |
| 3/4                     | 0.031  | 0.039   | 0.052   | 0.078   |          |
| 1                       | 0.039  | 0.064   | 0.077   | 0.100   |          |
| 1 1/4                   | 0.063  | 0.103   | 0.130   | 0.177   |          |
| 1 1/2                   | 0.085  | 0.142   | 0.184   | 0.254   |          |
| 2                       | 0.143  | 0.239   | 0.326   | 0.471   |          |
| 2 1/2                   | 0.288  | 0.406   | 0.683   | 0.897   |          |
| 3                       | 0.405  | 0.571   | 1.02    | 1.44    |          |
| 3 1/2                   | 0.540  | 0.764   | 1.41    | 2.05    |          |
| 4                       | 0.696  | 0.985   | 1.92    | 2.81    |          |
| 5                       | 1.42   | 1.71    | 3.24    | 4.85    |          |
| 6                       | 2.01   | 2.43    | 4.97    | 8.02    |          |
| 8                       | 3.53   | 5.01    | 10.10   | 16.22   |          |
| 10                      | 6.62   | 7.77    | 17.70   | 25.37   |          |
| 12                      | 11.10  | 12.50   | 28.10   | 36.69   |          |
| 14                      | 23.33  | 28.63   | 32.93   | 42.40   |          |
| 16                      | 30.44  | 34.84   | 43.31   | 55.76   |          |
| 18                      | 38.10  | 44.21   | 54.99   | 70.94   |          |
| 20                      | 48.11  | 68.13   | 68.13   | 88.02   |          |
| 22                      | 57.26  | 81.13   | 81.13   | 104.87  |          |
| 24                      | 68.95  | 98.57   | 98.57   | 127.52  |          |
| 26                      | 91.78  | 147.63  | 114.11  | 147.68  |          |
| 28                      | 106.68 | 171.82  | 132.73  | 171.82  |          |
| 30                      | 122.72 | 197.78  | 152.75  | 197.78  |          |
| 32                      | 139.89 | 225.59  | 156.24  | 225.59  |          |
| 34                      | 158.14 | 255.22  | 196.96  | 255.13  |          |
| 36                      | 177.58 | 286.66  | 221.20  | 286.57  |          |
| 38                      |        |         | 248.15  | 321.58  |          |
| 40                      |        |         | 257.35  | 356.69  |          |
| 42                      |        |         | 301.34  | 391.85  |          |
| 44                      |        |         | 333.68  | 432.66  |          |
| 46                      |        |         | 365.07  | 473.25  |          |
| 48                      |        |         | 397.69  | 515.74  |          |

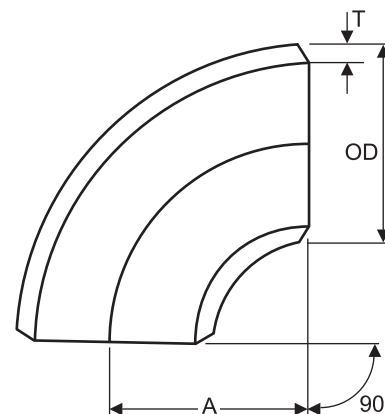
# STAINLESS STEEL FITTINGS



90° LONG RADIUS ELBOWS

| Dimensions              |                         |                    |         |         |         | Unit: inch            | Weights                 |        |         |         |         | Unit: kg |
|-------------------------|-------------------------|--------------------|---------|---------|---------|-----------------------|-------------------------|--------|---------|---------|---------|----------|
| Nominal<br>Pipe<br>Size | Outside<br>Dia.<br>(OD) | Wall Thickness (T) |         |         |         | Center<br>to<br>End A | Nominal<br>Pipe<br>Size | Sch.5S | Sch.10S | Sch.40S | Sch.80S |          |
|                         |                         | Sch.5S             | Sch.10S | Sch.40S | Sch.80S |                       |                         |        |         |         |         |          |
| 1/2                     | 0.840                   | 0.065              | 0.083   | 0.109   | 0.147   | 1.50                  | 1/2                     | 0.049  | 0.061   | 0.078   | 0.100   |          |
| 3/4                     | 1.050                   | 0.065              | 0.083   | 0.113   | 0.154   | 1.12                  | 3/4                     | 0.062  | 0.078   | 0.103   | 0.104   |          |
| 1                       | 1.315                   | 0.065              | 0.109   | 0.133   | 0.179   | 1.50                  | 1                       | 0.078  | 0.128   | 0.154   | 0.204   |          |
| 1 1/4                   | 1.660                   | 0.065              | 0.109   | 0.140   | 0.191   | 1.88                  | 1 1/4                   | 0.125  | 0.206   | 0.260   | 0.349   |          |
| 1 1/2                   | 1.900                   | 0.065              | 0.109   | 0.145   | 0.200   | 2.25                  | 1 1/2                   | 0.170  | 0.283   | 0.368   | 0.507   |          |
| 2                       | 2.375                   | 0.065              | 0.109   | 0.154   | 0.218   | 3.00                  | 2                       | 0.286  | 0.478   | 0.652   | 0.938   |          |
| 2 1/2                   | 2.875                   | 0.083              | 0.120   | 0.203   | 0.276   | 3.75                  | 2 1/2                   | 0.576  | 0.812   | 1.37    | 1.79    |          |
| 3                       | 3.500                   | 0.083              | 0.120   | 0.216   | 0.300   | 4.50                  | 3                       | 0.810  | 1.14    | 2.04    | 2.87    |          |
| 3 1/2                   | 4.000                   | 0.083              | 0.120   | 0.226   | 0.318   | 5.25                  | 3 1/2                   | 1.08   | 1.53    | 2.82    | 4.10    |          |
| 4                       | 4.500                   | 0.083              | 0.120   | 0.237   | 0.337   | 6.00                  | 4                       | 1.39   | 1.97    | 3.84    | 5.62    |          |
| 5                       | 5.563                   | 0.109              | 0.134   | 0.258   | 0.375   | 7.50                  | 5                       | 2.83   | 3.42    | 6.48    | 9.69    |          |
| 6                       | 6.625                   | 0.109              | 0.134   | 0.280   | 0.432   | 9.00                  | 6                       | 4.03   | 4.87    | 9.94    | 15.99   |          |
| 8                       | 8.625                   | 0.109              | 0.148   | 0.322   | 0.500   | 12.00                 | 8                       | 7.06   | 10.00   | 20.10   | 32.16   |          |
| 10                      | 10.750                  | 0.134              | 0.165   | 0.365   | 0.500   | 15.00                 | 10                      | 13.20  | 15.50   | 35.40   | 49.83   |          |
| 12                      | 12.750                  | 0.156              | 0.180   | 0.375   | 0.500   | 18.00                 | 12                      | 22.30  | 25.90   | 56.20   | 72.48   |          |
| 14                      | 14.000                  | 0.250              | 0.312   | 0.375   | 0.500   | 21                    | 14                      | 46.98  | 57.58   | 66.36   | 85.53   |          |
| 16                      | 16.000                  | 0.250              | 0.312   | 0.375   | 0.500   | 24                    | 16                      | 61.56  | 70.03   | 87.11   | 112.34  |          |
| 18                      | 18.000                  | 0.250              | 0.312   | 0.375   | 0.500   | 27                    | 18                      | 78.14  | 89.01   | 110.67  | 142.79  |          |
| 20                      | 20.000                  | 0.250              | 0.375   | 0.375   | 0.500   | 30                    | 20                      | 96.67  | 136.99  | 136.99  | 176.90  |          |
| 22                      | 22.000                  | 0.250              | 0.375   | 0.375   | 0.500   | 33                    | 22                      | 114.93 | 163.03  | 163.03  | 210.83  |          |
| 24                      | 24.000                  | 0.250              | 0.375   | 0.375   | 0.500   | 36                    | 24                      | 139.57 | 198.05  | 198.05  | 256.13  |          |
| 26                      | 26.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 39                    | 26                      | 184.10 | 296.76  | 229.17  | 296.53  |          |
| 28                      | 28.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 42                    | 28                      | 214.04 | 345.05  | 266.50  | 344.96  |          |
| 30                      | 30.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 45                    | 30                      | 246.16 | 397.15  | 306.59  | 396.74  |          |
| 32                      | 32.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 48                    | 32                      | 280.59 | 453.00  | 349.53  | 452.68  |          |
| 34                      | 34.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 51                    | 34                      | 317.19 | 512.16  | 395.24  | 511.98  |          |
| 36                      | 36.000                  | 0.312              | 0.500   | 0.375   | 0.500   | 54                    | 36                      | 356.01 | 575.13  | 443.71  | 574.95  |          |
| 38                      | 38.000                  |                    |         | 0.375   | 0.500   | 57                    | 38                      |        |         | 497.71  | 644.75  |          |
| 40                      | 40.000                  |                    |         | 0.375   | 0.500   | 60                    | 40                      |        |         | 551.98  | 715.51  |          |
| 42                      | 42.000                  |                    |         | 0.375   | 0.500   | 63                    | 42                      |        |         | 606.16  | 785.77  |          |
| 44                      | 44.000                  |                    |         | 0.375   | 0.500   | 66                    | 44                      |        |         | 669.04  | 867.40  |          |
| 46                      | 46.000                  |                    |         | 0.375   | 0.500   | 69                    | 46                      |        |         | 731.69  | 948.81  |          |
| 48                      | 48.000                  |                    |         | 0.375   | 0.500   | 72                    | 48                      |        |         | 797.28  | 1033.84 |          |

# STAINLESS STEEL FITTINGS



90° SHORT RADIUS ELBOWS (24" and Smaller)

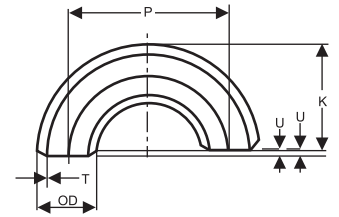
| Dimensions                    |                         |                    |         |         |         | Unit: inch            |
|-------------------------------|-------------------------|--------------------|---------|---------|---------|-----------------------|
| Nominal<br>Pipe<br>Size       | Outside<br>Dia.<br>(OD) | Wall Thickness (T) |         |         |         | Center<br>to<br>End A |
|                               |                         | Sch.5S             | Sch.10S | Sch.40S | Sch.80S |                       |
| 1                             | 1.315                   | 0.065              | 0.109   | 0.133   | 0.179   | 1.00                  |
| 1 <sup>1</sup> / <sub>4</sub> | 1.660                   | 0.065              | 0.109   | 0.140   | 0.191   | 1.25                  |
| 1 <sup>1</sup> / <sub>2</sub> | 1.900                   | 0.065              | 0.109   | 0.145   | 0.200   | 1.50                  |
| 2                             | 2.375                   | 0.065              | 0.109   | 0.154   | 0.218   | 2.00                  |
| 2 <sup>1</sup> / <sub>2</sub> | 2.875                   | 0.083              | 0.120   | 0.203   | 0.276   | 2.50                  |
| 3                             | 3.500                   | 0.083              | 0.120   | 0.216   | 0.300   | 3.00                  |
| 3 <sup>1</sup> / <sub>2</sub> | 4.000                   | 0.083              | 0.120   | 0.226   | 0.318   | 3.50                  |
| 4                             | 4.500                   | 0.083              | 0.120   | 0.237   | 0.337   | 4.00                  |
| 5                             | 5.563                   | 0.109              | 0.134   | 0.258   | 0.375   | 5.00                  |
| 6                             | 6.625                   | 0.109              | 0.134   | 0.280   | 0.432   | 6.00                  |
| 8                             | 8.625                   | 0.109              | 0.148   | 0.322   | 0.500   | 8.00                  |
| 10                            | 10.750                  | 0.134              | 0.165   | 0.365   | 0.500   | 10.00                 |
| 12                            | 12.750                  | 0.156              | 0.180   | 0.375   | 0.500   | 12.00                 |
| 14                            | 14.000                  | 0.156              | 0.188   | *       | **      | 14.00                 |
| 16                            | 16.000                  | 0.156              | 0.188   | *       | **      | 16.00                 |
| 18                            | 18.000                  | 0.156              | 0.188   | *       | **      | 18.00                 |
| 20                            | 20.000                  | 0.188              | 0.218   | *       | **      | 20.00                 |
| 22                            | 22.000                  | 0.188              | 0.218   | *       | **      | 22.00                 |
| 24                            | 24.000                  | 0.218              | 0.250   | *       | **      | 24.00                 |

| Weights                       |        |         |         |          | Unit: kg |
|-------------------------------|--------|---------|---------|----------|----------|
| Nominal<br>Pipe<br>Size       | Sch.5S | Sch.10S | Sch.40S | Sch.80S  |          |
| 1                             | 0.077  | 0.094   | 0.118   | 0.122    |          |
| 1 <sup>1</sup> / <sub>4</sub> | 0.101  | 0.152   | 0.204   | 0.217    |          |
| 1 <sup>1</sup> / <sub>2</sub> | 0.114  | 0.189   | 0.245   | 0.317    |          |
| 2                             | 0.191  | 0.318   | 0.434   | 0.593    |          |
| 2 <sup>1</sup> / <sub>2</sub> | 0.383  | 0.541   | 0.909   | 1.13     |          |
| 3                             | 0.540  | 0.761   | 1.36    | 1.82     |          |
| 3 <sup>1</sup> / <sub>2</sub> | 0.719  | 1.02    | 1.88    | 2.61     |          |
| 4                             | 0.928  | 1.31    | 2.56    | 3.56     |          |
| 5                             | 1.89   | 2.28    | 4.32    | 6.20     |          |
| 6                             | 2.68   | 3.24    | 6.63    | 10.26    |          |
| 8                             | 4.71   | 6.68    | 13.40   | 20.84    |          |
| 10                            | 8.83   | 10.40   | 23.60   | 32.94    |          |
| 12                            | 14.80  | 16.70   | 37.50   | 47.30    |          |
| 14                            | 15.21  | 24.55   | 48.33*  | 59.25**  |          |
| 16                            | 25.60  | 32.00   | 63.37*  | 81.81**  |          |
| 18                            | 40.40  | 40.40   | 73.57*  | 103.96** |          |
| 20                            | 50.42  | 60.25   | 99.66*  | 128.79** |          |
| 22                            | 61.06  | 73.02   | 120.83* | 156.29** |          |
| 24                            | 86.98  | 101.56  | 144.06* | 186.41** |          |

\* std = 0.375

\*\*XS = 0.500

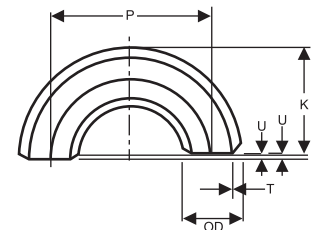
# STAINLESS STEEL FITTINGS



180° LONG RADIUS RETURNS (12" and Smaller)

| Dimensions        |                   | Unit: inch         |         |         |         |                    |                |
|-------------------|-------------------|--------------------|---------|---------|---------|--------------------|----------------|
| Nominal Pipe Size | Outside Dia. (OD) | Wall Thickness (T) |         |         |         | Center to Center P | Back to Face K |
|                   |                   | Sch.5S             | Sch.10S | Sch.40S | Sch.80S |                    |                |
| 1/2               | 0.840             | 0.065              | 0.083   | 0.109   | 0.147   | 3.00               | 1.88           |
| 3/4               | 1.050             | 0.065              | 0.083   | 0.113   | 0.154   | 2.25               | 1.69           |
| 1                 | 1.315             | 0.065              | 0.109   | 0.133   | 0.179   | 3.00               | 2.19           |
| 1 1/4             | 1.660             | 0.065              | 0.109   | 0.140   | 0.191   | 3.75               | 2.75           |
| 1 1/2             | 1.900             | 0.065              | 0.109   | 0.145   | 0.200   | 4.50               | 3.25           |
| 2                 | 2.375             | 0.065              | 0.109   | 0.154   | 0.218   | 6.00               | 4.19           |
| 2 1/2             | 2.875             | 0.083              | 0.120   | 0.203   | 0.276   | 7.50               | 5.19           |
| 3                 | 3.500             | 0.083              | 0.120   | 0.216   | 0.300   | 9.00               | 6.25           |
| 3 1/2             | 4.000             | 0.083              | 0.120   | 0.226   | 0.318   | 10.50              | 7.25           |
| 4                 | 4.500             | 0.083              | 0.120   | 0.237   | 0.337   | 12.00              | 8.25           |
| 5                 | 5.563             | 0.109              | 0.134   | 0.258   | 0.375   | 15.00              | 10.31          |
| 6                 | 6.625             | 0.109              | 0.134   | 0.280   | 0.432   | 18.00              | 12.31          |
| 8                 | 8.625             | 0.109              | 0.148   | 0.322   | 0.500   | 24.00              | 16.31          |
| 10                | 10.750            | 0.134              | 0.165   | 0.365   | 0.500   | 30.00              | 20.38          |
| 12                | 12.750            | 0.156              | 0.180   | 0.375   | 0.500   | 36.00              | 24.38          |

| Weights           |  | Unit: kg |         |         |         |
|-------------------|--|----------|---------|---------|---------|
| Nominal Pipe Size |  | Sch.5S   | Sch.10S | Sch.40S | Sch.80S |
| 1/2               |  | 0.091    | 0.118   | 0.156   | 0.199   |
| 3/4               |  | 0.091    | 0.118   | 0.207   | 0.208   |
| 1                 |  | 0.163    | 0.268   | 0.307   | 0.408   |
| 1 1/4             |  | 0.227    | 0.426   | 0.519   | 0.698   |
| 1 1/2             |  | 0.327    | 0.580   | 0.735   | 1.02    |
| 2                 |  | 0.562    | 0.989   | 1.30    | 1.88    |
| 2 1/2             |  | 1.097    | 1.651   | 2.73    | 3.56    |
| 3                 |  | 1.578    | 2.431   | 4.07    | 5.74    |
| 3 1/2             |  | 2.113    | 3.256   | 5.65    | 12.72   |
| 4                 |  | 2.730    | 4.190   | 7.67    | 15.76   |
| 5                 |  | 2.533    | 7.256   | 13.00   | 19.39   |
| 6                 |  | 8.118    | 10.431  | 19.90   | 31.98   |
| 8                 |  | 13.968   | 20.135  | 40.30   | 64.33   |
| 10                |  | 26.530   | 35.010  | 70.80   | 99.66   |
| 12                |  | 47.164   | 53.966  | 111.44  | 144.96  |

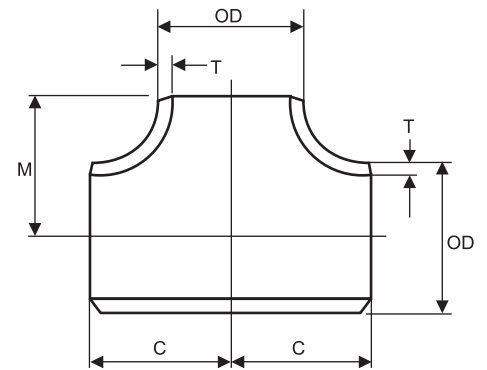


180° SHORT RADIUS RETURNS (12" and Smaller)

| Dimensions        |                   | Unit: inch         |         |         |         |                    |                |
|-------------------|-------------------|--------------------|---------|---------|---------|--------------------|----------------|
| Nominal Pipe Size | Outside Dia. (OD) | Wall Thickness (T) |         |         |         | Center to Center P | Back to Face K |
|                   |                   | Sch.5S             | Sch.10S | Sch.40S | Sch.80S |                    |                |
| 1                 | 1.315             | 0.065              | 0.109   | 0.133   | 0.179   | 2.00               | 1.63           |
| 1 1/4             | 1.660             | 0.065              | 0.109   | 0.140   | 0.191   | 2.50               | 2.06           |
| 1 1/2             | 1.900             | 0.065              | 0.109   | 0.145   | 0.200   | 3.00               | 2.44           |
| 2                 | 2.375             | 0.065              | 0.109   | 0.154   | 0.218   | 4.00               | 3.19           |
| 2 1/2             | 2.875             | 0.083              | 0.120   | 0.203   | 0.276   | 5.00               | 3.94           |
| 3                 | 3.500             | 0.083              | 0.120   | 0.216   | 0.300   | 6.00               | 4.75           |
| 3 1/2             | 4.000             | 0.083              | 0.120   | 0.226   | 0.318   | 7.00               | 5.50           |
| 4                 | 4.500             | 0.083              | 0.120   | 0.237   | 0.337   | 8.00               | 6.25           |
| 5                 | 5.563             | 0.109              | 0.134   | 0.258   | 0.375   | 10.00              | 7.75           |
| 6                 | 6.625             | 0.109              | 0.134   | 0.280   | 0.432   | 12.00              | 9.31           |
| 8                 | 8.625             | 0.109              | 0.148   | 0.322   | 0.500   | 16.00              | 12.31          |
| 10                | 10.750            | 0.134              | 0.165   | 0.365   | 0.500   | 20.00              | 15.38          |
| 12                | 12.750            | 0.156              | 0.180   | 0.375   | 0.500   | 24.00              | 18.38          |

| Weights           |  | Unit: kg |         |         |         |
|-------------------|--|----------|---------|---------|---------|
| Nominal Pipe Size |  | Sch.5S   | Sch.10S | Sch.40S | Sch.80S |
| 1                 |  | 0.104    | 0.168   | 0.205   | 0.262   |
| 1 1/4             |  | 0.168    | 0.272   | 0.347   | 0.456   |
| 1 1/2             |  | 0.231    | 0.376   | 0.490   | 0.655   |
| 2                 |  | 0.390    | 0.634   | 0.869   | 1.19    |
| 2 1/2             |  | 0.747    | 1.07    | 1.82    | 2.38    |
| 3                 |  | 1.10     | 1.57    | 2.71    | 3.65    |
| 3 1/2             |  | 1.47     | 2.10    | 3.77    | 5.21    |
| 4                 |  | 1.89     | 2.71    | 5.11    | 7.15    |
| 5                 |  | 3.84     | 4.69    | 8.64    | 12.20   |
| 6                 |  | 5.50     | 6.74    | 13.30   | 20.00   |
| 8                 |  | 9.60     | 12.97   | 26.80   | 40.70   |
| 10                |  | 18.39    | 22.57   | 47.20   | 74.90   |
| 12                |  | 30.49    | 35.10   | 71.94   | 94.91   |

# STAINLESS STEEL FITTINGS

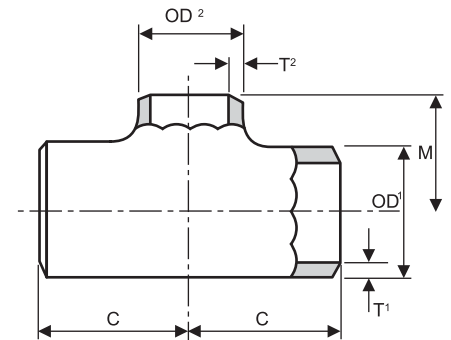


## STRAIGHT TEES (24" and Smaller)

| Dimensions        |        | Unit: inch        |                    |         |         |         |                   |
|-------------------|--------|-------------------|--------------------|---------|---------|---------|-------------------|
| Nominal Pipe Size |        | Outside Dia. (OD) | Wall Thickness (T) |         |         |         | Center to End C,M |
| Run               | Outlet |                   | Sch.5S             | Sch.10S | Sch.40S | Sch.80S |                   |
| 1/2               | 1/2    | 0.840             | 0.065              | 0.083   | 0.109   | 0.147   | 1.00              |
| 3/4               | 3/4    | 1.050             | 0.065              | 0.083   | 0.113   | 0.154   | 1.13              |
| 1                 | 1      | 1.315             | 0.065              | 0.109   | 0.133   | 0.179   | 1.50              |
| 1 1/4             | 1 1/4  | 1.660             | 0.065              | 0.109   | 0.140   | 0.191   | 1.88              |
| 1 1/2             | 1 1/2  | 1.900             | 0.065              | 0.109   | 0.145   | 0.200   | 2.25              |
| 2                 | 2      | 2.375             | 0.065              | 0.109   | 0.154   | 0.218   | 2.50              |
| 2 1/2             | 2 1/2  | 2.875             | 0.083              | 0.120   | 0.203   | 0.276   | 3.00              |
| 3                 | 3      | 3.500             | 0.083              | 0.120   | 0.216   | 0.300   | 3.38              |
| 4                 | 4      | 4.500             | 0.083              | 0.120   | 0.237   | 0.337   | 4.13              |
| 5                 | 5      | 5.563             | 0.109              | 0.134   | 0.258   | 0.375   | 4.88              |
| 6                 | 6      | 6.625             | 0.109              | 0.134   | 0.280   | 0.432   | 5.63              |
| 8                 | 8      | 8.625             | 0.109              | 0.148   | 0.322   | 0.500   | 7.00              |
| 10                | 10     | 10.750            | 0.134              | 0.165   | 0.365   | 0.500   | 8.50              |
| 12                | 12     | 12.750            | 0.156              | 0.180   | 0.375   | 0.500   | 10.00             |
| 14                | 14     | 14.000            | 0.156              | 0.188   | *       | **      | 11.00             |
| 16                | 16     | 16.000            | 0.165              | 0.188   | *       | **      | 12.00             |
| 18                | 18     | 18.000            | 0.165              | 0.188   | *       | **      | 13.50             |
| 20                | 20     | 20.000            | 0.188              | 0.218   | *       | **      | 15.00             |
| 22                | 22     | 22.000            | 0.188              | 0.218   | *       | **      | 16.50             |
| 24                | 24     | 25.000            | 0.218              | 0.250   | *       | **      | 17.00             |

| Weights           |        |         |         |          | Unit: kg |  |
|-------------------|--------|---------|---------|----------|----------|--|
| Nominal Pipe Size | Sch.5S | Sch.10S | Sch.40S | Sch.80S  |          |  |
| 1/2               | 0.090  | 0.090   | 0.181   | 0.204    |          |  |
| 3/4               | 0.129  | 0.129   | 0.258   | 0.294    |          |  |
| 1                 | 0.223  | 0.223   | 0.256   | 0.399    |          |  |
| 1 1/4             | 0.356  | 0.356   | 0.463   | 0.725    |          |  |
| 1 1/2             | 0.497  | 0.497   | 0.648   | 1.02     |          |  |
| 2                 | 0.681  | 0.681   | 0.893   | 1.81     |          |  |
| 2 1/2             | 1.20   | 1.20    | 1.98    | 3.17     |          |  |
| 3                 | 1.57   | 1.57    | 2.61    | 3.85     |          |  |
| 4                 | 2.46   | 2.46    | 4.76    | 7.11     |          |  |
| 5                 | 4.04   | 4.04    | 7.84    | 11.78    |          |  |
| 6                 | 5.50   | 5.50    | 9.54    | 18.12    |          |  |
| 8                 | 11.07  | 11.07   | 19.56   | 33.98    |          |  |
| 10                | 19.87  | 19.87   | 32.61   | 47.57    |          |  |
| 12                | 27.86  | 27.86   | 56.65   | 2.48     |          |  |
| 14                | 28.74  | 35.98   | 71.21*  | 91.78**  |          |  |
| 16                | 35.69  | 44.69   | 88.56*  | 114.23** |          |  |
| 18                | 45.21  | 56.62   | 112.33* | 144.98** |          |  |
| 20                | 69.97  | 83.69   | 138.99* | 179.66** |          |  |
| 22                | 84.72  | 101.35  | 170.07* | 217.46** |          |  |
| 24                | 113.23 | 132.24  | 192.66* | 243.13** |          |  |

# STAINLESS STEEL FITTINGS



## REDUCING OUTLET TEES (24" and Smaller)

| Dimensions        |         |                 |                 |                           |                |                |                |                |                |                |                | Unit: inch    |               |
|-------------------|---------|-----------------|-----------------|---------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|
| Nominal Pipe Size |         | Out-side        | Out-side        | Wall Thickness (T 1, T 2) |                |                |                |                |                |                |                | Center to End | Center to End |
| Run               | Out-let | Dia.            | Dia.            | Sch.5S                    |                | Sch.10S        |                | Sch.40S        |                | Sch.80S        |                | C             | M             |
|                   |         | OD <sub>1</sub> | OD <sub>2</sub> | T <sub>1</sub>            | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> |               |               |
| 3/4               | 1/2     | 1.050           | 0.840           | 0.065                     | 0.065          | 0.083          | 0.083          | 0.113          | 0.109          | 0.154          | 0.147          | 1.13          | 1.13          |
| 1                 | 1/2     | 1.310           | 0.840           | 0.065                     | 0.065          | 0.109          | 0.083          | 0.133          | 0.109          | 0.179          | 0.147          | 1.50          | 1.50          |
| 1                 | 3/4     | 1.315           | 1.050           | 0.065                     | 0.065          | 0.109          | 0.083          | 0.133          | 0.113          | 0.179          | 0.154          | 1.50          | 1.50          |
| 1 1/4             | 3/4     | 1.660           | 1.050           | 0.065                     | 0.065          | 0.109          | 0.083          | 0.140          | 0.113          | 0.191          | 0.154          | 1.88          | 1.88          |
| 1 1/4             | 1       | 1.660           | 1.315           | 0.065                     | 0.065          | 0.109          | 0.109          | 0.140          | 0.133          | 0.191          | 0.179          | 1.88          | 1.88          |
| 1 1/2             | 1/2     | 1.900           | 0.840           | 0.065                     | 0.065          | 0.109          | 0.083          | 0.145          | 0.109          | 0.200          | 0.147          | 2.25          | 2.25          |
| 1 1/2             | 3/4     | 1.900           | 1.050           | 0.065                     | 0.065          | 0.109          | 0.083          | 0.145          | 0.113          | 0.200          | 0.154          | 2.25          | 2.25          |
| 1 1/2             | 1       | 1.900           | 1.315           | 0.065                     | 0.065          | 0.109          | 0.109          | 0.145          | 0.133          | 0.200          | 0.179          | 2.25          | 2.25          |
| 1 1/2             | 1 1/4   | 1.900           | 1.660           | 0.065                     | 0.065          | 0.109          | 0.109          | 0.145          | 0.140          | 0.200          | 0.191          | 2.25          | 2.25          |
| 2                 | 3/4     | 2.375           | 1.050           | 0.065                     | 0.065          | 0.109          | 0.033          | 0.154          | 0.113          | 0.218          | 0.154          | 2.50          | 1.75          |
| 2                 | 1       | 2.375           | 1.315           | 0.065                     | 0.065          | 0.109          | 0.109          | 0.154          | 0.133          | 0.218          | 0.179          | 2.50          | 2.00          |
| 2                 | 1 1/4   | 2.375           | 1.660           | 0.065                     | 0.065          | 0.109          | 0.109          | 0.154          | 0.140          | 0.218          | 0.191          | 2.50          | 2.25          |
| 2                 | 1 1/2   | 2.375           | 1.900           | 0.065                     | 0.065          | 0.109          | 0.109          | 0.154          | 0.145          | 0.218          | 0.200          | 2.50          | 2.38          |
| 2 1/2             | 1 1/2   | 2.875           | 1.900           | 0.083                     | 0.065          | 0.120          | 0.109          | 0.203          | 0.145          | 0.276          | 0.200          | 3.00          | 2.63          |
| 2 1/2             | 2       | 2.875           | 2.375           | 0.083                     | 0.065          | 0.120          | 0.109          | 0.203          | 0.154          | 0.276          | 0.218          | 3.00          | 2.75          |
| 3                 | 1 1/2   | 3.500           | 1.900           | 0.083                     | 0.065          | 0.120          | 0.109          | 0.216          | 0.145          | 0.300          | 0.200          | 3.13          | 2.88          |
| 3                 | 2       | 3.500           | 2.375           | 0.083                     | 0.065          | 0.120          | 0.109          | 0.216          | 0.154          | 0.300          | 0.218          | 3.38          | 3.00          |
| 3                 | 2 1/2   | 3.500           | 2.875           | 0.083                     | 0.083          | 0.120          | 0.120          | 0.216          | 0.203          | 0.300          | 0.276          | 3.38          | 3.25          |
| 4                 | 1 1/2   | 4.500           | 1.900           | 0.083                     | 0.065          | 0.120          | 0.109          | 0.237          | 0.145          | 0.337          | 0.200          | 4.13          | 3.38          |
| 4                 | 2       | 4.500           | 2.375           | 0.083                     | 0.065          | 0.120          | 0.109          | 0.237          | 0.154          | 0.337          | 0.218          | 4.13          | 3.50          |
| 4                 | 2 1/2   | 4.500           | 2.875           | 0.083                     | 0.083          | 0.120          | 0.120          | 0.237          | 0.203          | 0.337          | 0.276          | 4.13          | 3.75          |
| 4                 | 3       | 4.500           | 3.500           | 0.083                     | 0.083          | 0.120          | 0.120          | 0.237          | 0.216          | 0.337          | 0.300          | 4.13          | 3.88          |
| 5                 | 3       | 5.563           | 3.500           | 0.109                     | 0.083          | 0.134          | 0.120          | 0.258          | 0.216          | 0.375          | 0.300          | 4.88          | 4.38          |
| 5                 | 4       | 5.563           | 4.500           | 0.109                     | 0.083          | 0.134          | 0.120          | 0.258          | 0.237          | 0.375          | 0.337          | 4.88          | 4.63          |
| 6                 | 2 1/2   | 6.625           | 2.875           | 0.109                     | 0.083          | 0.134          | 0.210          | 0.280          | 0.203          | 0.432          | 0.276          | 5.63          | 4.75          |
| 6                 | 3       | 6.625           | 3.500           | 0.109                     | 0.083          | 0.134          | 0.120          | 0.280          | 0.216          | 0.432          | 0.300          | 5.63          | 4.88          |
| 6                 | 4       | 6.625           | 4.500           | 0.109                     | 0.083          | 0.134          | 0.120          | 0.280          | 0.237          | 0.432          | 0.337          | 5.63          | 5.13          |
| 6                 | 5       | 6.625           | 5.563           | 0.109                     | 0.109          | 0.134          | 0.134          | 0.280          | 0.258          | 0.432          | 0.375          | 5.63          | 5.38          |

| Weights           |         |        |         |         |         | Unit: kg |  |
|-------------------|---------|--------|---------|---------|---------|----------|--|
| Nominal Pipe Size |         | Sch.5S | Sch.10S | Sch.40S | Sch.80S |          |  |
| Run               | Out-let |        |         |         |         |          |  |
| 3/4               | 1/2     | 0.123  | 0.123   | 0.123   | 0.186   |          |  |
| 1                 | 1/2     | 0.204  | 0.204   | 0.234   | 0.349   |          |  |
| 1                 | 3/4     | 0.212  | 0.212   | 0.244   | 0.353   |          |  |
| 1 1/4             | 3/4     | 0.327  | 0.327   | 0.423   | 0.612   |          |  |
| 1 1/4             | 1       | 0.339  | 0.339   | 0.441   | 0.643   |          |  |
| 1 1/2             | 1/2     | 0.431  | 0.431   | 0.561   | 0.897   |          |  |
| 1 1/2             | 3/4     | 0.446  | 0.446   | 0.578   | 0.901   |          |  |
| 1 1/2             | 1       | 0.461  | 0.461   | 0.600   | 0.929   |          |  |
| 1 1/2             | 1 1/4   | 0.482  | 0.482   | 0.629   | 0.978   |          |  |
| 2                 | 3/4     | 0.625  | 0.702   | 1.11    | 1.44    |          |  |
| 2                 | 1       | 0.643  | 0.725   | 1.14    | 1.46    |          |  |
| 2                 | 1 1/4   | 0.680  | 0.747   | 1.18    | 1.51    |          |  |
| 2                 | 1 1/2   | 0.734  | 0.766   | 1.23    | 1.57    |          |  |
| 2 1/2             | 1 1/2   | 0.824  | 1.18    | 2.08    | 2.61    |          |  |
| 2 1/2             | 2       | 0.838  | 1.21    | 2.12    | 2.66    |          |  |
| 3                 | 1 1/2   | 1.33   | 1.60    | 2.98    | 3.84    |          |  |
| 3                 | 2       | 1.42   | 1.63    | 3.01    | 3.86    |          |  |
| 3                 | 2 1/2   | 1.58   | 1.69    | 3.15    | 3.97    |          |  |
| 4                 | 1 1/2   | 2.45   | 2.51    | 5.04    | 6.63    |          |  |
| 4                 | 2       | 2.47   | 2.53    | 5.08    | 6.70    |          |  |
| 4                 | 2 1/2   | 2.52   | 2.59    | 5.22    | 6.79    |          |  |
| 4                 | 3       | 2.55   | 2.61    | 5.27    | 6.93    |          |  |
| 5                 | 3       | 3.38   | 4.17    | 8.13    | 10.95   |          |  |
| 5                 | 4       | 3.43   | 4.24    | 8.36    | 11.18   |          |  |
| 6                 | 2 1/2   | 4.62   | 5.69    | 11.87   | 17.05   |          |  |
| 6                 | 3       | 4.67   | 5.75    | 11.92   | 17.07   |          |  |
| 6                 | 4       | 4.75   | 5.85    | 12.10   | 17.19   |          |  |
| 6                 | 5       | 4.80   | 5.92    | 12.34   | 17.47   |          |  |

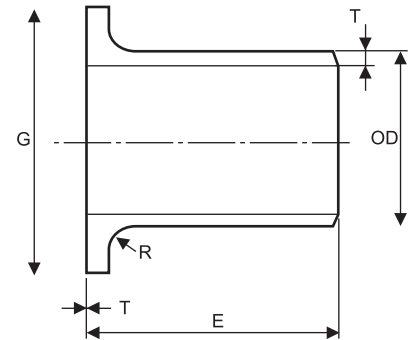


# STAINLESS STEEL FITTINGS

## REDUCING OUTLET TEES (24" and Smaller)

| Dimensions        |         |                 |                 |                         |                |                |                |                |                |                |                | Unit: inch    |               | Weights           |         |        |         |         |         | Unit: kg |  |
|-------------------|---------|-----------------|-----------------|-------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|-------------------|---------|--------|---------|---------|---------|----------|--|
| Nominal Pipe Size |         | Out-side        | Out-side        | Wall Thickness (T1, T2) |                |                |                |                |                |                |                | Center to End | Center to End | Nominal Pipe Size |         | Sch.5S | Sch.10S | Sch.40S | Sch.80S |          |  |
| Run               | Out-let | Dia.            | Dia.            | Sch.5S                  |                | Sch.10S        |                | Sch.40S        |                | Sch.80S        |                | C             | M             | Run               | Out-let |        |         |         |         |          |  |
|                   |         | OD <sub>1</sub> | OD <sub>2</sub> | T <sub>1</sub>          | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> |               |               |                   |         |        |         |         |         |          |  |
| 8                 | 4       | 8.625           | 4.500           | 0.109                   | 0.083          | 0.148          | 0.120          | 0.322          | 0.237          | 0.500          | 0.337          | 7.00          | 6.13          | 8                 | 4       | 7.35   | 9.80    | 21.26   | 30.76   |          |  |
| 8                 | 5       | 8.625           | 5.563           | 0.109                   | 0.109          | 0.148          | 0.134          | 0.322          | 0.258          | 0.500          | 0.375          | 7.00          | 6.38          | 8                 | 5       | 7.41   | 9.83    | 21.35   | 30.85   |          |  |
| 8                 | 6       | 8.625           | 6.625           | 0.109                   | 0.109          | 0.148          | 0.134          | 0.322          | 0.280          | 0.500          | 0.432          | 7.00          | 6.63          | 8                 | 6       | 7.44   | 9.90    | 21.62   | 31.25   |          |  |
| 10                | 5       | 10.750          | 5.563           | 0.134                   | 0.109          | 0.165          | 0.134          | 0.365          | 0.258          | 0.500          | 0.375          | 8.50          | 7.50          | 10                | 5       | 13.43  | 16.47   | 36.14   | 47.47   |          |  |
| 10                | 6       | 10.750          | 6.625           | 0.134                   | 0.109          | 0.165          | 0.134          | 0.365          | 0.280          | 0.500          | 0.432          | 8.50          | 7.63          | 10                | 6       | 13.49  | 16.58   | 36.27   | 47.95   |          |  |
| 10                | 8       | 10.750          | 8.625           | 0.134                   | 0.109          | 0.165          | 0.148          | 0.365          | 0.322          | 0.500          | 0.500          | 8.50          | 8.00          | 10                | 8       | 13.54  | 16.69   | 36.85   | 48.39   |          |  |
| 12                | 6       | 12.750          | 6.625           | 0.156                   | 0.109          | 0.180          | 0.134          | 0.375          | 0.280          | 0.500          | 0.432          | 10.00         | 8.63          | 12                | 6       | 22.93  | 25.10   | 51.96   | 67.48   |          |  |
| 12                | 8       | 12.750          | 8.625           | 0.156                   | 0.109          | 0.180          | 0.148          | 0.375          | 0.322          | 0.500          | 0.500          | 10.00         | 9.00          | 12                | 8       | 23.01  | 25.32   | 52.38   | 67.92   |          |  |
| 12                | 10      | 12.750          | 10.750          | 0.156                   | 0.104          | 0.180          | 0.165          | 0.375          | 0.365          | 0.500          | 0.500          | 10.00         | 9.50          | 12                | 10      | 23.13  | 25.48   | 53.72   | 68.81   |          |  |
| 14                | 8       | 14.000          | 8.620           | 0.188                   | 0.148          | 0.312          | 0.250          | 0.375          | 0.322          | 0.500          | 0.500          | 11.00         | 9.75          | 14                | 8       | 28.79  | 52.00   | 57.88   | 75.22   |          |  |
| 14                | 10      | 14.000          | 10.750          | 0.188                   | 0.165          | 0.312          | 0.250          | 0.375          | 0.365          | 0.500          | 0.500          | 11.00         | 10.13         | 14                | 10      | 28.96  | 52.37   | 58.34   | 75.89   |          |  |
| 14                | 12      | 14.000          | 12.750          | 0.188                   | 0.180          | 0.312          | 0.250          | 0.375          | 0.375          | 0.500          | 0.500          | 11.00         | 10.63         | 14                | 12      | 29.14  | 52.73   | 58.81   | 76.57   |          |  |
| 16                | 10      | 16.000          | 10.750          | 0.188                   | 0.165          | 0.312          | 0.250          | 0.375          | 0.365          | 0.500          | 0.500          | 12.00         | 11.13         | 16                | 10      | 34.26  | 61.89   | 68.88   | 89.51   |          |  |
| 16                | 12      | 16.000          | 12.750          | 0.188                   | 0.180          | 0.312          | 0.250          | 0.375          | 0.375          | 0.500          | 0.500          | 12.00         | 11.63         | 16                | 12      | 34.47  | 62.32   | 69.43   | 90.32   |          |  |
| 16                | 14      | 16.000          | 14.000          | 0.188                   | 0.188          | 0.312          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 12.00         | 12.00         | 16                | 14      | 34.67  | 62.76   | 69.98   | 91.13   |          |  |
| 18                | 12      | 18.000          | 12.750          | 0.188                   | 0.180          | 0.312          | 0.250          | 0.375          | 0.375          | 0.500          | 0.500          | 13.50         | 12.63         | 18                | 12      | 43.37  | 78.33   | 87.17   | 113.28  |          |  |
| 18                | 14      | 18.000          | 14.000          | 0.188                   | 0.188          | 0.312          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 13.50         | 13.00         | 18                | 14      | 43.62  | 78.88   | 87.87   | 114.30  |          |  |
| 18                | 16      | 18.000          | 16.000          | 0.188                   | 0.188          | 0.312          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 13.50         | 13.00         | 18                | 16      | 43.89  | 79.43   | 88.57   | 115.33  |          |  |
| 20                | 14      | 20.000          | 14.000          | 0.218                   | 0.188          | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 15.00         | 14.00         | 20                | 14      | 64.47  | 107.63  | 107.63  | 139.86  |          |  |
| 20                | 16      | 20.000          | 16.000          | 0.218                   | 0.188          | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 15.00         | 14.00         | 20                | 16      | 64.85  | 108.38  | 108.38  | 141.12  |          |  |
| 20                | 18      | 20.000          | 18.000          | 0.218                   | 0.188          | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 15.00         | 14.50         | 20                | 18      | 65.24  | 109.14  | 109.14  | 142.39  |          |  |
| 22                | 16      | 22.000          | 16.000          | 0.218                   | 0.188          | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 16.50         | 15.00         | 22                | 16      | 78.00  | 130.23  | 130.23  | 169.22  |          |  |
| 22                | 18      | 22.000          | 18.000          | 0.218                   | 0.188          | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 16.50         | 15.50         | 22                | 18      | 78.47  | 131.14  | 131.14  | 170.74  |          |  |
| 22                | 20      | 22.000          | 20.000          | 0.218                   | 0.118          | 0.375          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 16.50         | 16.00         | 22                | 20      | 78.94  | 132.06  | 132.06  | 172.28  |          |  |
| 24                | 16      | 24.000          | 16.000          | 0.250                   | 0.188          | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 17.00         | 16.00         | 24                | 16      | 96.84  | 138.24  | 138.24  | 179.64  |          |  |
| 24                | 18      | 24.000          | 18.000          | 0.250                   | 0.188          | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 17.00         | 16.50         | 24                | 18      | 97.69  | 139.21  | 139.21  | 181.25  |          |  |
| 24                | 20      | 24.000          | 20.000          | 0.250                   | 0.218          | 0.375          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 17.00         | 17.00         | 24                | 20      | 98.00  | 140.18  | 140.18  | 182.89  |          |  |
| 24                | 22      | 24.000          | 22.000          | 0.250                   | 0.218          | 0.375          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 17.00         | 17.00         | 24                | 22      | 98.59  | 141.16  | 141.16  | 184.53  |          |  |

# STAINLESS STEEL FITTINGS



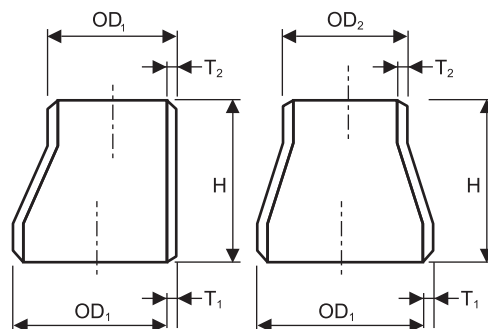
## LAP-JOINT STUB ENDS (24" and Smaller)

| Dimensions              |                         |                    |         |         |         |                      |          | Unit: inch |                 |                 |
|-------------------------|-------------------------|--------------------|---------|---------|---------|----------------------|----------|------------|-----------------|-----------------|
| Nominal<br>Pipe<br>Size | Outside<br>Dia.<br>(OD) | Wall Thickness (T) |         |         |         | Dia<br>of Lap<br>(G) | Length F |            | R               |                 |
|                         |                         | Sch.5S             | Sch.10S | Sch.40S | Sch.80S |                      | ANSI     | MSS        | Type A<br>(Max) | Type B<br>(Max) |
| 1½                      | 0.840                   | 0.065              | 0.083   | 0.109   | 0.147   | 1.375                | 3.0      | 2.0        | 0.13            | 0.03            |
| ¾                       | 1.050                   | 0.065              | 0.083   | 0.113   | 0.154   | 1.688                | 3.0      | 2.0        | 0.13            | 0.03            |
| 1                       | 1.315                   | 0.065              | 0.109   | 0.133   | 0.179   | 2.000                | 4.0      | 2.0        | 0.13            | 0.03            |
| 1¼                      | 1.660                   | 0.065              | 0.109   | 0.140   | 0.191   | 2.500                | 4.0      | 2.0        | 0.19            | 0.03            |
| 1½                      | 1.900                   | 0.065              | 0.109   | 0.145   | 0.200   | 2.875                | 4.0      | 2.0        | 0.25            | 0.03            |
| 2                       | 2.375                   | 0.065              | 0.109   | 0.154   | 0.218   | 3.625                | 6.0      | 2.5        | 0.31            | 0.03            |
| 2½                      | 2.875                   | 0.083              | 0.120   | 0.203   | 0.276   | 4.125                | 6.0      | 2.5        | 0.31            | 0.03            |
| 3                       | 3.500                   | 0.083              | 0.120   | 0.216   | 0.300   | 5.000                | 6.0      | 2.5        | 0.38            | 0.03            |
| 4                       | 4.500                   | 0.083              | 0.120   | 0.237   | 0.337   | 6.188                | 6.0      | 3.0        | 0.44            | 0.03            |
| 5                       | 5.563                   | 0.109              | 0.134   | 0.258   | 0.375   | 7.313                | 8.0      | 3.0        | 0.44            | 0.06            |
| 6                       | 6.625                   | 0.109              | 0.134   | 0.280   | 0.432   | 8.500                | 8.0      | 3.5        | 0.50            | 0.06            |
| 8                       | 8.625                   | 0.109              | 0.148   | 0.322   | 0.500   | 10.625               | 8.0      | 4.0        | 0.50            | 0.06            |
| 10                      | 10.750                  | 0.134              | 0.165   | 0.365   | 0.500   | 12.750               | 10.0     | 5.0        | 0.50            | 0.06            |
| 12                      | 12.750                  | 0.156              | 0.180   | 0.375   | 0.500   | 15.000               | 10.0     | 6.0        | 0.50            | 0.06            |
| 14                      | 14.000                  | 0.156              | 0.188   | *       | **      | 16.250               | 12.0     | 6.0        | 0.50            | 0.06            |
| 16                      | 16.000                  | 0.165              | 0.188   | *       | **      | 18.500               | 12.0     | 6.0        | 0.50            | 0.06            |
| 18                      | 18.000                  | 0.165              | 0.188   | *       | **      | 21.000               | 12.0     | 6.0        | 0.50            | 0.06            |
| 20                      | 20.000                  | 0.188              | 0.218   | *       | **      | 23.000               | 12.0     | 6.0        | 0.50            | 0.06            |
| 22                      | 22.000                  | 0.188              | 0.218   | *       | **      | 25.250               | 12.0     | -          | 0.50            | 0.06            |
| 24                      | 24.000                  | 0.118              | 0.250   | *       | **      | 27.250               | 12.0     | 6.0        | 0.50            | 0.06            |

\* Std 0.375 \*\* XS 0.500

| Weights                 |      |        |         |         |         | Unit: kg |  |
|-------------------------|------|--------|---------|---------|---------|----------|--|
| Nominal<br>Pipe<br>Size | F    | Sch.5S | Sch.10S | Sch.40S | Sch.80S |          |  |
| 1/2                     | ANSI | 0.068  | 0.091   | 0.136   | 0.115   |          |  |
|                         | MSS  | 0.059  | 0.077   | 0.118   | 0.127   |          |  |
| 3/4                     | ANSI | 0.077  | 0.100   | 0.181   | 0.199   |          |  |
|                         | MSS  | 0.068  | 0.086   | 0.154   | 0.168   |          |  |
| 1                       | ANSI | 0.145  | 0.227   | 0.299   | 0.385   |          |  |
|                         | MSS  | 0.091  | 0.140   | 0.186   | 0.240   |          |  |
| 1 1/4                   | ANSI | 0.204  | 0.326   | 0.412   | 0.548   |          |  |
|                         | MSS  | 0.131  | 0.208   | 0.263   | 0.349   |          |  |
| 1 1/2                   | ANSI | 0.231  | 0.367   | 0.553   | 0.675   |          |  |
|                         | MSS  | 0.159  | 0.249   | 0.376   | 0.458   |          |  |
| 2                       | ANSI | 0.449  | 0.693   | 0.992   | 1.364   |          |  |
|                         | MSS  | 0.245  | 0.376   | 0.539   | 0.743   |          |  |
| 2 1/2                   | ANSI | 0.684  | 0.924   | 1.563   | 2.079   |          |  |
|                         | MSS  | 0.349  | 0.471   | 0.797   | 1.060   |          |  |
| 3                       | ANSI | 0.879  | 1.205   | 2.134   | 2.840   |          |  |
|                         | MSS  | 0.467  | 0.638   | 1.133   | 1.508   |          |  |
| 4                       | ANSI | 1.191  | 1.640   | 3.035   | 4.227   |          |  |
|                         | MSS  | 0.711  | 0.978   | 1.812   | 2.523   |          |  |
| 5                       | ANSI | 2.188  | 2.582   | 5.300   | 7.520   |          |  |
|                         | MSS  | 1.046  | 1.237   | 2.537   | 3.601   |          |  |
| 6                       | ANSI | 3.01   | 3.61    | 6.89    | 10.42   |          |  |
|                         | MSS  | 1.62   | 1.95    | 3.72    | 5.57    |          |  |
| 8                       | ANSI | 2.96   | 3.76    | 10.42   | 15.86   |          |  |
|                         | MSS  | 2.45   | 3.10    | 5.89    | 10.12   |          |  |
| 10                      | ANSI | 7.02   | 8.45    | 18.12   | 24.28   |          |  |
|                         | MSS  | 4.04   | 4.86    | 10.42   | 13.95   |          |  |
| 12                      | ANSI | 9.11   | 10.51   | 22.20   | 29.81   |          |  |
|                         | MSS  | 6.60   | 7.11    | 14.95   | 19.93   |          |  |
| 14                      | ANSI | 11.78  | 14.19   | 28.31*  | 37.80** |          |  |
|                         | MSS  | 6.43   | 7.75    | 15.46*  | 20.62** |          |  |
| 16                      | ANSI | 14.38  | 16.38   | 32.68*  | 43.57** |          |  |
|                         | MSS  | 7.91   | 9.02    | 17.98*  | 23.98** |          |  |
| 18                      | ANSI | 16.51  | 18.81   | 37.53*  | 50.03** |          |  |
|                         | MSS  | 9.24   | 10.53   | 21.00*  | 28.00** |          |  |
| 20                      | ANSI | 20.89  | 24.22   | 41.66*  | 55.55** |          |  |
|                         | MSS  | 11.68  | 13.54   | 23.30*  | 31.07** |          |  |
| 22                      | ANSI | 23.20  | 26.90   | 46.28*  | 61.07** |          |  |
|                         | MSS  |        |         | *       | **      |          |  |
| 24                      | ANSI | 29.33  | 33.63   | 50.45*  | 67.27** |          |  |
|                         | MSS  | 16.52  | 18.94   | 28.41*  | 37.88** |          |  |

# STAINLESS STEEL FITTINGS

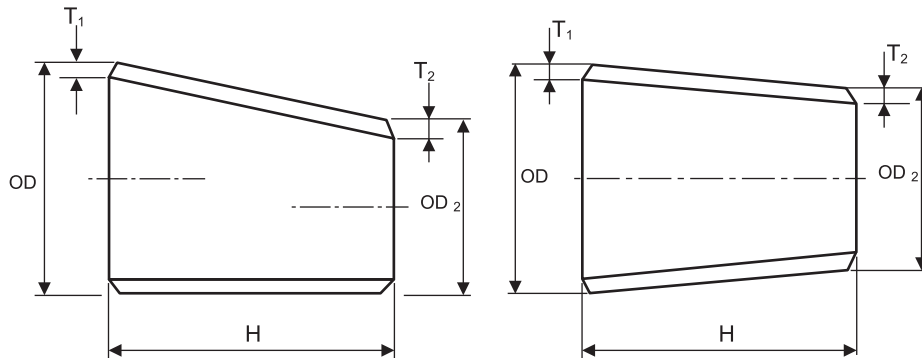


## ECC., CON. REDUCERS (12" and Smaller)

| Dimensions              |                      |                      |                 |                 |                |                |                |                |                |                | Unit: inch |
|-------------------------|----------------------|----------------------|-----------------|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------|
| Nominal<br>Pipe<br>Size | Out-<br>side<br>Dia. | Out-<br>side<br>Dia. | Wall Thickness  |                 |                |                |                |                |                |                | Length     |
|                         |                      |                      | Sch.5S          |                 | Sch.10S        |                | Sch.40S        |                | Sch.80S        |                |            |
|                         |                      |                      | OD <sub>1</sub> | OD <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> |            |
| 3/4 X 1/2               | 1.050                | 0.840                | 0.065           | 0.065           | 0.083          | 0.083          | 0.113          | 0.109          | 0.154          | 0.147          | 1.50       |
| 1 X 1/2                 | 1.315                | 0.840                | 0.065           | 0.065           | 0.109          | 0.083          | 0.133          | 0.109          | 0.179          | 0.147          | 2.00       |
| 1 X 3/4                 | 1.315                | 1.050                | 0.065           | 0.065           | 0.109          | 0.083          | 0.133          | 0.113          | 0.179          | 0.154          | 2.00       |
| 1 1/4 X 1               | 1.660                | 1.315                | 0.065           | 0.065           | 0.109          | 0.109          | 0.140          | 0.133          | 0.191          | 0.179          | 2.00       |
| 1 1/2 X 1/2             | 1.900                | 0.840                | 0.065           | 0.065           | 0.109          | 0.083          | 0.145          | 0.109          | 0.200          | 0.147          | 2.50       |
| 1 1/2 X 3/4             | 1.900                | 1.050                | 0.065           | 0.065           | 0.109          | 0.083          | 0.145          | 0.113          | 0.200          | 0.154          | 2.50       |
| 1 1/2 X 1               | 1.905                | 1.315                | 0.065           | 0.065           | 0.109          | 0.109          | 0.145          | 0.133          | 0.200          | 0.179          | 2.50       |
| 1 1/2 X 1 1/4           | 1.900                | 1.660                | 0.065           | 0.065           | 0.109          | 0.109          | 0.145          | 0.140          | 0.200          | 0.191          | 2.50       |
| 2 X 3/4                 | 2.375                | 1.050                | 0.065           | 0.065           | 0.109          | 0.083          | 0.154          | 0.113          | 0.218          | 0.154          | 3.00       |
| 2 X 1                   | 2.375                | 1.315                | 0.065           | 0.065           | 0.109          | 0.109          | 0.154          | 0.133          | 0.218          | 0.179          | 3.00       |
| 2 X 1 1/4               | 2.375                | 1.660                | 0.065           | 0.065           | 0.109          | 0.109          | 0.154          | 0.140          | 0.218          | 0.191          | 3.00       |
| 2 X 1 1/2               | 2.375                | 1.900                | 0.065           | 0.065           | 0.109          | 0.109          | 0.154          | 0.145          | 0.218          | 0.200          | 3.00       |
| 2 1/2 X 1               | 2.875                | 1.315                | 0.083           | 0.065           | 0.120          | 0.109          | 0.203          | 0.133          | 0.276          | 0.179          | 3.50       |
| 2 1/2 X 1 1/4           | 2.875                | 1.660                | 0.083           | 0.065           | 0.120          | 0.109          | 0.203          | 0.140          | 0.276          | 0.191          | 3.50       |
| 2 1/2 X 1 1/2           | 2.875                | 1.900                | 0.083           | 0.065           | 0.120          | 0.109          | 0.203          | 0.145          | 0.276          | 0.200          | 3.50       |
| 2 1/2 X 2               | 2.875                | 2.375                | 0.083           | 0.065           | 0.120          | 0.109          | 0.203          | 0.154          | 0.276          | 0.218          | 3.50       |
| 3 X 1                   | 3.500                | 1.315                | 0.083           | 0.065           | 0.120          | 0.109          | 0.216          | 0.133          | 0.300          | 0.179          | 3.50       |
| 3 X 1 1/4               | 3.500                | 1.660                | 0.083           | 0.065           | 0.120          | 0.109          | 0.216          | 0.140          | 0.300          | 0.191          | 3.50       |
| 3 X 1 1/2               | 3.500                | 1.900                | 0.083           | 0.065           | 0.120          | 0.109          | 0.216          | 0.145          | 0.300          | 0.200          | 3.50       |
| 3 X 2                   | 3.500                | 2.375                | 0.083           | 0.065           | 0.120          | 0.109          | 0.216          | 0.154          | 0.300          | 0.218          | 3.50       |
| 3 X 2 1/2               | 3.500                | 2.875                | 0.083           | 0.083           | 0.120          | 0.120          | 0.216          | 0.203          | 0.300          | 0.276          | 3.50       |
| 4 X 1 1/2               | 4.500                | 1.900                | 0.083           | 0.065           | 0.120          | 0.109          | 0.237          | 0.145          | 0.337          | 0.200          | 4.00       |
| 4 X 2                   | 4.500                | 2.375                | 0.083           | 0.065           | 0.120          | 0.109          | 0.237          | 0.154          | 0.337          | 0.218          | 4.00       |
| 4 X 2 1/2               | 4.500                | 2.875                | 0.083           | 0.083           | 0.120          | 0.120          | 0.237          | 0.203          | 0.337          | 0.276          | 4.00       |
| 4 X 3                   | 4.500                | 3.500                | 0.083           | 0.083           | 0.120          | 0.120          | 0.237          | 0.216          | 0.337          | 0.300          | 4.00       |
| 5 X 3                   | 5.563                | 3.500                | 0.109           | 0.083           | 0.134          | 0.120          | 0.258          | 0.216          | 0.375          | 0.300          | 5.00       |
| 5 X 4                   | 5.563                | 4.500                | 0.109           | 0.083           | 0.134          | 0.120          | 0.258          | 0.237          | 0.375          | 0.337          | 5.00       |
| 6 X 2 1/2               | 6.625                | 2.875                | 0.109           | 0.083           | 0.134          | 0.120          | 0.280          | 0.203          | 0.432          | 0.276          | 5.50       |
| 6 X 3                   | 6.625                | 3.500                | 0.109           | 0.083           | 0.134          | 0.120          | 0.280          | 0.216          | 0.432          | 0.300          | 5.50       |
| 6 X 4                   | 6.625                | 4.500                | 0.109           | 0.083           | 0.134          | 0.130          | 0.280          | 0.237          | 0.432          | 0.337          | 5.50       |
| 6 X 5                   | 6.625                | 5.563                | 0.109           | 0.109           | 0.134          | 0.134          | 0.280          | 0.258          | 0.432          | 0.375          | 5.50       |
| 8 X 4                   | 8.625                | 4.500                | 0.109           | 0.083           | 0.148          | 0.120          | 0.322          | 0.237          | 0.500          | 0.337          | 6.00       |
| 8 X 5                   | 8.625                | 5.563                | 0.109           | 0.109           | 0.148          | 0.134          | 0.322          | 0.258          | 0.500          | 0.375          | 6.00       |
| 8 X 6                   | 8.625                | 6.625                | 0.109           | 0.109           | 0.148          | 0.134          | 0.322          | 0.280          | 0.500          | 0.432          | 6.00       |
| 10 X 5                  | 10.750               | 5.563                | 0.134           | 0.109           | 0.165          | 0.134          | 0.365          | 0.258          | 0.500          | 0.375          | 7.00       |
| 10 X 6                  | 10.750               | 6.625                | 0.134           | 0.109           | 0.165          | 0.134          | 0.365          | 0.280          | 0.500          | 0.432          | 7.00       |
| 10 X 8                  | 10.750               | 8.625                | 0.134           | 0.109           | 0.165          | 0.148          | 0.365          | 0.322          | 0.500          | 0.500          | 7.00       |
| 12 X 6                  | 12.750               | 6.625                | 0.156           | 0.109           | 0.180          | 0.134          | 0.375          | 0.280          | 0.500          | 0.432          | 8.00       |
| 12 X 8                  | 12.750               | 8.625                | 0.156           | 0.109           | 0.180          | 0.148          | 0.375          | 0.322          | 0.500          | 0.500          | 8.00       |
| 12 X 10                 | 12.750               | 10.750               | 0.156           | 0.134           | 0.180          | 0.165          | 0.375          | 0.365          | 0.500          | 0.500          | 8.00       |

| Weights                 |            |             |             |             | Unit: kg |
|-------------------------|------------|-------------|-------------|-------------|----------|
| Nominal<br>Pipe<br>Size | Sch.<br>5S | Sch.<br>10S | Sch.<br>40S | Sch.<br>80S |          |
| 3/4 X 1/2               | 0.054      | 0.090       | 0.105       | 0.159       |          |
| 1 X 1/2                 | 0.060      | 0.089       | 0.105       | 0.163       |          |
| 1 X 3/4                 | 0.060      | 0.098       | 0.116       | 0.168       |          |
| 1 1/4 X 1               | 0.066      | 0.125       | 0.157       | 0.245       |          |
| 1 1/2 X 1/2             | 0.091      | 0.154       | 0.204       | 0.276       |          |
| 1 1/2 X 3/4             | 0.095      | 0.156       | 0.201       | 0.294       |          |
| 1 1/2 X 1               | 0.103      | 0.170       | 0.219       | 0.308       |          |
| 1 1/2 X 1 1/4           | 0.118      | 0.195       | 0.263       | 0.331       |          |
| 2 X 3/4                 | 0.121      | 0.214       | 0.298       | 0.458       |          |
| 2 X 1                   | 0.143      | 0.237       | 0.322       | 0.458       |          |
| 2 X 1 1/4               | 0.156      | 0.258       | 0.352       | 0.525       |          |
| 2 X 1 1/2               | 0.164      | 0.273       | 0.372       | 0.544       |          |
| 2 1/2 X 1               | 0.453      | 0.349       | 0.793       | 0.793       |          |
| 2 1/2 X 1 1/4           | 0.276      | 0.394       | 0.670       | 0.825       |          |
| 2 1/2 X 1 1/2           | 0.281      | 0.396       | 0.661       | 0.856       |          |
| 2 1/2 X 2               | 0.307      | 0.432       | 0.724       | 0.938       |          |
| 3 X 1                   | 0.281      | 0.408       | 0.716       | 0.888       |          |
| 3 X 1 1/4               | 0.299      | 0.430       | 0.775       | 1.037       |          |
| 3 X 1 1/2               | 0.315      | 0.444       | 0.783       | 1.105       |          |
| 3 X 2                   | 0.339      | 0.478       | 0.846       | 1.178       |          |
| 3 X 2 1/2               | 0.376      | 0.548       | 0.983       | 1.291       |          |
| 4 X 1 1/2               | 0.435      | 0.630       | 1.241       | 1.730       |          |
| 4 X 2                   | 0.464      | 0.656       | 1.27        | 1.780       |          |
| 4 X 2 1/2               | 0.499      | 0.706       | 1.37        | 2.007       |          |
| 4 X 3                   | 0.528      | 0.748       | 1.45        | 2.129       |          |
| 5 X 3 1/2               | 0.991      | 1.20        | 2.27        | 3.547       |          |
| 5 X 4                   | 1.107      | 1.32        | 2.50        | 3.769       |          |
| 6 X 2 1/2               | 1.12       | 1.35        | 2.74        | 4.51        |          |
| 6 X 3                   | 1.24       | 1.50        | 3.04        | 5.03        |          |
| 6 X 4                   | 1.34       | 1.62        | 3.30        | 5.44        |          |
| 6 X 5                   | 1.45       | 1.75        | 3.57        | 5.71        |          |
| 8 X 4                   | 1.80       | 2.56        | 5.10        | 8.43        |          |
| 8 X 5                   | 1.90       | 2.70        | 5.40        | 8.83        |          |
| 8 X 6                   | 2.01       | 2.85        | 5.71        | 9.24        |          |
| 10 X 5                  | 2.82       | 3.30        | 8.21        | 13.00       |          |
| 10 X 6                  | 3.30       | 3.87        | 8.78        | 13.50       |          |
| 10 X 8                  | 3.59       | 4.21        | 9.58        | 14.22       |          |
| 12 X 6                  | 4.97       | 5.73        | 12.40       | 18.39       |          |
| 12 X 8                  | 5.45       | 6.11        | 13.70       | 19.07       |          |
| 12 X 10                 | 5.83       | 6.55        | 14.70       | 19.48       |          |

# STAINLESS STEEL FITTINGS



## ECC., CON. REDUCES (14" and Larger)

| Dimensions              |    |                      |                      |                 |                 |                |                |                |                |                |                | Unit: inch |
|-------------------------|----|----------------------|----------------------|-----------------|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------|
| Nominal<br>Pipe<br>Size |    | Out-<br>side<br>Dia. | Out-<br>side<br>Dia. | Wall Thickness  |                 |                |                |                |                |                |                | Length     |
|                         |    |                      |                      | Sch.10S         |                 | Sch.20         |                | Std.           |                | XS             |                |            |
|                         |    |                      |                      | OD <sub>1</sub> | OD <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> |            |
| 14                      | 12 | 14.000               | 12.750               | 0.188           | 0.180           | 0.312          | 0.250          | 0.375          | 0.375          | 0.500          | 0.500          | 13.000     |
| 14                      | 10 | 14.000               | 10.750               | 0.188           | 0.165           | 0.312          | 0.250          | 0.375          | 0.365          | 0.500          | 0.500          | 13.000     |
| 14                      | 8  | 14.000               | 8.625                | 0.188           | 0.148           | 0.312          | 0.250          | 0.375          | 0.322          | 0.500          | 0.500          | 13.000     |
| 14                      | 6  | 14.000               | 6.625                | 0.188           | 0.134           | 0.312          | -              | 0.375          | 0.280          | 0.500          | 0.432          | 13.000     |
| 16                      | 14 | 16.000               | 14.000               | 0.188           | 0.188           | 0.312          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 14.000     |
| 16                      | 12 | 16.000               | 12.750               | 0.188           | 0.180           | 0.312          | 0.250          | 0.375          | 0.375          | 0.500          | 0.500          | 14.000     |
| 16                      | 10 | 16.000               | 10.750               | 0.188           | 0.165           | 0.312          | 0.250          | 0.375          | 0.365          | 0.500          | 0.500          | 14.000     |
| 16                      | 8  | 16.000               | 8.625                | 0.188           | 0.148           | 0.312          | 0.250          | 0.375          | 0.322          | 0.500          | 0.500          | 14.000     |
| 18                      | 16 | 18.000               | 16.000               | 0.188           | 0.188           | 0.312          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 15.000     |
| 18                      | 14 | 18.000               | 14.000               | 0.188           | 0.188           | 0.312          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 15.000     |
| 18                      | 12 | 18.000               | 12.750               | 0.188           | 0.180           | 0.312          | 0.250          | 0.375          | 0.375          | 0.500          | 0.500          | 15.000     |
| 18                      | 10 | 18.000               | 10.750               | 0.188           | 0.165           | 0.312          | 0.250          | 0.375          | 0.365          | 0.500          | 0.500          | 15.000     |
| 20                      | 18 | 20.000               | 18.000               | 0.218           | 0.188           | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 20                      | 16 | 20.000               | 16.000               | 0.218           | 0.188           | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 20                      | 14 | 20.000               | 14.000               | 0.218           | 0.188           | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 20                      | 12 | 20.000               | 12.750               | 0.218           | 0.180           | 0.375          | 0.250          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 22                      | 20 | 22.000               | 20.000               | 0.218           | 0.218           | 0.375          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 22                      | 18 | 22.000               | 18.000               | 0.218           | 0.188           | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 22                      | 16 | 22.000               | 16.000               | 0.218           | 0.188           | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 22                      | 14 | 22.000               | 14.000               | 0.218           | 0.188           | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 24                      | 22 | 24.000               | 22.000               | 0.250           | 0.218           | 0.375          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 24                      | 20 | 24.000               | 20.000               | 0.250           | 0.218           | 0.375          | 0.375          | 0.375          | 0.350          | 0.500          | 0.500          | 20.000     |
| 24                      | 18 | 24.000               | 18.000               | 0.250           | 0.188           | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 24                      | 16 | 24.000               | 16.000               | 0.250           | 0.188           | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 24                      | 14 | 24.000               | 14.000               | 0.250           | 0.188           | 0.375          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 20.000     |
| 26                      | 24 | 26.000               | 24.000               |                 | 0.250           | 0.500          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 26                      | 20 | 26.000               | 20.000               |                 | 0.218           | 0.500          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 26                      | 18 | 26.000               | 18.000               |                 | 0.188           | 0.500          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 26                      | 16 | 26.000               | 16.000               |                 | 0.188           | 0.500          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 28                      | 26 | 28.000               | 26.000               |                 |                 | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 28                      | 24 | 28.000               | 24.000               |                 | 0.250           | 0.500          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 28                      | 20 | 28.000               | 20.000               |                 | 0.218           | 0.500          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 28                      | 18 | 28.000               | 18.000               |                 | 0.188           | 0.500          | 0.312          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |

| Weights Unit: kg        |    |                |            |        |        |
|-------------------------|----|----------------|------------|--------|--------|
| Nominal<br>Pipe<br>Size |    | Wall Thickness |            |        |        |
|                         |    | Sch.<br>10S    | Sch.<br>20 | Std.   | XS     |
| 14                      | 12 | 14.27          | 25.65      | 28.55  | 37.18  |
| 14                      | 10 | 13.21          | 23.73      | 26.41  | 34.40  |
| 14                      | 8  | 12.07          | 21.70      | 24.15  | 31.44  |
| 14                      | 6  | 11.01          | 19.78      | 22.01  | 28.66  |
| 16                      | 14 | 17.24          | 30.98      | 34.48  | 44.90  |
| 16                      | 12 | 16.52          | 29.69      | 33.05  | 43.03  |
| 16                      | 10 | 15.37          | 27.62      | 30.75  | 40.04  |
| 16                      | 8  | 14.15          | 25.43      | 28.30  | 36.86  |
| 18                      | 16 | 20.94          | 37.62      | 41.87  | 54.52  |
| 18                      | 14 | 19.71          | 35.41      | 39.41  | 51.32  |
| 18                      | 12 | 18.94          | 34.02      | 37.87  | 49.32  |
| 18                      | 10 | 17.70          | 31.81      | 35.41  | 46.11  |
| 20                      | 18 | 38.01          | 62.40      | 62.40  | 81.25  |
| 20                      | 16 | 36.01          | 59.11      | 59.11  | 76.97  |
| 20                      | 14 | 33.98          | 55.83      | 55.83  | 72.70  |
| 20                      | 12 | 32.76          | 53.78      | 53.78  | 70.02  |
| 22                      | 20 | 42.01          | 68.96      | 68.96  | 89.80  |
| 22                      | 18 | 40.01          | 65.68      | 65.68  | 85.53  |
| 22                      | 16 | 38.01          | 62.40      | 62.40  | 81.25  |
| 22                      | 14 | 36.01          | 59.08      | 59.08  | 76.97  |
| 24                      | 22 | 52.91          | 75.53      | 75.53  | 98.36  |
| 24                      | 20 | 50.61          | 72.25      | 72.25  | 94.08  |
| 24                      | 18 | 48.31          | 68.96      | 68.96  | 89.80  |
| 24                      | 16 | 46.01          | 65.68      | 65.68  | 85.53  |
| 24                      | 14 | 43.71          | 62.40      | 62.40  | 81.25  |
| 26                      | 24 |                | 128.28     | 98.53  | 128.03 |
| 26                      | 20 |                | 118.02     | 90.65  | 117.79 |
| 26                      | 18 |                | 112.88     | 86.70  | 112.66 |
| 26                      | 16 |                | 107.76     | 82.76  | 107.54 |
| 28                      | 26 |                | 138.54     | 106.41 | 138.27 |
| 28                      | 24 |                | 133.41     | 102.47 | 133.15 |
| 28                      | 20 |                | 123.15     | 94.59  | 122.91 |
| 28                      | 18 |                | 118.02     | 90.65  | 117.78 |

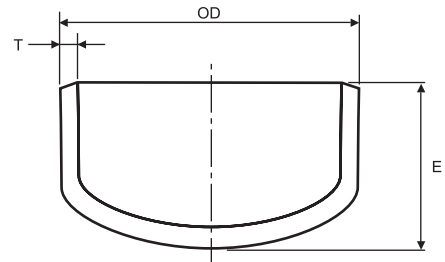
# STAINLESS STEEL FITTINGS

ECC., CON. REDUCES (14" and Larger)

| Dimensions              |    |                      |                      |                |                |                |                |                |                |                |                | Unit: inch |
|-------------------------|----|----------------------|----------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------|
| Nominal<br>Pipe<br>Size |    | Out-<br>side<br>Dia. | Out-<br>side<br>Dia. | Wall Thickness |                |                |                |                |                |                |                | Length     |
|                         |    |                      |                      | Sch.10S        |                | Sch.20         |                | Std.           |                | XS             |                |            |
|                         |    | OD <sub>1</sub>      | OD <sub>2</sub>      | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | T <sub>1</sub> | T <sub>2</sub> | H          |
| 30                      | 28 | 30.000               | 28.000               | 0.312          |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 30                      | 26 | 30.000               | 26.000               | 0.312          |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 30                      | 24 | 30.000               | 24.000               | 0.312          | 0.250          | 0.500          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 30                      | 20 | 30.000               | 20.000               | 0.312          | 0.218          | 0.500          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 32                      | 30 | 32.000               | 30.000               |                | 0.312          | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 32                      | 28 | 32.000               | 28.000               |                |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 32                      | 26 | 32.000               | 26.000               |                |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 32                      | 24 | 32.000               | 24.000               |                | 0.250          | 0.500          | 0.375          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 34                      | 32 | 34.000               | 32.000               |                |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 34                      | 30 | 34.000               | 30.000               |                | 0.312          | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 34                      | 28 | 34.000               | 28.000               |                |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 34                      | 26 | 34.000               | 26.000               |                |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 36                      | 34 | 36.000               | 34.000               |                |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 36                      | 32 | 36.000               | 32.000               |                |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 36                      | 30 | 36.000               | 30.000               |                | 0.312          | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 36                      | 28 | 36.000               | 28.000               |                |                | 0.500          | 0.500          | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 38                      | 36 | 38.000               | 36.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 38                      | 34 | 38.000               | 34.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 38                      | 32 | 38.000               | 32.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 38                      | 30 | 38.000               | 30.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 38                      | 28 | 38.000               | 28.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 38                      | 26 | 38.000               | 26.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 40                      | 38 | 40.000               | 38.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 40                      | 36 | 40.000               | 36.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 40                      | 34 | 40.000               | 34.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 40                      | 32 | 40.000               | 32.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 40                      | 30 | 40.000               | 30.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 42                      | 40 | 42.000               | 40.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 42                      | 38 | 42.000               | 38.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 42                      | 36 | 42.000               | 36.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 42                      | 34 | 42.000               | 34.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 42                      | 32 | 42.000               | 32.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 42                      | 30 | 42.000               | 30.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 44                      | 42 | 44.000               | 42.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 44                      | 40 | 44.000               | 40.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 44                      | 38 | 44.000               | 38.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 44                      | 36 | 44.000               | 36.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 46                      | 44 | 46.000               | 44.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 46                      | 42 | 46.000               | 42.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 46                      | 40 | 46.000               | 40.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 46                      | 38 | 46.000               | 38.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 46                      | 36 | 46.000               | 36.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 48                      | 44 | 48.000               | 44.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 48                      | 42 | 48.000               | 42.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |
| 48                      | 40 | 48.000               | 40.000               |                |                |                |                | 0.375          | 0.375          | 0.500          | 0.500          | 24.000     |

| Weights                 |    |                |            |        |    | Unit: kg |
|-------------------------|----|----------------|------------|--------|----|----------|
| Nominal<br>Pipe<br>Size |    | Wall Thickness |            |        |    |          |
|                         |    | Sch.<br>10S    | Sch.<br>20 | Std.   | XS |          |
| 30                      | 28 |                | 148.80     | 114.29 |    | 148.51   |
| 30                      | 26 |                | 143.67     | 110.35 |    | 143.39   |
| 30                      | 24 |                | 138.54     | 106.41 |    | 138.27   |
| 30                      | 20 |                | 128.28     | 98.53  |    | 128.03   |
| 32                      | 30 |                | 159.07     | 122.17 |    | 158.75   |
| 32                      | 28 |                | 153.93     | 118.23 |    | 153.63   |
| 32                      | 26 |                | 148.80     | 114.29 |    | 148.51   |
| 32                      | 24 |                | 143.67     | 110.40 |    | 143.39   |
| 34                      | 32 |                | 169.33     | 130.06 |    | 168.99   |
| 34                      | 30 |                | 164.20     | 126.12 |    | 163.88   |
| 34                      | 28 |                | 159.07     | 122.17 |    | 158.75   |
| 34                      | 26 |                | 153.93     | 118.51 |    | 153.63   |
| 36                      | 34 |                | 179.59     | 137.94 |    | 179.24   |
| 36                      | 32 |                | 174.46     | 133.99 |    | 174.12   |
| 36                      | 30 |                | 169.33     | 130.06 |    | 168.99   |
| 36                      | 28 |                | 164.20     | 126.12 |    | 163.88   |
| 38                      | 36 |                |            | 145.82 |    | 189.48   |
| 38                      | 34 |                |            | 141.88 |    | 184.36   |
| 38                      | 32 |                |            | 137.94 |    | 179.24   |
| 38                      | 30 |                |            | 133.99 |    | 174.12   |
| 38                      | 28 |                |            | 130.06 |    | 168.99   |
| 38                      | 26 |                |            | 126.12 |    | 163.88   |
| 40                      | 38 |                |            | 153.70 |    | 199.72   |
| 40                      | 36 |                |            | 149.76 |    | 194.60   |
| 40                      | 34 |                |            | 145.82 |    | 189.46   |
| 40                      | 32 |                |            | 141.88 |    | 184.36   |
| 40                      | 30 |                |            | 137.94 |    | 179.24   |
| 42                      | 40 |                |            | 161.59 |    | 209.97   |
| 42                      | 38 |                |            | 157.64 |    | 204.85   |
| 42                      | 36 |                |            | 153.70 |    | 199.72   |
| 42                      | 34 |                |            | 149.76 |    | 194.60   |
| 42                      | 32 |                |            | 145.82 |    | 189.48   |
| 42                      | 30 |                |            | 141.88 |    | 184.36   |
| 44                      | 42 |                |            | 169.47 |    | 220.21   |
| 44                      | 40 |                |            | 165.53 |    | 215.09   |
| 44                      | 38 |                |            | 161.59 |    | 209.97   |
| 44                      | 36 |                |            | 157.64 |    | 204.85   |
| 46                      | 44 |                |            | 177.35 |    | 230.45   |
| 46                      | 42 |                |            | 173.41 |    | 225.33   |
| 46                      | 40 |                |            | 169.47 |    | 220.21   |
| 46                      | 38 |                |            | 165.53 |    | 215.09   |
| 46                      | 36 |                |            | 165.53 |    | 215.09   |
| 48                      | 44 |                |            | 181.29 |    | 235.57   |
| 48                      | 42 |                |            | 177.35 |    | 230.45   |
| 48                      | 40 |                |            | 173.41 |    | 225.33   |

# STAINLESS STEEL FITTINGS



## CAPS (24" and Smaller)

| Dimensions              |                         |                    |         |         |         | Unit: inch  | Weights                 |        |         |         | Unit: kg |
|-------------------------|-------------------------|--------------------|---------|---------|---------|-------------|-------------------------|--------|---------|---------|----------|
| Nominal<br>Pipe<br>Size | Outside<br>Dia.<br>(OD) | Wall Thickness (T) |         |         |         | Length<br>E | Nominal<br>Pipe<br>Size | Sch.5S | Sch.10S | Sch.40S | Sch.80S  |
|                         |                         | Sch.5S             | Sch.10S | Sch.40S | Sch.80S |             |                         |        |         |         |          |
| 1/2                     | 0.840                   | 0.065              | 0.083   | 0.109   | 0.147   | 1.00        | 1/2                     | 0.036  | 0.040   | 0.046   | 0.045    |
| 3/4                     | 1.050                   | 0.065              | 0.083   | 0.113   | 0.154   | 1.00        | 3/4                     | 0.049  | 0.052   | 0.059   | 0.086    |
| 1                       | 1.315                   | 0.065              | 0.109   | 0.133   | 0.179   | 1.50        | 1                       | 0.053  | 0.090   | 0.109   | 0.109    |
| 1 1/4                   | 1.660                   | 0.065              | 0.109   | 0.140   | 0.191   | 1.50        | 1 1/4                   | 0.066  | 0.113   | 0.145   | 0.159    |
| 1 1/2                   | 1.900                   | 0.065              | 0.109   | 0.145   | 0.200   | 1.50        | 1 1/2                   | 0.076  | 0.130   | 0.171   | 0.222    |
| 2                       | 2.375                   | 0.065              | 0.109   | 0.154   | 0.218   | 1.50        | 2                       | 0.099  | 0.168   | 0.234   | 0.344    |
| 2 1/2                   | 2.875                   | 0.083              | 0.120   | 0.203   | 0.276   | 2.00        | 2 1/2                   | 0.169  | 0.242   | 0.420   | 0.512    |
| 3                       | 3.500                   | 0.083              | 0.120   | 0.216   | 0.300   | 2.50        | 3                       | 0.253  | 0.362   | 0.664   | 0.888    |
| 4                       | 4.500                   | 0.083              | 0.120   | 0.237   | 0.337   | 3.00        | 4                       | 0.409  | 0.585   | 1.17    | 1.51     |
| 5                       | 5.563                   | 0.109              | 0.134   | 0.258   | 0.375   | 3.50        | 5                       | 0.805  | 0.978   | 1.90    | 2.89     |
| 6                       | 6.625                   | 0.109              | 0.134   | 0.280   | 0.432   | 4.00        | 6                       | 1.11   | 1.35    | 2.83    | 4.24     |
| 8                       | 8.625                   | 0.109              | 0.148   | 0.322   | 0.500   | 5.00        | 8                       | 1.74   | 2.49    | 5.11    | 7.76     |
| 10                      | 10.750                  | 0.134              | 0.165   | 0.375   | 0.500   | 6.00        | 10                      | 3.26   | 3.84    | 8.92    | 13.11    |
| 12                      | 12.750                  | 0.156              | 0.180   | 0.375   | 0.500   | 7.00        | 12                      | 5.47   | 6.15    | 14.10   | 17.94    |
| 14                      | 14.000                  | 0.156              | 0.188   | 0.375*  | 0.500** | 7.50        | 14                      | 7.19   | 7.85    | 16.35   | 21.11    |
| 16                      | 16.000                  | 0.165              | 0.188   | 0.375*  | 0.500** | 8.00        | 16                      | 10.99  | 9.56    | 18.60   | 25.73    |
| 18                      | 18.000                  | 0.165              | 0.188   | 0.375*  | 0.500** | 9.00        | 18                      | 13.92  | 12.10   | 23.54   | 32.57    |
| 20                      | 20.000                  | 0.188              | 0.218   | 0.375*  | 0.500** | 10.00       | 20                      | 17.19  | 17.90   | 29.06   | 40.21    |
| 24                      | 24.000                  | 0.218              | 0.250   | 0.375*  | 0.500** | 12.00       | 24                      | 27.43  | 30.15   | 41.85   | 57.86    |

\*Std \*\*XS

For fittings made from seamless and welded pipes ASTM A312 shall be applied as follows.

| Grade   | Carbon<br>max,<br>per<br>cent | Mang-<br>anese,<br>max,<br>percent | Phos-<br>phorus<br>per<br>cent | Sulfur,<br>max,<br>per<br>cent | Silicon,<br>max,<br>per<br>cent | Chromium,<br>percent | Nickel,<br>percent | Other Elements            | T.S<br>Min<br>ksi | Y.S<br>Min<br>ksi |
|---------|-------------------------------|------------------------------------|--------------------------------|--------------------------------|---------------------------------|----------------------|--------------------|---------------------------|-------------------|-------------------|
| TP 304  | 0.08                          | 2.00                               | 0.040                          | 0.030                          | 0.75                            | 18.0~20.0            | 8.00~11.0          |                           | 75                | 30                |
| TP 304L | 0.035                         | 2.00                               | 0.040                          | 0.030                          | 0.75                            | 18.0~20.0            | 8.00~13.0          |                           | 70                | 25                |
| TP 316  | 0.08                          | 2.00                               | 0.040                          | 0.030                          | 0.75                            | 16.0~18.0            | 11.00~14.0         | Mo 2.0~3.0                | 75                | 30                |
| TP 316L | 0.035                         | 2.00                               | 0.040                          | 0.030                          | 0.75                            | 16.0~18.0            | 10.00~15.0         | Mo 2.0~3.0                | 70                | 25                |
| TP 321  | 0.08                          | 2.00                               | 0.040                          | 0.030                          | 0.75                            | 17.0~20.0            | 9.00~13.0          | Ti min. CX5 max. 0.60     | 75                | 30                |
| TP 347  | 0.08                          | 2.00                               | 0.040                          | 0.030                          | 0.75                            | 17.0~20.0            | 9.00~13.0          | Cb+Ta min. CX10 max. 1.00 | 75                | 30                |



## MACHINERY STEEL PRODUCTS

HARDEN & TEMPERED STEEL - CARBON STEEL

248

PRE-HARDEN & TEMPERED ALLOY STEEL 249

COLD FINISHED STEEL BAR

251



YICK HOE



# HARDEN & TEMPERED STEEL

## CARBON STEEL KS D3752, JIS G4051

| KS<br>JIS | AISI | Chemical Composition % |              |                     |                |                | Mechanical Properties |                    |     |    |        |          |
|-----------|------|------------------------|--------------|---------------------|----------------|----------------|-----------------------|--------------------|-----|----|--------|----------|
|           |      | C                      | Si           | Mn                  | P              | S              | YP                    | TS                 | EL  | RA | Impact | Hardness |
|           |      |                        |              |                     |                |                | kg/mm <sup>2</sup>    | kg/mm <sup>2</sup> | %   | %  | charpy | HB       |
| S10C      | 1010 | .08/.13<br>.08/.13     | .15/.35<br>- | .30/.60<br>.30/.60  | ≤0.03<br>≤.035 | ≤.035<br>≤.045 | ≥21                   | ≥32                | ≥33 |    |        | 109/159  |
| S12C      | 1012 | .10/.15<br>.10/.15     | .15/.35<br>- | .30/.60<br>.30/.60  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥24                   | ≥38                | ≥30 | -  | -      | 111/167  |
| S15C      | 1015 | .13/.18<br>.13/.18     | .15/.35<br>- | .30/.60<br>.30/.60  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥24                   | ≥38                | ≥30 | -  | -      | 111/167  |
| S17C      | 1017 | .15/.20<br>.15/.20     | .15/.35<br>- | .30/.60<br>.30/.60  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥25                   | ≥41                | ≥28 | -  | -      | 116/174  |
| S20C      | 1020 | .18/.23<br>.18/.23     | .15/.35<br>- | .30/.60<br>.30/.60  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥25                   | ≥41                | ≥28 | -  | -      | 116/174  |
| S22C      | 1023 | .20/.25<br>.20/.25     | .15/.35<br>- | .30/.60<br>.30/.60  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥27                   | ≥45                | ≥27 | -  | -      | 123/183  |
| S25C      | 1025 | .22/.28<br>.22/.28     | .15/.35<br>- | .30/.60<br>.30/.60  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥27                   | ≥45                | ≥27 | -  | -      | 123/183  |
| S28C      | 1029 | .25/.31<br>.25/.31     | .15/.35<br>- | .60/.90<br>.60/.90  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥29                   | ≥48                | ≥25 | -  | -      | 137/197  |
| S30C      | 1030 | .27/.33<br>.28/.34     | .15/.35<br>- | .60/.90<br>.60/.90  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥29                   | ≥48                | ≥25 | -  | -      | 137/197  |
| S33C      |      | .30/.36                | .15/.35      | .60/.90             | ≤.030          | ≤.035          | ≥31                   | ≥52                | ≥23 | -  | -      | 149/207  |
| S35C      | 1035 | .32/.36<br>.32/.38     | .15/.35<br>- | .60/.90<br>.60/.90  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥31                   | ≥52                | ≥23 | -  | -      | 149/207  |
| S38C      | 1038 | .35/.41<br>.35/.42     | .15/.35<br>- | .60/.90<br>.60/.90  | ≤.030<br>≤.030 | ≤.035<br>≤.045 | ≥33                   | ≥55                | ≥22 | -  | -      | 156/217  |
| S40C      | 1040 | .37/.43<br>.37/.44     | .15/.35<br>- | .60/.90<br>.60/.90  | ≤.030<br>≤.030 | ≤.035<br>≤.045 | ≥33                   | ≥55                | ≥22 | -  | -      | 156/217  |
| S43C      | 1043 | .40/.46<br>.40/.47     | .15/.35<br>- | .60/.90<br>.70/1.00 | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥35                   | ≥58                | ≥20 | -  | -      | 167/229  |
| S45C      | 1045 | .42/.48<br>.43/.50     | .15/.35<br>- | .60/.90<br>.60/.90  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥35                   | ≥58                | ≥20 | -  | -      | 167/229  |
| S48C      |      | .45/.51                | .15/.35      | .60/.90             | ≤.030          | ≤.035          | ≥37                   | ≥62                | ≥18 | -  | -      | 179/235  |
| S50C      | 1049 | .47/.53<br>.46/.53     | .15/.35<br>- | .60/.90<br>.60/.90  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥37                   | ≥62                | ≥18 | -  | -      | 179/235  |
| S53C      | 1053 | .50/.56<br>.48/.55     | .15/.35<br>- | .60/.90<br>.70/1.00 | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥40                   | ≥66                | ≥15 | -  | -      | 183/255  |
| S55C      | 1055 | .52/.58<br>.50/.60     | .15/.35<br>- | .60/.90<br>.60/.90  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥40                   | ≥66                | ≥15 | -  | -      | 183/255  |
| S58C      | 1060 | .55/.61<br>.55/.65     | .15/.35<br>- | .60/.90<br>.60/.90  | ≤.030<br>≤.035 | ≤.035<br>≤.045 | ≥40                   | ≥66                | ≥15 | -  | -      | 183/255  |

# PRE-HARDEN & TEMPERED ALLOY STEEL

## ALLOY STEEL KS D3707 D3711, JIS G4104 G4105

| KS<br>JIS | AISI | Chemical Composition % |         |          |       |       |           |         | Mechanical Properties |                    |     |     |        |          |
|-----------|------|------------------------|---------|----------|-------|-------|-----------|---------|-----------------------|--------------------|-----|-----|--------|----------|
|           |      | C                      | Si      | Mn       | P     | S     | Cr        | Mo      | YP                    | TS                 | EL  | RA  | Impact | Hardness |
|           |      |                        |         |          |       |       |           |         | kg/mm <sup>2</sup>    | kg/mm <sup>2</sup> | %   | %   | charpy | HB       |
| SCR415    |      | .13/.18                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | -       | -                     | ≥80                | ≥15 | ≥40 | ≥6     | 217/302  |
| SCR420    | 5120 | .18/.23                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | -       | -                     | ≥85                | ≥14 | ≥35 | ≥5     | 235/321  |
|           |      | .17/.22                | .15/.30 | .70/.90  | ≤.035 | ≤.040 | .70/.90   |         |                       |                    |     |     |        |          |
| SCR430    | 5130 | .28/.33                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | -       | ≥65                   | ≥80                | ≥18 | ≥55 | ≥9     | 229/285  |
|           |      | .28/.33                | .15/.30 | .70/.90  | ≤.035 | ≤.040 | .80/1.10  |         |                       |                    |     |     |        |          |
| SCR435    | 5135 | .33/.38                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | -       | ≥75                   | ≥90                | ≥15 | ≥50 | ≥7     | 255/311  |
|           |      | .33/.38                | .15/.30 | .60/.80  | ≤.035 | ≤.030 | .80/1.05  |         |                       |                    |     |     |        |          |
| SCR440    | 5140 | .38/.43                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | -       | ≥85                   | ≥100               | ≥12 | ≥40 | ≥5     | 285/341  |
|           |      | .38/.43                | .15/.30 | .70/.90  | ≤.035 | ≤.040 | .70/.90   |         |                       |                    |     |     |        |          |
| SCR445    | 5147 | .43/.48                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | -       | ≥85                   | ≥100               | ≥12 | ≥40 | ≥5     | 285/341  |
|           |      | .46/.51                | .15/.30 | .70/.95  | ≤.035 | ≤.030 | .85/1.15  |         |                       |                    |     |     |        |          |
| SCM415    |      | .13/.18                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | .15/.30 | -                     | ≥85                | ≥16 | ≥40 | ≥7     | 235/321  |
| SCM418    | 4118 | .16/.21                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | .15/.30 | -                     | ≥90                | ≥15 | ≥40 | ≥7     | 248/331  |
|           |      | .18/.23                | .15/.30 | .70/.90  | ≤.035 | ≤.040 | .40/.60   | .08/.15 |                       |                    |     |     |        |          |
| SCM420    |      | .18/.23                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | .15/.30 | -                     | ≥95                | ≥14 | ≥35 | ≥6     | 285/363  |
| SCM421    |      | .17/.23                | .15/.35 | .70/1.00 | ≤.030 | ≤.030 | .90/1.20  | .15/.30 | -                     | ≥100               | ≥14 | ≥35 | ≥6     | 285/363  |
| SCM822    |      | .20/.25                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | .35/.45 | -                     | ≥105               | ≥12 | ≥30 | ≥6     | 302/415  |
| SCM430    | 4130 | .28/.33                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | .15/.30 | ≥70                   | ≥85                | ≥18 | ≥55 | ≥11    | 241/293  |
|           |      | .28/.33                | .15/.30 | .40/.60  | ≤.035 | ≤.040 | .80/1.10  | .15/.25 |                       |                    |     |     |        |          |
| SCM432    | 4135 | .27/.37                | .15/.35 | .30/.60  | ≤.030 | ≤.030 | 1.00/1.50 | .15/.30 | ≥75                   | ≥90                | ≥16 | ≥50 | ≥9     | 225/321  |
|           |      | .33/.38                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | .15/.30 | ≥80                   | ≥95                | ≥15 | ≥50 | ≥8     | 269/321  |
| SCM435    | 4140 | .33/.38                | .20/.35 | .70/.90  | ≤.035 | ≤.040 | .80/1.10  | .15/.25 |                       |                    |     |     |        |          |
|           |      | .38/.43                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | .15/.30 | ≥85                   | ≥100               | ≥12 | ≥45 | ≥6     | 285/341  |
| SCM440    | 4145 | .38/.43                | .15/.30 | .75/1.00 | ≤.035 | ≤.040 | .80/1.10  | .15/.25 |                       |                    |     |     |        |          |
|           |      | .43/.48                | .15/.35 | .60/.85  | ≤.030 | ≤.030 | .90/1.20  | .15/.30 | ≥90                   | ≥105               | ≥12 | ≥40 | ≥4     | 302/363  |
| SCM445    |      | .43/.48                | .15/.30 | .75/1.00 | ≤.035 | ≤.040 | .80/1.10  | .15/.25 |                       |                    |     |     |        |          |

# PRE-HARDEN & TEMPERED ALLOY STEEL

## ALLOY STEEL KS D3708 D3709, JIS G4102 G4103

| KS<br>JIS | AISI    | Chemical Composition % |         |          |       |       |           |           | Mechanical Properties |                          |                          |         |         |                  |                |
|-----------|---------|------------------------|---------|----------|-------|-------|-----------|-----------|-----------------------|--------------------------|--------------------------|---------|---------|------------------|----------------|
|           |         | C                      | Si      | Mn       | P     | S     | Ni        | Cr        | Mo                    | YP<br>kg/mm <sup>2</sup> | TS<br>kg/mm <sup>2</sup> | EL<br>% | RA<br>% | Impact<br>charpy | Hardness<br>HB |
| SNC415    |         | .12/.18                | .15/.35 | .35/.65  | ≤.030 | ≤.030 | 2.00/2.50 | .20/.50   | -                     | -                        | ≤80                      | ≥17     | ≥45     | ≥9               | 235/341        |
| SNC815    |         | .12/.18                | .15/.35 | .35/.65  | ≤.030 | ≤.030 | 3.00/3.50 | .70/1.00  | -                     | -                        | ≤100                     | ≥12     | ≥45     | ≥8               | 285/388        |
| SNC631    |         | .27/.35                | .15/.35 | .35/.65  | ≤.030 | ≤.030 | 2.50/3.00 | .60/1.00  | -                     | ≥70                      | ≥85                      | ≥18     | ≥50     | ≥12              | 248/302        |
| SNC236    |         | .32/.40                | .15/.35 | .35/.65  | ≤.030 | ≤.030 | 2.50/3.00 | .60/1.00  | -                     | ≥70                      | ≥85                      | ≥18     | ≥50     | ≥12              | 248/302        |
|           | 3135    | .33/.38                | .20/.35 | .60/.80  | ≤.040 | ≤.040 | 1.10/1.40 | .55/.75   | -                     |                          |                          |         |         |                  |                |
| SNC836    |         | .32/.40                | .15/.35 | .35/.65  | ≤.030 | ≤.030 | 3.00/3.50 | .60/1.00  | -                     | ≥80                      | ≥95                      | ≥19     | ≥49     | ≥8               | 269/321        |
| SNCM415   |         | .12/.18                | .15/.35 | .40/.70  | ≤.030 | ≤.030 | 1.60/2.00 | .40/.65   | .15/.30               | -                        | ≥90                      | ≥16     | ≥45     | ≥7               | 255/341        |
| SNCM815   | .12/.18 | .15/.35                | .30/.60 | ≤.030    | ≤.030 | ≤.030 | 4.00/4.52 | .70/1.00  | .15/.30               | -                        | ≥110                     | ≥12     | ≥40     | ≥7               | 311/375        |
| SNCM616   |         | .13/.20                | .15/.35 | .80/1.20 | ≤.030 | ≤.030 | 2.80/3.20 | 1.40/1.80 | .40/.60               | -                        | ≥120                     | ≥14     | ≥40     | ≥8               | 341/415        |
| SNCM220   |         | .17/.23                | .15/.35 | .60/.90  | ≤.030 | ≤.030 | .40/.70   | .40/.60   | .15/.30               | -                        | ≥85                      | ≥17     | ≥40     | ≥6               | 248/341        |
| SNCM420   |         | .17/.23                | .13/.35 | .40/.70  | ≤.030 | ≤.030 | 1.60/2.00 | .40/.65   | .15/.30               | -                        | ≥100                     | ≥15     | ≥40     | ≥7               | 293/375        |
|           | 4320    | .17/.22                | .15/.30 | .45/.65  | ≤.035 | ≤.040 | 1.65/2.00 | .40/.60   | .20/.30               |                          |                          |         |         |                  |                |
| SNCM625   |         | .20/.30                | .15/.35 | .35/.60  | ≤.030 | ≤.030 | 3.00/3.50 | 1.00/1.50 | .15/.30               | ≥85                      | ≥95                      | ≥18     | ≥50     | ≥8               | 269/321        |
| SNCM630   |         | .25/.35                | .15/.35 | .35/.60  | ≤.030 | ≤.030 | 2.50/3.50 | 2.50/3.50 | .50/.70               | ≥90                      | ≥110                     | ≥15     | ≥45     | ≥8               | 302/352        |
| SNCM431   |         | .27/.35                | .15/.35 | .60/.90  | ≤.030 | ≤.030 | 1.60/2.00 | .60/1.00  | .15/.30               | ≥70                      | ≥85                      | ≥20     | ≥55     | ≥10              | 248/302        |
|           | 4337    | .35/.40                | .20/.35 | .60/.90  | ≤.040 | ≤.040 | 1.65/2.00 | .70/.90   | .20/.30               |                          |                          |         |         |                  |                |
| SNCM439   |         | .36/.43                | .15/.35 | .60/.90  | ≤.030 | ≤.030 | 1.60/2.00 | .60/1.00  | .15/.30               | ≥90                      | ≥100                     | ≥16     | ≥45     | ≥7               | 293/352        |
|           | 4340    | .38/.43                | .15/.30 | .60/.80  | ≤.035 | ≤.030 | 1.65/2.00 | .70/.90   | .20/.30               |                          |                          |         |         |                  |                |
| SNCM240   |         | .38/.43                | .15/.35 | .70/1.00 | ≤.030 | ≤.030 | .40/.70   | .40/.65   | .15/.30               | ≥80                      | ≥90                      | ≥17     | ≥50     | ≥7               | 255/311        |

YICK HOE

# COLD FINISHED STEEL BAR

## FREE CUTTING STEEL KS D3567, JIS G4804

| KS<br>JIS | AISI | Chemical Composition % |    |           |         |         |    | Mechanical Properties |                    |      |      |        |          |
|-----------|------|------------------------|----|-----------|---------|---------|----|-----------------------|--------------------|------|------|--------|----------|
|           |      | C                      | Si | Mn        | P       | S       | Pb | YP                    | TS                 | EL   | RA   | Impact | Hardness |
|           |      |                        |    |           |         |         |    | kg/mm <sup>2</sup>    | kg/mm <sup>2</sup> | %    | %    | charpy | HB       |
| SUM11     |      | .08/.13                | -  | .30/.60   | ≤.04    | .08/.13 | -  |                       |                    |      |      |        |          |
|           | 1109 | .08/.13                | -  | .60/.90   | ≤.04    | .08/.13 | -  |                       |                    |      |      |        |          |
| SUM12     |      | .08/.13                | -  | .60/.90   | ≤.04    | .08/.13 | -  |                       |                    |      |      |        |          |
| SUM32     |      | .12/.20                | -  | .60/1.10  | ≤.04    | .10/.20 | -  |                       |                    |      |      |        |          |
| SUM31     |      | .14/.20                | -  | 1.00/1.30 | ≤.04    | .08/.13 | -  |                       |                    |      |      |        |          |
|           | 1117 | .14/.20                | -  | 1.00/1.30 | ≤.04    | .08/.13 | -  | 28.4                  | 43.7               | 32.8 | 58   | 69     | 121      |
| SUM41     |      | .32/.39                | -  | 1.35/1.65 | ≤.04    | .08/.13 | -  |                       |                    |      |      |        |          |
|           | 1137 | .32/.39                | -  | 1.35/1.65 | ≤.04    | .08/.13 | -  | 35.1                  | 59.5               | 26.8 | 53.9 | 36.8   | 174      |
| SUM42     |      | .37/.45                | -  | 1.35/1.65 | ≤.04    | .08/.13 | -  |                       |                    |      |      |        |          |
|           | 1141 | .37/.45                | -  | 1.35/1.65 | ≤.04    | .08/.13 | -  |                       |                    |      |      |        |          |
| SUM43     |      | .37/.45                | -  | 1.35/1.65 | ≤.04    | .24/.33 | -  |                       |                    |      |      |        |          |
|           | 1144 | .40/.48                | -  | 1.35/1.65 | ≤.04    | .24/.33 | -  | 35.2                  | 59.5               | 24.8 | 47.3 | 48     | 167      |
| SUM23     |      | ≤.09                   | -  | .75/1.05  | .04/.09 | .26/.35 | -  |                       |                    |      |      |        |          |
| SUM21     |      | ≤.13                   | -  | .70/1.00  | .07/.12 | .16/.23 | -  |                       |                    |      |      |        |          |
|           | 1213 | ≤.13                   | -  | .70/1.00  | .07/.12 | .24/.33 | -  |                       |                    |      |      |        |          |
| SUM22     |      | ≤.13                   | -  | .70/1.00  | .07/.12 | .24/.33 | -  |                       |                    |      |      |        |          |

YICK HOE



## NON-FERROUS METALS

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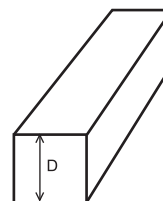
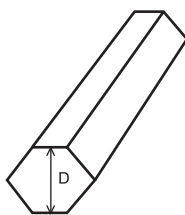
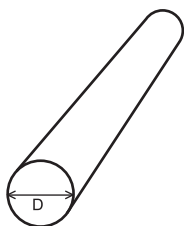
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YICK HOE

# COPPER

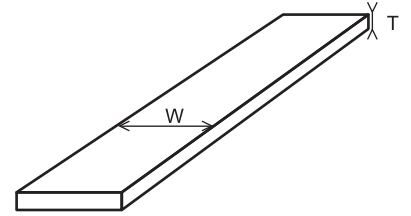
## ROUND BARS, HEXAGON BARS & SQUARE BARS



WEIGHTS CHART (Imperial Units)

| Dia.<br>in | Round Bar |       |         | Hexagon Bar |        |         | Square Bar |       |         |
|------------|-----------|-------|---------|-------------|--------|---------|------------|-------|---------|
|            | kg/ft     | lb/ft | kati/ft | kg/ft       | lb/ft  | kati/ft | kg/ft      | lb/ft | kati/ft |
| 1/8        | 0.022     | 0.048 | 0.036   | 0.024       | 0.052  | 0.040   | 0.027      | 0.060 | 0.045   |
| 3/16       | 0.049     | 0.107 | 0.081   | 0.054       | 0.118  | 0.089   | 0.062      | 0.136 | 0.103   |
| 1/4        | 0.086     | 0.189 | 0.142   | 0.095       | 0.209  | 0.157   | 0.109      | 0.241 | 0.180   |
| 5/16       | 0.134     | 0.296 | 0.222   | 0.148       | 0.326  | 0.245   | 0.171      | 0.377 | 0.283   |
| 3/8        | 0.193     | 0.426 | 0.319   | 0.213       | 0.470  | 0.352   | 0.246      | 0.543 | 0.407   |
| 7/16       | 0.263     | 0.580 | 0.435   | 0.290       | 0.640  | 0.480   | 0.335      | 0.739 | 0.554   |
| 1/2        | 0.344     | 0.758 | 0.569   | 0.379       | 0.835  | 0.627   | 0.438      | 0.965 | 0.724   |
| 9/16       | 0.435     | 0.959 | 0.719   | 0.479       | 1.057  | 0.792   | 0.554      | 1.221 | 0.916   |
| 5/8        | 0.537     | 1.184 | 0.888   | 0.592       | 1.305  | 0.979   | 0.684      | 1.507 | 1.131   |
| 11/16      | 0.650     | 1.432 | 1.075   | 0.716       | 1.579  | 1.184   | 0.827      | 1.824 | 1.368   |
| 3/4        | 0.773     | 1.705 | 1.279   | 0.853       | 1.880  | 1.411   | 0.984      | 2.170 | 1.628   |
| 13/16      | 0.907     | 2.000 | 1.500   | 1.002       | 2.210  | 1.657   | 1.157      | 2.550 | 1.914   |
| 7/8        | 1.052     | 2.320 | 1.740   | 1.160       | 2.558  | 1.919   | 1.340      | 2.954 | 2.216   |
| 15/16      | 1.211     | 2.670 | 2.003   | 1.334       | 2.940  | 2.206   | 1.542      | 3.400 | 2.550   |
| 1          | 1.374     | 3.030 | 2.273   | 1.515       | 3.341  | 2.506   | 1.750      | 3.858 | 2.895   |
| 1 1/8      | 1.740     | 3.835 | 2.878   | 1.918       | 4.229  | 3.172   | 2.215      | 4.883 | 3.664   |
| 1 1/4      | 2.148     | 4.735 | 3.553   | 2.368       | 5.221  | 3.917   | 2.735      | 6.029 | 4.524   |
| 1 3/8      | 2.599     | 5.729 | 4.299   | 2.865       | 6.317  | 4.739   | 3.309      | 7.295 | 5.473   |
| 1 1/2      | 3.093     | 6.818 | 5.116   | 3.410       | 7.518  | 5.640   | 3.938      | 8.681 | 6.513   |
| 1 5/8      | 3.630     | 8.002 | 6.004   | 4.003       | 8.824  | 6.621   | 4.622      | 10.19 | 7.645   |
| 1 3/4      | 4.210     | 9.281 | 6.963   | 4.636       | 10.22  | 7.668   | 5.362      | 11.82 | 8.869   |
| 1 7/8      | 4.831     | 10.65 | 7.990   | 5.330       | 11.750 | 8.816   | 6.151      | 13.56 | 10.174  |
| 2          | 5.498     | 12.12 | 9.094   | 6.065       | 13.37  | 10.032  | 6.999      | 15.43 | 11.576  |
| 2 1/8      | 6.214     | 13.70 | 10.278  |             |        |         |            |       |         |
| 2 1/4      | 6.958     | 15.34 | 11.509  | 7.675       | 16.92  | 12.694  | 8.859      | 19.53 | 14.653  |
| 2 3/8      | 7.766     | 17.12 | 12.845  |             |        |         |            |       |         |
| 2 1/2      | 8.605     | 18.97 | 14.233  | 9.448       | 20.83  | 15.627  | 10.941     | 24.12 | 18.096  |
| 2 5/8      | 9.485     | 20.91 | 15.688  |             |        |         |            |       |         |
| 2 3/4      | 10.41     | 22.95 | 17.218  | 11.462      | 25.27  | 18.958  | 13.236     | 29.18 | 21.892  |
| 2 7/8      | 11.381    | 25.09 | 18.824  |             |        |         |            |       |         |
| 3          | 12.388    | 27.31 | 20.490  | 13.662      | 30.12  | 22.597  | 15.753     | 34.73 | 26.055  |
| 3 1/4      | 14.542    | 32.06 | 24.052  | 16.007      | 35.29  | 26.476  | 18.484     | 40.75 | 30.573  |
| 3 1/2      | 16.865    | 37.18 | 27.895  | 18.566      | 40.93  | 30.708  | 21.442     | 47.27 | 35.465  |
| 3 3/4      | 19.360    | 42.68 | 32.021  | 21.315      | 46.99  | 35.255  | 24.612     | 54.26 | 40.708  |
| 4          | 22.027    | 48.56 | 36.433  | 24.249      | 53.46  | 40.108  | 28.001     | 61.73 | 46.314  |
| 4 1/4      | 24.866    | 54.82 | 41.128  |             |        |         |            |       |         |
| 4 1/2      | 27.874    | 61.45 | 46.104  |             |        |         | 35.462     | 78.18 | 58.654  |

## FLAT BARS



SIZES AND UNIT WEIGHTS (Approx. Wt. in lbs/ft)

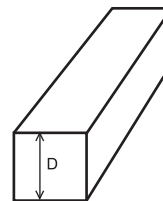
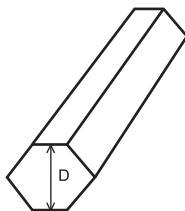
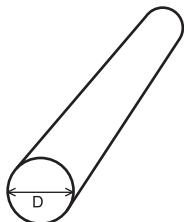
| Width          | Thickness      |               |                |               |                |               |               |               |               |               |       |
|----------------|----------------|---------------|----------------|---------------|----------------|---------------|---------------|---------------|---------------|---------------|-------|
| in             | $\frac{1}{16}$ | $\frac{1}{8}$ | $\frac{3}{16}$ | $\frac{1}{4}$ | $\frac{5}{16}$ | $\frac{3}{8}$ | $\frac{1}{2}$ | $\frac{5}{8}$ | $\frac{3}{4}$ | $\frac{7}{8}$ | 1     |
| $\frac{1}{2}$  | 0.12           | 0.24          | 0.36           | 0.48          | 0.60           | 0.72          |               |               |               |               |       |
| $\frac{5}{8}$  | 0.15           | 0.30          | 0.45           | 0.60          | 0.75           | 0.91          | 1.21          |               |               |               |       |
| $\frac{3}{4}$  | 0.18           | 0.36          | 0.54           | 0.72          | 0.91           | 1.09          | 1.45          | 1.81          |               |               |       |
| $\frac{7}{8}$  |                | 0.42          | 0.63           | 0.85          | 1.06           | 1.27          | 1.69          | 2.11          | 2.54          |               |       |
| 1              | 0.24           | 0.48          | 0.72           | 0.97          | 1.21           | 1.45          | 1.93          | 2.42          | 2.90          | 3.38          |       |
| $1\frac{1}{4}$ |                | 0.60          | 0.91           | 1.21          | 1.51           | 1.81          | 2.42          | 3.02          | 3.62          | 4.23          | 4.83  |
| $1\frac{1}{2}$ | 0.36           | 0.72          | 1.09           | 1.45          | 1.81           | 2.17          | 2.90          | 3.62          | 4.35          | 5.07          | 5.80  |
| $1\frac{3}{4}$ |                | 0.85          | 1.27           | 1.69          | 2.11           | 2.54          | 3.38          | 4.23          | 5.07          | 5.92          | 6.76  |
| 2              |                | 0.97          | 1.45           | 1.93          | 2.42           | 2.90          | 3.86          | 4.83          | 5.80          | 6.76          | 7.73  |
| $2\frac{1}{4}$ |                | 1.09          | 1.63           | 2.17          |                | 3.26          | 4.35          | 5.43          | 6.52          |               | 8.69  |
| $2\frac{1}{2}$ |                | 1.21          | 1.81           | 2.42          | 3.02           | 3.62          | 4.83          | 6.04          | 7.25          | 8.45          | 9.66  |
| $2\frac{3}{4}$ |                | 1.33          |                | 2.66          |                |               |               |               |               |               |       |
| 3              |                | 1.45          | 2.17           | 2.90          | 3.62           | 4.35          | 5.80          | 7.25          | 8.69          |               | 11.59 |
| $3\frac{1}{4}$ |                |               |                | 3.14          | 3.92           |               |               |               | 9.42          |               |       |
| $3\frac{1}{2}$ |                | 1.69          | 2.54           | 3.38          |                | 5.07          | 6.76          | 8.45          |               |               |       |
| 4              |                | 1.93          | 2.90           | 3.86          | 4.83           | 5.80          | 7.73          | 9.66          | 11.59         |               | 15.46 |
| $4\frac{1}{2}$ |                |               |                | 4.35          |                |               | 8.69          | 10.87         |               |               |       |
| 5              |                |               | 3.62           | 4.83          |                |               | 9.66          |               |               |               |       |
| 6              |                |               |                | 5.80          | 7.25           | 8.69          | 11.59         |               | 17.39         |               | 23.18 |



# BRASS

## ROUND BARS, HEXAGON BARS & SQUARE BARS

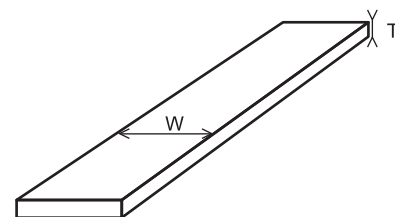
JIS H3250 C3604B/BS2874 CZ121/122



WEIGHTS CHART (Imperial Units)

| Dia.<br>in | Round Bar |        |         | Hexagon Bar |        |         | Square Bar |        |         |
|------------|-----------|--------|---------|-------------|--------|---------|------------|--------|---------|
|            | kg/ft     | lb/ft  | kati/ft | kg/ft       | lb/ft  | kati/ft | kg/ft      | lb/ft  | kati/ft |
| 1/8        | 0.020     | 0.045  | 0.033   | 0.023       | 0.050  | 0.038   | 0.026      | 0.058  | 0.043   |
| 3/16       | 0.046     | 0.102  | 0.076   | 0.051       | 0.112  | 0.084   | 0.059      | 0.130  | 0.098   |
| 1/4        | 0.082     | 0.181  | 0.136   | 0.091       | 0.200  | 0.151   | 0.105      | 0.231  | 0.174   |
| 5/16       | 0.128     | 0.283  | 0.212   | 0.142       | 0.312  | 0.235   | 0.163      | 0.360  | 0.270   |
| 3/8        | 0.185     | 0.407  | 0.306   | 0.204       | 0.449  | 0.337   | 0.235      | 0.519  | 0.389   |
| 7/16       | 0.252     | 0.555  | 0.417   | 0.278       | 0.612  | 0.460   | 0.320      | 0.706  | 0.529   |
| 1/2        | 0.328     | 0.724  | 0.543   | 0.362       | 0.799  | 0.599   | 0.418      | 0.922  | 0.691   |
| 9/16       | 0.416     | 0.917  | 0.688   | 0.459       | 1.011  | 0.759   | 0.529      | 1.167  | 0.875   |
| 5/8        | 0.513     | 1.132  | 0.849   | 0.566       | 1.248  | 0.936   | 0.654      | 1.441  | 1.082   |
| 11/16      | 0.621     | 1.369  | 1.027   | 0.685       | 1.510  | 1.133   | 0.805      | 1.774  | 1.331   |
| 3/4        | 0.739     | 1.630  | 1.222   | 0.815       | 1.797  | 1.348   | 0.941      | 2.075  | 1.556   |
| 13/16      | 0.864     | 1.905  | 1.429   | 0.953       | 2.100  | 1.576   | 1.098      | 2.420  | 1.816   |
| 7/8        | 1.006     | 2.218  | 1.664   | 1.109       | 2.446  | 1.834   | 1.281      | 2.824  | 2.119   |
| 15/16      | 1.150     | 2.535  | 1.902   | 1.266       | 2.790  | 2.094   | 1.465      | 3.230  | 2.423   |
| 1          | 1.314     | 2.897  | 2.173   | 1.449       | 3.195  | 2.397   | 1.673      | 3.689  | 2.767   |
| 1 1/8      | 1.663     | 3.667  | 2.751   | 1.834       | 4.043  | 3.033   | 2.118      | 4.669  | 3.503   |
| 1 1/4      | 2.053     | 4.527  | 3.396   | 2.264       | 4.992  | 3.745   | 2.615      | 5.764  | 4.325   |
| 1 3/8      | 2.485     | 5.478  | 4.110   | 2.740       | 6.040  | 4.032   | 3.163      | 6.974  | 5.232   |
| 1 1/2      | 2.957     | 6.519  | 4.891   | 3.260       | 7.188  | 5.392   | 3.765      | 8.300  | 6.227   |
| 1 5/8      | 3.470     | 7.651  | 5.739   | 3.827       | 8.436  | 6.330   | 4.418      | 9.741  | 7.307   |
| 1 3/4      | 4.025     | 8.873  | 6.657   | 4.438       | 9.784  | 7.340   | 5.126      | 11.300 | 8.478   |
| 1 7/8      | 4.622     | 10.190 | 7.645   | 5.094       | 11.230 | 8.425   | 5.883      | 12.970 | 9.730   |
| 2          | 5.257     | 11.590 | 8.695   | 5.797       | 12.780 | 9.688   | 6.695      | 14.760 | 11.074  |
| 2 1/8      | 5.910     | 13.029 | 9.775   |             |        |         |            |        |         |
| 2 1/4      | 6.654     | 14.670 | 11.006  | 7.335       | 16.170 | 12.132  | 8.473      | 18.680 | 14.014  |
| 2 3/8      | 7.380     | 16.270 | 12.207  |             |        |         |            |        |         |
| 2 1/2      | 8.215     | 18.110 | 13.588  | 9.058       | 19.970 | 14.982  | 10.460     | 23.060 | 17.301  |
| 2 5/8      | 9.010     | 19.870 | 14.903  |             |        |         |            |        |         |
| 2 3/4      | 9.938     | 21.910 | 16.437  | 10.959      | 24.160 | 18.126  | 12.655     | 27.900 | 20.931  |
| 2 7/8      | 10.810    | 23.840 | 17.880  |             |        |         |            |        |         |
| 3          | 11.830    | 26.080 | 19.567  | 13.041      | 28.750 | 21.670  | 15.059     | 33.200 | 24.908  |
| 3 1/4      | 13.820    | 30.460 | 22.858  |             |        |         | 17.595     | 38.790 | 29.102  |
| 3 1/2      | 16.148    | 35.600 | 26.709  |             |        |         | 20.602     | 45.420 | 34.076  |
| 3 3/4      | 18.400    | 40.560 | 30.434  |             |        |         | 23.424     | 51.640 | 38.743  |
| 4          | 20.930    | 46.140 | 34.618  |             |        |         | 26.649     | 58.750 | 44.077  |

## FLAT BARS



SIZES AND UNIT WEIGHTS (Approx. Wt. in lbs/ft)

| Width | Thickness |      |      |      |      |      |       |       |       |      |       |
|-------|-----------|------|------|------|------|------|-------|-------|-------|------|-------|
| in    | 1/16      | 1/8  | 3/16 | 1/4  | 5/16 | 3/8  | 1/2   | 5/8   | 3/4   | 7/8  | 1     |
| 1/2   | 0.11      | 0.23 | 0.34 | 0.46 | 0.57 | 0.69 |       |       |       |      |       |
| 5/8   | 0.14      | 0.29 | 0.43 | 0.57 | 0.71 | 0.86 | 1.14  |       |       |      |       |
| 3/4   | 0.17      | 0.34 | 0.51 | 0.69 | 0.86 | 1.03 | 1.37  | 1.72  |       |      |       |
| 7/8   |           | 0.40 | 0.60 | 0.80 | 1.00 | 1.20 | 1.60  | 2.00  | 2.40  |      |       |
| 1     | 0.23      | 0.46 | 0.69 | 0.92 | 1.14 | 1.37 | 1.83  | 2.29  | 2.75  | 3.20 |       |
| 1 1/4 |           | 0.57 | 0.86 | 1.14 | 1.43 | 1.72 | 2.29  | 2.86  | 3.43  | 4.00 | 4.58  |
| 1 1/2 | 0.34      | 0.69 | 1.03 | 1.37 | 1.72 | 2.06 | 2.75  | 3.43  | 4.12  | 4.80 | 5.49  |
| 1 3/4 |           | 0.80 | 1.20 | 1.60 | 2.00 | 2.40 | 3.20  | 4.00  | 4.80  | 5.60 | 6.41  |
| 2     |           | 0.92 | 1.37 | 1.83 | 2.29 | 2.75 | 3.66  | 4.58  | 5.49  | 6.41 | 7.32  |
| 2 1/4 |           | 1.03 | 1.54 | 2.06 |      | 3.09 | 4.12  | 5.15  | 6.18  |      | 8.24  |
| 2 1/2 |           | 1.14 | 1.72 | 2.29 | 2.86 | 3.43 | 4.58  | 5.72  | 6.86  | 8.01 | 9.15  |
| 2 3/4 |           | 1.26 |      | 2.52 |      |      |       |       |       |      |       |
| 3     |           | 1.37 | 2.06 | 2.75 | 3.43 | 4.12 | 5.49  | 6.86  | 8.24  |      | 10.98 |
| 3 1/4 |           |      |      |      | 3.72 |      | 5.95  |       | 8.92  |      |       |
| 3 1/2 |           | 1.60 | 2.40 | 3.20 |      | 4.80 | 6.41  | 8.01  |       |      |       |
| 4     |           | 1.83 | 2.75 | 3.66 | 4.58 | 5.49 | 7.32  | 9.15  | 10.98 |      | 14.64 |
| 4 1/2 |           |      |      | 4.12 |      |      | 8.24  | 10.29 |       |      |       |
| 5     |           |      | 3.42 | 4.58 |      |      | 9.15  |       |       |      |       |
| 6     |           |      |      | 5.49 | 6.86 | 8.24 | 10.98 |       | 16.47 |      | 21.96 |

# BRASS

## SHEETS

JIS H3100 C2801P 1/2 HARD

### IMPORTED BRASS PLATE FROM JAPAN - SAMBO

| Gauge | Thickness | Width | Length | Weight |
|-------|-----------|-------|--------|--------|
|       | mm        |       |        |        |
| 6 OZ  | 0.22      | 14"   | 48"    | 0.83   |
| 8 OZ  | 0.30      | 14"   | 48"    | 1.13   |
| 10 OZ | 0.35      | 14"   | 48"    | 1.31   |
| 12 OZ | 0.54      | 14"   | 48"    | 1.54   |
| 14 OZ | 0.50      | 14"   | 48"    | 1.88   |
| 16 OZ | 0.55      | 14"   | 48"    | 2.06   |
| 16 G  | 1.50      | 2'    | 4'     | 9.64   |
| 18 G  | 1.20      | 2'    | 4'     | 7.72   |
| 20 G  | 0.90      | 2'    | 4'     | 5.79   |
| 22 G  | 0.70      | 2'    | 4'     | 4.50   |
| 24 G  | 0.50      | 2'    | 4'     | 3.21   |
|       | 1.00      | 2'    | 4'     | 6.43   |
|       | 2.00      | 2'    | 4'     | 12.86  |
|       | 2.30      | 2'    | 4'     | 14.79  |
|       | 3.00      | 2'    | 4'     | 19.29  |
|       | 4.50      | 2'    | 4'     | 30.61  |
|       | 6.00      | 2'    | 4'     | 40.83  |
|       | 9.00      | 2'    | 4'     | 61.24  |
|       | 12.00     | 2'    | 4'     | 81.65  |
|       | 0.50      | 4'    | 8'     | 13.61  |
|       | 0.70      | 4'    | 8'     | 18.00  |
|       | 0.90      | 4'    | 8'     | 23.15  |
|       | 1.20      | 4'    | 8'     | 30.86  |
|       | 1.50      | 4'    | 8'     | 38.85  |
|       | 2.00      | 4'    | 8'     | 51.44  |
|       | 3.00      | 4'    | 8'     | 77.16  |
|       | 4.50      | 4'    | 8'     | 122.43 |
|       | 6.00      | 4'    | 8'     | 163.32 |
|       | 9.00      | 4'    | 8'     | 244.98 |
|       | 12.00     | 4'    | 8'     | 326.64 |

## PRINCIPLE OF CONTINUOUS CASTING

Molten bronze is poured from a furnace into a crucible which is cemented to the neck of a graphite die. The lower half of the die is water cooled, the metal sets in this section and is withdrawn through two sets of rollers. A flying saw mounted below the roller cuts the bar to the required length. Continuous cast rod has greatly improved physical properties compared with normal sand-cast material. Because of the bottom pouring conditions and rapid cooling the metal receives in the cooling zone, inclusions are completely eliminated, grain structure is tight and even giving excellent pressure tightness qualities. Tensile strength and wearing qualities are increased by up to 35%.

## SPECIFICATIONS

Our standard stock bronze is listed below. Other specifications can be supplied on request provided sufficient quantity is ordered.

### Material Specification:

|                   |                     |
|-------------------|---------------------|
| British Standard  | -----BS.1400 LG2-C  |
| Japanese Standard | -----JIS H5111 BC6C |
| American Standard | -----ASTM SAE 40    |
|                   | ASTM B145-52.4A     |
|                   | ASTM 584-85a C83600 |

### Approximate Chemical Composition

-----Copper 85%  
 -----Tin 5%  
 -----Zinc 5%  
 -----Lead 5%

### Typical Mechanical Properties : U.T.S.

-----18 Tons/in<sup>2</sup>  
 -----28 kg/mm<sup>2</sup>  
 -----17 Tons/in<sup>2</sup> BS.1400(min)

### Yield Stress

-----10 Tons/in<sup>2</sup>  
 -----15 kg/mm<sup>2</sup>

### Elongation

-----25%  
 -----15% BS.1400(min)

### Hardness (Brinell No.)

-----80

## ADVANTAGES

Constant research and improved production techniques ensure top quality material with many advantages :-

- ☐ Reduced machining allowance - less waste
- ☐ Reduced tool wear
- ☐ Higher machining speeds
- ☐ Excellent mechanical properties
- ☐ Freedom from gas and porosity
- ☐ Close grain structure provides good pressure resistance

## APPLICATION

General purpose bearings, Automotive and Machine

- Hydraulic Components
- Gears, Worm Wheels
- Valve seats
- Slide valves
- Rock drill parts
- Crusher bearings
- Hose nozzles
- Irrigation equipment
- Rollers
- Washers
- Wear rings

# BRONZE

## STOCK SIZES & WEIGHTS

| Outside Dia.<br>mm(in) | Inside Dia.<br>mm(in) | Solid<br>kg/ft | Weight<br>kg/ft |
|------------------------|-----------------------|----------------|-----------------|
| 22(7/8)                | 12.5(1/2)             | 1.2            | .93             |
| 25.5(1)                | 12.5(1/2)             | 1.54           | 1.28            |
|                        | 16(5/8)               |                | 1.10            |
|                        | 19(3/4)               |                | 0.89            |
| 28.5(1 1/8)            | 12.5(1/2)             | 1.93           | 1.66            |
|                        | 16(5/8)               |                | 1.49            |
|                        | 19(3/4)               |                | 1.27            |
|                        | 22(7/8)               |                | 1.01            |
| 31(1 1/4)              | 12.5(1/2)             | 2.34           | 2.09            |
|                        | 16(5/8)               |                | 1.91            |
|                        | 19(3/4)               |                | 1.74            |
|                        | 22(7/8)               |                | 1.44            |
|                        | 25.5(1)               |                | 1.14            |
| 35(1 3/8)              | 16(5/8)               | 2.82           | 2.39            |
|                        | 19(3/4)               |                | 2.17            |
|                        | 22(7/8)               |                | 1.71            |
|                        | 25.5(1)               |                | 1.61            |
| 38(1 1/2)              | 12.5(1/2)             | 3.34           | 3.07            |
|                        | 19(3/4)               |                | 2.68            |
|                        | 25.5(1)               |                | 2.13            |
|                        | 31.5(1 1/4)           |                | 1.40            |
| 41(1 5/8)              | 25.5(1)               | 3.89           | 2.68            |
|                        | 31.5(1 1/4)           |                | 1.95            |
| 44.5(1 3/4)            | 12.5(1/2)             | 4.49           | 4.23            |
|                        | 19(3/4)               |                | 3.84            |
|                        | 22(7/8)               |                | 3.60            |
|                        | 25.5(1)               |                | 3.29            |
|                        | 31.5(1 1/4)           |                | 2.55            |
|                        | 38(1 1/2)             |                | 1.65            |
| 51(2)                  | 12.5(1/2)             | 5.81           | 5.55            |
|                        | 19(3/4)               |                | 5.17            |
|                        | 25.5(1)               |                | 4.61            |
|                        | 31.5(1 1/4)           |                | 3.88            |
|                        | 38(1 1/2)             |                | 2.98            |
| 57(2 1/4)              | 19(3/4)               | 7.31           | 6.67            |
|                        | 25.5(1)               |                | 6.11            |
|                        | 31.5(1 1/4)           |                | 5.38            |
|                        | 38(1 1/2)             |                | 4.48            |
|                        | 44.5(1 3/4)           |                | 3.40            |
|                        | 51(2)                 |                | 2.16            |
| 63.5(2 1/2)            | 19(3/4)               | 8.98           | 8.34            |
|                        | 25.5(1)               |                | 7.78            |
|                        | 31.5(1 1/4)           |                | 7.05            |
|                        | 38(1 1/2)             |                | 6.14            |
|                        | 44.5(1 3/4)           |                | 5.07            |
|                        | 51(2)                 |                | 3.82            |
| 69.5(2 3/4)            | 25.5(1)               | 10.83          | 9.62            |
|                        | 31.5(1 1/4)           |                | 8.89            |
|                        | 38(1 1/2)             |                | 7.90            |
|                        | 44.5(1 3/4)           |                | 6.91            |
|                        | 51(2)                 |                | 5.67            |
|                        | 57(2 1/4)             |                | 4.25            |
| 76(3)                  | 25.5(1)               | 12.84          | 11.64           |
|                        | 31.5(1 1/4)           |                | 10.90           |
|                        | 38(1 1/2)             |                | 10.00           |
|                        | 44.5(1 3/4)           |                | 8.93            |
|                        | 51(2)                 |                | 7.68            |
|                        | 57(2 1/4)             |                | 6.27            |
|                        | 63.5(2 1/2)           |                | 4.68            |
| 82.5(3 1/4)            | 25.5(1)               | 15.02          | 13.82           |
|                        | 31.5(1 1/4)           |                | 13.09           |
|                        | 38(1 1/2)             |                | 12.19           |

| Outside Dia.<br>mm(in) | Inside Dia.<br>mm(in) | Solid<br>kg/ft | Weight<br>kg/ft |
|------------------------|-----------------------|----------------|-----------------|
| 82.5(3 1/4)            | 44.5(1 3/4)           | 15.02          | 11.11           |
|                        | 51(2)                 |                | 9.87            |
|                        | 57(2 1/4)             |                | 8.45            |
|                        | 63.5(2 1/2)           |                | 6.87            |
|                        | 69.5(2 3/4)           |                | 5.11            |
| 89(3 1/2)              | 25.5(1)               | 17.38          | 16.18           |
|                        | 31.5(1 1/4)           |                | 15.44           |
|                        | 38(1 1/2)             |                | 14.54           |
|                        | 44.5(1 3/4)           |                | 13.47           |
|                        | 51(2)                 |                | 12.22           |
|                        | 57(2 1/4)             |                | 10.81           |
|                        | 63.5(2 1/2)           |                | 9.22            |
|                        | 69.5(2 3/4)           |                | 7.46            |
|                        | 76(3)                 |                | 5.53            |
| 96(3 3/4)              | 31.5(1 1/4)           | 19.91          | 17.97           |
|                        | 51(2)                 |                | 14.76           |
|                        | 63.5(2 1/2)           |                | 11.75           |
|                        | 69.5(2 3/4)           |                | 9.99            |
|                        | 76(3)                 |                | 8.06            |
| 101(4)                 | 25.5(1)               | 22.6           | 21.40           |
|                        | 38(1 1/2)             |                | 19.76           |
|                        | 51(2)                 |                | 17.45           |
|                        | 63.5(2 1/2)           |                | 14.45           |
|                        | 69.5(2 3/4)           |                | 12.69           |
|                        | 76(3)                 |                | 10.76           |
|                        | 82.5(3 1/4)           |                | 9.23            |
|                        | 89(3 1/2)             |                | 6.99            |
| 108(4 1/4)             | 63.5(2 1/2)           | 25.47          | 17.32           |
|                        | 69.5(2 3/4)           |                | 15.56           |
|                        | 76(3)                 |                | 13.63           |
|                        | 82.5(3 1/4)           |                | 12.09           |
| 115(4 1/2)             | 38(1 1/2)             | 28.51          | 26.40           |
|                        | 51(2)                 |                | 23.36           |
|                        | 63.5(2 1/2)           |                | 20.36           |
|                        | 76(3)                 |                | 16.67           |
|                        | 82.5(3 1/4)           |                | 15.13           |
|                        | 89(3 1/2)             |                | 12.90           |
|                        | 96(3 3/4)             |                | 10.50           |
|                        | 101(4)                |                | 7.93            |
| 121(4 3/4)             | 51(2)                 | 31.73          | 26.58           |
|                        | 63.5(2 1/2)           |                | 23.57           |
|                        | 69.5(2 3/4)           |                | 21.81           |
|                        | 76(3)                 |                | 19.88           |
|                        | 89(3 1/2)             |                | 16.12           |
|                        | 101(4)                |                | 11.14           |
| 127(5)                 | 38(1 1/2)             | 35.11          | 33.50           |
|                        | 51(2)                 |                | 29.96           |
|                        | 63.5(2 1/2)           |                | 26.96           |
|                        | 76(3)                 |                | 23.27           |
|                        | 89(3 1/2)             |                | 19.50           |
|                        | 101(4)                |                | 14.53           |
|                        | 108(4 1/4)            |                | 11.78           |
|                        | 115(4 1/2)            |                | 8.89            |
| 139.5(5 1/2)           | 51(2)                 | 42.40          | 37.24           |
|                        | 57(2 1/4)             |                | 35.83           |
|                        | 63.5(2 1/2)           |                | 34.24           |
|                        | 76(3)                 |                | 30.55           |
|                        | 89(3 1/2)             |                | 26.78           |
|                        | 96(3 3/4)             |                | 24.38           |
|                        | 101(4)                |                | 21.81           |
|                        | 108(4 1/4)            |                | 19.07           |
|                        | 115(4 1/2)            |                | 16.15           |

Continuously Cast ☐  
Centrifugally Cast ☐

| Outside Dia.<br>mm(in) | Inside Dia.<br>mm(in) | Solid<br>kg/ft | Weight<br>kg/ft |
|------------------------|-----------------------|----------------|-----------------|
| 152.5(6)               | 51(2)                 | 50.36          | 45.21           |
|                        | 63.5(2½)              |                | 42.21           |
|                        | 76(3)                 |                | 38.52           |
|                        | 89(3½)                |                | 34.75           |
|                        | 101(4)                |                | 29.78           |
|                        | 115(4½)               |                | 24.12           |
|                        | 127(5)                |                | 17.78           |
| 165(6½)                | 139.5(5½)             | 60.19          | 10.75           |
|                        | 76(3)                 |                | 48.35           |
|                        | 89(3½)                |                | 44.57           |
|                        | 101(4)                |                | 39.61           |
|                        | 115(4½)               |                | 33.95           |
| 178(7)                 | 127(5)                | 69.60          | 27.61           |
|                        | 139.5(5½)             |                | 20.58           |
|                        | 76(3)                 |                | 57.78           |
|                        | 101(4)                |                | 49.03           |
|                        | 115(4½)               |                | 43.37           |
| 190(7½)                | 127(5)                | 79.70          | 37.03           |
|                        | 139.5(5½)             |                | 30.00           |
|                        | 152.5(6)              |                | 22.29           |
|                        | 76(3)                 |                | 67.00           |
|                        | 101(4)                |                | 59.15           |
|                        | 127(5)                |                | 47.15           |
| 203(8)                 | 139.5(5½)             | 90.50          | 40.12           |
|                        | 152.5(6)              |                | 32.41           |
|                        | 165(6½)               |                | 24.01           |
|                        | 76(3)                 |                | 78.69           |
|                        | 101(4)                |                | 69.95           |
|                        | 127(5)                |                | 57.95           |
| 228(9)                 | 152.5(6)              | 112.56         | 43.20           |
|                        | 178(7)                |                | 34.80           |
|                        | 203(8)                |                | 25.72           |
|                        | 101(4)                |                | 93.60           |
|                        | 127(5)                |                | 81.60           |
| 254(10)                | 152.5(6)              | 138.82         | 66.86           |
|                        | 178(7)                |                | 49.38           |
|                        | 203(8)                |                | 29.14           |
|                        | 127(5)                |                | 108.00          |
|                        | 152.5(6)              |                | 93.26           |
| 279(11)                | 178(7)                | 167.78         | 75.78           |
|                        | 203(8)                |                | 55.55           |
|                        | 228(9)                |                | 32.58           |
|                        | 127(5)                |                | 138.00          |
|                        | 152.5(6)              |                | 123.00          |
| 304(12)                | 178(7)                | 200.00         | 105.00          |
|                        | 203(8)                |                | 85.00           |
|                        | 228(9)                |                | 60.67           |
|                        | 127(5)                |                | 170.00          |
|                        | 152.5(6)              |                | 155.00          |
| 330(13)                | 178(7)                |                | 137.00          |
|                        | 203(8)                |                | 117.00          |
|                        | 228(9)                |                | 94.00           |
|                        | 254(10)               |                | 68.00           |
|                        | 127(5)                |                | 152.00          |
|                        | 228(9)                |                | 129.00          |
|                        | 254(10)               |                | 103.00          |
|                        | 279(11)               |                | 76.00           |
|                        | 304(12)               |                | 43.00           |
|                        |                       |                |                 |

| Outside Dia.<br>mm(in) | Inside Dia.<br>mm(in) | Solid<br>kg/ft | Weight<br>kg/ft |
|------------------------|-----------------------|----------------|-----------------|
| 355(14)                | 203(8)                |                | 189.00          |
|                        | 228(9)                |                | 166.00          |
|                        | 254(10)               |                | 140.00          |
|                        | 279(11)               |                | 112.00          |
|                        | 304(12)               |                | 81.00           |
|                        | 330(13)               |                | 47.00           |
| 381(15)                | 203(8)                |                | 229.00          |
|                        | 228(9)                |                | 206.00          |
|                        | 254(10)               |                | 181.00          |
|                        | 279(11)               |                | 152.00          |
|                        | 304(12)               |                | 121.00          |
|                        | 330(13)               |                | 87.00           |
| 406(16)                | 355(14)               |                | 50.00           |
|                        | 228(9)                |                | 249.00          |
|                        | 254(10)               |                | 223.00          |
|                        | 279(11)               |                | 195.00          |
|                        | 304(12)               |                | 164.00          |
|                        | 330(13)               |                | 130.00          |
| 432(17)                | 355(14)               |                | 93.00           |
|                        | 381(15)               |                | 54.00           |
|                        | 254(10)               |                | 264.00          |
|                        | 279(11)               |                | 235.00          |
|                        | 304(12)               |                | 203.00          |
|                        | 330(13)               |                | 167.00          |
| 457(18)                | 355(14)               |                | 130.00          |
|                        | 381(15)               |                | 90.00           |
|                        | 406(16)               |                | 56.58           |
|                        | 279(11)               |                | 283.00          |
|                        | 304(12)               |                | 251.00          |
|                        | 330(13)               |                | 216.00          |
| 483(19)                | 355(14)               |                | 179.00          |
|                        | 381(15)               |                | 138.00          |
|                        | 406(16)               |                | 95.00           |
|                        | 432(17)               |                | 49.00           |
|                        | 304(12)               |                | 313.00          |
|                        | 330(13)               |                | 268.00          |
| 508(20)                | 355(14)               |                | 230.00          |
|                        | 381(15)               |                | 190.00          |
|                        | 406(16)               |                | 146.00          |
|                        | 432(17)               |                | 101.00          |
|                        | 457(18)               |                | 51.00           |
|                        | 330(13)               |                | 329.00          |
| 533(21)                | 355(14)               |                | 292.00          |
|                        | 381(15)               |                | 253.00          |
|                        | 406(16)               |                | 210.00          |
|                        | 432(17)               |                | 165.00          |
|                        | 457(18)               |                | 118.00          |
|                        | 355(14)               |                | 348.00          |
| 559(22)                | 381(15)               |                | 309.00          |
|                        | 406(16)               |                | 267.00          |
|                        | 432(17)               |                | 222.00          |
|                        | 457(18)               |                | 174.00          |
|                        | 483(19)               |                | 124.00          |
|                        | 381(15)               |                | 369.00          |
|                        | 406(16)               |                | 326.00          |
|                        | 432(17)               |                | 281.00          |
|                        | 457(18)               |                | 234.00          |
|                        | 483(19)               |                | 183.00          |
|                        | 508(20)               |                | 130.00          |

SOLID  
Machining Allowance O/D

|          |     |         |         |
|----------|-----|---------|---------|
| <152.5mm | <6" | + 1.5mm | + .060" |
| >152.5mm | >6" | + 3.0mm | + .125" |

CORED  
Machining Allowance I/D

|       |     |        |        |
|-------|-----|--------|--------|
| <76mm | <3" | -1.5mm | -.060" |
| >76mm | >3" | -3.0mm | -.125" |





APPENDIX

GAUGE TABLE

264

CONVERSION FACTORS

265





# APPENDIX

## GAUGE TABLE

| S.W.G.<br>B.G.<br>U.S.G. | Standard Wire Gauge<br>Birmingham Gauge<br>US. government Standard Gauge |   |        |      | B.W.G.<br>B.S. | Birmingham (or Stubs Iron) Wire Gauge<br>American Brown and Sharp's Gauge |    |        |        |       |   |        |       |       |
|--------------------------|--------------------------------------------------------------------------|---|--------|------|----------------|---------------------------------------------------------------------------|----|--------|--------|-------|---|--------|-------|-------|
| No.                      | S                                                                        | W | G      | B    | W              | GB                                                                        | G  | B      | S      | U     | S | G      |       |       |
|                          | inch                                                                     |   | mm     |      | inch           |                                                                           | mm | inch   | mm     | inch  |   | mm     | inch  | mm    |
| 0/7                      | .500                                                                     |   | 12.700 | -    | -              |                                                                           |    | .666   | 16.932 | -     | - |        | .5000 | 12.70 |
| 0/6                      | .464                                                                     |   | 11.785 | -    | -              |                                                                           |    | .625   | 15.875 | .580  |   | 14.730 | .4688 | 11.91 |
| 0/5                      | .432                                                                     |   | 10.973 | .500 |                | 12.700                                                                    |    | .588   | 14.943 | .516  |   | 13.120 | .4375 | 11.11 |
| 0/4                      | .400                                                                     |   | 10.160 | .454 |                | 11.532                                                                    |    | .542   | 13.757 | .460  |   | 11.684 | .4063 | 10.32 |
| 0/3                      | .372                                                                     |   | 9.449  | .425 |                | 10.795                                                                    |    | .500   | 12.700 | .410  |   | 10.404 | .3750 | 9.53  |
| 0/2                      | .348                                                                     |   | 8.839  | .380 |                | 9.652                                                                     |    | .445   | 11.308 | .365  |   | 9.266  | .3438 | 8.73  |
| 0                        | .324                                                                     |   | 8.23   | .340 |                | 8.636                                                                     |    | .396   | 10.068 | .325  |   | 8.252  | .3125 | 7.94  |
| 1                        | .300                                                                     |   | 7.620  | .300 |                | 7.620                                                                     |    | .353   | 8.971  | .289  |   | 7.348  | .2813 | 7.15  |
| 2                        | .276                                                                     |   | 7.010  | .294 |                | 7.214                                                                     |    | .315   | 7.993  | .258  |   | 6.543  | .2656 | 6.75  |
| 3                        | .252                                                                     |   | 6.401  | .259 |                | 6.579                                                                     |    | .280   | 7.122  | .229  |   | 5.827  | .2500 | 6.35  |
| 4                        | .232                                                                     |   | 5.893  | .238 |                | 6.045                                                                     |    | .250   | 6.850  | .204  |   | 5.189  | .2344 | 5.95  |
| 5                        | .212                                                                     |   | 5.385  | .220 |                | 5.588                                                                     |    | .223   | 5.651  | .182  |   | 4.620  | .2188 | 5.56  |
| 6                        | .192                                                                     |   | 4.877  | .203 |                | 5.156                                                                     |    | .198   | 5.032  | .162  |   | 4.115  | .2031 | 5.16  |
| 7                        | .176                                                                     |   | 4.470  | .180 |                | 4.572                                                                     |    | .176   | 4.480  | .144  |   | 3.665  | .1875 | 4.76  |
| 8                        | .160                                                                     |   | 4.064  | .165 |                | 4.191                                                                     |    | .157   | 3.988  | .129  |   | 3.264  | .1719 | 4.37  |
| 9                        | .144                                                                     |   | 3.658  | .148 |                | 3.759                                                                     |    | .140   | 3.551  | .114  |   | 2.906  | .1563 | 3.97  |
| 10                       | .128                                                                     |   | 3.251  | .134 |                | 3.404                                                                     |    | .125   | 3.157  | .102  |   | 2.588  | .1406 | 3.57  |
| 11                       | .116                                                                     |   | 2.946  | .120 |                | 3.048                                                                     |    | .111   | 2.827  | .091  |   | 2.304  | .1250 | 3.18  |
| 12                       | .104                                                                     |   | 2.642  | .109 |                | 2.769                                                                     |    | .099   | 2.517  | .081  |   | 2.052  | .1094 | 2.78  |
| 13                       | .092                                                                     |   | 2.337  | .095 |                | 2.413                                                                     |    | .088   | 2.240  | .072  |   | 1.829  | .0938 | 2.38  |
| 14                       | .080                                                                     |   | 2.032  | .083 |                | 2.108                                                                     |    | .079   | 1.994  | .064  |   | 1.628  | .0781 | 1.98  |
| 15                       | .072                                                                     |   | 1.829  | .072 |                | 1.829                                                                     |    | .070   | 1.775  | .057  |   | 1.450  | .0703 | 1.79  |
| 16                       | .064                                                                     |   | 1.626  | .065 |                | 1.651                                                                     |    | .063   | 1.587  | .051  |   | 1.290  | .0625 | 1.59  |
| 17                       | .056                                                                     |   | 1.422  | .058 |                | 1.473                                                                     |    | .056   | 1.412  | .045  |   | 1.151  | .0563 | 1.43  |
| 18                       | .048                                                                     |   | 1.219  | .049 |                | 1.245                                                                     |    | .050   | 1.257  | .040  |   | 1.024  | .0500 | 1.27  |
| 19                       | .0400                                                                    |   | 1.016  | .042 |                | 1.067                                                                     |    | .0440  | 1.118  | .0359 |   | .912   | .0438 | 1.11  |
| 20                       | .0360                                                                    |   | .914   | .035 |                | .889                                                                      |    | .0392  | .996   | .0320 |   | .813   | .0375 | .953  |
| 21                       | .0320                                                                    |   | .813   | .032 |                | .813                                                                      |    | .0349  | .886   | .0285 |   | .724   | .0344 | .873  |
| 22                       | .0280                                                                    |   | .711   | .028 |                | .711                                                                      |    | .0313  | .794   | .0253 |   | .643   | .0313 | .794  |
| 23                       | .0240                                                                    |   | .610   | .025 |                | .635                                                                      |    | .0278  | .707   | .0226 |   | .574   | .0281 | .714  |
| 24                       | .0220                                                                    |   | .559   | .022 |                | .559                                                                      |    | .2248  | .629   | .0201 |   | .511   | .0250 | .635  |
| 25                       | .0200                                                                    |   | .508   | .020 |                | .508                                                                      |    | .0220  | .560   | .0179 |   | .455   | .0219 | .556  |
| 26                       | .0180                                                                    |   | .457   | .018 |                | .457                                                                      |    | .0196  | .498   | .0159 |   | .404   | .0188 | .478  |
| 27                       | .0164                                                                    |   | .417   | .016 |                | .406                                                                      |    | .0175  | .443   | .0142 |   | .361   | .0172 | .437  |
| 28                       | .0148                                                                    |   | .376   | .014 |                | .356                                                                      |    | .0156  | .397   | .0126 |   | .320   | .0156 | .496  |
| 29                       | .0136                                                                    |   | .345   | .013 |                | .330                                                                      |    | .0139  | .353   | .0113 |   | .287   | .0141 | .358  |
| 30                       | .0124                                                                    |   | .315   | .012 |                | .305                                                                      |    | .0123  | .312   | .0100 |   | .254   | .0125 | .318  |
| 31                       | .0116                                                                    |   | .295   | .010 |                | .254                                                                      |    | .0110  | .279   | .0089 |   | .227   | .0109 | .278  |
| 32                       | .0108                                                                    |   | .274   | .009 |                | .228                                                                      |    | .0098  | .249   | .0080 |   | .202   | .0102 | .258  |
| 33                       | .0100                                                                    |   | .254   | .008 |                | .203                                                                      |    | .0087  | .221   | .0071 |   | .180   | .0094 | .238  |
| 34                       | .0092                                                                    |   | .234   | .007 |                | .178                                                                      |    | .0077  | .196   | .0063 |   | .160   | .0086 | .218  |
| 35                       | .0084                                                                    |   | .213   | .005 |                | .127                                                                      |    | .0069  | .175   | .0056 |   | .143   | .0078 | .198  |
| 36                       | .0076                                                                    |   | .193   | .004 |                | .101                                                                      |    | .0061  | .155   | .0050 |   | .127   | .0070 | .179  |
| 37                       | .0068                                                                    |   | .173   | -    |                | -                                                                         |    | .0054  | .137   | .0045 |   | .113   | .0066 | .169  |
| 38                       | .0060                                                                    |   | .152   | -    |                | -                                                                         |    | .0048  | .122   | .0040 |   | .101   | .0063 | .159  |
| 39                       | .0052                                                                    |   | .132   | -    |                | -                                                                         |    | .0043  | .109   | .0035 |   | .090   | .0059 | .149  |
| 40                       | .0048                                                                    |   | .122   | -    |                | -                                                                         |    | .00386 | .098   | .0031 |   | .080   | .0055 | .139  |

## CONVERSION FACTORS

## LINEAR MEASURE

|                 | Millimeter<br>mm | Centimeter<br>cm | Meter<br>m | Inch<br>in | Foot<br>ft | Yard<br>yd | Mile<br>mi            |
|-----------------|------------------|------------------|------------|------------|------------|------------|-----------------------|
| Millimeter (mm) | 1                | 0.1              | 0.001      | 0.03937    | 0.0032808  | 0.0010936  | 0.0 <sub>6</sub> 214  |
| Centimeter (cm) | 10               | 1                | 0.01       | 0.3937     | 0.032808   | 0.010936   | 0.0 <sub>5</sub> 6214 |
| Meter (m)       | 1,000            | 100              | 1          | 39.37      | 3.28083    | 1.0936     | 0.0 <sub>3</sub> 6214 |
| Inch (in)       | 25.4             | 2.54             | 0.0254     | 1          | 0.0833     | 0.02778    | 0.0 <sub>4</sub> 1578 |
| Foot (ft)       | 304.8            | 30.48            | 0.3048     | 12         | 1          | 0.3333     | 0.0 <sub>3</sub> 1894 |
| Yard (yd)       | 914.4            | 91.44            | 0.9144     | 36         | 3          | 1          | 0.0 <sub>3</sub> 5682 |
| Mile (mi)       | 1,609,347        | 160,934.7        | 1,609.35   | 63,360     | 5,280      | 1,760      | 1                     |

## SQUARE MEASURE

|                                      | Square<br>Millimeter<br>mm <sup>2</sup> | Square<br>Centimeter<br>cm <sup>2</sup> | Square<br>Meter<br>m <sup>2</sup> | Square<br>Inch<br>in <sup>2</sup> | Square<br>Foot<br>ft <sup>2</sup> | Square<br>Yard<br>yd <sup>2</sup> |
|--------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| Square millimeter (mm <sup>2</sup> ) | 1                                       | 0.01                                    | 0.0 <sub>1</sub>                  | 0.00155                           | 0.0 <sub>4</sub> 10764            | 0.0 <sub>5</sub> 119599           |
| Square centimeter (cm <sup>2</sup> ) | 100                                     | 1                                       | 0.0001                            | 0.154999                          | 0.0010764                         | 0.0 <sub>3</sub> 119599           |
| Square meter (m <sup>2</sup> )       | 1,000,000                               | 10,000                                  | 1                                 | 1,549.99                          | 10.7639                           | 1.19599                           |
| Square inch (in <sup>2</sup> )       | 645.2                                   | 6.452                                   | 0.0 <sub>6</sub> 452              | 1                                 | 0.006944                          | 0.0 <sub>3</sub> 7616             |
| Square foot (ft <sup>2</sup> )       | 92,900                                  | 929                                     | 0.0929                            | 144                               | 1                                 | 0.11111                           |
| Square yard (yd <sup>2</sup> )       | 836,100                                 | 8,361                                   | 0.8361                            | 1,296                             | 9                                 | 1                                 |

## CUBE MEASURE

|                                     | Cubic<br>Centimeter<br>cm <sup>3</sup> | Cubic<br>Meter<br>m <sup>3</sup> | Cubic<br>Inch<br>in <sup>3</sup> | Cubic<br>Foot<br>ft <sup>3</sup> | Cubic<br>Yard<br>yd <sup>3</sup> |
|-------------------------------------|----------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Cubic centimeter (cm <sup>3</sup> ) | 1                                      | 0.0 <sub>1</sub>                 | 0.06102                          | 0.0 <sub>4</sub> 3531            | 0.0 <sub>5</sub> 1308            |
| Cubic meter (m <sup>3</sup> )       | 1,000,000                              | 1                                | 61,023                           | 35.31                            | 1.308                            |
| Cubic inch (in <sup>3</sup> )       | 16.39                                  | 0.0 <sub>4</sub> 1639            | 1                                | 0.0 <sub>3</sub> 5787            | 0.0 <sub>4</sub> 2143            |
| Cubic foot (ft <sup>3</sup> )       | 28,317                                 | 0.028317                         | 1,728                            | 1                                | 0.03704                          |
| Cubic yard (yd <sup>3</sup> )       | 764,500                                | 0.7645                           | 46,660                           | 27                               | 1                                |

## WEIGHT

|                | Kilogram<br>kg | Ounce<br>oz | Pound<br>lb | Net ton<br>(2,000 lbs)<br>nt | Gross ton<br>(2,240 lbs)<br>gt | Metric ton<br>(1,000 kg)<br>t |
|----------------|----------------|-------------|-------------|------------------------------|--------------------------------|-------------------------------|
| Kilogram (kg)  | 1              | 35.274      | 2.20462     | 0.001102                     | 0.0 <sub>3</sub> 9842          | 0.001                         |
| Ounce (oz)     | 0.02835        | 1           | 0.0625      | 0.0 <sub>4</sub> 3125        | 0.0 <sub>4</sub> 2790          | 0.0 <sub>4</sub> 2835         |
| Pound (lb)     | 0.45359        | 16          | 1           | 0.0005                       | 0.0 <sub>3</sub> 4464          | 0.0 <sub>3</sub> 4536         |
| Net ton (nt)   | 907.185        | 32,000      | 2,000       | 1                            | 0.89286                        | 0.90719                       |
| Gross ton (gt) | 1,016.05       | 35,840      | 2,240       | 1.12                         | 1                              | 1.01605                       |
| Metric ton (t) | 1,000          | 35,274      | 2,204.62    | 1.10231                      | 0.98421                        | 1                             |

Note:

The small subnumeral following a zero indicates that the zero is to be repeated that number of time, thus 0.0<sub>3</sub>4 = 0.0004

# APPENDIX

## WEIGHT PER LINEAR UNIT

|                            | Gram per<br>centimeter<br>g/cm | Kilogram per<br>meter<br>kg/m | Pound per<br>inch<br>lb/in | Pound per<br>foot<br>lb/ft | Pound per<br>yard<br>lb/yd |
|----------------------------|--------------------------------|-------------------------------|----------------------------|----------------------------|----------------------------|
| Gram per centimeter (g/cm) | 1 0.1                          | 0.5056                        | 0.0672                     | 0.20159                    |                            |
| Kilogram per meter (kg/m)  | 10                             | 1                             | 0.056                      | 0.67197                    | 2.0159                     |
| Pound per inch (lb/in)     | 178.5                          | 17.8579                       | 1                          | 12                         | 36                         |
| Pound per foot (lb/ft)     | 148,816                        | 1.48816                       | 0.08333                    | 1                          | 3                          |
| Pound per yard (lb/yd)     | 4.96054                        | 0.49605                       | 0.02778                    | 0.3333                     | 1                          |

## WEIGHT PER UNIT AREA

|                                                      | Kilogram per<br>square centimeter<br>kg/cm <sup>2</sup> | Kilogram per<br>square meter<br>kg/m <sup>2</sup> | Metric ton per<br>square meter<br>t/m <sup>2</sup> |
|------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------|----------------------------------------------------|
| Kilogram per square centimeter (kg/cm <sup>2</sup> ) | 1                                                       | 10,000                                            | 10                                                 |
| Kilogram per square meter (kg/m <sup>2</sup> )       | 0.0001                                                  | 1                                                 | 0.001                                              |
| Metric ton per square meter (t/m <sup>2</sup> )      | 0.1                                                     | 1,000                                             | 1                                                  |
| Pound per square inch (lb/in <sup>2</sup> )          | 0.0703067                                               | 703.06686                                         | 0.7031                                             |
| Pound per square foot (lb/ft <sup>2</sup> )          | 0.0004882                                               | 4.8824087                                         | 0.004882                                           |

## WEIGHT PER UNIT VOLUME

|                                                     | Kilogram per<br>cubic centimeter<br>kg/cm <sup>3</sup> | Kilogram per<br>cubic meter<br>kg/m <sup>3</sup> | Metric ton per<br>cubic meter<br>t/m <sup>3</sup> |
|-----------------------------------------------------|--------------------------------------------------------|--------------------------------------------------|---------------------------------------------------|
| Kilogram per cubic centimeter (kg/cm <sup>3</sup> ) | 1                                                      | 1,000,000                                        | 1,000                                             |
| Kilogram per cubic meter (kg/m <sup>3</sup> )       | 0.000001                                               | 1                                                | 0.001                                             |
| Metric ton per cubic meter (t/m <sup>3</sup> )      | 0.001                                                  | 1,000                                            | 1                                                 |
| Pound per cubic inch (lb/in <sup>3</sup> )          | 0.02768                                                | 27,680.4                                         | 27.6804                                           |
| Pound per cubic foot (lb/ft <sup>3</sup> )          | 0.000016                                               | 16.0196                                          | 0.01602                                           |

## ENERGY

|                     | B.t.u.                | Calories              | Ft-lb      | Kg-m      | Hp-hr                 | Kw-hr                 | Joules      |
|---------------------|-----------------------|-----------------------|------------|-----------|-----------------------|-----------------------|-------------|
| B.t.u. (mean)       | 1                     | 0.252                 | 778        | 107.563   | 0.0 <sub>3</sub> 3929 | 0.0 <sub>3</sub> 2931 | 1,054.8     |
| Calories (mean)     | 3.968                 | 1                     | 3,091.36   | 426.84    | 0.001559              | 0.001163              | 4,185       |
| Ft-lb               | 0.001285              | 0.0 <sub>3</sub> 3239 | 1          | 0.1383    | 0.0 <sub>6</sub> 505  | 0.0 <sub>6</sub> 3767 | 1.355       |
| Ft-ton              | 2.571                 | 0.6478                | 2,000      | 276.511   | 0.00101               | 0.0 <sub>3</sub> 7535 | 2,712.59    |
| Kg-m                | 0.009297              | 0.002343              | 7.23301    | --        | 0.0 <sub>5</sub> 3653 | 0.0 <sub>5</sub> 2725 | 9.806       |
| Hp-hr               | 2,544.99              | 641.327               | 1,980,000  | 273,747   | 1                     | 0.746                 | 2,685,600   |
| Kw-hr               | 3,411.57              | 859.702               | 2,654,200  | 366,959   | 1.34041               | 1                     | 3,600,000   |
| Joules (absolute)   | 0.0 <sub>3</sub> 9477 | 0.0 <sub>3</sub> 2389 | 0.73735    | 0.101937  | 0.0 <sub>6</sub> 3725 | 0.0 <sub>6</sub> 2778 | 1           |
| Lb C                | 14,544                | 3,665                 | 11,315,000 | 1,564,396 | 5.714                 | 4.263                 | 153,470,000 |
| Lb H <sub>2</sub> O | 970.4                 | 244.537               | 754,971    | 104,379   | 0.38127               | 0.284424              | 1,023,966   |

Note.  
The small subnumeral following a zero indicates that the zero is to be repeated that number of time, thus 0.0<sub>3</sub>4 = 0.0004.